Linear Algebra

MIT 18.06SC · after Gilbert Strang's lectures, MIT OpenCourseWare

Study notes

"The best way to learn is to do; the worst way to teach is to talk."

— Paul Halmos

October 2026

Contents

I $A\boldsymbol{x} = \boldsymbol{b}$ and the four subspaces **5**

1 The geometry of linear equations **5**

2 An overview of key ideas **10**

3 Elimination with matrices **16**

4 Multiplication and inverse matrices **22**

5 Factorization into $A = LU$ **28**

6 Transposes, permutations, vector spaces **33**

7 Column space and nullspace **39**

8 Solving $A\boldsymbol{x} = \boldsymbol{0}$: pivot variables, special solutions **44**

9 Solving $A\boldsymbol{x} = \boldsymbol{b}$: row reduced form R **50**

10 Independence, basis and dimension **55**

11 The four fundamental subspaces **60**

12 Matrix spaces; rank one; small world graphs **64**

13 Graphs, networks, incidence matrices **70**

14 Exam 1 review **75**

15 Exam 1 **80**

II Least squares, determinants and eigenvalues **85**

16 Orthogonal vectors and subspaces **85**

17 Projections onto subspaces **90**

18 Projection matrices and least squares **95**

19 Orthogonal matrices and Gram–Schmidt **100**

20 Properties of determinants **105**

21 Determinant formulas and cofactors **109**

22 Cramer's rule, inverse matrix and volume **114**

23 Eigenvalues and eigenvectors **120**

24 Diagonalization and powers of A **125**

25 Differential equations and e^{At} **131**

26 Markov matrices; Fourier series **137**

27 Exam 2 review **142**

28 Exam 2 **147**

III Positive definite matrices and applications **150**

29 Symmetric matrices and positive definiteness **150**

30 Complex matrices; fast Fourier transform (FFT) **157**

31 Positive definite matrices and minima **163**

32 Similar matrices and Jordan form **168**

33 Singular value decomposition **174**

34 Linear transformations and their matrices **179**

35 Change of basis; image compression **184**

36 Left and right inverses; pseudoinverse **189**

37 Exam 3 review **196**

38 Exam 3 **201**

IV Review **205**

39 Final course review **206**

40 Final exam **210**

41 Final 18.06 lecture (2023) **216**

A Formula sheet **219**

About these notes

These notes follow MIT 18.06SC (Fall 2011), Gilbert Strang's first course in linear algebra, session by session. Part I solves $A\boldsymbol{x} = \boldsymbol{b}$ and builds the four fundamental subspaces; Part II adds orthogonality, least squares, determinants and eigenvalues; Part III turns to symmetric and positive definite matrices, the SVD and linear transformations. High-school algebra is the only prerequisite. Each section ends with exercises and full solutions, and the exam sections collect solved practice problems.

Notation

Symbol	Meaning
A, $m \times n$	a matrix with m rows and n columns
$\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{b} \in \mathbb{R}^m$	the unknown vector and the right-hand side in $A\boldsymbol{x} = \boldsymbol{b}$
$\boldsymbol{a}_j$	column j of A
A^{T}	the transpose: $(A^{\mathsf{T}})_{ij} = A_{ji}$
A^{-1}	the inverse of a square invertible matrix: $A^{-1}A = I$
I	the identity matrix
E, P, L, U	elimination, permutation, lower and upper triangular matrices ($A = LU$, $PA = LU$)
R	the reduced row echelon form of A
$r = \operatorname{rank} A$	the rank: the number of pivots
$\mathbf{C}(A)$, $\mathbf{C}(A^{\mathsf{T}})$	the column space and the row space
$\mathbf{N}(A)$, $\mathbf{N}(A^{\mathsf{T}})$	the nullspace and the left nullspace
$\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$	all linear combinations of the vectors
$\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{v}$	dot product; $\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v}^{\mathsf{T}}\boldsymbol{v}}$ is length
P, $\hat{\boldsymbol{x}}$, $\boldsymbol{e}$	projection matrix, least-squares solution, error $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$
Q	a matrix with orthonormal columns ($Q^{\mathsf{T}}Q = I$)
$\det A$	the determinant (signed volume factor)
λ, $\boldsymbol{x}$	an eigenvalue and eigenvector: $A\boldsymbol{x} = \lambda\boldsymbol{x}$
X, Λ	eigenvector matrix (the 2011 lectures write S) and diagonal eigenvalue matrix: $A = X\Lambda X^{-1}$
$\operatorname{tr} A$	the trace: the sum of the diagonal entries (= sum of eigenvalues)
σ_i; U, Σ, V	singular values and the SVD $A = U\Sigma V^{\mathsf{T}}$
A^{+}	the pseudoinverse

Part I

$A\boldsymbol{x} = \boldsymbol{b}$ and the four subspaces

1 The geometry of linear equations

The central problem of linear algebra is to solve a system of linear equations, and the course starts with the friendliest case: n equations in n unknowns. This section looks at one small system in three ways. The *row picture* draws one equation at a time and looks for the point where the graphs meet. The *column picture* reads the same system as a question about combining vectors. The *matrix form* packs everything into the single equation $A\boldsymbol{x} = \boldsymbol{b}$. The column picture is the one that survives in ten or a thousand dimensions, and it leads directly to the question that drives the first part of the course: for which right-hand sides $\boldsymbol{b}$ can the system be solved at all?

1.1 One system, three readings

Our first example is the system

$$\begin{aligned} 2x - y &= 0, \\ -x + 2y &= 3. \end{aligned} \tag{1}$$

All the information sits in the numbers multiplying the unknowns and in the numbers on the right, so we collect them.

Definition 1.1 (Matrix, coefficient matrix, $A\boldsymbol{x} = \boldsymbol{b}$)**.** A *matrix* is a rectangular array of numbers. An $m \times n$ matrix has m rows and n columns, and a_{ij} denotes the entry in row i and column j. For a system of m linear equations in n unknowns $x_1, \ldots, x_n$, the *coefficient matrix* A holds the coefficient of x_j in equation i as its entry a_{ij}; the unknowns form the vector $\boldsymbol{x} = (x_1, \ldots, x_n)$ and the right-hand sides form the vector $\boldsymbol{b} = (b_1, \ldots, b_m)$. The system is written $A\boldsymbol{x} = \boldsymbol{b}$.

Vectors are written as columns; to save space we also write them as lists in parentheses, so $\boldsymbol{b} = (0, 3)$ means the column with entries 0 and 3. For the system (1),

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \qquad \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

At this point $A\boldsymbol{x}$ is shorthand for "the left-hand sides". In Section 1.4 we make it a genuine product of a matrix and a vector.

1.2 The row picture: lines and planes that meet

In the row picture each equation is drawn separately in the xy-plane. All points satisfying $2x - y = 0$ lie on a straight line (the word *linear* comes from *line*). It passes through the origin, because $(0, 0)$ satisfies it, and through $(1, 2)$ and $(\frac{1}{2}, 1)$. The second line $-x + 2y = 3$ does *not* pass through the origin, since $x = y = 0$ gives $0 \neq 3$; two of its points are $(-3, 0)$ and $(-1, 1)$. A solution of the system must lie on both lines, so it is their crossing point. The left panel of Figure 1 shows the lines meeting at $(x, y) = (1, 2)$, and substituting confirms it: $2 \cdot 1 - 2 = 0$ and $-1 + 2 \cdot 2 = 3$.

Two lines in a plane can do three things. They can cross at one point (one solution), they can be parallel and distinct (no solution), or they can coincide (infinitely many solutions). The nice case

n equations, n unknowns, one solution, is the first; the other two are the singular cases that the course will learn to detect.

In three unknowns each equation describes a *plane* in xyz-space. Take

$$\begin{aligned} 2x - y \quad\quad &= 0, \\ -x + 2y - z &= -1, \\ -3y + 4z &= 4. \end{aligned} \tag{2}$$

The second equation is not satisfied by the origin; it contains $(1,0,0)$, $(0,0,1)$ and $(0,-\frac{1}{2},0)$, and those three points determine its plane. The first equation is another plane (it does not involve z, so it is "vertical"). Two non-parallel planes meet in a line, and the third plane cuts that line in a single point, the solution. The picture is already hard to draw for three planes, and in four dimensions it is hopeless. We need a better view.

1.3 The column picture: combinations of columns

Read the system (1) a column at a time. The unknown x multiplies the first column of A and y multiplies the second:

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}. \tag{3}$$

Vectors are added component by component and multiplied by a number (a *scalar*) in every component, so (3) is exactly the two equations again. Now the question is different: which amounts of the two column vectors add up to the vector $\boldsymbol{b}$?

Definition 1.2 (Linear combination)**.** Given vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ in $\mathbb{R}^m$ and scalars $c_1, \ldots, c_n$, the vector

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n$$

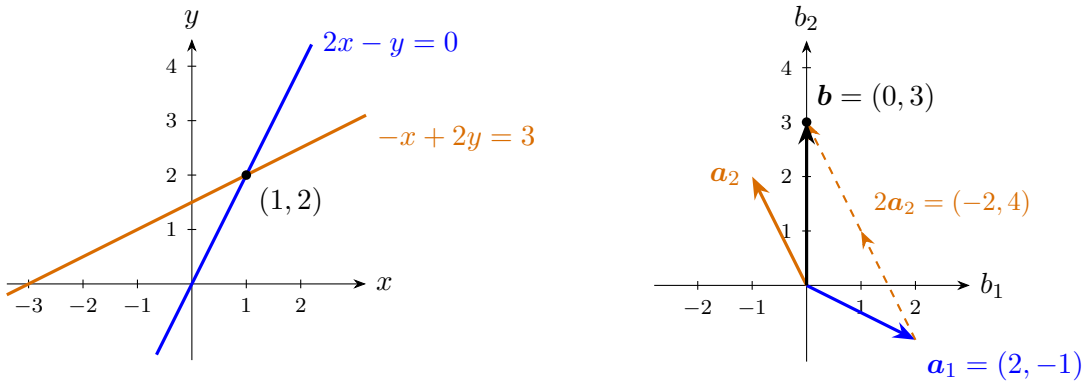
is a *linear combination* of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$. The scalars c_j are its *weights* (or coefficients).

Taking linear combinations is the basic operation of the whole subject. The column picture says: solving $A\boldsymbol{x} = \boldsymbol{b}$ means finding the linear combination of the columns of A that produces $\boldsymbol{b}$, and the weights in that combination are the unknowns.

Example 1.3 (Two pictures of one system)**.** Solve (1) in both pictures. The columns of A are $\boldsymbol{a}_1 = (2,-1)$ and $\boldsymbol{a}_2 = (-1,2)$, and the target is $\boldsymbol{b} = (0,3)$.

1. *Row picture.* The lines $y = 2x$ and $-x + 2y = 3$ cross where $-x + 2(2x) = 3$, that is $3x = 3$. So $x = 1$, $y = 2$, and indeed $-1 + 2 \cdot 2 = 3$.
2. *Column picture.* Take $x = 1$ copy of $\boldsymbol{a}_1$: the arrow from the origin to $(2,-1)$.
3. Take $y = 2$ copies of $\boldsymbol{a}_2$: $2(-1,2) = (-2,4)$.
4. Add tip to tail: $(2,-1) + (-2,4) = (2-2,\ -1+4) = (0,3)$, which is exactly $\boldsymbol{b}$.

The same two numbers $x = 1$, $y = 2$ are the crossing point of the rows and the recipe $1\,\boldsymbol{a}_1 + 2\,\boldsymbol{a}_2 = \boldsymbol{b}$ for the columns. Figure 1 shows both.



row picture: (x, y)-plane column picture: $1\,\boldsymbol{a}_1 + 2\,\boldsymbol{a}_2 = \boldsymbol{b}$

Figure 1: The system $2x - y = 0$, $-x + 2y = 3$ two ways. Left: each equation is a line and the solution is the crossing point $(1, 2)$. Right: the columns $\boldsymbol{a}_1, \boldsymbol{a}_2$ are arrows in the plane of right-hand sides; one $\boldsymbol{a}_1$ followed by two copies of $\boldsymbol{a}_2$ (dashed, tip to tail) lands on $\boldsymbol{b} = (0, 3)$. The weights $1, 2$ are the coordinates of the crossing point.

The two panels live in different planes. In the row picture the axes are the unknowns x and y, and the answer is a point. In the column picture the axes are the components b_1, b_2 of the right-hand side (the first and second equations), and the answer is a pair of weights.

What do *all* combinations $x\boldsymbol{a}_1 + y\boldsymbol{a}_2$ give? Multiples of $\boldsymbol{a}_1$ fill a line through the origin, multiples of $\boldsymbol{a}_2$ fill another line, and because the two lines point in different directions, adding a point of one to a point of the other reaches every point of the plane. So for this A the system can be solved for *every* right-hand side $\boldsymbol{b}$ (Exercise 1.12 finds the weights explicitly).

The column picture of the 3×3 system (2) is

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

This right-hand side was chosen to equal the third column, so the combination is visible at once: $x = 0$, $y = 0$, $z = 1$. That is the point where the three planes of the row picture meet, found without drawing a single plane. If we change the right-hand side to $\boldsymbol{b} = (1, 1, -3)$, which is column 1 plus column 2, the solution becomes $(x, y, z) = (1, 1, 0)$. In the row picture all three planes moved; in the column picture the three columns stayed put and only the target changed. Usually the combination is not visible by eye, and the systematic method of finding it, elimination, is the subject of Section 3.

1.4 Multiplying a matrix by a vector

The column picture tells us what $A\boldsymbol{x}$ should mean.

Definition 1.4 (Matrix times vector)**.** Let A be an $m \times n$ matrix with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ (each in $\mathbb{R}^m$), and let $\boldsymbol{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$. The product $A\boldsymbol{x}$ is the combination of the columns with weights from $\boldsymbol{x}$:

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n \in \mathbb{R}^m.$$

The product is defined only when the number of columns of A equals the number of components of $\boldsymbol{x}$. The familiar "row times vector" rule is the same computation organised differently.

Proposition 1.5 (Row form of $A\boldsymbol{x}$)**.** *For A and $\boldsymbol{x}$ as in Definition 1.4, component i of $A\boldsymbol{x}$ is the dot product of row i of A with $\boldsymbol{x}$:*

$$(A\boldsymbol{x})_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{j=1}^{n} a_{ij}x_j.$$

Proof. Component i of the vector $x_j\boldsymbol{a}_j$ is x_j times component i of $\boldsymbol{a}_j$, which is $a_{ij}x_j$. Components of a sum are sums of components, so component i of $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$ is $a_{i1}x_1 + \cdots + a_{in}x_n$, and this is row i of A dotted with $\boldsymbol{x}$. □

So the row picture corresponds to computing $A\boldsymbol{x}$ by rows (each equation is one dot product), and the column picture corresponds to computing $A\boldsymbol{x}$ by columns. With Definition 1.4, the shorthand $A\boldsymbol{x} = \boldsymbol{b}$ for a linear system is now a true equation between vectors.

Example 1.6 (Both ways)**.** Compute $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. By columns, take one of the first column and two of the second:

$$1\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+10 \\ 1+6 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

By rows, $(2,5)\cdot(1,2) = 2+10 = 12$ and $(1,3)\cdot(1,2) = 1+6 = 7$. The same four multiplications happen in a different order. In particular, if the right-hand side were $\boldsymbol{b} = (12,7)$, the solution of $A\boldsymbol{x} = \boldsymbol{b}$ would be $\boldsymbol{x} = (1,2)$.

1.5 Can every right-hand side be reached?

For a given square matrix A, can we solve $A\boldsymbol{x} = \boldsymbol{b}$ for *every* $\boldsymbol{b}$? In the language of the column picture the question becomes: do the linear combinations of the columns fill the whole space $\mathbb{R}^n$? For the 2×2 example the answer was yes, and for the 3×3 matrix of (2) it is also yes. Such matrices are called *nonsingular* or *invertible*; Section 4 defines the inverse matrix that undoes A. When the answer is no, A is *singular*.

Here is how the answer can fail in three dimensions. If the three columns happen to lie in one plane through the origin, every combination of them lies in that plane too, and no right-hand side off the plane can be produced.

Proposition 1.7 (A dependent column adds nothing)**.** *Let A have columns $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ in $\mathbb{R}^3$ and suppose $\boldsymbol{a}_3 = c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2$. Then every vector $A\boldsymbol{x}$ is a combination of $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ alone.*

Proof. For any $\boldsymbol{x} = (x_1, x_2, x_3)$,

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + x_3(c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2) = (x_1 + c_1x_3)\,\boldsymbol{a}_1 + (x_2 + c_2x_3)\,\boldsymbol{a}_2.$$ □

The combinations of two vectors in $\mathbb{R}^3$ fill at most a plane through the origin, so a matrix with a dependent column is singular: some $\boldsymbol{b}$ are out of reach. (That two vectors cannot fill $\mathbb{R}^3$ is intuitively clear; the precise statement, that a space of dimension 3 needs three independent vectors, is proved in Section 10.)

Example 1.8 (Three columns in a plane)**.** Let $\boldsymbol{w}_1 = (1, 2, 3)$, $\boldsymbol{w}_2 = (4, 5, 6)$, $\boldsymbol{w}_3 = (7, 8, 9)$. Find a combination $x_1\boldsymbol{w}_1 + x_2\boldsymbol{w}_2 + x_3\boldsymbol{w}_3$ that gives the zero vector, and a vector $\boldsymbol{b}$ that is not a combination of them.

The requirement is three equations:

$$x_1 + 4x_2 + 7x_3 = 0, \qquad 2x_1 + 5x_2 + 8x_3 = 0, \qquad 3x_1 + 6x_2 + 9x_3 = 0.$$

Subtract twice the first equation from the second: $-3x_2 - 6x_3 = 0$. Subtract three times the first from the third: $-6x_2 - 12x_3 = 0$, which is twice the previous equation and so tells us nothing new. Choose $x_3 = 1$; then $x_2 = -2$ and $x_1 = -4x_2 - 7x_3 = 8 - 7 = 1$. Hence

$$\boldsymbol{w}_1 - 2\boldsymbol{w}_2 + \boldsymbol{w}_3 = (1 - 8 + 7,\ 2 - 10 + 8,\ 3 - 12 + 9) = (0, 0, 0).$$

The vectors are *dependent*: $\boldsymbol{w}_2 = \frac{1}{2}(\boldsymbol{w}_1 + \boldsymbol{w}_3)$ lies in the plane of $\boldsymbol{w}_1$ and $\boldsymbol{w}_3$, so all three lie in one plane through the origin. Which plane? Each $\boldsymbol{w}_j = (w_1, w_2, w_3)$ satisfies $w_1 - 2w_2 + w_3 = 0$ ($1 - 4 + 3 = 0$, $4 - 10 + 6 = 0$, $7 - 16 + 9 = 0$), and that equation is preserved by adding vectors and multiplying by scalars. So every combination $\boldsymbol{b}$ satisfies $b_1 - 2b_2 + b_3 = 0$. The vector $\boldsymbol{b} = (1, 0, 0)$ gives $1 \neq 0$, so $W\boldsymbol{x} = (1, 0, 0)$ has no solution for the matrix W with columns $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$, and W is not invertible.

The same thinking works in any dimension, even where nothing can be drawn. Nine equations in nine unknowns give nine columns, each a vector in $\mathbb{R}^9$. A matrix with random entries almost certainly has columns whose combinations fill all of $\mathbb{R}^9$. But if, say, the ninth column equals the eighth, it contributes nothing new, and the combinations fill only an eight-dimensional "plane" through the origin inside $\mathbb{R}^9$. Learning to think about such flat pieces of high-dimensional space is the heart of the course.

Common trap. Do not mix up the two pictures. In the row picture the point $(1, 2)$ is a *location* in the plane of unknowns; in the column picture the same numbers are *weights*, and the arrows live in the plane of right-hand sides, where the target is $\boldsymbol{b} = (0, 3)$, not $(1, 2)$. Likewise, $A\boldsymbol{x}$ is not only "rows dotted with $\boldsymbol{x}$": it is a combination of the columns of A, and that is the view that keeps working when no lines or planes can be drawn.

1.6 Exercises

Exercise 1.9. Compute $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ by rows, and again as a combination of columns.

Solution. By rows: $1{\cdot}3 + 2{\cdot}(-2) + 0{\cdot}1 = -1$, then $2{\cdot}3 + 0{\cdot}(-2) + 3{\cdot}1 = 9$, then $4{\cdot}3 + 1{\cdot}(-2) + 1{\cdot}1 = 11$. By columns:

$$3\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 - 4 + 0 \\ 6 - 0 + 3 \\ 12 - 2 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 9 \\ 11 \end{bmatrix}. \qquad \square$$

Exercise 1.10. True or false: a 3×2 matrix A times a 2×3 matrix B is a 3×3 matrix AB. State the general rule for the shapes. (Think of AB as A multiplying each column of B; Section 4 develops this.)

Solution. True. Each column of B has 2 components, which matches the 2 columns of A, so A can multiply it and produces a vector with 3 components. B has 3 columns, so AB has 3 columns of length 3. In general an $m \times n$ matrix times an $n \times p$ matrix is $m \times p$: the inner dimensions must agree and the outer ones give the shape. □

Exercise 1.11. Solve $2x + y = 3$, $x - 2y = -1$. Sketch the row picture (find two points on each line) and the column picture, and write the system in matrix form.

Solution. From the second equation $x = 2y-1$; substituting, $2(2y-1)+y = 3$, so $5y = 5$, $y = 1$ and $x = 1$. Row picture: $2x + y = 3$ passes through $(0, 3)$ and $(1, 1)$; $x - 2y = -1$ passes through $(0, \frac{1}{2})$ and $(1, 1)$; the lines cross at $(1, 1)$. Column picture: the columns are $\boldsymbol{v}_1 = (2, 1)$ and $\boldsymbol{v}_2 = (1, -2)$, and one of each gives the diagonal of the parallelogram they span, $(2 + 1,\ 1 - 2) = (3, -1) = \boldsymbol{b}$. Matrix form: $\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. □

Exercise 1.12. With the columns $\boldsymbol{a}_1 = (2, -1)$ and $\boldsymbol{a}_2 = (-1, 2)$ of Example 1.3, which combination reaches $\boldsymbol{b} = (1, 1)$? Then find the weights x, y with $x\boldsymbol{a}_1 + y\boldsymbol{a}_2 = (b_1, b_2)$ for an arbitrary right-hand side, confirming that every $\boldsymbol{b}$ is reachable.

Solution. $\boldsymbol{a}_1 + \boldsymbol{a}_2 = (2 - 1,\ -1 + 2) = (1, 1)$, so $x = y = 1$. In general we need $2x - y = b_1$ and $-x + 2y = b_2$. Adding twice the first equation to the second gives $3x = 2b_1 + b_2$; adding the first to twice the second gives $3y = b_1 + 2b_2$. So

$$x = \frac{2b_1 + b_2}{3}, \qquad y = \frac{b_1 + 2b_2}{3}.$$

Check: $2x - y = \frac{4b_1+2b_2-b_1-2b_2}{3} = b_1$ and $-x + 2y = \frac{-2b_1-b_2+2b_1+4b_2}{3} = b_2$. Weights exist for every $\boldsymbol{b}$, so the columns fill the plane. For $\boldsymbol{b} = (1, 1)$ the formula gives $x = y = 1$ again. □

Exercise 1.13. The matrix A has columns $(1, 0, 1)$, $(0, 1, 1)$ and $(1, 1, c)$. For $c = 2$, show that the columns lie in a plane, find an equation of that plane, and give a right-hand side $\boldsymbol{b}$ for which $A\boldsymbol{x} = \boldsymbol{b}$ has no solution. For $c = 3$, solve $A\boldsymbol{x} = (0, 0, 1)$.

Solution. For $c = 2$ the third column is $(1, 0, 1) + (0, 1, 1)$, so by Proposition 1.7 every $A\boldsymbol{x}$ is a combination $s(1, 0, 1) + t(0, 1, 1) = (s,\ t,\ s + t)$. These vectors satisfy $b_3 = b_1 + b_2$, the equation of the plane. The vector $\boldsymbol{b} = (0, 0, 1)$ has $b_3 = 1 \neq 0 = b_1 + b_2$, so it is unreachable.

For $c = 3$ we need $x_1 + x_3 = 0$, $x_2 + x_3 = 0$ and $x_1 + x_2 + 3x_3 = 1$. The first two give $x_1 = x_2 = -x_3$, and the third becomes $-x_3 - x_3 + 3x_3 = x_3 = 1$. So $\boldsymbol{x} = (-1, -1, 1)$, and indeed $-(1, 0, 1) - (0, 1, 1) + (1, 1, 3) = (0, 0, 1)$. The column $(1, 1, 3)$ is off the plane $b_3 = b_1 + b_2$, which is why the target off that plane can now be reached. □

2 An overview of key ideas

Before the course settles into elimination, it is worth seeing the whole road at once. Linear algebra moves from *vectors* to *matrices* to *subspaces*: what you do with vectors is take combinations; a matrix packages those combinations as a single operation $\boldsymbol{x} \mapsto A\boldsymbol{x}$; and the set of all combinations is a subspace. This section, adapted from Strang's opening review in 18.085, follows two 3×3 matrices that differ in a single column. One is invertible and its inverse can be written down

by hand; the other has dependent columns, solves $C\boldsymbol{x} = \boldsymbol{b}$ only for special $\boldsymbol{b}$, and shows what a subspace is. Almost every idea of Part I appears here in preview, and later sections make each one precise.

2.1 Vectors: take their combinations

Work in $\mathbb{R}^3$, the space of vectors with three real components, with

$$\boldsymbol{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \qquad \boldsymbol{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The only operations we need are multiplying a vector by a scalar and adding vectors, which together produce linear combinations $x_1\boldsymbol{u} + x_2\boldsymbol{v} + x_3\boldsymbol{w}$ (Definition 1.2). All multiples $c\,\boldsymbol{u}$ fill a line through the origin. All multiples of $\boldsymbol{v}$ fill a second line. Because $\boldsymbol{v}$ is not a multiple of $\boldsymbol{u}$, the combinations $x_1\boldsymbol{u} + x_2\boldsymbol{v}$ fill something bigger than either line: a plane through the origin. Whether adding $\boldsymbol{w}$ fills all of $\mathbb{R}^3$ is the question the rest of the section answers.

2.2 Matrices: the difference matrix and its inverse

Put the three vectors into the columns of a matrix. Then the combination with weights x_1, x_2, x_3 is exactly the matrix-vector product of Definition 1.4:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1\boldsymbol{u} + x_2\boldsymbol{v} + x_3\boldsymbol{w} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix}. \tag{4}$$

The calculation is the same as before but the point of view has changed. Before, numbers x_j multiplied vectors; now the matrix A acts on the vector $\boldsymbol{x}$ and produces an output. This A is a *difference matrix*: each output component is a difference of consecutive inputs (with $x_0 = 0$).

Example 2.1 (Differences of squares)**.** Feed in the squares $\boldsymbol{x} = (1, 4, 9)$:

$$A \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 4-1 \\ 9-4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

Differences of consecutive squares are the odd numbers, since $(k+1)^2 - k^2 = 2k + 1$.

The deeper question runs backwards: given the output $\boldsymbol{b}$, find the input $\boldsymbol{x}$ with $A\boldsymbol{x} = \boldsymbol{b}$. The three equations are

$$x_1 = b_1, \qquad x_2 - x_1 = b_2, \qquad x_3 - x_2 = b_3.$$

Because A is lower triangular they can be solved from the top down: $x_1 = b_1$, then $x_2 = b_1 + b_2$, then $x_3 = b_1 + b_2 + b_3$. Read the answer as a matrix times $\boldsymbol{b}$: the coefficients of b_1 form the first column, and so on.

$$\boldsymbol{x} = \begin{bmatrix} b_1 \\ b_1 + b_2 \\ b_1 + b_2 + b_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = S\boldsymbol{b}. \tag{5}$$

The matrix S takes running sums. It undoes A, and A undoes S:

Proposition 2.2 (The sum matrix inverts the difference matrix)**.** *For every $\boldsymbol{x} \in \mathbb{R}^3$, $S(A\boldsymbol{x}) = \boldsymbol{x}$, and for every $\boldsymbol{b} \in \mathbb{R}^3$, $A(S\boldsymbol{b}) = \boldsymbol{b}$. Consequently $A\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution for every $\boldsymbol{b}$, namely $\boldsymbol{x} = S\boldsymbol{b}$, and $A\boldsymbol{x} = \mathbf{0}$ only for $\boldsymbol{x} = \mathbf{0}$.*

Proof. Running sums of differences telescope:

$$S(A\boldsymbol{x}) = S\begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 + (x_2 - x_1) \\ x_1 + (x_2 - x_1) + (x_3 - x_2) \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Differences of running sums return the original entries: $A(S\boldsymbol{b}) = \big(b_1,\ (b_1 + b_2) - b_1,\ (b_1 + b_2 + b_3) - (b_1 + b_2)\big) = (b_1, b_2, b_3)$. The second identity shows that $\boldsymbol{x} = S\boldsymbol{b}$ solves $A\boldsymbol{x} = \boldsymbol{b}$. If $\boldsymbol{x}$ is any solution, then $\boldsymbol{x} = S(A\boldsymbol{x}) = S\boldsymbol{b}$ by the first identity, so the solution is unique. With $\boldsymbol{b} = \mathbf{0}$ this unique solution is $S\mathbf{0} = \mathbf{0}$. □

A matrix that undoes A in this way is called the *inverse* of A and written A^{-1}; here $A^{-1} = S$. Section 4 defines inverses in general and shows how to compute them. Think of A as a transform taking $\boldsymbol{x}$ to $\boldsymbol{b}$ and of S as the inverse transform taking $\boldsymbol{b}$ back to $\boldsymbol{x}$. For example $S(1, 3, 5) = (1, 1 + 3, 1 + 3 + 5) = (1, 4, 9)$ recovers the squares. Calculus has the same structure: differences are the discrete version of the derivative, running sums the discrete version of the integral, and Proposition 2.2 is a small fundamental theorem of calculus.

2.3 The cyclic difference matrix

Keep $\boldsymbol{u}$ and $\boldsymbol{v}$ but replace $\boldsymbol{w}$ by $\boldsymbol{w}^* = (-1, 0, 1)$. The new matrix

$$C\boldsymbol{x} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} \tag{6}$$

still takes differences, but now cyclically: the first row compares x_1 with x_3, as if the three entries sat on a circle. Only one column changed, yet C behaves completely differently from A.

Zero output from a nonzero input. If $x_1 = x_2 = x_3 = c$, every difference is zero, so $C(c, c, c) = \mathbf{0}$. A whole line of inputs, all multiples of $(1, 1, 1)$, is sent to zero.

Proposition 2.3 (No inverse when a nonzero vector goes to zero)**.** *If $C\boldsymbol{x} = \mathbf{0}$ for some $\boldsymbol{x} \neq \mathbf{0}$, then there is no matrix T with $T(C\boldsymbol{y}) = \boldsymbol{y}$ for all $\boldsymbol{y}$; that is, C has no inverse.*

Proof. Such a T would give $\boldsymbol{x} = T(C\boldsymbol{x}) = T\mathbf{0} = \mathbf{0}$, since any matrix times the zero vector is the zero vector. This contradicts $\boldsymbol{x} \neq \mathbf{0}$. Once C has crushed $\boldsymbol{x}$ to zero, nothing can bring it back. □

Solvable only for special right-hand sides. Write out $C\boldsymbol{x} = \boldsymbol{b}$ and add the three equations. The left sides cancel, $(x_1 - x_3) + (x_2 - x_1) + (x_3 - x_2) = 0$, so the right sides must satisfy $0 = b_1 + b_2 + b_3$. This condition is not only necessary but sufficient.

Proposition 2.4 (When the cyclic system is solvable)**.** *$C\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if $b_1 + b_2 + b_3 = 0$. In that case the solutions are exactly*

$$\boldsymbol{x} = \begin{bmatrix} 0 \\ b_2 \\ b_2 + b_3 \end{bmatrix} + c\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad c \in \mathbb{R}.$$

Proof. Necessity was shown above by adding the equations. Conversely, suppose $b_1 + b_2 + b_3 = 0$ and let $\boldsymbol{x}_p = (0,\, b_2,\, b_2 + b_3)$. Then

$$C\boldsymbol{x}_p = \begin{bmatrix} 0 - (b_2 + b_3) \\ b_2 - 0 \\ (b_2 + b_3) - b_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

using $-(b_2 + b_3) = b_1$. Adding $c(1,1,1)$ does not change $C\boldsymbol{x}$, because $C(c,c,c) = \mathbf{0}$. Finally, if $\boldsymbol{x}$ is any solution, then $\boldsymbol{y} = \boldsymbol{x} - \boldsymbol{x}_p$ satisfies $C\boldsymbol{y} = \boldsymbol{b} - \boldsymbol{b} = \mathbf{0}$, so $y_2 - y_1 = 0$ and $y_3 - y_2 = 0$; hence $\boldsymbol{y} = c(1,1,1)$ for some c, and every solution has the stated form. □

In a physical model where x_1, x_2, x_3 are positions of masses joined in a ring by springs and $\boldsymbol{b}$ holds the applied forces, the condition $b_1 + b_2 + b_3 = 0$ says the forces must balance; the free constant c says the whole ring can be shifted without changing any spring.

Example 2.5 (Solving and failing with C)**.** Take $\boldsymbol{b} = (1, 2, -3)$. Its components sum to 0, so the system is solvable, and Proposition 2.4 gives $\boldsymbol{x}_p = (0,\, 2,\, 2 - 3) = (0, 2, -1)$. Check: $C\boldsymbol{x}_p = (0 - (-1),\; 2 - 0,\; -1 - 2) = (1, 2, -3)$. All solutions are $(0, 2, -1) + c(1, 1, 1)$, for instance $(1, 3, 0)$ with $c = 1$. For $\boldsymbol{b} = (1, 1, 1)$ the components sum to $3 \neq 0$, and $C\boldsymbol{x} = \boldsymbol{b}$ has no solution at all.

2.4 Subspaces: what all the combinations fill

Which vectors $\boldsymbol{b}$ are combinations of the columns $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}^*$ of C? Each column has components adding to zero, and that property survives addition and scalar multiplication, so every combination $C\boldsymbol{x}$ lies on the plane $b_1 + b_2 + b_3 = 0$. Proposition 2.4 says that every vector on that plane is reached. So *all vectors* $C\boldsymbol{x}$ form exactly the plane $b_1 + b_2 + b_3 = 0$. In fact $\boldsymbol{w}^* = -(\boldsymbol{u} + \boldsymbol{v})$ lies in the plane of $\boldsymbol{u}$ and $\boldsymbol{v}$ and adds nothing new (Proposition 1.7). For A, Proposition 2.2 says every $\boldsymbol{b}$ in $\mathbb{R}^3$ is reached, so the combinations of $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ fill all of $\mathbb{R}^3$. Figure 2 shows the difference.

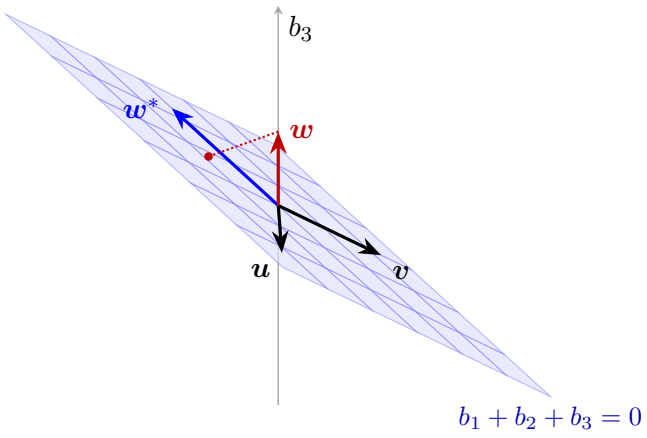


Figure 2: The columns $\boldsymbol{u} = (1, -1, 0)$, $\boldsymbol{v} = (0, 1, -1)$ and $\boldsymbol{w}^* = (-1, 0, 1)$ of the cyclic matrix C all lie in the plane $b_1 + b_2 + b_3 = 0$ (shaded), so every $C\boldsymbol{x}$ stays in that plane. The third column $\boldsymbol{w} = (0, 0, 1)$ of the difference matrix A points out of the plane (the dotted segment drops from its tip to the nearest point of the plane), so the combinations of $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ fill all of $\mathbb{R}^3$.

Here are the words for what we have seen; Sections 6, 7 and 10 develop them in full.

Definition 2.6 (Independent and dependent vectors)**.** Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are *dependent* if some combination $c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$ with weights not all zero equals the zero vector. Otherwise they are *independent*: the only combination giving $\mathbf{0}$ is the one with all weights zero.

In matrix language, the columns of a matrix are independent exactly when $A\boldsymbol{x} = \mathbf{0}$ forces $\boldsymbol{x} = \mathbf{0}$. So $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are independent (Proposition 2.2), while $\boldsymbol{u}+\boldsymbol{v}+\boldsymbol{w}^* = C(1,1,1) = \mathbf{0}$ shows that $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}^*$ are dependent.

Definition 2.7 (Vector space, subspace, basis)**.** A *vector space* is a collection of vectors closed under linear combinations: any combination of vectors in the collection is again in it. A *subspace* is a vector space contained in a larger one. A *basis* of $\mathbb{R}^n$ is a set of n independent vectors in $\mathbb{R}^n$.

The plane $b_1 + b_2 + b_3 = 0$ is a subspace of $\mathbb{R}^3$: it is the set of all combinations of $\boldsymbol{u}$ and $\boldsymbol{v}$, and a combination of combinations is a combination. A filled triangle with corners $\mathbf{0}, \boldsymbol{u}, \boldsymbol{v}$ is *not* a subspace, because $2\boldsymbol{u}$ leaves it. Taking all combinations of some vectors is the standard way to build a subspace, and it always produces a subspace. The subspaces of $\mathbb{R}^3$ are: the origin alone, the lines through the origin, the planes through the origin, and $\mathbb{R}^3$ itself, of dimensions $0, 1, 2, 3$. The zero vector belongs to every one of them, since $0 \cdot \boldsymbol{v} = \mathbf{0}$.

For n vectors in $\mathbb{R}^n$ there are three equivalent ways to say "basis", which this section has seen side by side for $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$: (i) the vectors are independent; (ii) their combinations fill $\mathbb{R}^n$; (iii) the $n \times n$ matrix with those columns is invertible. The vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}^*$ fail all three. The proof that the three statements always agree comes in Section 10.

Example 2.8 (A basis of the plane from Section 1)**.** The columns $\boldsymbol{a}_1 = (2, -1)$ and $\boldsymbol{a}_2 = (-1, 2)$ of the first example of the course form a basis of $\mathbb{R}^2$. They are independent: if $c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2 = \mathbf{0}$ then $2c_1 - c_2 = 0$ and $-c_1 + 2c_2 = 0$; the first gives $c_2 = 2c_1$, and the second becomes $-c_1 + 4c_1 = 3c_1 = 0$, so $c_1 = c_2 = 0$. Their combinations fill $\mathbb{R}^2$: we found $1\,\boldsymbol{a}_1 + 2\,\boldsymbol{a}_2 = (0, 3)$, and $\boldsymbol{a}_1 + \boldsymbol{a}_2 = (1, 1)$, and Exercise 1.12 gives the weights for every $\boldsymbol{b}$. And the matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is invertible, since each $\boldsymbol{b}$ has exactly one pair of weights.

Common trap. Dependence is not the same as "two columns are parallel". No column of C is a multiple of another, yet the three columns are dependent because $\boldsymbol{u} + \boldsymbol{v} + \boldsymbol{w}^* = \mathbf{0}$. To test three vectors you must look for a combination of all of them, not compare them in pairs.

2.5 Looking ahead: rectangular matrices

The habit to build is to look at a matrix and ask what it *does*. A takes differences and can be undone; C takes cyclic differences, crushes the constant vectors and reaches only a plane. Matrices need not be square: a 7×3 matrix sends $\mathbb{R}^3$ into $\mathbb{R}^7$, can reach at most a three-dimensional part of $\mathbb{R}^7$, and has no inverse. A matrix that keeps appearing with rectangular A is $A^{\mathsf{T}}A$: it is square (3×3 here) and symmetric, and it will be the key to least squares in Part II (Section 17).

2.6 Exercises

Exercise 2.9. A matrix A has the property that *all* solutions of $A\boldsymbol{x} = (1, 4, 1, 1)$ are

$$\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \qquad c \in \mathbb{R}.$$

Say everything you can about the columns $\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3$ of A.

Solution. Since $\boldsymbol{x}$ has three components, A has three columns, and since $\boldsymbol{b} = (1, 4, 1, 1)$ has four, each column lies in $\mathbb{R}^4$. Taking $c = 0$ gives $A(0, 1, 1) = \boldsymbol{b}$, that is, $\boldsymbol{c}_2 + \boldsymbol{c}_3 = \boldsymbol{b}$. Taking $c = 1$ gives $A(0, 3, 2) = \boldsymbol{b}$; subtracting, $A(0, 2, 1) = \boldsymbol{0}$, that is, $2\boldsymbol{c}_2 + \boldsymbol{c}_3 = \boldsymbol{0}$. Subtract the first relation from the second: $\boldsymbol{c}_2 = -\boldsymbol{b}$, and then $\boldsymbol{c}_3 = \boldsymbol{b} - \boldsymbol{c}_2 = 2\boldsymbol{b}$. So $\boldsymbol{c}_2 = (-1, -4, -1, -1)$ and $\boldsymbol{c}_3 = (2, 8, 2, 2)$.

About $\boldsymbol{c}_1$: it cannot be a multiple of $\boldsymbol{b}$. If $\boldsymbol{c}_1 = k\boldsymbol{b}$, then $A(1, k, 0) = k\boldsymbol{b} - k\boldsymbol{b} = \boldsymbol{0}$, and $(0, 1, 1) + (1, k, 0)$ would be another solution of $A\boldsymbol{x} = \boldsymbol{b}$ whose first component is 1, which is not of the given form. Conversely, if $\boldsymbol{c}_1$ is not a multiple of $\boldsymbol{b}$, then $A\boldsymbol{x} = x_1\boldsymbol{c}_1 + (2x_3 - x_2)\boldsymbol{b} = \boldsymbol{0}$ forces $x_1 = 0$ and $x_2 = 2x_3$ (two non-parallel vectors combine to zero only with zero weights), so the solutions of $A\boldsymbol{x} = \boldsymbol{0}$ are exactly the multiples of $(0, 2, 1)$, as required. Hence: $\boldsymbol{c}_1$ is any vector of $\mathbb{R}^4$ that is not a multiple of $\boldsymbol{b}$ (in particular not $\boldsymbol{0}$), and nothing more can be said. □

Exercise 2.10. Let A_4 be the 4×4 difference matrix, $A_4\boldsymbol{x} = (x_1,\ x_2 - x_1,\ x_3 - x_2,\ x_4 - x_3)$. Write A_4 and its inverse S_4, and use S_4 to solve $A_4\boldsymbol{x} = (1, 3, 5, 7)$.

Solution. Solving from the top, $x_1 = b_1$, $x_2 = b_1 + b_2$, $x_3 = b_1 + b_2 + b_3$, $x_4 = b_1 + b_2 + b_3 + b_4$, so

$$A_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \qquad S_4 = A_4^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Then $\boldsymbol{x} = S_4(1, 3, 5, 7) = (1,\ 4,\ 9,\ 16)$: running sums of odd numbers are the squares. Check: $A_4(1, 4, 9, 16) = (1, 3, 5, 7)$. □

Exercise 2.11. For the cyclic matrix C of (6), decide which of $\boldsymbol{b} = (1, 0, -1)$, $(2, -3, 1)$ and $(1, 1, 0)$ lie in the plane of all $C\boldsymbol{x}$, and solve $C\boldsymbol{x} = \boldsymbol{b}$ when it is possible.

Solution. Use Proposition 2.4. For $(1, 0, -1)$ the sum is 0; the particular solution is $(0,\ b_2,\ b_2 + b_3) = (0, 0, -1)$, and indeed $C(0, 0, -1) = (0 + 1,\ 0 - 0,\ -1 - 0) = (1, 0, -1)$. All solutions: $(0, 0, -1) + c(1, 1, 1)$. For $(2, -3, 1)$ the sum is 0; the particular solution is $(0, -3, -2)$, and $C(0, -3, -2) = (0 + 2,\ -3 - 0,\ -2 + 3) = (2, -3, 1)$. All solutions: $(0, -3, -2) + c(1, 1, 1)$. For $(1, 1, 0)$ the sum is $2 \neq 0$, so this $\boldsymbol{b}$ is off the plane and there is no solution. □

Exercise 2.12. Which of these subsets of $\mathbb{R}^3$ are subspaces? (a) all $\boldsymbol{b}$ with $b_1 + b_2 + b_3 = 0$; (b) all $\boldsymbol{b}$ with $b_1 + b_2 + b_3 = 1$; (c) all multiples of $(1, 1, 1)$; (d) all $\boldsymbol{b}$ with $b_1 \geq 0$; (e) the zero vector alone.

Solution. (a) Yes: it is the plane of all combinations of $\boldsymbol{u}$ and $\boldsymbol{v}$, and directly, if two vectors have component sums 0, so does $c\boldsymbol{b} + d\boldsymbol{b}'$. (b) No: the zero vector has sum $0 \neq 1$, and every subspace contains $\boldsymbol{0}$. (c) Yes: a line through the origin; $c(1, 1, 1) + d(1, 1, 1) = (c + d)(1, 1, 1)$. It is also the set of $\boldsymbol{x}$ with $C\boldsymbol{x} = \boldsymbol{0}$. (d) No: $(1, 0, 0)$ is in the set but $-1 \cdot (1, 0, 0)$ is not; the set is closed under addition but not under multiplication by negative numbers. (e) Yes: $\boldsymbol{0} + \boldsymbol{0} = \boldsymbol{0}$ and $c\,\boldsymbol{0} = \boldsymbol{0}$; this is the smallest subspace. □

3 Elimination with matrices

Section 1 asked which combination of the columns produces $\boldsymbol{b}$ but had no systematic way to find it. *Gaussian elimination* is that way, and it is how essentially every software package solves $A\boldsymbol{x} = \boldsymbol{b}$. It subtracts multiples of equations from other equations until the system is triangular, then solves the triangular system from the bottom up. This section does elimination by hand, sees how it can fail, and then rewrites each step as multiplication by a simple matrix. That last move turns a procedure into algebra, and it is the starting point for the factorization $A = LU$ in Section 5.

3.1 Pivots and multipliers

Our example has three equations in three unknowns:

$$\begin{aligned} x + 2y + z &= 2 \\ 3x + 8y + z &= 12 \\ 4y + z &= 2 \end{aligned} \qquad A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}. \tag{7}$$

Elimination works on the rows of A (the equations), so we write only the numbers.

Definition 3.1 (Pivot, multiplier, upper triangular)**.** Elimination clears the columns of A from left to right. At step k the *pivot* is the entry in position (k, k) of the current matrix; it must be nonzero. To clear an entry a_{ik} below the pivot ($i > k$), subtract

$$\ell_{ik} = \frac{\text{entry to eliminate}}{\text{pivot}}$$

times the pivot row k from row i. The number ℓ_{ik} is the *multiplier*. When all columns are cleared the result U is *upper triangular*: every entry below the main diagonal is zero, and the pivots sit on the diagonal.

Example 3.2 (Forward elimination on a 3×3 system)**.** Eliminate below the pivots of the matrix A in (7).

- *Step* $(2, 1)$. The first pivot is 1. The entry below it is 3, so $\ell_{21} = 3/1 = 3$: subtract 3 times row 1 from row 2, giving $(3, 8, 1) - 3(1, 2, 1) = (0, 2, -2)$.
- *Step* $(3, 1)$. The entry in position $(3, 1)$ is already 0, so $\ell_{31} = 0$ and nothing changes. (Software still checks this position.)
- *Step* $(3, 2)$. The second pivot is the 2 now in position $(2, 2)$. The entry below it is 4, so $\ell_{32} = 4/2 = 2$: subtract 2 times the *new* row 2 from row 3, giving $(0, 4, 1) - 2(0, 2, -2) = (0, 0, 5)$.

The third pivot is 5, and

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}.$$

The pivots $1, 2, 5$ already tell us a great deal about A. For instance, their product $1 \cdot 2 \cdot 5 = 10$ is the determinant of A (Section 20).

The right-hand side must receive the same operations. The convenient bookkeeping is the *augmented matrix* $[\,A \;\; \boldsymbol{b}\,]$, with $\boldsymbol{b}$ tacked on as an extra column. Subtracting 3 times row 1 changes 12 into $12 - 3 \cdot 2 = 6$; subtracting 2 times row 2 changes 2 into $2 - 2 \cdot 6 = -10$. Elimination turns $A\boldsymbol{x} = \boldsymbol{b}$ into $U\boldsymbol{x} = \boldsymbol{c}$ with $\boldsymbol{c} = (2, 6, -10)$; Figure 3 records the whole computation.

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array}\right] \xrightarrow[\text{row}_2 - 3\,\text{row}_1]{E_{21}} \left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}\right] \xrightarrow[\text{row}_3 - 2\,\text{row}_2]{E_{32}} \left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 0 & \boxed{5} & -10 \end{array}\right]$$

$$[A \;\; \boldsymbol{b}] \qquad\qquad [E_{21}A \;\; E_{21}\boldsymbol{b}] \qquad\qquad [U \;\; \boldsymbol{c}]$$

Figure 3: Forward elimination on the augmented matrix. Boxed entries are pivots. Each arrow is one row operation, equivalently one multiplication on the left by an elimination matrix; the step E_{31} is skipped because the $(3,1)$ entry is already zero. The right-hand column changes from $\boldsymbol{b} = (2, 12, 2)$ to $\boldsymbol{c} = (2, 6, -10)$.

3.2 Back substitution

The triangular system $U\boldsymbol{x} = \boldsymbol{c}$ reads

$$x + 2y + z = 2, \qquad 2y - 2z = 6, \qquad 5z = -10.$$

Solve it in reverse order. The last equation gives $z = -2$. The second becomes $2y + 4 = 6$, so $y = 1$. The first becomes $x + 2 - 2 = 2$, so $x = 2$. Checking in the original equations (7): $2 + 2 - 2 = 2$, $6 + 8 - 2 = 12$, $4 - 2 = 2$.

Proposition 3.3 (Back substitution)**.** *Let U be an $n \times n$ upper triangular matrix whose diagonal entries $u_{11}, \ldots, u_{nn}$ are all nonzero. Then for every $\boldsymbol{c}$ the system $U\boldsymbol{x} = \boldsymbol{c}$ has exactly one solution, given by*

$$x_n = \frac{c_n}{u_{nn}}, \qquad x_i = \frac{1}{u_{ii}}\Big(c_i - \sum_{j=i+1}^{n} u_{ij}x_j\Big) \quad (i = n-1, \ldots, 1).$$

Proof. Equation i of $U\boldsymbol{x} = \boldsymbol{c}$ is $u_{ii}x_i + \sum_{j>i} u_{ij}x_j = c_i$, because the entries left of the diagonal are zero. Equation n involves only x_n and forces $x_n = c_n/u_{nn}$. Suppose $x_{i+1}, \ldots, x_n$ are already forced. Then equation i has exactly one unknown left, x_i, with the nonzero coefficient u_{ii}, so it forces the stated value. Working upward, every component is forced, which proves uniqueness; and the values so obtained satisfy every equation, which proves existence. □

3.3 When elimination breaks down

Zero is never allowed as a pivot. If a zero appears in the pivot position, there are two cases.

Temporary failure. If some row *below* has a nonzero entry in that column, exchange the two rows and continue.

Example 3.4 (A row exchange rescues elimination)**.** Change the 8 in A to a 6. The first step gives row $2 = (3, 6, 1) - 3(1, 2, 1) = (0, 0, -2)$, a zero in the $(2,2)$ pivot position. Row 3 is $(0, 4, 1)$, with a nonzero entry there, so exchange rows 2 and 3:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -2 \\ 0 & 4 & 1 \end{bmatrix} \xrightarrow{\text{exchange rows 2, 3}} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & -2 \end{bmatrix},$$

which is already triangular, with pivots $1, 4, -2$. The matrix is fine; only the order of the equations was unlucky.

Complete failure. If the pivot position holds a zero and every entry below it is zero too, no exchange helps.

Example 3.5 (Elimination fails: a singular matrix)**.** Change the $(3,3)$ entry of A from 1 to -4. The first step gives row $2 = (0,2,-2)$ and leaves row $3 = (0,4,-4)$; subtracting 2 times row 2 leaves row $3 = (0,0,0)$. There is no third pivot. The matrix is *singular*: depending on $\boldsymbol{b}$, the system has no solution or infinitely many, never exactly one. (In the column picture, column 3 has become $(1,1,-4) = 3\,(1,3,0) - (2,8,4)$, a combination of the first two columns.)

In summary, elimination on an $n \times n$ matrix proceeds as follows.

1. For $k = 1, \ldots, n$: if the (k,k) entry is zero, exchange row k with a lower row that has a nonzero entry in column k; if there is none, stop: A is singular.
2. For each row $i > k$, compute $\ell_{ik} = a_{ik}/a_{kk}$ (current entries) and subtract ℓ_{ik} times row k from row i, in the augmented column as well.
3. When all n pivots are found, solve $U\boldsymbol{x} = \boldsymbol{c}$ by back substitution.

Common trap. The multipliers use the *current* entries, not the original ones: in step $(3,2)$ the multiplier is $4/2$, where 2 is the pivot created by step $(2,1)$, not the original entry 8. And a zero in a pivot position is not yet a verdict of singularity: look below it first. Only a zero with nothing nonzero underneath means failure.

3.4 Each elimination step is a matrix

To put elimination into matrix language we need products of matrices. Multiplying a matrix by a column gives a combination of its columns (Definition 1.4). The mirror image is a row times a matrix.

Proposition 3.6 (Row times matrix)**.** *If $\boldsymbol{y}^{\mathsf{T}} = (y_1, \ldots, y_m)$ is a row vector and A is $m \times n$ with rows $\boldsymbol{r}_1, \ldots, \boldsymbol{r}_m$, then $\boldsymbol{y}^{\mathsf{T}} A = y_1\boldsymbol{r}_1 + \cdots + y_m\boldsymbol{r}_m$, a combination of the rows of A.*

Proof. Entry j of $\boldsymbol{y}^{\mathsf{T}}A$ is the row $\boldsymbol{y}^{\mathsf{T}}$ times column j of A, namely $\sum_i y_i a_{ij}$. Entry j of $\sum_i y_i \boldsymbol{r}_i$ is also $\sum_i y_i a_{ij}$, since a_{ij} is entry j of row $\boldsymbol{r}_i$. □

For example, $\begin{bmatrix}1 & 2 & 7\end{bmatrix} A = 1\,(1,2,1) + 2\,(3,8,1) + 7\,(0,4,1) = (7,46,10)$. For a matrix E with m columns and a matrix A with m rows, the product EA is the matrix whose column j is E times column j of A; its (i,j) entry is row i of E times column j of A. By Proposition 3.6, *row i of EA is the combination of the rows of A given by row i of E.* (Section 4 collects all the ways to see a matrix product.) So to subtract 3 times row 1 from row 2 and leave the other rows alone, the rows of E must be $(1,0,0)$, $(-3,1,0)$ and $(0,0,1)$.

Definition 3.7 (Elimination matrix)**.** For $i \neq j$, the *elimination matrix* E_{ij} with multiplier ℓ is the identity matrix with the entry $-\ell$ placed in row i, column j. Multiplying on the left, $E_{ij}A$ subtracts ℓ times row j of A from row i and leaves the other rows unchanged.

Example 3.8 (Elimination as matrix multiplication)**.** For the example (7), the two nontrivial steps are

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Then

$$E_{21}A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}, \qquad E_{32}(E_{21}A) = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = U.$$

Spot-check one entry: the $(2,3)$ entry of $E_{21}A$ is row 2 of E_{21} times column 3 of A, $(-3)(1) + (1)(1) + (0)(1) = -2$. The step E_{31} is the identity, because $\ell_{31} = 0$.

The whole of forward elimination is the single line $E_{32}(E_{31}(E_{21}A)) = U$. Can we multiply the E's together first, to get one matrix that does everything? Yes, because matrix multiplication is associative.

Proposition 3.9 (Associative law)**.** *If the shapes allow the products,* $(EF)A = E(FA)$.

Proof. Compare the (i,j) entries, writing e, f, a for the entries of E, F, A:

$$\big((EF)A\big)_{ij} = \sum_k \Big(\sum_l e_{il}f_{lk}\Big)a_{kj} = \sum_l e_{il}\Big(\sum_k f_{lk}a_{kj}\Big) = \big(E(FA)\big)_{ij}.$$

Both are the sum of $e_{il}f_{lk}a_{kj}$ over all k and l, added in a different order. □

So we may move parentheses freely, as long as the order of the factors is kept. For the example, $E = E_{32}E_{21}$ does all of elimination at once:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 6 & -2 & 1 \end{bmatrix}, \qquad EA = U, \quad E\boldsymbol{b} = \boldsymbol{c}.$$

The entry 6 appears because 3 of row 1 went into row 2 and then 2 of that row 2 went into row 3. Section 5 shows how to avoid such entries by recording elimination the other way around.

The order of the factors cannot be changed: the *commutative* law fails for matrices. Row exchanges show this vividly.

Definition 3.10 (Permutation matrix)**.** A *permutation matrix* P is the identity matrix with its rows reordered. Multiplying on the left, PA reorders the rows of A in the same way.

To exchange the two rows of a 2×2 matrix, exchange the rows of I:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}, \qquad \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}.$$

On the left, P acts on rows; on the right, the same P acts on columns (each column of AP is a combination of the columns of A). So $PA \neq AP$ in general. In elimination we use row operations, so the elimination and permutation matrices always multiply from the left.

3.5 Undoing a step: inverse matrices

Every elimination step can be undone. If E_{21} subtracts 3 times row 1 from row 2, the step that brings row 2 back adds 3 times row 1 to row 2:

$$E_{21}^{-1}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

For example, E_{21} turns $\boldsymbol{b} = (2, 12, 2)$ into $(2, 6, 2)$, and E_{21}^{-1} turns $(2, 6, 2)$ back into $(2, 12, 2)$. The matrix I is the *identity*: $I\boldsymbol{x} = \boldsymbol{x}$ for every $\boldsymbol{x}$, the matrix analogue of the number 1.

Proposition 3.11 (Inverse of an elimination matrix)**.** *Let $E_{ij}(\ell)$ denote the elimination matrix with $-\ell$ in position (i, j). Then $E_{ij}(-\ell)\, E_{ij}(\ell) = I = E_{ij}(\ell)\, E_{ij}(-\ell)$. A permutation matrix that exchanges two rows is its own inverse.*

Proof. Apply $E_{ij}(-\ell)E_{ij}(\ell)$ to any matrix M. The first factor replaces row i by (row i) $-\ell$ (row j) and leaves row j untouched, since $j \neq i$. The second factor adds ℓ (row j) back to row i, restoring M. A matrix that leaves every M unchanged is I (take $M = I$). The other order is the same argument with ℓ replaced by $-\ell$. Exchanging the same two rows twice restores every matrix. □

This is why elimination is legitimate: with E the product of all the step matrices, $A\boldsymbol{x} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{c}$ have the same solutions. If $A\boldsymbol{x} = \boldsymbol{b}$ then $EA\boldsymbol{x} = E\boldsymbol{b}$, that is, $U\boldsymbol{x} = \boldsymbol{c}$. Conversely, each step can be reversed, so multiplying $U\boldsymbol{x} = \boldsymbol{c}$ on the left by the inverse steps returns $A\boldsymbol{x} = \boldsymbol{b}$. The answer $\boldsymbol{x} = (2, 1, -2)$ found by back substitution is therefore the solution of the original system.

Common trap. Row operations multiply on the *left*; a matrix on the right of A combines columns. Writing AE_{21} instead of $E_{21}A$ subtracts 3 times column 2 from column 1, a different operation entirely. Also keep the factors in order: elimination is $E_{32}E_{21}A$, not $E_{21}E_{32}A$.

3.6 Exercises

Exercise 3.12. For the system $2x + 3y = 5$, $6x + 15y = 12$: what multiple of equation 1 should be subtracted from equation 2? Write the system as $A\boldsymbol{x} = \boldsymbol{b}$, eliminate (what are the pivots?), and solve by back substitution. Check your answer in both equations.

Solution. Subtract $\ell_{21} = 6/2 = 3$ times equation 1. In matrix form,

$$\begin{bmatrix} 2 & 3 \\ 6 & 15 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix},$$

since $15 - 3 \cdot 3 = 6$ and $12 - 3 \cdot 5 = -3$. The pivots are 2 and 6. Back substitution: $6y = -3$ gives $y = -\frac{1}{2}$; then $2x - \frac{3}{2} = 5$ gives $x = \frac{13}{4}$. Check: $2 \cdot \frac{13}{4} + 3(-\frac{1}{2}) = \frac{13}{2} - \frac{3}{2} = 5$ and $6 \cdot \frac{13}{4} + 15(-\frac{1}{2}) = \frac{39}{2} - \frac{15}{2} = 12$. □

Exercise 3.13. Solve by elimination, using a row exchange when needed:

$$x - y - z + u = 0, \qquad 2x + 2z = 8, \qquad -y - 2z = -8, \qquad 3x - 3y - 2z + 4u = 7.$$

Solution. The augmented matrix and the elimination steps:

$$\left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 2 & 0 & 2 & 0 & 8 \\ 0 & -1 & -2 & 0 & -8 \\ 3 & -3 & -2 & 4 & 7 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 0 & 2 & 4 & -2 & 8 \\ 0 & -1 & -2 & 0 & -8 \\ 0 & 0 & 1 & 1 & 7 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 0 & 2 & 4 & -2 & 8 \\ 0 & 0 & 0 & -1 & -4 \\ 0 & 0 & 1 & 1 & 7 \end{array}\right].$$

The first arrow subtracts 2 times row 1 from row 2 and 3 times row 1 from row 4. The second adds $\frac{1}{2}$ of row 2 to row 3 (multiplier $\ell_{32} = -\frac{1}{2}$). Now the $(3, 3)$ position holds 0, but row 4 has a 1 below it, so exchange rows 3 and 4; the matrix is then upper triangular with pivots $1, 2, 1, -1$. Back

substitution: $-u=-4$ gives $u=4$; $z+u=7$ gives $z=3$; $2y+4z-2u=8$ gives $2y=8-12+8=4$, $y=2$; $x-y-z+u=0$ gives $x=2+3-4=1$. So $(x,y,z,u)=(1,2,3,4)$. Check the last equation: $3-6-6+16=7$. □

Exercise 3.14. Find the lower triangular E that reduces the 4×4 Pascal matrix to a smaller Pascal matrix:

$$E\begin{bmatrix}1&0&0&0\\1&1&0&0\\1&2&1&0\\1&3&3&1\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&1&1&0\\0&1&2&1\end{bmatrix}.$$

Which matrix M, a product of several such E's, reduces the Pascal matrix all the way to I?

Solution. Subtracting each row from the row below it uses Pascal's rule $\binom{i}{j}-\binom{i-1}{j}=\binom{i-1}{j-1}$ and produces the smaller Pascal matrix:

$$E=\begin{bmatrix}1&0&0&0\\-1&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix}.$$

(Here row i is replaced by row i minus the *original* row $i-1$; as a sequence of elimination steps, do row 4 first, then row 3, then row 2.) The same idea one level down clears the second column with E_2 (rows 3 and 4 minus the rows above them, leaving rows 1, 2 alone), and then E_3 subtracts row 3 from row 4:

$$E_2=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix},\quad E_3=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&-1&1\end{bmatrix},\quad M=E_3E_2E=\begin{bmatrix}1&0&0&0\\-1&1&0&0\\1&-2&1&0\\-1&3&-3&1\end{bmatrix}.$$

Since M times the Pascal matrix is I, M is its inverse: Pascal's triangle with alternating signs. □

Exercise 3.15. For which numbers d does elimination on $A=\begin{bmatrix}2&1&0\\4&d&1\\0&1&2\end{bmatrix}$ meet a zero in a pivot position? For each such d, say whether a row exchange rescues it.

Solution. The first pivot is 2; subtracting 2 times row 1 from row 2 gives row $2=(0,\,d-2,\,1)$. If $d=2$ the $(2,2)$ position is zero, but row $3=(0,1,2)$ has a 1 below it; exchanging rows 2 and 3 gives pivots $2,1,1$ (the matrix becomes $\begin{bmatrix}2&1&0\\0&1&2\\0&0&1\end{bmatrix}$), so this failure is only temporary. If $d\neq 2$, the second pivot is $d-2$; the multiplier is $\ell_{32}=1/(d-2)$, and the third pivot is $2-\frac{1}{d-2}$. It is zero when $d-2=\frac{1}{2}$, that is $d=\frac{5}{2}$, and then nothing lies below it: complete failure, and A is singular. For every other d, elimination succeeds without exchanges. □

4 Multiplication and inverse matrices

We have been multiplying matrices since Section 3, where elimination steps were matrices acting on the left. This section sets down the rule for the product AB and then looks at the same product in four more ways: by columns, by rows, as a sum of column-times-row pieces, and by blocks. All five give the same answer, and each is the right one in some argument later on. The second half answers the most important question about a square matrix: does it have an inverse? A singular matrix sends some nonzero vector to zero, which no inverse can undo; for an invertible matrix, Gauss–Jordan elimination computes A^{-1}.

4.1 The rule: row times column

Definition 4.1 (Matrix product)**.** Let A be $m \times n$ and B be $n \times p$. The product $C = AB$ is the $m \times p$ matrix with entries

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj},$$

the dot product of row i of A with column j of B. The product is defined only when the number of columns of A equals the number of rows of B.

For instance, the entry c_{34} comes from row 3 of A and column 4 of B: $c_{34} = a_{31}b_{14} + a_{32}b_{24} + \cdots + a_{3n}b_{n4}$. The shapes follow the pattern $(m \times n)(n \times p) = (m \times p)$: the inner numbers must match and disappear, the outer numbers give the shape. When B is a single column ($p = 1$), this is the row form of $A\boldsymbol{x}$ from Proposition 1.5.

4.2 Columns, rows, and columns times rows

The entry rule hides what the product does to whole vectors. The next three views each look at whole columns or rows.

Proposition 4.2 (Three more views of AB)**.** *Let A be $m \times n$ with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ and let B be $n \times p$ with columns $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_p$ and rows $\boldsymbol{r}_1^{\mathsf{T}}, \ldots, \boldsymbol{r}_n^{\mathsf{T}}$.*

(a) Columns: column j of AB is $A\boldsymbol{b}_j$. Every column of AB is a combination of the columns of A.

(b) Rows: row i of AB is (row i of A) times B. Every row of AB is a combination of the rows of B.

(c) Columns times rows: $AB = \boldsymbol{a}_1\boldsymbol{r}_1^{\mathsf{T}} + \boldsymbol{a}_2\boldsymbol{r}_2^{\mathsf{T}} + \cdots + \boldsymbol{a}_n\boldsymbol{r}_n^{\mathsf{T}}$, a sum of n matrices, each an $m \times 1$ column times a $1 \times p$ row.

Proof. (a) The entries of $A\boldsymbol{b}_j$ are $\sum_k a_{ik}b_{kj}$ for $i = 1, \ldots, m$ (Proposition 1.5), which is column j of AB. By Definition 1.4, $A\boldsymbol{b}_j = b_{1j}\boldsymbol{a}_1 + \cdots + b_{nj}\boldsymbol{a}_n$. (b) Entry j of (row i of A) B is $\sum_k a_{ik}b_{kj} = c_{ij}$, and by Proposition 3.6 that row is $a_{i1}\boldsymbol{r}_1^{\mathsf{T}} + \cdots + a_{in}\boldsymbol{r}_n^{\mathsf{T}}$. (c) The column-times-row product $\boldsymbol{a}_k\boldsymbol{r}_k^{\mathsf{T}}$ is the $m \times p$ matrix with (i, j) entry $a_{ik}b_{kj}$. Summing over k gives $\sum_k a_{ik}b_{kj} = c_{ij}$. □

Example 4.3 (One product, four ways)**.** Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (2×2) and $B = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$ (2×3), so AB is 2×3.

- *Entries.* $c_{23} = (3,4) \cdot (2,0) = 6$; similarly for the other five entries.
- *Columns.* $A(1,1) = (1+2,\ 3+4) = (3,7)$, $A(0,1) = (2,4)$, $A(2,0) = (2,6)$.
- *Rows.* $(1,2)B = 1\,(1,0,2) + 2\,(1,1,0) = (3,2,2)$ and $(3,4)B = 3\,(1,0,2) + 4\,(1,1,0) = (7,4,6)$.
- *Columns times rows.*

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 0 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 2 & 0 \\ 4 & 4 & 0 \end{bmatrix}.$$

Every view gives $AB = \begin{bmatrix} 3 & 2 & 2 \\ 7 & 4 & 6 \end{bmatrix}$. The same twelve multiplications are being done each time; only the grouping changes.

A single column times a single row is a very special matrix:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}.$$

Its columns are multiples of $(2,3,4)$ and its rows are multiples of $(1,6)$, as views (a) and (b) predict. Plotted as points, the rows $(2,12)$, $(3,18)$, $(4,24)$ all lie on the line through $(1,6)$, and both columns lie on the line through $(2,3,4)$. In later language (Section 11), its row space and column space are both lines: it is a matrix of rank one. View (c) says that every product AB is a sum of such rank-one pieces. For instance, if A has columns $(2,3,4)$ and $(7,8,9)$ and B has rows $(1,6)$ and $(0,0)$, the second piece is zero and AB is the matrix above.

4.3 Block multiplication

Matrices can be cut into blocks, and if the cuts match, blocks multiply like numbers (keeping their order). With A and B cut into 2×2 arrays of blocks,

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1B_1 + A_2B_3 & A_1B_2 + A_2B_4 \\ A_3B_1 + A_4B_3 & A_3B_2 + A_4B_4 \end{bmatrix},$$

provided the columns of A are cut the same way as the rows of B, so that every product A_iB_j is defined.

Proposition 4.4 (Block multiplication)**.** *If the column partition of A matches the row partition of B, then block (I,J) of AB is $\sum_K A_{IK}B_{KJ}$, a sum of block products.*

Proof. Take an entry c_{ij} inside block (I,J). Its sum $\sum_k a_{ik}b_{kj}$ runs over all column indices k of A. Split this sum according to the column blocks K: the part with k in block K is exactly the (i,j) entry of $A_{IK}B_{KJ}$. Adding over K gives the claim. □

Example 4.5 (Blocks behave like numbers)**.** Let $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and let I, 0 be the 2×2 identity and zero matrices. By block multiplication,

$$\begin{bmatrix} I & 0 \\ C & I \end{bmatrix} \begin{bmatrix} I & 0 \\ -C & I \end{bmatrix} = \begin{bmatrix} I \cdot I + 0 \cdot (-C) & I \cdot 0 + 0 \cdot I \\ C \cdot I + I \cdot (-C) & C \cdot 0 + I \cdot I \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

So the 4×4 matrix $\begin{bmatrix} I & 0 \\ C & I \end{bmatrix}$, which adds combinations of rows 1–2 to rows 3–4, is undone by subtracting the same combinations, exactly like the elimination matrices of Proposition 3.11.

Matrix multiplication obeys the associative law (Proposition 3.9) and the distributive laws $A(B + C) = AB + AC$, $(A + B)C = AC + BC$ (Exercise 4.14), but in general $AB \neq BA$.

4.4 Inverse matrices

Definition 4.6 (Inverse, invertible, singular)**.** A square $n \times n$ matrix A is *invertible* (or *nonsingular*) if there is an $n \times n$ matrix A^{-1} with

$$A^{-1}A = I \qquad \text{and} \qquad AA^{-1} = I.$$

A square matrix with no inverse is *singular*.

An inverse, when it exists, is unique: if $BA = I$ and $AC = I$, then $B = B(AC) = (BA)C = C$. For square matrices it is enough to find a one-sided inverse (Corollary 4.13 below). (A rectangular matrix can have an inverse on one side only, as Section 36 explains.)

Whether A is invertible is the first question to ask about a square matrix. The clearest way to see that a matrix is *not* invertible is to find a vector it sends to zero.

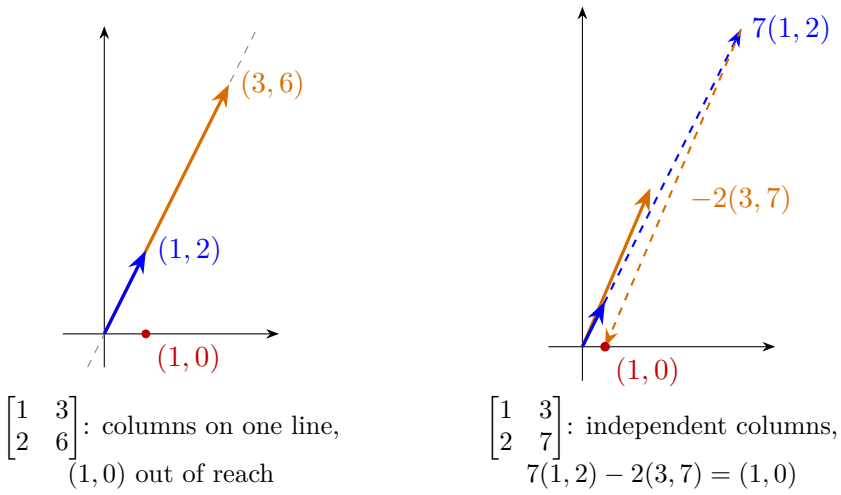
Theorem 4.7 (A nonzero solution of $A\boldsymbol{x} = \mathbf{0}$ rules out an inverse)**.** *If A is square and $A\boldsymbol{x} = \mathbf{0}$ for some $\boldsymbol{x} \neq \mathbf{0}$, then A is singular.*

Proof. Suppose A^{-1} existed. Multiplying $A\boldsymbol{x} = \mathbf{0}$ on the left by A^{-1} gives $\boldsymbol{x} = A^{-1}A\boldsymbol{x} = A^{-1}\mathbf{0} = \mathbf{0}$, contradicting $\boldsymbol{x} \neq \mathbf{0}$. □

Example 4.8 (A singular matrix, three ways)**.** Take $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$. *Null vector:* $\boldsymbol{x} = (3, -1)$ gives $A\boldsymbol{x} = 3(1, 2) - 1(3, 6) = (0, 0)$, so A is singular by Theorem 4.7. *Columns:* column 2 is 3 times column 1, so both columns lie on the line through $(1, 2)$. By Proposition 4.2(a), every column of any product AB lies on that line too, so AB can never have the column $(1, 0)$ of I. *Elimination:* subtracting 2 times row 1 from row 2 leaves $(0, 0)$, so there is no second pivot. (For readers who know determinants: $1 \cdot 6 - 3 \cdot 2 = 0$.)

4.5 Gauss–Jordan elimination computes A^{-1}

Now change the 6 to a 7: $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$. Its columns point in different directions (Figure 4), and we look for $A^{-1} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ with $AA^{-1} = I$. By the column view, this is two systems with the same



$\begin{bmatrix}1 & 3\\ 2 & 6\end{bmatrix}$: columns on one line, $(1,0)$ out of reach

$\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}$: independent columns, $7(1,2) - 2(3,7) = (1,0)$

Figure 4: Left: the columns of the singular matrix lie on one line (dashed), so every combination stays on it and the column $(1,0)$ of I cannot be produced. Right: the columns $(1,2)$ and $(3,7)$ of the invertible matrix (solid, near the origin) point in different directions, so their combinations fill the plane. Seven copies of the first column followed by -2 copies of the second (dashed) land on $(1,0)$; the weights $7, -2$ form the first column of A^{-1}. They are large because the two columns are nearly parallel.

matrix:

$$A\begin{bmatrix}a\\ b\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix}, \qquad A\begin{bmatrix}c\\ d\end{bmatrix} = \begin{bmatrix}0\\ 1\end{bmatrix}.$$

In general, column j of A^{-1} solves $A\boldsymbol{x} = \boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is column j of I. Finding A^{-1} means solving n systems at once. Gauss's elimination would solve them one at a time; Jordan's idea is to carry all n right-hand sides along as extra columns and keep eliminating *upwards* until the left half is I.

Example 4.9 (Gauss–Jordan on a 2×2 matrix)**.**

$$\left[\begin{array}{cc|cc}1 & 3 & 1 & 0\\ 2 & 7 & 0 & 1\end{array}\right] \xrightarrow{\text{row}_2 - 2\,\text{row}_1} \left[\begin{array}{cc|cc}1 & 3 & 1 & 0\\ 0 & 1 & -2 & 1\end{array}\right] \xrightarrow{\text{row}_1 - 3\,\text{row}_2} \left[\begin{array}{cc|cc}1 & 0 & 7 & -3\\ 0 & 1 & -2 & 1\end{array}\right].$$

The first step is ordinary elimination (Gauss stops at the upper triangular form). The second clears the entry *above* the pivot (Jordan continues). The right half is $A^{-1} = \begin{bmatrix}7 & -3\\ -2 & 1\end{bmatrix}$. Check: $A^{-1}A = \begin{bmatrix}7-6 & 21-21\\ -2+2 & -6+7\end{bmatrix} = I$.

Theorem 4.10 (Why Gauss–Jordan works)**.** *Suppose a sequence of row operations (elimination steps, row exchanges, and division of rows by nonzero pivots) turns* $[\,A \;\; I\,]$ *into* $[\,I \;\; E\,]$*. Then* $E = A^{-1}$*: both* $EA = I$ *and* $AE = I$*.*

Proof. Each row operation is multiplication on the left by an invertible matrix (an elimination matrix, a permutation, or a diagonal matrix with one nonzero entry changed), so the whole sequence is multiplication by their product, call it M. By block multiplication $M[\,A \;\; I\,] = [\,MA \;\; M\,]$, and this equals $[\,I \;\; E\,]$. So $M = E$ and $EA = I$. Moreover M is invertible, its inverse being the product of the inverse steps in reverse order. Hence $A = M^{-1}(MA) = M^{-1}$, and $AE = M^{-1}M = I$. □

Example 4.11 (A 3×3 inverse)**.** Invert $K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. Forward elimination (multipliers $-\frac{1}{2}$, then $-\frac{2}{3}$):

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array}\right].$$

Upward elimination: add $\frac{3}{4}$ of row 3 to row 2, then $\frac{2}{3}$ of the new row 2 to row 1; finally divide by the pivots $2, \frac{3}{2}, \frac{4}{3}$:

$$\to \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{array}\right].$$

So $K^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. Notice that K is symmetric and so is K^{-1}, and that K^{-1} has no zero entries although K has two.

Putting the pieces together gives the basic test for invertibility of a square matrix.

Theorem 4.12 (Invertibility test)**.** *For an $n \times n$ matrix A the following are equivalent:*

(a) A is invertible;

(b) $A\boldsymbol{x} = \mathbf{0}$ has only the solution $\boldsymbol{x} = \mathbf{0}$;

(c) elimination, with row exchanges allowed, produces n nonzero pivots.

Proof. (a)⇒(b) is Theorem 4.7. (c)⇒(a): with n pivots, continue upward and divide by the pivots; this is Gauss–Jordan, and Theorem 4.10 produces A^{-1}. (b)⇒(c): suppose elimination stops at column k because the entries of the current matrix M in column k, from row k down, are all zero. The first $k-1$ columns of M have pivots in rows $1, \dots, k-1$ and zeros below, so column k of M is a combination of them: back substitution in the top $(k-1) \times (k-1)$ triangle (Proposition 3.3) gives weights $y_1, \dots, y_{k-1}$. Then $\boldsymbol{x} = (-y_1, \dots, -y_{k-1}, 1, 0, \dots, 0) \neq \mathbf{0}$ satisfies $M\boldsymbol{x} = \mathbf{0}$. Since M is A multiplied on the left by invertible step matrices, also $A\boldsymbol{x} = \mathbf{0}$, contradicting (b). □

Corollary 4.13 (One side is enough)**.** *If A and B are $n \times n$ and $BA = I$, then A is invertible, $B = A^{-1}$, and $AB = I$.*

Proof. If $A\boldsymbol{x} = \mathbf{0}$ then $\boldsymbol{x} = BA\boldsymbol{x} = B\mathbf{0} = \mathbf{0}$, so (b) of Theorem 4.12 holds and A^{-1} exists. Then $B = B(AA^{-1}) = (BA)A^{-1} = A^{-1}$, and $AB = AA^{-1} = I$. (If instead $AB = I$, apply this to the pair B, A.) □

Common trap. The entry c_{ij} of AB is *not* $a_{ij}b_{ij}$; it uses a whole row of A and a whole column of B. And the right half of $[\,A \;\; I\,]$ becomes A^{-1} only if the left half really reaches I; stopping at the triangular form U leaves a matrix that is not the inverse. Finally, a matrix with no zero entries can still be singular, as $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ shows.

4.6 Exercises

Exercise 4.14. With $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$, $B = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$ and $C = \begin{bmatrix}0 & 0\\ 5 & 6\end{bmatrix}$, compute $AB+AC$ and $A(B+C)$. Then prove $A(B+C) = AB + AC$ in general.

Solution. $AB = \begin{bmatrix}1 & 0\\ 3 & 0\end{bmatrix}$ (column 1 of A in column 1) and $AC = \begin{bmatrix}10 & 12\\ 20 & 24\end{bmatrix}$ (rows of AC are $2(5,6)$ and $4(5,6)$), so $AB + AC = \begin{bmatrix}11 & 12\\ 23 & 24\end{bmatrix}$. Also $B + C = \begin{bmatrix}1 & 0\\ 5 & 6\end{bmatrix}$ and $A(B+C) = \begin{bmatrix}1+10 & 0+12\\ 3+20 & 0+24\end{bmatrix} = \begin{bmatrix}11 & 12\\ 23 & 24\end{bmatrix}$, the same. In general, the (i,j) entry of $A(B+C)$ is $\sum_k a_{ik}(b_{kj}+c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj}$, which is the (i,j) entry of $AB+AC$. □

Exercise 4.15. Use Gauss–Jordan elimination on $[\,U \;\; I\,]$ to find the inverse of the upper triangular matrix $U = \begin{bmatrix}1 & a & b\\ 0 & 1 & c\\ 0 & 0 & 1\end{bmatrix}$.

Solution. U is already triangular with pivots $1,1,1$, so only upward steps are needed. Subtract a times row 2 from row 1 and c times row 3 from row 2:

$$\left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0\\ 0 & 1 & c & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0\\ 0 & 1 & 0 & 0 & 1 & -c\\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Then subtract $b-ac$ times row 3 from row 1:

$$\to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & ac-b\\ 0 & 1 & 0 & 0 & 1 & -c\\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right], \qquad U^{-1} = \begin{bmatrix}1 & -a & ac-b\\ 0 & 1 & -c\\ 0 & 0 & 1\end{bmatrix}.$$

Check the $(1,3)$ entry of UU^{-1}: $1\cdot(ac-b) + a\cdot(-c) + b\cdot 1 = 0$. The inverse of an upper triangular matrix is upper triangular. □

Exercise 4.16. For which numbers a and b is $A = \begin{bmatrix}a & b & b\\ a & a & b\\ a & a & a\end{bmatrix}$ invertible? Find A^{-1} when it exists.

Solution. Quick tests first: if $a = 0$ the first column is zero, and if $a = b$ all rows are equal; either way A is singular. Gauss–Jordan: subtract row 1 from rows 2 and 3, then the new row 2 from row 3:

$$\left[\begin{array}{ccc|ccc} a & b & b & 1 & 0 & 0\\ a & a & b & 0 & 1 & 0\\ a & a & a & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} a & b & b & 1 & 0 & 0\\ 0 & a-b & 0 & -1 & 1 & 0\\ 0 & 0 & a-b & 0 & -1 & 1 \end{array}\right].$$

The pivots are a, $a-b$, $a-b$, so A is invertible exactly when $a \neq 0$ and $a \neq b$. Then divide rows 2 and 3 by $a-b$, divide row 1 by a, and subtract $\frac{b}{a}$ times rows 2 and 3 from row 1. The right half becomes

$$A^{-1} = \frac{1}{a-b}\begin{bmatrix}1 & 0 & -b/a\\ -1 & 1 & 0\\ 0 & -1 & 1\end{bmatrix}.$$

For example, the $(1,1)$ entry is $\frac{1}{a} + \frac{b}{a} \cdot \frac{1}{a-b} = \frac{(a-b)+b}{a(a-b)} = \frac{1}{a-b}$. Check row 3 of A times column 1 of A^{-1}: $\frac{a\cdot 1 + a\cdot(-1) + a\cdot 0}{a-b} = 0$. □

Exercise 4.17. Write $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}$ as a sum of two column-times-row matrices, and check the result against the entry rule.

Solution. By Proposition 4.2(c),

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}5 & 6\end{bmatrix} + \begin{bmatrix}2\\4\end{bmatrix}\begin{bmatrix}7 & 8\end{bmatrix} = \begin{bmatrix}5 & 6\\ 15 & 18\end{bmatrix} + \begin{bmatrix}14 & 16\\ 28 & 32\end{bmatrix} = \begin{bmatrix}19 & 22\\ 43 & 50\end{bmatrix}.$$

Entry rule: $c_{11} = 1\cdot 5 + 2\cdot 7 = 19$, $c_{12} = 1\cdot 6 + 2\cdot 8 = 22$, $c_{21} = 3\cdot 5 + 4\cdot 7 = 43$, $c_{22} = 3\cdot 6 + 4\cdot 8 = 50$. □

5 Factorization into $A = LU$

Section 3 wrote elimination as $EA = U$, with E a product of elimination matrices. This section turns that statement around into $A = LU$: a lower triangular L times an upper triangular U. The surprise is that L needs no computation at all; it simply stores the multipliers. On the way we need the rules for inverting and transposing a product, and afterwards we count how much work elimination costs (about $\frac{1}{3}n^3$ operations) and see what row exchanges do to the factorization.

5.1 Inverses and transposes of products

Theorem 5.1 (Inverse of a product)**.** *If A and B are invertible $n\times n$ matrices, then AB is invertible and*

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Proof. Using the associative law to regroup, $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$, and similarly $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$. □

The order reverses for the same reason as in everyday life: you put on socks, then shoes; to undo it you take off the shoes first. The same holds for any number of factors: $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Definition 5.2 (Transpose)**.** The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^{T} whose rows are the columns of A: $(A^{\mathsf{T}})_{ij} = a_{ji}$. For example $\begin{bmatrix}2 & 1\\ 8 & 7\end{bmatrix}^{\mathsf{T}} = \begin{bmatrix}2 & 8\\ 1 & 7\end{bmatrix}$.

Theorem 5.3 (Transpose of a product)**.** *If AB is defined, then $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$. If A is invertible, then so is A^{T}, and $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$.*

Proof. Compare entries: $\big((AB)^{\mathsf{T}}\big)_{ij} = (AB)_{ji} = \sum_k a_{jk}b_{ki} = \sum_k (B^{\mathsf{T}})_{ik}(A^{\mathsf{T}})_{kj} = (B^{\mathsf{T}}A^{\mathsf{T}})_{ij}$. For the second statement, transpose $AA^{-1} = I$ using the first: $(A^{-1})^{\mathsf{T}}A^{\mathsf{T}} = I^{\mathsf{T}} = I$. A one-sided inverse of a square matrix is a true inverse (Corollary 4.13), so $(A^{-1})^{\mathsf{T}}$ is the inverse of A^{T}. □

So for a single matrix, inverting and transposing can be done in either order.

5.2 From $EA = U$ to $A = LU$

Example 5.4 (LU and LDU for a 2×2 matrix)**.** The first pivot of $A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$ is 2 and the multiplier is $\ell_{21} = 8/2 = 4$:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}.$$

Multiply both sides on the left by E_{21}^{-1}, which *adds* back 4 times row 1:

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}.$$

L is lower triangular with ones on its diagonal and the multiplier 4 below; U is upper triangular with the pivots $2, 3$ on its diagonal. (Had the 7 been a 4, the second pivot would have been 0.) Sometimes it is useful to divide the pivots out of U into a diagonal matrix D, leaving ones on the diagonal of the last factor as well:

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \qquad (A = LDU).$$

For a 3×3 matrix with no row exchanges, elimination is $E_{32}E_{31}E_{21}A = U$, and inverting the three factors in reverse order gives

$$A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U = LU.$$

Why prefer $L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ to $E = E_{32}E_{31}E_{21}$?

Example 5.5 (E mixes the multipliers, L keeps them)**.** Take $\ell_{21} = 2$, $\ell_{31} = 0$, $\ell_{32} = 5$. Then

$$E = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix}}_{E_{32}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix}, \qquad L = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}^{-1}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}}_{E_{32}^{-1}} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

The 10 in E appears because 2 of row 1 went into row 2 and then 5 of that row 2 went into row 3, so row 1 reached row 3 with weight 10. In L the multipliers do not interfere: they sit in their places untouched.

This is true in general.

Theorem 5.6 (The multipliers go into L)**.** *Suppose elimination on the $n \times n$ matrix A succeeds without row exchanges, with multipliers ℓ_{ij} $(i > j)$ and upper triangular result U. Then $A = LU$, where L is the lower triangular matrix with ones on the diagonal and entry ℓ_{ij} in position (i, j) below it.*

Proof. Let $\boldsymbol{u}_j^{\mathsf{T}}$ denote row j of U. When elimination subtracts ℓ_{ij} times the pivot row j from row i, the pivot row is already finished (all its own eliminations used rows above it and happened earlier),

so what is subtracted is $\ell_{ij}\boldsymbol{u}_j^{\mathsf{T}}$. Row i undergoes these subtractions for $j = 1, \ldots, i-1$ and what remains is $\boldsymbol{u}_i^{\mathsf{T}}$:

$$\boldsymbol{u}_i^{\mathsf{T}} = (\text{row } i \text{ of } A) - \sum_{j<i} \ell_{ij}\boldsymbol{u}_j^{\mathsf{T}}, \qquad \text{so} \qquad \text{row } i \text{ of } A = \sum_{j<i} \ell_{ij}\boldsymbol{u}_j^{\mathsf{T}} + 1 \cdot \boldsymbol{u}_i^{\mathsf{T}}.$$

The right side is the combination of the rows of U with weights from row i of L, which by the row view (Proposition 4.2(b)) is row i of LU. □

Example 5.7 (LU for the elimination example)**.** For $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}$ of Section 3 the multipliers were $\ell_{21} = 3$, $\ell_{31} = 0$, $\ell_{32} = 2$ and the pivots $1, 2, 5$. So

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 2 & \\ & & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Check row 3 as in the proof: $0\,(1,2,1) + 2\,(0,2,-2) + 1\,(0,0,5) = (0,4,1)$, which is row 3 of A. To solve $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{b} = (2, 12, 2)$, solve two triangular systems:

$$\begin{aligned} L\boldsymbol{c} = \boldsymbol{b}: &\quad c_1 = 2, \quad c_2 = 12 - 3c_1 = 6, \quad c_3 = 2 - 0 \cdot c_1 - 2c_2 = -10; \\ U\boldsymbol{x} = \boldsymbol{c}: &\quad 5z = -10, \quad 2y - 2z = 6, \quad x + 2y + z = 2 \quad \Longrightarrow \quad (x, y, z) = (2, 1, -2). \end{aligned}$$

The vector $\boldsymbol{c}$ is exactly the right-hand side that forward elimination produced in Section 3: solving $L\boldsymbol{c} = \boldsymbol{b}$ *is* elimination applied to $\boldsymbol{b}$.

In a computer, A can be overwritten as elimination proceeds: once row i is finished, the new row of U and the multipliers ℓ_{ij} (which fill exactly the positions that became zero) are all that must be kept. Nothing else about A is needed again.

Common trap. L is not E with its signs flipped. The matrix $E = L^{-1}$ can have extra entries, like the 10 above; only L holds the plain multipliers. Also, "$A = LU$ exists" is about the *order of the rows*, not about invertibility: $\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ is invertible but has no LU factorization (its first pivot position holds 0), while a singular matrix whose last pivot is 0 still has one (Exercise 5.11).

5.3 How expensive is elimination?

How long does a computer need to solve n equations? Count the operations on a full $n \times n$ matrix, taking one "operation" to be a multiplication or division, each paired with at most one subtraction. The first stage clears column 1: for each of the $n-1$ rows below the pivot it computes a multiplier and updates the other $n-1$ entries of the row, about n^2 operations in all. The second stage does the same on the remaining $(n-1) \times (n-1)$ block, about $(n-1)^2$ operations, and so on down to the last 1×1 block. The total is about

$$n^2 + (n-1)^2 + \cdots + 2^2 + 1^2 \;\approx\; \int_0^n x^2\,dx = \frac{n^3}{3}.$$

Figure 5 compares the sum with the integral. The exact count is just as easy.

Proposition 5.8 (Operation count)**.** *Without row exchanges, reducing an $n \times n$ matrix A to U takes exactly $\frac{1}{3}(n^3 - n)$ operations (multiplications and divisions). Applying the same steps to one right-hand side and back substituting takes n^2 operations.*

Proof. At the stage with a $k \times k$ active block ($k = n, n-1, \dots, 2$), each of the $k-1$ rows below the pivot costs one division (its multiplier) and $k-1$ multiply-subtracts, so the stage costs $(k-1)k$. Summing,

$$\sum_{k=1}^{n}(k^2 - k) = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} = \frac{n(n+1)(2n-2)}{6} = \frac{(n-1)n(n+1)}{3} = \frac{n^3 - n}{3},$$

using the standard formulas for $\sum k^2$ and $\sum k$. For the right-hand side, the stage with a $k \times k$ block updates $k-1$ entries of $\boldsymbol{b}$, a total of $\sum_k (k-1) = \frac{1}{2}n(n-1)$. Back substitution for x_i uses $n-i$ multiply-subtracts and one division, a total of $\frac{1}{2}n(n-1)+n$. Together that is $n(n-1)+n = n^2$. □

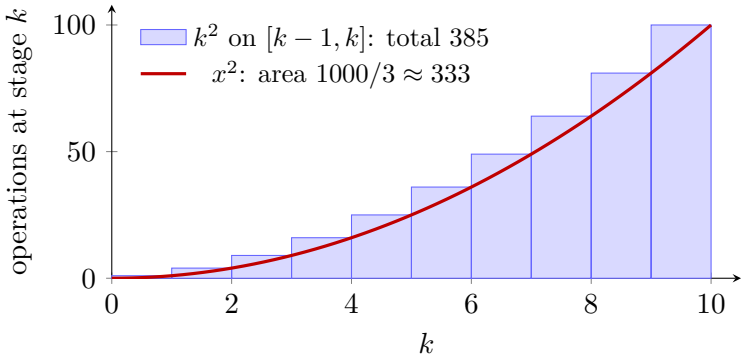


Figure 5: The work of elimination for $n = 10$. Stage k works on a $k \times k$ block and costs about k^2 operations (bars). The bars and the parabola x^2 enclose nearly the same area, so the total is about $\int_0^n x^2\,dx = n^3/3$. The exact count $\frac{1}{3}(n^3 - n) = 330$ is even closer, because the true cost of stage k is $k^2 - k$.

The cube matters. For $n = 100$ elimination costs $\frac{1}{3}(10^6 - 100) = 333{,}300$ operations; for $n = 1000$ about 3.3×10^8. Doubling n multiplies the work by 8. Each additional right-hand side costs only n^2, so a computer factors $A = LU$ once (the expensive part) and then solves for many vectors $\boldsymbol{b}$ cheaply. Matrices with many zeros in good positions (band matrices, for instance) cost far less, since zero entries need no elimination. By contrast, the determinant formula with $n!$ terms (Section 21) would be hopeless for even $n = 25$.

5.4 Row exchanges and permutation matrices

If a zero appears in a pivot position, elimination needs a row exchange, which is multiplication by a permutation matrix (Definition 3.10). For 3×3 matrices there are exactly six permutation matrices, one for each ordering of the rows of I:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The first four leave rows alone or exchange one pair; the last two move all three rows in a cycle (the fifth sends row 2 up to row 1, row 3 up to row 2 and row 1 down to row 3). The family is closed: a product of two of them reorders rows twice, which is again a reordering, and the inverse of each one is also in the list. The exchanges P_{12}, P_{13}, P_{23} are their own inverses, and the two cycles are inverses of each other. In every case the inverse is the transpose, $P^{-1} = P^{\mathsf{T}}$, a fact proved in Section 6. There are $n!$ permutation matrices of size n: 6 for $n = 3$ and 24 for $n = 4$.

With exchanges allowed, $A = LU$ becomes $PA = LU$: put the rows in a good order first, then eliminate. The smallest example is $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$. Its first pivot position holds 0, so exchange the rows: $P_{12}A = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$, which is already upper triangular, so $L = I$ and $U = P_{12}A$. Section 6 completes the story with a 3×3 example.

5.5 Exercises

Exercise 5.9. Which matrix E puts $A = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 2 & 0 & 1 \end{bmatrix}$ into triangular form $EA = U$? Multiply by $E^{-1} = L$ to factor $A = LU$.

Solution. The multipliers are $\ell_{21} = 2$ (row 2 becomes $(0, -2, 0)$), $\ell_{31} = 2$ (row 3 becomes $(0, -6, 1)$) and $\ell_{32} = -6/-2 = 3$ (row 3 becomes $(0, 0, 1)$). So $U = \begin{bmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and

$$E = E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{bmatrix}.$$

(The 4 is $-2 - 3 \cdot (-2)$: row 3 lost 2 of row 1 directly and gained 6 of it through row 2.) The inverse just collects the multipliers: $L = E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}$, and one checks $LU = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 6-2 & 0 \\ 2 & 6-6 & 1 \end{bmatrix} = A$. □

Exercise 5.10. Compute L and U for the symmetric matrix $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$. Find four conditions on a, b, c, d for $A = LU$ with four pivots.

Solution. Subtract row 1 from rows 2, 3 and 4 (multipliers 1): the rows become $(0, b-a, b-a, b-a)$, $(0, b-a, c-a, c-a)$, $(0, b-a, c-a, d-a)$. Subtract the new row 2 from rows 3 and 4 (multipliers 1): $(0, 0, c-b, c-b)$ and $(0, 0, c-b, d-b)$. Subtract row 3 from row 4 (multiplier 1): $(0, 0, 0, d-c)$. Hence

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}.$$

Four nonzero pivots, with no row exchanges, require $a \neq 0$, $b \neq a$, $c \neq b$ and $d \neq c$. □

Common trap. The posted solution to this problem lists the last condition as $d = c$. That is a slip: with $d = c$ the fourth pivot $d - c$ is zero. The four conditions are all inequalities.

Exercise 5.11. Find the LU factorization of $A = \begin{bmatrix} 1 & 0 & 1 \\ a & a & a \\ b & b & a \end{bmatrix}$ when it exists. For which real numbers a, b does it exist?

Solution. Step $(2,1)$: $\ell_{21} = a$, row 2 becomes $(0, a, 0)$. Step $(3,1)$: $\ell_{31} = b$, row 3 becomes $(0, b, a-b)$. Step $(3,2)$ needs the pivot a to be nonzero; then $\ell_{32} = b/a$ and row 3 becomes $(0, 0, a - b)$. So, for $a \neq 0$,

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & b/a & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & 0 & a-b \end{bmatrix}.$$

If $a = 0$, row 2 of the partly reduced matrix is zero while row 3 is $(0, b, -b)$; for $b \neq 0$ a row exchange is required, and for $b = 0$ there is no pivot in column 2 at all. Either way there is no LU factorization with a nonzero second pivot, so the condition is $a \neq 0$. The third pivot $a - b$ may be zero: for $a = b \neq 0$ the matrix is singular, yet $A = LU$ exists, because no row exchange was needed to reach U. Always check by multiplying: row 3 of LU is $b(1,0,1) + \frac{b}{a}(0, a, 0) + (0, 0, a - b) = (b, b, a)$. □

Exercise 5.12. A computer performs 10^9 operations per second. Using Proposition 5.8, estimate the time to factor a full 1000×1000 matrix, the time for each additional right-hand side, and the time to factor a $10{,}000 \times 10{,}000$ matrix.

Solution. For $n = 1000$: $\frac{1}{3}(10^9 - 10^3) \approx 3.3 \times 10^8$ operations, about 0.33 seconds. Each right-hand side then costs $n^2 = 10^6$ operations, one millisecond. For $n = 10^4$ the factorization costs about $\frac{1}{3} \times 10^{12}$ operations, roughly 333 seconds or five and a half minutes: ten times the size, a thousand times the work. (Real machines are faster, and sparse matrices are much cheaper, but the n^3 growth is the same.) □

6 Transposes, permutations, vector spaces

This section closes the chapter on elimination and opens the one on vector spaces. First we finish two loose ends: permutation matrices, which complete $A = LU$ to $PA = LU$, and transposes, which produce the symmetric matrices that dominate applications. Then comes the step that turns a toolbox of computations into linear algebra proper: instead of individual vectors we study *spaces* of vectors, and especially the subspaces that live inside $\mathbb{R}^n$. The first such subspace attached to a matrix is its column space.

6.1 Permutations complete elimination: $PA = LU$

A permutation matrix is the identity with its rows reordered (Definition 3.10); PA puts the rows of A in the same new order. There is one permutation matrix for each ordering of n rows. The first row of P can be any of the n rows of I, the second any of the remaining $n - 1$, and so on, so there are $n(n-1)\cdots 2 \cdot 1 = n!$ permutation matrices of size n: 6 when $n = 3$, 24 when $n = 4$, 120 when $n = 5$.

Theorem 6.1 (The inverse of a permutation is its transpose)**.** *Every permutation matrix satisfies $P^{\mathsf{T}}P = I$, so $P^{-1} = P^{\mathsf{T}}$.*

Proof. The rows of P are the rows of I in a new order, so each column of P contains exactly one 1 and the columns of P are the standard vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ in some order, each appearing once. The (i, j) entry of $P^{\mathsf{T}}P$ is (column i of P)$\cdot$(column j of P). For $i = j$ this is $\boldsymbol{e}_k \cdot \boldsymbol{e}_k = 1$; for $i \neq j$ it is the dot product of two different standard vectors, which is 0. Hence $P^{\mathsf{T}}P = I$, and by Corollary 4.13 P^{T} is the inverse of P. □

Section 5 factored $A = LU$ when elimination needs no row exchanges. Most matrices need none, but some do: a zero can land in a pivot position while a nonzero sits below it. The cure is to do all the necessary exchanges first.

Theorem 6.2 ($PA = LU$)**.** *If A is an invertible $n \times n$ matrix, there is a permutation matrix P such that elimination on PA needs no row exchanges, so that $PA = LU$ with L lower triangular with unit diagonal and U upper triangular with the (nonzero) pivots on its diagonal.*

Sketch. Run elimination on A, exchanging rows whenever a pivot position holds zero; since A is invertible a nonzero entry is always available below (Theorem 4.12). Let P be the product of all the exchanges made. A row exchange only swaps two rows that are not yet pivot rows, and elimination never mixes a row into the rows above it, so performing all the exchanges at the very start puts every row where it eventually ends up. Elimination on PA then meets the same pivots in the same places and needs no exchange, and Theorem 5.6 gives $PA = LU$. (In the bookkeeping, a later exchange of rows i and k also exchanges the multipliers already recorded in those rows of L.) □

Example 6.3 (A 3×3 factorization $PA = LU$)**.** Let $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 3 \end{bmatrix}$. The $(1,1)$ position holds 0, so exchange rows 1 and 2:

$$P_{12}A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}.$$

Now eliminate without further exchanges: $\ell_{21} = 0$; $\ell_{31} = 2$ turns row 3 into $(2, 0, 3) - 2(1, 1, 1) = (0, -2, 1)$; then $\ell_{32} = -2/2 = -1$ turns it into $(0, -2, 1) + (0, 2, 1) = (0, 0, 2)$. Therefore

$$\underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{P} \underbrace{\begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 3 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}}_{U}.$$

Check row 3: $2\,(1, 1, 1) - 1\,(0, 2, 1) + 1\,(0, 0, 2) = (2, 0, 3)$, which is row 3 of PA. The pivots are $1, 2, 2$, and A itself has no LU factorization because its first pivot position holds zero.

In practice, software exchanges rows more often than algebra requires. MATLAB's elimination chooses, in each column, the largest available entry as the pivot (*partial pivoting*): dividing by a tiny pivot would magnify rounding errors. The algebra is the same, $PA = LU$; only P changes.

6.2 Transposes and symmetric matrices

The transpose (Definition 5.2) flips a matrix across its main diagonal, $(A^{\mathsf{T}})_{ij} = a_{ji}$; the rows of A become the columns of A^{T}. A 3×2 matrix becomes 2×3:

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^{\mathsf{T}} = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}.$$

Definition 6.4 (Symmetric matrix)**.** A square matrix S is *symmetric* if $S^{\mathsf{T}} = S$, that is, $s_{ij} = s_{ji}$ for all i, j.

The diagonal of a symmetric matrix is unrestricted, and each entry above the diagonal is mirrored below it, as in $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix}$. Symmetric matrices are the most important special family in applications, and the most common way they arise is this.

Theorem 6.5 ($R^{\mathsf{T}}R$ is symmetric)**.** *For every $m \times n$ matrix R, the $n \times n$ matrix $R^{\mathsf{T}}R$ is symmetric, and so is the $m \times m$ matrix RR^{T}.*

Proof. By the rule for transposes of products (Theorem 5.3) and $(R^{\mathsf{T}})^{\mathsf{T}} = R$,

$$(R^{\mathsf{T}}R)^{\mathsf{T}} = R^{\mathsf{T}}(R^{\mathsf{T}})^{\mathsf{T}} = R^{\mathsf{T}}R.$$

The same computation with R^{T} in place of R shows $(RR^{\mathsf{T}})^{\mathsf{T}} = RR^{\mathsf{T}}$. □

Example 6.6 (Building symmetric matrices from a rectangular one)**.** Take the 2×3 matrix $R = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$. Then

$$R^{\mathsf{T}}R = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 7 \\ 11 & 13 & 11 \\ 7 & 11 & 17 \end{bmatrix}, \qquad RR^{\mathsf{T}} = \begin{bmatrix} 21 & 13 \\ 13 & 19 \end{bmatrix}.$$

The $(1,2)$ entry of $R^{\mathsf{T}}R$ is row 1 of R^{T} times column 2 of R, namely $(1,3) \cdot (2,3) = 11$; the $(2,1)$ entry is $(2,3) \cdot (1,3)$, the same dot product taken in the other order. In general $(R^{\mathsf{T}}R)_{ij}$ is the dot product of columns i and j of R, which is why the matrix must be symmetric. The product $R^{\mathsf{T}}R$ will be central to least squares (Section 17).

6.3 Vector spaces

The two operations we perform on vectors are adding them and multiplying them by numbers (scalars). A *space* of vectors is a collection in which these operations can always be carried out without leaving the collection.

Definition 6.7 (The space $\mathbb{R}^n$)**.** $\mathbb{R}^n$ is the set of all column vectors with n real components, with addition and scalar multiplication done component by component.

$\mathbb{R}^2$ contains $(3,2)$, $(0,0)$, (π,e) and every other pair of real numbers; drawn as arrows from the origin (or as their endpoints), it is the whole xy-plane. The vector $(3,2,0)$ belongs to $\mathbb{R}^3$, not to $\mathbb{R}^2$: it has three components, one of which happens to be zero.

Definition 6.8 (Vector space)**.** A *vector space* is a nonempty set V of objects (vectors) with an addition and a multiplication by real scalars such that V is *closed* under both: if $\boldsymbol{v}, \boldsymbol{w} \in V$ and $c \in \mathbb{R}$, then $\boldsymbol{v} + \boldsymbol{w} \in V$ and $c\boldsymbol{v} \in V$. The operations must obey the usual rules: $\boldsymbol{v} + \boldsymbol{w} = \boldsymbol{w} + \boldsymbol{v}$; $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$; there is a zero vector with $\boldsymbol{v} + \mathbf{0} = \boldsymbol{v}$; each $\boldsymbol{v}$ has a negative with $\boldsymbol{v} + (-\boldsymbol{v}) = \mathbf{0}$; $1\boldsymbol{v} = \boldsymbol{v}$; $c(d\boldsymbol{v}) = (cd)\boldsymbol{v}$; $c(\boldsymbol{v} + \boldsymbol{w}) = c\boldsymbol{v} + c\boldsymbol{w}$; and $(c + d)\boldsymbol{v} = c\boldsymbol{v} + d\boldsymbol{v}$.

In $\mathbb{R}^n$ the eight rules hold automatically, because they hold for each component; the question in practice is always closure. Closure under both operations is the same as closure under linear combinations $c\boldsymbol{v} + d\boldsymbol{w}$. Vector spaces need not consist of columns: the set of all 2×2 matrices, with the usual addition and scalar multiplication, is a vector space too (Section 12 studies such spaces).

Proposition 6.9 (Every vector space contains zero)**.** *If V is a vector space, then $\mathbf{0} \in V$.*

Proof. V is nonempty; pick any $\boldsymbol{v} \in V$. Closure under scalar multiplication puts $0 \cdot \boldsymbol{v}$ in V, and $0 \cdot \boldsymbol{v} = (0 + 0)\boldsymbol{v} = 0 \cdot \boldsymbol{v} + 0 \cdot \boldsymbol{v}$; adding the negative of $0 \cdot \boldsymbol{v}$ to both sides gives $0 \cdot \boldsymbol{v} = \mathbf{0}$. □

So the xy-plane with the origin removed is not a vector space. Neither is the quarter-plane Q of vectors with both components nonnegative: adding two vectors of Q stays in Q, but $-1 \cdot (3,2) = (-3,-2)$ does not. Q is closed under addition but not under multiplication by all real numbers (right panel of Figure 6).

6.4 Subspaces

Definition 6.10 (Subspace)**.** A *subspace* of a vector space V is a nonempty subset $S \subseteq V$ that is itself a vector space with the same operations. Equivalently, S is nonempty and closed under addition and scalar multiplication: $\boldsymbol{v}, \boldsymbol{w} \in S$ and $c \in \mathbb{R}$ imply $\boldsymbol{v} + \boldsymbol{w} \in S$ and $c\boldsymbol{v} \in S$.

The two descriptions agree because the eight rules are inherited from V, and the zero vector and negatives lie in S by closure ($0\boldsymbol{v} = \mathbf{0}$, $(-1)\boldsymbol{v} = -\boldsymbol{v}$). By Proposition 6.9 every subspace contains $\mathbf{0}$.

If a subspace of $\mathbb{R}^2$ contains a nonzero vector $\boldsymbol{v}$, it contains every multiple $c\boldsymbol{v}$: the whole line through $\mathbf{0}$ and $\boldsymbol{v}$. That line $L = \{c\boldsymbol{v} : c \in \mathbb{R}\}$ is a subspace, since $c\boldsymbol{v} + d\boldsymbol{v} = (c + d)\boldsymbol{v}$ and $k(c\boldsymbol{v}) = (kc)\boldsymbol{v}$ stay on it. A line that misses the origin is never a subspace: multiplying one of its vectors by 0 leaves the line (left panel of Figure 6).

Proposition 6.11 (The subspaces of $\mathbb{R}^2$)**.** *Every subspace of $\mathbb{R}^2$ is one of: the zero subspace $Z = \{\mathbf{0}\}$; a line through the origin; all of $\mathbb{R}^2$.*

Proof. Let S be a subspace. If $S = \{\mathbf{0}\}$ we are done. Otherwise S contains some $\boldsymbol{v} \neq \mathbf{0}$ and hence the line L of its multiples. If $S = L$ we are done. Otherwise S contains a vector $\boldsymbol{w}$ off that line, so $\delta = v_1 w_2 - v_2 w_1 \neq 0$ (if $\delta = 0$, $\boldsymbol{w}$ would be a multiple of $\boldsymbol{v}$). For any $\boldsymbol{b} \in \mathbb{R}^2$, the weights $x = (b_1 w_2 - b_2 w_1)/\delta$ and $y = (v_1 b_2 - v_2 b_1)/\delta$ satisfy $x\boldsymbol{v} + y\boldsymbol{w} = \boldsymbol{b}$, as direct substitution confirms. By closure $\boldsymbol{b} \in S$, so $S = \mathbb{R}^2$. □

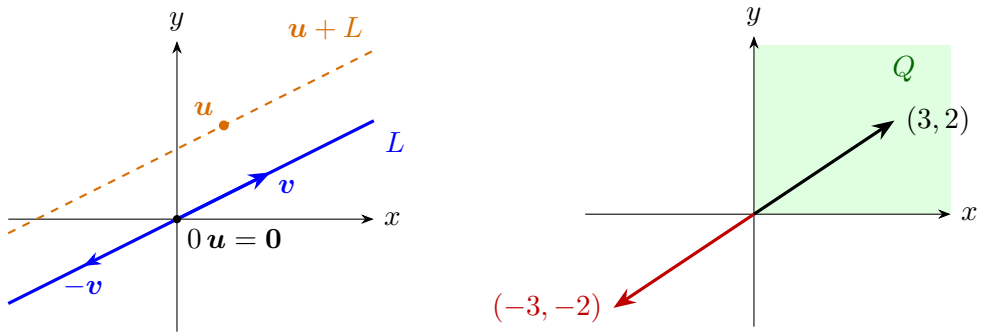


Figure 6: Left: the line L through the origin (all multiples of $\boldsymbol{v}$) is a subspace of $\mathbb{R}^2$. The parallel dashed line through $\boldsymbol{u} = (1, 2)$ is not: $0\,\boldsymbol{u} = \boldsymbol{0}$ is not on it. Right: the quarter-plane Q is closed under addition but not under scalar multiplication, since $-1 \cdot (3, 2)$ leaves it.

Similarly, the subspaces of $\mathbb{R}^3$ are Z, the lines through the origin, the planes through the origin, and $\mathbb{R}^3$ itself; the proof uses the idea of dimension (Section 10). A line through the origin in $\mathbb{R}^2$ looks like $\mathbb{R}^1$, but it is not $\mathbb{R}^1$: its vectors have two components.

Common trap. $\mathbb{R}^2$ is *not* a subspace of $\mathbb{R}^3$: its vectors have two components, so it is not even a subset of $\mathbb{R}^3$. The plane of vectors $(x, y, 0)$ *is* a subspace of $\mathbb{R}^3$; it looks like $\mathbb{R}^2$ but is a different set. Also, closure under addition alone proves nothing (the quarter-plane passes that test); always check scalar multiplication by negative numbers and by zero.

6.5 The column space of a matrix

Subspaces of $\mathbb{R}^m$ come naturally from matrices. Put two vectors in the columns of a matrix; a subspace containing them must contain all their combinations, and the set of all combinations is already a subspace.

Definition 6.12 (Column space)**.** The *column space* $\mathbf{C}(A)$ of an $m \times n$ matrix A is the set of all linear combinations of the columns of A. Equivalently, $\mathbf{C}(A) = \{A\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^n\}$. It is a subspace of $\mathbb{R}^m$.

It is a subspace because $A\boldsymbol{x}+A\boldsymbol{y} = A(\boldsymbol{x}+\boldsymbol{y})$ and $c(A\boldsymbol{x}) = A(c\boldsymbol{x})$ are again of the form A(something).

Example 6.13 (A plane in $\mathbb{R}^3$)**.** Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$. Its columns $(1, 2, 4)$ and $(3, 3, 1)$ are vectors in $\mathbb{R}^3$ that do not lie on one line, so their combinations fill a plane through the origin: $\mathbf{C}(A)$ is a plane in $\mathbb{R}^3$. It contains $\boldsymbol{0}$, both columns, their sum $(4, 5, 5)$, and, e.g., $(1, 2, 4) + 3(3, 3, 1) = (10, 11, 7)$. Which plane? A vector $\boldsymbol{b}$ is in $\mathbf{C}(A)$ when $x_1(1, 2, 4) + x_2(3, 3, 1) = \boldsymbol{b}$ can be solved. Eliminate on the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 3 & b_1 \\ 2 & 3 & b_2 \\ 4 & 1 & b_3 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 3 & b_1 \\ 0 & -3 & b_2 - 2b_1 \\ 0 & -11 & b_3 - 4b_1 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 3 & b_1 \\ 0 & -3 & b_2 - 2b_1 \\ 0 & 0 & b_3 - 4b_1 - \frac{11}{3}(b_2 - 2b_1) \end{array}\right].$$

The last row says $0 = \frac{10}{3}b_1 - \frac{11}{3}b_2 + b_3$. So

$$\mathbf{C}(A) = \{\boldsymbol{b} \in \mathbb{R}^3 : 10b_1 - 11b_2 + 3b_3 = 0\}.$$

Check: $(1,2,4)$ gives $10-22+12=0$, $(3,3,1)$ gives $30-33+3=0$, $(4,5,5)$ gives $40-55+15=0$. The vector $(1,0,0)$ gives $10\neq 0$; it is not in the column space, and $A\boldsymbol{x}=(1,0,0)$ has no solution.

The same construction works in any dimension: five vectors in $\mathbb{R}^{10}$, each with ten components, have combinations that fill a subspace of $\mathbb{R}^{10}$, at most five-dimensional and smaller if the vectors are dependent. How the column space decides which systems $A\boldsymbol{x}=\boldsymbol{b}$ are solvable is the subject of Section 7.

6.6 Exercises

Exercise 6.14. (a) Find a 3×3 permutation matrix with $P^3=I$ but $P\neq I$. (b) Find a 4×4 permutation matrix $\widehat{P}$ with $\widehat{P}^4\neq I$.

Solution. (a) Move the rows in a cycle: row 1 to position 2, row 2 to position 3, row 3 to position 1.

$$P=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix},\qquad P^2=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix},\qquad P^3=I.$$

Indeed $P(1,2,3)=(3,1,2)$, and applying the cycle three times returns every row home. (b) Use the block matrix $\widehat{P}=\begin{bmatrix}1&0\\0&P\end{bmatrix}$, which leaves row 1 alone and cycles rows 2, 3, 4. By block multiplication $\widehat{P}^3=I$, so $\widehat{P}^4=\widehat{P}\neq I$. □

Exercise 6.15. How many entries of a 4×4 matrix A can be chosen independently if (a) A is symmetric; (b) A is skew-symmetric, $A^{\mathsf{T}}=-A$?

Solution. (a) The 4 diagonal entries and the 6 entries above the diagonal are free; the 6 below are copies. So 10 entries: $A=\begin{bmatrix}a&e&f&g\\e&b&h&i\\f&h&c&j\\g&i&j&d\end{bmatrix}$. (b) Now $a_{ii}=-a_{ii}$ forces a zero diagonal, and the entries below the diagonal are the negatives of those above. So 6 entries: $A=\begin{bmatrix}0&-a&-b&-c\\a&0&-d&-e\\b&d&0&-f\\c&e&f&0\end{bmatrix}$. □

Exercise 6.16. In the vector space M of all $n\times n$ matrices, decide which of these are subspaces: (a) the symmetric matrices; (b) the skew-symmetric matrices; (c) the unsymmetric matrices ($A^{\mathsf{T}}\neq A$).

Solution. (a) Yes. If $A^{\mathsf{T}}=A$ and $B^{\mathsf{T}}=B$, then $(A+B)^{\mathsf{T}}=A^{\mathsf{T}}+B^{\mathsf{T}}=A+B$ and $(cA)^{\mathsf{T}}=cA^{\mathsf{T}}=cA$. (b) Yes. If $A^{\mathsf{T}}=-A$ and $B^{\mathsf{T}}=-B$, then $(A+B)^{\mathsf{T}}=-A-B=-(A+B)$ and $(cA)^{\mathsf{T}}=-cA$. (c) No. The zero matrix is symmetric, so it is not in the set; and the set is not closed under addition: $\begin{bmatrix}1&1\\0&0\end{bmatrix}+\begin{bmatrix}0&0\\1&1\end{bmatrix}=\begin{bmatrix}1&1\\1&1\end{bmatrix}$, two unsymmetric matrices with a symmetric sum. □

Exercise 6.17. Let $\boldsymbol{x}_1=(1,2,3)$ and $\boldsymbol{x}_2=(1,3,0)$ in $\mathbb{R}^3$. Let V_1 and V_2 be the smallest subspaces containing $\boldsymbol{x}_1$, respectively $\boldsymbol{x}_2$, and V_3 the smallest subspace containing both. (a) Describe V_1, V_2 and $V_1\cap V_2$. (b) Is $V_1\cup V_2$ a subspace? Is it V_3? (c) Find a subspace S of V_3 containing neither $\boldsymbol{x}_1$ nor $\boldsymbol{x}_2$. (d) Describe $V_3\cap\{\text{the } xy\text{-plane}\}$.

Solution. (a) A subspace containing $\boldsymbol{x}_1$ contains all its multiples, and the multiples already form a subspace, so V_1 is the line through $(1,2,3)$; likewise V_2 is the line through $(1,3,0)$. If $c(1,2,3) = d(1,3,0)$, the third components give $3c = 0$, so $c = 0 = d$: the lines meet only at the origin, and $V_1 \cap V_2 = \{\mathbf{0}\}$, the zero subspace. (b) No. $\boldsymbol{x}_1 + \boldsymbol{x}_2 = (2,5,3)$ is on neither line ($(2,5,3)$ is not a multiple of $(1,2,3)$ since $5 \neq 2 \cdot 2$, and its third component is not 0), so the union is not closed under addition and cannot equal V_3. Instead V_3 is the plane of all combinations $s\boldsymbol{x}_1 + t\boldsymbol{x}_2 = (s+t,\, 2s+3t,\, 3s)$. Eliminating $s = b_3/3$ and $t = b_1 - b_3/3$ from $b_2 = 2s + 3t$ gives its equation $9b_1 - 3b_2 - b_3 = 0$. (c) The line through $\boldsymbol{x}_1 + \boldsymbol{x}_2 = (2,5,3)$ lies in V_3 and contains neither $\boldsymbol{x}_1$ nor $\boldsymbol{x}_2$ (neither is a multiple of $(2,5,3)$). Other choices, such as the line through $2\boldsymbol{x}_1 + \boldsymbol{x}_2$, work too. (d) Points of the xy-plane have $b_3 = 0$; on V_3 this forces $9b_1 = 3b_2$, i.e. $b_2 = 3b_1$. So the intersection is the line through $(1,3,0)$, which is V_2. This makes sense: $\boldsymbol{x}_2$ lies in both planes, and two different planes through the origin meet in a line. □

7 Column space and nullspace

Every matrix carries two subspaces that answer the two basic questions about $A\boldsymbol{x} = \boldsymbol{b}$. The *column space* $\mathbf{C}(A)$ answers "for which $\boldsymbol{b}$ is there a solution?": exactly those $\boldsymbol{b}$ that are combinations of the columns. The *nullspace* $\mathbf{N}(A)$ answers "how many solutions are there?": any two solutions differ by a vector that A sends to zero. The two subspaces are built in opposite ways. The column space is *spanned*: we are handed some vectors and take all their combinations. The nullspace is *cut out by equations*: we are told the conditions $A\boldsymbol{x} = \mathbf{0}$ and must find which vectors satisfy them. This section works both out for small examples by inspection; Section 8 gives the systematic method by elimination.

7.1 Unions and intersections of subspaces

Recall that a subspace is a nonempty set of vectors closed under addition and scalar multiplication (Definition 6.10); equivalently, for any $\boldsymbol{v}, \boldsymbol{w}$ in the subspace and any numbers c, d, the combination $c\boldsymbol{v} + d\boldsymbol{w}$ is in it. In $\mathbb{R}^3$ a plane P through the origin and a line L through the origin are subspaces. What about combining them?

The union is usually not a subspace. Take P to be the x_1x_2-plane (vectors $(x_1, x_2, 0)$) and L the x_3-axis. Then $(1,0,0) \in P$ and $(0,0,1) \in L$, but their sum $(1,0,1)$ is in neither, so $P \cup L$ is not closed under addition.

Theorem 7.1 (Intersections of subspaces are subspaces)**.** *If S and T are subspaces of a vector space V, then $S \cap T$, the set of vectors lying in both, is a subspace of V.*

Proof. $\mathbf{0}$ lies in both S and T (Proposition 6.9), so $S \cap T$ is nonempty. Let $\boldsymbol{v}, \boldsymbol{w} \in S \cap T$ and $c, d \in \mathbb{R}$. Since $\boldsymbol{v}, \boldsymbol{w} \in S$ and S is a subspace, $c\boldsymbol{v} + d\boldsymbol{w} \in S$. Since $\boldsymbol{v}, \boldsymbol{w} \in T$ and T is a subspace, $c\boldsymbol{v} + d\boldsymbol{w} \in T$. So $c\boldsymbol{v} + d\boldsymbol{w} \in S \cap T$. □

For the plane and line above, $P \cap L = \{\mathbf{0}\}$, the zero subspace. Intersecting adds requirements, so the intersection is usually smaller, but it is always a subspace. (Exercise 7.9 shows that the right way to combine two subspaces into a bigger one is their sum $S + T$, not their union.)

7.2 The column space decides solvability

Our example is the 4×3 matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}, \qquad A\boldsymbol{x} = x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}. \tag{8}$$

Its columns are vectors in $\mathbb{R}^4$, so its column space $\mathbf{C}(A) = \{A\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^3\}$ (Definition 6.12) is a subspace of $\mathbb{R}^4$. These are four equations in only three unknowns, and we should not expect a solution for every $\boldsymbol{b}$: the combinations of three vectors cannot fill four-dimensional space. But some right-hand sides are fine, and they are exactly the vectors in the column space.

Theorem 7.2 (Solvability)**.** *$A\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if $\boldsymbol{b}$ is in the column space $\mathbf{C}(A)$.*

Proof. $A\boldsymbol{x}$ is the combination of the columns with weights $x_1, \ldots, x_n$. If a solution $\boldsymbol{x}$ exists, $\boldsymbol{b} = A\boldsymbol{x}$ is such a combination, so $\boldsymbol{b} \in \mathbf{C}(A)$. Conversely, if $\boldsymbol{b}$ is a combination $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$, those weights form a solution $\boldsymbol{x}$. □

A good way to find vectors in $\mathbf{C}(A)$ is to choose $\boldsymbol{x}$ first and compute $\boldsymbol{b}$. The right side $\boldsymbol{b} = \mathbf{0}$ is always reached (by $\boldsymbol{x} = \mathbf{0}$); $\boldsymbol{b} = (1, 2, 3, 4)$ is column 1, reached by $\boldsymbol{x} = (1, 0, 0)$; $\boldsymbol{b} = (1, 1, 1, 1)$ is column 2, reached by $\boldsymbol{x} = (0, 1, 0)$.

How big is $\mathbf{C}(A)$? Column 3 is the sum of columns 1 and 2, so it contributes nothing new: by Proposition 1.7 every combination of all three columns is already a combination of the first two. Columns 1 and 2 are not on one line, so their combinations form a plane. Hence $\mathbf{C}(A)$ *is a two-dimensional subspace (a plane) of* $\mathbb{R}^4$. We could equally have thrown away column 1, since column 1 = column 3 − column 2; the convention (made precise in Section 8) is to keep the earliest columns that are not combinations of earlier ones, the *pivot columns*.

Example 7.3 (Equations for the column space)**.** Which $\boldsymbol{b} \in \mathbb{R}^4$ lie in $\mathbf{C}(A)$? We need $\boldsymbol{b} = x_1(1, 2, 3, 4) + x_2(1, 1, 1, 1)$, that is, $b_i = i\,x_1 + x_2$ for $i = 1, 2, 3, 4$: the components of $\boldsymbol{b}$ must form an arithmetic progression. That is two equations:

$$\mathbf{C}(A) = \{\boldsymbol{b} \in \mathbb{R}^4 : \ b_1 - 2b_2 + b_3 = 0 \text{ and } b_2 - 2b_3 + b_4 = 0\}.$$

Indeed each column passes both tests (for column 3: $2 - 6 + 4 = 0$ and $3 - 8 + 5 = 0$), and the tests are preserved by combinations. Conversely, if $\boldsymbol{b}$ passes both tests then the differences $b_2 - b_1 = b_3 - b_2 = b_4 - b_3$ share a common value d, and $\boldsymbol{b} = d\,(1, 2, 3, 4) + (b_1 - d)\,(1, 1, 1, 1)$. So $A\boldsymbol{x} = (2, 3, 4, 5)$ is solvable (it is column 3), while $A\boldsymbol{x} = (1, 2, 3, 5)$ is not, because $b_2 - 2b_3 + b_4 = 2 - 6 + 5 = 1 \neq 0$.

7.3 The nullspace

Definition 7.4 (Nullspace)**.** The *nullspace* of an $m \times n$ matrix A is the set of all solutions of $A\boldsymbol{x} = \mathbf{0}$:

$$\mathbf{N}(A) = \{\boldsymbol{x} \in \mathbb{R}^n : \ A\boldsymbol{x} = \mathbf{0}\}.$$

The column space contains right-hand sides $\boldsymbol{b} \in \mathbb{R}^m$; the nullspace contains inputs $\boldsymbol{x} \in \mathbb{R}^n$. For the 4×3 example, $\mathbf{C}(A) \subseteq \mathbb{R}^4$ while $\mathbf{N}(A) \subseteq \mathbb{R}^3$.

Theorem 7.5 (The nullspace is a subspace)**.** *For every $m \times n$ matrix A, $\mathbf{N}(A)$ is a subspace of $\mathbb{R}^n$.*

Proof. $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \mathbf{N}(A)$. If $A\boldsymbol{v} = \mathbf{0}$ and $A\boldsymbol{w} = \mathbf{0}$, then by the distributive law $A(\boldsymbol{v} + \boldsymbol{w}) = A\boldsymbol{v} + A\boldsymbol{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}$, and for any scalar c, $A(c\boldsymbol{v}) = c\,A\boldsymbol{v} = c\,\mathbf{0} = \mathbf{0}$. So $\mathbf{N}(A)$ is closed under addition and scalar multiplication. □

Example 7.6 (A nullspace that is a line)**.** For A in (8), the relation column 1 + column 2 − column 3 = $\mathbf{0}$ says that $(1, 1, -1) \in \mathbf{N}(A)$, and so is every multiple $c(1, 1, -1)$. Are there other solutions? Write $\boldsymbol{a}_1, \boldsymbol{a}_2$ for the first two columns. Since $\boldsymbol{a}_3 = \boldsymbol{a}_1 + \boldsymbol{a}_2$,

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + x_3(\boldsymbol{a}_1 + \boldsymbol{a}_2) = (x_1 + x_3)\,\boldsymbol{a}_1 + (x_2 + x_3)\,\boldsymbol{a}_2.$$

Because $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ are not parallel, this combination is zero only when both weights are zero: $x_1 = -x_3$ and $x_2 = -x_3$. So $\boldsymbol{x} = -x_3(1, 1, -1)$, and

$$\mathbf{N}(A) = \{c\,(1, 1, -1) : c \in \mathbb{R}\},$$

a line through the origin in $\mathbb{R}^3$.

7.4 The solutions of $A\boldsymbol{x} = \boldsymbol{b}$ are a shifted nullspace

Now take a right-hand side that is not zero, say $\boldsymbol{b} = (1, 2, 3, 4)$. One solution is $\boldsymbol{x} = (1, 0, 0)$; another is $(0, -1, 1)$, since −column 2 + column 3 = $(1, 2, 3, 4)$. The solutions do *not* form a subspace: $\boldsymbol{x} = \mathbf{0}$ is not a solution, because $A\mathbf{0} = \mathbf{0} \neq \boldsymbol{b}$. Their structure is described next.

Proposition 7.7 (All solutions)**.** *Suppose $A\boldsymbol{x}_p = \boldsymbol{b}$. Then the solutions of $A\boldsymbol{x} = \boldsymbol{b}$ are exactly the vectors $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ with $\boldsymbol{x}_n \in \mathbf{N}(A)$.*

Proof. If $\boldsymbol{x}_n \in \mathbf{N}(A)$, then $A(\boldsymbol{x}_p + \boldsymbol{x}_n) = \boldsymbol{b} + \mathbf{0} = \boldsymbol{b}$. Conversely, if $A\boldsymbol{x} = \boldsymbol{b}$, then $A(\boldsymbol{x} - \boldsymbol{x}_p) = \boldsymbol{b} - \boldsymbol{b} = \mathbf{0}$, so $\boldsymbol{x}_n = \boldsymbol{x} - \boldsymbol{x}_p$ is in $\mathbf{N}(A)$ and $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$. □

For our example the solutions of $A\boldsymbol{x} = (1, 2, 3, 4)$ are $(1, 0, 0) + c(1, 1, -1)$: a line parallel to the nullspace but shifted away from the origin (Figure 7); $c = -1$ gives $(0, -1, 1)$. Section 9 builds on this to find the complete solution of any system.

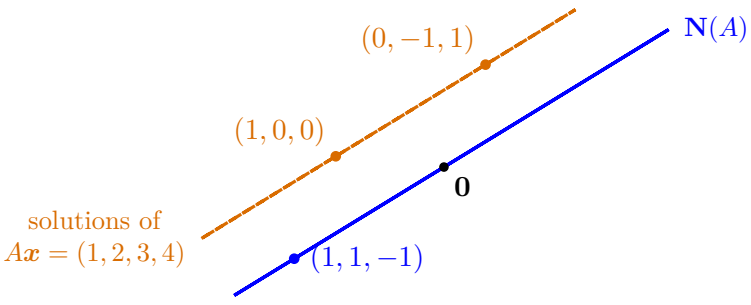


Figure 7: For the 4×3 matrix A of (8), the nullspace is the line through $(1, 1, -1)$ (solid). The solutions of $A\boldsymbol{x} = (1, 2, 3, 4)$ form the parallel line through $(1, 0, 0)$ and $(0, -1, 1)$ (dashed): one particular solution plus the whole nullspace. It does not pass through the origin, so it is not a subspace.

The two examples show the two ways of describing a subspace. For $\mathbf{C}(A)$ we were *given* vectors (the columns) and took all their combinations. For $\mathbf{N}(A)$ we were given *equations* ($A\boldsymbol{x} = \mathbf{0}$) and had to discover which vectors satisfy them; the answer turned out to be all combinations of one vector. Example 7.3 went the other way, describing $\mathbf{C}(A)$ by equations. Turning one description into the other is a recurring task of the course.

Common trap. Keep the spaces apart: for an $m \times n$ matrix, $\mathbf{C}(A)$ lies in $\mathbb{R}^m$ (vectors as long as a column) while $\mathbf{N}(A)$ lies in $\mathbb{R}^n$ (vectors with one entry per column). And the solution set of $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{b} \neq \mathbf{0}$ is never a subspace, even when it looks like a line or plane: it is a copy of $\mathbf{N}(A)$ shifted off the origin.

7.5 What a matrix keeps and what it crushes

The next example previews the main theorem of Part I. A matrix sends some input directions to zero; those directions form the nullspace, and the information they carry is lost.

Example 7.8 (What the matrix crushes)**.** Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix}$: two outputs, three inputs ($m = 2$, $n = 3$).

1. *Eliminate.* Row 2 $-$ 2 row 1 $= (0, 0, 0)$, leaving one pivot. So $A\boldsymbol{x} = \mathbf{0}$ is the single equation $x_1 + 2x_2 + 2x_3 = 0$. (The number of pivots, here $r = 1$, is the *rank*; see Section 8.)
2. *The nullspace is a plane.* Choosing x_2, x_3 freely and setting $x_1 = -2x_2 - 2x_3$ gives every solution: $\boldsymbol{x} = x_2(-2, 1, 0) + x_3(-2, 0, 1)$. The two *special solutions* $\boldsymbol{s}_1 = (-2, 1, 0)$ and $\boldsymbol{s}_2 = (-2, 0, 1)$ satisfy $A\boldsymbol{s}_1 = A\boldsymbol{s}_2 = (0, 0)$, so $\mathbf{N}(A)$ is the plane of their combinations, of dimension $2 = n - r$.
3. *The row is perpendicular to that plane.* The row $(1, 2, 2)$ has dot product $-2 + 2 + 0 = 0$ with $\boldsymbol{s}_1$ and $-2 + 0 + 2 = 0$ with $\boldsymbol{s}_2$. The *row space*, all multiples of $(1, 2, 2)$, is a line (dimension $r = 1$) perpendicular to $\mathbf{N}(A)$.
4. *The nullspace part is crushed.* Take $\boldsymbol{x} = (-1, 3, 2) = (1, 2, 2) + \boldsymbol{s}_1$. Then $A\boldsymbol{x} = (-1 + 6 + 4,\ -2 + 12 + 8) = (9, 18)$, exactly $A(1, 2, 2) = (1 + 4 + 4,\ 2 + 8 + 8) = (9, 18)$: adding $\boldsymbol{s}_1$ changed nothing.
5. *Outputs.* Every column is a multiple of $(1, 2)$, so $\mathbf{C}(A)$ is the line through $(1, 2)$ in $\mathbb{R}^2$ (dimension $r = 1$). The perpendicular line through $(2, -1)$ is the *left nullspace* (dimension $m - r = 1$): it contains the vectors $\boldsymbol{y}$ with $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$, since $(2, -1) \cdot (1, 2) = 0$.

The input space splits as $\mathbb{R}^3$ = row-space line + nullspace plane, $1 + 2 = 3$; the output space splits as $\mathbb{R}^2$ = column-space line + left-nullspace line, $1 + 1 = 2$. Figure 8 draws both. Sections 10 and 11 prove that these dimension counts, $r + (n - r) = n$ and $r + (m - r) = m$, hold for every matrix.

7.6 Exercises

Exercise 7.9. Let S and T be subspaces of a vector space V. Their *sum* $S + T$ is the set of all vectors $\boldsymbol{s} + \boldsymbol{t}$ with $\boldsymbol{s} \in S$ and $\boldsymbol{t} \in T$. (a) Show that $S + T$ is a subspace. (b) If S and T are two different lines through the origin in $\mathbb{R}^m$, how do $S + T$ and $S \cup T$ differ? Explain why the set of all combinations of vectors in $S \cup T$ is $S + T$.

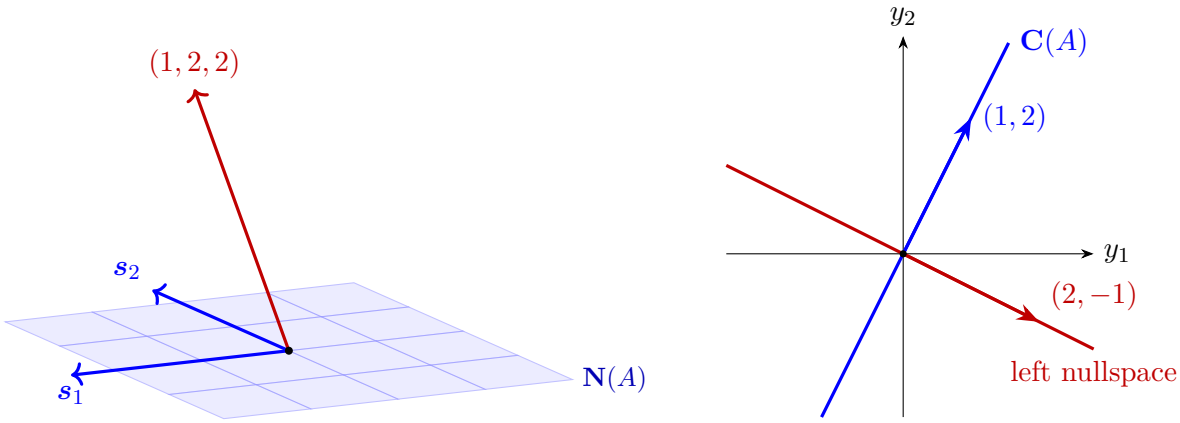


Figure 8: The matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix}$. Left (inputs, $\mathbb{R}^3$): the nullspace is the plane spanned by $\boldsymbol{s}_1 = (-2, 1, 0)$ and $\boldsymbol{s}_2 = (-2, 0, 1)$; the row vector $(1, 2, 2)$ is perpendicular to it. Right (outputs, $\mathbb{R}^2$): every $A\boldsymbol{x}$, including $A(-1, 3, 2) = (9, 18)$, lies on the column-space line through $(1, 2)$; the perpendicular line through $(2, -1)$ is the left nullspace.

Solution. (a) $\mathbf{0} = \mathbf{0} + \mathbf{0} \in S + T$. If $\boldsymbol{s} + \boldsymbol{t}$ and $\boldsymbol{s}' + \boldsymbol{t}'$ are in $S + T$ and c is a scalar, then $(\boldsymbol{s} + \boldsymbol{t}) + (\boldsymbol{s}' + \boldsymbol{t}') = (\boldsymbol{s} + \boldsymbol{s}') + (\boldsymbol{t} + \boldsymbol{t}')$ and $c(\boldsymbol{s} + \boldsymbol{t}) = c\boldsymbol{s} + c\boldsymbol{t}$, with $\boldsymbol{s} + \boldsymbol{s}', c\boldsymbol{s} \in S$ and $\boldsymbol{t} + \boldsymbol{t}', c\boldsymbol{t} \in T$. So $S + T$ is closed under both operations. (b) $S \cup T$ is just the two lines, while $S + T$ is the whole plane containing them. Every combination of vectors from $S \cup T$ can be grouped as (a vector of S) + (a vector of T), so it lies in $S + T$; conversely each $\boldsymbol{s} + \boldsymbol{t}$ is such a combination. Hence the combinations of $S \cup T$ are exactly $S + T$, the smallest subspace containing both lines. □

Exercise 7.10. The plane $x - 3y - z = 12$ is parallel to the plane $x - 3y - z = 0$, and $(12, 0, 0)$ lies on it. Write every point of the plane in the form $(x, y, z) = (\,\cdot\,, 0, 0) + y(\,\cdot\,, 1, 0) + z(\,\cdot\,, 0, 1)$. Which part is a particular solution and which part is the nullspace of $\begin{bmatrix} 1 & -3 & -1 \end{bmatrix}$?

Solution. Solve for x: $x = 12 + 3y + z$. So

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 + 3y + z \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

The vector $(12, 0, 0)$ is a particular solution of $x - 3y - z = 12$; the combinations of $(3, 1, 0)$ and $(1, 0, 1)$ form the plane $x - 3y - z = 0$, the nullspace of $\begin{bmatrix} 1 & -3 & -1 \end{bmatrix}$. This is Proposition 7.7 for a 1×3 matrix. □

Exercise 7.11. If $C = \begin{bmatrix} A \\ B \end{bmatrix}$ is the matrix formed by placing the rows of B below the rows of A (both with n columns), how is $\mathbf{N}(C)$ related to $\mathbf{N}(A)$ and $\mathbf{N}(B)$?

Solution. By block multiplication $C\boldsymbol{x} = \begin{bmatrix} A\boldsymbol{x} \\ B\boldsymbol{x} \end{bmatrix}$, which is zero exactly when $A\boldsymbol{x} = \mathbf{0}$ and $B\boldsymbol{x} = \mathbf{0}$. So $\mathbf{N}(C) = \mathbf{N}(A) \cap \mathbf{N}(B)$, the vectors in both nullspaces. (This is a subspace, as Theorem 7.1 guarantees: stacking more equations can only shrink the nullspace.) □

Exercise 7.12. Which of these subsets of $\mathbb{R}^3$ are subspaces? (a) all $\boldsymbol{b}$ with $b_1 + b_2 - b_3 = 0$; (b) all $\boldsymbol{b}$ with $b_3 = b_1 b_2$; (c) all $\boldsymbol{b} = c_1(1,0,-1) + c_2(1,0,1) + (1,0,0)$; (d) all $\boldsymbol{b} = c_1(1,0,-1) + c_2(1,0,1) + (0,1,0)$, where c_1, c_2 range over $\mathbb{R}$.

Solution. (a) Yes: it is the nullspace of the 1×3 matrix $\begin{bmatrix}1 & 1 & -1\end{bmatrix}$. (b) No: $(1,1,1)$ is in the set $(1 = 1 \cdot 1)$, but $2(1,1,1) = (2,2,2)$ is not $(2 \neq 2 \cdot 2)$. (c) Yes: $(1,0,0) = \frac{1}{2}(1,0,-1) + \frac{1}{2}(1,0,1)$ already lies in the plane spanned by the two vectors, so $\boldsymbol{b} = (c_1 + \frac{1}{2})(1,0,-1) + (c_2 + \frac{1}{2})(1,0,1)$ runs over exactly that plane (the plane $b_2 = 0$), a subspace. (d) No: every vector in the span of $(1,0,-1)$ and $(1,0,1)$ has second component 0, so every $\boldsymbol{b}$ in the set has $b_2 = 1$. The zero vector is not in the set. □

Exercise 7.13. Change the last entry of Example 7.8: $A = \begin{bmatrix}1 & 2 & 2\\ 2 & 4 & 5\end{bmatrix}$. How many pivots are there now, what is $\mathbf{N}(A)$, and what is $\mathbf{C}(A)$?

Solution. Row $2 - 2$ row $1 = (0,0,1)$, so there are two pivots, in columns 1 and 3 ($r = 2$). The equations $x_1 + 2x_2 + 2x_3 = 0$ and $x_3 = 0$ give $x_1 = -2x_2$, so $\mathbf{N}(A)$ is the line of multiples of $(-2,1,0)$, of dimension $3 - 2 = 1$. Columns 1 and 3, $(1,2)$ and $(2,5)$, are not parallel, so their combinations fill the plane: $\mathbf{C}(A) = \mathbb{R}^2$, and $A\boldsymbol{x} = \boldsymbol{b}$ is solvable for every $\boldsymbol{b}$. □

8 Solving $A\boldsymbol{x} = \mathbf{0}$: pivot variables, special solutions

Section 7 defined the nullspace $\mathbf{N}(A)$ as the set of all solutions of $A\boldsymbol{x} = \mathbf{0}$ and showed that it is a subspace of $\mathbb{R}^n$. A definition is not yet a description: we want to *list* what is in $\mathbf{N}(A)$. This section turns elimination, which so far solved square invertible systems, into an algorithm that works for every $m \times n$ matrix. The algorithm produces one number, the rank r, and a short list of $n - r$ *special solutions* whose combinations are exactly the nullspace.

8.1 Elimination on a rectangular matrix

The running example of the lecture is the 3×4 matrix

$$A = \begin{bmatrix}1 & 2 & 2 & 2\\ 2 & 4 & 6 & 8\\ 3 & 6 & 8 & 10\end{bmatrix}.$$

Two facts are visible before any computation: column 2 is twice column 1, and row 3 is row 1 plus row 2. Elimination should discover both. The first question is whether elimination is allowed to change the problem.

Proposition 8.1 (Row operations keep the nullspace)**.** *If E is an invertible $m \times m$ matrix, then $\mathbf{N}(EA) = \mathbf{N}(A)$. In particular, subtracting a multiple of one row from another, exchanging two rows, or dividing a row by a nonzero number never changes the nullspace.*

Proof. If $A\boldsymbol{x} = \mathbf{0}$ then $EA\boldsymbol{x} = E\mathbf{0} = \mathbf{0}$. Conversely, if $EA\boldsymbol{x} = \mathbf{0}$, multiply on the left by E^{-1} to get $A\boldsymbol{x} = \mathbf{0}$. Each elimination step is multiplication on the left by an invertible elementary or permutation matrix (Sections 3 and 6), and a product of invertible matrices is invertible. □

Because the right-hand side is $\mathbf{0}$ and stays $\mathbf{0}$ under every row operation, we do not need to carry it along. Elimination proceeds column by column as in the square case, with one new rule: if the pivot position and every entry below it are zero, there is nothing to do in that column, so we move to the next column (staying in the same row) instead of stopping.

Example 8.2 (Elimination with a missing pivot)**.** Subtract 2×row 1 from row 2 and 3×row 1 from row 3:

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}.$$

In column 2 the pivot position $(2, 2)$ holds a zero and so does the entry below it, so no row exchange can help: column 2 gets no pivot. The next pivot is the 2 in row 2, column 3. Subtracting row 2 from row 3 finishes the job:

$$U = \begin{bmatrix} \boxed{1} & 2 & 2 & 2 \\ 0 & 0 & \boxed{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The row of zeros records that row 3 of A was a combination of rows 1 and 2; elimination knocked it out.

Definition 8.3 (Echelon form, pivots, rank)**.** A matrix is in *echelon form* if every zero row lies below all nonzero rows and the first nonzero entry of each nonzero row (its *pivot*) lies strictly to the right of the pivot in the row above. Elimination brings every matrix A to an echelon form U. The *rank* of A, written $r = \operatorname{rank} A$, is the number of pivots. Columns that contain a pivot are *pivot columns*; the remaining columns are *free columns*.

A row holds at most one pivot and so does a column, so always

$$r \le m \qquad \text{and} \qquad r \le n.$$

In Example 8.2 the rank is 2, the pivot columns are 1 and 3, and the free columns are 2 and 4. (That the rank does not depend on the order of the elimination steps follows from Section 10, where r is identified with the dimension of the column space, a number defined without elimination.)

8.2 Pivot variables, free variables and special solutions

The system $U\boldsymbol{x} = \mathbf{0}$ of Example 8.2 is

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 0, \\ 2x_3 + 4x_4 &= 0. \end{aligned}$$

The unknowns that multiply pivot columns, x_1 and x_3, are *pivot variables*; the others, x_2 and x_4, are *free variables*. The free variables may be given any values. Once they are chosen, back substitution determines the pivot variables uniquely, because each nonzero row, read from the bottom up, contains exactly one new pivot variable with a nonzero coefficient.

Definition 8.4 (Special solutions)**.** For each free variable, the *special solution* is the solution of $A\boldsymbol{x} = \mathbf{0}$ in which that free variable equals 1, every other free variable equals 0, and the pivot variables are found by back substitution.

Example 8.5 (The two special solutions)**.** Take $x_2 = 1$, $x_4 = 0$. The second equation gives $2x_3 = 0$, so $x_3 = 0$; the first gives $x_1 + 2 = 0$, so $x_1 = -2$. Next take $x_2 = 0$, $x_4 = 1$. Then $2x_3 + 4 = 0$ gives $x_3 = -2$, and $x_1 + 2(-2) + 2 = 0$ gives $x_1 = 2$:

$$\boldsymbol{s}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{s}_2 = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

Each special solution is a dependency among the columns of A. The first says $-2(\text{col } 1) + (\text{col } 2) = \mathbf{0}$: column 2 is twice column 1. The second says $2(\text{col } 1) - 2(\text{col } 3) + (\text{col } 4) = \mathbf{0}$, and indeed $2(1,2,3) - 2(2,6,8) + (2,8,10) = (0,0,0)$. The whole nullspace is

$$\mathbf{N}(A) = \{\, c\,\boldsymbol{s}_1 + d\,\boldsymbol{s}_2 : c, d \in \mathbb{R} \,\},$$

a two-dimensional plane through the origin in $\mathbb{R}^4$.

Theorem 8.6 (Special solutions describe the nullspace)**.** *Let A be $m \times n$ with rank r. There are exactly $n - r$ special solutions $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_{n-r}$, and $\mathbf{N}(A)$ consists of all their linear combinations. More precisely, if $\boldsymbol{x} \in \mathbf{N}(A)$ and its free variables have the values $c_1, \ldots, c_{n-r}$, then $\boldsymbol{x} = c_1\boldsymbol{s}_1 + \cdots + c_{n-r}\boldsymbol{s}_{n-r}$.*

Proof. There are r pivot columns, hence $n - r$ free columns and one special solution for each. The special solutions lie in $\mathbf{N}(A)$, which is a subspace, so all their combinations do too.

Conversely, let $\boldsymbol{x} \in \mathbf{N}(A)$ have free variables $c_1, \ldots, c_{n-r}$ and put $\boldsymbol{y} = \boldsymbol{x} - \sum_j c_j \boldsymbol{s}_j$. Then $\boldsymbol{y} \in \mathbf{N}(A) = \mathbf{N}(U)$ by Proposition 8.1. Since $\boldsymbol{s}_j$ has a 1 in the jth free position and 0 in the other free positions, every free variable of $\boldsymbol{y}$ is 0. Now solve $U\boldsymbol{y} = \mathbf{0}$ from the last nonzero row upward. In that row every entry to the right of the pivot multiplies a free variable, which is zero, so the pivot variable is zero. In each row above, the entries to the right of the pivot multiply free variables or pivot variables already shown to be zero, so that row's pivot variable is zero as well. Hence $\boldsymbol{y} = \mathbf{0}$, that is, $\boldsymbol{x} = \sum_j c_j \boldsymbol{s}_j$. □

Corollary 8.7 (More unknowns than equations)**.** *If $n > m$, then $A\boldsymbol{x} = \mathbf{0}$ has a solution $\boldsymbol{x} \neq \mathbf{0}$.*

Proof. Since $r \leq m < n$, there is at least one free variable, and its special solution has a 1 in that position. □

8.3 The reduced row echelon form R

Elimination can go further. Working upward, we can also clear the entries *above* each pivot and then divide each pivot row by its pivot.

Definition 8.8 (Reduced row echelon form)**.** A matrix is in *reduced row echelon form* if it is in echelon form, every pivot equals 1, and every pivot is the only nonzero entry in its column. Elimination brings every matrix A to such a matrix, and (a standard fact we use without proof) the result does not depend on the order of the steps; it is denoted $R = \text{rref}(A)$ (MATLAB: `rref(A)`).

Example 8.9 (From U to R)**.** In U from Example 8.2, subtract row 2 from row 1 to clear the 2 above the second pivot, then divide row 2 by its pivot 2:

$$U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The equations $R\boldsymbol{x} = \mathbf{0}$ are $x_1 = -2x_2 + 2x_4$ and $x_3 = -2x_4$: the pivot variables are written directly in terms of the free ones. The free columns of R contain $(2, 0)$ and $(-2, 2)$ in the pivot rows, and these numbers reappear in $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$ with their signs reversed: $(x_1, x_3) = (-2, 0)$ and $(2, -2)$.

The sign reversal is no accident. Suppose for the moment that the pivot columns come first (this is only a renaming of the unknowns). Then R has an identity block in its pivot rows and pivot columns, a block F in its pivot rows and free columns, and zero rows below:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}, \qquad I \text{ is } r \times r, \quad F \text{ is } r \times (n - r).$$

Proposition 8.10 (The nullspace matrix)**.** *With R in this block form, the $n \times (n - r)$ matrix*

$$N = \begin{bmatrix} -F \\ I \end{bmatrix} \qquad (\textit{here } I \textit{ is } (n - r) \times (n - r))$$

satisfies $RN = 0$, and its columns are the special solutions.

Proof. Block multiplication gives $RN = \begin{bmatrix} I(-F) + F\,I \\ 0 \end{bmatrix} = 0$. Column j of N has free part equal to column j of the identity, a 1 in the jth free position and zeros in the others, so it is the solution whose free variables are those of the jth special solution. Since back substitution determines the pivot variables uniquely, it *is* the jth special solution. Equivalently: $R\boldsymbol{x} = \mathbf{0}$ says $\boldsymbol{x}_{\text{pivot}} + F\boldsymbol{x}_{\text{free}} = \mathbf{0}$, so $\boldsymbol{x}_{\text{pivot}} = -F\boldsymbol{x}_{\text{free}}$. □

When the pivot columns are not first, put the rows of $-F$ in the pivot positions and the rows of I in the free positions. In Example 8.9, $F = \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$ (rows 1–2, columns 2 and 4), so rows 1 and 3 of N come from $-F$ and rows 2 and 4 from I:

$$N = \begin{bmatrix} -2 & 2 \\ 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{s}_1 & \boldsymbol{s}_2 \end{bmatrix}.$$

Figure 9 records the shapes; this is exactly what the teaching code `null(A)` does after computing R.

8.4 Two more examples

Example 8.11 (The transpose needs a row exchange)**.** Take $A^{\mathsf{T}} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix}$. Its third column is the sum of the first two, so we expect a free third column. Subtract 2×row 1 from rows 2, 3

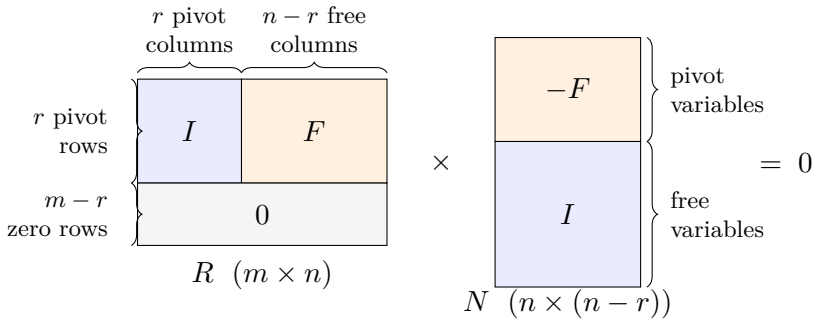


Figure 9: Block shapes in $RN = 0$ when the pivot columns come first. Each column of N is a special solution: the identity block sets one free variable to 1 and the others to 0, and $-F$ supplies the pivot variables, so that $I(-F) + F\,I = 0$.

and 4; column 2 then has a zero in the pivot position but a 2 below it, so exchange rows 2 and 3; finally subtract 2×row 2 from row 4:

$$\begin{bmatrix}1&2&3\\0&0&0\\0&2&2\\0&4&4\end{bmatrix} \longrightarrow \begin{bmatrix}1&2&3\\0&2&2\\0&0&0\\0&4&4\end{bmatrix} \longrightarrow U = \begin{bmatrix}1&2&3\\0&2&2\\0&0&0\\0&0&0\end{bmatrix}.$$

Column 3 has nothing in the pivot position or below, so the rank is 2, the same as the rank of A; Section 11 proves that $\operatorname{rank} A^{\mathsf{T}} = \operatorname{rank} A$ always. With the free variable $x_3 = 1$, the second row gives $2x_2 + 2 = 0$, so $x_2 = -1$, and the first gives $x_1 - 2 + 3 = 0$, so $x_1 = -1$. The nullspace is the line $\{c(-1,-1,1)\}$ in $\mathbb{R}^3$. Going on to $R = \begin{bmatrix}1&0&1\\0&1&1\\0&0&0\\0&0&0\end{bmatrix}$ shows $F = \begin{bmatrix}1\\1\end{bmatrix}$ and $N = \begin{bmatrix}-F\\1\end{bmatrix} = (-1,-1,1)$, as it must.

Example 8.12 (What a matrix crushes)**.** Let $A = \begin{bmatrix}1&2&2\\2&4&4\end{bmatrix}$, which takes $n = 3$ inputs to $m = 2$ outputs. Row 2 minus 2×row 1 is $(0,0,0)$, so $R = \begin{bmatrix}1&2&2\\0&0&0\end{bmatrix}$ has one pivot and $r = 1$. The free columns 2 and 3 give

$$\boldsymbol{s}_1 = \begin{bmatrix}-2\\1\\0\end{bmatrix}, \qquad \boldsymbol{s}_2 = \begin{bmatrix}-2\\0\\1\end{bmatrix}, \qquad A\boldsymbol{s}_1 = \begin{bmatrix}-2+2\\-4+4\end{bmatrix} = \mathbf{0}, \quad A\boldsymbol{s}_2 = \begin{bmatrix}-2+2\\-4+4\end{bmatrix} = \mathbf{0}.$$

So $\mathbf{N}(A)$ is a plane in $\mathbb{R}^3$ of dimension $n - r = 2$: a whole plane of inputs is sent to zero. Both special solutions are perpendicular to the row $(1,2,2)$, since $-2+2+0 = 0$ and $-2+0+2 = 0$. That is no coincidence: $A\boldsymbol{x} = \mathbf{0}$ says precisely that $\boldsymbol{x}$ has zero dot product with every row. Section 11 completes this example with the other three subspaces.

Common trap. Row operations preserve the nullspace but *not* the column space. In Example 8.2, column 3 of U is $(2,2,0)$. Every vector $A\boldsymbol{x}$ has third component equal to the sum of the first two (row 3 of A is row 1 plus row 2), and $2+2 \neq 0$, so $(2,2,0)$ is not in $\mathbf{C}(A)$. Use U or R to find $\mathbf{N}(A)$ and to locate the pivot columns, but take those pivot columns from the original A when you want

the column space. A second slip: setting *all* free variables to zero produces only $\boldsymbol{x} = \mathbf{0}$; a special solution needs a 1.

8.5 Exercises

Exercise 8.13. Let $A = \begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$. (a) Find the reduced row echelon form of A. (b) What is the rank of A? (c) Find the special solutions of $A\boldsymbol{x} = \mathbf{0}$.

Solution. (a) Subtract 2×row 1 from row 3, giving row 3 $= (0, -12, -3, -21)$. Divide row 2 by 4: $(0, 1, \frac{1}{4}, \frac{7}{4})$. Add 12×(new row 2) to row 3, giving a zero row. Subtract 5×row 2 from row 1: $7 - \frac{5}{4} = \frac{23}{4}$ and $9 - \frac{35}{4} = \frac{1}{4}$. Hence

$$R = \begin{bmatrix} 1 & 0 & \frac{23}{4} & \frac{1}{4} \\ 0 & 1 & \frac{1}{4} & \frac{7}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(b) There are two pivots, so $\operatorname{rank} A = 2$. (c) The free variables are x_3, x_4, and $F = \begin{bmatrix} 23/4 & 1/4 \\ 1/4 & 7/4 \end{bmatrix}$. The columns of $\begin{bmatrix} -F \\ I \end{bmatrix}$ are $\boldsymbol{s}_1 = \left(-\frac{23}{4}, -\frac{1}{4}, 1, 0\right)$ and $\boldsymbol{s}_2 = \left(-\frac{1}{4}, -\frac{7}{4}, 0, 1\right)$. Check one entry: row 2 of A times $\boldsymbol{s}_1$ is $4(-\frac{1}{4}) + 1 = 0$. □

Exercise 8.14. Change one entry of Example 8.12: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{bmatrix}$. Find the rank, the special solutions and the dimension of the nullspace.

Solution. Row 2 minus 2×row 1 is now $(0, 0, 1)$, so there are pivots in columns 1 and 3 and $r = 2$. Clearing above the second pivot gives $R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The only free variable is x_2; setting $x_2 = 1$ gives $x_3 = 0$ and $x_1 = -2$. The nullspace is the line through $(-2, 1, 0)$, of dimension $3 - 2 = 1$: the plane of Example 8.12 has shrunk to a line. □

Exercise 8.15. Find a 2×4 matrix R in reduced row echelon form whose special solutions are $(-3, 1, 0, 0)$ and $(-2, 0, -6, 1)$.

Solution. In a special solution the free variables are 1 and 0; here the pattern $(x_2, x_4) = (1, 0)$ and $(0, 1)$ shows that x_2, x_4 are free and x_1, x_3 are pivot variables. By Proposition 8.10 the pivot entries of the special solutions are the columns of $-F$: $(x_1, x_3) = (-3, 0)$ and $(-2, -6)$, so F has columns $(3, 0)$ and $(2, 6)$. Placing the identity in columns 1 and 3 and F in columns 2 and 4,

$$R = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 6 \end{bmatrix}.$$

Check: $R(-3, 1, 0, 0) = (-3 + 3,\ 0) = \mathbf{0}$ and $R(-2, 0, -6, 1) = (-2 + 2,\ -6 + 6) = \mathbf{0}$. □

Exercise 8.16. Let $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Find 2×2 matrices A_1 and A_2 with $\operatorname{rank}(A_1 B) = 1$ and $\operatorname{rank}(A_2 B) = 0$, and describe all possible A_2.

Solution. $A_1 = I$ works: $A_1 B = B$ has equal rows, so elimination leaves one pivot. $A_2 = 0$ works trivially. In general, a row (a, b) of A_2 multiplies B to give $(a + b, a + b)$, which is zero exactly when $b = -a$. So $A_2 B = 0$ precisely when every row of A_2 is a multiple of $(1, -1)$, for example $A_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$. □

9 Solving $A\boldsymbol{x} = \boldsymbol{b}$: row reduced form R

Section 8 found every solution of $A\boldsymbol{x} = \boldsymbol{0}$. Now the right-hand side is a general vector $\boldsymbol{b} \in \mathbb{R}^m$, and three questions arise: *is there* a solution, *how do we find one*, and *what are all of them*? Elimination answers all three. A solution exists only when $\boldsymbol{b}$ passes a test that elimination produces; every solution is one particular solution plus a vector from the nullspace; and the rank r alone predicts whether there are 0, 1 or infinitely many solutions.

9.1 When is $A\boldsymbol{x} = \boldsymbol{b}$ solvable?

Keep the matrix of Section 8, whose third row is the sum of the first two:

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}.$$

Adding the first two equations of $A\boldsymbol{x} = \boldsymbol{b}$ gives the left side of the third, so a solution can exist only if $b_1 + b_2 = b_3$. With $\boldsymbol{b} = (1, 5, 17)$ there is no solution; with $b_1 = 1$, $b_2 = 5$ the only acceptable third component is $b_3 = 6$.

Elimination discovers this condition by itself. We attach $\boldsymbol{b}$ as an extra column, forming the *augmented matrix* $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$, and do to the last column whatever we do to the rows of A:

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array}\right].$$

The last row is the equation $0 = b_3 - b_2 - b_1$. If a row of A is wiped out, the same combination of the entries of $\boldsymbol{b}$ must vanish. For $\boldsymbol{b} = (1, 5, 6)$ the last column becomes $(1, 3, 0)$ and the system is consistent.

There are now two descriptions of the good right-hand sides. Section 7 said that $A\boldsymbol{x} = \boldsymbol{b}$ is solvable exactly when $\boldsymbol{b} \in \mathbf{C}(A)$, because $A\boldsymbol{x}$ is the combination of the columns with weights $x_1, \ldots, x_n$. Elimination says: whenever a combination of the rows gives the zero row, the same combination of the b_i must give zero. These are not obviously the same, but they are.

Theorem 9.1 (Solvability)**.** *For an $m \times n$ matrix A and $\boldsymbol{b} \in \mathbb{R}^m$ the following are equivalent:*

(a) $A\boldsymbol{x} = \boldsymbol{b}$ has a solution;

(b) $\boldsymbol{b} \in \mathbf{C}(A)$;

(c) $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{b} = 0$ for every $\boldsymbol{y} \in \mathbb{R}^m$ with $\boldsymbol{y}^{\mathsf{T}} A = \boldsymbol{0}^{\mathsf{T}}$ (every combination of rows that gives the zero row also gives zero when applied to $\boldsymbol{b}$).

Proof. (a)⇔(b): $A\boldsymbol{x} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$, so $A\boldsymbol{x} = \boldsymbol{b}$ asks exactly for weights that produce $\boldsymbol{b}$ from the columns.

(a)⇒(c): if $A\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{y}^{\mathsf{T}}A = \boldsymbol{0}^{\mathsf{T}}$, then $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{b} = \boldsymbol{y}^{\mathsf{T}}A\boldsymbol{x} = \boldsymbol{0}^{\mathsf{T}}\boldsymbol{x} = 0$.

(c)⇒(a): let E be the invertible matrix that records the elimination steps, so that $EA = R$ is the reduced row echelon form, and let $r = \operatorname{rank} A$. Then $A\boldsymbol{x} = \boldsymbol{b}$ is equivalent to $R\boldsymbol{x} = E\boldsymbol{b}$. For $i > r$, row i of R is zero, so the ith row $\boldsymbol{y}_i^{\mathsf{T}}$ of E satisfies $\boldsymbol{y}_i^{\mathsf{T}}A = \boldsymbol{0}^{\mathsf{T}}$, and by (c) the ith entry of $E\boldsymbol{b}$, namely $\boldsymbol{y}_i^{\mathsf{T}}\boldsymbol{b}$, is 0. Hence the zero rows of $R\boldsymbol{x} = E\boldsymbol{b}$ read $0 = 0$. In the first r rows, set every free variable to 0; since the pivot columns of R are columns of the identity, the ith pivot variable equals the ith entry of $E\boldsymbol{b}$. This $\boldsymbol{x}$ solves $R\boldsymbol{x} = E\boldsymbol{b}$, hence $A\boldsymbol{x} = E^{-1}R\boldsymbol{x} = E^{-1}E\boldsymbol{b} = \boldsymbol{b}$. □

The vectors $\boldsymbol{y}$ in (c) form the left nullspace $\mathbf{N}(A^{\mathsf{T}})$, the fourth of the subspaces in Section 11. In the example, $\boldsymbol{y} = (-1, -1, 1)$ gives the condition $-b_1 - b_2 + b_3 = 0$.

9.2 The complete solution $\boldsymbol{x}_p + \boldsymbol{x}_n$

The proof just given also finds a solution.

Definition 9.2 (Particular solution)**.** When $A\boldsymbol{x} = \boldsymbol{b}$ is solvable, the *particular solution* $\boldsymbol{x}_p$ is the solution whose free variables are all 0. Its pivot variables come from back substitution, or directly from the last column of the reduced augmented matrix $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$.

Example 9.3 (A particular solution)**.** For $\boldsymbol{b} = (1, 5, 6)$, set the free variables $x_2 = x_4 = 0$. The two remaining equations are $x_1 + 2x_3 = 1$ and $2x_3 = 3$, so $x_3 = \frac{3}{2}$ and $x_1 = 1 - 3 = -2$. Carrying the elimination on to the reduced form gives the same numbers without back substitution:

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 0 & -2 & -2 \\ 0 & 0 & 1 & 2 & \frac{3}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = \begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}, \qquad \boldsymbol{x}_p = \begin{bmatrix} -2 \\ 0 \\ \frac{3}{2} \\ 0 \end{bmatrix}.$$

The entries of $\boldsymbol{d}$ sit in the pivot positions of $\boldsymbol{x}_p$ and zeros sit in the free positions. Check: $A\boldsymbol{x}_p = (-2 + 3,\ -4 + 9,\ -6 + 12) = (1, 5, 6)$.

Theorem 9.4 (Complete solution)**.** *Suppose $A\boldsymbol{x}_p = \boldsymbol{b}$. Then $\boldsymbol{x}$ solves $A\boldsymbol{x} = \boldsymbol{b}$ if and only if $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ for some $\boldsymbol{x}_n \in \mathbf{N}(A)$.*

Proof. If $\boldsymbol{x}_n \in \mathbf{N}(A)$ then $A(\boldsymbol{x}_p + \boldsymbol{x}_n) = A\boldsymbol{x}_p + A\boldsymbol{x}_n = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$. Conversely, if $A\boldsymbol{x} = \boldsymbol{b}$ then $A(\boldsymbol{x} - \boldsymbol{x}_p) = \boldsymbol{b} - \boldsymbol{b} = \boldsymbol{0}$, so $\boldsymbol{x}_n = \boldsymbol{x} - \boldsymbol{x}_p$ lies in $\mathbf{N}(A)$ and $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$. □

Example 9.5 (All solutions for $\boldsymbol{b} = (1, 5, 6)$)**.** The nullspace was found in Example 8.5. Therefore

$$\boldsymbol{x}_{\text{complete}} = \begin{bmatrix} -2 \\ 0 \\ \frac{3}{2} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \qquad c_1, c_2 \in \mathbb{R}.$$

There is no free constant in front of $\boldsymbol{x}_p$: tripling it would give right-hand side $3\boldsymbol{b}$. The constants belong to the nullspace part, because only there do multiples and sums keep the right-hand side at $\boldsymbol{0}$.

Geometrically, $\mathbf{N}(A)$ is a two-dimensional subspace of $\mathbb{R}^4$, and the solutions of $A\boldsymbol{x} = \boldsymbol{b}$ form a copy of it shifted away from the origin so that it passes through $\boldsymbol{x}_p$. That shifted copy is *not* a subspace (it does not contain $\mathbf{0}$ when $\boldsymbol{b} \neq \mathbf{0}$). Figure 10 shows the same structure in $\mathbb{R}^3$.

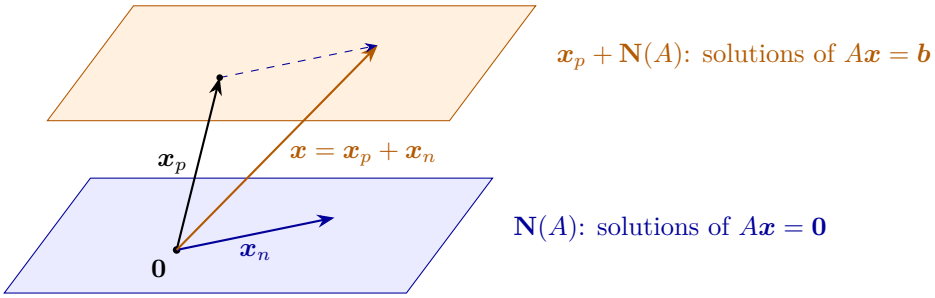


Figure 10: The complete solution. The nullspace is a subspace through $\mathbf{0}$; the solutions of $A\boldsymbol{x} = \boldsymbol{b}$ form the parallel copy through $\boldsymbol{x}_p$. For the 1×3 matrix $A = \begin{bmatrix} 1 & -5 & 2 \end{bmatrix}$ and $b = 9$ these are the parallel planes $x - 5y + 2z = 0$ and $x - 5y + 2z = 9$, and $\boldsymbol{x}_p = (9, 0, 0)$ (Exercise 9.13).

9.3 The rank decides the number of solutions

Let A be $m \times n$ of rank r, so $r \leq m$ and $r \leq n$. The two extreme cases, *full column rank* $r = n$ and *full row rank* $r = m$, behave in opposite ways.

Theorem 9.6 (Full rank). *(a) $r = n$ if and only if $\mathbf{N}(A) = \{\mathbf{0}\}$, if and only if $A\boldsymbol{x} = \boldsymbol{b}$ has at most one solution for every $\boldsymbol{b}$. Then $R = \begin{bmatrix} I \\ 0 \end{bmatrix}$.*

(b) $r = m$ if and only if $A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution for every $\boldsymbol{b} \in \mathbb{R}^m$. Then $R = \begin{bmatrix} I & F \end{bmatrix}$ (with the columns of I possibly interleaved with those of F) has no zero rows, and there are $n - m$ free variables.

Proof. (a) If $r = n$ there are no free variables, hence no special solutions, and Theorem 8.6 gives $\mathbf{N}(A) = \{\mathbf{0}\}$; by Theorem 9.4 two solutions differ by a nullspace vector, so there is at most one. If $r < n$, a special solution $\boldsymbol{s} \neq \mathbf{0}$ exists, and $\mathbf{0}$ and $\boldsymbol{s}$ are two solutions of $A\boldsymbol{x} = \mathbf{0}$. With all n columns pivot columns, R is the $n \times n$ identity above $m - n$ zero rows.

(b) If $r = m$, the matrix R has no zero rows, so condition (c) of Theorem 9.1 is never tested and the construction in its proof solves $A\boldsymbol{x} = \boldsymbol{b}$ for every $\boldsymbol{b}$. If $r < m$, take $\boldsymbol{y}^{\mathsf{T}}$ to be the last row of E (where $EA = R$). It is nonzero because E is invertible, and $\boldsymbol{y}^{\mathsf{T}}A$ is the last row of R, which is zero. For $\boldsymbol{b} = \boldsymbol{y}$ we get $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{b} = \|\boldsymbol{y}\|^2 \neq 0$, so this $\boldsymbol{b}$ has no solution. □

Example 9.7 (Full column rank: tall and thin). $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix}$ has two columns pointing in different directions, so $r = n = 2$ and $R = \begin{bmatrix} I \\ 0 \end{bmatrix}$ with two zero rows. For $\boldsymbol{b} = (4, 3, 7, 6)$, the sum of the columns, the unique solution is $\boldsymbol{x} = (1, 1)$. For $\boldsymbol{b} = (1, 0, 0, 0)$ the first two equations force $x_1 = -\frac{1}{5}$, $x_2 = \frac{2}{5}$, and then the third equation gives $6(-\frac{1}{5}) + \frac{2}{5} = -\frac{4}{5} \neq 0$: no solution. A random $\boldsymbol{b}$ in $\mathbb{R}^4$ has no solution, because $\mathbf{C}(A)$ is only a plane.

Example 9.8 (Full row rank: short and wide)**.** Transpose the last matrix: $A = \begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix}$, again of rank 2, now $r = m = 2 < n = 4$. Elimination gives

$$R = \begin{bmatrix} 1 & 0 & -\frac{4}{5} & -\frac{3}{5} \\ 0 & 1 & \frac{17}{5} & \frac{14}{5} \end{bmatrix},$$

with no zero rows, so every $\boldsymbol{b} \in \mathbb{R}^2$ is reachable, and there are $n - r = 2$ free variables. For $\boldsymbol{b} = (3, 4)$, the sum of the first two columns, $\boldsymbol{x}_p = (1, 1, 0, 0)$, and after scaling the special solutions by 5,

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 4 \\ -17 \\ 5 \\ 0 \end{bmatrix} + d \begin{bmatrix} 3 \\ -14 \\ 0 \\ 5 \end{bmatrix}.$$

(Check: $A(4, -17, 5, 0) = (4 - 34 + 30,\ 12 - 17 + 5) = (0, 0)$.) Infinitely many solutions, for every $\boldsymbol{b}$.

When $r = m = n$, both parts of Theorem 9.6 apply: A is square and invertible, $R = I$, and $\boldsymbol{x} = A^{-1}\boldsymbol{b}$ is the one and only solution. For instance $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ has rank 2 and $A^{-1} = \frac{1}{5}\begin{bmatrix} -1 & 2 \\ 3 & -1 \end{bmatrix}$. In the remaining case $r < m$ and $r < n$, both failures occur: some $\boldsymbol{b}$ are unreachable (zero rows in R), and whenever a solution exists there are infinitely many (free variables). The running example of this section is of that kind. Table 1 summarises.

	$r = m = n$	$r = n < m$	$r = m < n$	$r < m,\ r < n$
shape of R	I	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
number of solutions	1	0 or 1	∞	0 or ∞
example	$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$	Example 9.7	Example 9.8	Example 9.5

Table 1: The rank tells everything about the *number* of solutions of $A\boldsymbol{x} = \boldsymbol{b}$ (with the pivot columns of R moved first).

9.4 A system with parameters

Example 9.9 (Solvability depends on $\boldsymbol{b}$)**.** Solve $x - 2y - 2z = b_1$, $2x - 5y - 4z = b_2$, $4x - 9y - 8z = b_3$. Eliminate on the augmented matrix (row 2 $-$ 2 row 1, row 3 $-$ 4 row 1, then row 3 $-$ row 2):

$$\left[\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 2 & -5 & -4 & b_2 \\ 4 & -9 & -8 & b_3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & -1 & 0 & b_2 - 2b_1 \\ 0 & -1 & 0 & b_3 - 4b_1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & -1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 - 2b_1 \end{array}\right].$$

If $b_3 - b_2 - 2b_1 \neq 0$ there is no solution. If $b_3 = 2b_1 + b_2$, continue to the reduced form: multiply row 2 by -1 and add twice the new row 2 to row 1, giving

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 5b_1 - 2b_2 \\ 0 & 1 & 0 & 2b_1 - b_2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The pivot variables are x, y and the free variable is z. Setting $z = 0$ gives $\boldsymbol{x}_p = (5b_1 - 2b_2,\ 2b_1 - b_2,\ 0)$; setting $z = 1$ in $R\boldsymbol{x} = \boldsymbol{0}$ gives the special solution $(2, 0, 1)$. So

$$\boldsymbol{x} = \begin{bmatrix} 5b_1 - 2b_2 \\ 2b_1 - b_2 \\ 0 \end{bmatrix} + c \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \qquad \text{provided } b_3 = 2b_1 + b_2.$$

Here $m = n = 3$ and $r = 2$: the case "0 or ∞".

Common trap. The complete solution is *not* "any combination of $\boldsymbol{x}_p$ and $\boldsymbol{x}_n$". The coefficient of $\boldsymbol{x}_p$ must be exactly 1; only the nullspace part carries free constants. Relatedly, "the" particular solution is just a convenient choice: $\boldsymbol{x}_p + \boldsymbol{x}_n$ is an equally good particular solution for any $\boldsymbol{x}_n \in \mathbf{N}(A)$. And an invertible A still has a nullspace, namely $\{\boldsymbol{0}\}$; it is not empty.

9.5 Exercises

Exercise 9.10. Explain why each statement is false. (a) The complete solution is any linear combination of $\boldsymbol{x}_p$ and $\boldsymbol{x}_n$. (b) A system $A\boldsymbol{x} = \boldsymbol{b}$ has at most one particular solution. (c) If A is invertible, there is no solution $\boldsymbol{x}_n$ in the nullspace.

Solution. (a) The coefficient of $\boldsymbol{x}_p$ must be 1: $A(2\boldsymbol{x}_p) = 2\boldsymbol{b} \neq \boldsymbol{b}$ unless $\boldsymbol{b} = \boldsymbol{0}$. (b) If $\boldsymbol{x}_p$ is a particular solution and $\boldsymbol{x}_n \in \mathbf{N}(A)$, then $\boldsymbol{x}_p + \boldsymbol{x}_n$ is another one; the choice "free variables = 0" is a convention. (c) $\boldsymbol{x}_n = \boldsymbol{0}$ is always in the nullspace. □

Exercise 9.11. Let $U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ and $\boldsymbol{c} = (5, 8)$. Use Gauss–Jordan elimination to reduce $\begin{bmatrix} U & \boldsymbol{0} \end{bmatrix}$ and $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$ to $\begin{bmatrix} R & \boldsymbol{0} \end{bmatrix}$ and $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$, solve $R\boldsymbol{x} = \boldsymbol{0}$ and $R\boldsymbol{x} = \boldsymbol{d}$, and check the answers in $U\boldsymbol{x} = \boldsymbol{0}$ and $U\boldsymbol{x} = \boldsymbol{c}$.

Solution. Divide row 2 by 4, then subtract 3×row 2 from row 1:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right],$$

so $R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\boldsymbol{d} = (-1, 2)$ (the zero column stays zero). $R\boldsymbol{x} = \boldsymbol{0}$ says $x_1 = -2x_2$, $x_3 = 0$: the nullspace is all $c(-2, 1, 0)$. $R\boldsymbol{x} = \boldsymbol{d}$ has $\boldsymbol{x}_p = (-1, 0, 2)$, so all solutions are $(-1, 0, 2) + c(-2, 1, 0)$. Checks: $U(-2, 1, 0) = (-2 + 2, 0) = \boldsymbol{0}$ and $U(-1, 0, 2) = (-1 + 6,\ 8) = (5, 8) = \boldsymbol{c}$. □

Exercise 9.12. Suppose $A\boldsymbol{x} = \boldsymbol{b}$ and $C\boldsymbol{x} = \boldsymbol{b}$ have the same complete solutions for every $\boldsymbol{b}$. Must $A = C$?

Solution. Yes. Take any vector $\boldsymbol{v}$ of the right size and set $\boldsymbol{b} = A\boldsymbol{v}$. Then $\boldsymbol{v}$ solves $A\boldsymbol{x} = \boldsymbol{b}$, so by hypothesis it also solves $C\boldsymbol{x} = \boldsymbol{b}$: $C\boldsymbol{v} = A\boldsymbol{v}$. Choosing $\boldsymbol{v}$ to be the jth column of the identity shows that column j of C equals column j of A, for every j. □

Exercise 9.13. Let S be the set of points (x, y, z) with $x - 5y + 2z = 9$ and S_0 the set with $x - 5y + 2z = 0$. (a) Describe S and S_0 and how they are related. (b) Fill in the blanks so that every point of S has the form $(x, y, z) = (_, 0, 0) + c_1(_, 1, 0) + c_2(_, 0, 1)$.

Solution. (a) Both are planes in $\mathbb{R}^3$ (one linear equation removes one of three degrees of freedom). Subtracting the equations would give $0 = 9$, so they never meet: they are parallel. S_0 is the nullspace of $A = \begin{bmatrix} 1 & -5 & 2 \end{bmatrix}$ and passes through the origin; S is the solution set of $A\boldsymbol{x} = 9$. (b) The particular solution with free variables $y = z = 0$ is $x = 9$. The pivot variable is x and the special solutions of $A\boldsymbol{x} = 0$ are $(5, 1, 0)$ (from $x - 5 = 0$) and $(-2, 0, 1)$ (from $x + 2 = 0$). Hence $(x, y, z) = (9, 0, 0) + c_1(5, 1, 0) + c_2(-2, 0, 1)$: go from S_0 to S by the vector $(9, 0, 0)$, exactly the structure of Figure 10. □

10 Independence, basis and dimension

Sections 8 and 9 spoke loosely of a "two-dimensional nullspace". This section gives precise meaning to four words: vectors are *independent*, vectors *span* a space, vectors form a *basis*, and a space has a *dimension*. The words attach to different objects. Independence, spanning and being a basis are properties of a list of vectors; dimension is a property of a space; rank is a property of a matrix. The payoff is the pair of formulas $\dim \mathbf{C}(A) = r$ and $\dim \mathbf{N}(A) = n - r$.

10.1 Linear independence

Definition 10.1 (Independence)**.** Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ in a vector space are *linearly independent* if the only combination that gives the zero vector is the zero combination:

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n = \mathbf{0} \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_n = 0.$$

Otherwise they are *dependent*: some combination with not all $c_i = 0$ gives $\mathbf{0}$.

Two quick cases. The vectors $\boldsymbol{v}$ and $2\boldsymbol{v}$ are dependent, since $2\boldsymbol{v} - (2\boldsymbol{v}) = \mathbf{0}$. Any list that contains the zero vector is dependent: if $\boldsymbol{v}_2 = \mathbf{0}$, take $0 \cdot \boldsymbol{v}_1 + 6 \cdot \boldsymbol{v}_2 = \mathbf{0}$. Two vectors are independent when they do not lie on one line; three vectors are independent when they do not lie in one plane.

Example 10.2 (Three vectors in the plane)**.** Let $\boldsymbol{v}_1 = (2, 1)$, $\boldsymbol{v}_2 = (1, 2)$, $\boldsymbol{v}_3 = (\frac{5}{2}, -1)$. Put them in the columns of the 2×3 matrix $A = \begin{bmatrix} 2 & 1 & \frac{5}{2} \\ 1 & 2 & -1 \end{bmatrix}$. It has more columns than rows, so Corollary 8.7 guarantees a nonzero $\boldsymbol{c}$ in $\mathbf{N}(A)$. Elimination replaces row 2 by row 2 minus $\frac{1}{2}$ row 1, which is $(0, \frac{3}{2}, -\frac{9}{4})$. Choosing the free variable $c_3 = 2$, the second row gives $c_2 = 3$ and the first gives $c_1 = -4$:

$$-4\boldsymbol{v}_1 + 3\boldsymbol{v}_2 + 2\boldsymbol{v}_3 = (-8 + 3 + 5,\ -4 + 6 - 2) = (0, 0).$$

Figure 11 draws this: go out along $3\boldsymbol{v}_2$, then along $2\boldsymbol{v}_3$, then back along $-4\boldsymbol{v}_1$, and you are home.

When the vectors are columns, independence is a statement about a nullspace, and elimination decides it.

Theorem 10.3 (Independence test)**.** *The columns of an $m \times n$ matrix A are independent if and only if $\mathbf{N}(A) = \{\mathbf{0}\}$, if and only if $\operatorname{rank} A = n$ (every column is a pivot column, no free variables).*

Proof. $A\boldsymbol{c} = c_1\boldsymbol{a}_1 + \cdots + c_n\boldsymbol{a}_n$, so a combination of the columns that gives $\mathbf{0}$ is exactly a vector $\boldsymbol{c}$ in $\mathbf{N}(A)$; the columns are independent precisely when the only such vector is $\boldsymbol{c} = \mathbf{0}$. By Theorem 8.6, $\mathbf{N}(A) = \{\mathbf{0}\}$ exactly when there are no special solutions, i.e. no free columns, i.e. $r = n$. □

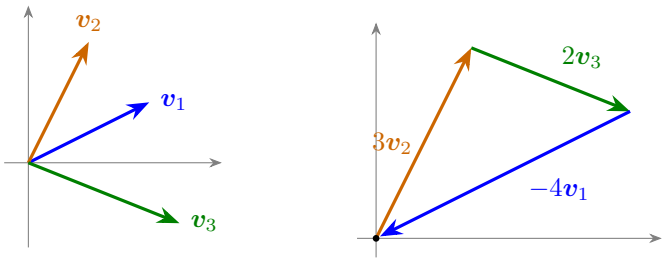


Figure 11: Left: three vectors in $\mathbb{R}^2$. Right: the combination $3\boldsymbol{v}_2 + 2\boldsymbol{v}_3 - 4\boldsymbol{v}_1$ traced head to tail returns to the origin, so the three vectors are dependent. Any three vectors in the plane admit such a closed path (Corollary 10.4).

Corollary 10.4 (Too many vectors)**.** *Any n vectors in $\mathbb{R}^m$ with $n > m$ are dependent.*

Proof. As columns of an $m \times n$ matrix they give $r \leq m < n$, so Theorem 10.3 applies. □

10.2 Spanning sets and bases

Definition 10.5 (Span)**.** Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ *span* a space V if V consists of all their linear combinations; we write $V = \text{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$. For example, the columns of A span $\mathbf{C}(A)$.

The span is the *smallest* subspace containing the vectors: it is a subspace (sums and multiples of combinations are combinations), and any subspace that contains $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ must contain all their combinations.

Definition 10.6 (Basis)**.** A *basis* for a vector space V is a list of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d$ that (i) are independent and (ii) span V.

Spanning means there are enough vectors; independence means there are not too many. A basis tells us everything about the space: every vector in it is a combination of the basis vectors, and, by the next proposition, in only one way.

Proposition 10.7 (Coordinates are unique)**.** *If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d$ is a basis of V, every $\boldsymbol{v} \in V$ can be written as $\boldsymbol{v} = c_1\boldsymbol{v}_1 + \cdots + c_d\boldsymbol{v}_d$ in exactly one way.*

Proof. Spanning gives at least one way. If $\sum c_i\boldsymbol{v}_i = \sum c_i'\boldsymbol{v}_i$ then $\sum(c_i - c_i')\boldsymbol{v}_i = \mathbf{0}$, and independence forces $c_i = c_i'$ for every i. □

Proposition 10.8 (Bases of $\mathbb{R}^n$)**.** *Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ in $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$ if and only if the $n \times n$ matrix with these columns is invertible.*

Proof. Let A be that matrix. Independence means $r = n$ (Theorem 10.3); spanning $\mathbb{R}^n$ means $A\boldsymbol{x} = \boldsymbol{b}$ is solvable for every $\boldsymbol{b}$, i.e. $r = m = n$ (Theorem 9.6). For a square matrix both say $r = n$, which is invertibility. □

Example 10.9 (Many bases of $\mathbb{R}^3$)**.** The standard basis $(1,0,0), (0,1,0), (0,0,1)$ consists of the columns of I, and $\mathbf{N}(I) = \{\mathbf{0}\}$. It is far from the only basis. The vectors $(1,1,2)$ and $(2,2,5)$ are independent (neither is a multiple of the other) but cannot span $\mathbb{R}^3$: as columns of a 3×2 matrix

they give $r \le 2 < 3$. They *are* a basis for the plane they span. To complete them to a basis of $\mathbb{R}^3$ we need a third vector outside that plane. $(3,3,7)$ fails, being their sum. The vector $(1,0,0)$ works: the matrix $\begin{bmatrix}1&2&1\\1&2&0\\2&5&0\end{bmatrix}$ is invertible (its determinant is 1; elimination finds three pivots).

Common trap. In the lecture, after rejecting $(3,3,7)$, the vector $(3,3,8)$ was offered as a third basis vector "sticking a little out of the plane". It does not: the matrix $\begin{bmatrix}1&2&3\\1&2&3\\2&5&8\end{bmatrix}$ has two equal rows, so it is not invertible and by Proposition 10.8 its columns are dependent. Indeed $(3,3,8) = -(1,1,2) + 2(2,2,5)$. Every vector in the plane of $(1,1,2)$ and $(2,2,5)$ has equal first and second components, and so does $(3,3,8)$. Checking the *rows* can expose a dependence among the columns that is invisible column by column; Section 11 explains why (row rank equals column rank).

10.3 Dimension

All bases of $\mathbb{R}^3$ above had three vectors. That is the central fact of the section.

Lemma 10.10 (Independent sets are no larger than spanning sets)**.** *If $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$ span V and $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n \in V$ are independent, then $n \le m$.*

Proof. Suppose $n > m$. Each $\boldsymbol{w}_j$ is a combination of the $\boldsymbol{v}$'s: $\boldsymbol{w}_j = a_{1j}\boldsymbol{v}_1 + \cdots + a_{mj}\boldsymbol{v}_m$. Collect the coefficients in the $m \times n$ matrix $A = (a_{ij})$; in matrix shorthand $W = VA$, where V and W have the $\boldsymbol{v}$'s and $\boldsymbol{w}$'s as columns. Since $n > m$, Corollary 8.7 gives $\boldsymbol{x} \neq \boldsymbol{0}$ with $A\boldsymbol{x} = \boldsymbol{0}$. Then $W\boldsymbol{x} = VA\boldsymbol{x} = \boldsymbol{0}$, i.e. $x_1\boldsymbol{w}_1 + \cdots + x_n\boldsymbol{w}_n = \boldsymbol{0}$ with not all $x_j = 0$, contradicting the independence of the $\boldsymbol{w}$'s. □

Theorem 10.11 (Every basis has the same size)**.** *Any two bases of a vector space V contain the same number of vectors.*

Proof. Let $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$ and $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$ be bases. The $\boldsymbol{v}$'s span and the $\boldsymbol{w}$'s are independent, so $n \le m$ by Lemma 10.10. Exchanging roles, $m \le n$. □

Definition 10.12 (Dimension)**.** The *dimension* of V, written $\dim V$, is the number of vectors in any basis of V. The space $\{\boldsymbol{0}\}$ has the empty basis and dimension 0.

So $\dim \mathbb{R}^n = n$, the plane spanned by $(1,1,2)$ and $(2,2,5)$ has dimension 2, and a line through the origin has dimension 1. Knowing the dimension saves half the work of checking a basis:

Proposition 10.13 (The right number of vectors)**.** *Let $\dim V = d$. (a) Any d independent vectors in V form a basis. (b) Any d vectors that span V form a basis. (c) Any independent list in V can be extended to a basis of V.*

Proof. First a remark: if $\boldsymbol{u}_1, \dots, \boldsymbol{u}_k$ are independent and $\boldsymbol{u} \notin \operatorname{span}\{\boldsymbol{u}_i\}$, then $\boldsymbol{u}_1, \dots, \boldsymbol{u}_k, \boldsymbol{u}$ are independent. (In a combination equal to $\boldsymbol{0}$ the coefficient of $\boldsymbol{u}$ must vanish, otherwise $\boldsymbol{u}$ would be a combination of the $\boldsymbol{u}_i$; then the other coefficients vanish too.)

(c) Keep adjoining a vector of V outside the current span. Each list stays independent, and by Lemma 10.10 (comparing with a basis of d vectors) no independent list has more than d vectors, so

the process stops; when it stops the list spans V. (a) If d independent vectors did not span V, the remark would produce $d+1$ independent vectors, contradicting Lemma 10.10. (b) If d spanning vectors were dependent, one of them would be a combination of the others and could be dropped, leaving $d-1$ spanning vectors; but a basis of V consists of d independent vectors, contradicting Lemma 10.10. □

10.4 Bases for the column space and the nullspace

Example 10.14 (Pivot columns and special solutions)**.** Take

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The four columns span $\mathbf{C}(A)$ by definition, but they are not independent: column 3 is column 1 plus column 2, and column 4 equals column 1. The pivot columns $(1,1,1)$ and $(2,1,2)$ form a basis of $\mathbf{C}(A)$, so $\dim \mathbf{C}(A) = 2 = \operatorname{rank} A$. Other bases are columns $1,3$, or columns $2,3$, or columns $2,4$, or vectors not among the columns at all such as $(2,2,2)$ and $(7,5,7)$ (two independent vectors in a two-dimensional space, Proposition 10.13(a)). The special solutions

$$\boldsymbol{s}_1 = (-1,-1,1,0), \qquad \boldsymbol{s}_2 = (-1,0,0,1)$$

record exactly those two dependencies, and they form a basis of $\mathbf{N}(A)$, which has dimension $4-2=2$.

Theorem 10.15 (Basis of the column space)**.** *The pivot columns of A (columns of A itself, not of R) form a basis of $\mathbf{C}(A)$. Hence $\dim \mathbf{C}(A) = \operatorname{rank} A$.*

Proof. Let $EA = R$ with E invertible; by Proposition 8.1, $A\boldsymbol{c} = \mathbf{0}$ if and only if $R\boldsymbol{c} = \mathbf{0}$, so the columns of A satisfy exactly the same linear relations as the columns of R.

Independent: a combination of the pivot columns of A equal to $\mathbf{0}$ is a vector $\boldsymbol{c}$ with $A\boldsymbol{c} = \mathbf{0}$ and nonzero entries only in pivot positions. Then $R\boldsymbol{c} = \mathbf{0}$. The pivot columns of R are the distinct identity columns $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_r$, so $R\boldsymbol{c} = \sum_i c_{p_i}\boldsymbol{e}_i = \mathbf{0}$ forces every $c_{p_i} = 0$.

Spanning: for a free column $\boldsymbol{a}_j$, the special solution $\boldsymbol{s}_j$ has 1 in position j, zeros in the other free positions, and $A\boldsymbol{s}_j = \mathbf{0}$. So $\boldsymbol{a}_j$ equals minus a combination of pivot columns. Every column, hence every vector of $\mathbf{C}(A)$, is a combination of the pivot columns. □

Theorem 10.16 (Basis of the nullspace)**.** *The $n-r$ special solutions form a basis of $\mathbf{N}(A)$. Hence $\dim \mathbf{N}(A) = n - r$.*

Proof. They span $\mathbf{N}(A)$ by Theorem 8.6. If $\sum_j c_j \boldsymbol{s}_j = \mathbf{0}$, look at the jth free position: only $\boldsymbol{s}_j$ is nonzero there, with entry 1, so the left side has entry c_j there. Hence every $c_j = 0$. □

Mind the grammar: a matrix has a rank, a space has a dimension. "The rank of A is the dimension of its column space" is correct; "the dimension of A" and "the rank of $\mathbf{C}(A)$" are not.

Example 10.17 (Basis and dimension of a span)**.** Find a basis and the dimension of the subspace of $\mathbb{R}^5$ spanned by

$$\boldsymbol{w}_1 = (1,1,-2,0,-1), \quad \boldsymbol{w}_2 = (1,2,0,-4,1), \quad \boldsymbol{w}_3 = (0,1,3,-3,2), \quad \boldsymbol{w}_4 = (2,3,0,-2,0).$$

As rows. Put the vectors in the rows of a matrix and eliminate: row 2 − row 1 and row 4 − 2 row 1 give $(0,1,2,-4,2)$ and $(0,1,4,-2,2)$; subtracting this new row 2 from rows 3 and 4 gives $(0,0,1,1,0)$ and $(0,0,2,2,0)$; finally row 4 − 2 row 3 $= \mathbf{0}$. Every step replaces a row by a combination of rows and can be undone, so the span of the rows never changes. The three nonzero rows

$$(1,1,-2,0,-1), \qquad (0,1,2,-4,2), \qquad (0,0,1,1,0)$$

are independent (look at the staircase of leading entries) and span the space: a basis, and the dimension is 3.

As columns. Put the vectors in the columns of a 5×4 matrix instead. Elimination finds pivots in columns $1,2,3$, so $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$ themselves form a basis (Theorem 10.15); the free column records $\boldsymbol{w}_4 = 3\boldsymbol{w}_1 - \boldsymbol{w}_2 + 2\boldsymbol{w}_3$. Here one must use the original columns: the columns of the eliminated matrix have zeros in their last two entries and span a different space. Both answers are correct; a space has many bases but one dimension.

10.5 Exercises

Exercise 10.18. Find the largest possible number of independent vectors among

$$\boldsymbol{v}_1 = (1,-1,0,0),\ \boldsymbol{v}_2 = (1,0,-1,0),\ \boldsymbol{v}_3 = (1,0,0,-1),$$
$$\boldsymbol{v}_4 = (0,1,-1,0),\ \boldsymbol{v}_5 = (0,1,0,-1),\ \boldsymbol{v}_6 = (0,0,1,-1).$$

Solution. Observe $\boldsymbol{v}_4 = \boldsymbol{v}_2 - \boldsymbol{v}_1$, $\boldsymbol{v}_5 = \boldsymbol{v}_3 - \boldsymbol{v}_1$ and $\boldsymbol{v}_6 = \boldsymbol{v}_3 - \boldsymbol{v}_2$, so the span is that of $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$. Eliminating $\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3\end{bmatrix}$ (row 2 + row 1, then row 3 + row 2, then row 4 + row 3) leaves pivots $1,1,1$ in the first three rows and a zero fourth row: three pivots, so $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are independent. The answer is 3, the rank of the 4×6 matrix of all six vectors. (All six vectors have components adding to zero, so they lie in a three-dimensional subspace of $\mathbb{R}^4$; Section 13 meets them again as the edges of a graph.) □

Exercise 10.19. Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbb{R}^3$, then a basis for its intersection with the xy-plane, then a basis for all vectors perpendicular to the plane.

Solution. The plane is $\mathbf{N}(\begin{bmatrix}1 & -2 & 3\end{bmatrix})$. The pivot variable is x; the special solutions are $\boldsymbol{v}_1 = (2,1,0)$ (from $y = 1, z = 0$) and $\boldsymbol{v}_2 = (-3,0,1)$ (from $y = 0, z = 1$), a basis of the plane. The intersection with the xy-plane ($z = 0$) contains $\boldsymbol{v}_1$ but not $\boldsymbol{v}_2$, so it is a line, with basis $\boldsymbol{v}_1$. The vector $\boldsymbol{v}_3 = (1,-2,3)$ satisfies $\boldsymbol{v}_3 \cdot \boldsymbol{v}_1 = 2 - 2 + 0 = 0$ and $\boldsymbol{v}_3 \cdot \boldsymbol{v}_2 = -3 + 0 + 3 = 0$, so it is perpendicular to the plane; the perpendicular vectors form a line, with basis $\boldsymbol{v}_3$. □

Exercise 10.20. Suppose $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$ are independent. Show that $\boldsymbol{v}_1 = \boldsymbol{w}_2 - \boldsymbol{w}_3$, $\boldsymbol{v}_2 = \boldsymbol{w}_1 - \boldsymbol{w}_3$, $\boldsymbol{v}_3 = \boldsymbol{w}_1 - \boldsymbol{w}_2$ are dependent, and that $\boldsymbol{u}_1 = \boldsymbol{w}_2 + \boldsymbol{w}_3$, $\boldsymbol{u}_2 = \boldsymbol{w}_1 + \boldsymbol{w}_3$, $\boldsymbol{u}_3 = \boldsymbol{w}_1 + \boldsymbol{w}_2$ are independent.

Solution. $\boldsymbol{v}_1 - \boldsymbol{v}_2 + \boldsymbol{v}_3 = (\boldsymbol{w}_2 - \boldsymbol{w}_3) - (\boldsymbol{w}_1 - \boldsymbol{w}_3) + (\boldsymbol{w}_1 - \boldsymbol{w}_2) = \mathbf{0}$ is a nonzero combination giving $\mathbf{0}$. For the $\boldsymbol{u}$'s, suppose $c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + c_3\boldsymbol{u}_3 = \mathbf{0}$, i.e. $(c_2 + c_3)\boldsymbol{w}_1 + (c_1 + c_3)\boldsymbol{w}_2 + (c_1 + c_2)\boldsymbol{w}_3 = \mathbf{0}$. Independence of the $\boldsymbol{w}$'s gives $c_2 + c_3 = c_1 + c_3 = c_1 + c_2 = 0$; adding, $c_1 + c_2 + c_3 = 0$, and subtracting each equation gives $c_1 = c_2 = c_3 = 0$. □

Exercise 10.21. Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ be a basis of $\mathbb{R}^n$ and B an invertible $n \times n$ matrix. Show that $B\boldsymbol{v}_1, \ldots, B\boldsymbol{v}_n$ is also a basis of $\mathbb{R}^n$.

Solution. If $\sum c_i B\boldsymbol{v}_i = \mathbf{0}$ then $B\big(\sum c_i \boldsymbol{v}_i\big) = \mathbf{0}$; multiplying by B^{-1} gives $\sum c_i \boldsymbol{v}_i = \mathbf{0}$, so all $c_i = 0$. The n vectors $B\boldsymbol{v}_i$ are therefore independent, and since $\dim \mathbb{R}^n = n$, Proposition 10.13(a) says they form a basis. □

11 The four fundamental subspaces

So far each matrix has come with two subspaces, the column space and the nullspace. Transposing the matrix gives two more, and the four together are the heart of this approach to linear algebra. This section finds a basis and the dimension of each, all from one elimination $EA = R$, and proves the first great count: the row space and the column space have the same dimension r. That fact already corrects the slip of Section 10, where $(3, 3, 8)$ was proposed as independent of $(1, 1, 2)$ and $(2, 2, 5)$: the matrix with those columns has two equal rows, so its row space is only two-dimensional, and therefore so is its column space.

11.1 Two subspaces in $\mathbb{R}^n$, two in $\mathbb{R}^m$

Definition 11.1 (The four fundamental subspaces)**.** Let A be an $m \times n$ matrix.

- The *column space* $\mathbf{C}(A) \subseteq \mathbb{R}^m$: all combinations of the columns.
- The *nullspace* $\mathbf{N}(A) \subseteq \mathbb{R}^n$: all solutions of $A\boldsymbol{x} = \mathbf{0}$.
- The *row space* $\mathbf{C}(A^{\mathsf{T}}) \subseteq \mathbb{R}^n$: all combinations of the rows of A, written as column vectors, i.e. the column space of A^{T}.
- The *left nullspace* $\mathbf{N}(A^{\mathsf{T}}) \subseteq \mathbb{R}^m$: all solutions of $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$.

Transposing $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$ gives $\boldsymbol{y}^{\mathsf{T}}A = \mathbf{0}^{\mathsf{T}}$: a row vector multiplying A from the *left* gives the zero row. That explains the name. A vector $\boldsymbol{y}$ in the left nullspace lists the coefficients of a combination of the rows of A that produces the zero row, just as a vector in $\mathbf{N}(A)$ lists a combination of the columns that produces the zero column. These are exactly the vectors that appeared in the solvability test of Theorem 9.1. Figure 12 shows where the four spaces live.

For the column space and the nullspace the work is already done (Theorems 10.15 and 10.16): the pivot columns of A are a basis of $\mathbf{C}(A)$, of dimension r, and the $n - r$ special solutions are a basis of $\mathbf{N}(A)$. One could treat A^{T} the same way, but that means a second elimination. The point is that the elimination already done on A suffices.

11.2 The row space

Theorem 11.2 (Row space)**.** *If E is invertible and $EA = R$, then A and R have the same row space. In particular the r nonzero rows of $R = \operatorname{rref}(A)$ form a basis of $\mathbf{C}(A^{\mathsf{T}})$, and $\dim \mathbf{C}(A^{\mathsf{T}}) = r$.*

Proof. Row i of $R = EA$ is $\sum_k e_{ik}$(row k of A), a combination of the rows of A; so the row space of R is contained in that of A. Since $A = E^{-1}R$, each row of A is likewise a combination of the rows of R, giving the reverse inclusion. The zero rows of R contribute nothing, so the r nonzero rows span. They are independent: the ith nonzero row has a 1 in the ith pivot column, where every other row of R has a 0, so in any combination equal to $\mathbf{0}$ the ith coefficient is 0. □

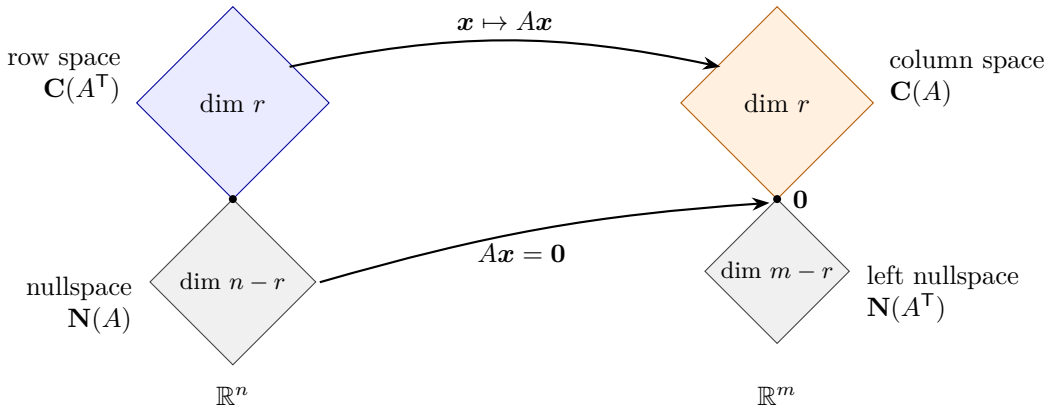


Figure 12: The four fundamental subspaces of an $m \times n$ matrix of rank r. The row space and nullspace live in $\mathbb{R}^n$ and their dimensions add to n; the column space and left nullspace live in $\mathbb{R}^m$ and their dimensions add to m. A sends every input into the column space and sends the nullspace to $\mathbf{0}$.

Corollary 11.3 (Row rank equals column rank)**.** *For every matrix,* $\dim \mathbf{C}(A^\mathsf{T}) = \dim \mathbf{C}(A)$*; that is,* $\operatorname{rank} A^\mathsf{T} = \operatorname{rank} A$*.*

Proof. Both dimensions equal the number r of pivots of A, by Theorems 11.2 and 10.15. Applying Theorem 10.15 to A^T shows that $\dim \mathbf{C}(A^\mathsf{T})$ is also the rank of A^T. □

This is a statement with content: there is no evident reason why the number of independent rows of a 3×7 matrix should equal its number of independent columns, yet it always does.

11.3 The left nullspace and the matrix E

To find $\mathbf{N}(A^\mathsf{T})$ without transposing, we keep a record of the elimination. Attach the $m \times m$ identity to A and apply to $\begin{bmatrix} A & I \end{bmatrix}$ the row operations that turn A into R:

$$\begin{bmatrix} A_{m\times n} & I_{m\times m} \end{bmatrix} \longrightarrow \begin{bmatrix} R_{m\times n} & E_{m\times m} \end{bmatrix}.$$

The row operations amount to multiplying on the left by some invertible matrix, and that matrix turned I into E, so it *is* E: we get $EA = R$. When A is square and invertible, $R = I$ and this is Gauss–Jordan: $E = A^{-1}$. In general E is not unique when $r < m$ (one may add multiples of the bottom rows to the others), but any such E works below.

Theorem 11.4 (Left nullspace)**.** *Let* $EA = R$ *with* E *invertible and* $R = \operatorname{rref}(A)$*. The last* $m - r$ *rows of* E *form a basis of* $\mathbf{N}(A^\mathsf{T})$*, and* $\dim \mathbf{N}(A^\mathsf{T}) = m - r$*.*

Proof. A^T is $n \times m$ with rank r (Corollary 11.3), so by Theorem 10.16 its nullspace has dimension $m - r$. The last $m - r$ rows of $R = EA$ are zero, so each of the last $m - r$ rows $\boldsymbol{y}^\mathsf{T}$ of E satisfies $\boldsymbol{y}^\mathsf{T} A = \mathbf{0}^\mathsf{T}$. These rows are independent because E is invertible (Theorem 10.3 applied to E^T). So they are $m - r$ independent vectors in an $(m - r)$-dimensional space, and Proposition 10.13(a) makes them a basis. □

Theorem 11.5 (Dimensions of the four subspaces)**.** *For an $m \times n$ matrix of rank r:*

$$\dim \mathbf{C}(A) = r, \qquad \dim \mathbf{N}(A) = n - r, \qquad \dim \mathbf{C}(A^{\mathsf{T}}) = r, \qquad \dim \mathbf{N}(A^{\mathsf{T}}) = m - r.$$

In $\mathbb{R}^n$: $r + (n - r) = n$. In $\mathbb{R}^m$: $r + (m - r) = m$.

Proof. Theorems 10.15, 10.16, 11.2 and 11.4. □

The count $r + (n - r) = n$ is the old count of pivot variables plus free variables; the count in $\mathbb{R}^m$ is the same statement for A^{T}, which is just as good a matrix as A.

Example 11.6 (All four subspaces at once)**.** Take $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$ from Example 10.14, so $m = 3$, $n = 4$, $r = 2$. Reduce $\begin{bmatrix} A & I \end{bmatrix}$: subtract row 1 from rows 2 and 3, multiply row 2 by -1, subtract 2×row 2 from row 1:

$$\left[\begin{array}{cccc|ccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|ccc} 1 & 0 & 1 & 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{array}\right].$$

Check $EA = R$ row by row: $-(1,2,3,1) + 2(1,1,2,1) = (1,0,1,1)$, $(1,2,3,1) - (1,1,2,1) = (0,1,1,0)$, $-(1,2,3,1) + (1,2,3,1) = \mathbf{0}$. Now read off:

- $\mathbf{C}(A)$ (dimension 2): pivot columns of A, $(1,1,1)$ and $(2,1,2)$.
- $\mathbf{N}(A)$ (dimension $4 - 2 = 2$): special solutions $(-1,-1,1,0)$ and $(-1,0,0,1)$.
- $\mathbf{C}(A^{\mathsf{T}})$ (dimension 2): nonzero rows of R, $(1,0,1,1)$ and $(0,1,1,0)$.
- $\mathbf{N}(A^{\mathsf{T}})$ (dimension $3 - 2 = 1$): last row of E, $\boldsymbol{y} = (-1,0,1)$; indeed $-(\text{row } 1) + (\text{row } 3)$ is the zero row.

Common trap. Elimination keeps the row space and the nullspace but changes the column space and the left nullspace. In Example 11.6, $(1,1,1)$ is a column of A, so it lies in $\mathbf{C}(A)$; but every column of R has third component 0, so $(1,1,1) \notin \mathbf{C}(R)$. Take the *rows of R* for the row space, but the *columns of A* for the column space. Likewise $\mathbf{N}(R^{\mathsf{T}})$ is spanned by $(0,0,1)$, not by the correct $(-1,0,1)$.

11.4 What a matrix keeps and what it crushes

Example 11.7 (The four subspaces of a rank-one matrix)**.** Return to $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix}$ from Example 8.12, with $m = 2$, $n = 3$ and $r = 1$.

- *Inputs,* $\mathbb{R}^3$*.* The row space is the line through $(1,2,2)$ (dimension $r = 1$). The nullspace is the plane spanned by $\boldsymbol{s}_1 = (-2,1,0)$ and $\boldsymbol{s}_2 = (-2,0,1)$ (dimension $n - r = 2$). Both special solutions are perpendicular to $(1,2,2)$. Count: $1 + 2 = 3$.
- *Outputs,* $\mathbb{R}^2$*.* Every column is a multiple of $(1,2)$, so the column space is the line through $(1,2)$ (dimension 1). Row 2 minus 2×row 1 is zero, so $\boldsymbol{y} = (-2,1)$, or equally $(2,-1)$, spans the left nullspace (dimension $m - r = 1$). Count: $1 + 1 = 2$.

Now take the input $\boldsymbol{x} = (-1, 3, 2)$. It splits as a row-space part plus a nullspace part,

$$\boldsymbol{x} = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \qquad A\boldsymbol{x} = \begin{bmatrix} -1+6+4 \\ -2+12+8 \end{bmatrix} = \begin{bmatrix} 9 \\ 18 \end{bmatrix} = A \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The nullspace part was crushed to $\mathbf{0}$; only the row-space part reached the output, landing at $(9, 18)$ on the column-space line. The information that A destroys is not scattered at random: it is exactly the nullspace component, and that subspace is perpendicular to the rows. (Section 16 proves that the nullspace is always perpendicular to the row space, and the left nullspace to the column space; here $(1, 2) \cdot (2, -1) = 0$.) Figure 13 draws both sides.

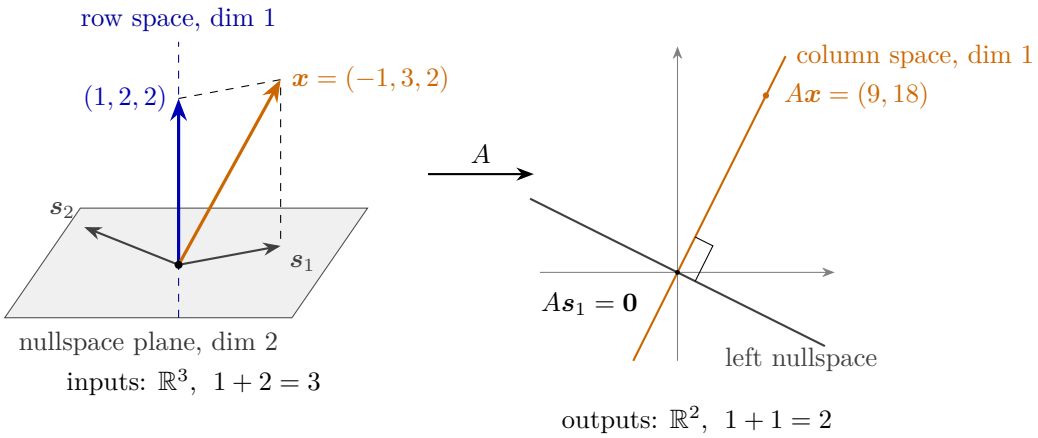


Figure 13: The matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix}$. Left (schematic): the input $\boldsymbol{x} = (-1, 3, 2)$ is the row-space vector $(1, 2, 2)$ plus the nullspace vector $\boldsymbol{s}_1$. Right (to scale): A crushes $\boldsymbol{s}_1$ to $\mathbf{0}$ and sends the row-space part to $(9, 18)$ on the column-space line, which is perpendicular to the left nullspace line through $(2, -1)$.

11.5 A new vector space: matrices

Nothing in the definition of a vector space requires the vectors to be columns of numbers. All we use is that vectors can be added and multiplied by scalars, following the usual rules.

Example 11.8 (The space of 3×3 matrices)**.** Let M be the set of all 3×3 real matrices. Matrices can be added and multiplied by numbers, there is a zero matrix, and all eight vector-space rules hold; that we can also *multiply* two matrices is irrelevant here. Three subspaces of M: all upper triangular matrices U; all symmetric matrices S; and their intersection $S \cap U$, which is the set D of diagonal matrices (symmetric with zeros below the diagonal forces zeros above it). Each is closed under addition and scalar multiplication. A basis of D is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix};$$

they are independent (a combination with weights a, b, c is $\operatorname{diag}(a, 3b, 7c)$, which is zero only when $a = b = c = 0$), and every diagonal matrix $\operatorname{diag}(d_1, d_2, d_3)$ is the combination with weights d_1, $d_2/3$, $d_3/7$. So $\dim D = 3$. Section 12 finds the dimensions of M, S and U.

11.6 Exercises

Exercise 11.9. A is $m \times n$ of rank r, and some right-hand sides $\boldsymbol{b}$ give no solution of $A\boldsymbol{x} = \boldsymbol{b}$. (a) Which inequalities ($<$ or $\leq$) must hold between m, n and r? (b) How do you know that $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{0}$ has solutions other than $\boldsymbol{y} = \boldsymbol{0}$?

Solution. (a) Always $r \leq m$ and $r \leq n$. Since some $\boldsymbol{b}$ is unreachable, $\mathbf{C}(A) \neq \mathbb{R}^m$, so $r < m$ (Theorem 9.6(b)). Nothing more can be said about r versus n. (b) $\dim \mathbf{N}(A^{\mathsf{T}}) = m - r > 0$, so the left nullspace contains nonzero vectors. □

Exercise 11.10. For which $\boldsymbol{d}$ is $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{d}$ solvable? When is the solution unique?

Solution. $A^{\mathsf{T}}\boldsymbol{y}$ is a combination of the columns of A^{T}, i.e. of the rows of A, so the system is solvable exactly when $\boldsymbol{d}$ is in the row space $\mathbf{C}(A^{\mathsf{T}})$. Two solutions differ by a vector in $\mathbf{N}(A^{\mathsf{T}})$, so the solution is unique when the left nullspace is $\{\boldsymbol{0}\}$, i.e. when $r = m$. □

Exercise 11.11. Let $B = LU$ with $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 5 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Find a basis and the dimension of each of the four subspaces of B, without multiplying out B.

Solution. U has two pivots, so $r = 2$ and all four dimensions are 2, $3-2=1$, 2, $3-2=1$. *Column space:* since L is invertible, the columns of $B = LU$ satisfy the same relations as the columns of U, so columns 1 and 2 of B are pivot columns. They are L times $(5,0,0)$ and $(0,1,0)$, i.e. $5(1,2,-1)$ and $(0,1,0)$; a basis is $(1,2,-1), (0,1,0)$, the first two columns of L. *Nullspace:* $\mathbf{N}(B) = \mathbf{N}(U)$; the free variable $x_3 = 1$ gives $x_2 = -1$ and $5x_1 + 3 = 0$, so the basis is $(-\frac{3}{5}, -1, 1)$. *Row space:* $L^{-1}B = U$ is a sequence of row operations, so the row space is spanned by the pivot rows of U: $(5,0,3)$ and $(0,1,1)$. *Left nullspace:* $L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ plays the role of E, and its last row $(1,0,1)$ is the basis: row 1 of B is $(5,0,3)$, row 3 is $-(5,0,3)$, and they cancel. □

Exercise 11.12. Find a 3×3 matrix whose column space contains $(1,1,0)$ and $(1,0,1)$ and whose nullspace contains $(1,1,1)$. What must its rank be?

Solution. The column space contains two independent vectors, so $r \geq 2$; the nullspace is at least a line, so $3 - r \geq 1$. Hence $r = 2$. The condition $A(1,1,1) = \boldsymbol{0}$ says the three columns add to zero. Take the first two columns to be the given vectors and the third their negative sum: $A = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$. Its column space is the plane spanned by $(1,1,0)$ and $(1,0,1)$, and $A(1,1,1) = (0,0,0)$. □

12 Matrix spaces; rank one; small world graphs

Everything in Sections 10 and 11 used only two operations: adding vectors and multiplying them by numbers. So the same ideas (subspace, basis, dimension) apply to any objects that can be added and scaled. This section works them out for spaces of matrices and a space of functions, introduces

two natural operations on subspaces (intersection and sum), shows that rank-one matrices are the building blocks of every matrix, and ends with the graphs that lead into Section 13.

12.1 Bases and dimensions of matrix spaces

Let M be the space of all 3×3 matrices (Example 11.8). Write E_{ij} for the matrix with a 1 in position (i, j) and zeros elsewhere.

Proposition 12.1 (Dimensions in M)**.** *(a) The nine matrices E_{ij} form a basis of M, so $\dim M = 9$.*

(b) The six E_{ij} with $i \le j$ form a basis of the upper triangular matrices U: $\dim U = 6$.

(c) $E_{11}, E_{22}, E_{33},\ E_{12}+E_{21},\ E_{13}+E_{31},\ E_{23}+E_{32}$ form a basis of the symmetric matrices S: $\dim S = 6$.

(d) E_{11}, E_{22}, E_{33} form a basis of the diagonal matrices $D = S \cap U$: $\dim D = 3$.

Proof. (a) $A = \sum_{i,j} a_{ij} E_{ij}$, and this combination is the zero matrix only if every $a_{ij} = 0$. (b) The same argument applies to matrices with $a_{ij} = 0$ for $i > j$. (c) A symmetric matrix is determined by its three diagonal entries and three entries above the diagonal: $A = \sum_i a_{ii} E_{ii} + \sum_{i<j} a_{ij}(E_{ij} + E_{ji})$, and the six coefficients are entries of A, so the combination vanishes only when they are all 0. (d) As in (a). □

So M is "practically $\mathbb{R}^9$", with the nine numbers written in a square. The bases of U and D happen to be subsets of the standard basis of M; the basis of S cannot be, because only three of the nine E_{ij} are symmetric. In general a subspace needs its own basis.

12.2 Intersections and sums of subspaces

Proposition 12.2 (Intersection)**.** *If V and W are subspaces of the same space, so is $V \cap W$.*

Proof. If $\boldsymbol{x}, \boldsymbol{y}$ lie in both V and W, then $\boldsymbol{x}+\boldsymbol{y}$ and $c\boldsymbol{x}$ lie in V (a subspace) and in W (a subspace). □

The union $V \cup W$ is usually *not* a subspace: two different lines through the origin in $\mathbb{R}^2$ are not closed under addition (Figure 14). To get a subspace we must fill in between them.

Definition 12.3 (Sum of subspaces)**.** The *sum* of subspaces V and W is $V + W = \{\boldsymbol{v} + \boldsymbol{w} : \boldsymbol{v} \in V,\ \boldsymbol{w} \in W\}$.

$V + W$ is a subspace: $(\boldsymbol{v}+\boldsymbol{w}) + (\boldsymbol{v}'+\boldsymbol{w}') = (\boldsymbol{v}+\boldsymbol{v}') + (\boldsymbol{w}+\boldsymbol{w}')$ and $c(\boldsymbol{v}+\boldsymbol{w}) = c\boldsymbol{v} + c\boldsymbol{w}$. It is the smallest subspace containing $V \cup W$, since any subspace containing both must contain all such sums.

Example 12.4 (Symmetric plus upper triangular)**.** $S + U = M$: every 3×3 matrix is a symmetric matrix plus an upper triangular one. Copy the part below the diagonal into a symmetric matrix and put the rest into the upper triangular one:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 7 \\ 4 & 0 & 8 \\ 7 & 8 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -2 & -4 \\ 0 & 5 & -2 \\ 0 & 0 & 9 \end{bmatrix}.$$

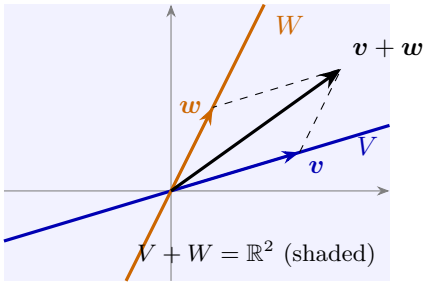


Figure 14: Two lines V and W through the origin. Their union is not a subspace: $\boldsymbol{v}+\boldsymbol{w}$ lies on neither line. Their sum $V+W$ is the whole plane, and their intersection is $\{\mathbf{0}\}$, so $\dim(V+W)+\dim(V\cap W)=2+0=1+1$.

The same recipe works for every matrix. Note the dimensions: $\dim S+\dim U = 6+6$ and $\dim(S+U)+\dim(S\cap U)=9+3$.

Theorem 12.5 (Dimension of a sum)**.** *For finite-dimensional subspaces V and W of the same space,*

$$\dim(V+W)+\dim(V\cap W)=\dim V+\dim W.$$

Proof. Let $\boldsymbol{u}_1,\dots,\boldsymbol{u}_k$ be a basis of $V\cap W$. By Proposition 10.13(c) it extends to a basis of V, say $\boldsymbol{u}_1,\dots,\boldsymbol{u}_k,\boldsymbol{v}_1,\dots,\boldsymbol{v}_p$, and to a basis of W, say $\boldsymbol{u}_1,\dots,\boldsymbol{u}_k,\boldsymbol{w}_1,\dots,\boldsymbol{w}_q$. We claim that all $k+p+q$ vectors $\boldsymbol{u}_i,\boldsymbol{v}_j,\boldsymbol{w}_l$ together form a basis of $V+W$; then $\dim(V+W)+\dim(V\cap W)=(k+p+q)+k=(k+p)+(k+q)$, as required.

They span $V+W$, since every $\boldsymbol{v}\in V$ and every $\boldsymbol{w}\in W$ is a combination of them. For independence, suppose $\sum a_i\boldsymbol{u}_i+\sum b_j\boldsymbol{v}_j+\sum c_l\boldsymbol{w}_l=\mathbf{0}$. Then $\boldsymbol{z}=\sum c_l\boldsymbol{w}_l=-\sum a_i\boldsymbol{u}_i-\sum b_j\boldsymbol{v}_j$ lies in W (left side) and in V (right side), hence in $V\cap W$, so $\boldsymbol{z}=\sum d_i\boldsymbol{u}_i$ for some d_i. Then $\sum c_l\boldsymbol{w}_l-\sum d_i\boldsymbol{u}_i=\mathbf{0}$, and independence of the basis of W gives all $c_l=0$ (and $d_i=0$). Now $\sum a_i\boldsymbol{u}_i+\sum b_j\boldsymbol{v}_j=\mathbf{0}$, and independence of the basis of V gives all $a_i=b_j=0$. □

Common trap. The dimension formula involves the *sum* $V+W$, not the union. "$\dim(S\cup U)$" has no meaning, because $S\cup U$ is not a vector space: the symmetric matrix $E_{12}+E_{21}$ plus the upper triangular matrix E_{12} is $2E_{12}+E_{21}$, which is neither symmetric nor upper triangular.

12.3 A space of functions

The solutions of the differential equation

$$\frac{d^2y}{dx^2}+y=0$$

form a vector space: if $y_1''+y_1=0$ and $y_2''+y_2=0$ then $(c_1y_1+c_2y_2)''+(c_1y_1+c_2y_2)=0$. It is the "nullspace" of the operation $y\mapsto y''+y$. Both $\cos x$ and $\sin x$ are in it.

Theorem 12.6 (Solutions of $y''+y=0$)**.** *$\cos x$ and $\sin x$ form a basis of the solution space, which therefore has dimension 2. The solution with $y(0)=a$ and $y'(0)=b$ is $y=a\cos x+b\sin x$.*

Proof. Independent: if $c_1\cos x+c_2\sin x=0$ for all x, then $x=0$ gives $c_1=0$ and $x=\pi/2$ gives $c_2=0$. *Spanning:* let y be any solution and put $z=y-y(0)\cos x-y'(0)\sin x$. Then z is a solution

with $z(0) = 0$ and $z'(0) = 0$. The quantity $E = z^2 + (z')^2$ satisfies $E' = 2zz' + 2z'z'' = 2z'(z + z'') = 0$, so E is constant, equal to $E(0) = 0$. Hence $z \equiv 0$ and $y = y(0)\cos x + y'(0)\sin x$. □

These two functions play the role of the special solutions: a second-order equation has a two-dimensional solution space, and solving it means finding a basis. Other bases exist; if complex scalars are allowed, e^{ix} and e^{-ix} form one (they are $\cos x \pm i \sin x$). For example, the solution with $y(0) = 2$ and $y'(0) = -1$ is $y = 2\cos x - \sin x$; check: $y'' = -2\cos x + \sin x = -y$.

12.4 Rank-one matrices

Example 12.7 (A column times a row)**.** In $A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix}$ the second row is twice the first, so the rank is 1. A basis of the row space is $(1, 4, 5)$, a basis of the column space is $(1, 2)$, and every column is a multiple of $(1, 2)$, with multipliers $1, 4, 5$. That is exactly the product

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}.$$

Theorem 12.8 (Rank one)**.** *A matrix has rank* 1 *if and only if it equals* $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ *for some nonzero column vectors* $\boldsymbol{u} \in \mathbb{R}^m$*,* $\boldsymbol{v} \in \mathbb{R}^n$*. Then* $\mathbf{C}(A)$ *is the line through* $\boldsymbol{u}$*,* $\mathbf{C}(A^{\mathsf{T}})$ *is the line through* $\boldsymbol{v}$*, and* $\mathbf{N}(A) = \{\boldsymbol{x} : \boldsymbol{v}^{\mathsf{T}}\boldsymbol{x} = 0\}$*.*

Proof. Column j of $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ is $v_j\boldsymbol{u}$, so every column lies on the line through $\boldsymbol{u}$, and some column is nonzero because $\boldsymbol{v} \neq \mathbf{0}$ and $\boldsymbol{u} \neq \mathbf{0}$; hence the column space is that line and the rank is 1. Conversely, if $\operatorname{rank} A = 1$, then $\mathbf{C}(A)$ has a basis consisting of one vector $\boldsymbol{u}$, so each column is $\boldsymbol{a}_j = v_j\boldsymbol{u}$ for some number v_j; that says $A = \boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$, and $\boldsymbol{v} \neq \mathbf{0}$ because $A \neq 0$. The same argument applied to the rows ($A^{\mathsf{T}} = \boldsymbol{v}\boldsymbol{u}^{\mathsf{T}}$) gives the row space. Finally $A\boldsymbol{x} = \boldsymbol{u}(\boldsymbol{v}^{\mathsf{T}}\boldsymbol{x}) = \mathbf{0}$ exactly when the number $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{x}$ is 0. □

Example 12.9 (The four subspaces of $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$)**.** The vectors $\boldsymbol{v} \in \mathbb{R}^4$ with $v_1 + v_2 + v_3 + v_4 = 0$ form a subspace, the nullspace of the rank-one matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$ ($m = 1$, $n = 4$, $r = 1$). The pivot variable is v_1, and the three special solutions

$$(-1, 1, 0, 0), \qquad (-1, 0, 1, 0), \qquad (-1, 0, 0, 1)$$

are a basis, so the dimension is $n - r = 3$. The row space is the line through $(1, 1, 1, 1)$ (dimension 1); $3 + 1 = 4$. The column space is all of $\mathbb{R}^1$ (dimension 1), and the left nullspace is $\{0\}$ (dimension 0, with the empty set as basis); $1 + 0 = 1$.

Rank-one matrices are the building blocks of all matrices.

Theorem 12.10 (A sum of rank-one pieces)**.** *Let* A *have rank* r*, let* C *be the* $m \times r$ *matrix of pivot columns of* A*, and let* R_0 *be the* $r \times n$ *matrix of nonzero rows of* $R = \operatorname{rref}(A)$*. Then* $A = CR_0$*; equivalently,* A *is the sum of the* r *rank-one matrices* (*pivot column* k)(*row* k *of* R_0)*.*

Proof. Column j of CR_0 is C times column j of R_0. If j is the kth pivot column, column j of R_0 is the kth standard basis vector and the product is the kth column of C, which is $\boldsymbol{a}_j$. If j is free, column j of R_0 holds the entries $f_{1j}, \ldots, f_{rj}$ of F in that column, and the special solution $\boldsymbol{s}_j$ (pivot entries $-f_{kj}$, entry 1 in position j) gives $A\boldsymbol{s}_j = \boldsymbol{a}_j - \sum_k f_{kj}$(pivot column k) $= \mathbf{0}$. So $\boldsymbol{a}_j = \sum_k f_{kj}$(pivot column k), which is again column j of CR_0. The column-row expansion of a product then writes CR_0 as a sum of r column-times-row terms. □

For the matrix of Example 11.6 this reads

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix},$$

two rank-one pieces for a rank-two matrix. In the same way a 5×17 matrix of rank 4 is a sum of four rank-one matrices.

Proposition 12.11 (Rank of a sum)**.** $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$ *for matrices of the same size.*

Proof. Every column of $A+B$ is a column of A plus a column of B, so $\mathbf{C}(A+B) \subseteq \mathbf{C}(A) + \mathbf{C}(B)$. By Theorem 12.5, $\dim(\mathbf{C}(A) + \mathbf{C}(B)) \le \dim \mathbf{C}(A) + \dim \mathbf{C}(B)$. □

Consequently the rank-one matrices do *not* form a subspace of M: $E_{11} + E_{22}$ has rank 2. Nor do the 5×17 matrices of rank at most 4 (even with the zero matrix included): with E_{ij} now 5×17, the matrices $E_{11} + E_{22} + E_{33} + E_{44}$ and $E_{22} + E_{33} + E_{44} + E_{55}$ have rank 4, but their sum has the five pivots $1, 2, 2, 2, 1$ and rank 5.

12.5 Small world graphs

Definition 12.12 (Graph, distance, diameter)**.** A *graph* $G = \{\text{nodes}, \text{edges}\}$ is a set of nodes together with edges, each edge joining two nodes. The *distance* between two nodes is the smallest number of edges in a path joining them; the *diameter* is the largest distance between two nodes.

Take one node for each person in a country and an edge between any two people who are friends. The phrase "six degrees of separation" is the claim that this graph has small diameter, roughly six; when two strangers on a plane find a mutual friend they remark that it is a small world. The World Wide Web, with sites as nodes and links as edges, raises the same question. The mathematical point is that a few long-range shortcuts make all distances drop dramatically.

Example 12.13 (A ring with shortcuts)**.** Put 12 people in a ring, each friends only with the two neighbours. The distance between nodes i and j is $\min(|i-j|,\ 12 - |i-j|)$, so the diameter is 6. From any node the other eleven are at distances $1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6$, which sum to 36: the average distance is $36/11 \approx 3.27$. Now add the long-range friendships 1–7, 3–9, 5–11 one at a time (Figure 15). Breadth-first search from every node gives

shortcuts added	none	1–7	+ 3–9	+ 5–11
diameter	6	6	5	3
average distance	3.27	2.74	2.39	2.18

Three extra edges out of fifteen halve the diameter. For instance, after all three shortcuts node 2 reaches node 8 in three steps ($2 \to 1 \to 7 \to 8$) instead of six.

To study such questions with linear algebra we need a matrix for every graph. Section 13 builds it, the incidence matrix, for a small directed graph.

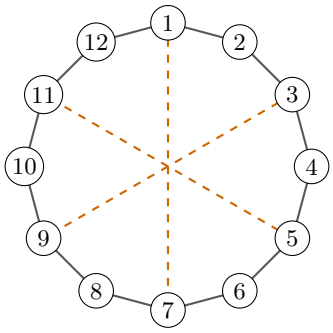


Figure 15: A ring of 12 friends (solid) with three long-range shortcuts (dashed). Without shortcuts the diameter is 6; with them it is 3.

12.6 Exercises

Exercise 12.14. Write the 3×3 identity as a combination of the other five 3×3 permutation matrices, and show that those five are linearly independent. (They form a basis of the subspace of 3×3 matrices whose row and column sums are all equal.)

Solution. The other five are P_{21}, P_{31}, P_{32} (exchanging two rows of I) and the cyclic shifts

$$P_{32}P_{21} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad P_{21}P_{32} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

$P_{21} + P_{31} + P_{32}$ is the all-ones matrix and $P_{32}P_{21} + P_{21}P_{32}$ has zeros on the diagonal and ones elsewhere, so $I = P_{21} + P_{31} + P_{32} - P_{32}P_{21} - P_{21}P_{32}$. For independence,

$$c_1P_{21} + c_2P_{31} + c_3P_{32} + c_4P_{32}P_{21} + c_5P_{21}P_{32} = \begin{bmatrix} c_3 & c_1 + c_4 & c_2 + c_5 \\ c_1 + c_5 & c_2 & c_3 + c_4 \\ c_2 + c_4 & c_3 + c_5 & c_1 \end{bmatrix}.$$

If this is zero, the diagonal gives $c_1 = c_2 = c_3 = 0$ and then the off-diagonal entries give $c_4 = c_5 = 0$. □

Exercise 12.15. Let M be the space of 3×3 matrices and $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$. Consider the operation $X \mapsto AX$ on M. (a) Which X give $AX = 0$? (b) Which matrices have the form AX? (c) What are the dimensions of these two subspaces of M, and why do they add to 9?

Solution. (a) A has rank 2 (the rows add to zero, and the first two are independent), and $\mathbf{N}(A)$ is the line through $(1, 1, 1)$. $AX = 0$ means every column of X lies in $\mathbf{N}(A)$, so $X = \begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix}$. (b) The columns of AX are combinations of the columns of A, all of which have components summing to 0; and since $\mathbf{C}(A)$ is the whole plane of such vectors (dimension 2), every matrix whose columns each sum to 0 is some AX. (c) The "nullspace" has dimension 3 (a, b, c free) and the "column space" has dimension 6 (each of three columns lies in a 2-dimensional plane). $3 + 6 = 9 = \dim M$: the count $(n - r) + r = n$ for an operation on a 9-dimensional input space, column by column $1 + 2 = 3$. □

Exercise 12.16. Show that the 2×3 matrices A with $(2,1,1)$ in their nullspace form a subspace of the space of 2×3 matrices, and find a basis and its dimension. Do the 2×3 matrices whose column space contains $(2,1)$ form a subspace?

Solution. If $A(2,1,1) = \mathbf{0}$ and $B(2,1,1) = \mathbf{0}$ then $(A+B)(2,1,1) = \mathbf{0}$ and $(cA)(2,1,1) = \mathbf{0}$, so it is a subspace. Each row (a,b,c) must satisfy $2a+b+c=0$, i.e. $c = -2a-b$, so each row is a combination of $(1,0,-2)$ and $(0,1,-1)$. A basis is

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix},$$

and the dimension is 4. The second set is not a subspace: the zero matrix has column space $\{\mathbf{0}\}$, which does not contain $(2,1)$, yet every subspace contains the zero vector. □

Exercise 12.17. Write $A = \begin{bmatrix} 2 & -4 & 6 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{bmatrix}$ as $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$, and give a basis for each of its four subspaces.

Solution. Every row is a multiple of $(1,-2,3)$, with multipliers $2,-1,3$, so $A = \boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$ with $\boldsymbol{u} = (2,-1,3)$ and $\boldsymbol{v} = (1,-2,3)$, and $r=1$. By Theorem 12.8: $\mathbf{C}(A)$ has basis $(2,-1,3)$; $\mathbf{C}(A^{\mathsf{T}})$ has basis $(1,-2,3)$; $\mathbf{N}(A)$ is the plane $x_1 - 2x_2 + 3x_3 = 0$, with special solutions $(2,1,0)$ and $(-3,0,1)$; and $\mathbf{N}(A^{\mathsf{T}})$ is the plane $2y_1 - y_2 + 3y_3 = 0$, with special solutions $(\frac{1}{2},1,0)$ and $(-\frac{3}{2},0,1)$. Dimensions $1,2,1,2$, consistent with $r=1$, $n-r=2$, $m-r=2$. □

13 Graphs, networks, incidence matrices

The matrices in the previous sections were invented on the spot. Matrices in applications come from somewhere and have structure, and whoever works with them uses that structure. The most important example in applied mathematics is the matrix of a graph. Here every one of the four subspaces has a physical meaning (potentials, potential differences, currents, and Kirchhoff's current law), and the dimension count of Section 11 proves Euler's formula for graphs.

13.1 The incidence matrix of a directed graph

A graph has nodes and edges (Definition 12.12). To measure currents we give each edge a direction, an arrow that says which way of flowing counts as positive; the choice of arrows is arbitrary.

Definition 13.1 (Incidence matrix)**.** Let a directed graph have n nodes and m edges. Its *incidence matrix* A is $m \times n$, with one row per edge and one column per node. If edge k goes from node i to node j, row k has -1 in column i, $+1$ in column j, and zeros elsewhere.

Example 13.2 (Four nodes, five edges)**.** The graph of the lecture (Figure 16, left) has $n=4$ nodes and $m=5$ edges: edge 1 from node 1 to 2, edge 2 from 2 to 3, edge 3 from 1 to 3, edge 4 from 1 to 4, and edge 5 from 3 to 4. Its incidence matrix is

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{matrix} \text{edge 1} \\ \text{edge 2} \\ \text{edge 3.} \\ \text{edge 4} \\ \text{edge 5} \end{matrix}$$

Each row has exactly two nonzeros, so a large graph gives a large but very sparse matrix ($2m$ nonzero entries out of mn). Edges $1, 2, 3$ form a loop, and the corresponding rows are dependent: row 1 + row 2 = $(-1, 0, 1, 0)$ = row 3. Loops will turn out to be exactly the source of dependent rows.

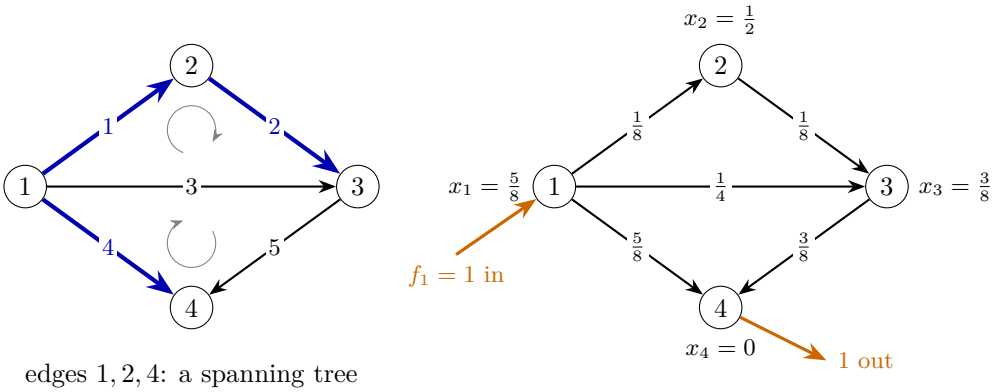


Figure 16: Left: the directed graph of Example 13.2; the two circular arrows mark the independent loops (edges $1, 2, 3$ and edges $3, 5, 4$), and the thick edges $1, 2, 4$ form a tree. Right: with unit conductances, one unit of current entering node 1 and leaving node 4 (Example 13.9): node potentials and edge currents, all flowing along the arrows.

13.2 Potentials and the nullspace

Multiply A by a vector $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ of *potentials* at the nodes:

$$A\boldsymbol{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix}.$$

Each component is a *potential difference* across an edge (head minus tail). Potential differences are what make current flow.

Proposition 13.3 (Nullspace of an incidence matrix)**.** *If the graph is connected (every two nodes are joined by a path), then* $\mathbf{N}(A)$ *is the line of constant vectors* $c(1, 1, \ldots, 1)$*, and* $\operatorname{rank} A = n - 1$.

Proof. A constant vector gives zero differences, so it lies in $\mathbf{N}(A)$. Conversely, $A\boldsymbol{x} = \mathbf{0}$ says $x_i = x_j$ whenever an edge joins i and j; following a path from node 1 to any node k gives $x_k = x_1$. So $\dim \mathbf{N}(A) = 1$ and $r = n - 1$. □

Physically, potentials are determined only up to an additive constant, like the constant of integration in calculus or the choice of zero on a temperature scale. One fixes it by *grounding* a node, say $x_4 = 0$. That removes column 4, and the remaining three columns are independent. In the example $r = 3$, and any three of the four columns are independent.

Which vectors $\boldsymbol{e}$ of edge values are potential differences, i.e. lie in $\mathbf{C}(A)$? Going around a loop, the differences telescope: on the loop of edges $1, 2, 3$, $(x_2 - x_1) + (x_3 - x_2) - (x_3 - x_1) = 0$. So every $\boldsymbol{e} = A\boldsymbol{x}$ satisfies $e_1 + e_2 - e_3 = 0$, and likewise $e_3 + e_5 - e_4 = 0$ on the other loop. This is *Kirchhoff's voltage law*, and by Theorem 9.1 these loop conditions are also sufficient once we know that the loop vectors span the left nullspace, which we prove next.

13.3 Currents and Kirchhoff's current law

Now let $\boldsymbol{y} = (y_1, \ldots, y_5)$ be *currents* on the edges, positive when flowing along the arrow. Then

$$A^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} -y_1 - y_3 - y_4 \\ y_1 - y_2 \\ y_2 + y_3 - y_5 \\ y_4 + y_5 \end{bmatrix}.$$

Row i of $A^{\mathsf{T}}\boldsymbol{y}$ is (current into node i) minus (current out of node i). The equation

$$A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0} \qquad \text{(Kirchhoff's current law)}$$

says that at every node, in equals out: charge does not accumulate. It is a balance equation, a conservation law, perhaps the most fundamental equation of applied mathematics. Its solutions form the left nullspace, of dimension $m - r = 5 - 3 = 2$.

Definition 13.4 (Loop vector)**.** For a loop in the graph traversed in a chosen direction, its *loop vector* has entry $+1$ for each edge traversed along its arrow, -1 for each edge traversed against its arrow, and 0 for edges not on the loop.

Theorem 13.5 (Loops span the left nullspace)**.** *Every loop vector satisfies $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$, and every solution of $A^{\mathsf{T}}\boldsymbol{y} = \mathbf{0}$ is a combination of loop vectors.*

Proof. A loop vector sends one unit of current around the loop. At each node on the loop, one unit arrives along one loop edge and one unit leaves along the next, so in minus out is 0; nodes off the loop see no current.

Conversely, let $\boldsymbol{y} \neq \mathbf{0}$ satisfy the current law. Pick an edge with $y_k \neq 0$ and walk along it in the direction the current flows. At the node reached, current arrives, so by the balance some other edge carries current away; walk along it in its direction of flow. Continue. With finitely many nodes, the walk eventually revisits a node, closing a loop on which every edge carries nonzero current in the walking direction. Let $t > 0$ be the smallest of these currents in absolute value and subtract t times this loop's vector from $\boldsymbol{y}$. The result still satisfies the current law and has at least one more zero entry. Repeating, after finitely many steps nothing is left, so $\boldsymbol{y}$ was a sum of multiples of loop vectors. □

Example 13.6 (Two independent loops)**.** Send one unit around the upper triangle $1 \to 2 \to 3 \to 1$: edges 1 and 2 are traversed along their arrows and edge 3 against it, giving $\boldsymbol{y} = (1, 1, -1, 0, 0)$. Check the four node equations: $-1 + 1 - 0 = 0$, $1 - 1 = 0$, $1 - 1 - 0 = 0$, $0 + 0 = 0$. The lower triangle $1 \to 3 \to 4 \to 1$ gives $(0, 0, 1, -1, 1)$. These two are independent (look at entries 1 and 5), and $\dim \mathbf{N}(A^{\mathsf{T}}) = 2$, so they form a basis. The big outer loop $1 \to 2 \to 3 \to 4 \to 1$ gives $(1, 1, 0, -1, 1)$, which is their sum: on edge 3 the two loop currents cancel. It is a loop, but not an independent one.

13.4 Trees, the row space and Euler's formula

The row space $\mathbf{C}(A^\mathsf{T})$ has dimension $r = 3$. Which three rows (edges) are independent? Row reduction of A^T finds pivots in columns $1, 2, 4$, the columns of A^T that belong to edges $1, 2$ and 4. Column 3 of A^T is column 1 plus column 2, which is the loop $1, 2, 3$ again. The pivot edges $1, 2, 4$ connect all four nodes and contain no loop (thick edges in Figure 16). A graph with no loops is a *tree*. Independent rows correspond to edges without loops; dependencies among rows are exactly loops.

Theorem 13.7 (Euler's formula)**.** *For a connected graph with n nodes, m edges and $\ell = \dim \mathbf{N}(A^\mathsf{T})$ independent loops,*

$$n - m + \ell = 1.$$

Proof. By Theorem 11.5, $\ell = \dim \mathbf{N}(A^\mathsf{T}) = m - r$, and $r = n - 1$ by Proposition 13.3. So $\ell = m - n + 1$. □

Nodes are zero-dimensional, edges one-dimensional, and loops enclose two-dimensional regions; the alternating count is always 1. For the example, $4-5+2 = 1$. For a graph with 5 nodes and 7 edges drawn in the plane with 3 independent loops, $5 - 7 + 3 = 1$. Linear algebra has proved a theorem of topology.

Common trap. Euler's formula counts *independent* loops. The graph of Figure 16 has three visible loops (two triangles and the outer quadrilateral), and $4-5+3 \neq 1$; the outer loop is the sum of the other two. Also, Proposition 13.3 needs a connected graph: if the graph falls into separate pieces, $\mathbf{N}(A)$ is larger (Exercise 13.13).

13.5 Ohm's law and $A^\mathsf{T}CA\boldsymbol{x} = \boldsymbol{f}$

The three steps of an electrical network chain together:

$$\underset{\text{potentials at nodes}}{\boldsymbol{x}} \xrightarrow{\boldsymbol{e} = A\boldsymbol{x}} \underset{\text{potential differences}}{\boldsymbol{e}} \xrightarrow[\text{Ohm}]{\boldsymbol{y} = -C\boldsymbol{e}} \underset{\text{currents on edges}}{\boldsymbol{y}} \xrightarrow[\text{Kirchhoff}]{A^\mathsf{T}\boldsymbol{y} = -\boldsymbol{f}} \underset{\text{current sources at nodes}}{\boldsymbol{f}}$$

Ohm's law says that the current on an edge is its conductance (one over its resistance) times the potential drop; C is the diagonal matrix of conductances. Current flows from high to low potential, hence the minus sign in $\boldsymbol{y} = -C\boldsymbol{e}$. If a current f_i is fed into node i from outside, the balance at that node becomes (in) − (out) $+f_i = 0$, i.e. $A^\mathsf{T}\boldsymbol{y} = -\boldsymbol{f}$. Combining the three equations, the signs cancel:

$$A^\mathsf{T}CA\,\boldsymbol{x} = \boldsymbol{f}.$$

(The lecture writes the chain as $\boldsymbol{e} = A\boldsymbol{x}$, $\boldsymbol{y} = C\boldsymbol{e}$, $A^\mathsf{T}\boldsymbol{y} = \boldsymbol{f}$, absorbing the signs into the conventions; the resulting equation is the same.) This is the basic equation of equilibrium for networks, and the matrix $A^\mathsf{T}CA$ is symmetric: $(A^\mathsf{T}CA)^\mathsf{T} = A^\mathsf{T}C^\mathsf{T}A = A^\mathsf{T}CA$.

Proposition 13.8 (The matrix $A^\mathsf{T}A$ of a graph)**.** *For an incidence matrix A, the diagonal entry $(A^\mathsf{T}A)_{ii}$ is the degree of node i (the number of edges touching it), the off-diagonal entry $(A^\mathsf{T}A)_{ij}$ is -1 if an edge joins i and j and 0 otherwise, and $\operatorname{tr}(A^\mathsf{T}A) = 2m$.*

Proof. $(A^{\mathsf{T}}A)_{ij}$ is the dot product of columns i and j of A. Column i has an entry ± 1 in each row whose edge touches node i, so its squared length is the degree. Two different columns share a nonzero row only for an edge joining i and j, where the entries are -1 and $+1$. The trace is the sum of the degrees, and each edge contributes to two degrees. □

Example 13.9 (Solving a network)**.** Take all conductances equal to 1 ($C = I$) in the graph of Example 13.2, feed one unit of current into node 1 and take it out at node 4: $\boldsymbol{f} = (1, 0, 0, -1)$. The degrees are $3, 2, 3, 2$, so

$$A^{\mathsf{T}}A = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}.$$

It is singular (its rows add to zero, reflecting $\mathbf{N}(A)$), so ground node 4: set $x_4 = 0$ and drop the fourth equation. The remaining equations $3x_1 - x_2 - x_3 = 1$, $-x_1 + 2x_2 - x_3 = 0$, $-x_1 - x_2 + 3x_3 = 0$ give, by elimination, $\boldsymbol{x} = (\frac{5}{8}, \frac{1}{2}, \frac{3}{8}, 0)$. Check the first: $\frac{15}{8} - \frac{4}{8} - \frac{3}{8} = 1$. The potential differences and currents are

$$\boldsymbol{e} = A\boldsymbol{x} = \left(-\tfrac{1}{8}, -\tfrac{1}{8}, -\tfrac{1}{4}, -\tfrac{5}{8}, -\tfrac{3}{8}\right), \qquad \boldsymbol{y} = -\boldsymbol{e} = \left(\tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{4}, \tfrac{5}{8}, \tfrac{3}{8}\right).$$

All currents are positive, so all flow along the arrows (Figure 16, right). Node 1 sends out $\frac{1}{8} + \frac{1}{4} + \frac{5}{8} = 1$, node 2 passes $\frac{1}{8}$ through, node 3 receives $\frac{1}{8} + \frac{1}{4}$ and sends $\frac{3}{8}$, and node 4 collects $\frac{5}{8} + \frac{3}{8} = 1$. The direct edge $1 \to 4$ carries the most current, because every other route to node 4 crosses two edges.

13.6 Exercises

Exercise 13.10. A square graph has nodes $1, 2, 3, 4$ and edges $1 \to 2$, $2 \to 3$, $4 \to 3$, $1 \to 4$ (in that order). Write its 4×4 incidence matrix A. Which vectors are in $\mathbf{N}(A)$? Why is $(1, 0, 0, 0)$ not in the row space of A?

Solution.

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \qquad A\boldsymbol{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_4 \\ x_4 - x_1 \end{bmatrix}.$$

$A\boldsymbol{x} = \boldsymbol{0}$ forces $x_1 = x_2 = x_3 = x_4$, so $\mathbf{N}(A)$ is all (a, a, a, a). Every row of A has entries adding to 0 (one -1 and one $+1$), so every combination of rows has entries adding to 0; the entries of $(1, 0, 0, 0)$ add to 1, so it is not in the row space. □

Exercise 13.11. For the square graph of Exercise 13.10 let the conductances be $C = \mathrm{diag}(1, 2, 2, 1)$. Find $A^{\mathsf{T}}CA$. For the current source $\boldsymbol{f} = (1, 0, -1, 0)$ (one unit into node 1, out of node 3), solve $A^{\mathsf{T}}CA\boldsymbol{x} = \boldsymbol{f}$ with node 3 grounded, and find the currents $\boldsymbol{y} = -CA\boldsymbol{x}$.

Solution. Multiplying out (or using Proposition 13.8 with each edge weighted by its conductance),

$$A^{\mathsf{T}}CA = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 4 & -2 \\ -1 & 0 & -2 & 3 \end{bmatrix}.$$

With $x_3 = 0$, the equations from nodes $1, 2, 4$ are $2x_1 - x_2 - x_4 = 1$, $-x_1 + 3x_2 = 0$, $-x_1 + 3x_4 = 0$. So $x_2 = x_4 = x_1/3$ and $2x_1 - \frac{2}{3}x_1 = 1$, giving $x_1 = \frac{3}{4}$ and $\boldsymbol{x} = (\frac{3}{4}, \frac{1}{4}, 0, \frac{1}{4})$. Then $A\boldsymbol{x} = (-\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{2})$ and $\boldsymbol{y} = -CA\boldsymbol{x} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$: half the current takes the route $1 \to 2 \to 3$ and half takes $1 \to 4 \to 3$, as symmetry demands (each route has conductances 1 and 2 in series). Check at node 1: $\frac{1}{2} + \frac{1}{2} = 1$ leaves. □

Exercise 13.12. A graph has 5 nodes and 6 edges: $1 \to 2$, $2 \to 3$, $1 \to 3$, $2 \to 4$, $4 \to 5$, $5 \to 3$. Without row reduction, find $\mathbf{N}(A)$, a basis of $\mathbf{N}(A^{\mathsf{T}})$, and $\operatorname{tr}(A^{\mathsf{T}}A)$.

Solution. The graph is connected, so $\mathbf{N}(A)$ is the line through $(1, 1, 1, 1, 1)$ and $r = 4$; then $\dim \mathbf{N}(A^{\mathsf{T}}) = 6 - 4 = 2$. Loop vectors: the triangle $1 \to 2 \to 3 \to 1$ gives $(1, 1, -1, 0, 0, 0)$, and the square $2 \to 4 \to 5 \to 3 \to 2$ gives $(0, -1, 0, 1, 1, 1)$ (edge 2 is traversed against its arrow). They are independent, so they form a basis; the outer loop $1 \to 2 \to 4 \to 5 \to 3 \to 1$ gives their sum $(1, 0, -1, 1, 1, 1)$. The degrees are $2, 3, 3, 2, 2$, so $\operatorname{tr}(A^{\mathsf{T}}A) = 12 = 2m$. □

Exercise 13.13. A graph consists of two separate triangles: nodes $1, 2, 3$ with edges $1 \to 2$, $2 \to 3$, $1 \to 3$, and nodes $4, 5, 6$ with edges $4 \to 5$, $5 \to 6$, $4 \to 6$. Find $\dim \mathbf{N}(A)$, the rank, and $\dim \mathbf{N}(A^{\mathsf{T}})$. What becomes of Euler's formula?

Solution. $A\boldsymbol{x} = \mathbf{0}$ forces equal potentials within each triangle but not across, so $\mathbf{N}(A)$ has basis $(1, 1, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 1, 1)$: dimension 2, rank $6 - 2 = 4$. Then $\dim \mathbf{N}(A^{\mathsf{T}}) = m - r = 6 - 4 = 2$, one loop per triangle. Now $n - m + \ell = 6 - 6 + 2 = 2$: in general the alternating count equals the number of connected pieces, and Euler's formula is the case of one piece. □

14 Exam 1 review

The first exam covers Part I, Sections 1–13, with the emphasis on rectangular matrices: nullspaces, ranks, complete solutions and the four subspaces. This section collects the essential facts in one place and then works through the review problems of the course, which test understanding by approaching familiar ideas from unfamiliar directions, for instance by giving the answer and asking for the matrix.

14.1 The essentials of Part I

- **Elimination** (Sections 3–5). Row operations are multiplications by elementary matrices: $E_{32}E_{31}E_{21}A = U$, so $A = LU$ with the multipliers in L (with row exchanges, $PA = LU$). Gauss–Jordan reduces $\begin{bmatrix} A & I \end{bmatrix}$ to $\begin{bmatrix} I & A^{-1} \end{bmatrix}$ when A is invertible.
- **Subspaces** (Sections 6–7). A subspace contains $\mathbf{0}$ and is closed under addition and scalar multiplication. Intersections of subspaces are subspaces; unions usually are not.
- **Rank and special solutions** (Section 8). r = number of pivots; $R = \operatorname{rref}(A)$ has the block form $\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$ (pivot columns first), and the special solutions are the columns of $N = \begin{bmatrix} -F \\ I \end{bmatrix}$.
- **Complete solution** (Section 9). $A\boldsymbol{x} = \boldsymbol{b}$ is solvable iff $\boldsymbol{b} \in \mathbf{C}(A)$; then $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$. Full column rank $r = n$: at most one solution; full row rank $r = m$: at least one (Table 1).

- **Basis and dimension** (Section 10). Independent and spanning; every basis of a space has the same size. Pivot columns of A are a basis of $\mathbf{C}(A)$.
- **Four subspaces** (Section 11). Dimensions r, $n-r$, r, $m-r$ for $\mathbf{C}(A)$, $\mathbf{N}(A)$, $\mathbf{C}(A^{\mathsf{T}})$, $\mathbf{N}(A^{\mathsf{T}})$. Row operations preserve the row space and the nullspace.
- **Beyond** $\mathbb{R}^n$ (Sections 12–13). Spaces of matrices and functions; rank one means $\boldsymbol{u}\boldsymbol{v}^{\mathsf{T}}$; for a connected graph, $\mathbf{N}(A)$ is the constants, $\mathbf{N}(A^{\mathsf{T}})$ is spanned by loops, and $n - m + \ell = 1$.

Figure 17 is the algorithm for $A\boldsymbol{x} = \boldsymbol{b}$ that runs through all of it.

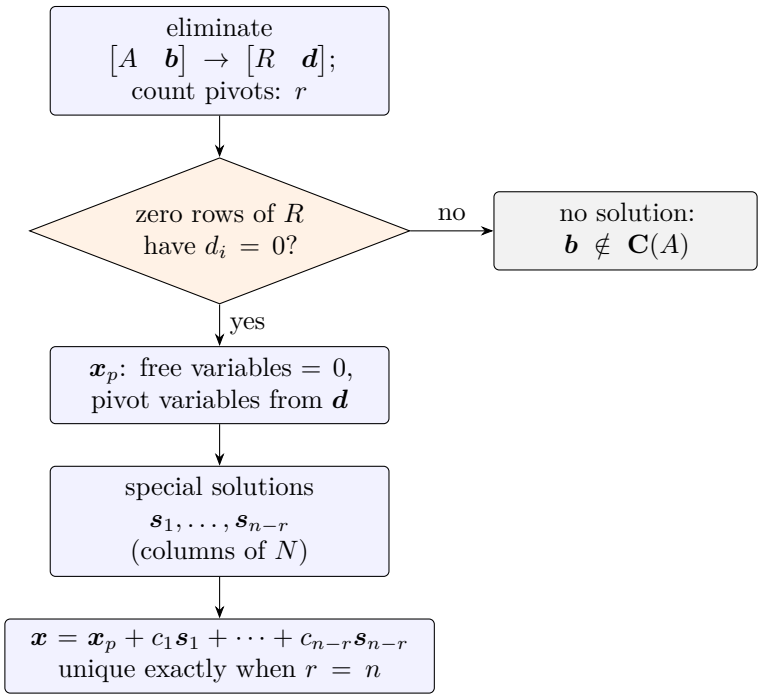


Figure 17: Solving $A\boldsymbol{x} = \boldsymbol{b}$ for any $m \times n$ matrix. When $r = m$ there are no zero rows and the test always passes; when $r = n$ there are no special solutions.

14.2 Solved review problems

Example 14.1 (The dimension of a span)**.** *Nonzero vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^7$ span a subspace. What are its possible dimensions?* The answer is 1, 2 or 3. A basis can be chosen from among the three spanning vectors, so the dimension is at most 3; it is at least 1 because the vectors are nonzero. All three values occur: $\boldsymbol{u} = \boldsymbol{v} = \boldsymbol{w}$ gives 1, and three independent vectors give 3.

Example 14.2 (Stacked echelon forms)**.** *A 5×3 matrix R in reduced echelon form has $r = 3$ pivots. (a) Find $\mathbf{N}(R)$. (b) Find the reduced echelon form and the rank of the 10×3 matrix $B = \begin{bmatrix} R \\ 2R \end{bmatrix}$. (c) Do the same for the 10×6 matrix $C = \begin{bmatrix} R & R \\ R & 0 \end{bmatrix}$. (d) Find $\dim \mathbf{N}(C^{\mathsf{T}})$.*

(a) Three pivots in three columns: no free variables, so $\mathbf{N}(R) = \{\mathbf{0}\}$. In fact $R = \begin{bmatrix} I_3 \\ 0 \end{bmatrix}$ is forced: three pivots equal to 1, zeros elsewhere in their columns, two zero rows.

(b) Subtract 2×(rows 1–5) from rows 6–10: $B \to \begin{bmatrix} R \\ 0 \end{bmatrix}$, which is already in reduced form. The rank is 3, not 6: the bottom rows of B are multiples of the top rows.

(c) Work with block rows. Subtract the top block row from the bottom one, then add the new bottom block row to the top one, then multiply the bottom by -1:

$$\begin{bmatrix} R & R \\ R & 0 \end{bmatrix} \to \begin{bmatrix} R & R \\ 0 & -R \end{bmatrix} \to \begin{bmatrix} R & 0 \\ 0 & -R \end{bmatrix} \to \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}.$$

Moving the zero rows (rows $4, 5$ and $9, 10$) to the bottom gives $\begin{bmatrix} I_6 \\ 0 \end{bmatrix}$. The rank is 6.

(d) C is 10×6 of rank 6, so $\dim \mathbf{N}(C^{\mathsf{T}}) = m - r = 10 - 6 = 4$ (and $\dim \mathbf{N}(C) = 6 - 6 = 0$).

Example 14.3 (From the complete solution back to the matrix)**.** *The complete solution of* $A\boldsymbol{x} = (2, 4, 2)$ *is*

$$\boldsymbol{x} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Find the shape of A*, its rank,* A *itself, and the* $\boldsymbol{b}$ *for which* $A\boldsymbol{x} = \boldsymbol{b}$ *is solvable.*

$\boldsymbol{x}$ and $\boldsymbol{b}$ both have three components, so A is 3×3. The two vectors multiplying c and d are independent and lie in $\mathbf{N}(A)$; nothing else does (otherwise there would be more free constants), so $\dim \mathbf{N}(A) = 2$ and $r = 3 - 2 = 1$. To recover A: $A(2, 0, 0) = 2\,\boldsymbol{a}_1 = (2, 4, 2)$ gives $\boldsymbol{a}_1 = (1, 2, 1)$; $A(0, 0, 1) = \boldsymbol{a}_3 = \boldsymbol{0}$; and $A(1, 1, 0) = \boldsymbol{a}_1 + \boldsymbol{a}_2 = \boldsymbol{0}$ gives $\boldsymbol{a}_2 = -\boldsymbol{a}_1$. Hence

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 1 & -1 & 0 \end{bmatrix}, \qquad A \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}.$$

$A\boldsymbol{x} = \boldsymbol{b}$ is solvable exactly when $\boldsymbol{b} \in \mathbf{C}(A)$, i.e. when $\boldsymbol{b}$ is a multiple of $(1, 2, 1)$. This is the case $r < m$, $r < n$: zero or infinitely many solutions.

Example 14.4 (A factored matrix)**.** *Let*

$$B = CD = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Without multiplying, find a basis of $\mathbf{N}(B)$ *and the complete solution of* $B\boldsymbol{x} = (1, 0, 1)$*.*

B is 3×4, so $\mathbf{N}(B) \subseteq \mathbb{R}^4$. The first factor C is invertible: in a combination of its columns giving $\boldsymbol{0}$, the second row forces the weight of column 2 to be 0, and columns 1 and 3 are clearly independent. By Proposition 8.1, $\mathbf{N}(CD) = \mathbf{N}(D)$, and D is already in reduced form with pivot columns $1, 2$ and $F = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$. The special solutions (columns of $\begin{bmatrix} -F \\ I \end{bmatrix}$) are

$$\boldsymbol{s}_1 = (1, -1, 1, 0), \qquad \boldsymbol{s}_2 = (-2, 1, 0, 1),$$

a basis of $\mathbf{N}(B)$ (and $\operatorname{rank} B = 2$). For a particular solution, note that the first column of B is C times the first column of D, which is the first column of C, namely $(1, 0, 1)$, exactly the right-hand

side. So $\boldsymbol{x}_p = (1,0,0,0)$ and

$$\boldsymbol{x} = \begin{bmatrix}1\\0\\0\\0\end{bmatrix} + c\begin{bmatrix}1\\-1\\1\\0\end{bmatrix} + d\begin{bmatrix}-2\\1\\0\\1\end{bmatrix}.$$

Any other solution would serve equally well as $\boldsymbol{x}_p$; this one is simply the most convenient.

Example 14.5 (A square system with a parameter)**.** *Let* $A = \begin{bmatrix}1&1&1\\1&2&3\\3&4&k\end{bmatrix}$ *and* $\boldsymbol{b} = (2,3,7)$*. (a) For which* k *does* $A\boldsymbol{x} = \boldsymbol{b}$ *have exactly one solution? (b) Infinitely many? (c) Find* $A = LU$*. (d) Find the complete solution for every* k*.*

Eliminate on the augmented matrix and record the steps, since part (c) needs them. E_{21} subtracts row 1 from row 2, E_{31} subtracts 3×row 1 from row 3, and E_{32} subtracts row 2 from row 3:

$$\left[\begin{array}{ccc|c}1&1&1&2\\1&2&3&3\\3&4&k&7\end{array}\right] \to \left[\begin{array}{ccc|c}1&1&1&2\\0&1&2&1\\0&1&k-3&1\end{array}\right] \to \left[\begin{array}{ccc|c}1&1&1&2\\0&1&2&1\\0&0&k-5&0\end{array}\right].$$

(a) The third pivot is $k-5$. For $k \neq 5$, A is invertible and the solution is unique. (b) For $k = 5$ the third row becomes $0 = 0$, consistent, and x_3 is free: infinitely many solutions. (Had the last entry been nonzero, $k = 5$ would give no solution at all.) (c) U is the final left block, and $L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ holds the multipliers $1, 3, 1$ in their positions:

$$A = \begin{bmatrix}1&0&0\\1&1&0\\3&1&1\end{bmatrix}\begin{bmatrix}1&1&1\\0&1&2\\0&0&k-5\end{bmatrix}.$$

(d) For $k \neq 5$: $x_3 = 0$, then $x_2 = 1$, then $x_1 = 2-1-0 = 1$, so $\boldsymbol{x} = (1,1,0)$. For $k = 5$: set $x_3 = c$; then $x_2 = 1-2c$ and $x_1 = 2-x_2-x_3 = 1+c$, so

$$\boldsymbol{x} = \begin{bmatrix}1\\1\\0\end{bmatrix} + c\begin{bmatrix}1\\-2\\1\end{bmatrix}.$$

Check: with $k = 5$, $A(1,-2,1) = (1-2+1,\ 1-4+3,\ 3-8+5) = \mathbf{0}$.

14.3 Quick checks

The following short questions test whether the definitions are understood. Each answer is followed by its reason.

1. *A is square and $\mathbf{N}(A) = \{\mathbf{0}\}$. What is $\mathbf{N}(A^{\mathsf{T}})$?* Also $\{\mathbf{0}\}$: $r = n = m$, so $\dim \mathbf{N}(A^{\mathsf{T}}) = m - r = 0$.
2. *Do the invertible 5×5 matrices form a subspace of the 25-dimensional space of 5×5 matrices?* No: the zero matrix is not invertible, and $I + (-I) = 0$. (The singular matrices fail too: $\mathrm{diag}(1,0) + \mathrm{diag}(0,1) = I$ in the 2×2 case.)
3. *If $B^2 = 0$, must $B = 0$?* No: $B = \begin{bmatrix}0&1\\0&0\end{bmatrix}$ has $B^2 = 0$.

4. *An $n \times n$ system with independent columns is solvable for every $\boldsymbol{b}$.* True: $r = n = m$, so A is invertible and $\boldsymbol{x} = A^{-1}\boldsymbol{b}$.
5. *If $m = n$, the row space equals the column space.* False: for $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ the row space is the line through $(0, 1)$ and the column space is the line through $(1, 0)$. Only the dimensions agree. (For a symmetric matrix they are equal.)
6. *A and $-A$ have the same four subspaces.* True: each subspace contains $-\boldsymbol{v}$ whenever it contains $\boldsymbol{v}$.
7. *If A and B have the same four subspaces, then A is a multiple of B.* False: every invertible 3×3 matrix has $\mathbf{C}(A) = \mathbf{C}(A^{\mathsf{T}}) = \mathbb{R}^3$ and both nullspaces $\{\mathbf{0}\}$, yet I and $\operatorname{diag}(1, 2, 3)$ are not multiples of each other. What *is* forced is equal rank.
8. *If two rows of A are exchanged, which subspaces stay the same?* The row space and the nullspace. The column space can change: exchanging the rows of $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ moves $\mathbf{C}(A)$ from the line through $(1, 0)$ to the line through $(0, 1)$.
9. *Why can $\boldsymbol{v} = (1, 2, 3)$ not be both a row of A and in $\mathbf{N}(A)$?* The entry of $A\boldsymbol{v}$ in that row would be $\boldsymbol{v} \cdot \boldsymbol{v} = 1 + 4 + 9 = 14 \neq 0$. In fact the row space and the nullspace meet only in $\mathbf{0}$, and they are perpendicular, the first topic of Part II (Section 16).

Common trap. Stacking copies of rows does not add rank. In Example 14.2 it is tempting to say $\operatorname{rank} \begin{bmatrix} R \\ 2R \end{bmatrix} = 6$ because the matrix "contains R twice"; but rank counts independent rows, and the rows of $2R$ are multiples of those of R. In $\begin{bmatrix} R & R \\ R & 0 \end{bmatrix}$, by contrast, the second block column breaks the dependence, and the rank does double.

14.4 Exercises

Exercise 14.6. A is 3×5 of rank 3. Give the dimensions of its four subspaces. Is $A\boldsymbol{x} = \boldsymbol{b}$ solvable for every $\boldsymbol{b}$? Can a solution be unique?

Solution. $\dim \mathbf{C}(A) = 3$, $\dim \mathbf{N}(A) = 5 - 3 = 2$, $\dim \mathbf{C}(A^{\mathsf{T}}) = 3$, $\dim \mathbf{N}(A^{\mathsf{T}}) = 3 - 3 = 0$. Since $r = m = 3$, $\mathbf{C}(A) = \mathbb{R}^3$ and every $\boldsymbol{b}$ is reachable. Since $r < n$, there are two free variables, so every solvable system has infinitely many solutions, never exactly one. □

Exercise 14.7. For which numbers k are $(1, 1, 0)$, $(0, 1, 1)$, $(1, 0, k)$ linearly independent? For the exceptional value, find the dependency.

Solution. Put them in the columns of $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & k \end{bmatrix}$ and eliminate: row 2 − row 1 gives $(0, 1, -1)$, then row 3 − row 2 gives $(0, 0, k + 1)$. Three pivots unless $k = -1$, so the vectors are independent exactly when $k \neq -1$. For $k = -1$ the free variable $x_3 = 1$ gives $x_2 = 1$ and $x_1 = -1$: $-(1, 1, 0) + (0, 1, 1) + (1, 0, -1) = \mathbf{0}$. □

Exercise 14.8. M is an invertible matrix. Which of the four subspaces of A are unchanged in MA (when M is $m \times m$)? Which are unchanged in AM (when M is $n \times n$)?

Solution. MA: the nullspace is unchanged ($MA\boldsymbol{x} = \boldsymbol{0} \iff A\boldsymbol{x} = \boldsymbol{0}$) and so is the row space (the rows of MA are combinations of the rows of A, and conversely via M^{-1}); this is Theorem 11.2. The column space and left nullspace may change, as row exchanges show. AM: transposing, $(AM)^{\mathsf{T}} = M^{\mathsf{T}}A^{\mathsf{T}}$ with M^{T} invertible, so by the first part A^{T} and $M^{\mathsf{T}}A^{\mathsf{T}}$ have the same nullspace and row space; that is, AM has the same left nullspace and column space as A. Its nullspace and row space may change. In every case the rank is unchanged. □

15 Exam 1

The first unit exam checks whether the machinery of Part I has become reflex: elimination written with elementary matrices E_{ij}, the factorization $A = LU$, the inverse, bases for the column space and the nullspace, the complete solution of $A\boldsymbol{x} = \boldsymbol{b}$, and what the reduced echelon form R says about the columns of A. This section collects the four exam problems as practice problems with complete solutions, preceded by the handful of facts they draw on and followed by variations to practise on. Try each problem before reading its solution.

15.1 The facts the exam draws on

Every problem on the exam is an application of a few statements from Sections 5–11. For an $m \times n$ matrix A of rank r:

- $A\boldsymbol{x} = \boldsymbol{b}$ is solvable exactly when $\boldsymbol{b} \in \mathbf{C}(A)$; it is solvable for *every* $\boldsymbol{b} \in \mathbb{R}^m$ exactly when $r = m$.
- A solution, when one exists, is unique exactly when $\mathbf{N}(A) = \{\boldsymbol{0}\}$, that is, when $r = n$. Otherwise the complete solution is $\boldsymbol{x}_p + \boldsymbol{x}_n$ with $\boldsymbol{x}_n$ ranging over the $(n - r)$-dimensional nullspace.
- The pivot columns of A (not of R) are a basis for $\mathbf{C}(A)$; the $n - r$ special solutions are a basis for $\mathbf{N}(A)$.
- Row operations do not change the nullspace, so A and $R = \mathrm{rref}(A)$ satisfy exactly the same column relations.
- The inverse of an elimination matrix E_{ij} (subtract ℓ times row j from row i) adds ℓ times row j back, and the inverse of a product reverses the order.

The first two items combine into the four cases of Table 2, which is the whole content of Problem 1 below.

Example 15.1 (Reading the case from two right-hand sides)**.** Take

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

Column 2 is twice column 1, so $r = 1$, which is smaller than both $m = 3$ and $n = 2$: the last row of the table. Check both possibilities. For $\boldsymbol{b} = (1, 2, 3)$, which is column 1, the system is solvable with $\boldsymbol{x}_p = (1, 0)$, and adding any multiple of the special solution $(-2, 1)$ gives infinitely many solutions

Rank	Shape of R	Solutions of $A\boldsymbol{x} = \boldsymbol{b}$	Example
$r = m = n$	I	exactly one, for every $\boldsymbol{b}$	square invertible
$r = n < m$	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	zero or one	tall, independent columns
$r = m < n$	$\begin{bmatrix} I & F \end{bmatrix}$	infinitely many, for every $\boldsymbol{b}$	wide, independent rows
$r < m,\ r < n$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$	zero or infinitely many	neither

Table 2: Rank decides how many solutions $A\boldsymbol{x} = \boldsymbol{b}$ can have (F is the free part of R; columns may need reordering).

$\boldsymbol{x} = (1 - 2t,\, t)$. For $\boldsymbol{b} = (1, 0, 0)$, elimination on the augmented matrix gives

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 3 & 6 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & -3 \end{array}\right],$$

and the equations $0 = -2$ and $0 = -3$ are impossible, so there is no solution. A single right-hand side with a unique solution would have forced $r = n$; a single unsolvable one forces $r < m$.

15.2 Problem 1: rank from solvability

Exercise 15.2 (Exam problem 1)**.** An $m \times n$ matrix A has the property that

$$A\boldsymbol{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ has no solution} \qquad \text{and} \qquad A\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ has exactly one solution.}$$

(a) Give all the information you can about m, n and the rank r.

(b) Find all solutions of $A\boldsymbol{x} = \boldsymbol{0}$ and explain.

(c) Write down an example of such a matrix A.

Solution. (a) The right-hand sides live in $\mathbb{R}^m$, so $m = 3$. Exactly one solution for $\boldsymbol{b} = (0, 1, 0)$ means there are no free variables: $\mathbf{N}(A) = \{\boldsymbol{0}\}$ and $r = n$. No solution for $\boldsymbol{b} = (1, 1, 1)$ means $\mathbf{C}(A) \neq \mathbb{R}^3$, so $r < m = 3$. Together, $r = n \leq 2$: either $n = r = 1$ or $n = r = 2$ (and $m = 3$ in both cases).

(b) Since $r = n$, the columns are independent and $A\boldsymbol{x} = \boldsymbol{0}$ has only the solution $\boldsymbol{x} = \boldsymbol{0}$. (The zero vector is always a solution; independence of the columns says no other combination of them vanishes.)

(c) Two examples, one for each possible n:

$$n = 1:\ A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad n = 2:\ A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For $n = 1$, $A\boldsymbol{x} = (0, 1, 0)$ is solved only by the scalar $x = 1$, and $(1, 1, 1)$ is not a multiple of $(0, 1, 0)$. For $n = 2$ the column space is the plane $x_3 = 0$ (Figure 18), which contains $(0, 1, 0) = A(0, 1)$ but not $(1, 1, 1)$; the two columns are independent, so the solution $(0, 1)$ is unique. □

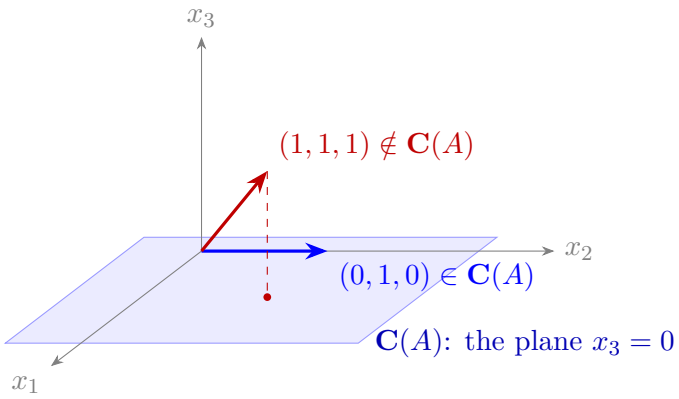


Figure 18: Problem 1(c) with $n = 2$: the column space of A is a plane through the origin in $\mathbb{R}^3$. Right-hand sides in the plane are reachable (uniquely, since the columns are independent); $(1, 1, 1)$ sits above the plane and is not.

Common trap. Each hypothesis controls one inequality: uniqueness for one $\boldsymbol{b}$ gives $r = n$ (no free variables), unsolvability for one $\boldsymbol{b}$ gives $r < m$. A frequent slip is to read "exactly one solution" as "A is square and invertible"; that would make every $\boldsymbol{b}$ solvable, contradicting the first hypothesis.

15.3 Problem 2: from elimination matrices to A^{-1}, A and L

Exercise 15.3 (Exam problem 2)**.** A 3×3 matrix A reduces to I by three row operations, in this order: E_{21} subtracts 4 times row 1 from row 2; E_{31} subtracts 3 times row 1 from row 3; E_{23} subtracts row 3 from row 2.

(a) Write A^{-1} in terms of the E's, then compute it.

(b) What is the original matrix A?

(c) What is the lower triangular factor L in $A = LU$?

Solution. (a) The operations give $E_{23}E_{31}E_{21}A = I$, so $A^{-1} = E_{23}E_{31}E_{21}$, with

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, \quad E_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Apply them to I in order. $E_{31}E_{21}$ places -4 and -3 in column 1; then E_{23} subtracts row 3 from row 2:

$$E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, \qquad A^{-1} = E_{23}(E_{31}E_{21}) = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{bmatrix}.$$

(b) Invert the product in reverse order, $A = E_{21}^{-1}E_{31}^{-1}E_{23}^{-1}$, where each inverse just flips the sign of its off-diagonal entry:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix}.$$

Check: multiplying this A by the A^{-1} of part (a) gives I.

(c) Forward elimination on A uses only E_{21} and E_{31}: subtracting 4 (row 1) from row 2 and 3 (row 1) from row 3 leaves a matrix that is already upper triangular. The step E_{23} is an *upward* step (it clears the entry above the last pivot), so it is not part of $A = LU$. The multipliers 4 and 3 go straight into L:

$$L = E_{21}^{-1}E_{31}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}, \qquad U = E_{23}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

and indeed the factorization in (b) already reads $A = LU$. □

Common trap. A^{-1} is the product of the E's in the order they were *applied* ($E_{23}E_{31}E_{21}$, last step on the left); A is the product of the inverses in the *opposite* order. Writing $A^{-1} = E_{21}E_{31}E_{23}$ gives a different matrix here, because E_{23} does not commute with E_{31}.

15.4 Problem 3: bases and the complete solution for a matrix with a parameter

Exercise 15.4 (Exam problem 3)**.** This 3×4 matrix depends on c:

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 3 & c & 2 & 8 \\ 0 & 0 & 2 & 2 \end{bmatrix}.$$

(a) For each c find a basis for the column space of A.

(b) For each c find a basis for the nullspace of A.

(c) For each c find the complete solution of $A\boldsymbol{x} = (1, c, 0)$.

Solution. Subtract 3 (row 1) from row 2:

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & c-3 & -4 & -4 \\ 0 & 0 & 2 & 2 \end{bmatrix}.$$

Everything depends on whether the entry $c - 3$ is a pivot.

Case $c \neq 3$. The pivots are in columns 1, 2, 3, so $r = 3$. Dividing and clearing upward gives

$$R = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

(a) The pivot columns of A, namely $(1, 3, 0)$, $(1, c, 0)$, $(2, 2, 2)$, are a basis for $\mathbf{C}(A)$. Since $r = m = 3$ this column space is all of $\mathbb{R}^3$, so the standard basis would do equally well. (b) One free variable, x_4. Setting $x_4 = 1$ in $R\boldsymbol{x} = \mathbf{0}$ gives $x_1 = -2$, $x_2 = 0$, $x_3 = -1$; the basis is $\boldsymbol{s} = (-2, 0, -1, 1)$.

Case $c = 3$. Row 2 becomes $(0, 0, -4, -4)$; adding half of it to row 3 kills row 3, so $r = 2$ with pivots in columns 1 and 3:

$$R = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(a) Basis for $\mathbf{C}(A)$: columns 1 and 3 of A, $(1,3,0)$ and $(2,2,2)$. (b) Free variables x_2 and x_4. The special solutions are $\boldsymbol{s}_1 = (-1,1,0,0)$ (from $x_2 = 1, x_4 = 0$) and $\boldsymbol{s}_2 = (-2,0,-1,1)$ (from $x_2 = 0, x_4 = 1$).

(c) The right-hand side $(1,c,0)$ is exactly column 2 of A, so $\boldsymbol{x}_p = (0,1,0,0)$ is a particular solution for every c. Adding the nullspace:

$$c \neq 3: \quad \boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}; \qquad c = 3: \quad \boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

When $c = 3$ the vector $(1,0,0,0)$ is another particular solution (column 1 equals $(1,3,0)$ too); it differs from $(0,1,0,0)$ by the null vector $\boldsymbol{s}_1$, so it describes the same solution set. □

Common trap. The basis for $\mathbf{C}(A)$ consists of columns of A, not of R. For $c = 3$ the pivot columns of R are $(1,0,0)$ and $(0,1,0)$, which span the plane $x_3 = 0$. But column 3 of A is $(2,2,2)$, which is not in that plane, so those two vectors are not a basis for $\mathbf{C}(A)$. Row operations preserve the *relations* among the columns, not the column space.

15.5 Problem 4: what rref(A) reveals

Exercise 15.5 (Exam problem 4)**.** (a) If A is 3×5, what do you know about its nullspace?

(b) Row operations on A lead to

$$R = \operatorname{rref}(A) = \begin{bmatrix} 1 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Write down everything you know about the columns of A.

(c) In the vector space M of all 3×3 matrices, which subspace S is spanned by all possible reduced row echelon forms R?

Solution. (a) $\mathbf{N}(A)$ is a subspace of $\mathbb{R}^5$ of dimension $5 - r$. Since $r \leq 3$, the dimension is at least 2: the nullspace contains at least two independent vectors, and every 3×5 matrix has nonzero solutions to $A\boldsymbol{x} = \mathbf{0}$.

(b) Write $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_5$ for the columns of A. The pivots of R are in columns 1, 4, 5, so $r = 3$ and:

- $\boldsymbol{a}_1, \boldsymbol{a}_4, \boldsymbol{a}_5$ are independent and form a basis for $\mathbf{C}(A) = \mathbb{R}^3$ (three pivots in three rows);
- $\boldsymbol{a}_2 = 4\boldsymbol{a}_1$, because $(-4,1,0,0,0) \in \mathbf{N}(R) = \mathbf{N}(A)$ means $-4\boldsymbol{a}_1 + \boldsymbol{a}_2 = \mathbf{0}$;
- $\boldsymbol{a}_3 = \mathbf{0}$, because $(0,0,1,0,0) \in \mathbf{N}(A)$.

Nothing else is determined: any three independent vectors in $\mathbb{R}^3$ can serve as $\boldsymbol{a}_1, \boldsymbol{a}_4, \boldsymbol{a}_5$.

(c) In an echelon matrix the pivot of row i sits in a column p_i with $p_1 < p_2 < \cdots$, so $p_i \geq i$ and everything to the left of a pivot is zero; zero rows are at the bottom. Hence every R is *upper triangular*, and so is every combination of R's. Conversely, the six matrices E_{ij} $(i \leq j)$ with a single 1 in position (i,j) are combinations of echelon forms: E_{11}, E_{12}, E_{13} are themselves echelon forms; E_{22} is the echelon form $\left[\begin{smallmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&0 \end{smallmatrix}\right]$ minus E_{11}; E_{23} is $\left[\begin{smallmatrix} 0&1&0 \\ 0&0&1 \\ 0&0&0 \end{smallmatrix}\right]$ minus E_{12}; E_{33} is I minus $\left[\begin{smallmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&0 \end{smallmatrix}\right]$. So S is the space of all 3×3 upper triangular matrices, of dimension 6. □

15.6 Variations for practice

Exercise 15.6. Suppose $A\boldsymbol{x} = \boldsymbol{b}$ has a solution for every $\boldsymbol{b} \in \mathbb{R}^2$, and $A\boldsymbol{x} = (1, 0)$ has infinitely many solutions. What can you say about m, n, r? Give an example.

Solution. $\boldsymbol{b} \in \mathbb{R}^2$ gives $m = 2$; solvable for every $\boldsymbol{b}$ gives $r = m = 2$; infinitely many solutions means free variables, so $n > r = 2$. Thus $m = r = 2 < n$. Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, where $A\boldsymbol{x} = (1, 0)$ is solved by $(1, 0, t)$ for every t. □

Exercise 15.7. Elimination on a 3×3 matrix A uses E_{21} (subtract 2 times row 1 from row 2) and then E_{32} (subtract 5 times row 2 from row 3), reaching $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Find L and A.

Solution. In the order E_{21}, E_{32} the multipliers go directly into L: $L = E_{21}^{-1}E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$.

Then

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 0 & 5 & 6 \end{bmatrix}.$$

Check: subtracting 2 (row 1) from row 2 gives $(0, 1, 1)$, then subtracting 5 times that from row 3 gives $(0, 0, 1)$. □

Part II

Least squares, determinants and eigenvalues

16 Orthogonal vectors and subspaces

Part I found the four fundamental subspaces of an $m \times n$ matrix and their dimensions: the row space $\mathbf{C}(A^\mathsf{T})$ and nullspace $\mathbf{N}(A)$ in $\mathbb{R}^n$, of dimensions r and $n - r$, and the column space $\mathbf{C}(A)$ and left nullspace $\mathbf{N}(A^\mathsf{T})$ in $\mathbb{R}^m$, of dimensions r and $m - r$ (Section 11). This section adds the geometry: the subspaces meet at right angles, in pairs. That fact is "part two" of the fundamental theorem of linear algebra (part one being the dimensions), and it is what makes the central problem of this part of the course solvable: finding the best $\boldsymbol{x}$ when $A\boldsymbol{x} = \boldsymbol{b}$ has no solution. The section ends with the matrix $A^\mathsf{T}A$ that will carry that problem.

16.1 Orthogonal vectors: the dot product test

Definition 16.1 (Length and orthogonality)**.** The *length* of $\boldsymbol{x} \in \mathbb{R}^n$ is $\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}^\mathsf{T}\boldsymbol{x}}$, where $\boldsymbol{x}^\mathsf{T}\boldsymbol{x} = x_1^2 + \cdots + x_n^2$. Two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ are *orthogonal* (perpendicular), written $\boldsymbol{x} \perp \boldsymbol{y}$, when

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n = 0.$$

Writing the dot product as $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$, a row times a column, is deliberate: it lets dot products sit inside matrix products. For real vectors $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{y}^{\mathsf{T}}\boldsymbol{x}$, and $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} > 0$ unless $\boldsymbol{x} = \mathbf{0}$. Why is "dot product zero" the right test for a right angle in n dimensions? Because it is exactly what Pythagoras requires.

Proposition 16.2 (Pythagoras)**.** *For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, $\|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 = \|\boldsymbol{x}+\boldsymbol{y}\|^2$ if and only if $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 0$.*

Proof. Expand the right side by the rules of matrix multiplication:

$$\|\boldsymbol{x}+\boldsymbol{y}\|^2 = (\boldsymbol{x}+\boldsymbol{y})^{\mathsf{T}}(\boldsymbol{x}+\boldsymbol{y}) = \boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} + \boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} + \boldsymbol{y}^{\mathsf{T}}\boldsymbol{x} + \boldsymbol{y}^{\mathsf{T}}\boldsymbol{y} = \|\boldsymbol{x}\|^2 + 2\,\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} + \|\boldsymbol{y}\|^2.$$

The two sides of the Pythagoras equation therefore differ by exactly $2\,\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$, which vanishes precisely when $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 0$. □

Example 16.3 (A right triangle in $\mathbb{R}^3$)**.** Let $\boldsymbol{x} = (1,2,3)$ and $\boldsymbol{y} = (2,-1,0)$. Then $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 2 - 2 + 0 = 0$, so the vectors are orthogonal. The lengths confirm it: $\|\boldsymbol{x}\|^2 = 1+4+9 = 14$, $\|\boldsymbol{y}\|^2 = 4+1+0 = 5$, and $\boldsymbol{x}+\boldsymbol{y} = (3,1,3)$ has $\|\boldsymbol{x}+\boldsymbol{y}\|^2 = 9+1+9 = 19 = 14+5$.

Two small consequences of the definition are worth stating. The zero vector is orthogonal to every vector, since $\mathbf{0}^{\mathsf{T}}\boldsymbol{y} = 0$; and it is the only vector orthogonal to itself, since $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = \|\boldsymbol{x}\|^2$.

16.2 Orthogonal subspaces

Definition 16.4 (Orthogonal subspaces)**.** Subspaces S and T of $\mathbb{R}^n$ are *orthogonal*, $S \perp T$, if every vector in S is orthogonal to every vector in T: $\boldsymbol{s}^{\mathsf{T}}\boldsymbol{t} = 0$ for all $\boldsymbol{s} \in S$, $\boldsymbol{t} \in T$.

The wall and the floor of a room are *not* orthogonal subspaces of $\mathbb{R}^3$, even though we call them perpendicular planes: the line along the crease where they meet lies in both, and a nonzero vector on that line is not orthogonal to itself. More generally:

Proposition 16.5. *If $S \perp T$, then $S \cap T = \{\mathbf{0}\}$.*

Proof. If $\boldsymbol{v} \in S \cap T$, then $\boldsymbol{v}$ is orthogonal to itself, so $\|\boldsymbol{v}\|^2 = \boldsymbol{v}^{\mathsf{T}}\boldsymbol{v} = 0$ and $\boldsymbol{v} = \mathbf{0}$. □

In the plane $\mathbb{R}^2$ the subspaces are $\{\mathbf{0}\}$, lines through the origin, and $\mathbb{R}^2$ itself. The zero subspace is orthogonal to everything; a line is never orthogonal to the whole plane (it is not orthogonal to itself); two lines are orthogonal exactly when they meet at 90°. The converse of Proposition 16.5 is false: two lines meeting at 45° share only $\mathbf{0}$ but are not orthogonal.

Checking orthogonality of subspaces does not require testing infinitely many pairs.

Lemma 16.6 (Spanning vectors suffice)**.** *If $S = \operatorname{span}\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k\}$ and $T = \operatorname{span}\{\boldsymbol{t}_1, \ldots, \boldsymbol{t}_\ell\}$, then $S \perp T$ if and only if $\boldsymbol{s}_i^{\mathsf{T}}\boldsymbol{t}_j = 0$ for all i, j.*

Proof. One direction is immediate. For the other, take $\boldsymbol{s} = \sum_i c_i \boldsymbol{s}_i$ and $\boldsymbol{t} = \sum_j d_j \boldsymbol{t}_j$; then $\boldsymbol{s}^{\mathsf{T}}\boldsymbol{t} = \sum_{i,j} c_i d_j\, \boldsymbol{s}_i^{\mathsf{T}}\boldsymbol{t}_j = 0$. □

16.3 The row space is orthogonal to the nullspace

Theorem 16.7 (Orthogonality of the fundamental subspaces)**.** *For every $m \times n$ matrix A, the row space is orthogonal to the nullspace in $\mathbb{R}^n$, and the column space is orthogonal to the left nullspace in $\mathbb{R}^m$:*

$$\mathbf{C}(A^\mathsf{T}) \perp \mathbf{N}(A), \qquad \mathbf{C}(A) \perp \mathbf{N}(A^\mathsf{T}).$$

Proof. Let $\boldsymbol{x} \in \mathbf{N}(A)$. Writing A by rows,

$$A\boldsymbol{x} = \begin{bmatrix} \text{row } 1 \\ \vdots \\ \text{row } m \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} (\text{row } 1) \cdot \boldsymbol{x} \\ \vdots \\ (\text{row } m) \cdot \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix},$$

so $\boldsymbol{x}$ is orthogonal to every row of A. The rows span the row space, so by Lemma 16.6 $\boldsymbol{x}$ is orthogonal to every vector in $\mathbf{C}(A^\mathsf{T})$. (Explicitly: $(c_1 \,\text{row } 1 + \cdots + c_m \,\text{row } m) \cdot \boldsymbol{x} = c_1 \cdot 0 + \cdots + c_m \cdot 0 = 0$.) The second statement is the first one applied to the matrix A^T, whose row space is $\mathbf{C}(A)$ and whose nullspace is $\mathbf{N}(A^\mathsf{T})$. □

There is more: the dimensions add up to the whole space ($r + (n - r) = n$ and $r + (m - r) = m$), so each subspace contains *all* the vectors orthogonal to its partner, not just some of them.

Definition 16.8 (Orthogonal complement)**.** The *orthogonal complement* of a subspace $V \subseteq \mathbb{R}^n$ is $V^\perp = \{\boldsymbol{w} \in \mathbb{R}^n : \ \boldsymbol{w}^\mathsf{T}\boldsymbol{v} = 0 \text{ for every } \boldsymbol{v} \in V\}$.

Theorem 16.9 (Fundamental theorem, part two)**.** *For every $m \times n$ matrix A,*

$$\mathbf{N}(A) = \mathbf{C}(A^\mathsf{T})^\perp, \quad \mathbf{C}(A^\mathsf{T}) = \mathbf{N}(A)^\perp \ \text{ in } \mathbb{R}^n, \qquad \mathbf{N}(A^\mathsf{T}) = \mathbf{C}(A)^\perp, \quad \mathbf{C}(A) = \mathbf{N}(A^\mathsf{T})^\perp \ \text{ in } \mathbb{R}^m.$$

Moreover, for any subspace $V \subseteq \mathbb{R}^n$, $\dim V + \dim V^\perp = n$.

Proof. $\mathbf{N}(A) = \mathbf{C}(A^\mathsf{T})^\perp$. Theorem 16.7 gives $\mathbf{N}(A) \subseteq \mathbf{C}(A^\mathsf{T})^\perp$. Conversely, if $\boldsymbol{w}$ is orthogonal to the whole row space, it is orthogonal to each row, so every component of $A\boldsymbol{w}$ is zero and $\boldsymbol{w} \in \mathbf{N}(A)$.

Dimension count. Given V with basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$, let B be the $k \times n$ matrix with these vectors as rows. Then $V = \mathbf{C}(B^\mathsf{T})$, so $V^\perp = \mathbf{N}(B)$ by the first step, and $\dim \mathbf{N}(B) = n - \operatorname{rank} B = n - k$.

$\mathbf{C}(A^\mathsf{T}) = \mathbf{N}(A)^\perp$. Theorem 16.7 gives $\mathbf{C}(A^\mathsf{T}) \subseteq \mathbf{N}(A)^\perp$. By the dimension count, $\dim \mathbf{N}(A)^\perp = n - (n - r) = r = \dim \mathbf{C}(A^\mathsf{T})$. A subspace contained in another of the same dimension equals it.

The statements in $\mathbb{R}^m$ follow by applying these to A^T. □

Corollary 16.10 (Splitting a vector)**.** *Every $\boldsymbol{x} \in \mathbb{R}^n$ can be written in exactly one way as $\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$ with $\boldsymbol{x}_r \in \mathbf{C}(A^\mathsf{T})$ and $\boldsymbol{x}_n \in \mathbf{N}(A)$. Likewise every $\boldsymbol{b} \in \mathbb{R}^m$ splits uniquely into a part in $\mathbf{C}(A)$ and a part in $\mathbf{N}(A^\mathsf{T})$.*

Proof. Put a basis of the row space (r vectors) together with a basis of the nullspace ($n - r$ vectors). If a combination of all n vectors is zero, move the nullspace terms to the other side: the vector $\sum c_i \boldsymbol{u}_i = -\sum d_j \boldsymbol{w}_j$ lies in both subspaces, hence is $\mathbf{0}$ by Proposition 16.5, and independence within each basis forces all c_i, d_j to vanish. So the n vectors are independent, hence a basis of $\mathbb{R}^n$, and every $\boldsymbol{x}$ is uniquely a combination of them. The same argument works in $\mathbb{R}^m$. □

Figure 19 is the picture to keep in mind for the rest of the course.

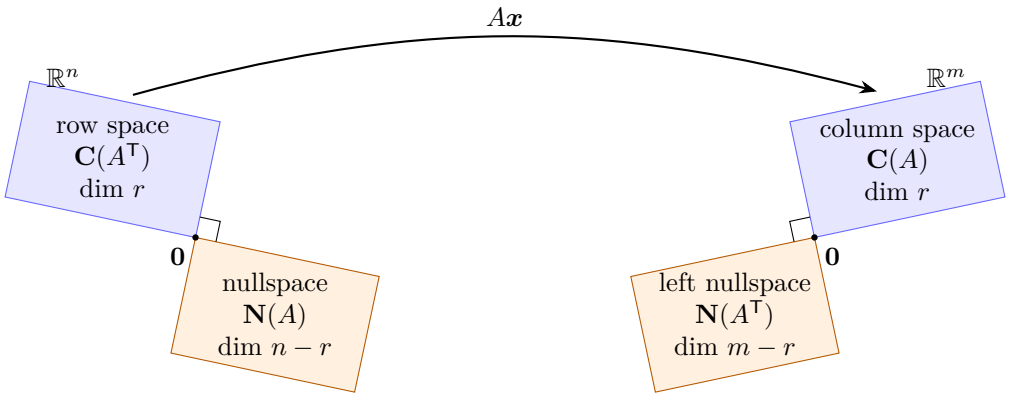


Figure 19: The four subspaces with their right angles. In $\mathbb{R}^n$ the row space and nullspace are orthogonal complements (dimensions $r + (n-r) = n$); in $\mathbb{R}^m$ the column space and left nullspace are (dimensions $r + (m-r) = m$). Every $A\boldsymbol{x}$ lands in the column space.

Example 16.11 (A plane perpendicular to a row)**.** Take $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}$. Row 2 is twice row 1, so $r = 1$ and the row space is the line through $(1, 2, 5)$. The nullspace has dimension $3 - 1 = 2$: it is the plane $x_1 + 2x_2 + 5x_3 = 0$, with special solutions $(-2, 1, 0)$ and $(-5, 0, 1)$. Both are orthogonal to $(1, 2, 5)$: $-2 + 2 + 0 = 0$ and $-5 + 0 + 5 = 0$. The second row adds nothing; the nullspace is the plane whose normal vector is $(1, 2, 5)$, exactly as in calculus. Notice what cannot happen: the row space and nullspace of a matrix with three columns can never be two perpendicular *lines* in $\mathbb{R}^3$, because their dimensions must add to 3.

Example 16.12 (Finding $S^\perp$ in $\mathbb{R}^4$)**.** Let $S = \text{span}\{(1, 2, 2, 3), (1, 3, 3, 2)\}$. A vector $\boldsymbol{x}$ is in $S^\perp$ when it is orthogonal to both spanning vectors (Lemma 16.6), that is, when $\boldsymbol{x} \in \mathbf{N}(B)$ for the matrix B with those vectors as rows. Elimination subtracts row 1 from row 2:

$$B = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix}.$$

The rank is 2, so $\dim S^\perp = 4 - 2 = 2$. With $x_3 = a$ and $x_4 = b$ free, row 2 gives $x_2 = -a + b$ and row 1 gives $x_1 = -2x_2 - 2x_3 - 3x_4 = -5b$. Hence

$$\boldsymbol{x} = a \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

and $(0, -1, 1, 0)$, $(-5, 1, 0, 1)$ form a basis for $S^\perp$ (check: each has zero dot product with both rows of B). By Corollary 16.10 the four vectors together are a basis of $\mathbb{R}^4$ (the 4×4 matrix with them as columns has determinant $53 \neq 0$), so every $\boldsymbol{v} \in \mathbb{R}^4$ splits uniquely into a piece of S plus a piece of $S^\perp$.

Example 16.13 (The left nullspace of a tall matrix)**.** The matrix of the best-fit problem in Sections 17 and 18 is

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its column space is a plane in $\mathbb{R}^3$ (the columns are independent), so its orthogonal complement $\mathbf{N}(A^\mathsf{T})$ is a line. Solving $A^\mathsf{T}\boldsymbol{y} = \mathbf{0}$, i.e. $y_1 + y_2 + y_3 = 0$ and $y_2 + 2y_3 = 0$, gives $y_2 = -2y_3$, $y_1 = y_3$: the line through $(1, -2, 1)$. Indeed $(1, -2, 1) \cdot (1, 1, 1) = 0$ and $(1, -2, 1) \cdot (0, 1, 2) = 0$. Any $\boldsymbol{b} \in \mathbb{R}^3$ that is not in the plane has a nonzero component along this line; that component will be the error vector of least squares.

16.4 The matrix $A^\mathsf{T}A$

The main problem of this part of the course is to solve $A\boldsymbol{x} = \boldsymbol{b}$ "as well as possible" when it has no solution. This is the normal situation for a tall matrix: a satellite's position measured a thousand times gives a thousand equations in a handful of unknowns, and noise in $\boldsymbol{b}$ pushes it off the column space. Throwing away equations until the system is square discards information; we want to use every measurement. The equation that does this, derived in Section 17, is

$$A^\mathsf{T}A\,\hat{\boldsymbol{x}} = A^\mathsf{T}\boldsymbol{b}.$$

Whatever A is, $A^\mathsf{T}A$ is square $(n \times n)$ and symmetric, since $(A^\mathsf{T}A)^\mathsf{T} = A^\mathsf{T}(A^\mathsf{T})^\mathsf{T} = A^\mathsf{T}A$. The question is when it is invertible.

Theorem 16.14 ($A^\mathsf{T}A$ has the same nullspace as A)**.** *For every $m \times n$ matrix A:* $\mathbf{N}(A^\mathsf{T}A) = \mathbf{N}(A)$ *and* $\operatorname{rank}(A^\mathsf{T}A) = \operatorname{rank} A$. *Consequently $A^\mathsf{T}A$ is invertible if and only if the columns of A are independent.*

Proof. If $A\boldsymbol{x} = \mathbf{0}$ then $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$. Conversely, suppose $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$ and multiply on the left by $\boldsymbol{x}^\mathsf{T}$:

$$0 = \boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = (A\boldsymbol{x})^\mathsf{T}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2,$$

so $A\boldsymbol{x} = \mathbf{0}$. Equal nullspaces in $\mathbb{R}^n$ give equal ranks, $n - \dim \mathbf{N}(A)$. The square matrix $A^\mathsf{T}A$ is invertible exactly when its nullspace is $\{\mathbf{0}\}$, that is, when $\mathbf{N}(A) = \{\mathbf{0}\}$, which means the columns of A are independent. □

Example 16.15 (Invertible and singular $A^\mathsf{T}A$)**.** For $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}$ the columns are independent and

$$A^\mathsf{T}A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 8 & 30 \end{bmatrix},$$

which is invertible ($3 \cdot 30 - 8 \cdot 8 = 26 \neq 0$). Changing the second column to $(3, 3, 3)$ makes the rank 1, and then $A^\mathsf{T}A = \begin{bmatrix} 3 & 9 \\ 9 & 27 \end{bmatrix}$ has rank 1 as well: the vector $(3, -1)$ is in the nullspace of both A and $A^\mathsf{T}A$. For the matrix of Example 16.13, $A^\mathsf{T}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, invertible as the theorem promises.

Common trap. Orthogonality of subspaces is much stronger than meeting only at $\mathbf{0}$. And "perpendicular" planes in the everyday sense, like a wall and the floor, are not orthogonal subspaces. When you test $S \perp T$, test every spanning vector of S against every spanning vector of T; when you claim $T = S^\perp$, also check that the dimensions add up to n.

16.5 Exercises

Exercise 16.16 (Fredholm's alternative)**.** Exactly one of these problems has a solution: $A\boldsymbol{x} = \boldsymbol{b}$, or $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{0}$ with $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{b} = 1$. (If $\boldsymbol{b} \notin \mathbf{C}(A)$, then $\boldsymbol{b}$ is not orthogonal to $\mathbf{N}(A^{\mathsf{T}})$.) Multiply the equations $x_1 - x_2 = 1$, $x_2 - x_3 = 1$, $x_1 - x_3 = 1$ by numbers y_1, y_2, y_3 chosen so that the equations add up to $0 = 1$.

Solution. Take $y_1 = 1$, $y_2 = 1$, $y_3 = -1$. The left sides add to $(x_1 - x_2) + (x_2 - x_3) - (x_1 - x_3) = 0$, while the right sides add to $1 + 1 - 1 = 1$. So the system implies $0 = 1$ and has no solution. In matrix language $\boldsymbol{y} = (1, 1, -1)$ satisfies $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{0}$ (it is in the left nullspace) and $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{b} = 1 \neq 0$, so $\boldsymbol{b}$ is not orthogonal to $\mathbf{N}(A^{\mathsf{T}})$ and therefore not in $\mathbf{C}(A) = \mathbf{N}(A^{\mathsf{T}})^{\perp}$. □

Exercise 16.17. You are given four nonzero vectors $\boldsymbol{r}, \boldsymbol{n}, \boldsymbol{c}, \boldsymbol{l}$ in $\mathbb{R}^2$. (a) What conditions must they satisfy to be bases for $\mathbf{C}(A^{\mathsf{T}})$, $\mathbf{N}(A)$, $\mathbf{C}(A)$, $\mathbf{N}(A^{\mathsf{T}})$ of some 2×2 matrix A? (b) Give one such matrix.

Solution. (a) Each subspace has a one-vector basis, so all four have dimension 1 and $r = 1$; the dimension counts $1 + 1 = 2$ hold automatically. What remains is orthogonality: $\boldsymbol{r}^{\mathsf{T}}\boldsymbol{n} = 0$ and $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{l} = 0$. These are also sufficient, by part (b). (b) Take $A = \boldsymbol{c}\,\boldsymbol{r}^{\mathsf{T}}$. Every column is a multiple of $\boldsymbol{c}$ and every row a multiple of $\boldsymbol{r}$, so $\mathbf{C}(A)$ is the line through $\boldsymbol{c}$ and $\mathbf{C}(A^{\mathsf{T}})$ the line through $\boldsymbol{r}$. The nullspaces are then the orthogonal complements, the lines through $\boldsymbol{n}$ and $\boldsymbol{l}$ (Theorem 16.9). □

Exercise 16.18. Is there a matrix whose row space contains $(1, 2, 1)$ and whose nullspace contains $(1, -2, 1)$?

Solution. No. The row space and the nullspace are orthogonal (Theorem 16.7), but $(1, 2, 1) \cdot (1, -2, 1) = 1 - 4 + 1 = -2 \neq 0$. □

Exercise 16.19. For $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ (Example 16.13), split $\boldsymbol{b} = (6, 0, 0)$ into a vector in $\mathbf{C}(A)$ plus a vector in $\mathbf{N}(A^{\mathsf{T}})$.

Solution. $\mathbf{N}(A^{\mathsf{T}})$ is the line through $\boldsymbol{n} = (1, -2, 1)$. Write $\boldsymbol{b} = \boldsymbol{p} + t\boldsymbol{n}$ with $\boldsymbol{p} \in \mathbf{C}(A)$; dotting with $\boldsymbol{n}$ kills $\boldsymbol{p}$ (Theorem 16.7), so $\boldsymbol{n}^{\mathsf{T}}\boldsymbol{b} = t\,\boldsymbol{n}^{\mathsf{T}}\boldsymbol{n}$, i.e. $6 = 6t$ and $t = 1$. Hence the left-nullspace part is $(1, -2, 1)$ and $\boldsymbol{p} = \boldsymbol{b} - (1, -2, 1) = (5, 2, -1)$. Check that $\boldsymbol{p}$ really is in the column space: $5(1, 1, 1) - 3(0, 1, 2) = (5, 2, -1)$. This is the projection computed again in Sections 17 and 18. □

17 Projections onto subspaces

Section 16 showed that $\mathbb{R}^m$ splits into the column space of A and the left nullspace, at right angles. Projection makes that split computable: given $\boldsymbol{b}$, find the point $\boldsymbol{p}$ of a subspace closest to $\boldsymbol{b}$. The answer is pinned down by a single right angle, the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ must be perpendicular to the subspace, and it leads to three formulas to remember: the coefficients $\hat{\boldsymbol{x}}$, the projection $\boldsymbol{p}$, and the projection matrix P. We do the one-dimensional case first and then repeat every step for a subspace of any dimension.

17.1 Projecting onto a line

Let $\boldsymbol{a} \neq \mathbf{0}$ and $\boldsymbol{b}$ be vectors in $\mathbb{R}^m$. The point of the line through $\boldsymbol{a}$ closest to $\boldsymbol{b}$ is some multiple $\boldsymbol{p} = \hat{x}\,\boldsymbol{a}$. Trigonometry would express $\hat{x}$ with $\cos\theta$; linear algebra only needs the right angle in Figure 20: the error $\boldsymbol{e} = \boldsymbol{b} - \hat{x}\boldsymbol{a}$ is perpendicular to $\boldsymbol{a}$.

$$\boldsymbol{a}^{\mathsf{T}}(\boldsymbol{b} - \hat{x}\boldsymbol{a}) = 0 \quad \Longrightarrow \quad \hat{x}\,\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a} = \boldsymbol{a}^{\mathsf{T}}\boldsymbol{b} \quad \Longrightarrow \quad \hat{x} = \frac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}, \qquad \boldsymbol{p} = \boldsymbol{a}\,\frac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}.$$

Two sanity checks: doubling $\boldsymbol{b}$ doubles $\boldsymbol{p}$; doubling $\boldsymbol{a}$ (or replacing it by $-\boldsymbol{a}$) does not change $\boldsymbol{p}$, because the line is the same: replacing $\boldsymbol{a}$ by $2\boldsymbol{a}$ multiplies the numerator and the denominator by 4.

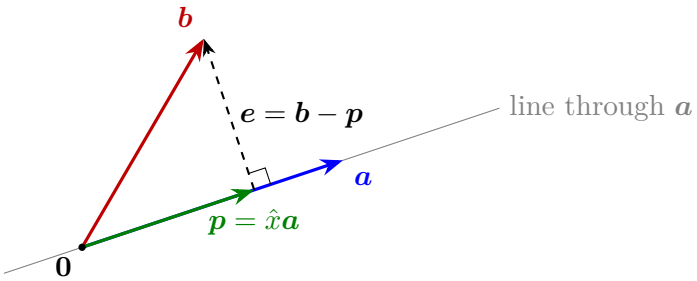


Figure 20: Projection onto a line. The closest point $\boldsymbol{p}$ is where the perpendicular from $\boldsymbol{b}$ meets the line; the error $\boldsymbol{e}$ is orthogonal to $\boldsymbol{a}$, which is the equation $\boldsymbol{a}^{\mathsf{T}}(\boldsymbol{b} - \hat{x}\boldsymbol{a}) = 0$.

Proposition 17.1 (The perpendicular foot is closest)**.** *For every number t, $\|\boldsymbol{b} - t\boldsymbol{a}\|^2 = \|\boldsymbol{e}\|^2 + (t - \hat{x})^2\|\boldsymbol{a}\|^2$. Hence $\boldsymbol{p} = \hat{x}\boldsymbol{a}$ is the unique point of the line closest to $\boldsymbol{b}$.*

Proof. Write $\boldsymbol{b} - t\boldsymbol{a} = \boldsymbol{e} + (\hat{x} - t)\boldsymbol{a}$. The two pieces are orthogonal, so Pythagoras (Section 16) adds their squared lengths. The second term is zero only for $t = \hat{x}$. □

The projection is a linear function of $\boldsymbol{b}$, so a matrix does it. Moving the parentheses in $\boldsymbol{p} = \boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b})/(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})$ gives $\boldsymbol{p} = P\boldsymbol{b}$ with:

Definition 17.2 (Projection matrix onto a line)**.** The matrix that projects onto the line through $\boldsymbol{a} \in \mathbb{R}^m$ is the $m \times m$ matrix

$$P = \frac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}.$$

The numerator is a column times a row, a full matrix; the denominator is a number.

Proposition 17.3. *The matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$ has column space the line through $\boldsymbol{a}$, rank 1, and satisfies $P^{\mathsf{T}} = P$ and $P^2 = P$.*

Proof. $P\boldsymbol{b}$ is always a multiple of $\boldsymbol{a}$, and $P\boldsymbol{a} = \boldsymbol{a}$, so $\mathbf{C}(P)$ is the line through $\boldsymbol{a}$ and the rank is 1. Transposing, $(\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}})^{\mathsf{T}} = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}$, so $P^{\mathsf{T}} = P$. Finally $P^2 = \boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})\boldsymbol{a}^{\mathsf{T}}/(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})^2 = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a} = P$: projecting a second time leaves a point already on the line where it is. □

Example 17.4 (Projecting onto the line through $(1,2,2)$)**.** Let $\boldsymbol{a} = (1,2,2)$ and $\boldsymbol{b} = (1,1,1)$. Then $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b} = 5$ and $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a} = 9$, so

$$\hat{x} = \tfrac{5}{9}, \qquad \boldsymbol{p} = \tfrac{5}{9}(1,2,2) = \left(\tfrac{5}{9}, \tfrac{10}{9}, \tfrac{10}{9}\right), \qquad \boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \left(\tfrac{4}{9}, -\tfrac{1}{9}, -\tfrac{1}{9}\right).$$

Check: $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{e} = \frac{4-2-2}{9} = 0$. The projection matrix is

$$P = \frac{1}{9}\begin{bmatrix}1\\2\\2\end{bmatrix}\begin{bmatrix}1 & 2 & 2\end{bmatrix} = \frac{1}{9}\begin{bmatrix}1 & 2 & 2\\2 & 4 & 4\\2 & 4 & 4\end{bmatrix},$$

symmetric and of rank one, and $P\boldsymbol{b} = \frac{1}{9}(5, 10, 10) = \boldsymbol{p}$.

17.2 Why project: the closest solvable problem

The equation $A\boldsymbol{x} = \boldsymbol{b}$ may have no solution: $A\boldsymbol{x}$ always lies in $\mathbf{C}(A)$, while $\boldsymbol{b}$, typically a list of measurements with errors, does not. The natural repair is to replace $\boldsymbol{b}$ by the closest vector $\boldsymbol{p}$ that *is* in the column space and solve $A\hat{\boldsymbol{x}} = \boldsymbol{p}$ instead. The hat on $\hat{\boldsymbol{x}}$ marks it as the best available coefficient vector rather than an exact solution. So we need to project onto a column space.

17.3 Projecting onto a subspace

Let the subspace be spanned by independent vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ in $\mathbb{R}^m$, and put them into the columns of the $m \times n$ matrix A, so the subspace is $\mathbf{C}(A)$. The columns need not be perpendicular, only independent. The projection is a combination $\boldsymbol{p} = \hat{x}_1\boldsymbol{a}_1 + \cdots + \hat{x}_n\boldsymbol{a}_n = A\hat{\boldsymbol{x}}$, and the error $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ must be perpendicular to the subspace, which means perpendicular to every column:

$$\boldsymbol{a}_1^{\mathsf{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = 0, \quad \ldots, \quad \boldsymbol{a}_n^{\mathsf{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = 0 \qquad \Longleftrightarrow \qquad A^{\mathsf{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \mathbf{0}.$$

The n dot products stack into one matrix equation because the rows of A^{T} are the columns of A. With one column this is the line equation $\boldsymbol{a}^{\mathsf{T}}(\boldsymbol{b} - \hat{x}\boldsymbol{a}) = 0$ again. The equation also says $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$, which by Section 16 is precisely the orthogonal complement of $\mathbf{C}(A)$: the algebra and the geometry agree.

Theorem 17.5 (Projection onto a column space)**.** *Let A be $m \times n$ with independent columns and $\boldsymbol{b} \in \mathbb{R}^m$. The vector $\hat{\boldsymbol{x}}$ for which $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ is orthogonal to $\mathbf{C}(A)$ is the unique solution of the normal equations*

$$A^{\mathsf{T}}A\,\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b},$$

and

$$\hat{\boldsymbol{x}} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}, \qquad \boldsymbol{p} = A\hat{\boldsymbol{x}} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}, \qquad P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}.$$

Proof. The orthogonality condition is $A^{\mathsf{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \mathbf{0}$, which rearranges to the normal equations. Because the columns of A are independent, $A^{\mathsf{T}}A$ is invertible (by $\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A)$, Section 16), so the solution exists, is unique, and equals $(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}$. Multiplying by A gives $\boldsymbol{p}$, and the matrix that multiplies $\boldsymbol{b}$ is P. □

When $n = 1$ the matrix $A^{\mathsf{T}}A$ is the number $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$ and "divide by it" becomes "multiply by its inverse"; the formulas reduce to those for a line.

Proposition 17.6 (Projection matrices are symmetric and idempotent)**.** $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ *satisfies $P^{\mathsf{T}} = P$ and $P^2 = P$.*

Proof. $A^{\mathsf{T}}A$ is symmetric, so its inverse is symmetric too, and $P^{\mathsf{T}} = (A^{\mathsf{T}})^{\mathsf{T}}\big((A^{\mathsf{T}}A)^{-1}\big)^{\mathsf{T}}A^{\mathsf{T}} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = P$. For the square, the middle factors cancel:

$$P^2 = A(A^{\mathsf{T}}A)^{-1}\underbrace{(A^{\mathsf{T}}A)(A^{\mathsf{T}}A)^{-1}}_{I}A^{\mathsf{T}} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = P. \qquad \square$$

Theorem 17.7 (The projection is the closest point)**.** *For every $\boldsymbol{x} \in \mathbb{R}^n$, $\|\boldsymbol{b} - A\boldsymbol{x}\|^2 = \|\boldsymbol{b} - \boldsymbol{p}\|^2 + \|A(\hat{\boldsymbol{x}} - \boldsymbol{x})\|^2$. Hence $\boldsymbol{p} = A\hat{\boldsymbol{x}}$ is the unique point of $\mathbf{C}(A)$ closest to $\boldsymbol{b}$.*

Proof. Split $\boldsymbol{b} - A\boldsymbol{x} = (\boldsymbol{b} - A\hat{\boldsymbol{x}}) + A(\hat{\boldsymbol{x}} - \boldsymbol{x}) = \boldsymbol{e} + A(\hat{\boldsymbol{x}} - \boldsymbol{x})$. The second piece lies in $\mathbf{C}(A)$ and $\boldsymbol{e}$ is orthogonal to $\mathbf{C}(A)$, so Pythagoras applies. The extra term vanishes only when $A\boldsymbol{x} = A\hat{\boldsymbol{x}}$, i.e. at $\boldsymbol{p}$. $\square$

Common trap. It is tempting to simplify $A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ to $AA^{-1}(A^{\mathsf{T}})^{-1}A^{\mathsf{T}} = I$. That is only legal when A is square and invertible, and then the answer $P = I$ is correct: the column space is all of $\mathbb{R}^m$ and every $\boldsymbol{b}$ is its own projection. For the tall matrices of interest, A has no inverse, $(A^{\mathsf{T}}A)^{-1}$ cannot be split, and $P \neq I$.

Example 17.8 (Projecting $(6,0,0)$ onto a plane)**.** Project $\boldsymbol{b} = (6,0,0)$ onto the plane spanned by $\boldsymbol{a}_1 = (1,1,1)$ and $\boldsymbol{a}_2 = (0,1,2)$ (this is the data of the line fit in Section 18). Here

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

The normal equations are $3\hat{x}_1 + 3\hat{x}_2 = 6$ and $3\hat{x}_1 + 5\hat{x}_2 = 0$. Subtracting, $2\hat{x}_2 = -6$, so $\hat{x}_2 = -3$ and then $\hat{x}_1 = 5$. Therefore

$$\boldsymbol{p} = 5\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 3\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}, \qquad \boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

The right angle checks out: $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{a}_1 = 1 - 2 + 1 = 0$ and $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{a}_2 = 0 - 2 + 2 = 0$. With $(A^{\mathsf{T}}A)^{-1} = \frac{1}{6}\left[\begin{smallmatrix} 5 & -3 \\ -3 & 3 \end{smallmatrix}\right]$ the projection matrix is

$$P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}, \qquad I - P = \frac{1}{6}\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}.$$

P is symmetric, $P\boldsymbol{b} = \frac{1}{6}(30, 12, -6) = (5, 2, -1)$, and $I - P$ is exactly $\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}}/\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e}$ with $\boldsymbol{e} = (1,-2,1)$: the projection onto the left nullspace, the line through $(1,-2,1)$ found in Section 16. Figure 21 shows the picture.

Different bases for the same subspace give the same P (Exercise 17.12). This is often useful, as in the next example: the formula needs *some* basis, and the complementary projection can be quicker.

Example 17.9 (Projection onto the plane $x + y - z = 0$)**.** Two independent vectors in the plane are $(1,-1,0)$ and $(1,0,1)$. With them as columns,

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A^{\mathsf{T}}A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad (A^{\mathsf{T}}A)^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix},$$

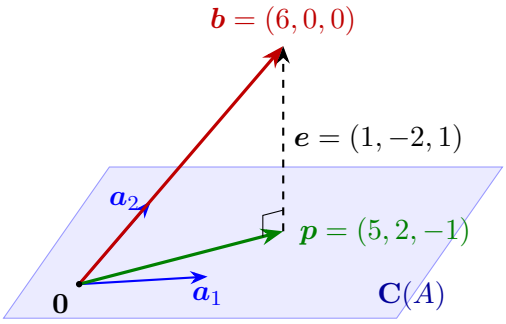


Figure 21: Projection onto the plane spanned by $\boldsymbol{a}_1 = (1,1,1)$ and $\boldsymbol{a}_2 = (0,1,2)$ (schematic). The error $\boldsymbol{e}$ is perpendicular to the whole plane, so it lies in $\mathbf{N}(A^{\mathsf{T}})$; $\boldsymbol{p}$ and $\boldsymbol{e}$ add back to $\boldsymbol{b}$.

and multiplying out,

$$P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

A quicker route: every $\boldsymbol{b}$ is its projection onto the plane plus its projection onto the normal line through $\boldsymbol{n} = (1,1,-1)$, so $I = P + P_{\boldsymbol{n}}$ and

$$P = I - \frac{\boldsymbol{n}\boldsymbol{n}^{\mathsf{T}}}{\boldsymbol{n}^{\mathsf{T}}\boldsymbol{n}} = I - \frac{1}{3}\begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix},$$

the same matrix. A good final check: $P\boldsymbol{n} = \mathbf{0}$, since the normal vector projects to the origin.

17.4 Exercises

Exercise 17.10. Let A be the 4×4 identity matrix with its last column removed, so A is 4×3. Project $\boldsymbol{b} = (1,2,3,4)$ onto $\mathbf{C}(A)$. What shape is P, and what is it?

Solution. P is 4×4: it takes vectors of $\mathbb{R}^4$ to vectors of $\mathbb{R}^4$. Here $A^{\mathsf{T}}A = I_3$, so $P = AA^{\mathsf{T}}$, which is the identity with its last diagonal entry replaced by 0:

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \boldsymbol{p} = P\boldsymbol{b} = (1,2,3,0).$$

Geometrically, $\mathbf{C}(A)$ is the coordinate subspace of the first three axes, and projecting onto it just zeroes the fourth coordinate. □

Exercise 17.11. If $P^2 = P$, show that $(I-P)^2 = I - P$. For the matrices of Exercise 17.10, P projects onto $\mathbf{C}(A)$; onto which subspace does $I - P$ project?

Solution. $(I-P)^2 = I - 2P + P^2 = I - 2P + P = I - P$. In Exercise 17.10, $I - P = \operatorname{diag}(0,0,0,1)$ projects onto the fourth axis, which is the left nullspace $\mathbf{N}(A^{\mathsf{T}})$: the vectors orthogonal to the first three columns of I. In general $I - P$ sends $\boldsymbol{b}$ to $\boldsymbol{e}$, the component in $\mathbf{N}(A^{\mathsf{T}})$. □

Exercise 17.12. Show that $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ does not change when A is replaced by AM for an invertible $n \times n$ matrix M (a different basis for the same column space).

Solution. Using $(XY)^{-1} = Y^{-1}X^{-1}$ for the square invertible factors M^{T}, $A^{\mathsf{T}}A$, M:

$$\begin{aligned} AM\big((AM)^{\mathsf{T}}AM\big)^{-1}(AM)^{\mathsf{T}} &= AM\big(M^{\mathsf{T}}(A^{\mathsf{T}}A)M\big)^{-1}M^{\mathsf{T}}A^{\mathsf{T}} \\ &= AM\,M^{-1}(A^{\mathsf{T}}A)^{-1}(M^{\mathsf{T}})^{-1}M^{\mathsf{T}}A^{\mathsf{T}} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}. \end{aligned}$$

So P depends only on the subspace, as the geometry demands. This is why the two different bases in Example 17.9 could not have given different answers. □

18 Projection matrices and least squares

This section puts the projection formulas of Section 17 to work on their most important application: fitting a straight line to data that no line passes through exactly. First we look once more at what $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ does to the two kinds of vectors it can meet, because that explains the whole least-squares picture: $\boldsymbol{b}$ splits into a part $\boldsymbol{p}$ that the model can explain and an error $\boldsymbol{e}$ that it cannot, at right angles.

18.1 What a projection does to $\boldsymbol{b}$

Proposition 18.1 (The two extreme cases)**.** *Let A have independent columns and $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$. If $\boldsymbol{b} \in \mathbf{C}(A)$ then $P\boldsymbol{b} = \boldsymbol{b}$; if $\boldsymbol{b} \in \mathbf{N}(A^{\mathsf{T}})$ (perpendicular to the column space) then $P\boldsymbol{b} = \mathbf{0}$.*

Proof. A vector in the column space has the form $\boldsymbol{b} = A\boldsymbol{x}$, and then $P\boldsymbol{b} = A(A^{\mathsf{T}}A)^{-1}(A^{\mathsf{T}}A)\boldsymbol{x} = A\boldsymbol{x} = \boldsymbol{b}$. A vector in $\mathbf{N}(A^{\mathsf{T}})$ has $A^{\mathsf{T}}\boldsymbol{b} = \mathbf{0}$, and the last factor of P kills it. □

A typical $\boldsymbol{b}$ is neither: it has a component in $\mathbf{C}(A)$ and a component in $\mathbf{N}(A^{\mathsf{T}})$ (Section 16), and the projection keeps the first and kills the second.

Proposition 18.2 (The complementary projection)**.** *$I - P$ is the projection matrix onto $\mathbf{N}(A^{\mathsf{T}})$: it is symmetric, $(I-P)^2 = I - P$, and $\boldsymbol{b} = P\boldsymbol{b} + (I-P)\boldsymbol{b} = \boldsymbol{p} + \boldsymbol{e}$ with $\boldsymbol{p} \in \mathbf{C}(A)$, $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$.*

Proof. $(I-P)^{\mathsf{T}} = I - P^{\mathsf{T}} = I - P$, and $(I-P)^2 = I - 2P + P^2 = I - P$ because $P^2 = P$. The vector $(I-P)\boldsymbol{b} = \boldsymbol{b} - A\hat{\boldsymbol{x}} = \boldsymbol{e}$ satisfies $A^{\mathsf{T}}\boldsymbol{e} = \mathbf{0}$ by the normal equations, so it lies in $\mathbf{N}(A^{\mathsf{T}})$. By Proposition 18.1, $I - P$ fixes vectors of $\mathbf{N}(A^{\mathsf{T}})$ and kills vectors of $\mathbf{C}(A)$; so it is the projection onto $\mathbf{N}(A^{\mathsf{T}})$. □

18.2 Fitting a line: the least-squares problem

We are given three measurements $(t, b) = (1, 1)$, $(2, 2)$, $(3, 2)$ and want the line $b = C + Dt$ that comes closest. If the line went through all three points we would have

$$\begin{aligned} C + D &= 1 \\ C + 2D &= 2 \\ C + 3D &= 2 \end{aligned} \qquad \text{i.e.} \qquad \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} C \\ D \end{bmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}}_{\boldsymbol{b}}.$$

The first two equations force $C = 0$, $D = 1$, and then the third fails ($3 \neq 2$). Three equations, two unknowns, no solution: $\boldsymbol{b}$ is not in the plane $\mathbf{C}(A)$. Any choice of C, D misses by the vector $A\boldsymbol{x} - \boldsymbol{b}$, whose components are the misses at the three data points.

Definition 18.3 (Least-squares solution)**.** A *least-squares solution* of $A\boldsymbol{x} = \boldsymbol{b}$ is a vector $\hat{\boldsymbol{x}}$ that minimizes the total squared error

$$E(\boldsymbol{x}) = \|A\boldsymbol{x} - \boldsymbol{b}\|^2 = e_1^2 + e_2^2 + \cdots + e_m^2.$$

Fitting a line this way is called *linear regression.*

There are two pictures of the same computation, and it pays to keep both. In the *data picture* (t horizontal, b vertical) the errors e_i are the *vertical* distances from the data points to the line, and $p_i = C + Dt_i$ are the heights of the line above the t_i. In the *vector picture* in $\mathbb{R}^m$, $\boldsymbol{b}$ is one vector, the line corresponds to a point $\boldsymbol{p} = A\hat{\boldsymbol{x}}$ of the column space, and $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is the perpendicular from $\boldsymbol{b}$ to that plane. The second picture tells us how to minimize.

Theorem 18.4 (Least squares = projection)**.** *If A has independent columns, the unique least-squares solution of $A\boldsymbol{x} = \boldsymbol{b}$ is the solution of the normal equations $A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$. The best fit is $\boldsymbol{p} = A\hat{\boldsymbol{x}} = P\boldsymbol{b}$, the minimum squared error is $\|\boldsymbol{e}\|^2$ with $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$, and $\boldsymbol{e}$ is orthogonal to every column of A.*

Proof. For any $\boldsymbol{x}$, $A\boldsymbol{x} - \boldsymbol{b} = A(\boldsymbol{x} - \hat{\boldsymbol{x}}) - \boldsymbol{e}$, where $A(\boldsymbol{x} - \hat{\boldsymbol{x}}) \in \mathbf{C}(A)$ and $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$ is orthogonal to it. By Pythagoras,

$$\|A\boldsymbol{x} - \boldsymbol{b}\|^2 = \|A(\boldsymbol{x} - \hat{\boldsymbol{x}})\|^2 + \|\boldsymbol{e}\|^2 \;\geq\; \|\boldsymbol{e}\|^2,$$

with equality only when $A(\boldsymbol{x} - \hat{\boldsymbol{x}}) = \mathbf{0}$, which (independent columns) means $\boldsymbol{x} = \hat{\boldsymbol{x}}$. □

Calculus reaches the same equations. For the three points above,

$$E(C, D) = (C + D - 1)^2 + (C + 2D - 2)^2 + (C + 3D - 2)^2,$$

and setting both partial derivatives to zero gives

$$\tfrac{1}{2}\frac{\partial E}{\partial C} = 3C + 6D - 5 = 0, \qquad \tfrac{1}{2}\frac{\partial E}{\partial D} = 6C + 14D - 11 = 0,$$

which are exactly the normal equations below. In general the gradient of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ is $2A^{\mathsf{T}}(A\boldsymbol{x} - \boldsymbol{b})$, and setting it to zero is $A^{\mathsf{T}}A\boldsymbol{x} = A^{\mathsf{T}}\boldsymbol{b}$. Because the error is a sum of *squares*, its derivatives are linear, which is why the minimum comes from a linear system.

Example 18.5 (The lecture's line through $(1,1), (2,2), (3,2)$)**.** Compute

$$A^{\mathsf{T}}A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 1+2+2 \\ 1+4+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}.$$

The matrix is symmetric and invertible, as Section 16 promised. Elimination: subtract twice the first equation $3\hat{C} + 6\hat{D} = 5$ from the second $6\hat{C} + 14\hat{D} = 11$ to get $2\hat{D} = 1$, so $\hat{D} = \frac{1}{2}$; back substitution gives $3\hat{C} + 3 = 5$, so $\hat{C} = \frac{2}{3}$. The best line is

$$b = \tfrac{2}{3} + \tfrac{1}{2}t.$$

Its heights at $t = 1, 2, 3$ and the errors are

$$\boldsymbol{p} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}, \qquad \boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix},$$

and indeed $\boldsymbol{p}+\boldsymbol{e} = \boldsymbol{b}$. Three checks of the geometry: $\boldsymbol{p}^{\mathsf{T}}\boldsymbol{e} = \frac{-7+20-13}{36} = 0$; $\boldsymbol{e}^{\mathsf{T}}(1,1,1) = -\frac{1}{6}+\frac{1}{3}-\frac{1}{6} = 0$; and $\boldsymbol{e}^{\mathsf{T}}(1,2,3) = \frac{-1+4-3}{6} = 0$. The error is perpendicular not only to $\boldsymbol{p}$ but to the whole column space. The minimum total squared error is $\|\boldsymbol{e}\|^2 = \frac{1+4+1}{36} = \frac{1}{6}$.

18.3 The best line leaves a perpendicular error

The following example is the one worked in the short video for this part of the course.

Example 18.6 (Fitting $(0,6), (1,0), (2,0)$)**.** Fit $b = C+Dt$ to the points $(t, b) = (0,6)$, $(1,0)$, $(2,0)$. Here

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} C \\ D \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}.$$

No exact line. The point at $t = 0$ forces $C = 6$. Then $C + D = 0$ needs $D = -6$, but $C + 2D = 0$ needs $D = -3$: contradiction.

Normal equations. $A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$ and $A^{\mathsf{T}}\boldsymbol{b} = (6, 0)$, so

$$3C + 3D = 6, \qquad 3C + 5D = 0.$$

Subtracting, $2D = -6$, so $D = -3$ and $C = 5$. The best line is $b = 5 - 3t$.

Projection and error. The line's heights at $t = 0, 1, 2$ are $\boldsymbol{p} = (5, 2, -1)$, so the errors are $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (1, -2, 1)$: the first and last points sit 1 above the line, the middle one 2 below (Figure 22, right).

Right-angle check. $\boldsymbol{e}^{\mathsf{T}}(1,1,1) = 1 - 2 + 1 = 0$ and $\boldsymbol{e}^{\mathsf{T}}(0,1,2) = 0 - 2 + 2 = 0$. The error is perpendicular to both columns, i.e. $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$, the line through $(1, -2, 1)$ found in Section 16.

Result. The total squared error is $1 + 4 + 1 = 6$. For comparison, the line $b = 6 - 3t$ through the first point has errors $(0, -3, 0)$ and squared error 9, and the horizontal line $b = 2$ (the mean of the data) has errors $(4, -2, -2)$ and squared error 24. The vector picture of this fit, with $\boldsymbol{b} = (6, 0, 0)$ dropping perpendicularly onto the plane of the columns, is drawn in Section 17.

Common trap. Least squares does *not* minimize the perpendicular distances from the points to the line. It minimizes the sum of squared *vertical* errors $e_i = b_i - (C + Dt_i)$, because those are the components of $A\boldsymbol{x}-\boldsymbol{b}$. The right angle is between $\boldsymbol{e}$ and $\mathbf{C}(A)$ in $\mathbb{R}^m$, not between the error segments and the line in the t–b plane. A second warning: squaring makes a single wild measurement (an *outlier*) count heavily, so one bad point can swing the whole line; statisticians check for outliers before trusting a least-squares fit.

18.4 When the normal equations can be solved

Everything above assumed that $A^{\mathsf{T}}A$ is invertible.

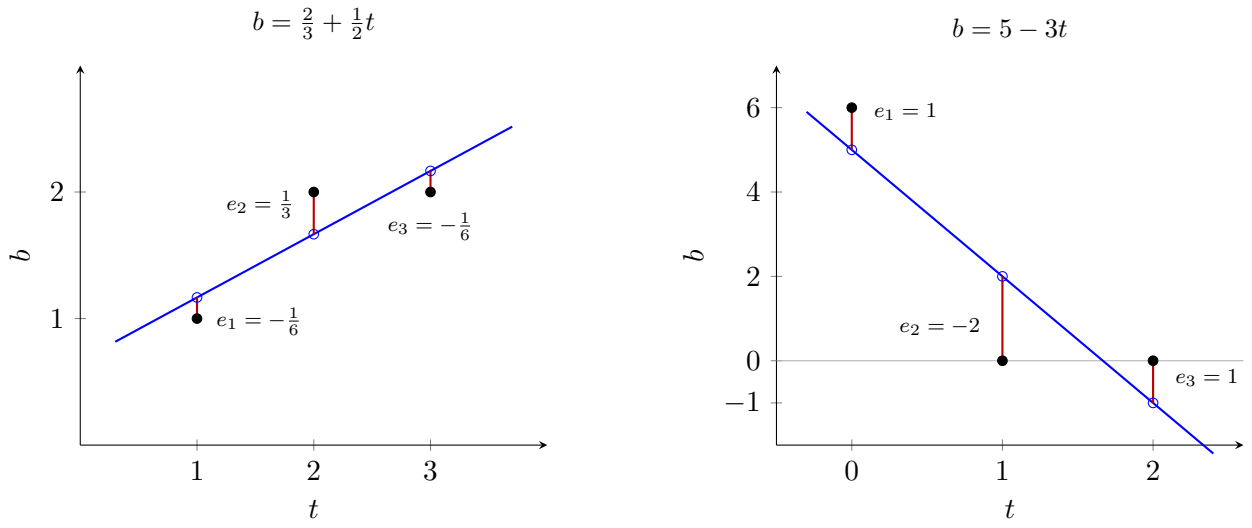


Figure 22: The data picture of two least-squares fits. Dots are the data b_i, circles the heights p_i of the best line, red segments the vertical errors $e_i = b_i - p_i$. Left: the lecture's points; right: the points $(0,6), (1,0), (2,0)$ with errors $(1,-2,1)$. The right angle of least squares is not visible here; it lives in $\mathbb{R}^3$, between $\boldsymbol{e}$ and the column space.

Theorem 18.7. *If A has independent columns, then $A^{\mathsf{T}}A$ is invertible.*

Proof. Suppose $A^{\mathsf{T}}A\boldsymbol{x} = \mathbf{0}$; we show $\boldsymbol{x} = \mathbf{0}$. Multiply by $\boldsymbol{x}^{\mathsf{T}}$: $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = (A\boldsymbol{x})^{\mathsf{T}}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2 = 0$, so $A\boldsymbol{x} = \mathbf{0}$. Independent columns mean the only solution of $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$. So $\mathbf{N}(A^{\mathsf{T}}A) = \{\mathbf{0}\}$ and the square matrix $A^{\mathsf{T}}A$ is invertible. (This is half of the equality $\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A)$ of Section 16.) □

For a line fit, the columns $(1,\dots,1)$ and $(t_1,\dots,t_m)$ are independent as soon as two of the t_i differ, so least squares always has a unique line unless all measurements are at the same time. Columns are certainly independent when they are *orthonormal*: perpendicular unit vectors such as $(1,0,0), (0,1,0), (0,0,1)$, or $(\cos\theta, \sin\theta)$ and $(-\sin\theta, \cos\theta)$. Then $A^{\mathsf{T}}A = I$ and the normal equations need no solving at all. Section 19 shows how to make the columns orthonormal.

Nothing in the method needs the model to be a straight line; it only needs the unknown coefficients to enter *linearly*. A parabola $C + Dt + Et^2$ gives a matrix with columns 1, t_i, t_i^2; a curve through the origin $ct + dt^2$ gives columns t_i and t_i^2 (Exercise 18.12).

18.5 Exercises

Exercise 18.8. Write down three equations for the line $b = C + Dt$ to go through $b = 7$ at $t = -1$, $b = 7$ at $t = 1$, and $b = 21$ at $t = 2$. Find the least-squares solution $\hat{\boldsymbol{x}} = (C, D)$, the projection $\boldsymbol{p} = A\hat{\boldsymbol{x}}$, and the error $\boldsymbol{e}$. Why is $P\boldsymbol{e} = \mathbf{0}$?

Solution. The equations are $C - D = 7$, $C + D = 7$, $C + 2D = 21$, so

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^{\mathsf{T}}A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, \quad A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 35 \\ 42 \end{bmatrix}.$$

Solving $3C + 2D = 35$, $2C + 6D = 42$: three times the first minus the second gives $7C = 63$, so $C = 9$ and $D = 4$. The closest line is $b = 9 + 4t$, with heights $\boldsymbol{p} = (5, 13, 17)$ and error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (2, -6, 4)$. Since $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$ (check: $2-6+4 = 0$ and $-2-6+8 = 0$), Proposition 18.1 gives $P\boldsymbol{e} = \mathbf{0}$; equivalently $P\boldsymbol{e} = P\boldsymbol{b} - P\boldsymbol{p} = \boldsymbol{p} - \boldsymbol{p} = \mathbf{0}$. □

Exercise 18.9. With the same times $t = -1, 1, 2$: (a) If the measurements are $\boldsymbol{b} = (2, -6, 4)$, find $\hat{\boldsymbol{x}}$ and the closest line. (b) If they are $\boldsymbol{b} = (5, 13, 17)$, find $\hat{\boldsymbol{x}}$ and $\boldsymbol{e}$. (c) Which of the four subspaces contain $\boldsymbol{e}$, $\boldsymbol{p}$, $\hat{\boldsymbol{x}}$? What is $\mathbf{N}(A)$?

Solution. (a) This $\boldsymbol{b}$ is the previous error, which is perpendicular to $\mathbf{C}(A)$; so $A^{\mathsf{T}}\boldsymbol{b} = \mathbf{0}$, $\hat{\boldsymbol{x}} = (0, 0)$, $\boldsymbol{p} = \mathbf{0}$, and the best line is $b = 0$. (b) This $\boldsymbol{b}$ is the previous projection, which lies in $\mathbf{C}(A)$; so $\hat{\boldsymbol{x}} = (9, 4)$ fits exactly and $\boldsymbol{e} = \mathbf{0}$. (c) $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$, $\boldsymbol{p} \in \mathbf{C}(A)$, $\hat{\boldsymbol{x}} \in \mathbf{C}(A^{\mathsf{T}}) = \mathbb{R}^2$ (the row space is all of $\mathbb{R}^2$ here), and $\mathbf{N}(A) = \{\mathbf{0}\}$ because the columns are independent. □

Exercise 18.10. Find the best line $C + Dt$ to fit $b = 4, 2, -1, 0, 0$ at times $t = -2, -1, 0, 1, 2$.

Solution. The column of t's is $(-2, -1, 0, 1, 2)$, which is orthogonal to the column of ones (the times are symmetric about 0), so $A^{\mathsf{T}}A$ is diagonal:

$$A^{\mathsf{T}}A = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}, \qquad A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 4+2-1+0+0 \\ -8-2+0+0+0 \end{bmatrix} = \begin{bmatrix} 5 \\ -10 \end{bmatrix}.$$

Hence $C = 1$, $D = -1$, and the best line is $b = 1 - t$. Its errors are $(1, 0, -2, 0, 1)$, which are orthogonal to both columns. □

Exercise 18.11. Fit a least-squares line through $(0, 0)$, $(1, 3)$, $(2, 0)$. What are C and D?

Solution. The matrix is the same as in Example 18.6, so $A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, and now $A^{\mathsf{T}}\boldsymbol{b} = (3, 3)$. The equations $3C + 3D = 3$ and $3C + 5D = 3$ give $D = 0$, $C = 1$: the flat line $b = 1$, with errors $(-1, 2, -1)$, again a multiple of $(1, -2, 1)$. □

Exercise 18.12. Find the curve $y = ct + dt^2$ through the origin that best fits $(1, 1)$, $(2, 5)$, $(-1, -2)$ in the least-squares sense.

Solution. Plugging each point into $ct + dt^2 = y$ gives

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ -1 & 1 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} c \\ d \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix},$$

with columns t_i and t_i^2. Then

$$A^{\mathsf{T}}A = \begin{bmatrix} 1+4+1 & 1+8-1 \\ 1+8-1 & 1+16+1 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 8 & 18 \end{bmatrix}, \qquad A^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 1+10+2 \\ 1+20-2 \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \end{bmatrix}.$$

The determinant is $6 \cdot 18 - 8 \cdot 8 = 44$, and solving gives $c = \frac{13 \cdot 18 - 8 \cdot 19}{44} = \frac{41}{22}$, $d = \frac{6 \cdot 19 - 8 \cdot 13}{44} = \frac{5}{22}$. The best curve is $y = \frac{41}{22}t + \frac{5}{22}t^2$, with errors $\boldsymbol{e} = \frac{1}{11}(-12, 4, -4)$; check that $A^{\mathsf{T}}\boldsymbol{e} = \mathbf{0}$. (If you watch the recitation video for this problem, note that the $(2, 2)$ entry of $A^{\mathsf{T}}A$ is $1 + 16 + 1 = 18$; using 10 there leads to a curve whose squared error, 56, is far larger than the true minimum $\frac{16}{11}$.) □

19 Orthogonal matrices and Gram–Schmidt

The last step in the story of orthogonality is an orthogonal *basis*. Section 18 ended with the observation that the normal equations become trivial when the columns of A are perpendicular unit vectors, because then $A^{\mathsf{T}}A = I$. This section collects such matrices, called Q, shows exactly which formulas simplify, and then gives the Gram–Schmidt process, which turns any independent columns into orthonormal ones. In matrix language Gram–Schmidt is a factorization $A = QR$, just as elimination was $A = LU$.

19.1 Orthonormal vectors and the matrix Q

Definition 19.1 (Orthonormal vectors)**.** Vectors $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ in $\mathbb{R}^m$ are *orthonormal* if

$$\boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{q}_j = \begin{cases} 0 & i \neq j \quad \text{(orthogonal)}, \\ 1 & i = j \quad \text{(normalized: unit length)}. \end{cases}$$

Put orthonormal vectors into the columns of an $m \times n$ matrix Q. Then $Q^{\mathsf{T}}Q$ is the matrix of all their dot products: row i of Q^{T} is $\boldsymbol{q}_i^{\mathsf{T}}$, column j of Q is $\boldsymbol{q}_j$.

Proposition 19.2. *Q has orthonormal columns if and only if $Q^{\mathsf{T}}Q = I$ (the $n \times n$ identity).*

Proof. The (i, j) entry of $Q^{\mathsf{T}}Q$ is $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_j$; Definition 19.1 says precisely that these are the entries of I. □

Q need not be square: two orthonormal columns in $\mathbb{R}^4$ give a 4×2 matrix with $Q^{\mathsf{T}}Q = I_2$ (but $QQ^{\mathsf{T}} \neq I_4$). Orthonormal columns are automatically independent (Exercise 19.11). This adds a new family to our list of special matrices (triangular, diagonal, permutation, symmetric, echelon, projection).

Definition 19.3 (Orthogonal matrix)**.** A *square* matrix with orthonormal columns is called an *orthogonal matrix*. For such Q, $Q^{\mathsf{T}}Q = I$ means $Q^{\mathsf{T}} = Q^{-1}$, and then also $QQ^{\mathsf{T}} = I$.

The name is a historical accident ("orthonormal matrix" would be more accurate), and it is reserved for the square case.

Example 19.4 (A gallery of orthogonal matrices)**.** Four families are worth knowing.

- Every permutation matrix, e.g. $Q = \left[\begin{smallmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right]$: its columns are distinct standard basis vectors, and Q^{T} (another permutation) is its inverse.
- Rotations $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$: both columns have length $\cos^2\theta + \sin^2\theta = 1$ and their dot product is $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$.
- $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ has orthogonal columns of length $\sqrt{2}$; dividing by $\sqrt{2}$ makes it orthogonal. Repeating the sign pattern in blocks gives

$$Q = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix},$$

a *Hadamard matrix*: entries ± 1 (scaled), columns of length 2 before scaling. Such matrices exist for sizes $2, 4, 8, 16, \ldots$ and many others, but which sizes allow them is still an open question.

- A rectangular example without square roots, and a square one extending it:

$$Q = \frac{1}{3}\begin{bmatrix} 1 & -2 \\ 2 & -1 \\ 2 & 2 \end{bmatrix}, \qquad Q' = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}.$$

  Each column has squared length $\frac{1+4+4}{9} = 1$, and the dot products vanish, e.g. $(1,2,2) \cdot (-2,-1,2) = -2-2+4 = 0$.

19.2 Orthonormal columns make projection easy

Suppose Q has orthonormal columns and we project onto its column space. The projection formula of Section 17 contains $Q^{\mathsf{T}}Q = I$, which disappears:

Proposition 19.5 (Projection with an orthonormal basis)**.** *If Q has orthonormal columns, the projection onto $\mathbf{C}(Q)$ is*

$$P = Q(Q^{\mathsf{T}}Q)^{-1}Q^{\mathsf{T}} = QQ^{\mathsf{T}},$$

the least-squares solution of $Q\boldsymbol{x} = \boldsymbol{b}$ is $\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b}$, that is $\hat{x}_i = \boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{b}$, and $\boldsymbol{p} = \boldsymbol{q}_1(\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b}) + \cdots + \boldsymbol{q}_n(\boldsymbol{q}_n^{\mathsf{T}}\boldsymbol{b})$. If Q is square, $P = I$.

Proof. Substitute $Q^{\mathsf{T}}Q = I$ into $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ and into the normal equations $Q^{\mathsf{T}}Q\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b}$. Writing $QQ^{\mathsf{T}}\boldsymbol{b}$ column by column gives the sum of one-dimensional projections. If Q is square, $QQ^{\mathsf{T}} = I$ by Definition 19.3. As a check, QQ^{T} is symmetric and $(QQ^{\mathsf{T}})(QQ^{\mathsf{T}}) = Q(Q^{\mathsf{T}}Q)Q^{\mathsf{T}} = QQ^{\mathsf{T}}$. □

No matrix needs to be inverted and no system solved: each coefficient is one dot product. This is the formula behind Fourier series and much of numerical linear algebra, where orthonormal vectors are prized because they never grow or shrink, so roundoff stays under control. For the 3×2 matrix of Example 19.4,

$$QQ^{\mathsf{T}} = \frac{1}{9}\begin{bmatrix} 5 & 4 & -2 \\ 4 & 5 & 2 \\ -2 & 2 & 8 \end{bmatrix},$$

the projection onto the plane spanned by $(1,2,2)$ and $(-2,-1,2)$.

19.3 The Gram–Schmidt process

Elimination aimed for a triangular matrix; Gram–Schmidt aims for orthonormal columns. Start with two independent vectors $\boldsymbol{a}$ and $\boldsymbol{b}$. Keep the direction of the first: $\boldsymbol{A} = \boldsymbol{a}$. The second is not perpendicular to $\boldsymbol{A}$, so remove its projection onto $\boldsymbol{A}$ and keep the rest, which is exactly the error vector $\boldsymbol{e}$ of Section 17:

$$\boldsymbol{B} = \boldsymbol{b} - \frac{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}}\,\boldsymbol{A}.$$

It is perpendicular to $\boldsymbol{A}$: $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{B} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{b} - \frac{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A} = 0$ (Figure 23). Finally divide by lengths: $\boldsymbol{q}_1 = \boldsymbol{A}/\|\boldsymbol{A}\|$, $\boldsymbol{q}_2 = \boldsymbol{B}/\|\boldsymbol{B}\|$. With a third vector $\boldsymbol{c}$, subtract its components along *both* earlier directions:

$$\boldsymbol{C} = \boldsymbol{c} - \frac{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{c}}{\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A}}\,\boldsymbol{A} - \frac{\boldsymbol{B}^{\mathsf{T}}\boldsymbol{c}}{\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B}}\,\boldsymbol{B}, \qquad \boldsymbol{q}_3 = \boldsymbol{C}/\|\boldsymbol{C}\|.$$

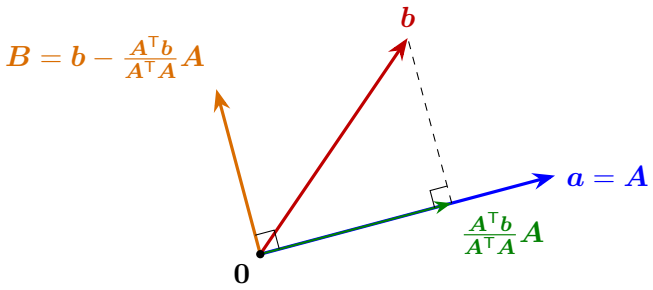


Figure 23: One Gram–Schmidt step. Subtracting from $\boldsymbol{b}$ its projection onto $\boldsymbol{A}$ leaves $\boldsymbol{B}$, perpendicular to $\boldsymbol{A}$ and parallel to the dashed error segment. The plane spanned by $\boldsymbol{A}, \boldsymbol{B}$ is the plane spanned by $\boldsymbol{a}, \boldsymbol{b}$.

Theorem 19.6 (Gram–Schmidt)**.** *Let $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$ be independent. Define recursively*

$$\boldsymbol{A}_k = \boldsymbol{a}_k - \sum_{i<k} (\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_k)\,\boldsymbol{q}_i, \qquad \boldsymbol{q}_k = \boldsymbol{A}_k/\|\boldsymbol{A}_k\|.$$

Then each $\boldsymbol{A}_k \neq \boldsymbol{0}$, the vectors $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ are orthonormal, and for every k

$$\operatorname{span}\{\boldsymbol{q}_1, \dots, \boldsymbol{q}_k\} = \operatorname{span}\{\boldsymbol{a}_1, \dots, \boldsymbol{a}_k\}.$$

(Since $\boldsymbol{q}_i = \boldsymbol{A}_i/\|\boldsymbol{A}_i\|$, the term $(\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_k)\boldsymbol{q}_i$ equals $\frac{\boldsymbol{A}_i^{\mathsf{T}}\boldsymbol{a}_k}{\boldsymbol{A}_i^{\mathsf{T}}\boldsymbol{A}_i}\boldsymbol{A}_i$, the form used above.)

Proof. Induction on k. Assume $\boldsymbol{q}_1, \dots, \boldsymbol{q}_{k-1}$ are orthonormal and span the same space V_{k-1} as $\boldsymbol{a}_1, \dots, \boldsymbol{a}_{k-1}$. The vector $\boldsymbol{A}_k$ is $\boldsymbol{a}_k$ minus an element of V_{k-1}. If $\boldsymbol{A}_k$ were $\boldsymbol{0}$, then $\boldsymbol{a}_k \in V_{k-1}$, contradicting independence; so $\boldsymbol{q}_k$ is defined. For $j < k$,

$$\boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{A}_k = \boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{a}_k - \sum_{i<k} (\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_k)\,\boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{q}_i = \boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{a}_k - \boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{a}_k = 0,$$

because only the $i = j$ term survives. So $\boldsymbol{q}_k$ is a unit vector orthogonal to the earlier $\boldsymbol{q}$'s. Finally $\boldsymbol{q}_k$ is a combination of $\boldsymbol{a}_k$ and V_{k-1}, and $\boldsymbol{a}_k = \|\boldsymbol{A}_k\|\boldsymbol{q}_k + \sum_{i<k}(\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_k)\boldsymbol{q}_i$ is a combination of $\boldsymbol{q}_1, \dots, \boldsymbol{q}_k$; hence the two spans agree. □

Example 19.7 (Two vectors in $\mathbb{R}^3$)**.** Let $\boldsymbol{a} = (1, 1, 1)$ and $\boldsymbol{b} = (1, 0, 2)$. Then $\boldsymbol{A} = \boldsymbol{a}$, $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{b} = 3$, $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A} = 3$, and

$$\boldsymbol{B} = \begin{bmatrix}1\\0\\2\end{bmatrix} - \frac{3}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\-1\\1\end{bmatrix}, \qquad \boldsymbol{A}^{\mathsf{T}}\boldsymbol{B} = 0 - 1 + 1 = 0.$$

Dividing by the lengths $\sqrt{3}$ and $\sqrt{2}$,

$$Q = \begin{bmatrix}\boldsymbol{q}_1 & \boldsymbol{q}_2\end{bmatrix} = \begin{bmatrix}1/\sqrt{3} & 0\\ 1/\sqrt{3} & -1/\sqrt{2}\\ 1/\sqrt{3} & 1/\sqrt{2}\end{bmatrix}.$$

The column space of Q is the same plane as the column space of $\left[\begin{smallmatrix}1&1\\1&0\\1&2\end{smallmatrix}\right]$; only the basis improved.

Common trap. When computing $\boldsymbol{C}$, subtract the projections of $\boldsymbol{c}$ onto the *new* orthogonal vectors $\boldsymbol{A}$ and $\boldsymbol{B}$, not onto the original $\boldsymbol{a}$ and $\boldsymbol{b}$; the original vectors are not perpendicular, and subtracting both of their projections over-corrects. And do not forget the final division by lengths: $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are orthogonal but generally not orthonormal. Doing the normalization last keeps square roots out of the intermediate arithmetic.

19.4 $A = QR$

Elimination is summarized by $A = LU$; Gram–Schmidt is summarized by a factorization that connects A to Q. Since $\mathbf{C}(A) = \mathbf{C}(Q)$, each column of A is a combination of the $\boldsymbol{q}$'s, and the coefficients form a matrix R.

Theorem 19.8 (QR factorization)**.** *If A ($m \times n$) has independent columns and Q comes from Gram–Schmidt, then*

$$A = QR, \qquad R = Q^{\mathsf{T}} A, \qquad r_{ij} = \boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{a}_j,$$

and R is upper triangular with positive diagonal entries $r_{kk} = \|\boldsymbol{A}_k\|$.

Proof. Rearranging the definition of $\boldsymbol{A}_k$, $\boldsymbol{a}_k = \sum_{i<k} (\boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{a}_k) \boldsymbol{q}_i + \|\boldsymbol{A}_k\| \, \boldsymbol{q}_k$. This says column k of A equals Q times the column $(r_{1k}, \ldots, r_{kk}, 0, \ldots, 0)$, with $r_{ik} = \boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{a}_k$ for $i < k$ and $r_{kk} = \|\boldsymbol{A}_k\| > 0$; note also $\boldsymbol{q}_k^{\mathsf{T}} \boldsymbol{a}_k = \|\boldsymbol{A}_k\|$ by dotting the same identity with $\boldsymbol{q}_k$. The entries below the diagonal are zero because $\boldsymbol{a}_k$ lies in the span of $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$, so $\boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{a}_k = 0$ for $i > k$: later $\boldsymbol{q}$'s are perpendicular to earlier $\boldsymbol{a}$'s. Multiplying $A = QR$ by Q^{T} gives $R = Q^{\mathsf{T}} A$. □

For two columns,

$$\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{q}_1^{\mathsf{T}} \boldsymbol{a}_1 & \boldsymbol{q}_1^{\mathsf{T}} \boldsymbol{a}_2 \\ 0 & \boldsymbol{q}_2^{\mathsf{T}} \boldsymbol{a}_2 \end{bmatrix},$$

and the zero is $\boldsymbol{q}_2^{\mathsf{T}} \boldsymbol{a}_1 = 0$ because $\boldsymbol{q}_2$ was built perpendicular to $\boldsymbol{a}_1 = \boldsymbol{A}$. In Example 19.7, $\boldsymbol{q}_1^{\mathsf{T}} \boldsymbol{a} = 3/\sqrt{3}$, $\boldsymbol{q}_1^{\mathsf{T}} \boldsymbol{b} = 3/\sqrt{3}$ and $\boldsymbol{q}_2^{\mathsf{T}} \boldsymbol{b} = 2/\sqrt{2}$, so

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}.$$

Example 19.9 (A lucky 3×3 with no square roots)**.** Orthonormalize the columns $\boldsymbol{a} = (1, 0, 0)$, $\boldsymbol{b} = (2, 0, 3)$, $\boldsymbol{c} = (4, 5, 6)$ of $A = \left[\begin{smallmatrix}1&2&4\\0&0&5\\0&3&6\end{smallmatrix}\right]$, working directly with unit vectors.

- $\boldsymbol{a}$ already has length 1: $\boldsymbol{q}_1 = (1, 0, 0)$.
- $\boldsymbol{b} - (\boldsymbol{b}^{\mathsf{T}} \boldsymbol{q}_1) \boldsymbol{q}_1 = (2, 0, 3) - 2(1, 0, 0) = (0, 0, 3)$, of length 3, so $\boldsymbol{q}_2 = (0, 0, 1)$.
- $\boldsymbol{c} - (\boldsymbol{c}^{\mathsf{T}} \boldsymbol{q}_1) \boldsymbol{q}_1 - (\boldsymbol{c}^{\mathsf{T}} \boldsymbol{q}_2) \boldsymbol{q}_2 = (4, 5, 6) - 4(1, 0, 0) - 6(0, 0, 1) = (0, 5, 0)$, of length 5, so $\boldsymbol{q}_3 = (0, 1, 0)$.

The numbers subtracted and the lengths are exactly the entries of R:

$$\underbrace{\begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}}_{R},$$

which reads column by column as $\boldsymbol{a} = 1\boldsymbol{q}_1$, $\boldsymbol{b} = 2\boldsymbol{q}_1 + 3\boldsymbol{q}_2$, $\boldsymbol{c} = 4\boldsymbol{q}_1 + 6\boldsymbol{q}_2 + 5\boldsymbol{q}_3$. Here Q happens to be a permutation matrix, and R is A with rows 2 and 3 exchanged.

Least squares with $A = QR$. Substituting $A = QR$ into the normal equations, $R^{\mathsf{T}}Q^{\mathsf{T}}QR\hat{\boldsymbol{x}} = R^{\mathsf{T}}Q^{\mathsf{T}}\boldsymbol{b}$ becomes $R^{\mathsf{T}}R\hat{\boldsymbol{x}} = R^{\mathsf{T}}Q^{\mathsf{T}}\boldsymbol{b}$, and since R is invertible,

$$R\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b},$$

a triangular system solved by back substitution. This is how least squares is computed in practice.

Example 19.10 (The best-fit line again, via QR)**.** For the data $(0,6),(1,0),(2,0)$ of Section 18, A has columns $\boldsymbol{a}_1 = (1,1,1)$ and $\boldsymbol{a}_2 = (0,1,2)$, and $\boldsymbol{b} = (6,0,0)$. Gram–Schmidt: $\boldsymbol{A}_2 = (0,1,2) - \frac{3}{3}(1,1,1) = (-1,0,1)$, so

$$\boldsymbol{q}_1 = \tfrac{1}{\sqrt{3}}(1,1,1), \quad \boldsymbol{q}_2 = \tfrac{1}{\sqrt{2}}(-1,0,1), \qquad R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}, \qquad Q^{\mathsf{T}}\boldsymbol{b} = \begin{bmatrix} 6/\sqrt{3} \\ -6/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ -3\sqrt{2} \end{bmatrix}.$$

Back substitution in $R\hat{\boldsymbol{x}} = Q^{\mathsf{T}}\boldsymbol{b}$: $\sqrt{2}\,D = -3\sqrt{2}$ gives $D = -3$, then $\sqrt{3}\,C + \sqrt{3}\,D = 2\sqrt{3}$ gives $C = 5$, the same line $b = 5 - 3t$. The projection is a sum of two independent one-dimensional projections: $\boldsymbol{p} = \boldsymbol{q}_1(\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b}) + \boldsymbol{q}_2(\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{b}) = 2(1,1,1) - 3(-1,0,1) = (5,2,-1)$.

19.5 Exercises

Exercise 19.11. Orthonormal vectors are automatically independent. Give a matrix proof: if Q has orthonormal columns, show that $Q\boldsymbol{x} = \boldsymbol{0}$ implies $\boldsymbol{x} = \boldsymbol{0}$. ($Q$ may be rectangular, so use Q^{T}, not Q^{-1}.)

Solution. Multiply $Q\boldsymbol{x} = \boldsymbol{0}$ by Q^{T}: $Q^{\mathsf{T}}Q\boldsymbol{x} = Q^{\mathsf{T}}\boldsymbol{0}$, i.e. $I\boldsymbol{x} = \boldsymbol{0}$, so $\boldsymbol{x} = \boldsymbol{0}$. Thus $\mathbf{N}(Q) = \{\boldsymbol{0}\}$ and no nontrivial combination of the columns vanishes. □

Exercise 19.12. Apply Gram–Schmidt to $\boldsymbol{a} = (1,-1,0,0)$, $\boldsymbol{b} = (0,1,-1,0)$, $\boldsymbol{c} = (0,0,1,-1)$ to get orthogonal vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$. Show that $\{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}\}$ and $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$ are both bases for the space of vectors perpendicular to $\boldsymbol{d} = (1,1,1,1)$.

Solution. $\boldsymbol{A} = \boldsymbol{a} = (1,-1,0,0)$. Then $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{b} = -1$, $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A} = 2$, and $\boldsymbol{B} = \boldsymbol{b} + \frac{1}{2}\boldsymbol{A} = (\frac{1}{2}, \frac{1}{2}, -1, 0)$. Next $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{c} = 0$, $\boldsymbol{B}^{\mathsf{T}}\boldsymbol{c} = -1$, $\boldsymbol{B}^{\mathsf{T}}\boldsymbol{B} = \frac{3}{2}$, so $\boldsymbol{C} = \boldsymbol{c} + \frac{2}{3}\boldsymbol{B} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)$. The three are mutually orthogonal (check the three dot products) and each has components summing to zero, so each is perpendicular to $\boldsymbol{d}$; so are $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$. The space $\boldsymbol{d}^{\perp}$ is the nullspace of the 1×4 matrix $\boldsymbol{d}^{\mathsf{T}}$, of dimension $4 - 1 = 3$. Three independent vectors in a 3-dimensional space form a basis. $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are independent because they are orthogonal and nonzero (normalize them and use Exercise 19.11). For $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$: the combination $x_1\boldsymbol{a} + x_2\boldsymbol{b} + x_3\boldsymbol{c} = (x_1,\, x_2 - x_1,\, x_3 - x_2,\, -x_3)$ is zero only when $x_1 = x_2 = x_3 = 0$. □

Exercise 19.13. Find the QR factorization of $A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 2 & 1 \end{bmatrix}$.

Solution. $\|\boldsymbol{a}_1\| = 3$, so $\boldsymbol{q}_1 = \frac{1}{3}(1,2,2)$ and $r_{11} = 3$. Next $r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = \frac{1+0+2}{3} = 1$, and $\boldsymbol{A}_2 = \boldsymbol{a}_2 - \boldsymbol{q}_1 = (1,0,1) - \frac{1}{3}(1,2,2) = \frac{1}{3}(2,-2,1)$, which already has length 1; so $\boldsymbol{q}_2 = \frac{1}{3}(2,-2,1)$ and $r_{22} = 1$. Hence

$$A = \frac{1}{3}\begin{bmatrix} 1 & 2 \\ 2 & -2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix},$$

and multiplying out reproduces the columns $(1,2,2)$ and $(1,0,1)$. □

Exercise 19.14. Show that multiplication by a matrix with orthonormal columns preserves lengths and dot products: $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$ and $(Q\boldsymbol{x})^{\mathsf{T}}(Q\boldsymbol{y}) = \boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$. What does this say about the rotation matrices of Example 19.4?

Solution. $(Q\boldsymbol{x})^{\mathsf{T}}(Q\boldsymbol{y}) = \boldsymbol{x}^{\mathsf{T}}Q^{\mathsf{T}}Q\boldsymbol{y} = \boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$; taking $\boldsymbol{y} = \boldsymbol{x}$ gives $\|Q\boldsymbol{x}\|^2 = \|\boldsymbol{x}\|^2$. Rotations therefore move vectors without stretching them and preserve the angles between them, as a rigid rotation of the plane should. □

20 Properties of determinants

Up to now most of our matrices were rectangular. The second half of the course concentrates on *square* matrices, and its two big tools are the determinant and the eigenvalues; the determinant comes first because eigenvalues will be found from it. The determinant of a square matrix A, written $\det A$ or $|A|$, is a single number that packs in a remarkable amount of information, above all: A is invertible exactly when $\det A \neq 0$. Rather than start from a formula (that comes in Section 21), we start from three properties. Everything else, including the familiar $ad - bc$ and an efficient way to compute, follows from them.

20.1 Three properties define the determinant

Definition 20.1 (The determinant)**.** The determinant is the function assigning a number $\det A$ to each $n \times n$ matrix such that:

1. $\det I = 1$.
2. Exchanging two rows reverses the sign of the determinant.
3. The determinant is linear in each row separately (the other rows held fixed):
   (a) multiplying one row by t multiplies the determinant by t;
   (b) if one row is a sum $\boldsymbol{r} + \boldsymbol{r}'$, the determinant is the sum of the two determinants with that row replaced by $\boldsymbol{r}$ and by $\boldsymbol{r}'$.

For 2×2 matrices, with vertical bars denoting determinants:

$$\text{(3a)}\quad \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t\begin{vmatrix} a & b \\ c & d \end{vmatrix}, \qquad \text{(3b)}\quad \begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}.$$

Properties 1 and 2 already give every permutation matrix: $\det P = +1$ if P comes from I by an even number of row exchanges and -1 if by an odd number. For instance $\left|\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right| = 1$ and $\left|\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right| = -1$. Hidden here is a fact that we take on trust until Section 21: a permutation can never be reached both

by an even and by an odd number of exchanges. Without it, property 2 could force $\det P = -\det P$, and the three properties would contradict each other.

Common trap. Property 3 is linearity in *one row at a time*. It does *not* say $\det(A+B) = \det A + \det B$, which is false: for $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 2 & 0 \\ 1 & 1 \end{smallmatrix}\right]$, $\det A = -2$ and $\det B = 2$, but $\det(A+B) = \left|\begin{smallmatrix} 3 & 2 \\ 4 & 5 \end{smallmatrix}\right| = 7$. Nor does it say $\det(2A) = 2\det A$: the factor 2 comes out of *every* row, so $\det(2A) = 2^n \det A$ for an $n \times n$ matrix, just as doubling the sides of a box in $\mathbb{R}^3$ multiplies its volume by $2^3 = 8$.

20.2 Seven consequences

Each of the following is derived from properties 1–3 alone. The proofs are short, and they are the reason the properties are worth learning: they explain why elimination computes determinants.

Proposition 20.2 (Property 4: equal rows)**.** *If two rows of A are equal, then* $\det A = 0$.

Proof. Exchanging the two equal rows leaves A unchanged, so the determinant is unchanged; by property 2 it also changes sign. So $\det A = -\det A$, and $\det A = 0$. □

Proposition 20.3 (Property 5: elimination does not change the determinant)**.** *Subtracting ℓ times row i from row k $(i \neq k)$ leaves* $\det A$ *unchanged.*

Proof. By linearity in row k (3b, then 3a), the new determinant is $\det A$ minus ℓ times the determinant of the matrix whose row k is a copy of row i. That matrix has two equal rows, so its determinant is 0 by property 4. In the 2×2 case:

$$\begin{vmatrix} a & b \\ c - \ell a & d - \ell b \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \ell \begin{vmatrix} a & b \\ a & b \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}. \qquad \square$$

Proposition 20.4 (Property 6: zero row)**.** *If A has a row of zeros, then* $\det A = 0$.

Proof. Multiply the zero row by $t = 0$: the matrix does not change, and by 3a the determinant is multiplied by 0. □

Proposition 20.5 (Property 7: triangular matrices)**.** *If U is triangular with diagonal entries $d_1, \ldots, d_n$, then* $\det U = d_1 d_2 \cdots d_n$.

Proof. Suppose U is upper triangular (the lower case is the same). If all $d_i \neq 0$, use each diagonal entry, from the bottom up, to eliminate the entries above it; by property 5 this does not change the determinant and it leaves the diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_n)$. Factor d_i out of row i for each i (property 3a) to get $d_1 \cdots d_n \det I$, and $\det I = 1$. If some $d_i = 0$, elimination produces a full row of zeros instead (take the lowest zero on the diagonal: the rows below it can clear everything in its row), and property 6 gives $\det U = 0 = d_1 \cdots d_n$. □

Theorem 20.6 (Property 8: the invertibility test)**.** $\det A = 0$ *exactly when A is singular. If elimination with k row exchanges reduces A to U with pivots $d_1, \ldots, d_n$, then*

$$\det A = (-1)^k \, d_1 d_2 \cdots d_n.$$

Proof. Elimination uses only subtractions of multiples of rows (no change, property 5) and row exchanges (a sign change each, property 2), so $\det A = (-1)^k \det U$. If A is singular, elimination produces a zero row and $\det U = 0$ (property 6). If A is invertible, there are n nonzero pivots and $\det U = d_1 \cdots d_n \neq 0$ (property 7). □

This "product of the pivots" is how software computes determinants: an $n \times n$ elimination costs about $n^3/3$ operations, while the formulas of Section 21 have $n!$ terms. The proof also shows that properties 1–3 leave no freedom: any function satisfying them must equal $(-1)^k d_1 \cdots d_n$ on every matrix. We use this uniqueness below.

Example 20.7 ($ad - bc$ from the pivots)**.** If $a \neq 0$, subtract c/a times row 1 from row 2:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ 0 & d - \frac{c}{a}b \end{vmatrix} = a\Big(d - \frac{cb}{a}\Big) = ad - bc.$$

If $a = 0$, exchange the rows first: $\left|\begin{smallmatrix} 0 & b \\ c & d \end{smallmatrix}\right| = -\left|\begin{smallmatrix} c & d \\ 0 & b \end{smallmatrix}\right| = -cb$, again $ad - bc$. (If also $c = 0$, the first column is zero, A is singular, and both sides are 0.) For instance $\left|\begin{smallmatrix} 0 & 2 \\ 3 & 4 \end{smallmatrix}\right| = -(3 \cdot 2) = -6$.

Example 20.8 (A 3×3 determinant by elimination)**.**

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix} = U.$$

The steps subtract 2 (row 1) from row 2, add row 1 to row 3, then add row 2 to row 3; none changes the determinant and there were no exchanges. So $\det A = 2 \cdot (-8) \cdot 1 = -16$.

Figure 24 shows what property 5 means geometrically for 2×2 matrices, where $|\det A|$ is the area of the parallelogram spanned by the rows (proved in Section 22).

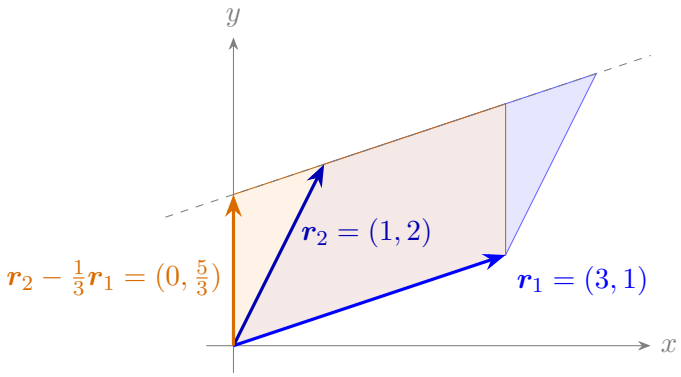


Figure 24: Subtracting a multiple of row 1 from row 2 slides $\boldsymbol{r}_2$ along the dashed line parallel to $\boldsymbol{r}_1$. Base and height are unchanged, so the area stays $\left|\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right| = 6 - 1 = 5$; the new matrix $\left[\begin{smallmatrix} 3 & 1 \\ 0 & 5/3 \end{smallmatrix}\right]$ is triangular, and its pivots give $3 \cdot \frac{5}{3} = 5$.

Theorem 20.9 (Property 9: the product rule)**.** *For $n \times n$ matrices,* $\det(AB) = (\det A)(\det B)$. *In particular* $\det A^{-1} = 1/\det A$ *when A is invertible, and* $\det A^2 = (\det A)^2$.

Proof. If $\det B = 0$, then B is singular: some $\boldsymbol{x} \neq \mathbf{0}$ has $B\boldsymbol{x} = \mathbf{0}$, so $AB\boldsymbol{x} = \mathbf{0}$, AB is singular, and both sides are 0. If $\det B \neq 0$, consider the function $D(A) = \det(AB)/\det B$ of the matrix A. It satisfies the three defining properties: $D(I) = 1$; exchanging two rows of A exchanges the same two rows of AB, so D changes sign; and row i of AB is (row i of A) times B, which depends linearly on row i of A, so D is linear in each row. By uniqueness (the remark after Theorem 20.6), $D(A) = \det A$, which is the product rule. Applying it to $A^{-1}A = I$ gives $\det A^{-1} \det A = 1$. □

For diagonal matrices the rule is plain: $A = \operatorname{diag}(2, 3)$ has $\det A = 6$ and $A^{-1} = \operatorname{diag}(\frac{1}{2}, \frac{1}{3})$ has determinant $\frac{1}{6}$.

Theorem 20.10 (Property 10: transpose)**.** $\det A^{\mathsf{T}} = \det A$.

Proof. If A is singular, so is A^{T} (same rank), and both determinants are 0. Otherwise elimination gives $PA = LU$ with P a permutation, L lower triangular with ones on the diagonal and U upper triangular. Since $P^{\mathsf{T}}P = I$, $A = P^{\mathsf{T}}LU$ and $A^{\mathsf{T}} = U^{\mathsf{T}}L^{\mathsf{T}}P$. By the product rule and property 7,

$$\det A = \det P^{\mathsf{T}} \cdot 1 \cdot \det U, \qquad \det A^{\mathsf{T}} = \det U^{\mathsf{T}} \cdot 1 \cdot \det P,$$

with $\det U^{\mathsf{T}} = \det U$ (same diagonal). Finally $\det P^{\mathsf{T}} \det P = \det(P^{\mathsf{T}}P) = 1$ and $\det P = \pm 1$, so $\det P^{\mathsf{T}} = \det P$. □

Property 10 turns every statement about rows into one about columns: a zero column gives $\det A = 0$, exchanging two columns reverses the sign, and the determinant is linear in each column separately.

Example 20.11 (Four quick determinants)**.** Each of these needs only one or two of the properties.

(a) Subtracting row 1 from row 2 and row 2 from row 3 (property 5) produces two rows equal to $(1, 1, 1)$, so by property 4

$$\begin{vmatrix} 101 & 201 & 301 \\ 102 & 202 & 302 \\ 103 & 203 & 303 \end{vmatrix} = \begin{vmatrix} 101 & 201 & 301 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

(b) The rank-one matrix $C = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -4 & 5 \end{bmatrix}$ has rows that are $1, 2, 3$ times the same row, so C is singular and $\det C = 0$ (property 8).

(c) The skew-symmetric matrix $D = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}$ has $D^{\mathsf{T}} = -D$. Then $\det D = \det D^{\mathsf{T}} = \det(-D) = (-1)^3 \det D$, so $\det D = 0$.

(d) $\det \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 1$: for even n the argument in (c) gives only $\det D = \det D$, and skew-symmetric matrices can be invertible.

20.3 Exercises

Exercise 20.12. If the entries in every row of a square matrix A add to zero, solve $A\boldsymbol{x} = \mathbf{0}$ to prove $\det A = 0$. If those entries add to one, show that $\det(A - I) = 0$. Does this mean $\det A = 1$?

Solution. If every row sums to zero, then $\boldsymbol{x} = (1, 1, \ldots, 1)$ satisfies $A\boldsymbol{x} = \mathbf{0}$, because each component of $A\boldsymbol{x}$ is a row sum. A nonzero nullspace means A is singular, so $\det A = 0$. If every row sums to one, every row of $A - I$ sums to zero, so $\det(A - I) = 0$. This does not force $\det A = 1$: the rows of $A = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ sum to one, but $\det A = -1$. □

Exercise 20.13 (Vandermonde determinant)**.** Use row operations and the properties to show

$$\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} = (b-a)(c-a)(c-b).$$

Solution. Subtract row 1 from rows 2 and 3 (no change), then factor $b - a$ out of row 2 and $c - a$ out of row 3 (property 3a), using $b^2 - a^2 = (b - a)(b + a)$:

$$\begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c-b \end{vmatrix},$$

where the last step subtracts row 2 from row 3. The matrix is now triangular with diagonal $1, 1, c-b$, so the determinant is $(b - a)(c - a)(c - b)$. It is zero exactly when two of a, b, c coincide, i.e. when two rows are equal. □

Exercise 20.14. A 3×3 matrix A has $\det A = 5$. Find $\det(2A)$, $\det(-A)$, $\det(A^2)$, $\det(A^{-1})$ and $\det(A^{\mathsf{T}})$.

Solution. $\det(2A) = 2^3 \cdot 5 = 40$; $\det(-A) = (-1)^3 \cdot 5 = -5$; $\det(A^2) = 5^2 = 25$; $\det(A^{-1}) = \frac{1}{5}$; $\det(A^{\mathsf{T}}) = 5$. □

Exercise 20.15. Show that every skew-symmetric matrix ($D^{\mathsf{T}} = -D$) of odd size n has $\det D = 0$.

Solution. $\det D = \det D^{\mathsf{T}} = \det(-D) = (-1)^n \det D = -\det D$ for odd n, using property 10 and property 3a applied to each of the n rows. Hence $\det D = 0$. Example 20.11(d) shows that the conclusion fails for even n. □

21 Determinant formulas and cofactors

Section 20 defined the determinant by three properties and computed it by elimination: $\det A$ is $\pm$ the product of the pivots. This section derives two explicit formulas from the same properties. The *big formula* writes $\det A$ as a sum of $n!$ signed products, one entry from each row and each column. The *cofactor formula* groups those products to express an $n \times n$ determinant through determinants of size $n-1$. We finish with a family of tridiagonal matrices whose determinants repeat with period six.

21.1 Splitting the rows: from 2×2 to the big formula

Linearity in each row (property 3) lets us split a row into pieces. For a 2×2 matrix split row 1 as $(a, b) = (a, 0) + (0, b)$, then split row 2 in each piece:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}.$$

The first and last pieces have a zero column, so they are singular and vanish. The second is diagonal, ad. The third is bc times the determinant of the row exchange $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, which is -1. So $\det A = ad - bc$, now derived rather than remembered.

For a 3×3 matrix the same splitting produces $3 \cdot 3 \cdot 3 = 27$ pieces, each with one entry from each row. A piece survives only if its three entries also lie in three *different columns*; otherwise it has a zero column. The survivors are the six patterns of Figure 25, and each equals the product of its entries times the determinant of a permutation matrix:

$$\det A = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}.$$

For example the term $a_{12}a_{23}a_{31}$ uses columns $(2, 3, 1)$; two exchanges (rows 1 and 3, then rows 2 and 3) restore the order $(1, 2, 3)$, so its sign is $+$.

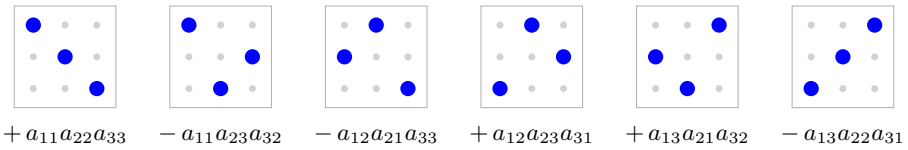


Figure 25: The six nonzero pieces of a 3×3 determinant: one entry from each row and each column, i.e. the positions of the 1's in a permutation matrix. The sign is the determinant of that permutation matrix: $+$ for an even number of row exchanges, $-$ for an odd number.

Definition 21.1 (Sign of a permutation)**.** A permutation σ of $\{1, \ldots, n\}$ corresponds to the permutation matrix P_σ whose row i has its 1 in column $\sigma(i)$. Its *sign* is $\operatorname{sgn}\sigma = \det P_\sigma$, which is $+1$ if P_σ is reached from I by an even number of row exchanges and -1 if by an odd number.

Theorem 21.2 (The big formula)**.** *For an $n \times n$ matrix A,*

$$\det A = \sum_\sigma \operatorname{sgn}(\sigma)\, a_{1\sigma(1)}a_{2\sigma(2)} \cdots a_{n\sigma(n)},$$

the sum running over all $n!$ permutations $\sigma = (\alpha, \beta, \ldots, \omega)$ of the column numbers.

Proof. Write row i of A as $\sum_j a_{ij}\boldsymbol{e}_j^{\mathsf{T}}$, where $\boldsymbol{e}_j^{\mathsf{T}}$ is row j of I. Using linearity in row 1, then in row 2, and so on, $\det A$ becomes a sum over all choices of column numbers $(j_1, \ldots, j_n)$:

$$\det A = \sum_{j_1,\ldots,j_n} a_{1j_1}a_{2j_2}\cdots a_{nj_n} \det \begin{bmatrix} \boldsymbol{e}_{j_1}^{\mathsf{T}} \\ \vdots \\ \boldsymbol{e}_{j_n}^{\mathsf{T}} \end{bmatrix}.$$

If two of the j's coincide, the last matrix has two equal rows and its determinant is 0. Otherwise $(j_1, \ldots, j_n)$ is a permutation σ and the matrix is P_σ, whose determinant is $\operatorname{sgn}\sigma$. □

Why $n!$ terms: the entry from row 1 can be chosen in n columns, the entry from row 2 in the $n-1$ columns still unused, and so on: 2 terms for $n = 2$, 6 for $n = 3$, 24 for $n = 4$. Half have each sign (for $n \geq 2$). The formula reproduces the properties; for instance $\det I = 1$ because only the identity permutation picks no zero entry of I. Conversely, the formula *defines* a function with properties 1–3, which settles the loose end of Section 20: the determinant exists, so a permutation cannot be both even and odd.

Common trap. The "down-right diagonals are +, down-left diagonals are −" picture of the 3×3 formula does not extend to 4×4. The 4×4 anti-diagonal permutation $(4,3,2,1)$ needs two exchanges (rows 1 and 4, rows 2 and 3), so its term carries a + sign. Always count exchanges, or use cofactors.

Example 21.3 (Most terms vanish)**.** For

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

look for ways to pick a nonzero entry in every row and column. Row 1 can use column 3 or column 4, and each choice forces the rest. If row 1 uses column 4, row 4 must use column 1, then row 3 column 2 and row 2 column 3: the permutation $(4,3,2,1)$, sign +, contributing $+1$. If row 1 uses column 3, row 2 must use column 2, row 3 column 1 and row 4 column 4: the permutation $(3,2,1,4)$, which one exchange (rows 1 and 3 of its permutation matrix) puts in order, so its sign is − and it contributes -1. Row 1 has no other nonzero entries, so these are the only two of the 24 terms, and $\det A = 1 - 1 = 0$. The matrix must be singular, and indeed row 1 − row 2 + row 3 − row 4 $= \mathbf{0}$.

21.2 Cofactors

Collect the terms of the big formula that contain a particular entry a_{1j}. What multiplies a_{1j} is built from the other rows and the other columns, one entry from each: it is (up to sign) a determinant of size $n-1$.

Definition 21.4 (Minor and cofactor)**.** The *minor* M_{ij} is the $(n-1)\times(n-1)$ matrix obtained from A by deleting row i and column j. The *cofactor* of a_{ij} is

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

The signs $(-1)^{i+j}$ form a checkerboard starting with + in the top-left corner.

Theorem 21.5 (Cofactor expansion)**.** *For each row i,*

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in},$$

and likewise down each column j: $\det A = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj}$.

Proof. Row 1. By linearity in row 1, $\det A = \sum_j a_{1j} \det A_j$, where A_j is A with row 1 replaced by $\boldsymbol{e}_j^{\mathsf{T}}$. In A_j subtract multiples of row 1 from the other rows to clear column j below the 1 (property 5, no change). Then move column j to the front by $j-1$ exchanges of neighbouring columns; each reverses the sign (exchanging columns is allowed because $\det A^{\mathsf{T}} = \det A$), and the other columns keep their order. The result is the block matrix

$$\begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & M_{1j} \end{bmatrix}.$$

Its determinant is $\det M_{1j}$, because as a function of the $(n-1)\times(n-1)$ matrix M, the determinant of $\left[\begin{smallmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & M \end{smallmatrix}\right]$ satisfies the three defining properties, so it equals $\det M$ by uniqueness (Section 20). Hence $\det A_j = (-1)^{j-1} \det M_{1j} = C_{1j}$.

Row i. Move row i to the top by $i-1$ exchanges of neighbouring rows, keeping the other rows in order. The determinant gains the factor $(-1)^{i-1}$, the new first row is row i of A, and its minors are the M_{ij}. So $\det A = (-1)^{i-1}\sum_j a_{ij}(-1)^{1+j}\det M_{ij} = \sum_j a_{ij}C_{ij}$.

Columns. Apply the row expansion to A^{T}. □

For a 2×2 matrix the cofactors along row 1 are $C_{11} = d$ and $C_{12} = -c$, giving $ad + b(-c)$. For 3×3 the expansion along row 1 is

$$\det A = a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(-a_{21}a_{33} + a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}),$$

which is the six-term formula regrouped; the middle parenthesis carries the minus sign of the $(1,2)$ position.

Example 21.6 (Expanding along a convenient row)**.** Take the matrix of Section 20 whose pivots $2, -8, 1$ gave $\det A = -16$:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

Along row 1:

$$C_{11} = +\begin{vmatrix} -6 & 0 \\ 7 & 2 \end{vmatrix} = -12, \quad C_{12} = -\begin{vmatrix} 4 & 0 \\ -2 & 2 \end{vmatrix} = -8, \quad C_{13} = +\begin{vmatrix} 4 & -6 \\ -2 & 7 \end{vmatrix} = 16,$$

so $\det A = 2(-12) + 1(-8) + 1(16) = -16$. Row 2 contains a zero, which saves one minor: with $C_{21} = -\left|\begin{smallmatrix} 1 & 1 \\ 7 & 2 \end{smallmatrix}\right| = 5$ and $C_{22} = +\left|\begin{smallmatrix} 2 & 1 \\ -2 & 2 \end{smallmatrix}\right| = 6$, $\det A = 4\cdot 5 + (-6)\cdot 6 + 0\cdot C_{23} = -16$ again.

There are now three ways to compute a determinant, and each has its use.

Method	Work for $n\times n$	Best when
Pivots (elimination)	about $n^3/3$ operations	any numerical matrix
Big formula	$n!$ products of n entries	very few nonzero entries
Cofactors	reduces size n to size $n-1$	a row or column with many zeros; patterns

The pivots have all the complexity built in efficiently; the big formula spreads it out completely; cofactors sit in between and are the tool for proving recursions.

21.3 Tridiagonal matrices of ones

Let A_n be the $n\times n$ matrix with 1's on the main diagonal and the two diagonals next to it, and 0 elsewhere:

$$A_1 = \begin{bmatrix} 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

Directly, $|A_1| = 1$ and $|A_2| = 0$. For A_3 the big formula has only three nonzero terms: $a_{11}a_{22}a_{33} = 1$, $-a_{11}a_{23}a_{32} = -1$ and $-a_{12}a_{21}a_{33} = -1$, so $|A_3| = -1$.

Theorem 21.7 (A three-term recursion)**.** *For $n \geq 3$, $|A_n| = |A_{n-1}| - |A_{n-2}|$. Consequently the sequence $|A_n|$ is*

$$1,\ 0,\ -1,\ -1,\ 0,\ 1,\ \ 1,\ 0,\ -1,\ -1,\ 0,\ 1,\ \ldots$$

and repeats with period 6*.*

Proof. Expand along row 1, whose only nonzero entries are $a_{11} = a_{12} = 1$. The cofactor C_{11} is the determinant of the lower-right block, which is A_{n-1}. The cofactor C_{12} is $-\det M_{12}$, where M_{12} (rows $2, \ldots, n$ and columns $1, 3, \ldots, n$) has first column $(1, 0, \ldots, 0)$. Expanding M_{12} down that column leaves its lower-right block, which is A_{n-2}. Hence $|A_n| = |A_{n-1}| - |A_{n-2}|$. Applying the recursion twice, $|A_{n+3}| = |A_{n+2}| - |A_{n+1}| = (|A_{n+1}| - |A_n|) - |A_{n+1}| = -|A_n|$, so $|A_{n+6}| = |A_n|$. Starting from $1, 0$ the recursion gives $-1, -1, 0, 1$ and then repeats. □

So $|A_4| = |A_3| - |A_2| = -1$, and for instance $|A_{61}| = |A_1| = 1$, a value no one would guess by looking at a 61×61 matrix. Figure 26 plots the pattern.

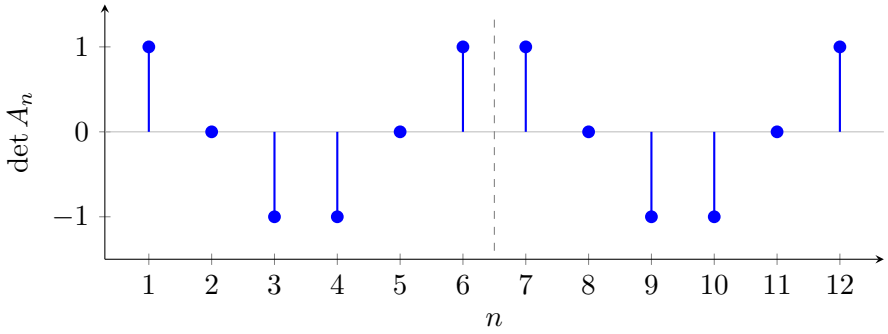


Figure 26: Determinants of the tridiagonal matrices of ones. The recursion $|A_n| = |A_{n-1}| - |A_{n-2}|$ from the cofactor expansion forces period 6 (dashed line: one period).

21.4 Exercises

Exercise 21.8. Compute the determinant of

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Which method do you prefer here, and why?

Solution. The big formula has a single nonzero term, from the permutation $(4, 1, 2, 3)$, with product 1. Moving row 1 to the bottom takes three exchanges (rows 1–2, 2–3, 3–4), which turns A into I; the permutation is odd, so $\det A = -1$. Cofactors along row 1 give the same with no searching: the only nonzero entry is $a_{14} = 1$, whose minor (rows 2–4, columns 1–3) is I_3, so $\det A = (-1)^{1+4} \cdot 1 = -1$. With a single 1 in each row, either method beats elimination. □

Exercise 21.9. The symmetric Pascal matrices have determinant 1; for instance $\det \left[\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{smallmatrix}\right] = 1$. Explain why subtracting 1 from the (n, n) entry (here $20 \to 19$) makes the determinant 0.

Solution. By linearity in the last row, the new determinant is the old one plus the determinant of the matrix whose last row is $(0, \dots, 0, -1)$. Expanding that matrix along its last row gives $(-1) \cdot C_{nn} = -\det(\text{Pascal matrix of size } n-1) = -1$, because $(-1)^{n+n} = +1$. So the determinant drops from 1 to $1 - 1 = 0$. In the example the cofactor is $\det \left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{smallmatrix}\right] = 1$. □

Exercise 21.10. Find the determinant of the 5×5 matrix with x on the diagonal, y just to the right of the diagonal in rows 1–4, y in position $(5, 1)$, and zeros elsewhere:

$$A = \begin{bmatrix} x & y & 0 & 0 & 0 \\ 0 & x & y & 0 & 0 \\ 0 & 0 & x & y & 0 \\ 0 & 0 & 0 & x & y \\ y & 0 & 0 & 0 & x \end{bmatrix}.$$

Solution. Expand down column 1, which has only $a_{11} = x$ and $a_{51} = y$. Deleting row 1 and column 1 leaves an upper triangular matrix with x's on its diagonal, determinant x^4. Deleting row 5 and column 1 leaves a lower triangular matrix with y's on its diagonal, determinant y^4, and the sign is $(-1)^{5+1} = +1$. Hence $\det A = x \cdot x^4 + y \cdot y^4 = x^5 + y^5$. □

Exercise 21.11. Find the determinant of the 5×5 matrix B with x on the diagonal and y in every other position.

Solution. Add rows 2–5 to row 1 (property 5, no change): every entry of row 1 becomes $x + 4y$. Factor it out (property 3a), leaving a row of ones. Now subtract y times this row of ones from each of rows 2–5. Row k had y in every position except x in position k, so it becomes 0 everywhere except $x - y$ in position k. The result is upper triangular with diagonal 1, $x-y$, $x-y$, $x-y$, $x-y$, so

$$\det B = (x + 4y)(x - y)^4.$$

Check: for $y = 0$ this is $x^5 = \det(xI)$, and for $x = y$ all rows are equal and the determinant is 0. □

22 Cramer's rule, inverse matrix and volume

The last two sections built the determinant from three properties and found two formulas for it: the big formula with $n!$ terms and the cofactor expansion (Section 21). This section puts the determinant to work. The cofactor expansion turns into an explicit formula for A^{-1}; applied to $\boldsymbol{b}$ that formula becomes Cramer's rule for the solution of $A\boldsymbol{x} = \boldsymbol{b}$; and the absolute value $|\det A|$ turns out to be a volume. Elimination remains the way to compute. What these formulas give instead is *algebra*: the answer written in terms of the entries, which shows how it depends on them.

22.1 The cofactor matrix gives A^{-1}

Start from the 2×2 inverse we already know:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

We divide by the determinant, which is exactly what fails when A is singular. The entries of the matrix on the right are cofactors: d is the cofactor of a (strike out row 1 and column 1), and $-b$ is the cofactor of c (strike out row 2 and column 1, and attach the sign $(-1)^{2+1}$). Notice that the cofactor of c, which sits in position $(2,1)$, appears in position $(1,2)$. The cofactors are transposed.

Definition 22.1 (Cofactor matrix)**.** Let A be $n \times n$ and let M_{ij} be the $(n-1) \times (n-1)$ matrix left when row i and column j of A are removed. The cofactor of a_{ij} is $C_{ij} = (-1)^{i+j} \det M_{ij}$, and the *cofactor matrix* is $C = (C_{ij})$. Its transpose C^{T} is sometimes called the adjugate of A.

Each cofactor is a sum of products of $n-1$ entries, while $\det A$ is a sum of products of n entries. So a formula $A^{-1} = C^{\mathsf{T}}/\det A$ has the right "degree": entries of A^{-1} behave like 1/(entry of A). For a general 3×3 matrix with rows (a,b,c), (d,e,f), (g,h,i), the $(1,1)$ entry of A^{-1} should be $(ei - fh)/\det A$. The theorem says this pattern holds for every entry and every n.

Theorem 22.2 (Inverse from cofactors)**.** *For every square matrix A,*

$$A\,C^{\mathsf{T}} = (\det A)\,I.$$

Consequently, if $\det A \neq 0$ *then* $A^{-1} = \dfrac{1}{\det A}\,C^{\mathsf{T}}$*, that is,* $(A^{-1})_{ij} = C_{ji}/\det A$*.*

Proof. The (i,k) entry of $A\,C^{\mathsf{T}}$ is row i of A times column k of C^{T}, which is row k of C:

$$(A\,C^{\mathsf{T}})_{ik} = \sum_{j=1}^{n} a_{ij}\,C_{kj}.$$

Diagonal entries $(k = i)$. The sum $\sum_j a_{ij}C_{ij}$ is the cofactor expansion of $\det A$ along row i (Section 21). So every diagonal entry equals $\det A$.

Off-diagonal entries $(k \neq i)$. Let A^* be the matrix A with row k replaced by a copy of row i. Then A^* has two equal rows, so $\det A^* = 0$. Expand $\det A^*$ along its row k. The entries of that row are $a_{i1}, \ldots, a_{in}$, and the cofactors of row k of A^* are the same as those of A, because they are computed after deleting row k, the only row where A and A^* differ. Hence

$$0 = \det A^* = \sum_{j} a_{ij}\,C_{kj} = (A\,C^{\mathsf{T}})_{ik}.$$

So $A\,C^{\mathsf{T}}$ has $\det A$ on its diagonal and zeros elsewhere. If $\det A \neq 0$, dividing by $\det A$ gives $A\,(C^{\mathsf{T}}/\det A) = I$, and a one-sided inverse of a square matrix is its inverse. □

In the 2×2 case the off-diagonal computation is visible: row 1 of A is (a,b), the cofactors of row 2 are $(-b,a)$, and $a(-b) + b(a) = 0$ is the determinant of the matrix with two equal rows $\left[\begin{smallmatrix} a & b \\ a & b \end{smallmatrix}\right]$.

Example 22.3 (A 3×3 inverse by cofactors)**.** Let

$$A = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix}.$$

The nine cofactors, each a signed 2×2 determinant, are

$$\begin{aligned}
&C_{11} = +(2 \cdot 5 - 2 \cdot 2) = 6, && C_{12} = -(1 \cdot 5 - 2 \cdot 1) = -3, && C_{13} = +(1 \cdot 2 - 2 \cdot 1) = 0,\\
&C_{21} = -(1 \cdot 5 - 4 \cdot 2) = 3, && C_{22} = +(1 \cdot 5 - 4 \cdot 1) = 1, && C_{23} = -(1 \cdot 2 - 1 \cdot 1) = -1,\\
&C_{31} = +(1 \cdot 2 - 4 \cdot 2) = -6, && C_{32} = -(1 \cdot 2 - 4 \cdot 1) = 2, && C_{33} = +(1 \cdot 2 - 1 \cdot 1) = 1.
\end{aligned}$$

Multiplying A by C^{T},

$$A\,C^{\mathsf{T}} = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 6 & 3 & -6 \\ -3 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3I,$$

so $\det A = 3$ and

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 6 & 3 & -6 \\ -3 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & -2 \\ -1 & \frac{1}{3} & \frac{2}{3} \\ 0 & -\frac{1}{3} & \frac{1}{3} \end{bmatrix}.$$

One more question the formula answers instantly: if the entry $a_{13} = 4$ is changed to 100, the determinant does not change, because $\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ and $C_{13} = 0$. (The inverse does change: the cofactors in rows 2 and 3 involve a_{13}.)

22.2 Cramer's rule

If A is invertible, the solution of $A\boldsymbol{x} = \boldsymbol{b}$ is $\boldsymbol{x} = A^{-1}\boldsymbol{b} = C^{\mathsf{T}}\boldsymbol{b}/\det A$. Cramer noticed that each component of $C^{\mathsf{T}}\boldsymbol{b}$ is itself a determinant.

Theorem 22.4 (Cramer's rule)**.** *Let A be $n \times n$ with $\det A \neq 0$, and let B_j be the matrix A with its column j replaced by $\boldsymbol{b}$. Then the solution of $A\boldsymbol{x} = \boldsymbol{b}$ is*

$$x_j = \frac{\det B_j}{\det A}, \qquad j = 1, \ldots, n.$$

Proof. By Theorem 22.2, $x_j = (C^{\mathsf{T}}\boldsymbol{b})_j/\det A$ and $(C^{\mathsf{T}}\boldsymbol{b})_j = \sum_i C_{ij}\, b_i$. Now expand $\det B_j$ down its column j, which holds $b_1, \ldots, b_n$. The cofactor of position (i, j) in B_j is computed after deleting column j, so it ignores $\boldsymbol{b}$ and equals the cofactor C_{ij} of A. Therefore $\det B_j = \sum_i b_i\, C_{ij} = (C^{\mathsf{T}}\boldsymbol{b})_j$. □

Example 22.5 (Cramer's rule on a 3×3 system)**.** Solve $A\boldsymbol{x} = \boldsymbol{b}$ with A from Example 22.3 and $\boldsymbol{b} = (5, 2, 5)$. We know $\det A = 3$. Replacing one column at a time,

$$\det B_1 = \begin{vmatrix} 5 & 1 & 4 \\ 2 & 2 & 2 \\ 5 & 2 & 5 \end{vmatrix} = 6, \qquad \det B_2 = \begin{vmatrix} 1 & 5 & 4 \\ 1 & 2 & 2 \\ 1 & 5 & 5 \end{vmatrix} = -3, \qquad \det B_3 = \begin{vmatrix} 1 & 1 & 5 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{vmatrix} = 3.$$

For instance, expanding $\det B_1$ along row 1 gives $5(10-4) - 1(10-10) + 4(4-10) = 30 - 0 - 24 = 6$. Hence $\boldsymbol{x} = (6/3,\ -3/3,\ 3/3) = (2, -1, 1)$. Check: $A(2, -1, 1) = (2 - 1 + 4,\ 2 - 2 + 2,\ 2 - 2 + 5) = (5, 2, 5)$. The same answer comes from $\boldsymbol{x} = C^{\mathsf{T}}\boldsymbol{b}/3$: $C^{\mathsf{T}}\boldsymbol{b} = (30 + 6 - 30,\ -15 + 2 + 10,\ 0 - 2 + 5) = (6, -3, 3)$.

Remark 22.6 (Formula versus algorithm). Cramer's rule needs $n + 1$ determinants of size n. Computed by elimination each costs about $n^3/3$ operations, so the rule costs about $n^4/3$, against $n^3/3$ for solving the system once by elimination; computed by the big formula it is hopeless. Software never uses it. Its value is theoretical: it shows that each x_j is a ratio of polynomials in the entries of A and $\boldsymbol{b}$, so the solution depends smoothly on the data wherever $\det A \neq 0$, and it shows exactly how $\det A$ appears in every denominator.

Common trap. The inverse is $C^{\mathsf{T}}/\det A$, not $C/\det A$. The cofactors of *row* i of A fill *column* i of A^{-1}. A quick test: for $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ the $(1, 2)$ entry of the inverse must be $-b/(ad - bc)$, which is the cofactor of c, not of b. Also remember the checkerboard signs $(-1)^{i+j}$; forgetting them is the other common error.

22.3 The determinant is a volume

Take the rows of an $n \times n$ matrix A as n edges leaving the origin and complete them to a box: the parallelogram ($n = 2$), the parallelepiped ($n = 3$), or its analogue in $\mathbb{R}^n$, consisting of all points $t_1\boldsymbol{r}_1 + \cdots + t_n\boldsymbol{r}_n$ with $0 \le t_i \le 1$. The claim of this subsection is that the volume of that box is $|\det A|$.

We use only these geometric facts about volume. (V1) A rectangular box whose edges point along the coordinate axes, with lengths $\ell_1, \ldots, \ell_n$, has volume $\ell_1 \cdots \ell_n$. (V2) Listing the edges in another order gives the same box. (V3) Volume is base times height: the box sits on the base spanned by $n-1$ of its edges, and its height is the distance of the last edge's tip from that base. (V4) If the edges are dependent, the box lies in a hyperplane and has volume 0.

Theorem 22.7 (Determinant and volume)**.** *The box whose edges are the rows of A has volume $|\det A|$. The same holds for the box built on the columns, since $\det A^{\mathsf{T}} = \det A$.*

Proof. If A is singular its rows are dependent, so the volume is $0 = |\det A|$ by (V4). Suppose A is invertible. Elimination with row exchanges, followed by elimination upward, reduces A to a diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_n)$ with nonzero entries, using two kinds of steps (no row is ever rescaled).

Subtract c times row j from row i. The determinant is unchanged (property 5 in Section 20). The box is sheared but its volume is unchanged: take as base the face spanned by the other $n-1$ rows, which do not move. Row i changes by a multiple of row j, which lies in that base, so the tip of row i moves parallel to the base and the height stays the same. By (V3) the volume is unchanged.

Exchange two rows. The determinant changes sign, so $|\det A|$ is unchanged, and by (V2) the box is the same.

So both $|\det A|$ and the volume are unchanged at every step, and at the end both equal $|d_1 d_2 \cdots d_n|$: the determinant of the diagonal matrix D is the product of its diagonal entries, and the box of D is a rectangular box with edge lengths $|d_1|, \ldots, |d_n|$ along the axes (V1). □

The lecture argues differently but equivalently: the volume of the box satisfies the three defining properties of $|\det|$. The unit cube ($A = I$) has volume 1; exchanging two edges does not change the box; stretching one edge by t multiplies the volume by $|t|$; and splitting one edge as a sum splits the volume (when the two pieces lie on the same side of the base). The shear step in the proof is the geometric heart of the linearity property. The sign of $\det A$ also means something: it records whether the edges form a right-handed or a left-handed system, and exchanging two edges reverses the handedness.

A good test case is an orthogonal matrix Q. Its rows are perpendicular unit vectors, so its box is a unit cube turned in space, with volume 1. The determinant agrees: from $Q^{\mathsf{T}}Q = I$ and the product rule, $(\det Q^{\mathsf{T}})(\det Q) = (\det Q)^2 = 1$, so $\det Q = \pm 1$.

22.4 Areas and volumes from coordinates

Theorem 22.7 is a formula that needs no lengths, angles or heights, only coordinates. In the plane, the parallelogram with edges (a, b) and (c, d) has area $|ad - bc|$, and the triangle with vertices $(0, 0)$, (a, b), (c, d) has half that area (Figure 27). For a triangle placed anywhere:

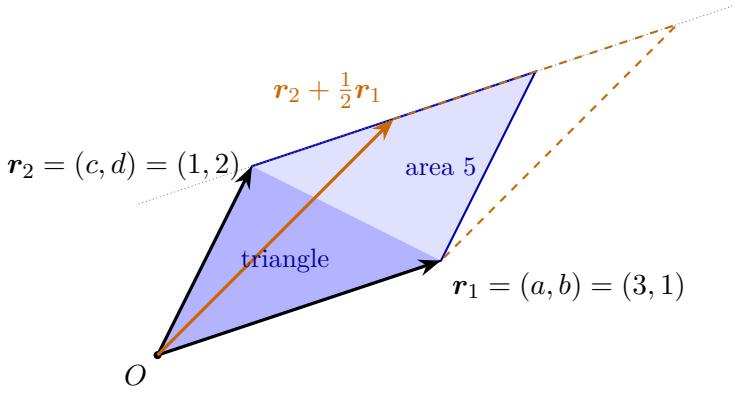


Figure 27: The parallelogram with edges $\boldsymbol{r}_1 = (3, 1)$ and $\boldsymbol{r}_2 = (1, 2)$ has area $|ad-bc| = |3\cdot 2-1\cdot 1| = 5$, and the shaded triangle is half of it. Adding $\frac{1}{2}\boldsymbol{r}_1$ to $\boldsymbol{r}_2$ (a row operation) moves the tip along the dotted line parallel to $\boldsymbol{r}_1$: the dashed parallelogram has the same base and height, hence the same area, just as the determinant is unchanged.

Proposition 22.8 (Area of a triangle)**.** *The triangle with vertices* (x_1, y_1), (x_2, y_2), (x_3, y_3) *has area*

$$\frac{1}{2}\left|\det\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}\right|.$$

Proof. Subtract row 1 from rows 2 and 3; the determinant does not change. The last column becomes $(1, 0, 0)$, and expanding along it leaves

$$\det\begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix}.$$

The row operations translated the triangle so that its first vertex is at the origin; its other vertices are now the two edge vectors, and half the parallelogram area is the triangle area. □

For example, the triangle with vertices $(1, 1)$, $(4, 2)$, $(2, 5)$ has edges $(3, 1)$ and $(1, 4)$ from the first vertex, so its area is $\frac{1}{2}|3 \cdot 4 - 1 \cdot 1| = \frac{11}{2}$. A zero determinant means the three points are collinear.

Example 22.9 (Volume of a tetrahedron)**.** A tetrahedron T has one vertex at the origin O and the others at $A_1 = (2, 2, -1)$, $A_2 = (1, 3, 0)$ and $A_3 = (-1, 1, 4)$. Find its volume. Then keep A_1 and A_2, move the apex to $A_3' = (-201, -199, 104)$, and find the new volume.

Relate T to a box. Let P be the parallelepiped on the edges OA_1, OA_2, OA_3. Take the triangle OA_1A_2 as the base of T and the parallelogram on OA_1, OA_2 as the base of P; the parallelogram has twice the area of the triangle, and both solids have the same height h (the distance from A_3 to that plane). So

$$\text{vol}(T) = \tfrac{1}{3}(\text{area of triangle})\, h = \tfrac{1}{6}(\text{area of parallelogram})\, h = \tfrac{1}{6}\,\text{vol}(P).$$

Compute. By Theorem 22.7,

$$\text{vol}(P) = \left|\det\begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & 0 \\ -1 & 1 & 4 \end{bmatrix}\right| = \big|2(12-0) - 2(4-0) + (-1)(1+3)\big| = |24 - 8 - 4| = 12,$$

so $\mathrm{vol}(T) = 12/6 = 2$.

Move the apex. Notice that $A_3' = A_3 - 100A_1$: indeed $(-1,1,4) - 100(2,2,-1) = (-201,-199,104)$. In the matrix this is the row operation "subtract 100 times row 1 from row 3", which leaves the determinant at 12. So the long, needle-like tetrahedron still has volume 2. Geometrically, the apex moved parallel to the edge OA_1, which lies in the base plane, so the height did not change.

22.5 Exercises

Exercise 22.10 (Spherical coordinates)**.** Spherical coordinates (ρ, ϕ, θ) are defined by

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

Find the 3×3 matrix of partial derivatives of (x,y,z) with respect to (ρ,ϕ,θ) and show that its determinant is $J = \rho^2\sin\phi$. Explain why $\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$ is the volume of a small coordinate box.

Solution. The rows are the partial derivatives of x, y, z:

$$J = \det\begin{bmatrix} \sin\phi\cos\theta & \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \\ \cos\phi & -\rho\sin\phi & 0 \end{bmatrix}.$$

Expand along the bottom row (cofactor signs $+,-,+$; the last entry is 0):

$$\begin{aligned} J &= \cos\phi\left(\rho^2\cos\phi\sin\phi\cos^2\theta + \rho^2\cos\phi\sin\phi\sin^2\theta\right) + \rho\sin\phi\left(\rho\sin^2\phi\cos^2\theta + \rho\sin^2\phi\sin^2\theta\right) \\ &= \rho^2\cos^2\phi\sin\phi + \rho^2\sin^3\phi = \rho^2\sin\phi\,(\cos^2\phi + \sin^2\phi) = \rho^2\sin\phi. \end{aligned}$$

Small changes $d\rho$, $d\phi$, $d\theta$ move the point by the three columns of this matrix times $d\rho$, $d\phi$, $d\theta$. These three small vectors are the edges of the tiny coordinate box, so by Theorem 22.7 its volume is $|J|\,d\rho\,d\phi\,d\theta = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$ (here $0 \le \phi \le \pi$, so $\sin\phi \ge 0$). □

Exercise 22.11 (Integer inverses)**.** Suppose A has integer entries. Show that A^{-1} has integer entries if $\det A = \pm 1$, and that conversely, if both A and A^{-1} have integer entries, then $\det A = \pm 1$. Find the inverse of $\left[\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right]$.

Solution. Every cofactor is a sum of products of integers, so C^{T} has integer entries; if $\det A = \pm 1$ then $A^{-1} = \pm C^{\mathsf{T}}$ is an integer matrix. Conversely, $\det A$ and $\det A^{-1}$ are integers (sums of products of integers) and their product is $\det(AA^{-1}) = \det I = 1$, so both are ± 1. For the example, $\det A = 4 - 3 = 1$ and $A^{-1} = C^{\mathsf{T}} = \left[\begin{smallmatrix} 2 & -3 \\ -1 & 2 \end{smallmatrix}\right]$; indeed $\left[\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right]\left[\begin{smallmatrix} 2 & -3 \\ -1 & 2 \end{smallmatrix}\right] = I$. □

Exercise 22.12 (Determinant of the cofactor matrix)**.** Show that $\det C = (\det A)^{n-1}$ for an invertible $n\times n$ matrix A, and check it on Example 22.3.

Solution. Take determinants in $A\,C^{\mathsf{T}} = (\det A)I$. The right side is a diagonal matrix with n entries equal to $\det A$, so its determinant is $(\det A)^n$. The left side gives $\det A\cdot\det C^{\mathsf{T}} = \det A\cdot\det C$. Dividing by $\det A \neq 0$ gives $\det C = (\det A)^{n-1}$. In Example 22.3, $n = 3$ and $\det A = 3$, so $\det C$ should be 9. Directly, expanding $C = \left[\begin{smallmatrix} 6 & -3 & 0 \\ 3 & 1 & -1 \\ -6 & 2 & 1 \end{smallmatrix}\right]$ along row 1: $6(1+2)+3(3-6)+0 = 18-9 = 9$. □

Exercise 22.13 (Cramer's rule and collinear points)**.** (a) Solve $2x+y=3$, $x+3y=5$ by Cramer's rule. (b) Use Proposition 22.8 to decide whether $(1,2)$, $(3,5)$, $(7,11)$ lie on one line.

Solution. (a) $\det A = 2\cdot 3 - 1\cdot 1 = 5$, $\det B_1 = \left|\begin{smallmatrix} 3 & 1 \\ 5 & 3 \end{smallmatrix}\right| = 9-5 = 4$, $\det B_2 = \left|\begin{smallmatrix} 2 & 3 \\ 1 & 5 \end{smallmatrix}\right| = 10-3 = 7$. So $x = 4/5$, $y = 7/5$. Check: $8/5 + 7/5 = 3$ and $4/5 + 21/5 = 5$.

(b) Subtracting the first row as in the proof of Proposition 22.8 leaves $\det \left[\begin{smallmatrix} 2 & 3 \\ 6 & 9 \end{smallmatrix}\right] = 18 - 18 = 0$. The triangle has area 0, so the points are collinear: the edge $(6, 9)$ is 3 times the edge $(2, 3)$. □

23 Eigenvalues and eigenvectors

Up to now a matrix has mostly been the coefficient matrix of a static problem $A\boldsymbol{x} = \boldsymbol{b}$. From here on we also think of a square matrix as an operation that is applied again and again: $A\boldsymbol{u}$, $A^2\boldsymbol{u}$, $A^3\boldsymbol{u}$, or continuously in time. For that, the important vectors are the few whose direction A does not change. They and their stretching factors, the eigenvectors and eigenvalues, occupy most of the rest of the course. This section says what they are and how to find them; the next sections use them.

23.1 Vectors that keep their direction

A matrix acts like a function: in goes $\boldsymbol{x}$, out comes $A\boldsymbol{x}$. For most vectors the output points in a new direction. The eigenvectors are the exceptions.

Definition 23.1 (Eigenvalue and eigenvector)**.** Let A be $n \times n$. A number λ is an *eigenvalue* of A if there is a *nonzero* vector $\boldsymbol{x}$ with

$$A\boldsymbol{x} = \lambda\boldsymbol{x}.$$

Such an $\boldsymbol{x}$ is an *eigenvector* for λ. The eigenvectors for λ, together with $\mathbf{0}$, form the *eigenspace* $\mathbf{N}(A - \lambda I)$.

"Same direction" allows λ to be negative (the vector reverses), zero, or even complex. The vector $\mathbf{0}$ is excluded because $A\mathbf{0} = \lambda\mathbf{0}$ for every λ and would tell us nothing. Any nonzero multiple of an eigenvector is again an eigenvector, so "the" eigenvector really means a basis for the eigenspace.

Eigenvalue zero. $A\boldsymbol{x} = 0\boldsymbol{x}$ says $\boldsymbol{x}$ is a nonzero vector in the nullspace. So $\lambda = 0$ is an eigenvalue exactly when A is singular, and its eigenvectors are the nonzero vectors of $\mathbf{N}(A)$. Among eigenvalues, zero is no longer special.

Before any formulas, two matrices whose eigenvectors we can see from what they do.

Example 23.2 (A projection matrix)**.** Let P project $\mathbb{R}^3$ onto the plane $x + y + z = 0$. By the formula $P = I - \boldsymbol{n}\boldsymbol{n}^{\mathsf{T}}/\boldsymbol{n}^{\mathsf{T}}\boldsymbol{n}$ with normal $\boldsymbol{n} = (1, 1, 1)$,

$$P = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

A vector already in the plane is not moved: $P\boldsymbol{x} = \boldsymbol{x}$, so every nonzero vector of the plane is an eigenvector with $\lambda = 1$, for example $(1, -1, 0)$ and $(1, 0, -1)$. A vector along the normal is projected to zero: $P(1, 1, 1) = \mathbf{0}$, so $\boldsymbol{n}$ is an eigenvector with $\lambda = 0$. The eigenvalues are $1, 1, 0$, and the three eigenvectors are independent. A typical vector such as $\boldsymbol{b} = (1, 0, 0)$ is *not* an eigenvector: $P\boldsymbol{b} = \frac{1}{3}(2, -1, -1)$ points in another direction.

Example 23.3 (A permutation matrix)**.** $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ swaps the two components: $B(x_1, x_2) = (x_2, x_1)$. The vector $(1, 1)$ is unchanged, so it is an eigenvector with $\lambda = 1$. The vector $(1, -1)$ becomes $(-1, 1)$, its negative, so it is an eigenvector with $\lambda = -1$. A matrix with no negative entries can still have a negative eigenvalue.

23.2 The characteristic equation

The equation $A\boldsymbol{x} = \lambda\boldsymbol{x}$ has two unknowns multiplied together, so elimination cannot be applied to it directly. The trick is to find λ first.

Theorem 23.4 (Characteristic equation)**.** *λ is an eigenvalue of A if and only if*

$$\det(A - \lambda I) = 0.$$

The eigenvectors for λ are the nonzero vectors in $\mathbf{N}(A - \lambda I)$.

Proof. $A\boldsymbol{x} = \lambda\boldsymbol{x}$ is the same as $(A - \lambda I)\boldsymbol{x} = \mathbf{0}$. A nonzero solution exists exactly when $A - \lambda I$ is singular, and a square matrix is singular exactly when its determinant is zero (Section 20). □

By the big formula, $p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n in λ, the *characteristic polynomial*; its leading term comes from the product of the diagonal entries and is $(-\lambda)^n$. Over the complex numbers it factors as $p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda)$, so an $n \times n$ matrix has exactly n eigenvalues if repeated roots are counted with their multiplicity. The recipe is:

1. compute $\det(A - \lambda I)$ and find its n roots $\lambda_1, \ldots, \lambda_n$;
2. for each λ_i, find the nullspace of the singular matrix $A - \lambda_i I$ by elimination (give the free variable the value 1).

Example 23.5 (A symmetric 2×2 matrix)**.** Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Then

$$\det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{vmatrix} = (3 - \lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2),$$

so $\lambda_1 = 4$ and $\lambda_2 = 2$. For $\lambda_1 = 4$: $A - 4I = \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, whose nullspace is the line through $\boldsymbol{x}_1 = (1, 1)$. For $\lambda_2 = 2$: $A - 2I = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, whose nullspace is the line through $\boldsymbol{x}_2 = (-1, 1)$. Check: $A(1, 1) = (4, 4)$ and $A(-1, 1) = (-2, 2)$. Figure 28 shows the two eigenvector lines.

The matrix in Example 23.5 is $B + 3I$, with B the permutation of Example 23.3. Its eigenvalues $4, 2$ are those of B (namely $1, -1$) plus 3, and its eigenvectors are the same lines. This is a general fact.

Proposition 23.6 (Shifting by a multiple of I)**.** *If $A\boldsymbol{x} = \lambda\boldsymbol{x}$, then $(A + cI)\boldsymbol{x} = (\lambda + c)\boldsymbol{x}$. So $A + cI$ has the eigenvalues of A shifted by c, with the same eigenvectors.*

Proof. $(A + cI)\boldsymbol{x} = A\boldsymbol{x} + c\boldsymbol{x} = \lambda\boldsymbol{x} + c\boldsymbol{x} = (\lambda + c)\boldsymbol{x}$. □

Common trap. It is tempting to argue: if $A\boldsymbol{x} = \lambda\boldsymbol{x}$ and $B\boldsymbol{x} = \alpha\boldsymbol{x}$ then $(A + B)\boldsymbol{x} = (\lambda + \alpha)\boldsymbol{x}$, so eigenvalues add. The flaw is that A and B usually have *different* eigenvectors, and then nothing can be concluded. For $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$ all four eigenvalues are 0, yet $A + B$ has eigenvalues 1 and -1, and $AB = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ has eigenvalues 1 and 0. Eigenvalues are neither additive nor multiplicative; the shift $B = cI$ works only because every vector is an eigenvector of cI.

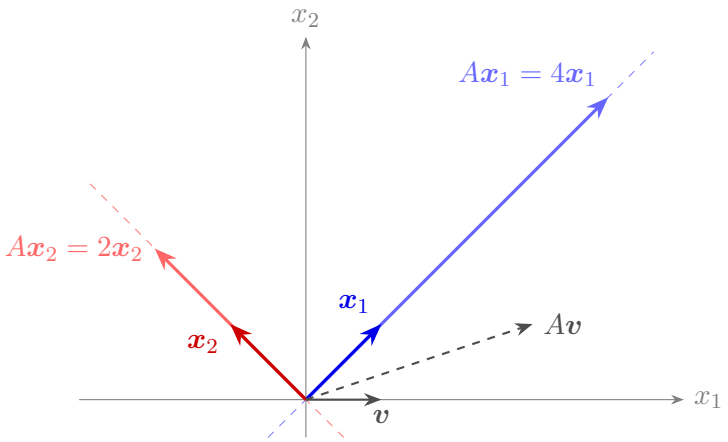


Figure 28: For $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$, vectors on the dashed line through $(1,1)$ are stretched by $\lambda_1 = 4$ and vectors on the line through $(-1,1)$ by $\lambda_2 = 2$, without turning. A typical vector such as $\boldsymbol{v} = (1,0)$ is turned: $A\boldsymbol{v} = (3,1)$ is not a multiple of $\boldsymbol{v}$.

23.3 Trace and determinant

In Example 23.5 the coefficient 6 in $\lambda^2 - 6\lambda + 8$ is $3+3$, the sum of the diagonal entries, and the constant 8 is $\det A$. That is no accident.

Theorem 23.7 (Sum and product of the eigenvalues)**.** *Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A, repeated according to multiplicity. Then*

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = a_{11} + a_{22} + \cdots + a_{nn} = \operatorname{tr} A, \qquad \lambda_1 \lambda_2 \cdots \lambda_n = \det A.$$

Proof. Compare two expressions for the characteristic polynomial,

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda).$$

Product. Set $\lambda = 0$: $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$.

Sum. Look at the coefficient of λ^{n-1}. On the right it is $(-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$. On the left, use the big formula: each term is a product of n entries of $A - \lambda I$, one from each row and each column. A term that uses an off-diagonal entry in position (i,j) cannot also use the diagonal entries of row i or of column j, so it contains at most $n-2$ factors with λ. Hence only the diagonal product $(a_{11} - \lambda) \cdots (a_{nn} - \lambda)$ reaches λ^{n-1}, with coefficient $(-1)^{n-1}(a_{11} + \cdots + a_{nn})$. Equate the two coefficients. □

For a 2×2 matrix this pins the characteristic equation down completely:

$$\lambda^2 - (\operatorname{tr} A)\,\lambda + \det A = 0.$$

Knowing one eigenvalue, the trace gives the other at once. The two facts are also the quickest check on a computation: add the eigenvalues and compare with the diagonal.

Example 23.8 (Two cities: finding eigenvalues by inspection)**.** Each year 80% of the people in city 1 stay and 20% move to city 2, while 70% of city 2 stays and 30% moves to city 1. The populations change by

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

Try $\boldsymbol{x}_1 = (600, 400)$: $A\boldsymbol{x}_1 = (480 + 120,\ 120 + 280) = (600, 400)$. So $\boldsymbol{x}_1$ is an eigenvector with $\lambda_1 = 1$. The trace is $0.8 + 0.7 = 1.5$, so $\lambda_2 = 1.5 - 1 = 0.5$. The determinant confirms it: $0.56 - 0.06 = 0.5 = \lambda_1\lambda_2$. For λ_2, $A - 0.5I = \left[\begin{smallmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{smallmatrix}\right]$ has nullspace spanned by $\boldsymbol{x}_2 = (1, -1)$, and indeed $A(1, -1) = (0.5, -0.5) = 0.5\,\boldsymbol{x}_2$. The two eigenvectors are the key to what happens to the populations year after year: Section 24 follows them.

23.4 Complex eigenvalues

The 90° rotation $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ turns every nonzero vector, so no real vector can be an eigenvector. The trace and determinant agree that something is strange: $\lambda_1 + \lambda_2 = 0$ but $\lambda_1\lambda_2 = 1$, impossible for real numbers. The characteristic equation settles it:

$$\det(Q - \lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0, \qquad \lambda_1 = i, \quad \lambda_2 = -i.$$

For $\lambda_1 = i$, the nullspace of $Q - iI$ contains $\boldsymbol{x}_1 = (1, -i)$; check: $Q(1, -i) = (i, 1) = i(1, -i)$. Similarly $\boldsymbol{x}_2 = (1, i)$ goes with $-i$. A perfectly real matrix can have complex eigenvalues and eigenvectors.

Proposition 23.9 (Conjugate pairs)**.** *If A is real and $A\boldsymbol{x} = \lambda\boldsymbol{x}$, then $A\bar{\boldsymbol{x}} = \bar{\lambda}\bar{\boldsymbol{x}}$. So the nonreal eigenvalues of a real matrix come in conjugate pairs $a \pm ib$, with conjugate eigenvectors.*

Proof. Take complex conjugates of every entry in $A\boldsymbol{x} = \lambda\boldsymbol{x}$. Since $\bar{A} = A$, this gives $A\bar{\boldsymbol{x}} = \bar{\lambda}\bar{\boldsymbol{x}}$, and $\bar{\boldsymbol{x}} \neq \boldsymbol{0}$. □

How far a matrix is from symmetric decides how complex its eigenvalues can be. Symmetric matrices ($A^{\mathsf{T}} = A$) have real eigenvalues and perpendicular eigenvectors; this is proved in Section 29, and the examples B and $\left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$ above show it. The rotation Q is antisymmetric ($Q^{\mathsf{T}} = -Q$), the opposite extreme, and its eigenvalues are purely imaginary.

23.5 Triangular matrices and repeated eigenvalues

Proposition 23.10 (Triangular matrices)**.** *The eigenvalues of a triangular matrix are its diagonal entries.*

Proof. $A - \lambda I$ is triangular with diagonal entries $a_{ii} - \lambda$, and the determinant of a triangular matrix is the product of its diagonal entries. So $\det(A - \lambda I) = (a_{11} - \lambda)\cdots(a_{nn} - \lambda)$, with roots $a_{11}, \ldots, a_{nn}$. □

Triangular matrices show the one real danger in this subject: a repeated eigenvalue can come with too few eigenvectors.

Example 23.11 (A shortage of eigenvectors)**.** $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ has $\det(A - \lambda I) = (3 - \lambda)^2$, so $\lambda_1 = \lambda_2 = 3$. The eigenvectors fill the nullspace of $A - 3I = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$, which has rank 1, so the nullspace is only the line through $(1, 0)$. There is no second independent eigenvector. Compare $3I$, which has the same eigenvalues $3, 3$ but for which every nonzero vector is an eigenvector. A repeated eigenvalue may or may not cost us eigenvectors; Section 24 explains why this matters.

23.6 Eigenvalues of powers, inverses and polynomials

Once the eigenvectors of A are known, many related matrices come for free.

Proposition 23.12 (Functions of A)**.** *Suppose $A\boldsymbol{x} = \lambda\boldsymbol{x}$. Then $A^k\boldsymbol{x} = \lambda^k\boldsymbol{x}$ for $k = 1, 2, \ldots$, and for any polynomial p, $p(A)\boldsymbol{x} = p(\lambda)\boldsymbol{x}$. If A is invertible, then $\lambda \neq 0$ and $A^{-1}\boldsymbol{x} = \lambda^{-1}\boldsymbol{x}$.*

Proof. $A^2\boldsymbol{x} = A(\lambda\boldsymbol{x}) = \lambda A\boldsymbol{x} = \lambda^2\boldsymbol{x}$, and induction gives $A^k\boldsymbol{x} = \lambda^k\boldsymbol{x}$. A polynomial $p(A) = c_0 I + c_1 A + \cdots + c_m A^m$ then sends $\boldsymbol{x}$ to $(c_0 + c_1\lambda + \cdots + c_m\lambda^m)\boldsymbol{x}$. If A is invertible it is not singular, so $\lambda \neq 0$; multiplying $A\boldsymbol{x} = \lambda\boldsymbol{x}$ by A^{-1} and dividing by λ gives $A^{-1}\boldsymbol{x} = \lambda^{-1}\boldsymbol{x}$. □

Example 23.13 (Eigenvalues of A^2 and $A^{-1} - I$ without computing them)**.** Find the eigenvalues and eigenvectors of A^2 and of $A^{-1} - I$ for

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 1 & 4 \end{bmatrix}.$$

By Proposition 23.12 it is enough to find those of A. Expand $\det(A - \lambda I)$ down the first column, which has a single nonzero entry:

$$\det(A - \lambda I) = (1 - \lambda)\begin{vmatrix} 1 - \lambda & -2 \\ 1 & 4 - \lambda \end{vmatrix} = (1 - \lambda)(\lambda^2 - 5\lambda + 6) = (1 - \lambda)(\lambda - 2)(\lambda - 3).$$

The eigenvalues are $1, 2, 3$ (trace check: $1 + 1 + 4 = 6 = 1 + 2 + 3$). The eigenvectors:

- $\lambda = 1$: $A - I = \left[\begin{smallmatrix} 0 & 2 & 3 \\ 0 & 0 & -2 \\ 0 & 1 & 3 \end{smallmatrix}\right]$ has a zero first column, so x_1 is free; the last two rows force $x_3 = 0$, then $x_2 = 0$. Eigenvector $(1, 0, 0)$.
- $\lambda = 2$: $A - 2I = \left[\begin{smallmatrix} -1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 1 & 2 \end{smallmatrix}\right]$. Row 2 gives $x_2 = -2x_3$; row 1 gives $x_1 = 2x_2 + 3x_3 = -x_3$. With $x_3 = 1$: $(-1, -2, 1)$.
- $\lambda = 3$: $A - 3I = \left[\begin{smallmatrix} -2 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 1 & 1 \end{smallmatrix}\right]$. Row 3 gives $x_2 = -x_3$; row 1 gives $2x_1 = 2x_2 + 3x_3 = x_3$. With $x_3 = 2$: $(1, -2, 2)$.

These three vectors are eigenvectors of A^2 with eigenvalues $1, 4, 9$, and of $A^{-1} - I$ with eigenvalues $\frac{1}{1} - 1 = 0$, $\frac{1}{2} - 1 = -\frac{1}{2}$ and $\frac{1}{3} - 1 = -\frac{2}{3}$. We never squared or inverted a 3×3 matrix.

23.7 Exercises

Exercise 23.14. A 3×3 matrix B has eigenvalues 0, 1, 2. Which of the following can be determined from this information? Give them where possible. (a) the rank of B; (b) $\det(B^{\mathsf{T}}B)$; (c) the eigenvalues of $B^{\mathsf{T}}B$; (d) the eigenvalues of $(B^2 + I)^{-1}$.

Solution. (a) The rank is 2. Since 0 is an eigenvalue, B is singular and the rank is at most 2. Eigenvectors $\boldsymbol{x}_1, \boldsymbol{x}_2$ for $\lambda = 1, 2$ lie in the column space, because $\boldsymbol{x} = B(\boldsymbol{x}/\lambda)$ when $\lambda \neq 0$. They are independent: if $\boldsymbol{x}_2 = c\,\boldsymbol{x}_1$ then $2\boldsymbol{x}_2 = B\boldsymbol{x}_2 = c\,\boldsymbol{x}_1 = \boldsymbol{x}_2$, impossible for $\boldsymbol{x}_2 \neq \mathbf{0}$. So the rank is at least 2.

(b) $\det(B^{\mathsf{T}}B) = \det B^{\mathsf{T}} \det B = 0 \cdot 0 = 0$, since $\det B = 0 \cdot 1 \cdot 2$.

(c) Not determined. For $B = \mathrm{diag}(0,1,2)$, $B^{\mathsf{T}}B = \mathrm{diag}(0,1,4)$ has eigenvalues $0,1,4$. For the triangular $B = \left[\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right]$, which also has eigenvalues $0,1,2$, we get $B^{\mathsf{T}}B = \left[\begin{smallmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right]$ with eigenvalues $0,2,4$.

(d) By Proposition 23.12, an eigenvector of B with eigenvalue λ is an eigenvector of $B^2 + I$ with eigenvalue $\lambda^2 + 1 \neq 0$, hence of its inverse with eigenvalue $1/(\lambda^2+1)$. The eigenvalues are 1, $\frac{1}{2}$, $\frac{1}{5}$. □

Exercise 23.15. Find the eigenvalues of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

Solution. A is triangular, so its eigenvalues are $1,4,6$ (Proposition 23.10). For B, expand along the middle row:

$$\det(B - \lambda I) = (2-\lambda)\begin{vmatrix} -\lambda & 1 \\ 3 & -\lambda \end{vmatrix} = (2-\lambda)(\lambda^2 - 3),$$

so the eigenvalues are 2, $\sqrt{3}$, $-\sqrt{3}$ (check: the sum is $2 = \mathrm{tr}\, B$ and the product -6 equals $\det B = -6$). C has rank 1, so its nullspace is a plane and $\lambda = 0$ is an eigenvalue twice; the trace 6 gives the third eigenvalue 6, with eigenvector $(1,1,1)$. Directly, $\det(C - \lambda I) = -\lambda^3 + 6\lambda^2 = -\lambda^2(\lambda - 6)$. □

Exercise 23.16. Show that every eigenvalue of a projection matrix ($P^2 = P$) is 0 or 1.

Solution. If $P\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$, then by Proposition 23.12 $\lambda^2\boldsymbol{x} = P^2\boldsymbol{x} = P\boldsymbol{x} = \lambda\boldsymbol{x}$, so $(\lambda^2 - \lambda)\boldsymbol{x} = \mathbf{0}$ and $\lambda(\lambda - 1) = 0$. Example 23.2 shows both values occurring: the plane gives $\lambda = 1$, the normal line $\lambda = 0$. □

Exercise 23.17. Find the eigenvalues and eigenvectors of the rotation through the angle θ,

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

and check Theorem 23.7.

Solution. $\det(R - \lambda I) = (\cos\theta - \lambda)^2 + \sin^2\theta = \lambda^2 - 2\lambda\cos\theta + 1$. The quadratic formula gives $\lambda = \cos\theta \pm \sqrt{\cos^2\theta - 1} = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$. For $\lambda = e^{i\theta}$, the vector $(1,-i)$ works: $R(1,-i) = (\cos\theta + i\sin\theta,\ \sin\theta - i\cos\theta) = e^{i\theta}(1,-i)$; the conjugate $(1,i)$ goes with $e^{-i\theta}$ (Proposition 23.9). The sum $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ is the trace and the product 1 is $\det R = \cos^2\theta + \sin^2\theta$. For $\theta = 90°$ this is the matrix Q above, with eigenvalues $\pm i$. Both eigenvalues have $|\lambda| = 1$: a rotation does not change lengths. □

24 Diagonalization and powers of A

Section 23 found eigenvalues and eigenvectors one at a time. Now we use all of them at once. Putting the eigenvectors into the columns of a matrix turns A into a diagonal matrix, and that single factorization makes powers of A easy: A^{100} costs no more than A. Powers of a matrix are what we need for any process that repeats the same linear step, such as populations that move every year or the Fibonacci numbers.

24.1 The eigenvector matrix diagonalizes A

Theorem 24.1 (Diagonalization)**.** *Suppose the $n \times n$ matrix A has n linearly independent eigenvectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, with $A\boldsymbol{x}_i = \lambda_i \boldsymbol{x}_i$. Put them in the columns of the eigenvector matrix X and the eigenvalues on the diagonal of the eigenvalue matrix Λ:*

$$X = \begin{bmatrix} \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}.$$

Then $AX = X\Lambda$, and since X is invertible,

$$X^{-1}AX = \Lambda \qquad \text{and} \qquad A = X\Lambda X^{-1}.$$

Conversely, if $A = X\Lambda X^{-1}$ with X invertible and Λ diagonal, then the columns of X are n independent eigenvectors of A and the diagonal entries of Λ are the matching eigenvalues.

Proof. Multiply a column at a time. Column j of AX is $A\boldsymbol{x}_j = \lambda_j \boldsymbol{x}_j$. Column j of $X\Lambda$ is X times column j of Λ, which is λ_j times the jth unit vector, so it is $\lambda_j \boldsymbol{x}_j$ as well. Hence $AX = X\Lambda$. The columns of X are independent, so X^{-1} exists; multiplying on the left or on the right by X^{-1} gives the two forms. Conversely, $A = X\Lambda X^{-1}$ gives $AX = X\Lambda$, whose column j reads $A\boldsymbol{x}_j = \lambda_j \boldsymbol{x}_j$; and $\boldsymbol{x}_j \neq \boldsymbol{0}$ because X is invertible. □

A matrix that can be written this way is called *diagonalizable*. The lectures call the eigenvector matrix S; these notes write X, as in the notation table. To remember the order of the factors, note that A multiplies its eigenvectors, so X stands immediately to the right of A in $AX = X\Lambda$. The factorization $A = X\Lambda X^{-1}$ joins $A = LU$ from elimination and $A = QR$ from Gram–Schmidt; unlike those two, it describes what A *does* when it is applied repeatedly.

Example 24.2 (Diagonalizing the two-city matrix)**.** In Section 23 the two-city matrix $A = \left[\begin{smallmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{smallmatrix}\right]$ had eigenvalues 1 and 0.5 with eigenvectors along $(600, 400)$ and $(1, -1)$. Scale the first to $(3, 2)$:

$$X = \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}, \qquad X^{-1} = \frac{1}{-5}\begin{bmatrix} -1 & -1 \\ -2 & 3 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}.$$

Check: $X\Lambda = \left[\begin{smallmatrix} 3 & 0.5 \\ 2 & -0.5 \end{smallmatrix}\right]$ and

$$X\Lambda X^{-1} = \frac{1}{5}\begin{bmatrix} 3+1 & 3-1.5 \\ 2-1 & 2+1.5 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} = A.$$

24.2 Powers of A

Squaring $A = X\Lambda X^{-1}$, the inner $X^{-1}X$ cancels: $A^2 = X\Lambda X^{-1}X\Lambda X^{-1} = X\Lambda^2 X^{-1}$.

Proposition 24.3 (Powers)**.** *If $A = X\Lambda X^{-1}$, then $A^k = X\Lambda^k X^{-1}$ for every $k \geq 1$, where $\Lambda^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)$. So A^k has the same eigenvectors as A and eigenvalues λ_i^k. If A is invertible the same holds for $k = -1$.*

Proof. In the product of k copies of $X\Lambda X^{-1}$, each of the $k-1$ inner pairs $X^{-1}X$ equals I, leaving $X\Lambda^k X^{-1}$. A power of a diagonal matrix is diagonal with the powers of its entries. For $k = -1$, $(X\Lambda X^{-1})^{-1} = X\Lambda^{-1}X^{-1}$, which exists when no λ_i is zero. □

Compare this with LU: multiplying LU by itself a hundred times gives a hundred factors that do not simplify. The eigenvector picture is the one that makes A^{100} transparent. For the two-city matrix of Example 24.2,

$$A^k = X\begin{bmatrix}1 & 0\\ 0 & 0.5^k\end{bmatrix}X^{-1} = \frac{1}{5}\begin{bmatrix}3+2(0.5)^k & 3-3(0.5)^k\\ 2-2(0.5)^k & 2+3(0.5)^k\end{bmatrix} \longrightarrow \begin{bmatrix}0.6 & 0.6\\ 0.4 & 0.4\end{bmatrix} \quad (k\to\infty).$$

At $k = 1$ this gives back A. The question "does A^k go to zero?" cannot be answered from the pivots, but the eigenvalues answer it at once.

Theorem 24.4 (When powers go to zero)**.** *Let A be diagonalizable with eigenvalues $\lambda_1,\dots,\lambda_n$. Then $A^k \to 0$ as $k\to\infty$ if and only if $|\lambda_i| < 1$ for every i.*

Proof. If every $|\lambda_i| < 1$, then each $\lambda_i^k \to 0$, so $\Lambda^k \to 0$ and $A^k = X\Lambda^k X^{-1} \to 0$ because X and X^{-1} do not change with k. Conversely, if some $|\lambda_j| \geq 1$, then $A^k\boldsymbol{x}_j = \lambda_j^k\boldsymbol{x}_j$ has length $|\lambda_j|^k\|\boldsymbol{x}_j\| \geq \|\boldsymbol{x}_j\| > 0$ for every k, so A^k does not go to zero. □

The absolute value matters because eigenvalues can be negative or complex. The theorem stays true for matrices that cannot be diagonalized (Exercise 24.15 shows one), but that proof needs the Jordan form of Section 32.

24.3 Which matrices can be diagonalized?

Everything above assumed n independent eigenvectors. Distinct eigenvalues guarantee them.

Theorem 24.5 (Distinct eigenvalues give independent eigenvectors)**.** *Eigenvectors $\boldsymbol{x}_1,\dots,\boldsymbol{x}_k$ that belong to distinct eigenvalues $\lambda_1,\dots,\lambda_k$ are linearly independent. In particular, an $n\times n$ matrix with n distinct eigenvalues is diagonalizable.*

Proof. Induction on k; one nonzero vector is independent. Suppose the statement holds for $k-1$ vectors, and $c_1\boldsymbol{x}_1 + \cdots + c_k\boldsymbol{x}_k = \boldsymbol{0}$. Multiply by $A - \lambda_k I$. Since $(A-\lambda_k I)\boldsymbol{x}_i = (\lambda_i - \lambda_k)\boldsymbol{x}_i$, the last term disappears:

$$c_1(\lambda_1-\lambda_k)\boldsymbol{x}_1 + \cdots + c_{k-1}(\lambda_{k-1}-\lambda_k)\boldsymbol{x}_{k-1} = \boldsymbol{0}.$$

By the induction hypothesis every coefficient is zero, and $\lambda_i \neq \lambda_k$, so $c_1 = \cdots = c_{k-1} = 0$. Then $c_k\boldsymbol{x}_k = \boldsymbol{0}$ forces $c_k = 0$. □

A random matrix has distinct eigenvalues (MATLAB's `eig(rand(10))` will show ten different numbers), so "most" matrices are diagonalizable. Trouble can come only from repeated eigenvalues, and then we must count.

Definition 24.6 (Algebraic and geometric multiplicity)**.** The *algebraic multiplicity* of an eigenvalue λ is the number of times it is a root of $\det(A-\lambda I)$. Its *geometric multiplicity* is the number of independent eigenvectors, $\dim \mathbf{N}(A-\lambda I)$.

The geometric multiplicity is at least 1 and never exceeds the algebraic one (a proof uses the similar matrices of Section 32). Eigenvectors from different eigenspaces combine independently, by the same argument as Theorem 24.5, so A is diagonalizable exactly when the geometric multiplicities add up to n, that is, when every eigenvalue has as many independent eigenvectors as its algebraic multiplicity.

Example 24.7 (Repeated eigenvalues, two outcomes)**.** The $n \times n$ identity has the eigenvalue 1 repeated n times, and every nonzero vector is an eigenvector; any invertible X gives $X^{-1}IX = I = \Lambda$. In contrast, $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ has $\det(A - \lambda I) = (2-\lambda)^2$, algebraic multiplicity 2, but $A - 2I = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has a one-dimensional nullspace spanned by $(1, 0)$: geometric multiplicity 1. There is no invertible X with $X^{-1}AX$ diagonal; indeed the only diagonal candidate is $\Lambda = 2I$, and $X(2I)X^{-1} = 2I \neq A$.

Common trap. Diagonalizable and invertible are unrelated properties. Invertibility is about the eigenvalues (none is 0); diagonalizability is about the eigenvectors (enough of them). $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ is invertible but not diagonalizable; $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ is diagonalizable but singular. Also, X is not unique: its columns can be rescaled and reordered, as long as Λ is reordered the same way.

24.4 Difference equations $\boldsymbol{u}_{k+1} = A\boldsymbol{u}_k$

Start with a vector $\boldsymbol{u}_0$ and apply the same step over and over: $\boldsymbol{u}_1 = A\boldsymbol{u}_0$, $\boldsymbol{u}_2 = A\boldsymbol{u}_1 = A^2\boldsymbol{u}_0$, and in general $\boldsymbol{u}_k = A^k\boldsymbol{u}_0$. This is a first-order system of difference equations. The compact formula $A^k\boldsymbol{u}_0$ does not show what happens; the eigenvectors do.

Proposition 24.8 (Solution by eigenvectors)**.** *Let A have independent eigenvectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, and write the start as $\boldsymbol{u}_0 = c_1\boldsymbol{x}_1 + \cdots + c_n\boldsymbol{x}_n$, that is, solve $X\boldsymbol{c} = \boldsymbol{u}_0$. Then*

$$\boldsymbol{u}_k = A^k\boldsymbol{u}_0 = c_1\lambda_1^k\boldsymbol{x}_1 + c_2\lambda_2^k\boldsymbol{x}_2 + \cdots + c_n\lambda_n^k\boldsymbol{x}_n = X\Lambda^k\boldsymbol{c}.$$

Proof. A^k is linear and $A^k\boldsymbol{x}_i = \lambda_i^k\boldsymbol{x}_i$, so $A^k\boldsymbol{u}_0 = \sum_i c_i A^k\boldsymbol{x}_i = \sum_i c_i\lambda_i^k\boldsymbol{x}_i$. In matrix form, $A^k\boldsymbol{u}_0 = X\Lambda^k X^{-1}\boldsymbol{u}_0 = X\Lambda^k\boldsymbol{c}$. □

Each eigenvector component goes its own way, multiplied by its own λ at every step. The three tasks are: find the eigenvalues, find the eigenvectors, find the c_i from $\boldsymbol{u}_0$. The eigenvalues alone already tell whether $\boldsymbol{u}_k$ grows, decays or settles.

Example 24.9 (Where do the 1000 people end up?)**.** A thousand people move between two cities by the rule of Example 24.2: each year 80% of city 1 stays and 20% moves, 70% of city 2 stays and 30% moves. All start in city 1: $\boldsymbol{u}_0 = (1000, 0)$.

Eigenvectors. $A(600, 400) = (480 + 120,\ 120 + 280) = (600, 400)$, so $\lambda_1 = 1$ with $\boldsymbol{x}_1 = (600, 400)$; and $A(1, -1) = (0.5, -0.5)$, so $\lambda_2 = 0.5$ with $\boldsymbol{x}_2 = (1, -1)$.

Split the start. $(1000, 0) = (600, 400) + 400\,(1, -1)$, so $c_1 = 1$ and $c_2 = 400$.

Follow each piece. The $\lambda = 1$ piece never changes; the $\lambda = 0.5$ piece halves every year:

$$\boldsymbol{u}_k = (600, 400) + 400\,(0.5)^k\,(1, -1).$$

year k	0	1	2	3	4	∞
city 1	1000	800	700	650	625	600
city 2	0	200	300	350	375	400
gap $400(0.5)^k$	400	200	100	50	25	0

A direct multiplication confirms one step: $A(800, 200) = (640 + 60,\ 160 + 140) = (700, 300)$.

Interpretation. The populations approach the steady state $(600, 400)$, the eigenvector for $\lambda = 1$, and the distance to it halves every year because the other eigenvalue is $\frac{1}{2}$ (Figure 29). Starting instead from $(0, 1000) = (600, 400) - 600\,(1, -1)$ gives $\boldsymbol{u}_1 = (300, 700)$ and the same limit $(600, 400)$: the start only decides how large the decaying piece is.

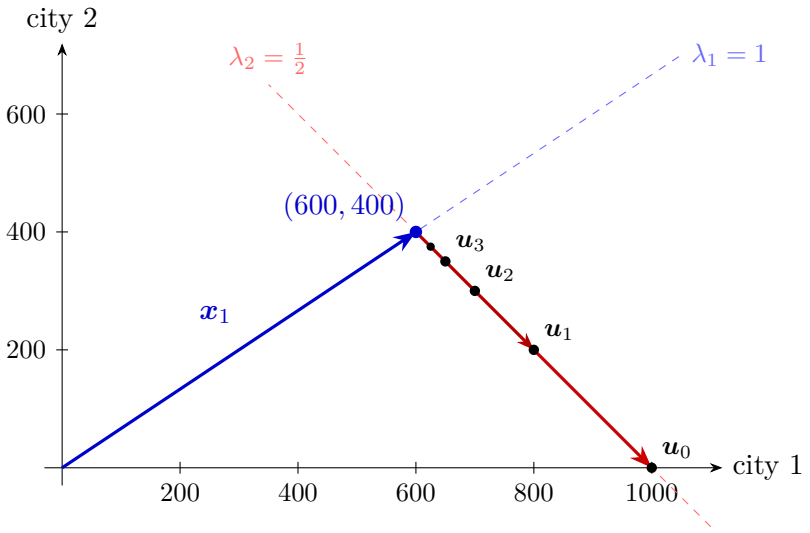


Figure 29: The two-city model of Example 24.9. Every state lies on the line $u_1 + u_2 = 1000$ through the steady state $(600, 400)$ in the direction $(1, -1)$. Each state is the fixed arrow $(600, 400)$ plus a sliding arrow $400(0.5)^k(1, -1)$; every year the sliding arrow halves and the state hops $\boldsymbol{u}_0 \to \boldsymbol{u}_1 \to \boldsymbol{u}_2 \to \dots$ toward $(600, 400)$.

24.5 Fibonacci numbers

The Fibonacci numbers $0, 1, 1, 2, 3, 5, 8, 13, \dots$ obey $F_{k+2} = F_{k+1} + F_k$. How fast do they grow, and what is F_{100}? The rule is a single second-order equation; a small trick makes it a first-order system. Let $\boldsymbol{u}_k = (F_{k+1}, F_k)$ and add the trivial equation $F_{k+1} = F_{k+1}$:

$$\begin{aligned} F_{k+2} &= F_{k+1} + F_k \\ F_{k+1} &= F_{k+1} \end{aligned} \qquad \Longleftrightarrow \qquad \boldsymbol{u}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \boldsymbol{u}_k.$$

Example 24.10 (A formula for F_k)**.** For $A = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ the trace is 1 and the determinant is -1, so

$$\det(A - \lambda I) = \lambda^2 - \lambda - 1 = 0, \qquad \lambda_1 = \frac{1 + \sqrt{5}}{2} \approx 1.618, \qquad \lambda_2 = \frac{1 - \sqrt{5}}{2} \approx -0.618.$$

The equation $\lambda^2 = \lambda + 1$ is the Fibonacci rule itself. The eigenvalues are distinct, so A is diagonalizable. For the eigenvectors, $A - \lambda I = \left[\begin{smallmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{smallmatrix}\right]$ kills $(\lambda, 1)$: the second row gives $\lambda - \lambda = 0$ and the first gives $(1 - \lambda)\lambda + 1 = -(\lambda^2 - \lambda - 1) = 0$. So $\boldsymbol{x}_1 = (\lambda_1, 1)$ and $\boldsymbol{x}_2 = (\lambda_2, 1)$.

The start is $\boldsymbol{u}_0 = (F_1, F_0) = (1, 0) = c_1\boldsymbol{x}_1 + c_2\boldsymbol{x}_2$. The second components give $c_1 + c_2 = 0$, the first $c_1\lambda_1 + c_2\lambda_2 = 1$, so $c_1 = -c_2 = 1/(\lambda_1 - \lambda_2) = 1/\sqrt{5}$. Reading off the second component of $\boldsymbol{u}_k = c_1\lambda_1^k\boldsymbol{x}_1 + c_2\lambda_2^k\boldsymbol{x}_2$:

$$F_k = \frac{1}{\sqrt{5}}\left(\frac{1 + \sqrt{5}}{2}\right)^k - \frac{1}{\sqrt{5}}\left(\frac{1 - \sqrt{5}}{2}\right)^k.$$

The second term is smaller than $\frac{1}{2}$ in absolute value for every $k \geq 0$ and shrinks fast, so F_k is the integer nearest to $\lambda_1^k/\sqrt{5}$. For $k = 10$: $\lambda_1^{10}/\sqrt{5} \approx 55.004$ and $F_{10} = 55$. For $k = 100$: $F_{100} \approx 3.54 \times 10^{20}$. The large eigenvalue controls the growth: each Fibonacci number is about 1.618 times the previous one.

24.6 Exercises

Exercise 24.11. Describe all matrices X that diagonalize $A = \left[\begin{smallmatrix} 4 & 0 \\ 1 & 2 \end{smallmatrix}\right]$. Then describe all matrices that diagonalize A^{-1}.

Solution. A is triangular, so $\lambda_1 = 4$, $\lambda_2 = 2$. For $\lambda_1 = 4$: $A - 4I = \left[\begin{smallmatrix} 0 & 0 \\ 1 & -2 \end{smallmatrix}\right]$, so $y = 2z$ and the eigenvectors are the nonzero multiples of $(2, 1)$. For $\lambda_2 = 2$: $A - 2I = \left[\begin{smallmatrix} 2 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, so $y = 0$ and the eigenvectors are the nonzero multiples of $(0, 1)$. The diagonalizing matrices are $X = \left[\begin{smallmatrix} 2a & 0 \\ a & b \end{smallmatrix}\right]$ and $X = \left[\begin{smallmatrix} 0 & 2a \\ b & a \end{smallmatrix}\right]$ with $a, b \neq 0$ (the eigenvalues in Λ then come in the order $4, 2$ or $2, 4$). Since $A^{-1} = X\Lambda^{-1}X^{-1}$, exactly the same matrices diagonalize A^{-1}. □

Exercise 24.12. Diagonalize $A = \left[\begin{smallmatrix} 0.6 & 0.9 \\ 0.4 & 0.1 \end{smallmatrix}\right]$. What is the limit of Λ^k as $k \to \infty$? What is the limit of $X\Lambda^k X^{-1}$, and what do you see in its columns?

Solution. Each column adds to 1, so (as Section 26 explains) $\lambda_1 = 1$; the trace 0.7 gives $\lambda_2 = -0.3$. From $A - I = \left[\begin{smallmatrix} -0.4 & 0.9 \\ 0.4 & -0.9 \end{smallmatrix}\right]$, $\boldsymbol{x}_1 = (9, 4)$; from $A + 0.3I = \left[\begin{smallmatrix} 0.9 & 0.9 \\ 0.4 & 0.4 \end{smallmatrix}\right]$, $\boldsymbol{x}_2 = (1, -1)$. So $X = \left[\begin{smallmatrix} 9 & 1 \\ 4 & -1 \end{smallmatrix}\right]$, $\Lambda = \text{diag}(1, -0.3)$, and $\Lambda^k \to \text{diag}(1, 0)$ because $|-0.3| < 1$. With $X^{-1} = \frac{1}{13}\left[\begin{smallmatrix} 1 & 1 \\ 4 & -9 \end{smallmatrix}\right]$,

$$A^k \to X \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} X^{-1} = \begin{bmatrix} 9 \\ 4 \end{bmatrix} \frac{1}{13} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{13} \begin{bmatrix} 9 & 9 \\ 4 & 4 \end{bmatrix}.$$

Both columns are the steady state $(9, 4)/13$: whatever the start (with components adding to 1), $A^k\boldsymbol{u}_0$ approaches it. □

Exercise 24.13. For numbers $a \neq b$, let $C = \left[\begin{smallmatrix} 2b-a & a-b \\ 2b-2a & 2a-b \end{smallmatrix}\right]$. Find the eigenvalues and eigenvectors of C, a formula for C^k, and C^{100} when $a = 1$, $b = -1$.

Solution. The trace is $a + b$ and the determinant is $(2b - a)(2a - b) - (a - b)(2b - 2a) = ab$, so $\det(C - \lambda I) = \lambda^2 - (a+b)\lambda + ab = (\lambda - a)(\lambda - b)$. For $\lambda = a$: $C - aI = \left[\begin{smallmatrix} 2b-2a & a-b \\ 2b-2a & a-b \end{smallmatrix}\right]$; its first column is -2 times its second, so $(1, 2)$ is in the nullspace. For $\lambda = b$: $C - bI = \left[\begin{smallmatrix} b-a & a-b \\ 2b-2a & 2a-2b \end{smallmatrix}\right]$, killed by $(1, 1)$. With $X = \left[\begin{smallmatrix} 1 & 1 \\ 2 & 1 \end{smallmatrix}\right]$ and $X^{-1} = \left[\begin{smallmatrix} -1 & 1 \\ 2 & -1 \end{smallmatrix}\right]$,

$$C^k = X \begin{bmatrix} a^k & 0 \\ 0 & b^k \end{bmatrix} X^{-1} = \begin{bmatrix} 2b^k - a^k & a^k - b^k \\ 2b^k - 2a^k & 2a^k - b^k \end{bmatrix}.$$

At $k = 1$ this is C, a useful check. For $a = 1$, $b = -1$ and $k = 100$, $a^k = b^k = 1$ and $C^{100} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right] = I$. (Here $C = \left[\begin{smallmatrix} -3 & 2 \\ -4 & 3 \end{smallmatrix}\right]$ already has $C^2 = I$.) □

Exercise 24.14. The Lucas numbers $2, 1, 3, 4, 7, 11, 18, \ldots$ follow the Fibonacci rule $L_{k+2} = L_{k+1} + L_k$ but start from $L_0 = 2$, $L_1 = 1$. Show that $L_k = \lambda_1^k + \lambda_2^k$ with $\lambda_{1,2} = (1 \pm \sqrt{5})/2$.

Solution. Use the same A and eigenvectors as in Example 24.10 with $\boldsymbol{u}_0 = (L_1, L_0) = (1, 2)$. The equations $c_1 + c_2 = 2$ and $c_1\lambda_1 + c_2\lambda_2 = 1$ are solved by $c_1 = c_2 = 1$, because $\lambda_1 + \lambda_2 = 1$ (the trace). The second component of $\boldsymbol{u}_k = \lambda_1^k\boldsymbol{x}_1 + \lambda_2^k\boldsymbol{x}_2$ is $L_k = \lambda_1^k + \lambda_2^k$. Check: $\lambda_1^2 + \lambda_2^2 = (\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2 = 1 + 2 = 3 = L_2$. □

Exercise 24.15. Let $N = \left[\begin{smallmatrix} 1/2 & 1 \\ 0 & 1/2 \end{smallmatrix}\right]$. Show that N is not diagonalizable, that $N^k = \left[\begin{smallmatrix} 2^{-k} & k\,2^{1-k} \\ 0 & 2^{-k} \end{smallmatrix}\right]$, and that $N^k \to 0$ anyway.

Solution. The eigenvalue $\frac{1}{2}$ is repeated, and $N - \frac{1}{2}I = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has only a one-dimensional nullspace, so there is just one independent eigenvector. The formula holds for $k = 1$; if it holds for k, then

$$N^{k+1} = N^k N = \begin{bmatrix} 2^{-k} \cdot \frac{1}{2} & 2^{-k} + k\,2^{1-k} \cdot \frac{1}{2} \\ 0 & 2^{-k} \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 2^{-(k+1)} & (k+1)\,2^{-k} \\ 0 & 2^{-(k+1)} \end{bmatrix},$$

which is the formula for $k+1$. Since $k\,2^{1-k} \to 0$, $N^k \to 0$. The off-diagonal entry decays more slowly than 2^{-k} (it is $1, 1, \frac{3}{4}, \frac{1}{2}, \dots$) because of the factor k, but the decay of $|\lambda|^k$ wins in the end. □

25 Differential equations and e^{At}

Section 24 solved $\boldsymbol{u}_{k+1} = A\boldsymbol{u}_k$, where time moves in steps and the solution is built from powers λ^k. Here time is continuous: we solve the system of linear, constant-coefficient differential equations

$$\frac{d\boldsymbol{u}}{dt} = A\boldsymbol{u}, \qquad \boldsymbol{u}(0) \text{ given.}$$

The key fact from calculus is that solutions of constant-coefficient linear equations are exponentials. Looking for an exponential solution turns the differential equation into the eigenvalue problem, and the whole section runs parallel to the previous one with $e^{\lambda t}$ in place of λ^k.

25.1 Pure exponential solutions

Try $\boldsymbol{u}(t) = e^{\lambda t}\boldsymbol{x}$ with a fixed vector $\boldsymbol{x}$. Then $d\boldsymbol{u}/dt = \lambda e^{\lambda t}\boldsymbol{x}$ while $A\boldsymbol{u} = e^{\lambda t}A\boldsymbol{x}$. These agree for all t exactly when $A\boldsymbol{x} = \lambda\boldsymbol{x}$. So every eigenvalue–eigenvector pair gives a *pure exponential solution* $e^{\lambda t}\boldsymbol{x}$, and since the equation is linear, combinations of solutions are solutions.

Theorem 25.1 (Solution by eigenvectors)**.** *Let A have n independent eigenvectors $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Given $\boldsymbol{u}(0)$, solve $X\boldsymbol{c} = \boldsymbol{u}(0)$ for the coefficients c_i. Then*

$$\boldsymbol{u}(t) = c_1 e^{\lambda_1 t}\boldsymbol{x}_1 + c_2 e^{\lambda_2 t}\boldsymbol{x}_2 + \cdots + c_n e^{\lambda_n t}\boldsymbol{x}_n$$

solves $d\boldsymbol{u}/dt = A\boldsymbol{u}$ with the given initial value.

Proof. Each term satisfies the equation: $\frac{d}{dt}\big(c_i e^{\lambda_i t}\boldsymbol{x}_i\big) = c_i\lambda_i e^{\lambda_i t}\boldsymbol{x}_i = c_i e^{\lambda_i t}A\boldsymbol{x}_i = A\big(c_i e^{\lambda_i t}\boldsymbol{x}_i\big)$. Adding the terms, the sum satisfies it too. At $t = 0$ every exponential equals 1, so $\boldsymbol{u}(0) = c_1\boldsymbol{x}_1 + \cdots + c_n\boldsymbol{x}_n = X\boldsymbol{c}$, which is the given vector. (That this is the *only* solution follows from the matrix exponential below.) □

Set the two formulas side by side: $\boldsymbol{u}_k = \sum c_i\lambda_i^k\boldsymbol{x}_i$ for steps and $\boldsymbol{u}(t) = \sum c_i e^{\lambda_i t}\boldsymbol{x}_i$ for continuous time. Each eigenvector component evolves on its own; A couples the components of $\boldsymbol{u}$, and the eigenvectors uncouple them.

Example 25.2 (Flow between two compartments)**.** Solve

$$\frac{du_1}{dt} = -u_1 + 2u_2, \qquad \frac{du_2}{dt} = u_1 - 2u_2, \qquad \boldsymbol{u}(0) = (1, 0).$$

Eigenvalues. $A = \left[\begin{smallmatrix} -1 & 2 \\ 1 & -2 \end{smallmatrix}\right]$ is singular (column 2 is -2 times column 1), so $\lambda_1 = 0$; the trace -3 gives $\lambda_2 = -3$. Directly, $\det(A - \lambda I) = \lambda^2 + 3\lambda = \lambda(\lambda + 3)$.

Eigenvectors. $A\boldsymbol{x}_1 = \mathbf{0}$ gives $\boldsymbol{x}_1 = (2, 1)$. $A + 3I = \left[\begin{smallmatrix} 2 & 2 \\ 1 & 1 \end{smallmatrix}\right]$ gives $\boldsymbol{x}_2 = (1, -1)$.

Coefficients. $X\boldsymbol{c} = \boldsymbol{u}(0)$ reads $c_1(2, 1) + c_2(1, -1) = (1, 0)$; adding the components, $3c_1 = 1$, and then $c_2 = 1 - 2c_1 = \frac{1}{3}$. So $c_1 = c_2 = \frac{1}{3}$ and

$$\boldsymbol{u}(t) = \frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3}e^{-3t}\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \text{i.e.} \quad u_1 = \tfrac{2}{3} + \tfrac{1}{3}e^{-3t}, \quad u_2 = \tfrac{1}{3} - \tfrac{1}{3}e^{-3t}.$$

Check. $u_1' = -e^{-3t}$ and $-u_1 + 2u_2 = -\frac{2}{3} - \frac{1}{3}e^{-3t} + \frac{2}{3} - \frac{2}{3}e^{-3t} = -e^{-3t}$.

Interpretation. Material flows out of compartment 1 into compartment 2. The e^{-3t} term dies out, and the $e^{0t} = 1$ term stays: the solution approaches the steady state $\boldsymbol{u}(\infty) = (\frac{2}{3}, \frac{1}{3})$, the eigenvector for $\lambda = 0$ scaled by c_1 (Figure 30). The total $u_1 + u_2$ stays equal to 1 throughout, because the columns of A add to zero, so $\frac{d}{dt}(u_1 + u_2) = 0$.

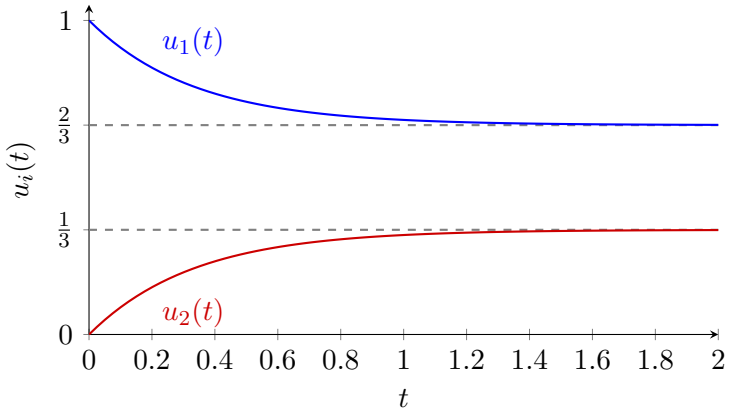


Figure 30: The solution $u_1 = \frac{2}{3} + \frac{1}{3}e^{-3t}$, $u_2 = \frac{1}{3} - \frac{1}{3}e^{-3t}$ of Example 25.2. The eigenvalue $\lambda_2 = -3$ makes the difference from the steady state decay like e^{-3t} (by $t = 1$ only 5% of it is left); the eigenvalue $\lambda_1 = 0$ supplies the steady state $(\frac{2}{3}, \frac{1}{3})$.

25.2 Stability and steady states

Which way the solution goes is decided by the eigenvalues alone. For a complex eigenvalue $\lambda = a + ib$,

$$e^{\lambda t} = e^{at}e^{ibt} = e^{at}(\cos bt + i \sin bt), \qquad |e^{\lambda t}| = e^{at},$$

since $\cos^2 bt + \sin^2 bt = 1$. The imaginary part only makes the solution oscillate; the real part decides growth or decay. For instance $\lambda = -3 + 6i$ gives $|e^{\lambda t}| = e^{-3t}$. For a diagonalizable A, Theorem 25.1 therefore gives three regimes:

1. *Stability*: $\boldsymbol{u}(t) \to \mathbf{0}$ for every start exactly when $\operatorname{Re}\lambda_i < 0$ for every i.
2. *Steady state*: if $\lambda_1 = 0$ and all other $\operatorname{Re}\lambda_i < 0$, then $\boldsymbol{u}(t) \to c_1\boldsymbol{x}_1$.
3. *Blow-up*: if some $\operatorname{Re}\lambda_i > 0$, every start with $c_i \neq 0$ produces a solution that grows without bound.

Changing A to $-A$ changes the sign of every eigenvalue: in Example 25.2 it would turn e^{-3t} into e^{3t} and the steady state into blow-up.

For 2×2 matrices there is a test that needs no eigenvalues at all.

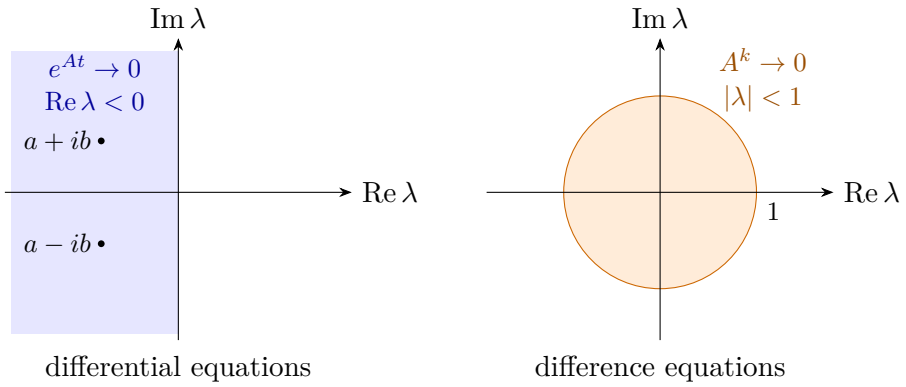


Figure 31: Where the eigenvalues must lie for the solutions to decay. For $d\boldsymbol{u}/dt = A\boldsymbol{u}$ it is the left half-plane $\operatorname{Re}\lambda < 0$; for $\boldsymbol{u}_{k+1} = A\boldsymbol{u}_k$ it is the inside of the unit circle $|\lambda| < 1$. The eigenvalue $\lambda = 0$ gives a steady state for differential equations; $\lambda = 1$ gives one for powers.

Proposition 25.3 (Stability of a 2×2 system)**.** *A real 2×2 matrix A has both eigenvalues with negative real part if and only if*

$$\operatorname{tr} A < 0 \qquad \textit{and} \qquad \det A > 0.$$

Proof. Recall $\operatorname{tr} A = \lambda_1 + \lambda_2$ and $\det A = \lambda_1\lambda_2$ (Section 23). If the eigenvalues are a complex pair $a \pm ib$ ($b \neq 0$), then $\operatorname{tr} A = 2a$ and $\det A = a^2 + b^2 > 0$, so the condition says exactly $a < 0$. If they are real, $\det A > 0$ means they have the same sign and $\operatorname{tr} A < 0$ means that sign is negative; conversely two negative numbers have negative sum and positive product. □

A negative trace alone is not enough: $\left[\begin{smallmatrix} -2 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ has trace -1 but eigenvalues -2 and 1, so e^t blows up. Its determinant -2 fails the test.

25.3 The matrix exponential

For a single equation $du/dt = au$ the solution is $u(t) = e^{at}u(0)$. We would like to write $\boldsymbol{u}(t) = e^{At}\boldsymbol{u}(0)$ for systems. The exponential of a matrix is defined by the same power series as e^x.

Definition 25.4 (Matrix exponential)**.** For a square matrix A and a number t,

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}.$$

The series always converges: if every entry of A has absolute value at most m, every entry of A^k has absolute value at most $(nm)^k$, and $\sum (nm|t|)^k/k!$ is finite. Compare the geometric series $(I - At)^{-1} = I + At + (At)^2 + \cdots$, which has no factorials and converges only when the eigenvalues of At are smaller than 1 in absolute value; for small t it says $(I - At)^{-1} \approx I + At$.

Proposition 25.5 (The exponential solves the system)**.** $\dfrac{d}{dt}e^{At} = Ae^{At}$ *and* $e^{A\cdot 0} = I$. *Hence* $\boldsymbol{u}(t) = e^{At}\boldsymbol{u}(0)$ *solves* $d\boldsymbol{u}/dt = A\boldsymbol{u}$ *with the given* $\boldsymbol{u}(0)$.

Proof. Differentiate the series term by term (allowed for power series): $\frac{d}{dt}\frac{t^k A^k}{k!} = \frac{t^{k-1}A^k}{(k-1)!} = A\,\frac{(At)^{k-1}}{(k-1)!}$. Summing over $k \geq 1$ gives $A\,e^{At}$. At $t = 0$ only the term I survives. □

The formula needs no eigenvectors, so it also covers matrices that are not diagonalizable. It also gives uniqueness: if $\boldsymbol{u}$ is any solution, then $\frac{d}{dt}\big(e^{-At}\boldsymbol{u}\big) = -Ae^{-At}\boldsymbol{u} + e^{-At}A\boldsymbol{u} = \boldsymbol{0}$ (the series show that A commutes with e^{-At}), so $e^{-At}\boldsymbol{u}(t)$ is constant, equal to $\boldsymbol{u}(0)$. And $e^{At}e^{-At} = I$, because this product has derivative $Ae^{At}e^{-At} - e^{At}Ae^{-At} = 0$ and equals I at $t = 0$; multiplying by e^{At} gives $\boldsymbol{u}(t) = e^{At}\boldsymbol{u}(0)$. When A *is* diagonalizable, the eigenvectors make the series easy to sum.

Theorem 25.6 (Exponential by diagonalization)**.** *If $A = X\Lambda X^{-1}$, then*

$$e^{At} = Xe^{\Lambda t}X^{-1}, \qquad e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix}.$$

Proof. As in Section 24, $(At)^k = X\Lambda^k X^{-1}t^k$. Write $I = XX^{-1}$ and factor X out on the left and X^{-1} on the right of every term:

$$e^{At} = X\Big(I + \Lambda t + \frac{(\Lambda t)^2}{2!} + \cdots\Big)X^{-1} = Xe^{\Lambda t}X^{-1}.$$

The series for the diagonal matrix Λt is diagonal, with the scalar series $\sum(\lambda_i t)^k/k! = e^{\lambda_i t}$ on the diagonal. □

Another way to see the same thing: substitute $\boldsymbol{u} = X\boldsymbol{v}$. Then $X\,d\boldsymbol{v}/dt = AX\boldsymbol{v}$, so $d\boldsymbol{v}/dt = X^{-1}AX\boldsymbol{v} = \Lambda\boldsymbol{v}$, which is n separate scalar equations $dv_i/dt = \lambda_i v_i$ with solutions $v_i(t) = e^{\lambda_i t}v_i(0)$. Going back, $\boldsymbol{u}(t) = Xe^{\Lambda t}X^{-1}\boldsymbol{u}(0)$, and with $\boldsymbol{c} = X^{-1}\boldsymbol{u}(0)$ this is the sum in Theorem 25.1. It also shows again that $e^{At} \to 0$ exactly when every $e^{\lambda_i t} \to 0$, that is, every $\operatorname{Re}\lambda_i < 0$.

Example 25.7 (Computing e^{At})**.** Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$. It is triangular: $\lambda_1 = 1$ with $\boldsymbol{x}_1 = (1, 0)$, and $\lambda_2 = 3$ with $\boldsymbol{x}_2 = (1, 2)$ from $A - 3I = \left[\begin{smallmatrix} -2 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. So

$$X = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, \qquad X^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}, \qquad e^{At} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} e^t & 0 \\ 0 & e^{3t} \end{bmatrix}\begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} e^t & \frac{1}{2}(e^{3t} - e^t) \\ 0 & e^{3t} \end{bmatrix}.$$

Check: at $t = 0$ this is I. Its derivative is $\left[\begin{smallmatrix} e^t & \frac{1}{2}(3e^{3t} - e^t) \\ 0 & 3e^{3t} \end{smallmatrix}\right]$, which at $t = 0$ equals $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 3 \end{smallmatrix}\right] = A$, as Proposition 25.5 requires.

Common trap. The exponential of a matrix is *not* the matrix of exponentials of its entries. In Example 25.7 the $(1, 2)$ entry of e^{At} is $\frac{1}{2}(e^{3t} - e^t)$, not e^t, and the zero entry gives 0, not $e^0 = 1$. Similarly $e^A e^B = e^{A+B}$ is true when $AB = BA$ but false in general. When A cannot be diagonalized, new kinds of terms appear: for $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ we have $A^2 = 0$, the series stops, and $e^{At} = I + At = \left[\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right]$ contains $t\,e^{0t}$, which is not a pure exponential.

25.4 Higher-order equations become first-order systems

The second-order equation $y'' + by' + ky = 0$ becomes a 2×2 system by the same trick as for Fibonacci: take $\boldsymbol{u} = (y', y)$ and add the trivial equation $y' = y'$,

$$\boldsymbol{u}' = \begin{bmatrix} y'' \\ y' \end{bmatrix} = \begin{bmatrix} -b & -k \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y' \\ y \end{bmatrix}.$$

Its characteristic polynomial is $\det \left[\begin{smallmatrix} -b-\lambda & -k \\ 1 & -\lambda \end{smallmatrix}\right] = \lambda^2 + b\lambda + k$, exactly what we get by substituting $y = e^{\lambda t}$ into the original equation. An equation of order m gives an $m \times m$ matrix with the coefficients (with signs changed) across the first row, 1's just below the diagonal, and zeros elsewhere. For such a matrix the vector $(\lambda^{m-1}, \ldots, \lambda, 1)$ is an eigenvector whenever λ is an eigenvalue: the rows below the first just say $\lambda^j = \lambda \cdot \lambda^{j-1}$, and the first row is the characteristic equation.

Example 25.8 (A third-order equation)**.** Solve $y''' + 2y'' - y' - 2y = 0$ and find the first column of e^{At} for the corresponding system.

With $\boldsymbol{u} = (y'', y', y)$, the equation $y''' = -2y'' + y' + 2y$ gives

$$\boldsymbol{u}' = A\boldsymbol{u}, \qquad A = \begin{bmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Expanding along the first row,

$$\det(A - \lambda I) = (-2-\lambda)\lambda^2 - 1 \cdot (-\lambda) + 2 \cdot 1 = -(\lambda^3 + 2\lambda^2 - \lambda - 2) = -(\lambda - 1)(\lambda + 1)(\lambda + 2),$$

so $\lambda = 1, -1, -2$, with eigenvectors $(\lambda^2, \lambda, 1)$: $\boldsymbol{x}_1 = (1, 1, 1)$, $\boldsymbol{x}_2 = (1, -1, 1)$, $\boldsymbol{x}_3 = (4, -2, 1)$. The last component of $\boldsymbol{u}(t) = C_1 e^t \boldsymbol{x}_1 + C_2 e^{-t} \boldsymbol{x}_2 + C_3 e^{-2t} \boldsymbol{x}_3$ is

$$y(t) = C_1 e^t + C_2 e^{-t} + C_3 e^{-2t}.$$

Because $\lambda = 1 > 0$, solutions blow up unless $C_1 = 0$.

To get the first column of $e^{At} = X e^{\Lambda t} X^{-1}$, only the first column of X^{-1} is needed. By the cofactor formula of Section 22 (X^{-1} is the transposed cofactor matrix divided by $\det X$), that column is the cofactors of the first row of X divided by $\det X$. For $X = \left[\begin{smallmatrix} 1 & 1 & 4 \\ 1 & -1 & -2 \\ 1 & 1 & 1 \end{smallmatrix}\right]$ the cofactors of the first row are

$$C_{11} = (-1)(1) - (-2)(1) = 1, \qquad C_{12} = -\big(1 \cdot 1 - (-2) \cdot 1\big) = -3, \qquad C_{13} = 1 \cdot 1 - (-1) \cdot 1 = 2,$$

and $\det X = 1 \cdot 1 + 1 \cdot (-3) + 4 \cdot 2 = 6$. So the first column of X^{-1} is $(\frac{1}{6}, -\frac{1}{2}, \frac{1}{3})$, and the first column of e^{At} is

$$\tfrac{1}{6} e^t \, \boldsymbol{x}_1 - \tfrac{1}{2} e^{-t} \, \boldsymbol{x}_2 + \tfrac{1}{3} e^{-2t} \, \boldsymbol{x}_3.$$

At $t = 0$ it equals $\frac{1}{6}(1,1,1) - \frac{1}{2}(1,-1,1) + \frac{1}{3}(4,-2,1) = (1,0,0)$, the first column of I, as it must.

25.5 Exercises

Exercise 25.9 (Skew-symmetric systems conserve length)**.** The matrix in

$$\frac{d\boldsymbol{u}}{dt} = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \boldsymbol{u}, \quad \text{that is,} \quad u_1' = cu_2 - bu_3, \quad u_2' = au_3 - cu_1, \quad u_3' = bu_1 - au_2,$$

is skew-symmetric ($A^{\mathsf{T}} = -A$). Compute $\frac{d}{dt}\|\boldsymbol{u}(t)\|^2$. What does this say about $\boldsymbol{u}(t)$?

Solution.

$$\frac{d}{dt}(u_1^2+u_2^2+u_3^2) = 2u_1u_1' + 2u_2u_2' + 2u_3u_3' = 2u_1(cu_2-bu_3) + 2u_2(au_3-cu_1) + 2u_3(bu_1-au_2) = 0,$$

since the six products cancel in pairs. So $\|\boldsymbol{u}(t)\| = \|\boldsymbol{u}(0)\|$ for all t: the solution stays on the sphere of radius $\|\boldsymbol{u}(0)\|$, neither decaying nor blowing up. In matrix form, $\frac{d}{dt}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u} = \boldsymbol{u}^{\mathsf{T}}(A^{\mathsf{T}}+A)\boldsymbol{u} = 0$ because $A^{\mathsf{T}} = -A$. Consistently, the eigenvalues of a real skew-symmetric matrix are purely imaginary (Section 23). □

Exercise 25.10 (The exponential of a rotation generator)**.** For $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ show that $e^{At} = \left[\begin{smallmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{smallmatrix}\right]$. Describe the solutions of $d\boldsymbol{u}/dt = A\boldsymbol{u}$.

Solution. $A^2 = -I$, so $A^3 = -A$, $A^4 = I$, and the powers repeat with period 4. Group the series:

$$e^{At} = \Big(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \cdots\Big)I + \Big(t - \frac{t^3}{3!} + \cdots\Big)A = (\cos t)\, I + (\sin t)\, A,$$

which is the stated rotation matrix. (Equivalently, the eigenvalues are $\pm i$ and $e^{\pm it} = \cos t \pm i \sin t$.) Every solution $\boldsymbol{u}(t) = e^{At}\boldsymbol{u}(0)$ rotates $\boldsymbol{u}(0)$ through the angle t: it circles the origin at constant length, the case $\operatorname{Re}\lambda = 0$ between decay and blow-up. □

Exercise 25.11 (Stability by trace and determinant)**.** Decide whether solutions of $d\boldsymbol{u}/dt = A\boldsymbol{u}$ decay, blow up, or neither, for

$$\text{(a) } \begin{bmatrix} -1 & 5 \\ -1 & -1 \end{bmatrix}, \quad \text{(b) } \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}, \quad \text{(c) } \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix}, \quad \text{(d) } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Solution. Use Proposition 25.3. (a) $\operatorname{tr} = -2 < 0$, $\det = 1 + 5 = 6 > 0$: stable. The eigenvalues are $-1 \pm i\sqrt{5}$, so solutions spiral into the origin. (b) $\operatorname{tr} = -6$, $\det = 9 - 1 = 8$: stable, with real eigenvalues -2 and -4. (c) $\operatorname{tr} = 0$, $\det = 4$: the eigenvalues are $\pm 2i$, with real part 0. Solutions oscillate like $\cos 2t$ and $\sin 2t$ forever, neither decaying nor growing. (This is $y'' + 4y = 0$ written as a system for (y, y').) (d) $\operatorname{tr} = 5$, $\det = 4 - 6 = -2 < 0$: one positive and one negative eigenvalue, $\frac{1}{2}(5 \pm \sqrt{33})$, so solutions blow up except along one eigenvector line. □

Exercise 25.12 (A damped oscillator)**.** Solve $y'' + 3y' + 2y = 0$ with $y(0) = 1$, $y'(0) = 0$ by writing it as a system for $\boldsymbol{u} = (y', y)$.

Solution. $A = \left[\begin{smallmatrix} -3 & -2 \\ 1 & 0 \end{smallmatrix}\right]$ and $\det(A - \lambda I) = \lambda^2 + 3\lambda + 2 = (\lambda+1)(\lambda+2)$. The eigenvectors are $(\lambda, 1)$: $\boldsymbol{x}_1 = (-1, 1)$ for $\lambda = -1$ and $\boldsymbol{x}_2 = (-2, 1)$ for $\lambda = -2$. The start $\boldsymbol{u}(0) = (y'(0), y(0)) = (0, 1) = c_1(-1, 1) + c_2(-2, 1)$ gives $-c_1 - 2c_2 = 0$ and $c_1 + c_2 = 1$, so $c_2 = -1$, $c_1 = 2$. The second component of $\boldsymbol{u}(t) = 2e^{-t}\boldsymbol{x}_1 - e^{-2t}\boldsymbol{x}_2$ is

$$y(t) = 2e^{-t} - e^{-2t}.$$

Check: $y(0) = 1$, $y'(t) = -2e^{-t} + 2e^{-2t}$ so $y'(0) = 0$, and $y'' + 3y' + 2y = (2e^{-t} - 4e^{-2t}) + (-6e^{-t} + 6e^{-2t}) + (4e^{-t} - 2e^{-2t}) = 0$. Both eigenvalues are negative (trace -3, determinant 2), so $y \to 0$ without oscillating. □

26 Markov matrices; Fourier series

This section gives two applications. The first uses eigenvalues: a Markov matrix moves probabilities or populations around without creating or destroying any, and its powers approach a steady state given by the eigenvector for $\lambda = 1$. The second uses projections: expanding a vector in an orthonormal basis is easy, and Fourier series are exactly that expansion carried out in a space of functions.

26.1 A and A^{T} have the same eigenvalues

Proposition 26.1 (Eigenvalues of the transpose)**.** *For every square matrix A,* $\det(A^{\mathsf{T}} - \lambda I) = \det(A - \lambda I)$. *So A^{T} has the same eigenvalues as A, with the same multiplicities.*

Proof. $(A - \lambda I)^{\mathsf{T}} = A^{\mathsf{T}} - \lambda I$, because $I^{\mathsf{T}} = I$; and a matrix and its transpose have the same determinant (property 10, Section 20). □

The eigen*vectors* are usually different: an eigenvector of A^{T} for λ lies in $\mathbf{N}(A^{\mathsf{T}} - \lambda I)$, the left nullspace of $A - \lambda I$, not in its nullspace.

26.2 Markov matrices

Definition 26.2 (Markov matrix)**.** A square matrix is a *Markov matrix* if every entry is nonnegative and every column adds to 1. With $\mathbf{1} = (1, 1, \ldots, 1)$, the column condition reads $\mathbf{1}^{\mathsf{T}} A = \mathbf{1}^{\mathsf{T}}$.

The two conditions come from probability. Entry a_{ij} is the probability (or fraction of the population) that moves from state j to state i in one step; probabilities are never negative, and everything in state j must go somewhere, so column j adds to 1. A typical example is

$$A = \begin{bmatrix} 0.1 & 0.01 & 0.3 \\ 0.2 & 0.99 & 0.3 \\ 0.7 & 0 & 0.4 \end{bmatrix}.$$

Powers of a Markov matrix are Markov matrices: products of nonnegative matrices are nonnegative, and $\mathbf{1}^{\mathsf{T}} A^2 = (\mathbf{1}^{\mathsf{T}} A)A = \mathbf{1}^{\mathsf{T}} A = \mathbf{1}^{\mathsf{T}}$. The same computation shows that the step $\boldsymbol{u}_{k+1} = A\boldsymbol{u}_k$ conserves the total: $\mathbf{1}^{\mathsf{T}} \boldsymbol{u}_{k+1} = \mathbf{1}^{\mathsf{T}} A \boldsymbol{u}_k = \mathbf{1}^{\mathsf{T}} \boldsymbol{u}_k$.

Theorem 26.3 (Eigenvalues of a Markov matrix)**.** *Let A be a Markov matrix. Then $\lambda = 1$ is an eigenvalue of A, and every eigenvalue satisfies $|\lambda| \le 1$.*

Proof. $\lambda = 1$. Every column of $A - I$ adds to 0, so adding all the rows of $A - I$ gives the zero row: $\mathbf{1}^{\mathsf{T}}(A - I) = \mathbf{0}^{\mathsf{T}}$. The rows are dependent, so $A - I$ is singular and 1 is an eigenvalue. (Equivalently, $A^{\mathsf{T}}\mathbf{1} = \mathbf{1}$, so $\mathbf{1}$ is an eigenvector of A^{T}, and Proposition 26.1 transfers the eigenvalue to A.)

$|\lambda| \le 1$. By Proposition 26.1 it suffices to treat A^{T}. Let $A^{\mathsf{T}}\boldsymbol{y} = \lambda \boldsymbol{y}$ with $\boldsymbol{y} \neq \mathbf{0}$, and choose i with $|y_i|$ as large as possible. Row i of A^{T} is column i of A, so

$$|\lambda|\,|y_i| = \Big|\sum_j a_{ji}\, y_j\Big| \le \sum_j a_{ji}\,|y_j| \le |y_i| \sum_j a_{ji} = |y_i|,$$

using $a_{ji} \ge 0$ and the column sum 1. Divide by $|y_i| > 0$. □

The eigenvector for $\lambda = 1$ is the interesting one. It is found in the nullspace of $A - I$ (not from $\mathbf{1}$, which lies in the left nullspace). Two more facts, which we state without proof (they are part of the Perron–Frobenius theorem): this eigenvector can be chosen with no negative components, and if every entry of A is positive then every other eigenvalue has $|\lambda| < 1$. Zero entries can allow other eigenvalues on the unit circle: the permutation $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ is a Markov matrix with $\lambda = -1$, and its powers swap two states forever without settling down.

Example 26.4 (Finding the steady-state eigenvector)**.** For the 3×3 matrix above,

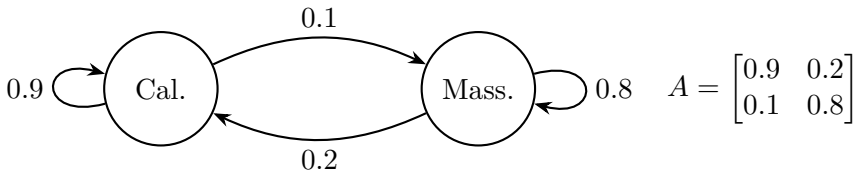
$$A - I = \begin{bmatrix} -0.9 & 0.01 & 0.3 \\ 0.2 & -0.01 & 0.3 \\ 0.7 & 0 & -0.6 \end{bmatrix}.$$

The last row says $0.7x_1 = 0.6x_3$, so try $x_1 = 0.6$, $x_3 = 0.7$. The first row then requires $-0.54 + 0.01x_2 + 0.21 = 0$, so $x_2 = 33$. Check the middle row: $0.12 - 0.33 + 0.21 = 0$. So $\boldsymbol{x}_1 = (0.6,\, 33,\, 0.7)$, with all components positive as promised. The middle state, which keeps 99% of its contents each step, ends up holding almost everything.

Now combine Theorem 26.3 with the solution of difference equations (Section 24). If A has independent eigenvectors, $\lambda_1 = 1$, and $|\lambda_i| < 1$ for $i \geq 2$, then

$$\boldsymbol{u}_k = c_1\, 1^k\, \boldsymbol{x}_1 + c_2\lambda_2^k \boldsymbol{x}_2 + \cdots + c_n\lambda_n^k \boldsymbol{x}_n \;\longrightarrow\; c_1\boldsymbol{x}_1.$$

The limit is the steady state: the $\boldsymbol{x}_1$-component of the start. Because the total is conserved, the multiple c_1 is fixed by $\mathbf{1}^{\mathsf{T}}(c_1\boldsymbol{x}_1) = \mathbf{1}^{\mathsf{T}}\boldsymbol{u}_0$. Steady states come from $\lambda = 0$ for differential equations (Section 25) but from $\lambda = 1$ for powers.



$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

Figure 32: A two-state Markov chain and its matrix. Column j of A lists where the population of state j goes in one year: the arrows leaving California carry 0.9 and 0.1, which form column 1 and add to 1.

Example 26.5 (California and Massachusetts)**.** Each year 90% of the people in California stay and 10% move to Massachusetts, while 80% of Massachusetts stays and 20% moves to California (Figure 32). Start with everyone in Massachusetts: $\boldsymbol{u}_0 = (u_{\text{Cal}}, u_{\text{Mass}}) = (0, 1000)$.

After one year $\boldsymbol{u}_1 = A\boldsymbol{u}_0 = 1000\,(0.2, 0.8) = (200, 800)$. For the long run we need the eigenvalues: $\lambda_1 = 1$ (Markov), and the trace 1.7 gives $\lambda_2 = 0.7$ (check: $\det A = 0.72 - 0.02 = 0.7$). The eigenvectors come from

$$A - I = \begin{bmatrix} -0.1 & 0.2 \\ 0.1 & -0.2 \end{bmatrix} \Rightarrow \boldsymbol{x}_1 = (2, 1), \qquad A - 0.7I = \begin{bmatrix} 0.2 & 0.2 \\ 0.1 & 0.1 \end{bmatrix} \Rightarrow \boldsymbol{x}_2 = (-1, 1).$$

Matching the start, $c_1(2,1) + c_2(-1,1) = (0, 1000)$ gives $c_1 = \frac{1000}{3}$ and $c_2 = \frac{2000}{3}$. Hence

$$\boldsymbol{u}_k = \frac{1000}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{2000}{3}(0.7)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

At $k = 1$ this gives $(\frac{2000-1400}{3}, \frac{1000+1400}{3}) = (200, 800)$, as computed directly; then $(340, 660)$ and $(438, 562)$. The populations approach $(666\frac{2}{3}, 333\frac{1}{3})$: two thirds in California, from any start with 1000 people. The gap shrinks by the factor 0.7 each year.

Example 26.6 (The two-city model as a Markov chain)**.** The matrix $A = \left[\begin{smallmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{smallmatrix}\right]$ of the two-city example of Section 24 is a Markov matrix, which is why $\lambda_1 = 1$ appeared there; $\lambda_2 = 1.5 - 1 = 0.5$ came from the trace. Its powers approach

$$A^k \to \begin{bmatrix} 0.6 & 0.6 \\ 0.4 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix},$$

a matrix whose columns are both the steady state $(0.6, 0.4)$. So $A^k \boldsymbol{u}_0 \to (0.6, 0.4)\,(u_1 + u_2)$ for every start: the long-run split does not depend on where the 1000 people begin. Starting from $(0, 1000) = (600, 400) - 600(1, -1)$, the first year gives $A(0, 1000) = (300, 700)$, and $\boldsymbol{u}_k = (600, 400) - 600(0.5)^k(1, -1)$ settles at $(600, 400)$, the same limit as from $(1000, 0)$. Only the size of the decaying $\lambda = 0.5$ part depends on the start.

Remark 26.7 (Rows instead of columns)*.* Many probability texts multiply a row vector by the matrix from the left, $\boldsymbol{p}_{k+1}^{\mathsf{T}} = \boldsymbol{p}_k^{\mathsf{T}} P$. Then $P = A^{\mathsf{T}}$ and it is the *rows* that add to 1. Everything above holds after transposing.

26.3 Expansion in an orthonormal basis

Let $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ be an orthonormal basis of $\mathbb{R}^n$. Every $\boldsymbol{v}$ is a combination $\boldsymbol{v} = x_1\boldsymbol{q}_1 + \cdots + x_n\boldsymbol{q}_n$, and orthonormality makes the coefficients trivial to find.

Proposition 26.8 (Coefficients in an orthonormal basis)**.** *If $\boldsymbol{v} = x_1\boldsymbol{q}_1 + \cdots + x_n\boldsymbol{q}_n$ with orthonormal $\boldsymbol{q}_i$, then $x_i = \boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{v}$. In matrix form, $Q\boldsymbol{x} = \boldsymbol{v}$ gives $\boldsymbol{x} = Q^{\mathsf{T}}\boldsymbol{v}$.*

Proof. Take the dot product of both sides with $\boldsymbol{q}_i$. Since $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_j = 0$ for $j \neq i$ and $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_i = 1$, every term on the right vanishes except $x_i\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_i = x_i$. In matrix language, Q is square with orthonormal columns, so $Q^{-1} = Q^{\mathsf{T}}$ (Section 19), and row i of $Q^{\mathsf{T}}\boldsymbol{v}$ is $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{v}$. □

Each term $x_i\boldsymbol{q}_i = (\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{v})\boldsymbol{q}_i$ is the projection of $\boldsymbol{v}$ onto the line through $\boldsymbol{q}_i$, so the expansion writes $\boldsymbol{v}$ as the sum of its projections onto n perpendicular lines. For instance, with $\boldsymbol{q}_1 = (1, 1)/\sqrt{2}$ and $\boldsymbol{q}_2 = (1, -1)/\sqrt{2}$, the vector $\boldsymbol{v} = (3, 1)$ has $x_1 = 4/\sqrt{2} = 2\sqrt{2}$ and $x_2 = 2/\sqrt{2} = \sqrt{2}$, and indeed $2\sqrt{2}\,\boldsymbol{q}_1 + \sqrt{2}\,\boldsymbol{q}_2 = (2, 2) + (1, -1) = (3, 1)$. No linear system had to be solved.

26.4 Fourier series: orthogonal functions

Fourier's idea was to do the same thing with functions. A periodic function $f(x) = f(x + 2\pi)$ is to be written as

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots .$$

The "vectors" are now functions, and the basis $1, \cos x, \sin x, \cos 2x, \sin 2x, \ldots$ is infinite: we have moved to an infinite-dimensional vector space. To speak of orthogonality we need an inner product. For vectors, $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{w} = v_1w_1 + \cdots + v_nw_n$ multiplies components and adds. A function has a continuum of "components" $f(x)$, so we multiply values and integrate over one period:

$$(f, g) = \int_0^{2\pi} f(x)\, g(x)\, dx, \qquad \|f\|^2 = \int_0^{2\pi} f(x)^2\, dx.$$

Proposition 26.9 (The Fourier basis is orthogonal)**.** *The functions* $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ *are orthogonal for this inner product, and* $\|1\|^2 = 2\pi$*,* $\|\cos kx\|^2 = \|\sin kx\|^2 = \pi$ *for* $k \geq 1$*.*

Proof. For an integer $m \neq 0$, $\int_0^{2\pi} \cos mx\, dx = \int_0^{2\pi} \sin mx\, dx = 0$, since each is a whole number of periods. The product formulas

$$\begin{aligned} \cos jx \cos kx &= \tfrac{1}{2}[\cos(j-k)x + \cos(j+k)x], \\ \sin jx \sin kx &= \tfrac{1}{2}[\cos(j-k)x - \cos(j+k)x], \\ \sin jx \cos kx &= \tfrac{1}{2}[\sin(j+k)x + \sin(j-k)x] \end{aligned}$$

turn every product of two different basis functions into such terms, so its integral is 0. (For example $\int_0^{2\pi} \sin x \cos x\, dx = \left[\tfrac{1}{2}\sin^2 x\right]_0^{2\pi} = 0$.) When $j = k \geq 1$, $\cos^2 kx = \tfrac{1}{2}(1 + \cos 2kx)$ and $\sin^2 kx = \tfrac{1}{2}(1 - \cos 2kx)$ integrate to π, and $\int_0^{2\pi} 1\, dx = 2\pi$. □

Theorem 26.10 (Fourier coefficients)**.** *If* $f(x) = a_0 + \sum_{k\geq 1}(a_k \cos kx + b_k \sin kx)$*, then*

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\, dx, \qquad a_k = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos kx\, dx, \qquad b_k = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin kx\, dx.$$

Proof. Exactly as in Proposition 26.8: take the inner product of both sides with $\cos kx$, that is, multiply by $\cos kx$ and integrate from 0 to 2π. By Proposition 26.9 every term on the right integrates to zero except $a_k \int_0^{2\pi} \cos^2 kx\, dx = \pi a_k$. Dividing by π gives a_k; the same steps with $\sin kx$ give b_k, and with the constant 1 (whose squared length is 2π) give a_0. (We integrate the infinite sum term by term; for the functions met in practice this is justified, and the series then converges to $f(x)$ wherever f is continuous.) □

Each coefficient is a projection: $a_k = (f, \cos kx)/(\cos kx, \cos kx)$ is the formula $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$ for projecting $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$ (Section 17), with functions in place of vectors. The basis is orthogonal but not orthonormal, which is why we divide by π or 2π. In particular, a_0 is the average value of f.

Example 26.11 (A square wave)**.** Let $f(x) = 1$ for $0 < x < \pi$ and $f(x) = -1$ for $\pi < x < 2\pi$, extended with period 2π. Its average is 0, so $a_0 = 0$. For $k \geq 1$,

$$a_k = \frac{1}{\pi}\Big(\int_0^{\pi} \cos kx\, dx - \int_{\pi}^{2\pi} \cos kx\, dx\Big) = \frac{1}{\pi}(0 - 0) = 0,$$

$$b_k = \frac{1}{\pi}\Big(\int_0^{\pi} \sin kx\, dx - \int_{\pi}^{2\pi} \sin kx\, dx\Big) = \frac{1}{\pi}\Big(\frac{1-\cos k\pi}{k} - \frac{\cos k\pi - 1}{k}\Big) = \frac{2\,(1-(-1)^k)}{\pi k}.$$

So $b_k = 4/(\pi k)$ for odd k and $b_k = 0$ for even k:

$$f(x) = \frac{4}{\pi}\Big(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\Big).$$

Only sines appear because f is odd about $x = \pi$, like the sines. At $x = \pi/2$, where $f = 1$, the series gives $1 = \frac{4}{\pi}\big(1 - \frac{1}{3} + \frac{1}{5} - \cdots\big)$, which is Leibniz's formula $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. Figure 33 shows partial sums approaching the square wave.

Common trap. The inner product of functions needs the integral over a full period, here $[0, 2\pi]$. Over other intervals the sines and cosines are generally *not* orthogonal (for example $\int_0^{\pi/2} \sin x \cos x\, dx = \frac{1}{2}$), and the coefficient formulas fail. Also remember the different normalization of a_0: the constant function has squared length 2π, not π.

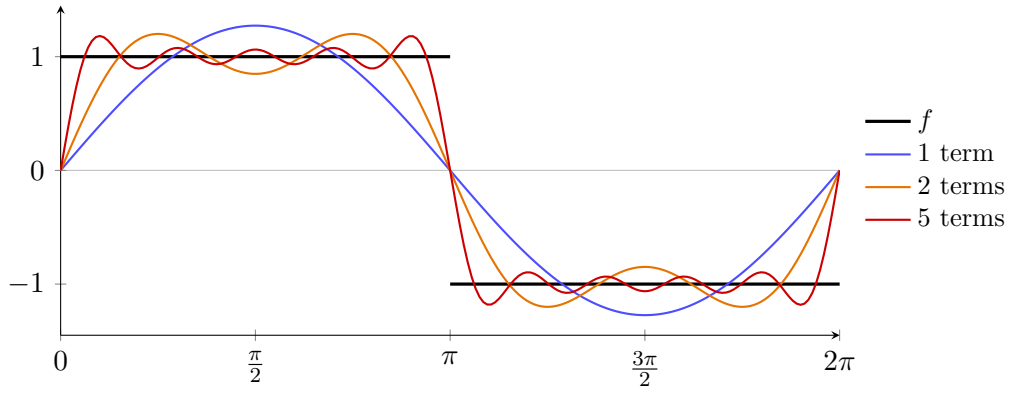


Figure 33: The square wave of Example 26.11 and the partial sums of its Fourier series with 1, 2 and 5 nonzero terms. Each added term is the projection of f onto one more basis function; the sums close in on f except near the jumps, where an overshoot of about 18% of the jump's half-height persists (the Gibbs phenomenon).

26.5 Exercises

Exercise 26.12 (A jumping particle)**.** A particle jumps between positions A and B. From A it stays with probability 0.6 and jumps to B with probability 0.4; from B it jumps to A with probability 0.2 and stays with probability 0.8. It starts at A. Find its probability vector after one step, after n steps, and in the limit.

Solution. With $\boldsymbol{p} = (\text{prob. at } A, \text{prob. at } B)$, the Markov matrix is $M = \left[\begin{smallmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{smallmatrix}\right]$ and $\boldsymbol{p}_0 = (1, 0)$. One step: $\boldsymbol{p}_1 = M\boldsymbol{p}_0 = (0.6, 0.4)$. The eigenvalues are 1 and (trace 1.4) 0.4, with eigenvectors $(1, 2)$ (from $M - I = \left[\begin{smallmatrix} -0.4 & 0.2 \\ 0.4 & -0.2 \end{smallmatrix}\right]$) and $(1, -1)$. Matching $(1, 0) = c_1(1, 2) + c_2(1, -1)$ gives $c_1 = \frac{1}{3}$, $c_2 = \frac{2}{3}$, so

$$\boldsymbol{p}_n = \frac{1}{3}\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2}{3}(0.4)^n \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 + 2(0.4)^n \\ 2 - 2(0.4)^n \end{bmatrix}.$$

For $n = 1$ this is $(0.6, 0.4)$ again. As $n \to \infty$, $\boldsymbol{p}_n \to (\frac{1}{3}, \frac{2}{3})$: the particle forgets where it started. □

Exercise 26.13 (Completing a Markov matrix)**.** Complete the first two rows $\left[\begin{smallmatrix} 0.7 & 0.1 & 0.2 \\ 0.1 & 0.6 & 0.3 \end{smallmatrix}\right]$ to a 3×3 Markov matrix and find its steady-state eigenvector. Why is $(1, 1, \ldots, 1)$ the steady state of every symmetric Markov matrix?

Solution. Each column must add to 1, so the third row is (0.2, 0.3, 0.5). The completed matrix is symmetric, so its rows also add to 1; then every row of $A - I$ adds to 0, which says $(A - I)(1, 1, 1) = \mathbf{0}$. The steady state is $(1, 1, 1)$, or $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ as probabilities. The same argument works for any symmetric Markov matrix: symmetry turns the column sums into row sums, so $A\mathbf{1} = \mathbf{1}$. □

Exercise 26.14 (Which kinds of matrix?)**.** Decide which of these properties hold for $A = \left[\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right]$ and $B = \frac{1}{3}\left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{smallmatrix}\right]$: invertible, orthogonal, projection, permutation, diagonalizable, Markov.

Solution. A: $\det A = -1$, so invertible; $AA^{\mathsf{T}} = I$, so orthogonal; $A^2 = I \neq A$, so not a projection; one 1 in each row and column, so a permutation; symmetric, so diagonalizable (eigenvalues $1, 1, -1$ with eigenvectors $(1, 0, 1)$, $(0, 1, 0)$, $(1, 0, -1)$); columns add to 1 and entries are nonnegative, so

Markov. B: rank 1, so singular and not orthogonal; $B^2 = B$ (each entry of B^2 is $3 \cdot \frac{1}{9}$) and $B = B^{\mathsf{T}}$, so it is the projection onto the line through $(1,1,1)$; not a permutation; diagonalizable (eigenvalues $1, 0, 0$ with eigenvectors $(1,1,1)$ and the plane $x + y + z = 0$); Markov, since its columns add to 1. □

Exercise 26.15 (Fourier series of a ramp)**.** Find the Fourier coefficients of $f(x) = x$ on $0 \le x < 2\pi$ (extended with period 2π).

Solution. $a_0 = \frac{1}{2\pi}\int_0^{2\pi} x\,dx = \frac{1}{2\pi} \cdot 2\pi^2 = \pi$, the average value. Integrating by parts with $k \ge 1$,

$$\int_0^{2\pi} x\cos kx\,dx = \Big[\frac{x\sin kx}{k}\Big]_0^{2\pi} - \int_0^{2\pi} \frac{\sin kx}{k}\,dx = 0 - 0 = 0,$$
$$\int_0^{2\pi} x\sin kx\,dx = \Big[-\frac{x\cos kx}{k}\Big]_0^{2\pi} + \int_0^{2\pi} \frac{\cos kx}{k}\,dx = -\frac{2\pi}{k} + 0.$$

So $a_k = 0$ and $b_k = -2/k$: $x = \pi - 2\big(\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \cdots\big)$ for $0 < x < 2\pi$. □

27 Exam 2 review

Part II is the heart of the course. It started from orthogonality and least squares, built the determinant, and arrived at eigenvalues and powers of a matrix. This section gathers the facts to have at your fingertips for the second exam, lists the quick checks that catch most errors, and then works through review problems of the kind the exam asks, most of them from old 18.06 exams. The exam itself covers Sections 16–24; differential equations and Markov matrices (Sections 25–26) build on the same ideas and are tested later.

27.1 What to know

Orthogonality and projections (Sections 16–18). The row space is orthogonal to the nullspace, and the column space is orthogonal to the left nullspace. Projecting $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$ gives $\boldsymbol{p} = \dfrac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}\,\boldsymbol{a}$ and $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}$. Projecting onto the column space of A (independent columns): solve the normal equations $A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$, then $\boldsymbol{p} = A\hat{\boldsymbol{x}}$ and $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$, with $P^2 = P = P^{\mathsf{T}}$. The error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is perpendicular to every column of A. Least squares chooses $\hat{\boldsymbol{x}}$ to minimize $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$, and the answer is exactly this projection.

Orthonormal columns and Gram–Schmidt (Section 19). Q has orthonormal columns exactly when $Q^{\mathsf{T}}Q = I$; then the projection onto its column space is QQ^{T}, and a square Q has $Q^{-1} = Q^{\mathsf{T}}$. Gram–Schmidt subtracts from each new vector its projections onto the earlier *orthogonal* directions, then normalizes; the result is $A = QR$ with R upper triangular.

Determinants (Sections 20–22). Three properties define det: $\det I = 1$; exchanging rows reverses the sign; det is linear in each row separately. Consequences: equal rows or a zero row give 0; subtracting a multiple of one row from another changes nothing; triangular matrices give the product of the diagonal; $\det A = 0$ exactly when A is singular; $\det AB = \det A \det B$; $\det A^{\mathsf{T}} = \det A$. The big formula has $n!$ signed terms, one entry from each row and column. The cofactor

expansion is $\det A = \sum_j a_{ij}C_{ij}$, and it leads to $A^{-1} = C^{\mathsf{T}}/\det A$, Cramer's rule and $|\det A| =$ volume.

Eigenvalues (Sections 23–24). Solve $\det(A - \lambda I) = 0$, then find $\mathbf{N}(A - \lambda I)$. The sum of the eigenvalues is the trace, the product is the determinant. With n independent eigenvectors, $A = X\Lambda X^{-1}$, $A^k = X\Lambda^k X^{-1}$, and $\boldsymbol{u}_k = A^k\boldsymbol{u}_0 = \sum c_i\lambda_i^k\boldsymbol{x}_i$ where $X\boldsymbol{c} = \boldsymbol{u}_0$. Distinct eigenvalues guarantee independent eigenvectors; a repeated eigenvalue may not.

27.2 Quick checks

An exam rewards answers that are checked as they are produced. The cheapest checks:

- eigenvalues: their sum must equal the trace and their product the determinant, and $A\boldsymbol{x} = \lambda\boldsymbol{x}$ can be verified in seconds;
- projections: $P^2 = P$, $P^{\mathsf{T}} = P$, $P\boldsymbol{a} = \boldsymbol{a}$ for vectors already in the subspace, and the error must be perpendicular to the subspace;
- Gram–Schmidt: each new vector must have zero dot product with the earlier ones;
- inverses and factorizations: substitute $k = 1$ into a formula for A^k, multiply $A\,C^{\mathsf{T}}$ to see $(\det A)I$, and expand a determinant along a second row or column.

27.3 Solved review problems

Example 27.1 (Projection onto a line, and its powers)**.** Let $\boldsymbol{a} = (2, 1, 2)$. (a) Find the projection matrix P onto the line through $\boldsymbol{a}$. (b) Find its rank, column space and eigenvalues, and an eigenvector for each eigenvalue. (c) Solve $\boldsymbol{u}_{k+1} = P\boldsymbol{u}_k$ with $\boldsymbol{u}_0 = (9, 9, 0)$.

(a) Here $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a} = 9$, so

$$P = \frac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}} = \frac{1}{9}\begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix}.$$

(b) Every column is a multiple of $\boldsymbol{a}$, so $\operatorname{rank} P = 1$ and $\mathbf{C}(P)$ is the line through $\boldsymbol{a}$. A rank-one 3×3 matrix has a two-dimensional nullspace, so $\lambda = 0$ is an eigenvalue twice, with eigenvectors such as $(1, -2, 0)$ and $(1, 0, -1)$, the vectors perpendicular to $\boldsymbol{a}$. The trace $\frac{4+1+4}{9} = 1$ gives the third eigenvalue $\lambda = 1$, with eigenvector $\boldsymbol{a}$ itself: $P\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})/9 = \boldsymbol{a}$.

(c) The first step projects $\boldsymbol{u}_0$ onto the line:

$$\boldsymbol{u}_1 = P\boldsymbol{u}_0 = \frac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{u}_0}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}\,\boldsymbol{a} = \frac{18 + 9 + 0}{9}\,\boldsymbol{a} = 3\boldsymbol{a} = (6, 3, 6).$$

Since $\boldsymbol{u}_1$ is already on the line, projecting again changes nothing: $P^2 = P$, so $\boldsymbol{u}_k = P^k\boldsymbol{u}_0 = P\boldsymbol{u}_0 = (6, 3, 6)$ for every $k \geq 1$. In the eigenvector language of Section 24, $\boldsymbol{u}_0 = 3\boldsymbol{a} + (3, 6, -6)$, where $(3, 6, -6)$ is perpendicular to $\boldsymbol{a}$ and has $\lambda = 0$; that part disappears after one step and the $\lambda = 1$ part $3\boldsymbol{a}$ stays forever. For a matrix that is not a projection one would find all eigenvalues and eigenvectors and use $\boldsymbol{u}_k = \sum c_i\lambda_i^k\boldsymbol{x}_i$.

Example 27.2 (Best line through the origin)**.** Fit the line $y = Dt$ to the data points $(t, y) = (1, 4)$, $(2, 5)$, $(3, 8)$, and say what vector is projected onto what line.

The three equations $1 \cdot D = 4$, $2 \cdot D = 5$, $3 \cdot D = 8$ form $A\boldsymbol{x} = \boldsymbol{b}$ with a single column $A = (1, 2, 3)$ and $\boldsymbol{b} = (4, 5, 8)$; they have no solution. The normal equation is

$$A^{\mathsf{T}} A\,\hat{D} = A^{\mathsf{T}}\boldsymbol{b}: \qquad 14\,\hat{D} = 4 + 10 + 24 = 38, \qquad \hat{D} = \frac{19}{7} \approx 2.71.$$

There are two pictures (Figure 34). In the ty-plane we fit a line to three points; the errors are the vertical distances. In $\mathbb{R}^3$ we project the vector $\boldsymbol{b} = (4, 5, 8)$ onto the line through $(1, 2, 3)$, the column space of A: $\boldsymbol{p} = \frac{19}{7}(1, 2, 3) = \frac{1}{7}(19, 38, 57)$ and $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \frac{1}{7}(9, -3, -1)$. Check: $\boldsymbol{e} \cdot (1, 2, 3) = \frac{1}{7}(9 - 6 - 3) = 0$. The components of $\boldsymbol{e}$ are exactly the vertical errors in the first picture.

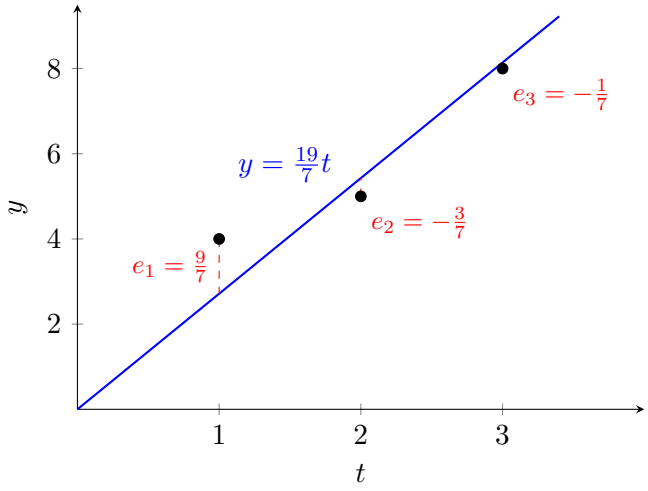


Figure 34: The least-squares line through the origin for Example 27.2. The vertical errors $e_i = y_i - \frac{19}{7}t_i$ are the components of the error vector $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$, which is perpendicular to the column $(1, 2, 3)$ in $\mathbb{R}^3$.

Example 27.3 (Gram–Schmidt in a plane)**.** The vectors $\boldsymbol{a}_1 = (1, 2, 3)$ and $\boldsymbol{a}_2 = (1, 1, 1)$ span a plane. Find two orthogonal vectors in it, then an orthonormal pair.

Keep $\boldsymbol{a}_1$. From $\boldsymbol{a}_2$ subtract its projection onto $\boldsymbol{a}_1$:

$$\boldsymbol{B} = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_1^{\mathsf{T}}\boldsymbol{a}_2}{\boldsymbol{a}_1^{\mathsf{T}}\boldsymbol{a}_1}\,\boldsymbol{a}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix} - \frac{6}{14}\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}1 - \frac{3}{7}\\1 - \frac{6}{7}\\1 - \frac{9}{7}\end{bmatrix} = \frac{1}{7}\begin{bmatrix}4\\1\\-2\end{bmatrix}.$$

Check: $\boldsymbol{a}_1 \cdot (4, 1, -2) = 4 + 2 - 6 = 0$. Dividing by the lengths $\sqrt{14}$ and $\sqrt{16 + 1 + 4} = \sqrt{21}$ gives the orthonormal pair $\boldsymbol{q}_1 = (1, 2, 3)/\sqrt{14}$, $\boldsymbol{q}_2 = (4, 1, -2)/\sqrt{21}$. The square roots appear only in this last step, so it pays to postpone it.

Example 27.4 (What the eigenvalues alone tell)**.** A 4×4 matrix A has eigenvalues $\lambda_1, \lambda_2, \lambda_3, \lambda_4$. When is it invertible? What are $\det A^{-1}$ and $\operatorname{tr}(A + I)$?

A is invertible exactly when no λ_i is 0: an eigenvalue 0 means a nonzero vector in the nullspace, and $\det A = \lambda_1\lambda_2\lambda_3\lambda_4$. The eigenvalues of A^{-1} are $1/\lambda_i$ (Section 23), so $\det A^{-1} = \dfrac{1}{\lambda_1\lambda_2\lambda_3\lambda_4}$.

Adding I adds 1 to every eigenvalue (Section 23), so $\operatorname{tr}(A+I) = \lambda_1+\lambda_2+\lambda_3+\lambda_4+4$, which is also $\operatorname{tr} A + 4$ directly.

Example 27.5 (Tridiagonal determinants repeat with period six)**.** Let A_n be the $n \times n$ tridiagonal matrix with 1's on the diagonal and on both neighbouring diagonals, and $D_n = \det A_n$. Find a recurrence for D_n and solve it.

Recurrence. Expand along the first row, which has 1's in columns 1 and 2 only. The $(1,1)$ cofactor is D_{n-1}. The $(1,2)$ cofactor is -1 times the determinant of A_n without row 1 and column 2; that matrix has first column $(1,0,\dots,0)$, and expanding down it leaves A_{n-2}. So

$$D_n = D_{n-1} - D_{n-2}, \qquad D_1 = 1, \quad D_2 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0.$$

As a matrix system. With $\boldsymbol{u}_n = (D_n, D_{n-1})$, $\boldsymbol{u}_n = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right] \boldsymbol{u}_{n-1}$. The characteristic equation is $\lambda^2 - \lambda + 1 = 0$, so

$$\lambda = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i = e^{\pm i\pi/3}.$$

Both have $|\lambda|^2 = \frac{1}{4} + \frac{3}{4} = 1$: the sequence neither grows nor decays. Since $\lambda^6 = e^{\pm 2\pi i} = 1$, the matrix $M = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ is diagonalizable with $\Lambda^6 = I$, so $M^6 = X\Lambda^6 X^{-1} = I$ and the sequence repeats every six steps. Running the recurrence:

$$D_1, \dots, D_6 = 1,\ 0,\ -1,\ -1,\ 0,\ 1, \qquad \text{then } D_7 = 1,\ D_8 = 0, \dots$$

so D_n depends only on n modulo 6.

Example 27.6 (A family of symmetric matrices)**.** Let A_n be symmetric with zeros on the diagonal and $1, 2, \dots, n-1$ just above and below it, for instance

$$A_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}, \qquad A_4 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{bmatrix}.$$

(a) Find the projection matrix onto $\mathbf{C}(A_3)$. (b) Find the eigenvalues and eigenvectors of A_3. (c) Find the projection onto $\mathbf{C}(A_4)$. (d) Show that A_n is singular for odd n and invertible for even n.

(a) Column 3 is twice column 1, so $\mathbf{C}(A_3)$ is the plane spanned by $(0,1,0)$ and $(1,0,2)$. These two columns are already orthogonal, so P is the sum of the two projections onto lines:

$$P = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & 0 & \frac{2}{5} \\ 0 & 1 & 0 \\ \frac{2}{5} & 0 & \frac{4}{5} \end{bmatrix}.$$

The formula $A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ with these two columns gives the same matrix. Check: P fixes all three columns of A_3.

(b) $\det(A_3 - \lambda I) = -\lambda^3 + 5\lambda = -\lambda(\lambda^2 - 5)$, so $\lambda = 0, \sqrt{5}, -\sqrt{5}$ (trace check: the sum is 0). The eigenvectors are $(-2,0,1)$ for 0 and $(1, \pm\sqrt{5}, 2)$ for $\pm\sqrt{5}$; for instance $A_3(1,\sqrt{5},2) = (\sqrt{5},\ 5,\ 2\sqrt{5}) = \sqrt{5}\,(1,\sqrt{5},2)$. The eigenvector $(-2,0,1)$ spans the nullspace, perpendicular to $\mathbf{C}(A_3)$, so $P = I - \frac{1}{5}(-2,0,1)(-2,0,1)^{\mathsf{T}}$ is another route to (a).

(c) If A_4 is invertible, its column space is all of $\mathbb{R}^4$ and the projection is $P = I$. Expanding along row 1 (only the entry 1 in column 2), then down the first column of the resulting 3×3 minor,

$$\det A_4 = -1 \cdot \begin{vmatrix} 1 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{vmatrix} = -1 \cdot 1 \cdot \begin{vmatrix} 0 & 3 \\ 3 & 0 \end{vmatrix} = -1 \cdot (-9) = 9 \neq 0.$$

So $P = I$.

(d) Expand $D_n = \det A_n$ along the last row, whose only nonzero entry is $n-1$ in column $n-1$. Its minor has last column $(0, \dots, 0, n-1)$; expanding down that column leaves A_{n-2}. With the sign $(-1)^{n+(n-1)} = -1$,

$$D_n = -(n-1)^2 \, D_{n-2}, \qquad D_1 = 0, \quad D_2 = -1.$$

So $D_n = 0$ for every odd n, and for even n the determinant is $\pm\big(1 \cdot 3 \cdot 5 \cdots (n-1)\big)^2 \neq 0$: for example $D_4 = -9D_2 = 9$ and $D_6 = -25D_4 = -225$.

Example 27.7 (The big formula when many entries are zero)**.** For the matrix below, find $\det A$ from the big formula, the cofactors $C_{11}, \dots, C_{14}$, and the first column of A^{-1}:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{bmatrix}.$$

Big formula. Of the 24 terms, only those avoiding the four zeros survive. Rows 3 and 4 must use columns 3 and 4, so rows 1 and 2 use columns 1 and 2: four terms in all. A term's sign is that of its column permutation:

$$\det A = +1 \cdot 6 \cdot 9 \cdot 12 - 1 \cdot 6 \cdot 10 \cdot 11 - 2 \cdot 5 \cdot 9 \cdot 12 + 2 \cdot 5 \cdot 10 \cdot 11 = 648 - 660 - 1080 + 1100 = 8.$$

(The permutations are $(1,2,3,4)$, $(1,2,4,3)$, $(2,1,3,4)$ and $(2,1,4,3)$, with 0, 1, 1 and 2 exchanges.)

Cofactors. $C_{11} = \det \left[\begin{smallmatrix} 6 & 7 & 8 \\ 0 & 9 & 10 \\ 0 & 11 & 12 \end{smallmatrix}\right] = 6(108 - 110) = -12$ and $C_{12} = -\det \left[\begin{smallmatrix} 5 & 7 & 8 \\ 0 & 9 & 10 \\ 0 & 11 & 12 \end{smallmatrix}\right] = -5(-2) = 10$, each expanded down its first column. $C_{13} = C_{14} = 0$ without computation: their minors have first two columns $(5,0,0)$ and $(6,0,0)$, which are dependent. Check: $\det A = 1 \cdot (-12) + 2 \cdot 10 + 0 + 0 = 8$.

First column of A^{-1}. By the formula $A^{-1} = C^{\mathsf{T}} / \det A$ of Section 22 it is the first row of cofactors divided by the determinant:

$$\frac{1}{\det A}\,(C_{11}, C_{12}, C_{13}, C_{14}) = \frac{1}{8}\,(-12, 10, 0, 0) = \Big(-\frac{3}{2}, \frac{5}{4}, 0, 0\Big).$$

A third check: A is block triangular, and $\det \left[\begin{smallmatrix} 1 & 2 \\ 5 & 6 \end{smallmatrix}\right] \det \left[\begin{smallmatrix} 9 & 10 \\ 11 & 12 \end{smallmatrix}\right] = (-4)(-2) = 8$.

27.4 Exercises

Exercise 27.8. Fit the best straight line $y = C + Dt$ to the points $(-1, 1)$, $(0, 2)$, $(1, 4)$, and find the error vector.

Solution. $A = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$, $\boldsymbol{b} = (1, 2, 4)$. Because the times add to zero, the columns of A are orthogonal and $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right]$, while $A^{\mathsf{T}}\boldsymbol{b} = (7, 3)$. So $\hat{C} = \frac{7}{3}$, $\hat{D} = \frac{3}{2}$. The fitted values are $\frac{7}{3} + \frac{3}{2}t = \frac{5}{6}, \frac{7}{3}, \frac{23}{6}$ at $t = -1, 0, 1$, and $\boldsymbol{e} = (1, 2, 4) - (\frac{5}{6}, \frac{14}{6}, \frac{23}{6}) = (\frac{1}{6}, -\frac{1}{3}, \frac{1}{6})$, which is perpendicular to both columns: $\frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0$ and $-\frac{1}{6} + 0 + \frac{1}{6} = 0$. □

Exercise 27.9. A 3×3 matrix A has eigenvalues $1, 2, 3$. Find $\det A$, $\operatorname{tr} A$, the eigenvalues of $A^2 - A$, and $\det(A^2 - A)$. Is A diagonalizable?

Solution. $\det A = 1\cdot 2\cdot 3 = 6$ and $\operatorname{tr} A = 6$. By Section 23 the eigenvalues of $A^2 - A$ are $\lambda^2 - \lambda = 0, 2, 6$, so $\det(A^2 - A) = 0$: the matrix $A^2 - A = A(A - I)$ is singular, as it must be since $A - I$ has the eigenvalue 0. A is diagonalizable because its three eigenvalues are distinct (Section 24). □

Exercise 27.10. Let T_n be tridiagonal with 2's on the diagonal and -1's on the two neighbouring diagonals. Use cofactors to show $\det T_n = 2\det T_{n-1} - \det T_{n-2}$, and find $\det T_n$.

Solution. Expanding along the first row exactly as in Example 27.5: the $(1,1)$ term gives $2\det T_{n-1}$; the $(1,2)$ entry -1 has cofactor -1 times a determinant whose first column is $(-1, 0, \dots, 0)$, which equals $-\det T_{n-2}$. Altogether $\det T_n = 2\det T_{n-1} + (-1)(-1)(-\det T_{n-2}) = 2\det T_{n-1} - \det T_{n-2}$. With $\det T_1 = 2$ and $\det T_2 = 4 - 1 = 3$, the recurrence gives $4, 5, 6, \dots$; by induction $\det T_n = n + 1$, since $2n - (n-1) = n + 1$. (In matrix form, $\left[\begin{smallmatrix} 2 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has the repeated eigenvalue 1 and only one eigenvector, which is why the answer grows linearly instead of like a power.) □

28 Exam 2

The second unit exam tests Sections 16–24 through four problems: determinants of matrices built from orthonormal vectors, a least-squares line, projections and the next Gram–Schmidt vector, and a determinant that is a polynomial in one repeated entry. Each problem is restated below and solved with the reasoning an examiner wants to see, followed by further practice problems of the same kind. Try each problem before reading its solution.

28.1 The exam problems

Example 28.1 (Determinants from orthonormal vectors)**.** Let $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ be orthonormal vectors in $\mathbb{R}^3$ and $Q = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{bmatrix}$. Find all possible values of (a) $\det Q$; (b) $\det \begin{bmatrix} \boldsymbol{q}_1 + \boldsymbol{q}_2 & \boldsymbol{q}_2 + \boldsymbol{q}_3 & \boldsymbol{q}_3 + \boldsymbol{q}_1 \end{bmatrix}$; (c) $\det Q$ times $\det \begin{bmatrix} \boldsymbol{q}_2 & \boldsymbol{q}_3 & \boldsymbol{q}_1 \end{bmatrix}$.

(a) $Q^{\mathsf{T}}Q = I$, so $(\det Q)^2 = \det Q^{\mathsf{T}} \det Q = \det I = 1$ and $\det Q = \pm 1$. Both values occur: $Q = I$ gives $+1$, and $Q = \operatorname{diag}(1, 1, -1)$ gives -1.

(b) The new columns are combinations of the old ones:

$$\begin{bmatrix} \boldsymbol{q}_1 + \boldsymbol{q}_2 & \boldsymbol{q}_2 + \boldsymbol{q}_3 & \boldsymbol{q}_3 + \boldsymbol{q}_1 \end{bmatrix} = Q\,M, \qquad M = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

since column j of QM is Q times column j of M. Expanding along the first row, $\det M = 1\cdot(1-0) - 0 + 1\cdot(1-0) = 2$, so the determinant is $\det Q \det M = \pm 2$. Linearity in each column gives the same answer: of the eight determinants obtained by splitting every column, all but two have a repeated column and vanish, leaving $\det[\boldsymbol{q}_1\ \boldsymbol{q}_2\ \boldsymbol{q}_3] + \det[\boldsymbol{q}_2\ \boldsymbol{q}_3\ \boldsymbol{q}_1] = 2\det Q$ by part (c).

(c) $[\boldsymbol{q}_2\ \boldsymbol{q}_3\ \boldsymbol{q}_1]$ is Q with its columns cycled, which takes two exchanges (swap columns 1 and 2, then columns 2 and 3). An even number of exchanges keeps the determinant, so the product is $(\det Q)^2 = 1$, whichever sign $\det Q$ has.

Example 28.2 (A least-squares line through 21 points)**.** Measurements are taken at the 21 times $t = -10, -9, \dots, 9, 10$. All are $b_i = 0$ except $b = 1$ at the middle time $t = 0$. (a) Find the least-squares line $C + Dt$. (b) Onto which subspace is $\boldsymbol{b}$ projected? Give a basis, and a nonzero vector perpendicular to it.

(a) Exact fitting would require the 21 equations $C + Dt_i = b_i$, that is $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{x} = (C, D)$ and A the 21×2 matrix whose columns are $\mathbf{1} = (1, \dots, 1)$ and $\boldsymbol{t} = (-10, \dots, 10)$. The normal equations need

$$\mathbf{1}^{\mathsf{T}}\mathbf{1} = 21, \qquad \mathbf{1}^{\mathsf{T}}\boldsymbol{t} = \sum t_i = 0, \qquad \boldsymbol{t}^{\mathsf{T}}\boldsymbol{t} = 2(1^2 + 2^2 + \cdots + 10^2) = 2 \cdot 385 = 770,$$

and $A^{\mathsf{T}}\boldsymbol{b} = (\mathbf{1}^{\mathsf{T}}\boldsymbol{b}, \boldsymbol{t}^{\mathsf{T}}\boldsymbol{b}) = (1, 0)$, since the only nonzero b_i sits at $t = 0$. So

$$\begin{bmatrix} 21 & 0 \\ 0 & 770 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \hat{C} = \frac{1}{21}, \quad \hat{D} = 0.$$

The best line is the horizontal line $y = \frac{1}{21}$ (Figure 35). Because the times are symmetric about 0, the two columns of A are orthogonal and the normal equations uncouple: $\hat{C}$ is the average of the b_i, and $\hat{D} = \boldsymbol{t}^{\mathsf{T}}\boldsymbol{b}/\boldsymbol{t}^{\mathsf{T}}\boldsymbol{t} = 0$. A symmetric data set cannot prefer an upward or a downward slope.

(b) $\boldsymbol{b} \in \mathbb{R}^{21}$ is projected onto the column space of A, a plane in $\mathbb{R}^{21}$ with basis $\mathbf{1}$ and $\boldsymbol{t}$. The error vector is perpendicular to it:

$$\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \boldsymbol{b} - \tfrac{1}{21}\mathbf{1} = \tfrac{1}{21}(\underbrace{-1, \dots, -1}_{10}, 20, \underbrace{-1, \dots, -1}_{10}).$$

Check: its components add to $\frac{1}{21}(20 - 20) = 0$, and $\boldsymbol{t}^{\mathsf{T}}\boldsymbol{e} = 0$ by symmetry. A simpler perpendicular vector is $(1, -2, 1, 0, \dots, 0)$: its components add to 0, and against $\boldsymbol{t}$ it gives $-10 + 18 - 8 = 0$.

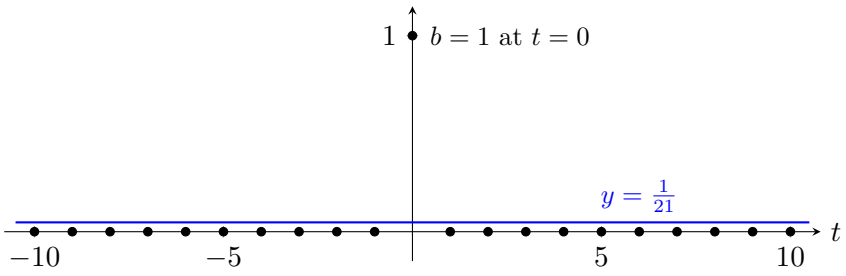


Figure 35: The data of Example 28.2 and its least-squares line. Twenty points sit at 0 and one at 1; the best line is the horizontal line at the average height $\frac{1}{21}$. Any tilt would help on one side of $t = 0$ exactly as much as it hurts on the other.

Example 28.3 (Projections from Gram–Schmidt and the next vector)**.** Gram–Schmidt turns independent vectors $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ in $\mathbb{R}^5$ into orthonormal $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$. Let A and Q be the 5×3 matrices with these columns. (a) Give formulas for the projection matrices P_A and P_Q onto their column spaces. (b) Is $P_Q = P_A$? What is $P_Q Q$? What is $\det P_Q$? (c) A fourth vector $\boldsymbol{a}_4$ is added, with $\boldsymbol{a}_1, \dots, \boldsymbol{a}_4$ independent. Which of these is the next Gram–Schmidt vector $\boldsymbol{q}_4$?

$$(1)\ \frac{P_Q\boldsymbol{a}_4}{\|P_Q\boldsymbol{a}_4\|}, \qquad (2)\ \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|} \text{ with } \boldsymbol{w} = \boldsymbol{a}_4 - \frac{\boldsymbol{a}_4^{\mathsf{T}}\boldsymbol{a}_1}{\boldsymbol{a}_1^{\mathsf{T}}\boldsymbol{a}_1}\boldsymbol{a}_1 - \frac{\boldsymbol{a}_4^{\mathsf{T}}\boldsymbol{a}_2}{\boldsymbol{a}_2^{\mathsf{T}}\boldsymbol{a}_2}\boldsymbol{a}_2 - \frac{\boldsymbol{a}_4^{\mathsf{T}}\boldsymbol{a}_3}{\boldsymbol{a}_3^{\mathsf{T}}\boldsymbol{a}_3}\boldsymbol{a}_3, \qquad (3)\ \frac{\boldsymbol{a}_4 - P_A\boldsymbol{a}_4}{\|\boldsymbol{a}_4 - P_A\boldsymbol{a}_4\|}.$$

(a) $P_A = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ and $P_Q = Q(Q^{\mathsf{T}}Q)^{-1}Q^{\mathsf{T}} = QQ^{\mathsf{T}}$, because $Q^{\mathsf{T}}Q = I$.

(b) Yes, $P_Q = P_A$. Gram–Schmidt never leaves the span: each $\boldsymbol{q}_j$ is a combination of $\boldsymbol{a}_1, \dots, \boldsymbol{a}_j$ and each $\boldsymbol{a}_j$ a combination of $\boldsymbol{q}_1, \dots, \boldsymbol{q}_j$, so $\mathbf{C}(Q) = \mathbf{C}(A)$, and the projection onto a subspace depends only on the subspace, not on the basis used to describe it. (Substituting $A = QR$ into $A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ and cancelling also works.) Next, $P_Q Q = QQ^{\mathsf{T}}Q = Q$: vectors already in the subspace are not moved. Finally $\det P_Q = 0$: P_Q is 5×5 with rank 3, and every vector perpendicular to $\mathbf{C}(Q)$ is sent to $\mathbf{0}$.

(c) Choice (3). The vector $\boldsymbol{a}_4 - P_A\boldsymbol{a}_4$ is the error of projecting $\boldsymbol{a}_4$ onto $\mathbf{C}(A)$, so it is perpendicular to $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$; it is nonzero because $\boldsymbol{a}_4$ is not in $\mathbf{C}(A)$; dividing by its length makes it a unit vector. This is exactly what Gram–Schmidt computes, usually written $\boldsymbol{a}_4 - (\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_4)\boldsymbol{q}_1 - (\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{a}_4)\boldsymbol{q}_2 - (\boldsymbol{q}_3^{\mathsf{T}}\boldsymbol{a}_4)\boldsymbol{q}_3$. Choice (1) is the projection itself, which lies *in* the old subspace, the opposite of what we want. Choice (2) is the tempting one: it would be right with the orthogonal $\boldsymbol{q}_i$ in place of the $\boldsymbol{a}_i$, but subtracting projections onto lines that are not perpendicular to each other does not leave a vector perpendicular to all of them.

The failure of choice (2) is easy to see in a small case. In $\mathbb{R}^3$ take $\boldsymbol{a}_1 = (1,0,0)$, $\boldsymbol{a}_2 = (1,1,0)$ and a new vector $\boldsymbol{a}_3 = (1,1,1)$. The formula of choice (2) gives $\boldsymbol{w} = (1,1,1) - \frac{1}{1}(1,0,0) - \frac{2}{2}(1,1,0) = (-1,0,1)$, and $\boldsymbol{w} \cdot \boldsymbol{a}_1 = -1 \neq 0$. The correct $\boldsymbol{a}_3 - P_A\boldsymbol{a}_3$ is $(0,0,1)$, perpendicular to the whole plane of $\boldsymbol{a}_1, \boldsymbol{a}_2$.

Example 28.4 (A determinant that is a polynomial)**.** A 4×4 matrix has the same number x in every position of its first row and first column (the exam writes $\times$ for it); the other nine entries are fixed numbers. (a) $\det A$ is a polynomial in x. What is its largest possible degree? (b) If the nine numbers form the 3×3 identity, find $\det A$ and the values of x that make it zero.

(a) Degree at most 2. Each term of the big formula takes exactly one entry from each row and each column. All the x's lie in row 1 or column 1, so a term can contain at most one x from row 1 and at most one from column 1; the corner entry $a_{11} = x$ uses up both. So each term has at most two factors x. Part (b) shows the degree 2 is actually reached.

(b) The matrix is

$$A = \begin{bmatrix} x & x & x & x \\ x & 1 & 0 & 0 \\ x & 0 & 1 & 0 \\ x & 0 & 0 & 1 \end{bmatrix}.$$

Subtract x times rows 2, 3 and 4 from row 1; this does not change the determinant and clears the last three entries of row 1, leaving $x - 3x^2$ in the corner. The result is lower triangular, so

$$\det A = (x - 3x^2) \cdot 1 \cdot 1 \cdot 1 = x(1 - 3x).$$

It is zero for $x = 0$ (a zero row) and for $x = \frac{1}{3}$. For $x = \frac{1}{3}$ the dependence is visible: rows 2, 3, 4 add to $(1,1,1,1)$, which is 3 times row 1.

Common trap. In Example 28.4(a) it is tempting to count the seven x's and guess degree 7, or to count four per row and guess 4. The big formula is the right tool for questions about the *form* of a determinant: one entry per row and per column limits how many x's can meet in one term.

28.2 Further practice

Exercise 28.5. With $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ orthonormal in $\mathbb{R}^3$, find the determinants

$$\det\begin{bmatrix}\boldsymbol{q}_1-\boldsymbol{q}_2 & \boldsymbol{q}_2-\boldsymbol{q}_3 & \boldsymbol{q}_3-\boldsymbol{q}_1\end{bmatrix} \qquad \text{and} \qquad \det\begin{bmatrix}\boldsymbol{q}_1+\boldsymbol{q}_2 & \boldsymbol{q}_1-\boldsymbol{q}_2 & \boldsymbol{q}_3\end{bmatrix}.$$

Solution. The three columns of the first matrix add to $\mathbf{0}$, so they are dependent and the determinant is 0. For the second, write it as QM with $M = \left[\begin{smallmatrix}1 & 1 & 0\\ 1 & -1 & 0\\ 0 & 0 & 1\end{smallmatrix}\right]$; $\det M = 1\cdot(-1) - 1\cdot 1 = -2$, so the determinant is $-2\det Q = \mp 2$. Geometrically, the first two columns are perpendicular with length $\sqrt{2}$ each, so the box has volume $\sqrt{2}\cdot\sqrt{2}\cdot 1 = 2$, in agreement with the volume interpretation of Section 22. □

Exercise 28.6. Measurements at $t = -2, -1, 0, 1, 2$ are $\boldsymbol{b} = (0, 0, 0, 0, 1)$. Find the least-squares line $C + Dt$ and check that the error is perpendicular to both columns of A.

Solution. The columns $\mathbf{1}$ and $\boldsymbol{t} = (-2, -1, 0, 1, 2)$ are orthogonal, with $\mathbf{1}^\mathsf{T}\mathbf{1} = 5$ and $\boldsymbol{t}^\mathsf{T}\boldsymbol{t} = 10$; also $\mathbf{1}^\mathsf{T}\boldsymbol{b} = 1$ and $\boldsymbol{t}^\mathsf{T}\boldsymbol{b} = 2$. So $\hat{C} = \frac{1}{5}$ and $\hat{D} = \frac{2}{10} = \frac{1}{5}$: the line is $y = \frac{1}{5}(1 + t)$. Its values at the five times are $\boldsymbol{p} = \frac{1}{5}(-1, 0, 1, 2, 3)$, so $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \frac{1}{5}(1, 0, -1, -2, 2)$. Check: $1 + 0 - 1 - 2 + 2 = 0$ and $-2 + 0 + 0 - 2 + 4 = 0$. Unlike Example 28.2, the lone nonzero measurement is off-centre, so the line tilts toward it. □

Exercise 28.7. Repeat Example 28.4(b) when the nine numbers form $2I$ instead of I.

Solution. Subtract $\frac{x}{2}$ times rows 2, 3, 4 from row 1. The corner becomes $x - 3\cdot\frac{x}{2}\cdot x = x - \frac{3}{2}x^2$, the rest of row 1 becomes 0, and the remaining diagonal is $2, 2, 2$. So $\det A = 8\big(x - \frac{3}{2}x^2\big) = 8x - 12x^2 = 4x(2 - 3x)$, which vanishes for $x = 0$ and $x = \frac{2}{3}$. The degree is again 2, as part (a) of the example predicts. □

Part III

Positive definite matrices and applications

29 Symmetric matrices and positive definiteness

Part II ended with the eigenvalue picture of a square matrix: when A has n independent eigenvectors, $A = X\Lambda X^{-1}$ (Section 24). A matrix with special structure should have special eigenvalues and eigenvectors; Markov matrices, for instance, always have $\lambda = 1$ (Section 26). This unit studies the most important special class, the *symmetric* matrices $A = A^\mathsf{T}$. Two facts carry the whole unit:

- the eigenvalues of a real symmetric matrix are **real**;
- its eigenvectors can be chosen **perpendicular**, so X becomes an orthogonal matrix Q.

Once eigenvalues are known to be real we can ask for their *signs*, and a surprising shortcut appears: the signs of the pivots are the signs of the eigenvalues. The symmetric matrices whose eigenvalues are all positive, the positive definite matrices, are where pivots, determinants and eigenvalues finally come together.

29.1 Real eigenvalues

Definition 29.1 (Symmetric matrix)**.** A square real matrix A is *symmetric* if $A^{\mathsf{T}} = A$, that is $a_{ij} = a_{ji}$ for all i, j.

A real matrix can have complex eigenvalues: the 90° rotation $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has $\lambda = \pm i$. So to prove that symmetric matrices do not, we must allow λ and $\boldsymbol{x}$ to be complex for a moment and see what symmetry forces. The tool is the complex conjugate: if $z = a + ib$ then $\bar{z} = a - ib$ and $\bar{z}z = a^2 + b^2 \geq 0$.

Lemma 29.2 (Conjugate length)**.** *For a complex vector* $\boldsymbol{x} = (x_1, \ldots, x_n)$*,*

$$\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = \bar{x}_1 x_1 + \cdots + \bar{x}_n x_n = |x_1|^2 + \cdots + |x_n|^2,$$

which is real, and positive unless $\boldsymbol{x} = \boldsymbol{0}$*.*

Proof. Each term is $\bar{x}_k x_k = (a_k - ib_k)(a_k + ib_k) = a_k^2 + b_k^2 = |x_k|^2$: the cross terms $\pm i a_k b_k$ cancel. A sum of such squares is zero only when every a_k and b_k is zero. □

Without the bar the lemma fails: for $\boldsymbol{x} = (1, i)$ we get $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = 1 + i^2 = 0$ although $\boldsymbol{x} \neq \boldsymbol{0}$. Section 30 builds the geometry of complex vectors on this observation.

Theorem 29.3 (Real eigenvalues)**.** *Every eigenvalue of a real symmetric matrix is real.*

Proof. Let $A\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \boldsymbol{0}$, where λ and $\boldsymbol{x}$ may be complex. Take complex conjugates of every factor. Because A is real, $\bar{A} = A$ and

$$A\bar{\boldsymbol{x}} = \bar{\lambda}\,\bar{\boldsymbol{x}}.$$

(This step alone shows that the complex eigenvalues of any real matrix come in conjugate pairs $\lambda, \bar{\lambda}$.) Now transpose and use $A^{\mathsf{T}} = A$, which is the only place symmetry enters:

$$\bar{\boldsymbol{x}}^{\mathsf{T}}A = \bar{\lambda}\,\bar{\boldsymbol{x}}^{\mathsf{T}}.$$

Multiply this on the right by $\boldsymbol{x}$, and multiply the original equation $A\boldsymbol{x} = \lambda\boldsymbol{x}$ on the left by $\bar{\boldsymbol{x}}^{\mathsf{T}}$:

$$\bar{\boldsymbol{x}}^{\mathsf{T}}A\boldsymbol{x} = \bar{\lambda}\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x}, \qquad \bar{\boldsymbol{x}}^{\mathsf{T}}A\boldsymbol{x} = \lambda\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x}.$$

The left sides agree, so $(\lambda - \bar{\lambda})\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = 0$. By Lemma 29.2, $\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} > 0$, so $\lambda = \bar{\lambda}$: the imaginary part of λ is zero. □

Once λ is real, $A - \lambda I$ is a real singular matrix, so its nullspace contains real nonzero vectors. Eigenvectors of a real symmetric matrix can therefore always be taken real.

Remark 29.4. If A is complex, the same proof goes through provided $\bar{A}^{\mathsf{T}} = A$: transposing $\bar{A}\bar{\boldsymbol{x}} = \bar{\lambda}\bar{\boldsymbol{x}}$ gives $\bar{\boldsymbol{x}}^{\mathsf{T}}\bar{A}^{\mathsf{T}} = \bar{\lambda}\bar{\boldsymbol{x}}^{\mathsf{T}}$, and we need $\bar{A}^{\mathsf{T}} = A$ to continue. So the right complex analogue of "symmetric" is "equal to its conjugate transpose". These Hermitian matrices are the subject of Section 30.

29.2 Perpendicular eigenvectors and the spectral theorem

Theorem 29.5 (Orthogonal eigenvectors)**.** *If A is real symmetric, $A\boldsymbol{x} = \lambda_1\boldsymbol{x}$ and $A\boldsymbol{y} = \lambda_2\boldsymbol{y}$ with $\lambda_1 \neq \lambda_2$, then $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 0$.*

Proof. Compute $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{y}$ in two ways. Using $A\boldsymbol{y} = \lambda_2\boldsymbol{y}$ gives $\lambda_2\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$. Using symmetry, $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{y} = (A^{\mathsf{T}}\boldsymbol{x})^{\mathsf{T}}\boldsymbol{y} = (A\boldsymbol{x})^{\mathsf{T}}\boldsymbol{y} = \lambda_1\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y}$. Hence $(\lambda_1 - \lambda_2)\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 0$, and since $\lambda_1 \neq \lambda_2$, $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{y} = 0$. □

When an eigenvalue is repeated there is a whole plane (or larger subspace) of eigenvectors, and not every pair in it is perpendicular. The identity matrix is the extreme case: every vector is an eigenvector. The correct statement is that the eigenvectors *can be chosen* perpendicular, and that there are always enough of them. That is the content of the main theorem.

Theorem 29.6 (Spectral theorem; principal axis theorem)**.** *Every real symmetric $n \times n$ matrix can be factored as*

$$A = Q\Lambda Q^{\mathsf{T}},$$

where Q is an orthogonal matrix ($Q^{\mathsf{T}}Q = I$) whose columns are orthonormal eigenvectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ of A, and Λ is the real diagonal matrix of eigenvalues. Conversely, every matrix of the form $Q\Lambda Q^{\mathsf{T}}$ with Λ real diagonal is symmetric.

Proof. The converse is one line: $(Q\Lambda Q^{\mathsf{T}})^{\mathsf{T}} = (Q^{\mathsf{T}})^{\mathsf{T}}\Lambda^{\mathsf{T}}Q^{\mathsf{T}} = Q\Lambda Q^{\mathsf{T}}$.

For the factorization we use induction on n; the case $n = 1$ is trivial. The characteristic polynomial of A has at least one root λ_1, which is real by Theorem 29.3, with a real unit eigenvector $\boldsymbol{q}_1$. Extend $\boldsymbol{q}_1$ to an orthonormal basis $\boldsymbol{q}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_n$ of $\mathbb{R}^n$ (Gram–Schmidt, Section 19) and let Q_1 be the orthogonal matrix with these columns. The first column of $Q_1^{\mathsf{T}}AQ_1$ is $Q_1^{\mathsf{T}}A\boldsymbol{q}_1 = \lambda_1 Q_1^{\mathsf{T}}\boldsymbol{q}_1 = \lambda_1\boldsymbol{e}_1$. The matrix $Q_1^{\mathsf{T}}AQ_1$ is symmetric, because $(Q_1^{\mathsf{T}}AQ_1)^{\mathsf{T}} = Q_1^{\mathsf{T}}A^{\mathsf{T}}Q_1 = Q_1^{\mathsf{T}}AQ_1$, so its first row is also $\lambda_1\boldsymbol{e}_1^{\mathsf{T}}$:

$$Q_1^{\mathsf{T}}AQ_1 = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & A_2 \end{bmatrix}, \qquad A_2 \text{ symmetric of size } n-1.$$

By induction $A_2 = Q_2\Lambda_2 Q_2^{\mathsf{T}}$ with Q_2 orthogonal. Put $Q = Q_1 \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & Q_2 \end{bmatrix}$, a product of orthogonal matrices and hence orthogonal. Then $Q^{\mathsf{T}}AQ = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \Lambda_2 \end{bmatrix} = \Lambda$, that is $A = Q\Lambda Q^{\mathsf{T}}$. Reading $AQ = Q\Lambda$ column by column shows $A\boldsymbol{q}_k = \lambda_k\boldsymbol{q}_k$. □

Compared with the general diagonalization $A = X\Lambda X^{-1}$, two things have improved: the eigenvalues in Λ are real, and the eigenvector matrix X has become an orthogonal Q, whose inverse is simply Q^{T}. Mathematicians call the set of eigenvalues the *spectrum*, hence the name; in mechanics the same statement is the principal axis theorem: in the right axes (the eigenvectors) a symmetric operation does not couple the directions together.

Corollary 29.7 (A sum of projections)**.** *If $A = Q\Lambda Q^{\mathsf{T}}$ then, multiplying columns times rows,*

$$A = \lambda_1\boldsymbol{q}_1\boldsymbol{q}_1^{\mathsf{T}} + \lambda_2\boldsymbol{q}_2\boldsymbol{q}_2^{\mathsf{T}} + \cdots + \lambda_n\boldsymbol{q}_n\boldsymbol{q}_n^{\mathsf{T}}.$$

Each $P_k = \boldsymbol{q}_k\boldsymbol{q}_k^{\mathsf{T}}$ is the projection matrix onto the line through $\boldsymbol{q}_k$; these projections satisfy $P_k^2 = P_k$, $P_jP_k = 0$ for $j \neq k$, and $P_1 + \cdots + P_n = QQ^{\mathsf{T}} = I$.

Proof. The product $Q(\Lambda Q^{\mathsf{T}})$ is a sum of (column k of Q) times (row k of ΛQ^{T}), and row k of ΛQ^{T} is $\lambda_k \boldsymbol{q}_k^{\mathsf{T}}$. For a unit vector the projection formula $\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$ of Section 17 reduces to $\boldsymbol{q}_k\boldsymbol{q}_k^{\mathsf{T}}$. Finally $P_j P_k = \boldsymbol{q}_j(\boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{q}_k)\boldsymbol{q}_k^{\mathsf{T}}$ is P_k when $j = k$ and 0 otherwise. □

So a symmetric matrix acts on $\boldsymbol{x}$ in three moves: split $\boldsymbol{x}$ into its pieces $(\boldsymbol{q}_k^{\mathsf{T}}\boldsymbol{x})\boldsymbol{q}_k$ along perpendicular axes, stretch piece k by λ_k, add.

Example 29.8 (A spectral decomposition, worked in full)**.** Take $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$. The trace is -1 and the determinant is $-2-4=-6$, so $\lambda^2+\lambda-6=(\lambda-2)(\lambda+3)=0$ and $\lambda_1=2$, $\lambda_2=-3$: real, as promised.

Eigenvectors. $A-2I = \left[\begin{smallmatrix} -1 & 2 \\ 2 & -4 \end{smallmatrix}\right]$ has nullspace spanned by $(2,1)$; $A+3I = \left[\begin{smallmatrix} 4 & 2 \\ 2 & 1 \end{smallmatrix}\right]$ has nullspace spanned by $(1,-2)$. Their dot product is $2-2=0$, as Theorem 29.5 predicts. Normalizing,

$$\boldsymbol{q}_1 = \tfrac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{q}_2 = \tfrac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad A = Q\Lambda Q^{\mathsf{T}} = \tfrac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}\tfrac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}.$$

Projections.

$$\boldsymbol{q}_1\boldsymbol{q}_1^{\mathsf{T}} = \tfrac{1}{5}\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, \qquad \boldsymbol{q}_2\boldsymbol{q}_2^{\mathsf{T}} = \tfrac{1}{5}\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}, \qquad 2\cdot\tfrac{1}{5}\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} - 3\cdot\tfrac{1}{5}\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} = \tfrac{1}{5}\begin{bmatrix} 5 & 10 \\ 10 & -10 \end{bmatrix} = A.$$

Action on a vector. For $\boldsymbol{x}=(1,1)$ the two pieces are $\boldsymbol{q}_1\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{x} = (1.2,\, 0.6)$ and $\boldsymbol{q}_2\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{x} = (-0.2,\, 0.4)$, which add back to $\boldsymbol{x}$. Stretching them by 2 and -3 gives $(2.4, 1.2)+(0.6,-1.2) = (3,0)$, and indeed $A\boldsymbol{x} = (1+2,\, 2-2) = (3,0)$. Figure 36 draws this.

Pivots. Elimination subtracts 2 times row 1 from row 2, leaving pivots 1 and $-2-4=-6$: one positive and one negative, exactly like the eigenvalues 2 and -3. Their products agree too: $1\cdot(-6) = 2\cdot(-3) = \det A$.

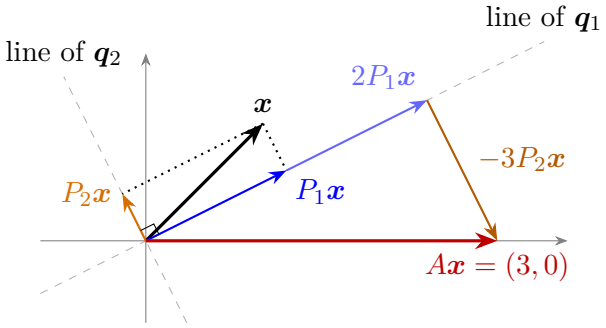


Figure 36: The spectral theorem at work for $A = \left[\begin{smallmatrix} \frac{1}{2} & 2 \\ 2 & -2 \end{smallmatrix}\right]$ and $\boldsymbol{x} = (1,1)$. The eigenvector lines are perpendicular; $\boldsymbol{x}$ splits into its projections $P_1\boldsymbol{x} = (1.2, 0.6)$ and $P_2\boldsymbol{x} = (-0.2, 0.4)$; A stretches them by $\lambda_1 = 2$ and $\lambda_2 = -3$ and adds, giving $A\boldsymbol{x} = (3,0)$.

29.3 Signs of pivots are signs of eigenvalues

Knowing the eigenvalues are real, the next question is whether they are positive or negative; for $d\boldsymbol{u}/dt = A\boldsymbol{u}$ that decides stability (Section 25). Finding the 50 eigenvalues of a 50×50 matrix

from $\det(A-\lambda I)=0$ is hopeless: the roots of a degree-50 polynomial are extremely sensitive to rounding. Pivots, on the other hand, are cheap and safe to compute. For symmetric matrices they carry the sign information we want.

If elimination on a symmetric A needs no row exchanges, it gives $A=LDU$ with L unit lower triangular, D the diagonal matrix of pivots and U unit upper triangular (Section 5). Transposing gives a second such factorization $A=A^{\mathsf{T}}=U^{\mathsf{T}}DL^{\mathsf{T}}$, and since the LDU factors are unique, $U=L^{\mathsf{T}}$:

$$A=LDL^{\mathsf{T}} \qquad (A \text{ symmetric, no row exchanges}).$$

Theorem 29.9 (Pivot signs)**.** *Let A be real symmetric with nonzero pivots $d_1,\dots,d_n$ (no row exchanges). Then the number of positive pivots equals the number of positive eigenvalues, and likewise for negative ones.*

Proof. Write $A=LDL^{\mathsf{T}}$ and, for $0\le t\le 1$, let $L_t=tL+(1-t)I$. Every L_t is lower triangular with ones on the diagonal, so $\det L_t=1$. The matrix $A_t=L_tDL_t^{\mathsf{T}}$ is symmetric, so its eigenvalues are real, and $\det A_t=\det D=d_1\cdots d_n\neq 0$, so 0 is never an eigenvalue of A_t. At $t=0$, $A_0=D$, whose eigenvalues are the pivots; at $t=1$, $A_1=A$. The eigenvalues of A_t are the roots of a polynomial whose coefficients depend continuously on t, so they move continuously. A real eigenvalue that changed sign would have to pass through 0, which never happens. Hence A and D have the same number of positive eigenvalues. □

The product of the pivots and the product of the eigenvalues are both $\det A$, but that is one number; the theorem gives n separate pieces of sign information. Combined with a shift it locates eigenvalues: the eigenvalues of $A-bI$ are λ_k-b, so the positive pivots of $A-bI$ count the eigenvalues of A that are larger than b.

Example 29.10 (Counting eigenvalues above a number)**.** Let $K=\begin{bmatrix}2&-1&0\\-1&2&-1\\0&-1&2\end{bmatrix}$. How many eigenvalues exceed $5/2$? Eliminate on

$$K-\tfrac{5}{2}I=\begin{bmatrix}-\frac{1}{2}&-1&0\\-1&-\frac{1}{2}&-1\\0&-1&-\frac{1}{2}\end{bmatrix}.$$

The first pivot is $-\frac{1}{2}$. Subtracting $2\times$row 1 from row 2 leaves $(0,\ \frac{3}{2},\ -1)$, so the second pivot is $\frac{3}{2}$. Adding $\frac{2}{3}\times$ the new row 2 to row 3 leaves $-\frac{1}{2}-\frac{2}{3}=-\frac{7}{6}$. The pivots $-\frac{1}{2},\ \frac{3}{2},\ -\frac{7}{6}$ have one positive sign, so exactly one eigenvalue of K is above $5/2$. Check: the eigenvalues of K are $2-\sqrt{2}\approx 0.59$, 2 and $2+\sqrt{2}\approx 3.41$ (Section 31), and only $2+\sqrt{2}$ exceeds 2.5.

Common trap. Pivots are not eigenvalues. For $A=\left[\begin{smallmatrix}5&2\\2&3\end{smallmatrix}\right]$ the pivots are 5 and $11/5$, the eigenvalues are $4\pm\sqrt{5}\approx 6.24,\ 1.76$. What they share is their signs and their product ($\det A=11$). Even the signs agree only for symmetric matrices: the nonsymmetric $\left[\begin{smallmatrix}1&-6\\1&-4\end{smallmatrix}\right]$ has positive pivots 1 and 2, but its eigenvalues are -1 and -2.

29.4 Positive definite matrices

Definition 29.11 (Positive definite)**.** A *positive definite* matrix is a real symmetric matrix whose eigenvalues are all positive.

Symmetric matrices are good; positive definite ones are the best. Testing the definition directly needs the eigenvalues, but pivots and determinants do the job without them.

Theorem 29.12 (Three equivalent tests)**.** *For a real symmetric $n \times n$ matrix A the following are equivalent:*

(a) all eigenvalues of A are positive;

(b) all n upper-left determinants $D_1 = a_{11},\ D_2,\ \ldots,\ D_n = \det A$ are positive, where D_k is the determinant of the top-left $k \times k$ submatrix A_k;

(c) elimination without row exchanges produces n pivots, all positive.

Proof. (a)⇒(b). Write $A = Q\Lambda Q^{\mathsf{T}}$ and $\boldsymbol{y} = Q^{\mathsf{T}}\boldsymbol{x}$. Then $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \boldsymbol{y}^{\mathsf{T}}\Lambda\boldsymbol{y} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$, which is positive for every $\boldsymbol{x} \neq \mathbf{0}$ (since then $\boldsymbol{y} \neq \mathbf{0}$). Now take $\boldsymbol{z} \in \mathbb{R}^k$, $\boldsymbol{z} \neq \mathbf{0}$, and pad it with zeros to $\boldsymbol{x} = (\boldsymbol{z}, 0, \ldots, 0)$; then $\boldsymbol{z}^{\mathsf{T}}A_k\boldsymbol{z} = \boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} > 0$. If μ is an eigenvalue of the symmetric matrix A_k with real eigenvector $\boldsymbol{z}$, this says $\mu\,\boldsymbol{z}^{\mathsf{T}}\boldsymbol{z} > 0$, so $\mu > 0$. Hence $D_k = \det A_k$, the product of the eigenvalues of A_k, is positive.

(b)⇒(c). Elimination on the first k rows only ever uses entries of A_k, so the first k pivots of A are the pivots of A_k and $D_k = d_1 d_2 \cdots d_k$. If every $D_k > 0$, then $d_1 = D_1 > 0$ and $d_k = D_k/D_{k-1} > 0$; in particular no zero pivot forces a row exchange.

(c)⇒(a). With all pivots positive, Theorem 29.9 says all n eigenvalues are positive. □

The quantity $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}$ that appeared in the first step will become the central character of Section 31, where it gives a fourth test and the link to minima.

Example 29.13 (The lecture's example)**.** $A = \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$ is symmetric. Its upper-left determinants are 5 and $15 - 4 = 11$, both positive. Its pivots are 5 and $11/5$ (the second pivot is $\det A$ divided by the first). So A is positive definite, and the eigenvalues must be positive without our computing them. Computing anyway: $\lambda^2 - 8\lambda + 11 = 0$ gives $\lambda = 4 \pm \sqrt{16 - 11} = 4 \pm \sqrt{5}$, and $4 - \sqrt{5} \approx 1.76 > 0$.

By contrast $B = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}$ has $\det B = 3 > 0$, yet its eigenvalues (and pivots) are -1 and -3. It fails test (b) at the first step: $D_1 = -1 < 0$.

Common trap. A positive determinant does not make a symmetric matrix positive definite: two negative eigenvalues also multiply to a positive number. Test *all* n upper-left determinants, not only the last one. Another quick necessary condition: every diagonal entry $a_{ii} = \boldsymbol{e}_i^{\mathsf{T}}A\boldsymbol{e}_i$ of a positive definite matrix is positive, so a single $a_{ii} \leq 0$ rules it out at once.

Some immediate consequences, each read off from the eigenvalues:

- a positive definite matrix is invertible, since $\det A = \lambda_1 \cdots \lambda_n > 0$;
- a diagonal matrix with positive diagonal entries is positive definite, since those entries are its eigenvalues;
- the inverse of a positive definite matrix is positive definite, with eigenvalues $1/\lambda_k > 0$ (more of this in Section 32).

29.5 Exercises

Exercise 29.14 (Where a false proof breaks)**.** Here is a "proof" that every real matrix has real eigenvalues: from $A\boldsymbol{x} = \lambda\boldsymbol{x}$ we get $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \lambda\,\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$, so $\lambda = \boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}/\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$, a ratio of real numbers. Test each step on the rotation $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ with $\lambda = i$ and $\boldsymbol{x} = (i, 1)$, and find the hidden assumption.

Solution. First $A\boldsymbol{x} = (0 \cdot i - 1,\ i + 0) = (-1,\, i) = i\,(i, 1)$, so $\lambda = i$ really is an eigenvalue. Next, $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = (i, 1) \cdot (-1, i) = -i + i = 0$ and $\lambda\,\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = i\,(i^2 + 1) = 0$, so the equation $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \lambda\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$ holds, as $0 = 0$. The final step divides by $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = i^2 + 1^2 = 0$. The hidden assumption is $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} \neq 0$, which is true for real vectors but false for complex ones; and the ratio is not obviously real either, since $\boldsymbol{x}$ may be complex. The correct version uses $\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = |x_1|^2 + |x_2|^2 > 0$ (Lemma 29.2) and needs symmetry. □

Exercise 29.15 (Which families are groups?)**.** A family of invertible matrices is a *group* if it contains AB and A^{-1} whenever it contains A and B. Which of these are groups?

(a) symmetric positive definite matrices;

(b) orthogonal matrices;

(c) all exponentials e^{tA} of one fixed matrix A, $t \in \mathbb{R}$;

(d) matrices with determinant 1.

Solution. (a) No. $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ (upper-left determinants $2, 1$) and $B = \left[\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right]$ (determinants $1, 3/4$) are positive definite, but $AB = \left[\begin{smallmatrix} 2.5 & 2 \\ 1.5 & 1.5 \end{smallmatrix}\right]$ is not even symmetric. (Inverses do stay positive definite.)

(b) Yes. If $A^{\mathsf{T}}A = I$ then $A^{-1} = A^{\mathsf{T}}$ and $(A^{\mathsf{T}})^{\mathsf{T}}A^{\mathsf{T}} = AA^{\mathsf{T}} = I$; if also $B^{\mathsf{T}}B = I$ then $(AB)^{\mathsf{T}}(AB) = B^{\mathsf{T}}A^{\mathsf{T}}AB = B^{\mathsf{T}}B = I$.

(c) Yes. $(e^{pA})^{-1} = e^{-pA}$ and $e^{pA}e^{qA} = e^{(p+q)A}$ are again of the form e^{tA}.

(d) Yes. $\det(A^{-1}) = 1/\det A = 1$ and $\det(AB) = \det A \det B = 1$.

□

Exercise 29.16 (Projections and determinants)**.** (a) Show that the only positive definite projection matrix is $P = I$.

(b) Show that $S = \left[\begin{smallmatrix} -3 & 1 \\ 1 & -2 \end{smallmatrix}\right]$ has $\det S > 0$ but is not positive definite, by exhibiting $\boldsymbol{x}$ with $\boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x} < 0$.

Solution. (a) A projection satisfies $P^2 = P$, so each eigenvalue satisfies $\lambda^2 = \lambda$: $\lambda \in \{0, 1\}$. Positive definite forces $\lambda > 0$, so all eigenvalues are 1. A projection matrix is symmetric, so $P = Q\Lambda Q^{\mathsf{T}} = QIQ^{\mathsf{T}} = I$.

(b) $\det S = 6 - 1 = 5 > 0$, but $\boldsymbol{x} = (1, 0)$ gives $\boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x} = s_{11} = -3 < 0$. In the language of Theorem 29.12, $D_1 = -3$ fails. In fact both eigenvalues, $(-5 \pm \sqrt{5})/2 \approx -1.38, -3.62$, are negative: S is negative definite.

□

Exercise 29.17 (Spectral decomposition and pivots)**.** For $A = \left[\begin{smallmatrix} 3 & 4 \\ 4 & -3 \end{smallmatrix}\right]$ find λ_1, λ_2, write $A = \lambda_1 P_1 + \lambda_2 P_2$ with projections $P_k = \boldsymbol{q}_k \boldsymbol{q}_k^{\mathsf{T}}$, and confirm that the pivot signs match the eigenvalue signs.

Solution. Trace 0 and determinant $-9 - 16 = -25$ give $\lambda^2 - 25 = 0$, so $\lambda = \pm 5$. From $A - 5I = \left[\begin{smallmatrix} -2 & 4 \\ 4 & -8 \end{smallmatrix}\right]$ the eigenvector is $(2, 1)$; from $A + 5I = \left[\begin{smallmatrix} 8 & 4 \\ 4 & 2 \end{smallmatrix}\right]$ it is $(1, -2)$; they are perpendicular. With $\boldsymbol{q}_1 = (2, 1)/\sqrt{5}$, $\boldsymbol{q}_2 = (1, -2)/\sqrt{5}$,

$$5 \cdot \tfrac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} - 5 \cdot \tfrac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}.$$

Elimination subtracts $\frac{4}{3}$ of row 1 from row 2, so the pivots are 3 and $-3 - \frac{16}{3} = -\frac{25}{3}$: one positive, one negative, like 5 and -5. Their product is $-25 = \det A$. □

30 Complex matrices; fast Fourier transform (FFT)

Complex numbers have slipped into the course through the back door: a real matrix such as a rotation can have complex eigenvalues and eigenvectors. This unit says once and for all what changes when vectors and matrices are complex. Only one rule is new: *whenever we transpose, we also take complex conjugates.* With that rule, symmetric matrices become Hermitian matrices and orthogonal matrices become unitary ones.

The payoff is the most important complex matrix of all, the Fourier matrix F_n. It has orthogonal columns, so it is easy to invert, and it has so much structure that it can be multiplied by a vector in about $\frac{1}{2} n \log_2 n$ steps instead of n^2. That algorithm, the fast Fourier transform, is a matrix factorization, and it changed scientific computing.

30.1 Length and inner product in $\mathbb{C}^n$

A vector $\boldsymbol{z} = (z_1, \ldots, z_n)$ with complex components lives in $\mathbb{C}^n$. The old length formula $\boldsymbol{z}^{\mathsf{T}}\boldsymbol{z} = z_1^2 + \cdots + z_n^2$ is no good: for $\boldsymbol{z} = (1, i)$ it gives $1 + i^2 = 0$, so a nonzero vector would have length zero. The cure is to conjugate one factor, because $\bar{z}z = |z|^2 \geq 0$ (Section 29).

Definition 30.1 (Conjugate transpose, length, inner product)**.** For a complex matrix or vector, $A^{\mathsf{H}} = \bar{A}^{\mathsf{T}}$ is the *conjugate transpose* (read "A Hermitian"; the H honours Charles Hermite). For $\boldsymbol{z}, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$,

$$\|\boldsymbol{z}\|^2 = \boldsymbol{z}^{\mathsf{H}}\boldsymbol{z} = |z_1|^2 + \cdots + |z_n|^2, \qquad \boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = \bar{y}_1 x_1 + \cdots + \bar{y}_n x_n.$$

The vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are *orthogonal* when $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = 0$.

The inner product $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x}$ is usually a complex number; only $\boldsymbol{z}^{\mathsf{H}}\boldsymbol{z}$ is guaranteed real and nonnegative. The definition is forced on us: the inner product of a vector with itself must be its length squared.

Proposition 30.2 (Rules for H)**.** *For complex vectors and matrices of compatible sizes: (a)* $\boldsymbol{x}^{\mathsf{H}}\boldsymbol{y} = \overline{\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x}}$*; (b)* $(AB)^{\mathsf{H}} = B^{\mathsf{H}}A^{\mathsf{H}}$*; (c)* $(A\boldsymbol{x})^{\mathsf{H}}\boldsymbol{y} = \boldsymbol{x}^{\mathsf{H}}(A^{\mathsf{H}}\boldsymbol{y})$*.*

Proof. (a) $\overline{\sum \bar{y}_k x_k} = \sum y_k \bar{x}_k = \boldsymbol{x}^{\mathsf{H}}\boldsymbol{y}$. (b) Conjugation respects products, $\overline{AB} = \bar{A}\bar{B}$, and transposing reverses the order: $(\bar{A}\bar{B})^{\mathsf{T}} = \bar{B}^{\mathsf{T}}\bar{A}^{\mathsf{T}}$. (c) By (b), $(A\boldsymbol{x})^{\mathsf{H}} = \boldsymbol{x}^{\mathsf{H}}A^{\mathsf{H}}$. □

Example 30.3 (Lengths and right angles in $\mathbb{C}^2$)**.** For $\boldsymbol{z} = (1, i)$: $\boldsymbol{z}^{\mathsf{H}}\boldsymbol{z} = \bar{1}\cdot 1 + \bar{i}\cdot i = 1 + (-i)(i) = 2$, so $\|\boldsymbol{z}\| = \sqrt{2}$, not 0. For $\boldsymbol{x} = (1, i)$ and $\boldsymbol{y} = (i, 1)$: $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = \bar{i}\cdot 1 + \bar{1}\cdot i = -i + i = 0$. These two vectors are orthogonal in $\mathbb{C}^2$, even though the unconjugated product $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{x} = i + i = 2i$ is not zero.

30.2 Hermitian matrices

Definition 30.4 (Hermitian matrix)**.** A square complex matrix is *Hermitian* if $A^{\mathsf{H}} = A$, that is $a_{ji} = \bar{a}_{ij}$ for all i, j. In particular each diagonal entry equals its own conjugate, so it is real.

For example $\begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$ is Hermitian: flipping across the diagonal and conjugating returns the same matrix. A real matrix is Hermitian exactly when it is symmetric, so everything below includes Section 29 as a special case.

Theorem 30.5 (Hermitian matrices are as good as symmetric ones)**.** *Let $A^{\mathsf{H}} = A$. Then*

(a) $\boldsymbol{z}^{\mathsf{H}}A\boldsymbol{z}$ is real for every $\boldsymbol{z} \in \mathbb{C}^n$;

(b) every eigenvalue of A is real;

(c) eigenvectors for different eigenvalues are orthogonal: $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = 0$.

Proof. (a) The number $\boldsymbol{z}^{\mathsf{H}}A\boldsymbol{z}$ is a 1×1 matrix, so it equals its transpose, and its conjugate is $(\boldsymbol{z}^{\mathsf{H}}A\boldsymbol{z})^{\mathsf{H}} = \boldsymbol{z}^{\mathsf{H}}A^{\mathsf{H}}\boldsymbol{z} = \boldsymbol{z}^{\mathsf{H}}A\boldsymbol{z}$. A number equal to its conjugate is real. (b) If $A\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \boldsymbol{0}$, then $\boldsymbol{x}^{\mathsf{H}}A\boldsymbol{x} = \lambda\,\boldsymbol{x}^{\mathsf{H}}\boldsymbol{x}$. The left side is real by (a) and $\boldsymbol{x}^{\mathsf{H}}\boldsymbol{x} > 0$ is real, so λ is real. (c) Let $A\boldsymbol{x} = \lambda\boldsymbol{x}$, $A\boldsymbol{y} = \mu\boldsymbol{y}$, $\lambda \neq \mu$, both real by (b). Then $\boldsymbol{y}^{\mathsf{H}}A\boldsymbol{x} = \lambda\,\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x}$, and also $\boldsymbol{y}^{\mathsf{H}}A\boldsymbol{x} = (A^{\mathsf{H}}\boldsymbol{y})^{\mathsf{H}}\boldsymbol{x} = (A\boldsymbol{y})^{\mathsf{H}}\boldsymbol{x} = \bar{\mu}\,\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = \mu\,\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x}$. Subtracting, $(\lambda - \mu)\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = 0$, so $\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x} = 0$. □

The proof of the spectral theorem in Section 29 goes through word for word with T replaced by H: every Hermitian matrix is $A = U\Lambda U^{\mathsf{H}}$ with Λ real and U unitary (defined next).

Example 30.6 (Diagonalizing a Hermitian matrix)**.** Let $A = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$. It is Hermitian: the diagonal is real and the off-diagonal entries are conjugates.

Eigenvalues. $\det(A - \lambda I) = (2-\lambda)(3-\lambda) - (1-i)(1+i) = \lambda^2 - 5\lambda + 6 - 2 = \lambda^2 - 5\lambda + 4$, because $(1-i)(1+i) = 1 - i^2 = 2$. So $(\lambda - 1)(\lambda - 4) = 0$: $\lambda_1 = 1$, $\lambda_2 = 4$, real although A is complex.

Eigenvectors. For $\lambda = 1$, $A - I = \left[\begin{smallmatrix} 1 & 1-i \\ 1+i & 2 \end{smallmatrix}\right]$. The second row is $(1 + i)$ times the first, because $(1+i)(1-i) = 2$; the rows are dependent, as they must be. The first row is solved by $\boldsymbol{v} = (1-i,\, -1)$: $1\cdot(1-i) + (1-i)(-1) = 0$. For $\lambda = 4$, $A - 4I = \left[\begin{smallmatrix} -2 & 1-i \\ 1+i & -1 \end{smallmatrix}\right]$; the second row gives $\boldsymbol{u} = (1,\, 1+i)$. Check: $A\boldsymbol{u} = (2 + (1-i)(1+i),\ 1 + i + 3(1+i)) = (4,\, 4+4i) = 4\boldsymbol{u}$.

Orthogonality. $\boldsymbol{v}^{\mathsf{H}}\boldsymbol{u} = \overline{(1-i)}\cdot 1 + \overline{(-1)}(1+i) = (1+i) - (1+i) = 0$.

Unitary eigenvector matrix. Both vectors have length $\sqrt{3}$: $|1-i|^2 + |-1|^2 = 2+1$ and $1 + |1+i|^2 = 1+2$. Dividing by $\sqrt{3}$,

$$A = U\Lambda U^{\mathsf{H}} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1-i & 1 \\ -1 & 1+i \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}\frac{1}{\sqrt{3}}\begin{bmatrix} 1+i & -1 \\ 1 & 1-i \end{bmatrix}.$$

Multiplying out: the $(1,1)$ entry is $\frac{1}{3}\big[(1-i)(1+i) + 4\big] = \frac{1}{3}(2+4) = 2$, the $(1,2)$ entry is $\frac{1}{3}\big[-(1-i) + 4(1-i)\big] = 1 - i$, and similarly for the rest, recovering A.

Common trap. A complex matrix that is merely *symmetric* has none of these properties. $\left[\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right]$ equals its transpose but not its conjugate transpose, and its eigenvalues $1 \pm i$ (eigenvectors $(1,1)$ and $(1,-1)$) are not real. For complex matrices, check $A^{\mathsf{H}} = A$, never only $A^{\mathsf{T}} = A$.

30.3 Unitary matrices

Definition 30.7 (Unitary matrix)**.** A square complex matrix U is *unitary* if $U^{\mathsf{H}}U = I$: its columns are orthonormal in the sense $\boldsymbol{u}_j^{\mathsf{H}}\boldsymbol{u}_k = 0$ for $j \neq k$ and $\boldsymbol{u}_k^{\mathsf{H}}\boldsymbol{u}_k = 1$. Then $U^{-1} = U^{\mathsf{H}}$.

Proposition 30.8 (Unitary matrices preserve lengths)**.** *If U is unitary then $(U\boldsymbol{x})^{\mathsf{H}}(U\boldsymbol{y}) = \boldsymbol{x}^{\mathsf{H}}\boldsymbol{y}$ and $\|U\boldsymbol{z}\| = \|\boldsymbol{z}\|$ for all vectors, and every eigenvalue of U has $|\lambda| = 1$.*

Proof. $(U\boldsymbol{x})^{\mathsf{H}}(U\boldsymbol{y}) = \boldsymbol{x}^{\mathsf{H}}U^{\mathsf{H}}U\boldsymbol{y} = \boldsymbol{x}^{\mathsf{H}}\boldsymbol{y}$; take $\boldsymbol{x} = \boldsymbol{y} = \boldsymbol{z}$ for lengths. If $U\boldsymbol{z} = \lambda\boldsymbol{z}$ with $\boldsymbol{z} \neq \boldsymbol{0}$, then $\|\boldsymbol{z}\| = \|U\boldsymbol{z}\| = |\lambda|\,\|\boldsymbol{z}\|$, so $|\lambda| = 1$. □

The translation from real to complex is mechanical; Table 3 collects it.

	real ($\mathbb{R}^n$)	complex ($\mathbb{C}^n$)
length squared	$\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = \sum x_k^2$	$\boldsymbol{z}^{\mathsf{H}}\boldsymbol{z} = \sum \lvert z_k\rvert^2$
inner product	$\boldsymbol{y}^{\mathsf{T}}\boldsymbol{x}$	$\boldsymbol{y}^{\mathsf{H}}\boldsymbol{x}$
transpose rule	$(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$	$(AB)^{\mathsf{H}} = B^{\mathsf{H}}A^{\mathsf{H}}$
good matrices	symmetric, $A^{\mathsf{T}} = A$	Hermitian, $A^{\mathsf{H}} = A$
orthonormal columns	orthogonal, $Q^{\mathsf{T}}Q = I$	unitary, $U^{\mathsf{H}}U = I$
spectral theorem	$A = Q\Lambda Q^{\mathsf{T}}$	$A = U\Lambda U^{\mathsf{H}}$

Table 3: The dictionary between $\mathbb{R}^n$ and $\mathbb{C}^n$: transpose becomes conjugate transpose.

30.4 The Fourier matrix

Electrical engineers number rows and columns from 0 to $n-1$, and for the Fourier matrix we follow them.

Definition 30.9 (Fourier matrix)**.** Let $w = w_n = e^{2\pi i/n} = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$. The $n \times n$ *Fourier matrix* F_n has entries $(F_n)_{jk} = w^{jk}$ for $j, k = 0, 1, \ldots, n-1$:

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)^2} \end{bmatrix}.$$

The number w sits on the unit circle at angle $2\pi/n$, one nth of the way round, and $w^n = 1$. Its powers $1, w, w^2, \ldots, w^{n-1}$ are the n different nth roots of 1, equally spaced on the circle (Figure 37). Powers are easy in the polar form: $w^k = e^{2\pi ik/n}$, which is why we never expand w into cosines and sines. Note that $F_n^{\mathsf{T}} = F_n$ (symmetric in the old sense), but F_n is not Hermitian.

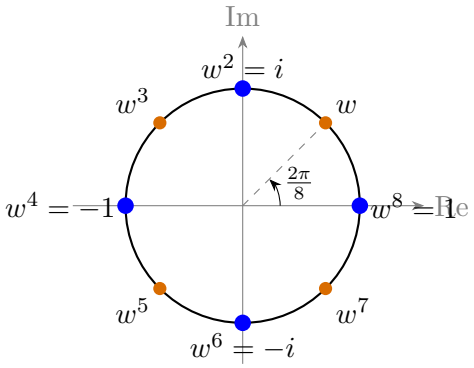


Figure 37: The eighth roots of unity, powers of $w = w_8 = e^{2\pi i/8}$. The even powers (blue) w^2, w^4, w^6, w^8 are exactly the fourth roots of unity $i, -1, -i, 1$: squaring w_8 gives w_4. This doubling of angles is what links F_8 to F_4 in the FFT.

For $n = 4$, $w = e^{2\pi i/4} = i$ and the exponent is always (row number)$\times$(column number):

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}.$$

Multiplying a four-component signal by F_4 (or by F_4^{-1}) is the four-point discrete Fourier transform.

Theorem 30.10 (The columns of F_n are orthogonal)**.** $F_n^{\mathsf{H}} F_n = nI$. *Hence* $\frac{1}{\sqrt{n}} F_n$ *is unitary and*

$$F_n^{-1} = \frac{1}{n} F_n^{\mathsf{H}} = \frac{1}{n} \bar{F}_n.$$

Proof. The (k, l) entry of $F_n^{\mathsf{H}} F_n$ is the inner product of columns k and l:

$$\sum_{j=0}^{n-1} \overline{w^{jk}}\, w^{jl} = \sum_{j=0}^{n-1} w^{j(l-k)} = \sum_{j=0}^{n-1} r^j, \qquad r = w^{l-k},$$

using $\bar{w} = w^{-1}$ (because $|w| = 1$). If $k = l$ then $r = 1$ and the sum is n. If $k \neq l$ then $0 < |l-k| < n$, so $r \neq 1$, while $r^n = (w^n)^{l-k} = 1$; the geometric series gives $\sum_{j=0}^{n-1} r^j = (1 - r^n)/(1 - r) = 0$. The last equality uses $F_n^{\mathsf{T}} = F_n$, so $F_n^{\mathsf{H}} = \bar{F}_n$. □

Example 30.11 (Orthogonality and an impulse)**.** Columns 1 and 3 of F_4 are $(1, i, -1, -i)$ and $(1, -i, -1, i)$. Without conjugating, their product is $1{+}1{+}1{+}1 = 4$, which looks far from orthogonal. With the conjugate, $\overline{(1, i, -1, -i)} \cdot (1, -i, -1, i) = 1{\cdot}1 + (-i)(-i) + (-1)(-1) + (i)(i) = 1 - 1 + 1 - 1 = 0$.

The impulse $\boldsymbol{x} = (1, 0, 0, 0)$ transforms to $F_4 \boldsymbol{x} = (1, 1, 1, 1)$, the first column: a single impulse contains all frequencies in equal amounts. Transforming back with $F_4^{-1} = \frac{1}{4}\bar{F}_4$ gives $\frac{1}{4}(1 + 1 + 1 + 1,\ 1 - i - 1 + i,\ 0,\ 0) = (1, 0, 0, 0)$.

Common trap. The inverse of F_n is $\frac{1}{n}\bar{F}_n$, not $\frac{1}{n}F_n$. For the impulse above both happen to give the same answer, but in general $F_4 F_4 = 4 \begin{bmatrix} 1&0&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \\ 0&1&0&0 \end{bmatrix}$: applying F_4 twice reverses components $1, 2, 3$ of the signal instead of restoring it.

30.5 The fast Fourier transform

Multiplying $F_n\boldsymbol{x}$ directly needs n^2 multiplications, since F_n has no zero entries. The fast Fourier transform (FFT) connects F_{2m} to two copies of F_m, using the fact seen in Figure 37:

$$w_{2m}^2 = \left(e^{2\pi i/2m}\right)^2 = e^{2\pi i/m} = w_m, \qquad w_{2m}^m = e^{\pi i} = -1.$$

Theorem 30.12 (FFT factorization)**.** *Let $D = \operatorname{diag}(1, w, w^2, \dots, w^{m-1})$ with $w = w_{2m}$, and let P be the $2m \times 2m$ permutation matrix that puts the even-numbered components first,*

$$P(x_0, x_1, \dots, x_{2m-1}) = (x_0, x_2, \dots, x_{2m-2},\ x_1, x_3, \dots, x_{2m-1}).$$

Then

$$F_{2m} = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_m & 0 \\ 0 & F_m \end{bmatrix} P.$$

Proof. Let $\boldsymbol{y} = F_{2m}\boldsymbol{x}$ and split the sum $y_j = \sum_{k=0}^{2m-1} w_{2m}^{jk} x_k$ into even $k = 2l$ and odd $k = 2l+1$:

$$y_j = \sum_{l=0}^{m-1} w_{2m}^{2jl} x_{2l} + w_{2m}^{j} \sum_{l=0}^{m-1} w_{2m}^{2jl} x_{2l+1} = y'_j + w_{2m}^{j}\, y''_j,$$

where $\boldsymbol{y}' = F_m(x_0, x_2, \dots)$ and $\boldsymbol{y}'' = F_m(x_1, x_3, \dots)$, because $w_{2m}^{2jl} = w_m^{jl}$. For $0 \le j < m$ this is the top half of the right side. For the bottom half, replace j by $j+m$: $w_m^{(j+m)l} = w_m^{jl}$ (as $w_m^m = 1$), so $\boldsymbol{y}'$ and $\boldsymbol{y}''$ are unchanged, while $w_{2m}^{j+m} = -w_{2m}^{j}$. Hence $y_{j+m} = y'_j - w_{2m}^{j} y''_j$, which is the bottom half. □

In words: separate $\boldsymbol{x}$ into evens and odds (free), do two half-size transforms, then recombine with m multiplications by the entries of D and $2m$ additions.

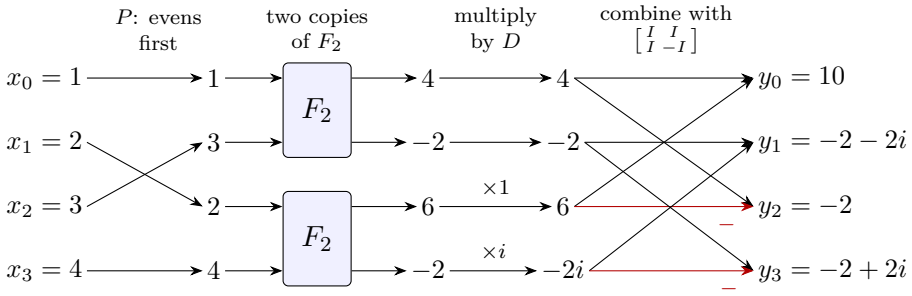


Figure 38: The factorization $F_4 = \left[\begin{smallmatrix} I & D \\ I & -D \end{smallmatrix}\right] \left[\begin{smallmatrix} F_2 & 0 \\ 0 & F_2 \end{smallmatrix}\right] P$ applied to $\boldsymbol{x} = (1, 2, 3, 4)$. Each "butterfly" on the right forms a sum and a difference (red lines carry the minus sign). Only the twiddle step $\times i$ costs a genuine multiplication.

Example 30.13 (A four-point FFT, step by step)**.** For F_4 we have $m = 2$, $w_4 = i$, $D = \operatorname{diag}(1, i)$ and $F_2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Take $\boldsymbol{x} = (1, 2, 3, 4)$.

1. Permute: evens $(x_0, x_2) = (1, 3)$, odds $(x_1, x_3) = (2, 4)$.
2. Half-size transforms: $\boldsymbol{y}' = F_2(1, 3) = (4, -2)$ and $\boldsymbol{y}'' = F_2(2, 4) = (6, -2)$.

3. Twiddle: $D\boldsymbol{y}'' = (1 \cdot 6,\ i \cdot (-2)) = (6, -2i)$.
4. Combine: top $\boldsymbol{y}' + D\boldsymbol{y}'' = (10,\ -2 - 2i)$, bottom $\boldsymbol{y}' - D\boldsymbol{y}'' = (-2,\ -2 + 2i)$.

So $F_4\boldsymbol{x} = (10,\ -2-2i,\ -2,\ -2+2i)$. Directly from the rows of F_4: $1+2+3+4 = 10$; $1+2i-3-4i = -2-2i$; $1-2+3-4 = -2$; $1-2i-3+4i = -2+2i$. The two agree. Figure 38 shows the same computation as a flow diagram.

Proposition 30.14 (Operation count)**.** *For $n = 2^\ell$, applying Theorem 30.12 recursively computes $F_n\boldsymbol{x}$ with $\frac{1}{2}n\log_2 n$ multiplications (counting the multiplications by entries of D).*

Proof. Let $C(n)$ be the count. A size-n transform costs two size-$n/2$ transforms plus $n/2$ multiplications by D, so $C(n) = 2C(n/2) + \frac{n}{2}$ with $C(1) = 0$. If $C(n/2) = \frac{1}{2} \cdot \frac{n}{2}(\ell - 1)$, then $C(n) = \frac{n}{2}(\ell - 1) + \frac{n}{2} = \frac{n}{2}\ell$, and induction on ℓ finishes the proof. □

Unrolling the recursion explains the count: there are $\log_2 n$ levels (for $n = 64$: $64 \to 32 \to 16 \to 8 \to 4 \to 2 \to 1$, six levels), each level's fix-up costs $n/2$ multiplications, and the permutations at all levels together just reorder the input (evens of evens, evens of odds, and so on). For $n = 1024 = 2^{10}$, direct multiplication costs $1024^2 \approx 10^6$, while the FFT costs $\frac{1}{2} \cdot 1024 \cdot 10 = 5120$, about 200 times less. Since $F_n^{-1} = \frac{1}{n}\bar{F}_n$, the inverse transform runs through the same algorithm with $\bar{w}$ in place of w. All of this rests on F_n being a very special matrix with orthogonal columns.

30.6 Exercises

Exercise 30.15 (The smallest Fourier matrix)**.** Compute F_2, check that $F_2^{\mathsf{H}}F_2 = 2I$, and explain why $F_2^{-1} = \frac{1}{2}F_2$ even though $F_n^{-1} \neq \frac{1}{n}F_n$ in general.

Solution. For $n = 2$, $w = e^{2\pi i/2} = e^{\pi i} = -1$, so $F_2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & w \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. It is real and symmetric, so $F_2^{\mathsf{H}}F_2 = F_2^2 = \left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right] = 2I$. In general $F_n^{-1} = \frac{1}{n}\bar{F}_n$; here $\bar{F}_2 = F_2$ because all entries are real, so the two formulas coincide. □

Exercise 30.16 (One step of the FFT for $n = 8$)**.** In $F_8 = \left[\begin{smallmatrix} I & D \\ I & -D \end{smallmatrix}\right] \left[\begin{smallmatrix} F_4 & 0 \\ 0 & F_4 \end{smallmatrix}\right] P$, write out D and describe P. How many multiplications does the full FFT need for $n = 8$, compared with n^2?

Solution. Here $w = w_8 = e^{2\pi i/8} = \frac{1+i}{\sqrt{2}}$, so $D = \operatorname{diag}(1, w, w^2, w^3) = \operatorname{diag}\big(1, \frac{1+i}{\sqrt{2}},\ i,\ \frac{-1+i}{\sqrt{2}}\big)$. These are the first four eighth roots of unity; $-D$ holds the other four. The permutation sends $(x_0, \ldots, x_7)$ to $(x_0, x_2, x_4, x_6, x_1, x_3, x_5, x_7)$. The count is $\frac{1}{2} \cdot 8 \cdot \log_2 8 = 12$ multiplications instead of 64. □

Exercise 30.17 (A unitary matrix)**.** Show that $U = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right]$ is unitary and that its eigenvalues have absolute value 1.

Solution. $U^{\mathsf{H}} = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & -i \\ -i & 1 \end{smallmatrix}\right]$ and $U^{\mathsf{H}}U = \frac{1}{2}\left[\begin{smallmatrix} 1+1 & i-i \\ -i+i & 1+1 \end{smallmatrix}\right] = I$. The matrix $\left[\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right]$ has eigenvectors $(1, 1)$ and $(1, -1)$ with eigenvalues $1 + i$ and $1 - i$, so U has eigenvalues $(1 \pm i)/\sqrt{2}$, and $|1 \pm i|/\sqrt{2} = \sqrt{2}/\sqrt{2} = 1$, as Proposition 30.8 requires. □

Exercise 30.18 (Skew-Hermitian matrices)**.** Suppose $A^{\mathsf{H}} = -A$. Show that iA is Hermitian and conclude that the eigenvalues of A are purely imaginary. Check this on the real rotation $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$.

Solution. $(iA)^{\mathsf{H}} = \bar{i}\,A^{\mathsf{H}} = (-i)(-A) = iA$, so iA is Hermitian and its eigenvalues are real (Theorem 30.5). If $A\boldsymbol{x} = \lambda\boldsymbol{x}$ then $(iA)\boldsymbol{x} = i\lambda\boldsymbol{x}$, so $i\lambda$ is real and λ is purely imaginary. The real antisymmetric rotation satisfies $A^{\mathsf{H}} = A^{\mathsf{T}} = -A$, and indeed $\det(A - \lambda I) = \lambda^2 + 1 = 0$ gives $\lambda = \pm i$. □

31 Positive definite matrices and minima

Section 29 defined a positive definite matrix as a symmetric matrix with positive eigenvalues and showed that positive pivots, or positive upper-left determinants, are equivalent tests. This unit adds the quantity that explains *why* these matrices matter: the quadratic expression $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}$. It gives a fourth test, which most books take as the definition, and it connects linear algebra to calculus (a minimum of a function of several variables) and to geometry (ellipses and ellipsoids). Pivots from the first week, determinants from the middle of the course and eigenvalues from the last weeks all meet here. All matrices in this unit are real and symmetric.

31.1 The energy $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}$ and four equivalent tests

Definition 31.1 (Quadratic form; positive (semi)definite)**.** The *quadratic form* of a symmetric $n \times n$ matrix A is $f(\boldsymbol{x}) = \boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \sum_{i,j} a_{ij}x_ix_j$. The matrix is *positive definite* if $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} > 0$ for every $\boldsymbol{x} \neq \mathbf{0}$, and *positive semidefinite* if $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} \geq 0$ for every $\boldsymbol{x}$.

For a 2×2 matrix the form is short enough to read off:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2.$$

The diagonal entries multiply the pure squares and each off-diagonal pair produces a cross term. The form is purely quadratic: no linear terms, no constant. The squares ax^2 and cy^2 are positive when $a, c > 0$; the cross term $2bxy$ can have either sign. Positive definiteness means the squares always win.

Theorem 31.2 (Four tests)**.** *For a symmetric matrix A the following are equivalent; any one of them makes A positive definite.*

(1) All eigenvalues are positive: $\lambda_i > 0$.

(2) All upper-left determinants are positive: $D_1 > 0, \ldots, D_n > 0$.

(3) All pivots are positive (elimination without row exchanges): $d_i > 0$.

(4) $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} > 0$ for every $\boldsymbol{x} \neq \mathbf{0}$.

For $A = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ these read: $\lambda_1, \lambda_2 > 0$; $a > 0$ and $ac - b^2 > 0$; pivots $a > 0$ and $(ac - b^2)/a > 0$; $ax^2 + 2bxy + cy^2 > 0$ unless $x = y = 0$.

Proof. Tests (1), (2), (3) were shown equivalent in Section 29. It remains to connect (4) with (1). Write $A = Q\Lambda Q^{\mathsf{T}}$ (spectral theorem) and $\boldsymbol{y} = Q^{\mathsf{T}}\boldsymbol{x}$. Then

$$\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \boldsymbol{y}^{\mathsf{T}}\Lambda\boldsymbol{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2.$$

If every $\lambda_i > 0$, this is positive whenever $\boldsymbol{y} \neq \mathbf{0}$, that is whenever $\boldsymbol{x} \neq \mathbf{0}$ (Q is invertible). Conversely, if (4) holds and $A\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$, then $0 < \boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \lambda\,\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$, so $\lambda > 0$. □

The same computation shows that A is positive *semi*definite exactly when all its eigenvalues are ≥ 0. A semidefinite matrix that is not definite is singular: it has the eigenvalue 0, fewer than n pivots, and a nonzero $\boldsymbol{x}$ with $\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} = 0$.

Example 31.3 (One matrix, three regimes)**.** For which c is $A_c = \begin{bmatrix} 2 & 6 \\ 6 & c \end{bmatrix}$ positive definite? We do not want eigenvalues; the determinant test is fastest. $D_1 = 2 > 0$ always, and $D_2 = 2c - 36 > 0$ exactly when $c > 18$. Its form is $f_c(x, y) = 2x^2 + 12xy + cy^2$.

- $c = 18$ is the borderline: $\det A_{18} = 0$, so A_{18} is singular with eigenvalues 0 and 20 (the trace). It has one pivot, 2, because row 2 is 3 times row 1. Its form is $2x^2+12xy+18y^2 = 2(x+3y)^2 \geq 0$, which vanishes along the line $x = -3y$, for instance at $(3, -1)$. A_{18} is positive semidefinite.
- $c = 7$ fails badly: $\det A_7 = 14 - 36 = -22$, the pivots are 2 and $-22/2 = -11$, and the eigenvalues (trace 9, determinant -22) are 11 and -2. The form goes negative: $f_7(1, -1) = 2 - 12 + 7 = -3$. Its graph is a *saddle*: up along some directions, down along others.
- $c = 20$ passes: $D_2 = 40 - 36 = 4 > 0$, the pivots are 2 and $4/2 = 2$, and the eigenvalues have product 4 and sum 22, so both are positive. The graph of f_{20} is a bowl.

Common trap. Checking $\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} > 0$ at a few convenient vectors proves nothing. The indefinite A_7 gives $f_7(1, 0) = 2$, $f_7(0, 1) = 7$ and $f_7(1, 1) = 21$, all positive, yet $f_7(1, -1) = -3$. Positive diagonal entries are necessary but far from sufficient: the cross term must be outweighed in every direction, which is what the determinant $ac - b^2 > 0$ measures.

31.2 Completing the square is elimination

How can we see that $f_{20} = 2x^2 + 12xy + 20y^2$ is positive everywhere except the origin? Write it as a sum of squares. Put all the x's into one square: $2(x + 3y)^2 = 2x^2 + 12xy + 18y^2$ produces the first two terms correctly and $18y^2$. We have $20y^2$, so two are left over:

$$2x^2 + 12xy + 20y^2 = 2(x + 3y)^2 + 2y^2,$$

a sum of squares with positive weights, zero only when $y = 0$ and $x + 3y = 0$. For $c = 7$ the same step leaves $7 - 18 = -11$:

$$2x^2 + 12xy + 7y^2 = 2(x + 3y)^2 - 11y^2,$$

negative for example at $(x, y) = (-3, 1)$, where it equals $0 - 11 = -11$. The number 18 is exactly the borderline found by the determinant test.

The numbers 2, 3, 2 are not new. Elimination on A_{20} subtracts 3 times row 1 from row 2:

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

The pivots stand outside the squares and the multiplier 3 sits inside.

Proposition 31.4 (Sum of squares from $A = LDL^{\mathsf{T}}$)**.** *If $A = LDL^{\mathsf{T}}$ (elimination without row exchanges, pivots $d_1, \ldots, d_n$), then*

$$\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} = d_1 (\boldsymbol{\ell}_1^{\mathsf{T}}\boldsymbol{x})^2 + d_2 (\boldsymbol{\ell}_2^{\mathsf{T}}\boldsymbol{x})^2 + \cdots + d_n (\boldsymbol{\ell}_n^{\mathsf{T}}\boldsymbol{x})^2,$$

where $\boldsymbol{\ell}_i$ is column i of L. In particular positive pivots give $\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} > 0$ for $\boldsymbol{x} \neq \mathbf{0}$, which proves test (3)⇒(4) directly.

Proof. $\boldsymbol{x}^{\mathsf{T}} L D L^{\mathsf{T}} \boldsymbol{x} = \boldsymbol{z}^{\mathsf{T}} D \boldsymbol{z}$ with $\boldsymbol{z} = L^{\mathsf{T}} \boldsymbol{x}$, and component i of $\boldsymbol{z}$ is $\boldsymbol{\ell}_i^{\mathsf{T}} \boldsymbol{x}$. So $\boldsymbol{z}^{\mathsf{T}} D \boldsymbol{z} = \sum d_i z_i^2$. If all $d_i > 0$ this vanishes only for $\boldsymbol{z} = \mathbf{0}$, and since L^{T} is invertible (ones on its diagonal) that means $\boldsymbol{x} = \mathbf{0}$. □

Example 31.5 (A 3×3 example in every test)**.** Take the second-difference matrix

$$K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

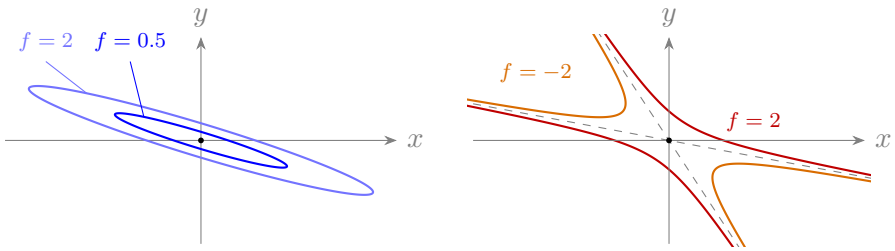
- *Determinants:* $D_1 = 2$, $D_2 = 4 - 1 = 3$, $D_3 = 2 \cdot 3 - (-1)(-2) = 4$. All positive.
- *Pivots:* D_k / D_{k-1} gives 2, $\frac{3}{2}$, $\frac{4}{3}$. Elimination uses multipliers $\ell_{21} = -\frac{1}{2}$ and $\ell_{32} = -\frac{2}{3}$.
- *Eigenvalues:* $2 - \sqrt{2}$, 2, $2 + \sqrt{2}$, all positive. Checks: their sum is $6 = \operatorname{tr} K$ and their product is $(4 - 2) \cdot 2 = 4 = \det K$.
- *Energy:* $\boldsymbol{x}^{\mathsf{T}} K \boldsymbol{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1 x_2 - 2x_2 x_3$, and completing squares with the pivots outside and the multipliers inside,

$$\boldsymbol{x}^{\mathsf{T}} K \boldsymbol{x} = 2\big(x_1 - \tfrac{1}{2} x_2\big)^2 + \tfrac{3}{2}\big(x_2 - \tfrac{2}{3} x_3\big)^2 + \tfrac{4}{3} x_3^2.$$

  Expanding: $2x_1^2 - 2x_1 x_2 + \frac{1}{2} x_2^2$, plus $\frac{3}{2} x_2^2 - 2x_2 x_3 + \frac{2}{3} x_3^2$, plus $\frac{4}{3} x_3^2$, which totals the form above.

Every test agrees: K is positive definite, and the graph of $\boldsymbol{x}^{\mathsf{T}} K \boldsymbol{x}$ over $\mathbb{R}^3$ is a bowl in four dimensions.

Figure 39 shows what separates a bowl from a saddle, by drawing curves $f(x, y) = k$ on which the form is constant.



$f_{20} = 2(x + 3y)^2 + 2y^2$: ellipses $f_7 = 2(x + 3y)^2 - 11y^2$: hyperbolas

Figure 39: Level curves of two quadratic forms that differ only in the corner entry. Left: for the positive definite A_{20} every level set $f = k > 0$ is an ellipse around the minimum at the origin. Right: for the indefinite A_7 the level sets $f = \pm 2$ are hyperbolas; along the dashed lines $x + 3y = \pm\sqrt{11/2}\, y$ the form is zero, and between them it changes sign: a saddle point.

31.3 Second derivatives decide a minimum

In one-variable calculus, $f'(x_0) = 0$ makes x_0 a critical point and $f''(x_0) > 0$ makes it a minimum: the slope is increasing as we pass through. For $f(x_1, \ldots, x_n)$ the first derivatives must all vanish, and the role of f'' is played by the *matrix of second derivatives* (the Hessian)

$$H = \left[\frac{\partial^2 f}{\partial x_i \, \partial x_j} \right], \qquad \text{for two variables} \quad H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

It is symmetric because mixed partial derivatives can be taken in either order: $f_{xy} = f_{yx}$.

Proposition 31.6 (Second-derivative test)**.** *Let f have continuous second derivatives near $\boldsymbol{x}_0$, with all first derivatives zero at $\boldsymbol{x}_0$. If $H(\boldsymbol{x}_0)$ is positive definite, $\boldsymbol{x}_0$ is a strict local minimum. If $H(\boldsymbol{x}_0)$ has a negative eigenvalue, $\boldsymbol{x}_0$ is not a minimum.*

Sketch. Taylor's formula gives $f(\boldsymbol{x}_0 + \boldsymbol{h}) = f(\boldsymbol{x}_0) + \frac{1}{2}\boldsymbol{h}^{\mathsf{T}} H(\boldsymbol{x}_0)\boldsymbol{h} + r(\boldsymbol{h})$ with $r(\boldsymbol{h})/\|\boldsymbol{h}\|^2 \to 0$ as $\boldsymbol{h} \to \boldsymbol{0}$; the linear term is absent because the gradient is zero. If H is positive definite with smallest eigenvalue $\lambda_{\min} > 0$, then $\frac{1}{2}\boldsymbol{h}^{\mathsf{T}} H\boldsymbol{h} \geq \frac{1}{2}\lambda_{\min}\|\boldsymbol{h}\|^2$ (expand $\boldsymbol{h}$ in the orthonormal eigenvectors), which beats $r(\boldsymbol{h})$ for small $\boldsymbol{h} \neq \boldsymbol{0}$, so $f(\boldsymbol{x}_0 + \boldsymbol{h}) > f(\boldsymbol{x}_0)$. If $H\boldsymbol{q} = \lambda\boldsymbol{q}$ with $\lambda < 0$, moving along $\boldsymbol{h} = t\boldsymbol{q}$ gives $f(\boldsymbol{x}_0 + t\boldsymbol{q}) - f(\boldsymbol{x}_0) = \frac{1}{2}\lambda t^2\|\boldsymbol{q}\|^2 + r(t\boldsymbol{q}) < 0$ for small $t \neq 0$. The full ε–δ details belong to multivariable calculus. □

For two variables the determinant test for H is the familiar calculus rule: a minimum when $f_{xx} > 0$ and $f_{xx}f_{yy} > f_{xy}^2$. Linear algebra extends it to any number of variables, where it is applied through pivots, determinants or eigenvalues of H. For a pure quadratic $f(\boldsymbol{x}) = \boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x}$ the Hessian is the constant matrix $2A$, so the local picture at the origin is the global one: a bowl when A is positive definite, a saddle when it is indefinite.

Example 31.7 (Classifying critical points)**.** Let $f(x, y) = x^3 + y^2 - 3xy$. Setting the first derivatives to zero,

$$f_x = 3x^2 - 3y = 0, \qquad f_y = 2y - 3x = 0.$$

The second equation gives $y = \frac{3}{2}x$; then $3x^2 - \frac{9}{2}x = 0$, so $x = 0$ or $x = \frac{3}{2}$. The critical points are $(0, 0)$ and $(\frac{3}{2}, \frac{9}{4})$. The Hessian is $H = \left[\begin{smallmatrix} 6x & -3 \\ -3 & 2 \end{smallmatrix}\right]$.

- At $(0,0)$: $H = \left[\begin{smallmatrix} 0 & -3 \\ -3 & 2 \end{smallmatrix}\right]$ has $\det H = -9 < 0$, so one eigenvalue is negative: a saddle point, not a minimum.
- At $(\frac{3}{2}, \frac{9}{4})$: $H = \left[\begin{smallmatrix} 9 & -3 \\ -3 & 2 \end{smallmatrix}\right]$ has $D_1 = 9 > 0$ and $D_2 = 18 - 9 = 9 > 0$: positive definite, so a local minimum, with value $f = \frac{27}{8} + \frac{81}{16} - \frac{81}{8} = -\frac{27}{16}$.

31.4 Ellipses, ellipsoids and principal axes

Cut the bowl $z = \boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x}$ at height 1. The cross-section $\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} = 1$ is an ellipse in the plane, an ellipsoid (a lopsided rugby ball) in $\mathbb{R}^3$. Cutting a saddle instead gives a hyperbola. The spectral theorem says exactly how the ellipse sits.

Theorem 31.8 (Principal axes)**.** *Let $A = Q\Lambda Q^{\mathsf{T}}$ be positive definite with orthonormal eigenvectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$. The set $\boldsymbol{x}^{\mathsf{T}} A\boldsymbol{x} = 1$ is an ellipsoid whose axes point along the eigenvectors $\boldsymbol{q}_i$, with half-lengths $1/\sqrt{\lambda_i}$.*

Proof. In the coordinates $\boldsymbol{y} = Q^{\mathsf{T}}\boldsymbol{x}$, which measure $\boldsymbol{x}$ along the perpendicular directions $\boldsymbol{q}_i$ (a rotation, possibly with a reflection, which changes no shapes), the equation becomes $\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1$, the standard ellipsoid $\sum y_i^2/(1/\sqrt{\lambda_i})^2 = 1$. Its axes are the y_i-axes, i.e. the $\boldsymbol{q}_i$ directions, and the axis along $\boldsymbol{q}_i$ reaches $y_i = \pm 1/\sqrt{\lambda_i}$. □

Large eigenvalues give short axes: the bowl rises steeply in that direction. Equal eigenvalues give a round cross-section; two equal out of three give a rugby ball; the identity gives a sphere.

Example 31.9 (An ellipse tilted at 45°)**.** For $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ the equation $\boldsymbol{x}^{\mathsf{T}} A \boldsymbol{x} = 1$ reads $5x^2 + 8xy + 5y^2 = 1$. The eigenvalues are 9 and 1 (trace 10, determinant 9) with eigenvectors $\boldsymbol{q}_1 = (1,1)/\sqrt{2}$ and $\boldsymbol{q}_2 = (1,-1)/\sqrt{2}$. So

$$5x^2 + 8xy + 5y^2 = 9\Big(\frac{x+y}{\sqrt{2}}\Big)^2 + 1\Big(\frac{x-y}{\sqrt{2}}\Big)^2,$$

and the ellipse has half-axis $1/\sqrt{9} = \frac{1}{3}$ along $(1,1)$ and $1/\sqrt{1} = 1$ along $(1,-1)$ (Figure 40). Check the endpoints: $(x,y) = \frac{1}{3\sqrt{2}}(1,1)$ gives $\frac{5+8+5}{18} = 1$, and $(x,y) = \frac{1}{\sqrt{2}}(1,-1)$ gives $\frac{5-8+5}{2} = 1$.

For K of Example 31.5, $\boldsymbol{x}^{\mathsf{T}} K \boldsymbol{x} = 1$ is an ellipsoid with three different axes: the longest, $1/\sqrt{2-\sqrt{2}} \approx 1.31$, along $(1, \sqrt{2}, 1)$; the middle one, $1/\sqrt{2} \approx 0.71$, along $(1, 0, -1)$; the shortest, $1/\sqrt{2+\sqrt{2}} \approx 0.54$, along $(1, -\sqrt{2}, 1)$.

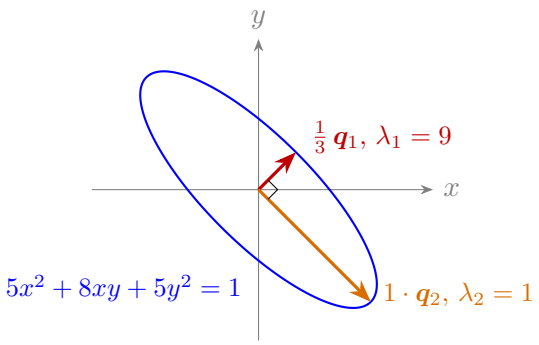


Figure 40: The principal axes of $\boldsymbol{x}^{\mathsf{T}} A \boldsymbol{x} = 1$ for $A = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$ point along the eigenvectors; the half-lengths are $1/\sqrt{\lambda}$, so the large eigenvalue 9 gives the short axis $\frac{1}{3}$.

31.5 Exercises

Exercise 31.10 (Products of positive definite matrices)**.** When A and B are symmetric positive definite, AB need not be symmetric, but its eigenvalues are still positive. Prove this: start from $AB\boldsymbol{x} = \lambda \boldsymbol{x}$ and take the dot product with $B\boldsymbol{x}$.

Solution. From $AB\boldsymbol{x} = \lambda \boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$, take the dot product with $B\boldsymbol{x}$:

$$(B\boldsymbol{x})^{\mathsf{T}} AB\boldsymbol{x} = \lambda\,(B\boldsymbol{x})^{\mathsf{T}} \boldsymbol{x} = \lambda\, \boldsymbol{x}^{\mathsf{T}} B \boldsymbol{x},$$

using $B^{\mathsf{T}} = B$. The left side is $\boldsymbol{y}^{\mathsf{T}} A \boldsymbol{y}$ with $\boldsymbol{y} = B\boldsymbol{x} \neq \mathbf{0}$ (B is invertible), which is positive because A is positive definite. On the right, $\boldsymbol{x}^{\mathsf{T}} B \boldsymbol{x} > 0$ because B is positive definite. Hence $\lambda = \boldsymbol{y}^{\mathsf{T}} A \boldsymbol{y} / \boldsymbol{x}^{\mathsf{T}} B \boldsymbol{x} > 0$. (Such a λ is automatically real: the quotient is real.) For instance $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, $B = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$ give the nonsymmetric $AB = \left[\begin{smallmatrix} 2.5 & 2 \\ 1.5 & 1.5 \end{smallmatrix}\right]$, whose eigenvalues $2 \pm \frac{\sqrt{13}}{2} \approx 3.80, 0.20$ are positive. □

Exercise 31.11 (The form of a nonsymmetric matrix)**.** Find the quadratic form $f(x,y)$ of $A = \left[\begin{smallmatrix} 1 & 5 \\ 7 & 9 \end{smallmatrix}\right]$. Is it always positive, always negative, or neither?

Solution. $f(x,y) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} x+5y \\ 7x+9y \end{bmatrix} = x^2 + 5xy + 7xy + 9y^2 = x^2 + 12xy + 9y^2$. Only the sum $5+7$ of the off-diagonal entries matters, so this is also the form of the symmetric matrix $\left[\begin{smallmatrix} 1 & 6 \\ 6 & 9 \end{smallmatrix}\right]$, whose determinant $9 - 36 = -27$ is negative. Hence f takes both signs: $f(1,0) = 1 > 0$ and $f(2,-2) = 4 - 48 + 36 = -8 < 0$. Completing the square, $f = (x+6y)^2 - 27y^2$. □

Exercise 31.12 (A parameter in the corner)**.** For which c is $B = \left[\begin{smallmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2+c \end{smallmatrix}\right]$ positive definite? Positive semidefinite? Confirm with pivots and by completing squares, and find $\boldsymbol{x} \neq \mathbf{0}$ with $\boldsymbol{x}^{\mathsf{T}} B \boldsymbol{x} = 0$ when $c = 0$.

Solution. Determinants. $D_1 = 2$, $D_2 = 4 - 1 = 3$, and expanding along row 1, $D_3 = 2\big(2(2+c) - 1\big) + 1\big(-(2+c) - 1\big) - 1\big(1+2\big) = (6+4c) - (3+c) - 3 = 3c$. So B is positive definite for $c > 0$ and positive semidefinite for $c \geq 0$ (for $c < 0$, $\det B < 0$ forces a negative eigenvalue).

Pivots. Add $\frac{1}{2}$ row 1 to rows 2 and 3: rows become $(0, \frac{3}{2}, -\frac{3}{2})$ and $(0, -\frac{3}{2}, \frac{3}{2} + c)$; add the new row 2 to row 3: $(0, 0, c)$. Pivots $2,\ \frac{3}{2},\ c$, which are the ratios $2,\ 3/2,\ 3c/3$ of the determinants.

Squares. $\boldsymbol{x}^{\mathsf{T}} B \boldsymbol{x} = 2x^2 + 2y^2 + (2+c)z^2 - 2xy - 2xz - 2yz = 2\big(x - \frac{1}{2}y - \frac{1}{2}z\big)^2 + \frac{3}{2}(y-z)^2 + cz^2$. For $c = 0$ this vanishes when $y = z$ and $x = \frac{1}{2}(y+z)$, for example at $\boldsymbol{x} = (1,1,1)$; indeed every row of B then sums to zero, so $(1,1,1)$ is in the nullspace. □

Exercise 31.13 (How big can the cross term be?)**.** For which b is $\left[\begin{smallmatrix} 1 & b \\ b & 9 \end{smallmatrix}\right]$ positive definite? For $b = 4$, find a vector that makes the form negative.

Solution. Need $1 > 0$ and $9 - b^2 > 0$, so $-3 < b < 3$. At $b = \pm 3$ the matrix is only semidefinite: $x^2 \pm 6xy + 9y^2 = (x \pm 3y)^2$. For $b = 4$, $x^2 + 8xy + 9y^2 = (x+4y)^2 - 7y^2$; choosing $x + 4y = 0$, $y = 1$, i.e. $(x,y) = (-4, 1)$, gives $16 - 32 + 9 = -7 < 0$. □

32 Similar matrices and Jordan form

This unit finishes the story of positive definite matrices by showing where they come from, $A^{\mathsf{T}}A$, and then turns to a new way of organizing all square matrices. Two matrices are *similar* when one is $M^{-1}AM$ for some invertible M. Similar matrices share their eigenvalues, so the square matrices fall into families, and in each family we look for the simplest member. When A has n independent eigenvectors that member is the diagonal Λ; when eigenvectors are missing it is the Jordan form. Diagonalization (Section 24) was the first example of similarity; Section 35 will show that similar matrices are the same linear transformation seen in different bases.

32.1 Where positive definite matrices come from

Positive definiteness behaves like positivity of numbers. Two quick facts show which of the four tests of Section 31 to reach for.

Proposition 32.1 (Inverses and sums)**.** *If A and B are symmetric positive definite, so are A^{-1} and $A + B$.*

Proof. A^{-1} is symmetric because $(A^{-1})^{\mathsf{T}} = (A^{\mathsf{T}})^{-1} = A^{-1}$, and its eigenvalues $1/\lambda_i$ are positive: the eigenvalue test is the natural one here, since we know nothing about the pivots of A^{-1}. For

$A + B$ we know neither eigenvalues nor pivots, but the energy test is immediate: for $\boldsymbol{x} \neq \mathbf{0}$, $\boldsymbol{x}^\mathsf{T}(A+B)\boldsymbol{x} = \boldsymbol{x}^\mathsf{T}A\boldsymbol{x} + \boldsymbol{x}^\mathsf{T}B\boldsymbol{x} > 0$. □

The most important source is a rectangular matrix. In least squares (Section 18) an $m \times n$ matrix A led to the square symmetric matrix $A^\mathsf{T}A$. Like the square of a number, it cannot be negative.

Theorem 32.2 ($A^\mathsf{T}A$ is positive definite)**.** *For any $m \times n$ matrix A, the matrix $A^\mathsf{T}A$ is symmetric positive semidefinite. It is positive definite exactly when A has independent columns (rank n).*

Proof. Symmetry: $(A^\mathsf{T}A)^\mathsf{T} = A^\mathsf{T}(A^\mathsf{T})^\mathsf{T} = A^\mathsf{T}A$. For the energy, move the parentheses:

$$\boldsymbol{x}^\mathsf{T}(A^\mathsf{T}A)\boldsymbol{x} = (A\boldsymbol{x})^\mathsf{T}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2 \geq 0.$$

This is zero only when $A\boldsymbol{x} = \mathbf{0}$. If the columns are independent, $\mathbf{N}(A) = \{\mathbf{0}\}$ and so $\boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \mathbf{0}$. If the columns are dependent, a nonzero $\boldsymbol{x}$ in the nullspace gives $\boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = 0$, so $A^\mathsf{T}A$ is only semidefinite. □

Example 32.3 (Two rectangular matrices)**.** For the line-fitting matrix $A = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, the columns are independent and $A^\mathsf{T}A = \left[\begin{smallmatrix} 3 & 3 \\ 3 & 5 \end{smallmatrix}\right]$ has upper-left determinants 3 and $15 - 9 = 6$: positive definite, so the normal equations $A^\mathsf{T}A\hat{\boldsymbol{x}} = A^\mathsf{T}\boldsymbol{b}$ always have a unique solution. For $B = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{smallmatrix}\right]$ the second column is twice the first, and $B^\mathsf{T}B = \left[\begin{smallmatrix} 14 & 28 \\ 28 & 56 \end{smallmatrix}\right]$ has determinant $784 - 784 = 0$: semidefinite only, with $\boldsymbol{x} = (2, -1)$ giving $\boldsymbol{x}^\mathsf{T}B^\mathsf{T}B\boldsymbol{x} = \|B\boldsymbol{x}\|^2 = 0$.

A practical bonus: elimination on a positive definite matrix never needs row exchanges, since all pivots are positive; there are no zeros, or dangerously small numbers, in the pivot positions. These are the best matrices to compute with.

32.2 Similar matrices share eigenvalues

Definition 32.4 (Similar matrices)**.** Two $n \times n$ matrices A and B are *similar* if $B = M^{-1}AM$ for some invertible matrix M.

If A has n independent eigenvectors in the columns of X, then $X^{-1}AX = \Lambda$ (Section 24): A is similar to Λ, with $M = X$. Any other invertible M also gives a similar matrix, just not a diagonal one.

Example 32.5 (A family with eigenvalues 3 and 1)**.** Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, with eigenvalues 3 and 1, so A is similar to $\Lambda = \left[\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right]$. Now choose any invertible M, say $M = \left[\begin{smallmatrix} 1 & 4 \\ 0 & 1 \end{smallmatrix}\right]$, whose inverse is $M^{-1} = \left[\begin{smallmatrix} 1 & -4 \\ 0 & 1 \end{smallmatrix}\right]$. Then

$$AM = \begin{bmatrix} 2 & 9 \\ 1 & 6 \end{bmatrix}, \qquad B = M^{-1}AM = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 9 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} -2 & -15 \\ 1 & 6 \end{bmatrix}.$$

Nothing about B looks special, but $\operatorname{tr} B = -2 + 6 = 4 = 3 + 1$ and $\det B = -12 + 15 = 3 = 3 \cdot 1$: its eigenvalues are again 3 and 1. Other members of the same family are the triangular matrices $\left[\begin{smallmatrix} 3 & 7 \\ 0 & 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 & 7 \\ 0 & 3 \end{smallmatrix}\right]$.

Theorem 32.6 (What similar matrices share)**.** *If $B = M^{-1}AM$, then A and B have the same characteristic polynomial, hence the same eigenvalues, trace and determinant. If $A\boldsymbol{x} = \lambda\boldsymbol{x}$ then $B(M^{-1}\boldsymbol{x}) = \lambda(M^{-1}\boldsymbol{x})$, and for each λ the two matrices have the same number of independent eigenvectors.*

Proof. Since $M^{-1}(\lambda I)M = \lambda I$,

$$\det(B - \lambda I) = \det\big(M^{-1}(A - \lambda I)M\big) = \det M^{-1}\,\det(A - \lambda I)\,\det M = \det(A - \lambda I).$$

For the eigenvectors, insert $MM^{-1} = I$: from $A\boldsymbol{x} = \lambda\boldsymbol{x}$ we get $AMM^{-1}\boldsymbol{x} = \lambda\boldsymbol{x}$, and multiplying on the left by M^{-1} gives $(M^{-1}AM)(M^{-1}\boldsymbol{x}) = \lambda(M^{-1}\boldsymbol{x})$. So M^{-1} carries the eigenvectors of A for λ onto eigenvectors of B for λ; it is invertible, so it maps independent vectors to independent vectors, and M carries them back. □

The eigenvalues stay; the eigenvectors move (by M^{-1}). Similarity is also an equivalence relation: $A = I^{-1}AI$; if $B = M^{-1}AM$ then $A = MBM^{-1} = (M^{-1})^{-1}B(M^{-1})$; and if also $C = N^{-1}BN$ then $C = (MN)^{-1}A(MN)$. So the $n \times n$ matrices split into disjoint families of mutually similar matrices.

Corollary 32.7 (Distinct eigenvalues)**.** *If A and B have the same n distinct eigenvalues, they are similar.*

Proof. Distinct eigenvalues give independent eigenvectors, so $A = X\Lambda X^{-1}$ and $B = Y\Lambda Y^{-1}$ with the same Λ. Then $B = YX^{-1}AXY^{-1} = M^{-1}AM$ with $M = XY^{-1}$. □

So when the eigenvalues are distinct, the family is simply "all matrices with these eigenvalues", and its best member is Λ.

32.3 Repeated eigenvalues: two families

Now let $\lambda_1 = \lambda_2 = 4$. The trouble with a repeated eigenvalue is that there may not be a full set of eigenvectors. The 2×2 matrices with eigenvalues $4, 4$ split into *two* families.

The loner. $4I$ is similar only to itself, because $M^{-1}(4I)M = 4M^{-1}M = 4I$ for every M.

Everyone else. Every other matrix with eigenvalues $4, 4$ is similar to the *Jordan block*

$$J = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}.$$

J is not diagonalizable: $J - 4I = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has rank 1, so there is only one independent eigenvector, $(1, 0)$. If J were diagonalizable it would be similar to $4I$, which is similar only to itself. To find more members of J's family, choose any trace 8 and determinant 16:

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 4 & 0 \\ 17 & 4 \end{bmatrix}, \qquad \begin{bmatrix} a & b \\ (8a - a^2 - 16)/b & 8 - a \end{bmatrix} \ (b \neq 0).$$

For the last one, the trace is $a + 8 - a = 8$ and the determinant is $a(8 - a) - (8a - a^2 - 16) = 16$. None of these is $4I$, so none is diagonalizable, and each has a single line of eigenvectors. Figure 41 pictures the families.

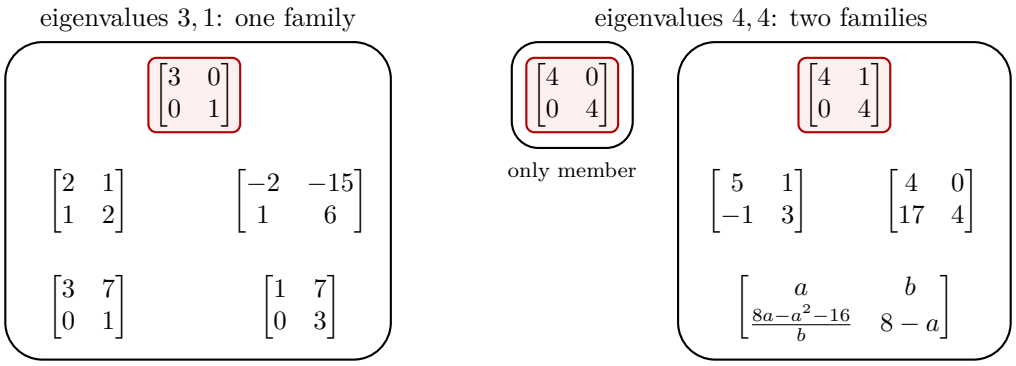


Figure 41: Families of similar 2×2 matrices. Within a family every pair is connected by some $M^{-1}AM$; the framed member is the simplest one (its Jordan form). With distinct eigenvalues $3, 1$ there is one family and its best member is Λ. With eigenvalues $4, 4$ the loner $4I$ forms a family by itself, and all the other matrices are similar to the Jordan block.

Example 32.8 (Bringing a matrix to its Jordan block)**.** Show directly that $B = \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$ is similar to $J = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$. The lectures do not compute Jordan forms in general, but for a 2×2 block the idea is short. We want $M = [\boldsymbol{x}_1 \ \boldsymbol{x}_2]$ with $BM = MJ$, i.e.

$$B\boldsymbol{x}_1 = 4\boldsymbol{x}_1, \qquad B\boldsymbol{x}_2 = \boldsymbol{x}_1 + 4\boldsymbol{x}_2 \quad \text{(equivalently } (B - 4I)\boldsymbol{x}_2 = \boldsymbol{x}_1\text{)}.$$

Here $B - 4I = \left[\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right]$ has rank 1; its nullspace gives the eigenvector $\boldsymbol{x}_1 = (1, -1)$, and $(B - 4I)\boldsymbol{x}_2 = \boldsymbol{x}_1$ is solved by $\boldsymbol{x}_2 = (1, 0)$. With $M = \left[\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right]$:

$$BM = \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ -4 & -1 \end{bmatrix}, \qquad MJ = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ -4 & -1 \end{bmatrix}.$$

They agree, M is invertible ($\det M = 1$), and so $M^{-1}BM = J$.

32.4 Jordan blocks and Jordan's theorem

Definition 32.9 (Jordan block, Jordan matrix)**.** A *Jordan block* J_i of size k has one eigenvalue λ_i repeated on the diagonal, ones just above the diagonal, and zeros elsewhere:

$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}.$$

Its only independent eigenvector is $\boldsymbol{e}_1$. A *Jordan matrix* is a block diagonal matrix whose diagonal blocks are Jordan blocks:

$$J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_d \end{bmatrix}.$$

Theorem 32.10 (Jordan form)**.** *Every square matrix A is similar to a Jordan matrix $J = M^{-1}AM$. The eigenvalues of A sit on the diagonal of J; the number of blocks d equals the number of independent eigenvectors of A, one eigenvector per block; and J is unique apart from the order of the blocks.*

The proof is long, and the course states the theorem without it; a proof is in the appendix on the Jordan form in Strang's *Introduction to Linear Algebra*, and in any advanced linear algebra text. Two cases summarize it. If A has n distinct eigenvalues, every block has size 1 and $J = \Lambda$: there are $d = n$ blocks and eigenvectors. If eigenvectors are missing, J has $n - d$ ones above the diagonal, one for each missing eigenvector. The uniqueness part, which is what separates families, can be proved with one invariant.

Proposition 32.11 (Ranks of powers are similarity invariants)**.** *If $B = M^{-1}AM$ then* $\operatorname{rank}(B - \lambda I)^k = \operatorname{rank}(A - \lambda I)^k$ *for every number λ and every $k \geq 1$.*

Proof. $B - \lambda I = M^{-1}(A - \lambda I)M$, and the inner factors MM^{-1} cancel in a power: $(B - \lambda I)^k = M^{-1}(A - \lambda I)^k M$. Multiplying by invertible matrices on either side does not change the rank. □

Counting eigenvectors uses only $k = 1$ (the nullity of $A - \lambda I$). Higher powers see the sizes of the blocks, as the next example shows.

Example 32.12 (Same eigenvalues, same eigenvector count, not similar)**.** Compare the 4×4 matrices

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Both are triangular with four zero eigenvalues. Both have rank 2, so each nullspace (the eigenvectors for $\lambda = 0$) has dimension $4 - 2 = 2$: two eigenvectors and two missing. But A consists of a 3×3 block and a 1×1 block, while C consists of two 2×2 blocks. The squares tell them apart: A^2 has a single 1 in position $(1, 3)$, so $\operatorname{rank} A^2 = 1$, whereas $C^2 = 0$. By Proposition 32.11, A and C are not similar.

Changing the $(1, 3)$ entry of A from 0 to 7 gives a matrix A' with the same ranks of powers ($\operatorname{rank} A' = 2$, $\operatorname{rank} A'^2 = 1$, $A'^3 = 0$); it is similar to A, and A is the Jordan form of that family: a 1 above the diagonal for every missing eigenvector, zeros elsewhere.

Common trap. Equal eigenvalues do not make matrices similar, and neither do equal eigenvalues plus equal numbers of eigenvectors. $\left[\begin{smallmatrix} 4 & 0 \\ 0 & 4 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 4 & 1 \\ 0 & 4 \end{smallmatrix}\right]$ share trace and determinant, and A, C above share even the eigenvector count. Only for distinct eigenvalues is "same eigenvalues" enough. Note also that the Jordan form, though perfect for algebra, is useless numerically: the slightest change in the entries splits repeated eigenvalues and changes the blocks.

32.5 Exercises

Exercise 32.13 (Jordan matrices with different block sizes)**.** Let J have Jordan blocks of sizes $2, 2$ and K blocks of sizes $3, 1$, all with eigenvalue 0:

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad K = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Both have two independent eigenvectors. Show that if $JM = MK$ then M is not invertible, so J and K are not similar.

Solution. Write $M = (m_{ij})$. Multiplying by J on the left shifts rows: row 1 of JM is row 2 of M, row 3 of JM is row 4 of M, rows 2 and 4 are zero. Multiplying by K on the right shifts columns: column 2 of MK is column 1 of M, column 3 is column 2, columns 1 and 4 are zero:

$$JM = \begin{bmatrix} m_{21} & m_{22} & m_{23} & m_{24} \\ 0 & 0 & 0 & 0 \\ m_{41} & m_{42} & m_{43} & m_{44} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad MK = \begin{bmatrix} 0 & m_{11} & m_{12} & 0 \\ 0 & m_{21} & m_{22} & 0 \\ 0 & m_{31} & m_{32} & 0 \\ 0 & m_{41} & m_{42} & 0 \end{bmatrix}.$$

Comparing entries: column 1 gives $m_{21} = m_{41} = 0$; row 2 gives $m_{21} = m_{22} = 0$ and row 4 gives $m_{41} = m_{42} = 0$; then entry $(1,2)$ gives $m_{11} = m_{22} = 0$ and entry $(3,2)$ gives $m_{31} = m_{42} = 0$. So the whole first column $(m_{11}, m_{21}, m_{31}, m_{41})$ is zero and M is singular. A similarity $K = M^{-1}JM$ would require an invertible M with $JM = MK$; none exists. (Ranks of squares give the same verdict: $J^2 = 0$ but $\operatorname{rank} K^2 = 1$.) □

Exercise 32.14 (True statements about similarity)**.** Explain why each statement is true.

(a) If A is similar to B then A^2 is similar to B^2.

(b) A^2 and B^2 can be similar when A and B are not.

(c) $\left[\begin{smallmatrix} 3 & 0 \\ 0 & 4 \end{smallmatrix}\right]$ is similar to $\left[\begin{smallmatrix} 3 & 1 \\ 0 & 4 \end{smallmatrix}\right]$.

(d) $\left[\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ is not similar to $\left[\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right]$.

(e) If B is obtained from A by exchanging rows 1 and 2 and then columns 1 and 2, then B is similar to A.

Solution. (a) If $A = M^{-1}BM$ then $A^2 = M^{-1}BMM^{-1}BM = M^{-1}B^2M$.

(b) Take $A = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. Then $A^2 = B^2 = 0$, which are certainly similar, but only the zero matrix is similar to the zero matrix, so A and B are not similar.

(c) The eigenvalues 3 and 4 are distinct, so both matrices are similar to $\operatorname{diag}(3,4)$ (Corollary 32.7). Explicitly, with $M = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $M^{-1}\left[\begin{smallmatrix} 3 & 1 \\ 0 & 4 \end{smallmatrix}\right]M = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 3 & 4 \\ 0 & 4 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 3 & 0 \\ 0 & 4 \end{smallmatrix}\right]$.

(d) For $3I$ every vector is an eigenvector; for the second matrix only the line through $(1,0)$. Similar matrices have the same number of independent eigenvectors.

(e) Exchanging rows 1 and 2 is left multiplication by the permutation matrix P obtained from I by exchanging its rows 1 and 2; exchanging columns 1 and 2 is right multiplication by the same P. So $B = PAP$, and since $P^{-1} = P$, $B = P^{-1}AP$.

□

Exercise 32.15 (Polynomials, distinct eigenvalues and Jordan blocks)**.** (a) If A and B are similar, show that $2A^3 + A - 3I$ is similar to $2B^3 + B - 3I$.

(b) If A and B are 3×3 matrices with eigenvalues $1, 0, -1$, must they be similar?

(c) Are $J_1 = \left[\begin{smallmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{smallmatrix}\right]$ and $J_2 = \left[\begin{smallmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right]$ similar?

Solution. (a) Let $B = MAM^{-1}$. Then $MA^3M^{-1} = (MAM^{-1})^3 = B^3$ and $MIM^{-1} = I$, so $M(2A^3 + A - 3I)M^{-1} = 2B^3 + B - 3I$. The same works for any polynomial in A.

(b) Yes. Three distinct eigenvalues make both diagonalizable with the same $\Lambda = \mathrm{diag}(1, 0, -1)$: $A = S\Lambda S^{-1}$ and $B = T\Lambda T^{-1}$, so $B = (TS^{-1})A(TS^{-1})^{-1}$. Similar to the same matrix means similar to each other.

(c) No. Both have the single eigenvalue -1, but $J_1 + I$ has rank 2 (a nullspace of dimension 1: one eigenvector), while $J_2 + I$ has rank 1 (a nullspace of dimension 2: two eigenvectors). Similar matrices have the same number of independent eigenvectors.

□

Exercise 32.16 (When is $A^\mathsf{T}A$ invertible?)**.** Let $A = \begin{bmatrix} 1 & 1 \\ 1 & t \\ 1 & t^2 \end{bmatrix}$. For which t is $A^\mathsf{T}A$ positive definite? Compute $A^\mathsf{T}A$ for $t = 1$ and exhibit $\boldsymbol{x} \neq \mathbf{0}$ with $\boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = 0$.

Solution. By Theorem 32.2, $A^\mathsf{T}A$ is positive definite exactly when the columns $(1, 1, 1)$ and $(1, t, t^2)$ are independent, i.e. unless $(1, t, t^2)$ is a multiple of $(1, 1, 1)$, which happens only for $t = 1$. So: positive definite for every $t \neq 1$. At $t = 1$ both columns are $(1, 1, 1)$ and $A^\mathsf{T}A = \left[\begin{smallmatrix} 3 & 3 \\ 3 & 3 \end{smallmatrix}\right]$; the vector $\boldsymbol{x} = (1, -1)$ gives $A\boldsymbol{x} = \mathbf{0}$, so $\boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = 3 - 3 - 3 + 3 = 0$. □

33 Singular value decomposition

The singular value decomposition, the SVD, is the final and best factorization of a matrix:

$$A = U\Sigma V^\mathsf{T}, \qquad U \text{ orthogonal}, \quad \Sigma \text{ diagonal}, \quad V \text{ orthogonal}.$$

Unlike every earlier factorization, it works for *any* matrix: rectangular, singular, not symmetric. We have met two relatives. A symmetric positive definite matrix is $Q\Lambda Q^\mathsf{T}$ (Section 29), which is already of this form with $U = V = Q$. A diagonalizable matrix is $X\Lambda X^{-1}$ (Section 24), but X is usually not orthogonal and many matrices have no such factorization at all. The SVD keeps the diagonal middle and the orthogonal outsides by paying one price: two different orthogonal matrices, U on the left and V on the right. The construction uses everything from the last units: $A^\mathsf{T}A$ is symmetric positive semidefinite (Section 32), so the spectral theorem applies to it, and the result produces orthonormal bases for all four fundamental subspaces (Section 11).

33.1 Every matrix turns a circle into an ellipse

Feed a 2×2 matrix every unit vector, the whole unit circle. The outputs always form an ellipse. The SVD explains why: some pair of perpendicular unit vectors goes to perpendicular vectors, and those are the axes of the ellipse.

Example 33.1 (Perpendicular in, perpendicular out)**.** Let $A = \begin{bmatrix} 6 & 8 \\ -4 & 3 \end{bmatrix}$ act on the unit circle.

The ordinary axes get skewed. $A(1, 0) = (6, -4)$ and $A(0, 1) = (8, 3)$. Their dot product is $48 - 12 = 36 \neq 0$: the right angle between $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ is lost.

A better pair. Take the perpendicular unit vectors $\boldsymbol{v}_1 = (0.6,\ 0.8)$ and $\boldsymbol{v}_2 = (-0.8,\ 0.6)$. Then

$$A\boldsymbol{v}_1 = (3.6 + 6.4,\ -2.4 + 2.4) = (10,\ 0) = 10\,(1, 0),$$
$$A\boldsymbol{v}_2 = (-4.8 + 4.8,\ 3.2 + 1.8) = (0,\ 5) = 5\,(0, 1).$$

The outputs are perpendicular too. So $\sigma_1 = 10$, $\sigma_2 = 5$, $\boldsymbol{u}_1 = (1, 0)$, $\boldsymbol{u}_2 = (0, 1)$, and

$$A = U\Sigma V^{\mathsf{T}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix}.$$

The unit circle becomes the ellipse with semi-axes 10 along $\boldsymbol{u}_1$ and 5 along $\boldsymbol{u}_2$ (Figure 42).

Checks. $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 6 & -4 \\ 8 & 3 \end{smallmatrix}\right] \left[\begin{smallmatrix} 6 & 8 \\ -4 & 3 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 52 & 36 \\ 36 & 73 \end{smallmatrix}\right]$ has trace $125 = 100 + 25$ and determinant $52 \cdot 73 - 36^2 = 3796 - 1296 = 2500 = 100 \cdot 25$, so its eigenvalues are $100 = 10^2$ and $25 = 5^2$. And the area factor is $|\det A| = |18 + 32| = 50 = \sigma_1\sigma_2$: the ellipse has area $\pi \cdot 10 \cdot 5 = 50\pi$, fifty times the circle's.

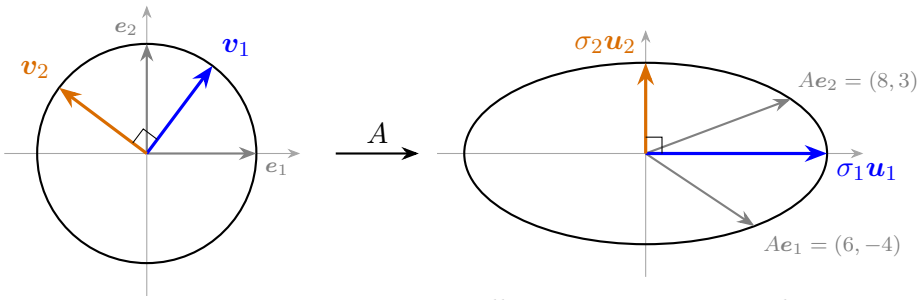


Figure 42: The SVD of $A = \left[\begin{smallmatrix} 6 & 8 \\ -4 & 3 \end{smallmatrix}\right]$ seen geometrically. The standard axes (grey) land on the skewed pair $(6, -4)$, $(8, 3)$. The perpendicular unit vectors $\boldsymbol{v}_1 = (0.6, 0.8)$, $\boldsymbol{v}_2 = (-0.8, 0.6)$ land on $(10, 0)$ and $(0, 5)$, still perpendicular: they are exactly the long and short axes of the image ellipse.

Common trap. Singular values are not eigenvalues. This A has $\operatorname{tr} A = 9$ and $\det A = 50$, so its eigenvalues solve $\lambda^2 - 9\lambda + 50 = 0$: $\lambda = \frac{1}{2}\left(9 \pm i\sqrt{119}\right)$, complex. No real direction keeps its line, yet the singular values 10 and 5 are real and positive. In general $A\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$ moves $\boldsymbol{v}_i$ to a different direction $\boldsymbol{u}_i$; only when $\boldsymbol{u}_i = \boldsymbol{v}_i$ is σ_i an eigenvalue.

33.2 The goal in matrix language

Think of A ($m \times n$, rank r) as taking each vector $\boldsymbol{v}$ of its row space to $A\boldsymbol{v}$ in its column space. Gram–Schmidt gives an orthonormal basis of the row space at once, but A will usually destroy its orthogonality. The SVD asks for a *special* orthonormal basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ of the row space whose images are orthogonal:

$$A\boldsymbol{v}_1 = \sigma_1\boldsymbol{u}_1, \quad A\boldsymbol{v}_2 = \sigma_2\boldsymbol{u}_2, \quad \ldots, \quad A\boldsymbol{v}_r = \sigma_r\boldsymbol{u}_r,$$

with $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ orthonormal in the column space and stretching factors $\sigma_i > 0$. In matrix form, with the $\boldsymbol{v}$'s and $\boldsymbol{u}$'s as columns,

$$A \begin{bmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_r \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}.$$

The nullspaces cause no difficulty. Complete $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ by an orthonormal basis $\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n$ of $\mathbf{N}(A)$ (so $A\boldsymbol{v}_j = \boldsymbol{0}$), and complete the $\boldsymbol{u}$'s by an orthonormal basis $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ of $\mathbf{N}(A^{\mathsf{T}})$. The extra columns are multiplied by zeros in Σ, and the equation becomes $AV = U\Sigma$ with square orthogonal V and U.

Definition 33.2 (Singular value decomposition)**.** A *singular value decomposition* of an $m \times n$ matrix A of rank r is a factorization

$$A = U\Sigma V^{\mathsf{T}},$$

where U ($m \times m$) and V ($n \times n$) are orthogonal and Σ ($m \times n$) is zero except for $\Sigma_{ii} = \sigma_i$, $i = 1, \dots, r$, with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$. The σ_i are the *singular values*; the columns $\boldsymbol{v}_i$ of V and $\boldsymbol{u}_i$ of U are the right and left *singular vectors*.

Multiplying $AV = U\Sigma$ on the right by $V^{-1} = V^{\mathsf{T}}$ gives $A = U\Sigma V^{\mathsf{T}}$; multiplying out columns times rows gives the SVD as a sum of r rank-one pieces,

$$A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathsf{T}} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathsf{T}} + \cdots + \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^{\mathsf{T}}.$$

Reading $A\boldsymbol{x} = U\big(\Sigma(V^{\mathsf{T}}\boldsymbol{x})\big)$ from the right: V^{T} rotates (or reflects), Σ stretches along the axes, U rotates (or reflects). That is the circle-to-ellipse picture in any dimension.

33.3 Finding V, Σ and U

Two orthogonal matrices are unknown, and we do not want to find them at once. The combination $A^{\mathsf{T}}A$ makes U disappear:

$$A^{\mathsf{T}}A = (V\Sigma^{\mathsf{T}}U^{\mathsf{T}})(U\Sigma V^{\mathsf{T}}) = V(\Sigma^{\mathsf{T}}\Sigma)V^{\mathsf{T}} = V \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix} V^{\mathsf{T}},$$

with $\sigma_i = 0$ for $i > r$. This is $Q\Lambda Q^{\mathsf{T}}$ for the symmetric matrix $A^{\mathsf{T}}A$: *the $\boldsymbol{v}$'s are eigenvectors of $A^{\mathsf{T}}A$ and the σ_i^2 are its eigenvalues.* In the opposite order, $AA^{\mathsf{T}} = U(\Sigma\Sigma^{\mathsf{T}})U^{\mathsf{T}}$: the $\boldsymbol{u}$'s are eigenvectors of AA^{T}, with the same nonzero eigenvalues σ_i^2. The construction turns this into a proof.

Theorem 33.3 (Every matrix has an SVD)**.** *Let A be $m \times n$ of rank r. Let $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ be orthonormal eigenvectors of $A^{\mathsf{T}}A$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$, and set $\sigma_i = \sqrt{\lambda_i}$. Then $\sigma_1, \dots, \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$; the vectors*

$$\boldsymbol{u}_i = \frac{A\boldsymbol{v}_i}{\sigma_i} \qquad (i = 1, \dots, r)$$

are orthonormal eigenvectors of AA^{T}; and completing them to an orthonormal basis of $\mathbb{R}^m$ gives $A = U\Sigma V^{\mathsf{T}}$.

Proof. $A^{\mathsf{T}}A$ is symmetric positive semidefinite (Section 32), so the spectral theorem supplies orthonormal eigenvectors with real eigenvalues $\lambda_i = \boldsymbol{v}_i^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{v}_i = \|A\boldsymbol{v}_i\|^2 \geq 0$. In particular $\|A\boldsymbol{v}_i\| = \sigma_i$. The number of nonzero eigenvalues is $\operatorname{rank}(A^{\mathsf{T}}A) = \operatorname{rank} A = r$ (because $\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A)$: if $A^{\mathsf{T}}A\boldsymbol{x} = \mathbf{0}$ then $\|A\boldsymbol{x}\|^2 = \boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = 0$).

For $i, j \leq r$,

$$\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{u}_j = \frac{\boldsymbol{v}_i^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{v}_j}{\sigma_i\sigma_j} = \frac{\sigma_j^2\,\boldsymbol{v}_i^{\mathsf{T}}\boldsymbol{v}_j}{\sigma_i\sigma_j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

so the $\boldsymbol{u}_i$ are orthonormal, and $AA^{\mathsf{T}}\boldsymbol{u}_i = A(A^{\mathsf{T}}A\boldsymbol{v}_i)/\sigma_i = \sigma_i^2\,A\boldsymbol{v}_i/\sigma_i = \sigma_i^2\boldsymbol{u}_i$. For $i > r$, $\|A\boldsymbol{v}_i\| = \sigma_i = 0$, so $A\boldsymbol{v}_i = \mathbf{0}$. Extend $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$ of $\mathbb{R}^m$. Column by column, $A\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$ for $i \leq r$ and $A\boldsymbol{v}_i = \mathbf{0}$ for $i > r$, which is $AV = U\Sigma$. Since V is orthogonal, $A = U\Sigma V^{\mathsf{T}}$. □

Common trap. Do not compute U separately from the eigenvectors of AA^{T} and then just multiply. Each eigenvector is determined only up to sign (more freedom still for repeated eigenvalues), and an independent choice of signs in U and V can give $U\Sigma V^{\mathsf{T}} \neq A$. This happened in the lecture itself (next example). Find V from $A^{\mathsf{T}}A$, then compute $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$; use AA^{T} only as a check.

Example 33.4 (The lecture's invertible example)**.** Take $A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$ (rank 2).

Step 1: $A^{\mathsf{T}}A$. $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 4 & -3 \\ 4 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 4 & 4 \\ -3 & 3 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 25 & 7 \\ 7 & 25 \end{smallmatrix}\right]$. Its eigenvectors are $(1,1)$ with eigenvalue $25+7=32$ and $(1,-1)$ with eigenvalue $25-7=18$. Normalizing, $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$, and $\sigma_1 = \sqrt{32} = 4\sqrt{2}$, $\sigma_2 = \sqrt{18} = 3\sqrt{2}$.

Step 2: $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$. $A\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(8,0) = (4\sqrt{2},0)$, so $\boldsymbol{u}_1 = (1,0)$. $A\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(0,-6) = (0,-3\sqrt{2})$, so $\boldsymbol{u}_2 = (0,-1)$. Therefore

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & 3\sqrt{2} \end{bmatrix}\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}.$$

Check with AA^{T}. $AA^{\mathsf{T}} = \left[\begin{smallmatrix} 32 & 0 \\ 0 & 18 \end{smallmatrix}\right]$, diagonal by luck, with the same eigenvalues 32 and 18 (the eigenvalues of AB and BA always agree). Its eigenvectors are $(1,0)$ and $(0,\pm 1)$. Choosing $\boldsymbol{u}_2 = (0,1)$ from AA^{T} alone, as the lecture first did, produces $\left[\begin{smallmatrix} 4 & 4 \\ 3 & -3 \end{smallmatrix}\right]$: the second row has the wrong sign. The formula $\boldsymbol{u}_2 = A\boldsymbol{v}_2/\sigma_2$ fixes the sign to $(0,-1)$.

Example 33.5 (A rank-one matrix with a nullspace)**.** Let $A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$. Every row is a multiple of $(4,3)$ and every column a multiple of $(1,2)$, so $r=1$: the row space is the line through $(4,3)$, the column space the line through $(1,2)$, and the nullspaces are the perpendicular lines.

- Row space: $\boldsymbol{v}_1 = (4,3)/5 = (0.8,\ 0.6)$. Nullspace: $\boldsymbol{v}_2 = (0.6,\ -0.8)$.
- Column space: $\boldsymbol{u}_1 = (1,2)/\sqrt{5}$. Left nullspace: $\boldsymbol{u}_2 = (2,-1)/\sqrt{5}$.
- $A^{\mathsf{T}}A = \left[\begin{smallmatrix} 80 & 60 \\ 60 & 45 \end{smallmatrix}\right]$ has rank 1, so one eigenvalue is 0 and the other equals the trace 125. So $\sigma_1 = \sqrt{125} = 5\sqrt{5}$.

Check the direction: $A\boldsymbol{v}_1 = (3.2+1.8,\ 6.4+3.6) = (5,10) = 5\sqrt{5}\cdot\frac{1}{\sqrt{5}}(1,2) = \sigma_1\boldsymbol{u}_1$. The SVD is

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0.8 & 0.6 \\ 0.6 & -0.8 \end{bmatrix},$$

and as a single rank-one piece, $\sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^{\mathsf{T}} = 5\sqrt{5}\cdot\frac{1}{\sqrt{5}}\left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right]\begin{bmatrix} 0.8 & 0.6 \end{bmatrix} = 5\left[\begin{smallmatrix} 0.8 & 0.6 \\ 1.6 & 1.2 \end{smallmatrix}\right] = A$. The nullspace vectors $\boldsymbol{v}_2$ and $\boldsymbol{u}_2$ only meet the zero in Σ.

33.4 The right bases for the four subspaces

Theorem 33.6 (SVD bases)**.** *In an SVD $A = U\Sigma V^{\mathsf{T}}$ of an $m \times n$ matrix of rank r:*

- *$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ is an orthonormal basis of the row space $\mathbf{C}(A^{\mathsf{T}})$;*
- *$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ is an orthonormal basis of the column space $\mathbf{C}(A)$;*
- *$\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n$ is an orthonormal basis of the nullspace $\mathbf{N}(A)$;*
- *$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ is an orthonormal basis of the left nullspace $\mathbf{N}(A^{\mathsf{T}})$;*

and in these bases A is diagonal: $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ and $A^{\mathsf{T}}\boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i$ for $i \le r$.

Proof. For $j > r$, $A\boldsymbol{v}_j = \boldsymbol{0}$, so $\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n$ are $n-r$ orthonormal vectors in $\mathbf{N}(A)$, whose dimension is $n-r$: they are a basis. The remaining $\boldsymbol{v}_1, \dots, \boldsymbol{v}_r$ are orthogonal to all of $\mathbf{N}(A)$, so they lie in its orthogonal complement, the row space (Section 16), which has dimension r. For $i \le r$, $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$ is in $\mathbf{C}(A)$, of dimension r, so $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ is a basis of it; the other $\boldsymbol{u}$'s are orthogonal to $\mathbf{C}(A)$, hence in $\mathbf{N}(A^{\mathsf{T}})$, of dimension $m-r$. Finally, transposing $A = U\Sigma V^{\mathsf{T}}$ gives $A^{\mathsf{T}} = V\Sigma^{\mathsf{T}}U^{\mathsf{T}}$, so $A^{\mathsf{T}}U = V\Sigma^{\mathsf{T}}$, i.e. $A^{\mathsf{T}}\boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i$. □

These are the "right" bases at last. Gram–Schmidt could already make bases orthonormal; the eigenvectors of $A^{\mathsf{T}}A$ make them orthonormal *and* uncouple the matrix: each $\boldsymbol{v}_i$ goes to a multiple of the matching $\boldsymbol{u}_i$ and nothing else (Figure 43). For a square matrix one more consequence follows: $|\det A| = |\det U| \det \Sigma |\det V| = \sigma_1\sigma_2\cdots\sigma_n$, because orthogonal matrices have determinant ± 1. The singular values measure how A stretches volume, as in Example 33.1.

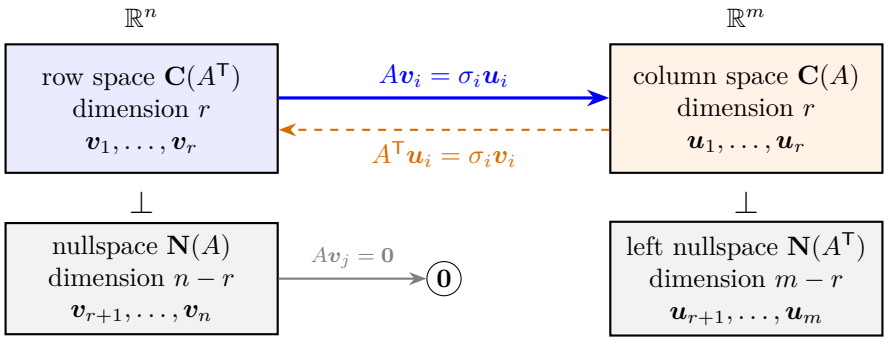


Figure 43: The SVD chooses orthonormal bases for the four fundamental subspaces so that A acts diagonally: the row space basis $\boldsymbol{v}_i$ goes to σ_i times the column space basis $\boldsymbol{u}_i$, and the nullspace basis goes to zero. The left nullspace basis $\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is never reached.

33.5 Exercises

Exercise 33.7 (Following the ellipse)**.** For $A = \left[\begin{smallmatrix} 6 & 8 \\ -4 & 3 \end{smallmatrix}\right]$ of Example 33.1, where does A send $2\boldsymbol{v}_1 + \boldsymbol{v}_2$? Answer once by direct multiplication and once using $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$.

Solution. $2\boldsymbol{v}_1 + \boldsymbol{v}_2 = (1.2 - 0.8,\ 1.6 + 0.6) = (0.4,\ 2.2)$. Directly, $A(0.4, 2.2) = (2.4 + 17.6,\ -1.6 + 6.6) = (20,\ 5)$. With the SVD, $A(2\boldsymbol{v}_1 + \boldsymbol{v}_2) = 2\sigma_1\boldsymbol{u}_1 + \sigma_2\boldsymbol{u}_2 = 2(10, 0) + (0, 5) = (20, 5)$. In the $\boldsymbol{v}$-coordinates the matrix just scales the coordinates 2 and 1 by 10 and 5. □

Exercise 33.8 (The Fibonacci matrix)**.** Show that the singular values of $A = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ are $\sigma_1 = \frac{1+\sqrt{5}}{2}$ and $\sigma_2 = \frac{\sqrt{5}-1}{2}$. How are they related to the eigenvalues of A?

Solution. $A^{\mathsf{T}}A = AA^{\mathsf{T}} = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, with characteristic polynomial $\lambda^2 - 3\lambda + 1$, so $\sigma_1^2 = \frac{3+\sqrt{5}}{2}$ and $\sigma_2^2 = \frac{3-\sqrt{5}}{2}$. Squaring the claimed values: $\left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{3+\sqrt{5}}{2}$ and $\left(\frac{\sqrt{5}-1}{2}\right)^2 = \frac{5-2\sqrt{5}+1}{4} = \frac{3-\sqrt{5}}{2}$. Since A is symmetric, its eigenvalues $\frac{1\pm\sqrt{5}}{2}$ are real and the singular values are their absolute values; the negative eigenvalue $\frac{1-\sqrt{5}}{2}$ becomes $\sigma_2 = \frac{\sqrt{5}-1}{2}$, and its sign moves into $\boldsymbol{u}_2 = -\boldsymbol{v}_2$. □

Exercise 33.9 (Orthogonal columns)**.** Suppose the n columns $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ of A are orthogonal, with lengths $\sigma_1 \geq \cdots \geq \sigma_n > 0$. Compute $A^{\mathsf{T}}A$ and find U, Σ, V.

Solution. The (i, j) entry of $A^{\mathsf{T}}A$ is $\boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{w}_j$, so $A^{\mathsf{T}}A = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. Its eigenvectors are the standard basis vectors, so $V = I$ and the singular values are the lengths σ_i. Then $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i = A\boldsymbol{e}_i/\sigma_i = \boldsymbol{w}_i/\sigma_i$: the columns of U are the unit vectors in the column directions (completed by an orthonormal basis of $\mathbf{N}(A^{\mathsf{T}})$ if $m > n$). □

Exercise 33.10 (An SVD from scratch)**.** Find the SVD of $C = \left[\begin{smallmatrix} 5 & 5 \\ -1 & 7 \end{smallmatrix}\right]$.

Solution. $C^{\mathsf{T}}C = \left[\begin{smallmatrix} 5 & -1 \\ 5 & 7 \end{smallmatrix}\right]\left[\begin{smallmatrix} 5 & 5 \\ -1 & 7 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 26 & 18 \\ 18 & 74 \end{smallmatrix}\right]$, with $\det(C^{\mathsf{T}}C - \lambda I) = \lambda^2 - 100\lambda + (26 \cdot 74 - 324) = \lambda^2 - 100\lambda + 1600 = (\lambda - 80)(\lambda - 20)$. For $\lambda = 80$: $C^{\mathsf{T}}C - 80I = \left[\begin{smallmatrix} -54 & 18 \\ 18 & -6 \end{smallmatrix}\right]$ gives $\boldsymbol{v}_1 = (1, 3)/\sqrt{10}$. For $\lambda = 20$: $\left[\begin{smallmatrix} 6 & 18 \\ 18 & 54 \end{smallmatrix}\right]$ gives $\boldsymbol{v}_2 = (-3, 1)/\sqrt{10}$. So $\sigma_1 = \sqrt{80} = 4\sqrt{5}$, $\sigma_2 = \sqrt{20} = 2\sqrt{5}$. Then

$$\boldsymbol{u}_1 = \frac{C\boldsymbol{v}_1}{\sigma_1} = \frac{(20, 20)}{\sqrt{10} \cdot 4\sqrt{5}} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{u}_2 = \frac{C\boldsymbol{v}_2}{\sigma_2} = \frac{(-10, 10)}{\sqrt{10} \cdot 2\sqrt{5}} = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

$$C = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4\sqrt{5} & 0 \\ 0 & 2\sqrt{5} \end{bmatrix}\frac{1}{\sqrt{10}}\begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}.$$

Check: $|\det C| = |35 + 5| = 40 = 4\sqrt{5} \cdot 2\sqrt{5}$. □

34 Linear transformations and their matrices

Many linear algebra courses begin where this unit begins: with linear transformations, described without coordinates. A projection onto a line or a rotation of the plane makes sense before anyone writes down a matrix, and some people (physicists especially) prefer to think that way, about the whole space at once. But as soon as we want to compute, we choose a basis, coordinates appear, and every linear transformation becomes a matrix. Everything we have learned about matrices (nullspaces, determinants, eigenvalues) then applies. This unit explains that passage from transformation to matrix, and why the eigenvector basis gives the best matrix of all.

34.1 Linearity without coordinates

Definition 34.1 (Linear transformation)**.** Let V and W be vector spaces. A *transformation* (or map) $T\colon V \to W$ assigns an output $T(\boldsymbol{v}) \in W$ to every input $\boldsymbol{v} \in V$. It is *linear* if, for all vectors $\boldsymbol{v}, \boldsymbol{w}$ and all scalars c,

$$T(\boldsymbol{v} + \boldsymbol{w}) = T(\boldsymbol{v}) + T(\boldsymbol{w}) \qquad \text{and} \qquad T(c\boldsymbol{v}) = cT(\boldsymbol{v}).$$

Equivalently, $T(c\boldsymbol{v} + d\boldsymbol{w}) = cT(\boldsymbol{v}) + dT(\boldsymbol{w})$: T respects the two operations on vectors, addition and scalar multiplication, and so respects every linear combination.

Proposition 34.2 (Zero goes to zero)**.** *Every linear transformation satisfies* $T(\mathbf{0}) = \mathbf{0}$.

Proof. Take $c = 0$ in $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$: $T(\mathbf{0}) = T(0\boldsymbol{v}) = 0\,T(\boldsymbol{v}) = \mathbf{0}$. □

Examples.

- *Projection* $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ onto a fixed line through the origin. Doubling $\boldsymbol{v}$ doubles its projection, reversing $\boldsymbol{v}$ reverses it, and the projection of a sum is the sum of the projections.
- *Rotation* of the plane by 45° about the origin. Rotating $\boldsymbol{v}+\boldsymbol{w}$ gives the same result as rotating $\boldsymbol{v}$ and $\boldsymbol{w}$ and then adding: the whole parallelogram rotates.
- *Multiplication by a matrix*, $T(\boldsymbol{v}) = A\boldsymbol{v}$. This is linear because $A(\boldsymbol{v}+\boldsymbol{w}) = A\boldsymbol{v} + A\boldsymbol{w}$ and $A(c\boldsymbol{v}) = c\,A\boldsymbol{v}$. Any 2×3 matrix gives a linear $T\colon \mathbb{R}^3 \to \mathbb{R}^2$.
- *Differentiation*, $T(f) = f'$, on a space of functions: $(f+g)' = f' + g'$ and $(cf)' = cf'$. This is why we can differentiate any combination of $\sin x$, $\cos x$, e^x, … once we know the derivatives of a short list.

Non-examples.

- *Shift the whole plane*, $T(\boldsymbol{v}) = \boldsymbol{v} + \boldsymbol{v}_0$ with a fixed $\boldsymbol{v}_0 \neq \boldsymbol{0}$. Then $T(2\boldsymbol{v}) = 2\boldsymbol{v} + \boldsymbol{v}_0$ but $2T(\boldsymbol{v}) = 2\boldsymbol{v} + 2\boldsymbol{v}_0$, and $T(\boldsymbol{0}) = \boldsymbol{v}_0 \neq \boldsymbol{0}$.
- *Length*, $T(\boldsymbol{v}) = \|\boldsymbol{v}\|$ from $\mathbb{R}^3$ to $\mathbb{R}$. It does send $\boldsymbol{0}$ to 0 and doubles when $\boldsymbol{v}$ doubles, but $T(-2\boldsymbol{v}) = 2\|\boldsymbol{v}\| \neq -2\|\boldsymbol{v}\|$.

The advantage of the coordinate-free view is the big picture. Instead of following one vector, watch what T does to a whole figure, such as the outline of a house (Figure 44). The matrix $\left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$ keeps each x-component and reverses each y-component, so it reflects the house across the x-axis and turns it upside down; the rotation turns it without distortion; the shift moves it, origin and all, which is how we see at once that it is not linear.

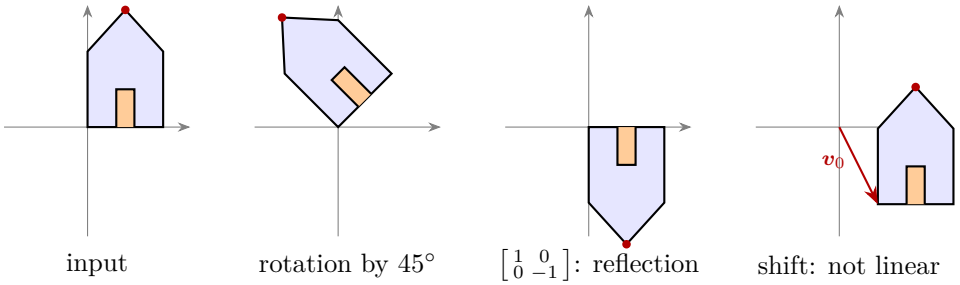


Figure 44: A linear transformation acts on the whole plane at once; following a house shows the effect. Rotation and the reflection matrix keep the origin fixed and carry straight lines to straight lines and parallelograms to parallelograms. The shift $T(\boldsymbol{v}) = \boldsymbol{v} + \boldsymbol{v}_0$ moves the corner at the origin to $\boldsymbol{v}_0$, violating $T(\boldsymbol{0}) = \boldsymbol{0}$.

34.2 A basis determines everything

How much must we be told about a linear T to know $T(\boldsymbol{v})$ for every input? Knowing $T(\boldsymbol{v}_1)$ gives $T(c\boldsymbol{v}_1) = cT(\boldsymbol{v}_1)$ on a whole line. Knowing $T(\boldsymbol{v}_1)$ and $T(\boldsymbol{v}_2)$ for independent $\boldsymbol{v}_1, \boldsymbol{v}_2$ gives $T(c\boldsymbol{v}_1 + d\boldsymbol{v}_2)$ on a whole plane. (If $\boldsymbol{v}_2$ were $6\boldsymbol{v}_1$, then $T(\boldsymbol{v}_2)$ would carry no new information.) A basis is enough.

Theorem 34.3 (Linear maps are determined by a basis)**.** *Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ be a basis of V. If T is linear, then for every $\boldsymbol{v} = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$,*

$$T(\boldsymbol{v}) = c_1T(\boldsymbol{v}_1) + c_2T(\boldsymbol{v}_2) + \cdots + c_nT(\boldsymbol{v}_n).$$

So T is completely known once $T(\boldsymbol{v}_1), \dots, T(\boldsymbol{v}_n)$ are known.

Proof. Because the $\boldsymbol{v}_i$ span V, every $\boldsymbol{v}$ is such a combination; linearity, applied repeatedly to the combination, gives the formula. □

Definition 34.4 (Coordinates)**.** If $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ is a basis of V and $\boldsymbol{v} = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$, the numbers $(c_1, \dots, c_n)$ are the *coordinates* of $\boldsymbol{v}$ in this basis. They are unique: if also $\boldsymbol{v} = \sum d_i\boldsymbol{v}_i$, then $\sum(c_i - d_i)\boldsymbol{v}_i = \boldsymbol{0}$, and independence forces every $c_i = d_i$.

Coordinates come from a basis. When someone writes $\boldsymbol{v} = (3, 2, 4)$ we silently assume the standard basis, $\boldsymbol{v} = 3\boldsymbol{e}_1 + 2\boldsymbol{e}_2 + 4\boldsymbol{e}_3$, and then the coordinates are the components. But another basis, for example the eigenvectors of some matrix, gives the same vector different coordinates.

34.3 The matrix of a transformation

To turn $T\colon V \to W$ into a matrix we choose *two* bases: an input basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ of V and an output basis $\boldsymbol{w}_1, \dots, \boldsymbol{w}_m$ of W. The matrix will take the input coordinates to the output coordinates.

Definition 34.5 (Matrix of T in chosen bases)**.** Expand the output of each input basis vector in the output basis:

$$T(\boldsymbol{v}_j) = a_{1j}\boldsymbol{w}_1 + a_{2j}\boldsymbol{w}_2 + \cdots + a_{mj}\boldsymbol{w}_m \qquad (j = 1, \dots, n).$$

The $m \times n$ matrix $A = (a_{ij})$ is the *matrix of T* in these bases: column j holds the output coordinates of $T(\boldsymbol{v}_j)$.

Theorem 34.6 (The matrix does what T does)**.** *If $\boldsymbol{v}$ has input coordinates $\boldsymbol{c} = (c_1, \dots, c_n)$, then $T(\boldsymbol{v})$ has output coordinates $A\boldsymbol{c}$.*

Proof. By Theorem 34.3 and the definition of the columns,

$$T(\boldsymbol{v}) = \sum_j c_j T(\boldsymbol{v}_j) = \sum_j c_j \sum_i a_{ij}\boldsymbol{w}_i = \sum_i \Big(\sum_j a_{ij}c_j\Big)\boldsymbol{w}_i,$$

and $\sum_j a_{ij}c_j$ is component i of $A\boldsymbol{c}$. As a sanity check, the input coordinates $(1, 0, \dots, 0)$ describe $\boldsymbol{v}_1$, and A times them is column 1, the coordinates of $T(\boldsymbol{v}_1)$. □

Example 34.7 (One projection, two matrices)**.** Let T project $\mathbb{R}^2$ onto the 45° line, using the same basis for inputs and outputs.

A good basis. Take $\boldsymbol{v}_1 = (1, 1)/\sqrt{2}$ on the line and $\boldsymbol{v}_2 = (-1, 1)/\sqrt{2}$ perpendicular to it. The projection leaves $\boldsymbol{v}_1$ alone and kills $\boldsymbol{v}_2$: $T(\boldsymbol{v}_1) = 1\boldsymbol{v}_1 + 0\boldsymbol{v}_2$ and $T(\boldsymbol{v}_2) = 0\boldsymbol{v}_1 + 0\boldsymbol{v}_2$. So

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad A \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ 0 \end{bmatrix}.$$

These basis vectors are the eigenvectors of the projection (eigenvalues 1 and 0), and the matrix came out as the diagonal eigenvalue matrix Λ.

The standard basis. With $\boldsymbol{v}_1 = \boldsymbol{w}_1 = (1, 0)$ and $\boldsymbol{v}_2 = \boldsymbol{w}_2 = (0, 1)$: the projection of $(1, 0)$ onto the line is $(\frac{1}{2}, \frac{1}{2})$, and so is the projection of $(0, 1)$. So

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}} \quad \text{with } \boldsymbol{a} = (1, 1),$$

the projection matrix of Section 17. It is symmetric with $P^2 = P$, but those facts are obvious only in the eigenvector basis, where the matrix is diagonal. The course has mostly used the handiest basis; the best basis is the eigenvector basis.

Example 34.8 (Differentiation as a matrix)**.** Let $T = \frac{d}{dx}$ take the quadratics $c_1 + c_2x + c_3x^2$ (input basis $1, x, x^2$) to the linear polynomials (output basis $1, x$): $T(c_1 + c_2x + c_3x^2) = c_2 + 2c_3x$. Column by column, $T(1) = 0$, $T(x) = 1$, $T(x^2) = 2x$, so

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \qquad A \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}.$$

For instance $p(x) = 5+3x-4x^2$ has coordinates $(5, 3, -4)$, and $A(5, 3, -4) = (3, -8)$, the coordinates of $p'(x) = 3 - 8x$. The 2×3 shape reflects a three-dimensional input space and a two-dimensional output space; the nullspace of A, spanned by $(1, 0, 0)$, is the constant polynomials, whose derivative is zero.

Example 34.9 (The transpose as a linear map)**.** On the four-dimensional space of 2×2 matrices let $T(X) = X^{\mathsf{T}}$. It is linear: $(X + Y)^{\mathsf{T}} = X^{\mathsf{T}} + Y^{\mathsf{T}}$ and $(cX)^{\mathsf{T}} = cX^{\mathsf{T}}$. Applying it twice gives back X, so $T^2 = I$ and $T^{-1} = T$.

- In the standard basis $\boldsymbol{v}_1 = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, $\boldsymbol{v}_2 = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$, $\boldsymbol{v}_3 = \left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, $\boldsymbol{v}_4 = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ we have $T(\boldsymbol{v}_1) = \boldsymbol{v}_1$, $T(\boldsymbol{v}_2) = \boldsymbol{v}_3$, $T(\boldsymbol{v}_3) = \boldsymbol{v}_2$, $T(\boldsymbol{v}_4) = \boldsymbol{v}_4$, so the matrix is the permutation

$$M_T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

  Check: $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ has coordinates $(1, 2, 3, 4)$ and $M_T(1, 2, 3, 4) = (1, 3, 2, 4)$, the coordinates of its transpose $\left[\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right]$.

- In the basis $\boldsymbol{w}_1 = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, $\boldsymbol{w}_2 = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, $\boldsymbol{w}_3 = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right]$, $\boldsymbol{w}_4 = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$, the first three are symmetric and the last is antisymmetric: $T(\boldsymbol{w}_i) = \boldsymbol{w}_i$ for $i \le 3$ and $T(\boldsymbol{w}_4) = -\boldsymbol{w}_4$. The matrix is $\operatorname{diag}(1, 1, 1, -1)$. These $\boldsymbol{w}$'s are eigenvectors of T: the symmetric matrices are its eigenspace for $\lambda = 1$ and the antisymmetric ones for $\lambda = -1$.

Common trap. A matrix of T means nothing until both bases are named, and the same T has many matrices (compare the two projection matrices above). Also, the coordinate vector of an output must be taken in the *output* basis: for $\frac{d}{dx}$, the column for x^2 is $(0, 2)$ because $2x = 0 \cdot 1 + 2 \cdot x$, not $(0, 0, 2)$ or $(2, 0)$.

34.4 Products of transformations are products of matrices

If $T_1 \colon U \to V$ and $T_2 \colon V \to W$ are linear, so is the composition $T_2 \circ T_1$ ($\boldsymbol{u} \mapsto T_2(T_1(\boldsymbol{u}))$). Matrix multiplication was defined to match.

Proposition 34.10 (Composition and inverses)**.** *Fix bases of U, V, W. If T_1 has matrix A_1 and T_2 has matrix A_2, then $T_2 \circ T_1$ has matrix A_2A_1. If $T \colon V \to V$ is invertible with matrix A (same basis in and out), then T^{-1} has matrix A^{-1}.*

Proof. Let $\boldsymbol{u}$ have coordinates $\boldsymbol{c}$. By Theorem 34.6, $T_1(\boldsymbol{u})$ has coordinates $A_1\boldsymbol{c}$, and then $T_2(T_1(\boldsymbol{u}))$ has coordinates $A_2(A_1\boldsymbol{c}) = (A_2A_1)\boldsymbol{c}$; a matrix that does this for every $\boldsymbol{c}$ is the matrix of the composition (its columns come from $\boldsymbol{c} = \boldsymbol{e}_j$). For the inverse, $T^{-1} \circ T$ is the identity, whose matrix is I, so the matrix B of T^{-1} satisfies $BA = I$. □

Example 34.11 (The second derivative)**.** Follow $\frac{d}{dx}$ from quadratics to linear polynomials (matrix A_1 above) by $\frac{d}{dx}$ from linear polynomials (basis $1, x$) to constants (basis 1), whose matrix is $A_2 = \begin{bmatrix}0 & 1\end{bmatrix}$. The product

$$A_2A_1 = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 2\end{bmatrix} = \begin{bmatrix}0 & 0 & 2\end{bmatrix}$$

sends (c_1, c_2, c_3) to $2c_3$, and indeed $\frac{d^2}{dx^2}(c_1 + c_2x + c_3x^2) = 2c_3$.

34.5 Exercises

Exercise 34.12 (Doubling every distance)**.** In polar coordinates let $T(r, \theta) = (2r, \theta)$: T doubles the distance from the origin and keeps the direction.

(a) Is T linear?

(b) Describe T in Cartesian coordinates and check that it doubles lengths.

(c) Find its matrix in the standard basis.

Solution. (a) Yes. As a map of vectors, $T(\boldsymbol{v}) = 2\boldsymbol{v}$, so $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = 2\boldsymbol{v}_1 + 2\boldsymbol{v}_2 = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ and $T(c\boldsymbol{v}) = 2c\boldsymbol{v} = cT(\boldsymbol{v})$.

(b) $T(x, y) = (2x, 2y)$, and $\|(2x, 2y)\| = \sqrt{4x^2 + 4y^2} = 2\sqrt{x^2 + y^2} = 2\|(x, y)\|$.

(c) $T(1, 0) = (2, 0)$ and $T(0, 1) = (0, 2)$, so the matrix is $\left[\begin{smallmatrix}2 & 0\\ 0 & 2\end{smallmatrix}\right] = 2I$.

□

Exercise 34.13 (Fixing zero is not enough)**.** Find a transformation of $\mathbb{R}^2$ with $T(\mathbf{0}) = \mathbf{0}$ that is not linear.

Solution. One option acts differently on different parts of the plane: $T(x, y) = (|x|, |y|)$. Then $T\big((1, 1) + (1, -1)\big) = T(2, 0) = (2, 0)$, but $T(1, 1) + T(1, -1) = (1, 1) + (1, 1) = (2, 2)$. Another uses a nonlinear formula: $T(x, y) = (x, y^2)$ gives $T(c(x, y)) = (cx, c^2y^2)$, which differs from $cT(x, y) = (cx, cy^2)$ when $c \neq 0, 1$ and $y \neq 0$. Both send $\mathbf{0}$ to $\mathbf{0}$, so $T(\mathbf{0}) = \mathbf{0}$ is a necessary condition for linearity, not a sufficient one. □

Exercise 34.14 (Rotations compose by adding angles)**.** Find the matrix R_θ of rotation by θ in the standard basis, and use Proposition 34.10 to show $R_\theta R_\varphi = R_{\theta+\varphi}$. What is R_θ^{-1}?

Solution. Rotating $(1, 0)$ gives $(\cos\theta, \sin\theta)$ and rotating $(0, 1)$ gives $(-\sin\theta, \cos\theta)$, so

$$R_\theta = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}.$$

Rotating by φ and then by θ is rotation by $\theta + \varphi$, so the product of the matrices must be $R_{\theta+\varphi}$. Multiplying out confirms it: the $(1, 1)$ entry is $\cos\theta\cos\varphi - \sin\theta\sin\varphi = \cos(\theta + \varphi)$ and the $(2, 1)$

entry is $\sin\theta\cos\varphi + \cos\theta\sin\varphi = \sin(\theta+\varphi)$: the matrix product contains the addition formulas of trigonometry. The inverse rotation is $R_{-\theta} = R_\theta^{\mathsf{T}}$. □

Exercise 34.15 (Integration undoes differentiation, on one side)**.** Let S take linear polynomials $c_1 + c_2x$ to quadratics by $S(p) = \int_0^x p(t)\,dt$, and let $D = \frac{d}{dx}$ take quadratics to linear polynomials (bases $1, x$ and $1, x, x^2$). Find the matrices of S and D, and compute DS and SD. Interpret.

Solution. $S(1) = x$ and $S(x) = \frac{1}{2}x^2$, so S has matrix $\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1/2 \end{bmatrix}$; D has matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ (Example 34.8). Then $DS = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$: differentiating an integral returns the original function. But $SD = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq I$: integrating a derivative loses the constant term c_1. So S is a right inverse of D but not a left inverse, a theme of Section 36. □

35 Change of basis; image compression

Section 34 showed that a linear transformation becomes a matrix once a basis is chosen, and that a better basis gives a better matrix. This unit makes the change from one basis to another explicit, for vectors (new coordinates) and for transformations (a new, similar matrix). It starts with the application where the choice of basis is worth billions: compressing images and video. Recording an image pixel by pixel uses the standard basis, which is a poor basis for real pictures; switching to a Fourier or wavelet basis lets us keep a few large coefficients and throw the rest away.

35.1 Images are vectors, and the standard basis is poor

A black-and-white image of 512×512 pixels records a gray level at every pixel, usually an integer from 0 to 255 (eight bits). The image is therefore a vector $\boldsymbol{x} \in \mathbb{R}^n$ with $n = 512^2 = 262{,}144$ components; a color image needs three times as many. Sending each frame of a video pixel by pixel would overload any channel.

Giving the value of each pixel is using the standard basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$: the coefficient of $\boldsymbol{e}_i$ is pixel i. This ignores the most important feature of real images, that neighboring pixels have nearly the same gray level. Before the lecture starts, the camera sees a blank blackboard: every pixel is (almost) the same. Pixel by pixel that image still costs n numbers. In a basis that contains the vector of all ones, it costs *one* coefficient. Other useful basis vectors are the checkerboard $(1, -1, 1, -1, \ldots)$ and the half-and-half vector $(1, \ldots, 1, -1, \ldots, -1)$, an image bright on the left and dark on the right. Which basis is best depends on the signal: the best basis for algebra lectures may not be the best for action films.

35.2 Transform, threshold, reconstruct

Lossy compression has three steps, only one of which loses information:

$$\underbrace{\boldsymbol{x}}_{\text{pixels}} \xrightarrow{\text{change basis (lossless)}} \underbrace{\boldsymbol{c}}_{\text{coefficients}} \xrightarrow{\text{threshold (lossy)}} \underbrace{\hat{\boldsymbol{c}}}_{\text{many zeros}} \xrightarrow{\text{reconstruct}} \hat{\boldsymbol{x}} = \sum_i \hat{c}_i \boldsymbol{w}_i.$$

The change of basis writes $\boldsymbol{x} = c_1\boldsymbol{w}_1 + \cdots + c_n\boldsymbol{w}_n$ in a new basis. *Thresholding* sets to zero every coefficient below a chosen size, small enough that the eye can hardly see the difference. A smooth

image has a large coefficient on the constant vector (zero frequency) and very little of the fast-alternating, high-frequency vectors. If 64 coefficients shrink to 3, that is compression by a factor of about 21.

The JPEG standard (Joint Photographic Experts Group) cuts the 512×512 image into 8×8 blocks of 64 pixels and changes basis separately in each block, using cosines: the real parts of the Fourier vectors of Section 30. Video adds a time dimension. Consecutive frames are very close, so a good coder predicts each frame from the previous one and encodes only the small correction. In space or in time, compression exploits the same fact: signals change smoothly, so neighboring values predict each other.

35.3 The Haar wavelet basis

The main competitor to the Fourier basis is a *wavelet* basis. The simplest wavelets, Haar's, in $\mathbb{R}^8$ are the columns $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_8$ of

$$W = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & -1 \end{bmatrix}:$$

the constant vector, one vector that is $+1$ on the first half and -1 on the second, two that do the same on each half, and four that do it on each quarter (Figure 45). Every entry is 0 or ± 1, so multiplying by W needs only additions and subtractions. JPEG2000 uses smoother wavelets built on the same idea.

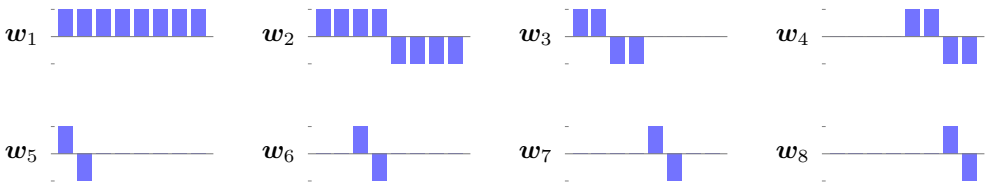


Figure 45: The eight Haar wavelets in $\mathbb{R}^8$, drawn as bar charts of their entries ($+1$ up, -1 down). Top row: the constant vector and the coarse wavelets on halves and quarters of the signal. Bottom row: the finest wavelets, each a single $+1, -1$ pair. Their sum $\boldsymbol{w}_5 + \boldsymbol{w}_6 + \boldsymbol{w}_7 + \boldsymbol{w}_8 = (1, -1, 1, -1, 1, -1, 1, -1)$ is the fastest-oscillating Fourier vector.

Finding the coefficients of $\boldsymbol{x}$ means solving $\boldsymbol{x} = W\boldsymbol{c}$, so $\boldsymbol{c} = W^{-1}\boldsymbol{x}$. The Haar vectors make this easy because they are orthogonal: any two of them overlap where one is constant, and there the other has as many $+1$'s as -1's, so the dot product is zero.

Proposition 35.1 (Coefficients in an orthogonal basis)**.** *If $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ is an orthogonal basis of $\mathbb{R}^n$ and $\boldsymbol{x} = \sum c_j \boldsymbol{w}_j$, then*

$$c_i = \frac{\boldsymbol{w}_i^{\mathsf{T}} \boldsymbol{x}}{\boldsymbol{w}_i^{\mathsf{T}} \boldsymbol{w}_i}.$$

In matrix form $W^{\mathsf{T}}W = D$ is diagonal and $W^{-1} = D^{-1}W^{\mathsf{T}}$. If the basis is orthonormal, $W^{-1} = W^{\mathsf{T}}$.

Proof. Take the dot product of $\boldsymbol{x} = \sum_j c_j \boldsymbol{w}_j$ with $\boldsymbol{w}_i$. Every term with $j \neq i$ vanishes, leaving $\boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{x} = c_i\, \boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{w}_i$. For the matrix form, the (i,j) entry of $W^{\mathsf{T}}W$ is $\boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{w}_j$, which is zero off the diagonal; so $(D^{-1}W^{\mathsf{T}})W = I$. □

For the Haar basis $\boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{w}_i$ is $8, 8, 4, 4, 2, 2, 2, 2$. Dividing each $\boldsymbol{w}_i$ by its length $\sqrt{8}$, $\sqrt{8}$, 2, 2, $\sqrt{2}$, $\sqrt{2}$, $\sqrt{2}$, $\sqrt{2}$ makes W orthogonal, with $W^{-1} = W^{\mathsf{T}}$.

Example 35.2 (Compressing eight pixels)**.** One row of an image has gray levels

$$\boldsymbol{p} = (100,\ 102,\ 104,\ 106,\ 60,\ 60,\ 62,\ 62)\ :$$

a bright region with a gentle ramp next to a dark region. Using Proposition 35.1, with $\boldsymbol{w}_i^{\mathsf{T}}\boldsymbol{w}_i = 8, 8, 4, 4, 2, 2, 2, 2$:

$$c_1 = \tfrac{1}{8}(100+102+104+106+60+60+62+62) = \tfrac{656}{8} = 82,$$
$$c_2 = \tfrac{1}{8}\big((100+102+104+106)-(60+60+62+62)\big) = \tfrac{412-244}{8} = 21,$$
$$c_3 = \tfrac{1}{4}\big((100+102)-(104+106)\big) = -2, \qquad c_4 = \tfrac{1}{4}\big((60+60)-(62+62)\big) = -1,$$
$$c_5 = \tfrac{1}{2}(100-102) = -1, \quad c_6 = \tfrac{1}{2}(104-106) = -1, \quad c_7 = \tfrac{1}{2}(60-60) = 0, \quad c_8 = \tfrac{1}{2}(62-62) = 0.$$

So $\boldsymbol{c} = (82, 21, -2, -1, -1, -1, 0, 0)$: the average gray level, the bright/dark contrast, and small details. *Threshold*: drop every coefficient with $|c_i| \leq 1$, keeping $\hat{\boldsymbol{c}} = (82, 21, -2, 0, 0, 0, 0, 0)$. *Reconstruct*:

$$\hat{\boldsymbol{x}} = 82\boldsymbol{w}_1 + 21\boldsymbol{w}_2 - 2\boldsymbol{w}_3 = (101, 101, 105, 105, 61, 61, 61, 61).$$

The error $\boldsymbol{p} - \hat{\boldsymbol{x}} = (-1, 1, -1, 1, -1, -1, 1, 1)$ is at most one gray level out of 255, invisible, and three numbers replace eight. A harsher threshold keeping only c_1 and c_2 gives $82\boldsymbol{w}_1 + 21\boldsymbol{w}_2 = (103, 103, 103, 103, 61, 61, 61, 61)$, with errors up to 3 (Figure 46). Had we kept three *pixels* instead, the other five would be lost entirely.

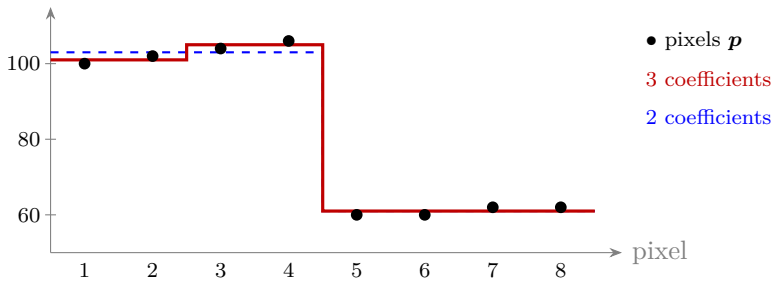


Figure 46: The pixel row of Example 35.2 (dots) and its reconstructions from the largest Haar coefficients. Three coefficients, $82\boldsymbol{w}_1 + 21\boldsymbol{w}_2 - 2\boldsymbol{w}_3$ (solid), miss no pixel by more than one gray level; two coefficients (dashed) keep the bright/dark split but flatten the ramp.

What makes a basis good for compression? Two properties, and the competition between bases is about both:

1. *Fast*: multiplying by W and by W^{-1} must be cheap. The Fourier basis has the FFT (Section 30); wavelets have a fast wavelet transform; orthonormal bases have $W^{-1} = W^{\mathsf{T}}$. The fastest choice of all is to change nothing, the standard basis, which fails the second test.
2. *Good compression*: a few basis vectors should come close to the signal, so that most coefficients are small and can be set to zero.

The ideal basis would be the eigenvectors of a matrix built from the image itself, but finding them costs far more than the savings; Fourier and wavelet bases are cheap and close.

35.4 Change of basis for vectors and for transformations

Vectors. Let the columns of W be new basis vectors $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$ of $\mathbb{R}^n$. A vector with standard coordinates $\boldsymbol{x}$ has new coordinates $\boldsymbol{c}$ related by

$$\boldsymbol{x} = W\boldsymbol{c}, \qquad \boldsymbol{c} = W^{-1}\boldsymbol{x}.$$

Transformations. Now let $T\colon V \to V$ be linear and use the same basis for inputs and outputs. In an old basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ it has a matrix A, in a new basis $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$ a matrix B. The same transformation should give closely related matrices, and it does.

Theorem 35.3 (Changing basis gives a similar matrix)**.** *Let M be the matrix whose column j holds the old coordinates of the new basis vector $\boldsymbol{w}_j$. Then for every vector, (old coordinates) $=$ M (new coordinates), and*

$$B = M^{-1}AM.$$

Proof. If $\boldsymbol{x} = \sum_j d_j \boldsymbol{w}_j$, substituting $\boldsymbol{w}_j = \sum_i M_{ij}\boldsymbol{v}_i$ gives $\boldsymbol{x} = \sum_i \big(\sum_j M_{ij} d_j\big)\boldsymbol{v}_i$, so the old coordinates are $M\boldsymbol{d}$. Since M converts coordinates of a basis into coordinates of a basis, it is invertible. Now follow Figure 47: start with new coordinates $\boldsymbol{d}$ of $\boldsymbol{x}$; $M\boldsymbol{d}$ are its old coordinates; $AM\boldsymbol{d}$ are the old coordinates of $T(\boldsymbol{x})$ (Section 34); $M^{-1}AM\boldsymbol{d}$ are the new coordinates of $T(\boldsymbol{x})$. A matrix that turns the new coordinates of every $\boldsymbol{x}$ into the new coordinates of $T(\boldsymbol{x})$ is the matrix B (apply it to $\boldsymbol{d} = \boldsymbol{e}_j$ to recover column j). So $B = M^{-1}AM$. □

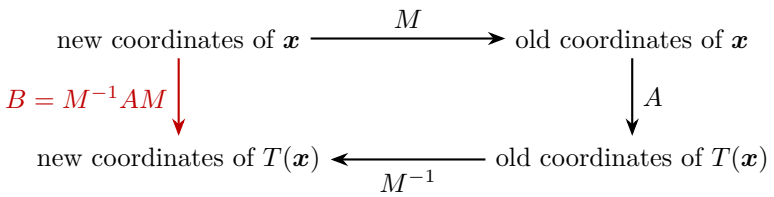


Figure 47: Two routes from the new coordinates of $\boldsymbol{x}$ to the new coordinates of $T(\boldsymbol{x})$: directly by B, or by translating to the old basis (M), applying A, and translating back (M^{-1}). Because both routes agree, $B = M^{-1}AM$: the two matrices of one transformation are similar.

This is where similar matrices (Section 32) really come from: they are the matrices of one transformation in different bases, which is why they share eigenvalues, trace and determinant, all of which belong to T itself. The best basis is again the eigenvector basis: if $T(\boldsymbol{v}_i) = \lambda_i \boldsymbol{v}_i$, then column i of the matrix is $\lambda_i \boldsymbol{e}_i$ and the matrix is Λ.

Example 35.4 (Diagonalizing a projection by a change of basis)**.** In the standard basis the projection onto the 45° line has matrix $A = \left[\begin{smallmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{smallmatrix}\right]$ (Section 34). Take the new basis $\boldsymbol{w}_1 = (1,1)$ on the line and $\boldsymbol{w}_2 = (-1,1)$ across it, so $M = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$ and $M^{-1} = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right]$. Then

$$AM = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \qquad B = M^{-1}AM = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \Lambda.$$

Example 35.5 (A basis known only by its values)**.** In the space of polynomials of degree at most 2, compare the basis $1, x, x^2$ with a basis $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$ known only through its values at $x = -1, 0, 1$:

	$x=-1$	$x=0$	$x=1$
$\boldsymbol{w}_1$	1	0	0
$\boldsymbol{w}_2$	0	1	0
$\boldsymbol{w}_3$	0	0	1

New coordinates are values. If $p = \alpha\boldsymbol{w}_1 + \beta\boldsymbol{w}_2 + \gamma\boldsymbol{w}_3$, evaluating at $-1, 0, 1$ gives $p(-1) = \alpha$, $p(0) = \beta$, $p(1) = \gamma$. So $y = 5 - x$, with values $6, 5, 4$, is $y = 6\boldsymbol{w}_1 + 5\boldsymbol{w}_2 + 4\boldsymbol{w}_3$.

Change of basis matrices. The new coordinates of 1, x, x^2 are their values: $(1,1,1)$, $(-1,0,1)$, $(1,0,1)$. These are the columns of M^{-1} (old $\to$ new), and inverting gives M (new $\to$ old), whose columns are the $\boldsymbol{w}$'s written in $1, x, x^2$:

$$M^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \qquad M = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \end{bmatrix},$$

that is $\boldsymbol{w}_1 = \frac{1}{2}(x^2 - x)$, $\boldsymbol{w}_2 = 1 - x^2$, $\boldsymbol{w}_3 = \frac{1}{2}(x^2 + x)$. (Check $\boldsymbol{w}_1$: values $\frac{1}{2}(1+1) = 1$, 0, $\frac{1}{2}(1-1) = 0$.)

The derivative in both bases. With $1, x, x^2$ for inputs and outputs, $T = \frac{d}{dx}$ has $A = \left[\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right]$. In the $\boldsymbol{w}$ basis,

$$B = M^{-1}AM = \begin{bmatrix} -\frac{3}{2} & 2 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & -2 & \frac{3}{2} \end{bmatrix}.$$

Column 1 should be the values of $\boldsymbol{w}_1' = x - \frac{1}{2}$ at $-1, 0, 1$: $(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2})$. It is. And $B(6,5,4) = (-9+10-2,\ -3+0+2,\ 3-10+6) = (-1,-1,-1)$, the values of $y' = -1$. Both A and B have trace 0 and only the eigenvalue 0, as similar matrices must.

Common trap. Keep the direction of M straight. Its columns are the *new* basis vectors written in the *old* coordinates, so M converts new coordinates to old ($\boldsymbol{x} = W\boldsymbol{c}$ in $\mathbb{R}^n$), and the new matrix is $M^{-1}AM$, not MAM^{-1}. A quick test: apply your formula for B to $\boldsymbol{e}_1$ and check that it gives the new coordinates of $T(\boldsymbol{w}_1)$.

35.5 Exercises

Exercise 35.6 (An orthonormal Haar basis)**.** Verify that the eight Haar vectors are orthogonal and rescale them to an orthonormal basis.

Solution. $\boldsymbol{w}_2, \ldots, \boldsymbol{w}_8$ each have equally many $+1$'s and -1's, so they are orthogonal to the constant $\boldsymbol{w}_1$. Each of $\boldsymbol{w}_3, \ldots, \boldsymbol{w}_8$ lives on one half of the signal, where $\boldsymbol{w}_2$ is constant, and has equally

many ± 1 there, so it is orthogonal to $\boldsymbol{w}_2$. The same argument on quarters handles $\boldsymbol{w}_3, \boldsymbol{w}_4$ against $\boldsymbol{w}_5, \dots, \boldsymbol{w}_8$, and vectors with disjoint supports (such as $\boldsymbol{w}_3$ and $\boldsymbol{w}_4$, or $\boldsymbol{w}_5$ and $\boldsymbol{w}_6$) have dot product 0. The lengths are $\sqrt{8}, \sqrt{8}, 2, 2, \sqrt{2}, \sqrt{2}, \sqrt{2}, \sqrt{2}$, so the orthonormal basis is $\boldsymbol{w}_1/\sqrt{8}$, $\boldsymbol{w}_2/\sqrt{8}$, $\boldsymbol{w}_3/2$, $\boldsymbol{w}_4/2$, $\boldsymbol{w}_5/\sqrt{2}, \dots, \boldsymbol{w}_8/\sqrt{2}$: entries $\pm 1/\sqrt{8}$, $\pm\frac{1}{2}$ and $\pm 1/\sqrt{2}$. □

Exercise 35.7 (Lossless compression)**.** Find the Haar coefficients of $\boldsymbol{p} = (4, 4, 4, 4, 2, 2, 0, 0)$. How many numbers are needed to store $\boldsymbol{p}$ exactly?

Solution. $c_1 = \frac{1}{8}(16 + 4) = 2.5$, $c_2 = \frac{1}{8}(16 - 4) = 1.5$, $c_3 = \frac{1}{4}(8 - 8) = 0$, $c_4 = \frac{1}{4}(4 - 0) = 1$, and $c_5 = c_6 = c_7 = c_8 = 0$ since each neighbouring pair is equal. So $\boldsymbol{p} = 2.5\boldsymbol{w}_1 + 1.5\boldsymbol{w}_2 + \boldsymbol{w}_4$, three numbers with no loss at all. Check the fifth entry: $2.5 - 1.5 + 1 = 2$. □

Exercise 35.8 (Bases for 2×2 matrices)**.** The 2×2 real matrices form a four-dimensional vector space. Give two different bases. Which is better for describing diagonal matrices? Triangular ones? Symmetric ones?

Solution. The standard basis is $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right]$. Another is $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$, $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$; viewed as vectors in $\mathbb{R}^4$ these are orthogonal, hence independent. The standard basis is best for diagonal matrices (two basis elements suffice) and for triangular ones (three). The second basis is also good for diagonal matrices, e.g. $\left[\begin{smallmatrix} 4 & 0 \\ 0 & 2 \end{smallmatrix}\right] = 3\left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right] + 1\left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$, and it is the best for symmetric matrices: they are exactly the combinations with coefficient 0 on the antisymmetric $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. □

Exercise 35.9 (A rotation in a reflected basis)**.** The rotation by 90° has standard matrix $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$. Find its matrix in the basis $\boldsymbol{w}_1 = (1, 1)$, $\boldsymbol{w}_2 = (1, -1)$, and explain the answer.

Solution. $M = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, $M^{-1} = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, $RM = \left[\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, and $B = M^{-1}RM = \frac{1}{2}\left[\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$. Directly: $R\boldsymbol{w}_1 = (-1, 1) = -\boldsymbol{w}_2$ and $R\boldsymbol{w}_2 = (1, 1) = \boldsymbol{w}_1$, which are the columns $(0, -1)$ and $(1, 0)$. The new matrix looks like rotation by $-90°$ because the basis $\boldsymbol{w}_1, \boldsymbol{w}_2$ has the opposite orientation ($\det M = -2 < 0$). Trace 0 and determinant 1 are unchanged, as they must be. □

36 Left and right inverses; pseudoinverse

We would like to "divide by A" and write $\boldsymbol{x} = A^{-1}\boldsymbol{b}$. Chapter-two matrices are square and of full rank, so A^{-1} exists. Every other matrix has a nonzero nullspace, a nonzero left nullspace, or both, and those nullspaces are exactly what block an inverse. This unit sorts all $m \times n$ matrices of rank r into four cases, finds the best available inverse in each, and ends with the pseudoinverse A^{+}, which exists for every matrix and is built from the SVD of Section 33. Nothing here is new machinery: it is the four subspaces of Section 11, the projections of Section 17 and the SVD, seen together.

36.1 Four cases, sorted by rank

Definition 36.1 (Two-sided, left and right inverses)**.** Let A be $m \times n$. A matrix B (necessarily $n \times m$) is a *left inverse* of A if $BA = I_n$, and a *right inverse* if $AB = I_m$. It is a *two-sided inverse*, written A^{-1}, if it is both.

The rank r never exceeds m or n, so there are four possibilities. Table 4 records what each one means for the nullspaces and for the number of solutions of $A\boldsymbol{x} = \boldsymbol{b}$.

Table 4: The four rank cases for an $m \times n$ matrix of rank r.

Case	Nullspaces	Solutions of $A\boldsymbol{x} = \boldsymbol{b}$	Inverse available
$r = m = n$	$\mathbf{N}(A) = \{\mathbf{0}\}$, $\mathbf{N}(A^\mathsf{T}) = \{\mathbf{0}\}$	exactly one	two-sided A^{-1}
$r = n < m$	$\mathbf{N}(A) = \{\mathbf{0}\}$	zero or one	left: $(A^\mathsf{T}A)^{-1}A^\mathsf{T}$
$r = m < n$	$\mathbf{N}(A^\mathsf{T}) = \{\mathbf{0}\}$	infinitely many ($n - m$ free)	right: $A^\mathsf{T}(AA^\mathsf{T})^{-1}$
$r < m$, $r < n$	both nonzero	zero or infinitely many	pseudoinverse A^+ only

Proposition 36.2 (Rectangular matrices have no two-sided inverse)**.** *If A is $m \times n$ with $m \neq n$, then A has no two-sided inverse.*

Proof. Suppose $n > m$. Then $r \leq m < n$, so A has $n - r > 0$ free variables and some $\boldsymbol{x} \neq \mathbf{0}$ has $A\boldsymbol{x} = \mathbf{0}$. A left inverse B would give $\boldsymbol{x} = BA\boldsymbol{x} = B\mathbf{0} = \mathbf{0}$, a contradiction. If $m > n$, apply the same argument to A^T: a two-sided inverse B of A would make B^T a two-sided inverse of A^T, and A^T has more columns than rows. □

36.2 Full column rank: a left inverse

When $r = n$ the columns are independent and $\mathbf{N}(A) = \{\mathbf{0}\}$. The key fact, already central to least squares in Section 18, is that $A^\mathsf{T}A$ inherits this.

Lemma 36.3 ($A^\mathsf{T}A$ is invertible exactly when the columns are independent)**.** *For any $m \times n$ matrix A, $\mathbf{N}(A^\mathsf{T}A) = \mathbf{N}(A)$. In particular the $n \times n$ symmetric matrix $A^\mathsf{T}A$ is invertible if and only if $r = n$.*

Proof. If $A\boldsymbol{x} = \mathbf{0}$ then $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$. Conversely, if $A^\mathsf{T}A\boldsymbol{x} = \mathbf{0}$, multiply on the left by $\boldsymbol{x}^\mathsf{T}$: $0 = \boldsymbol{x}^\mathsf{T}A^\mathsf{T}A\boldsymbol{x} = (A\boldsymbol{x})^\mathsf{T}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2$, so $A\boldsymbol{x} = \mathbf{0}$. A square matrix is invertible exactly when its nullspace is $\{\mathbf{0}\}$, which here means $\mathbf{N}(A) = \{\mathbf{0}\}$, that is $r = n$. □

Theorem 36.4 (Left inverse)**.** *If A has full column rank $r = n$, then $A^{-1}_{left} = (A^\mathsf{T}A)^{-1}A^\mathsf{T}$ is a left inverse of A, and*

$$A\,A^{-1}_{left} = A(A^\mathsf{T}A)^{-1}A^\mathsf{T} = P_{\mathbf{C}(A)}$$

is the $m \times m$ projection matrix onto the column space. It equals I_m only when $m = n$.

Proof. By Lemma 36.3 the inverse $(A^\mathsf{T}A)^{-1}$ exists, and $\big[(A^\mathsf{T}A)^{-1}A^\mathsf{T}\big]A = (A^\mathsf{T}A)^{-1}(A^\mathsf{T}A) = I_n$. The product in the other order is the projection formula $P = A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ of Section 17. A projection onto the n-dimensional subspace $\mathbf{C}(A)$ has rank n, so it is I_m only if $n = m$. □

There are other left inverses when $m > n$ (add to A^{-1}_{left} any matrix whose rows lie in $\mathbf{N}(A^\mathsf{T})$), but $(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ is the favourite: applied to $\boldsymbol{b}$ it gives the least-squares solution $\hat{\boldsymbol{x}} = (A^\mathsf{T}A)^{-1}A^\mathsf{T}\boldsymbol{b}$. The product AA^{-1}_{left} "tries to be the identity": it is the identity on $\mathbf{C}(A)$ and zero on $\mathbf{N}(A^\mathsf{T})$.

36.3 Full row rank: a right inverse

When $r = m$ the rows are independent, $\mathbf{N}(A^\mathsf{T}) = \{\mathbf{0}\}$, and $A\boldsymbol{x} = \boldsymbol{b}$ is solvable for every $\boldsymbol{b}$, with $n - m$ free variables. Now AA^T is the invertible $m \times m$ matrix (apply Lemma 36.3 to A^T), and everything above flips.

Theorem 36.5 (Right inverse)**.** *If A has full row rank $r = m$, then $A^{-1}_{right} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}$ satisfies $AA^{-1}_{right} = I_m$, and*

$$A^{-1}_{right}A = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}A = P_{\mathbf{C}(A^{\mathsf{T}})}$$

is the $n \times n$ projection onto the row space.

Proof. The first product is

$$A\big[A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}\big] = (AA^{\mathsf{T}})(AA^{\mathsf{T}})^{-1} = I_m.$$

Writing $M = A^{\mathsf{T}}$, the other product is $M(M^{\mathsf{T}}M)^{-1}M^{\mathsf{T}}$, the projection onto $\mathbf{C}(M) = \mathbf{C}(A^{\mathsf{T}})$. □

Example 36.6 (A right inverse, and the projection it leaves behind)**.** Take $A = \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$, of rank $2 = m$. Then

$$AA^{\mathsf{T}} = \begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}, \qquad (AA^{\mathsf{T}})^{-1} = \begin{bmatrix}1/2 & 0\\ 0 & 1\end{bmatrix}, \qquad A^{-1}_{\text{right}} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1} = \begin{bmatrix}1/2 & 0\\ 0 & 1\\ 1/2 & 0\end{bmatrix}.$$

Check: $AA^{-1}_{\text{right}} = \begin{bmatrix}1/2 + 1/2 & 0\\ 0 & 1\end{bmatrix} = I_2$. In the other order,

$$A^{-1}_{\text{right}}A = \begin{bmatrix}1/2 & 0 & 1/2\\ 0 & 1 & 0\\ 1/2 & 0 & 1/2\end{bmatrix},$$

which is symmetric, equals its own square, and fixes the rows $(1, 0, 1)$ and $(0, 1, 0)$ of A while sending the nullspace vector $(1, 0, -1)$ to $\mathbf{0}$: it is the projection onto the row space, as Theorem 36.5 predicts. The transpose A^{T} is 3×2 with full column rank, and its left inverse $(AA^{\mathsf{T}})^{-1}A = \begin{bmatrix}1/2 & 0 & 1/2\\ 0 & 1 & 0\end{bmatrix}$ is just the transpose of A^{-1}_{right}.

36.4 From row space to column space, A is invertible

In the general case $r < m$ and $r < n$ neither one-sided formula works, because both $A^{\mathsf{T}}A$ and AA^{T} are singular. The way forward is to stop asking A to be invertible on all of $\mathbb{R}^n$ and look only at the row space. Every $\boldsymbol{x} \in \mathbb{R}^n$ splits as $\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$ with $\boldsymbol{x}_r \in \mathbf{C}(A^{\mathsf{T}})$ and $\boldsymbol{x}_n \in \mathbf{N}(A)$, and $A\boldsymbol{x} = A\boldsymbol{x}_r$: the matrix kills the nullspace part, and the row-space part carries all the information.

Theorem 36.7 (A is one-to-one from row space onto column space)**.** *If $\boldsymbol{x}$ and $\boldsymbol{y}$ are in the row space $\mathbf{C}(A^{\mathsf{T}})$ and $\boldsymbol{x} \neq \boldsymbol{y}$, then $A\boldsymbol{x} \neq A\boldsymbol{y}$. Consequently $\boldsymbol{x} \mapsto A\boldsymbol{x}$ is an invertible linear map from the r-dimensional row space onto the r-dimensional column space.*

Proof. Suppose $A\boldsymbol{x} = A\boldsymbol{y}$. Then $A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$, so $\boldsymbol{x} - \boldsymbol{y} \in \mathbf{N}(A)$. But the row space is a subspace, so $\boldsymbol{x} - \boldsymbol{y}$ is also in $\mathbf{C}(A^{\mathsf{T}})$. The row space and nullspace are orthogonal complements (Section 16), so the only vector in both is $\mathbf{0}$, and $\boldsymbol{x} = \boldsymbol{y}$. Thus the map is one-to-one. Its image lies in $\mathbf{C}(A)$; it is all of $\mathbf{C}(A)$ because any $A\boldsymbol{z}$ equals $A\boldsymbol{z}_r$ with $\boldsymbol{z}_r$ the row-space part of $\boldsymbol{z}$. A one-to-one linear map between spaces of the same dimension r is invertible. □

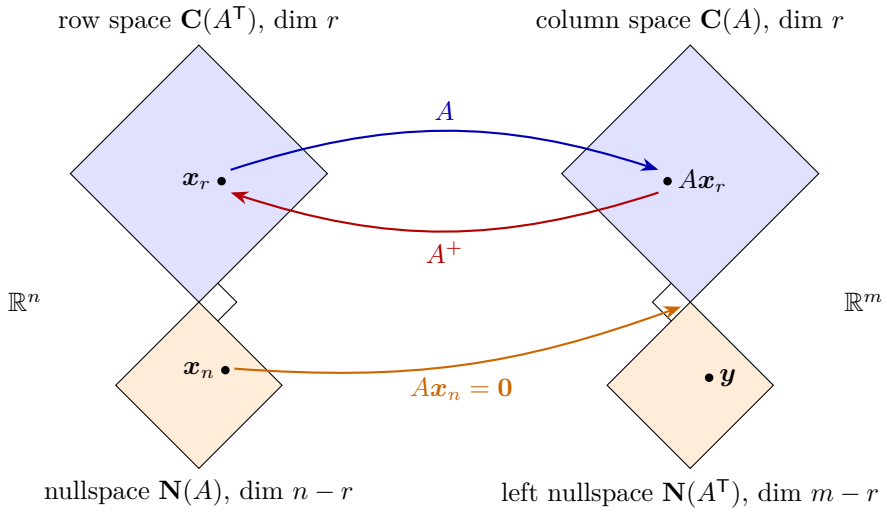


Figure 48: The big picture with inverses. Between the two r-dimensional spaces (blue) A is invertible, and the pseudoinverse A^+ is its inverse there. A sends the nullspace to $\mathbf{0}$, and A^+ sends every $\boldsymbol{y}$ in the left nullspace to $\mathbf{0}$.

Figure 48 is the whole unit in one picture. On the shaded spaces A is a perfect, invertible matrix; the trouble lives entirely in the two nullspaces. Statisticians meet this case constantly: a repeated or redundant experiment makes the columns of the design matrix dependent, $A^{\mathsf{T}}A$ becomes singular, and ordinary least squares breaks down. The inverse of the good part is what they need.

36.5 The pseudoinverse from the SVD

Definition 36.8 (Pseudoinverse)**.** The *pseudoinverse* of an $m \times n$ matrix A is the $n \times m$ matrix A^+ that inverts A from the row space to the column space and annihilates the left nullspace:

$$A^+(A\boldsymbol{x}) = \boldsymbol{x} \ \text{ for every } \boldsymbol{x} \in \mathbf{C}(A^{\mathsf{T}}), \qquad A^+\boldsymbol{y} = \mathbf{0} \ \text{ for every } \boldsymbol{y} \in \mathbf{N}(A^{\mathsf{T}}).$$

Since $\mathbb{R}^m = \mathbf{C}(A) \oplus \mathbf{N}(A^{\mathsf{T}})$ and Theorem 36.7 makes the first condition well defined, these two rules fix $A^+\boldsymbol{b}$ for every $\boldsymbol{b} \in \mathbb{R}^m$, so A^+ exists and is unique.

To compute it, start from the SVD $A = U\Sigma V^{\mathsf{T}}$: U $(m \times m)$ and V $(n \times n)$ are orthogonal, and Σ is $m \times n$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$ in its first r diagonal positions and zeros elsewhere. The orthogonal factors are trivial to invert; all the trouble is in Σ, where it is easy to see what to do. Define Σ^+ as the $n \times m$ matrix with $1/\sigma_1, \ldots, 1/\sigma_r$ in its first r diagonal positions and zeros elsewhere:

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \end{bmatrix}_{m \times n}, \qquad \Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & & \\ & \ddots & & \\ & & 1/\sigma_r & \\ & & & 0 \end{bmatrix}_{n \times m}.$$

Then $\Sigma\Sigma^+$ is the $m \times m$ diagonal matrix with r ones followed by zeros, and $\Sigma^+\Sigma$ is the $n \times n$ matrix of the same pattern. Neither is an identity unless $r = m$ or $r = n$; each is a projection.

Theorem 36.9 (Pseudoinverse via the SVD)**.** *If $A = U\Sigma V^{\mathsf{T}}$, then $A^+ = V\Sigma^+U^{\mathsf{T}}$. Moreover AA^+ is the projection onto the column space and A^+A is the projection onto the row space.*

Proof. The SVD says $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ for $i \le r$; the vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_r$ are an orthonormal basis of $\mathbf{C}(A^{\mathsf{T}})$ and $\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ an orthonormal basis of $\mathbf{N}(A^{\mathsf{T}})$. Let $B = V\Sigma^{+}U^{\mathsf{T}}$. Since $U^{\mathsf{T}}\boldsymbol{u}_i$ is the i-th standard basis vector,

$$B\boldsymbol{u}_i = V\Sigma^{+}\boldsymbol{e}_i = \begin{cases} \boldsymbol{v}_i/\sigma_i, & i \le r, \\ \mathbf{0}, & i > r. \end{cases}$$

So $B(A\boldsymbol{v}_i) = B(\sigma_i \boldsymbol{u}_i) = \boldsymbol{v}_i$ for $i \le r$, and by linearity $B(A\boldsymbol{x}) = \boldsymbol{x}$ for all $\boldsymbol{x}$ in the row space; and B kills a basis of $\mathbf{N}(A^{\mathsf{T}})$. These are the two rules of Definition 36.8, so $B = A^{+}$. Finally

$$AA^{+} = U\Sigma V^{\mathsf{T}} V\Sigma^{+}U^{\mathsf{T}} = U(\Sigma\Sigma^{+})U^{\mathsf{T}} = \sum_{i \le r} \boldsymbol{u}_i \boldsymbol{u}_i^{\mathsf{T}}, \qquad A^{+}A = V(\Sigma^{+}\Sigma)V^{\mathsf{T}} = \sum_{i \le r} \boldsymbol{v}_i \boldsymbol{v}_i^{\mathsf{T}},$$

the projections onto $\operatorname{span}\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\} = \mathbf{C}(A)$ and $\operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\} = \mathbf{C}(A^{\mathsf{T}})$. □

One could put nonzero numbers in the lower-right block of Σ^{+} and still get $\Sigma\Sigma^{+}$ with r ones, because they multiply zeros. The pseudoinverse is the choice with no extra entries, the smallest matrix that does the job. When an ordinary inverse exists, nothing changes: if $r = m = n$ then $\Sigma^{+} = \Sigma^{-1}$ and $A^{+} = A^{-1}$. The one-sided inverses are also special cases.

Proposition 36.10 (One-sided inverses are pseudoinverses)**.** *If $r = n$ then $A^{+} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$; if $r = m$ then $A^{+} = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}$.*

Proof. Let $r = n$ and $B = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$. Then $BA\boldsymbol{x} = \boldsymbol{x}$ for every $\boldsymbol{x}$ (here the row space is all of $\mathbb{R}^n$), and if $\boldsymbol{y} \in \mathbf{N}(A^{\mathsf{T}})$ then $B\boldsymbol{y} = (A^{\mathsf{T}}A)^{-1}(A^{\mathsf{T}}\boldsymbol{y}) = \mathbf{0}$. These are the defining rules, so $B = A^{+}$. The case $r = m$ is the transpose of this one: $(A^{\mathsf{T}})^{+} = (A^{+})^{\mathsf{T}}$ follows from $A^{+} = V\Sigma^{+}U^{\mathsf{T}}$. □

Example 36.11 (The pseudoinverse of a 1×2 matrix)**.** Let $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$. Shapes first: Σ has the shape of A (1×2), so U is 1×1 and V is 2×2. The only 1×1 orthogonal matrices are ± 1; take $U = [1]$. Next

$$A^{\mathsf{T}}A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

which is singular (row two is twice row one), so $\lambda = 0$ is an eigenvalue, and the trace gives the other, $\lambda = 5$. Unit eigenvectors: $\boldsymbol{v}_1 = \frac{1}{\sqrt{5}}(1, 2)$ for $\lambda = 5$ and $\boldsymbol{v}_2 = \frac{1}{\sqrt{5}}(-2, 1)$ for $\lambda = 0$. So $\sigma_1 = \sqrt{5}$ and

$$A = \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}, \qquad A^{+} = V\Sigma^{+}U^{\mathsf{T}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} \\ 0 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Because A has full row rank, Proposition 36.10 gives the same answer faster: $A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1} = \frac{1}{5}(1, 2)$. Now the two products:

$$AA^{+} = \tfrac{1}{5}(1 + 4) = [1], \qquad A^{+}A = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

AA^{+} is the 1×1 identity, since $\mathbf{C}(A) = \mathbb{R}^1$. For $\boldsymbol{x} = (-2, 1)$ in the nullspace, $A^{+}A\boldsymbol{x} = \frac{1}{5}(-2 + 2, -4 + 4) = \mathbf{0}$. For $\boldsymbol{x} = (1, 2)$ in the row space, $A^{+}A\boldsymbol{x} = \frac{1}{5}(1 + 4, 2 + 8) = (1, 2) = \boldsymbol{x}$. So $A^{+}A$ acts as the identity on the row space and as zero on the nullspace: it is the projection onto the line through $(1, 2)$.

Example 36.12 (A square matrix with no inverse of either kind)**.** Does $A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$ have a left inverse, a right inverse, a pseudoinverse? Row two is twice row one, so $r = 1$ and $\det A = 0$. If $AB = I$ then $\det A \det B = 1$, impossible; likewise $BA = I$. So there is no one-sided inverse. The pseudoinverse always exists. From $A^{\mathsf{T}}A = \begin{bmatrix} 80 & 60 \\ 60 & 45 \end{bmatrix}$, with eigenvalues 125 and 0, the SVD is

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.6 \\ 0.6 & -0.8 \end{bmatrix},$$

where $\boldsymbol{v}_1 = (0.8, 0.6)$ and $\boldsymbol{u}_1 = A\boldsymbol{v}_1/\sigma_1 = (5, 10)/\sqrt{125} = (1, 2)/\sqrt{5}$. Only the first singular pair survives in $V\Sigma^+U^{\mathsf{T}}$:

$$A^+ = \frac{1}{\sigma_1}\boldsymbol{v}_1\boldsymbol{u}_1^{\mathsf{T}} = \frac{1}{\sqrt{125}} \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 0.8 & 1.6 \\ 0.6 & 1.2 \end{bmatrix} = \frac{1}{125} \begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix}.$$

Check against the definition: $A^+\boldsymbol{u}_1 = \frac{1}{125} \cdot \frac{1}{\sqrt{5}}(4 + 16,\, 3 + 12) = \frac{1}{\sqrt{125}}(0.8, 0.6) = \boldsymbol{v}_1/\sigma_1$, and for $\boldsymbol{u}_2 = (2, -1)/\sqrt{5}$ we get $A^+\boldsymbol{u}_2 = \frac{1}{125\sqrt{5}}(8 - 8,\, 6 - 6) = \boldsymbol{0}$. Here $A^+ = A^{\mathsf{T}}/125$, and that is no accident: for any rank-one matrix $A = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^{\mathsf{T}}$ the pseudoinverse is $\boldsymbol{v}_1\boldsymbol{u}_1^{\mathsf{T}}/\sigma_1 = A^{\mathsf{T}}/\sigma_1^2$.

36.6 The shortest least-squares solution

Theorem 36.13 ($A^+\boldsymbol{b}$ is the minimum-length least-squares solution)**.** *For any A and $\boldsymbol{b}$, the vector $\boldsymbol{x}^+ = A^+\boldsymbol{b}$ minimizes $\|A\boldsymbol{x} - \boldsymbol{b}\|$, and among all minimizers it is the unique one of smallest length. If $A\boldsymbol{x} = \boldsymbol{b}$ is solvable, $\boldsymbol{x}^+$ is its shortest solution.*

Proof. By Theorem 36.9, $A\boldsymbol{x}^+ = AA^+\boldsymbol{b}$ is the projection $\boldsymbol{p}$ of $\boldsymbol{b}$ onto $\mathbf{C}(A)$, the closest point of the column space to $\boldsymbol{b}$; so $\boldsymbol{x}^+$ minimizes $\|A\boldsymbol{x} - \boldsymbol{b}\|$. Any other minimizer $\boldsymbol{x}$ also has $A\boldsymbol{x} = \boldsymbol{p}$ (the closest point is unique), so $\boldsymbol{x} = \boldsymbol{x}^+ + \boldsymbol{n}$ with $\boldsymbol{n} \in \mathbf{N}(A)$. The vector $\boldsymbol{x}^+ = V\Sigma^+U^{\mathsf{T}}\boldsymbol{b}$ is a combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$, so it lies in the row space and is orthogonal to $\boldsymbol{n}$. By Pythagoras $\|\boldsymbol{x}\|^2 = \|\boldsymbol{x}^+\|^2 + \|\boldsymbol{n}\|^2 \geq \|\boldsymbol{x}^+\|^2$, with equality only when $\boldsymbol{n} = \boldsymbol{0}$. □

Example 36.14 (Least squares with dependent columns)**.** With $A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$ from Example 36.12 and $\boldsymbol{b} = (5, 0)$, the normal equations $A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$ read

$$\begin{bmatrix} 80 & 60 \\ 60 & 45 \end{bmatrix} \hat{\boldsymbol{x}} = \begin{bmatrix} 20 \\ 15 \end{bmatrix},$$

which is the single equation $4\hat{x}_1 + 3\hat{x}_2 = 1$ written twice: a whole line of least-squares solutions, since $A^{\mathsf{T}}A$ is singular. The pseudoinverse picks one:

$$\boldsymbol{x}^+ = A^+\boldsymbol{b} = \frac{1}{125} \begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.12 \end{bmatrix}.$$

It lies on that line ($0.64 + 0.36 = 1$) and is a multiple of the row $(4, 3)$, so it is the point of the line closest to the origin. And $A\boldsymbol{x}^+ = (1, 2)$ is the projection of $(5, 0)$ onto the column line through $(1, 2)$, namely $\frac{5}{5}(1, 2)$.

Common trap. $(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ is a left inverse, not an inverse: $A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ is a projection, not I, whenever $m > n$. And for a rank-deficient matrix there is no $(A^{\mathsf{T}}A)^{-1}$ at all, so "$A^+ = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$" is meaningless there; go through the SVD. Finally, the pseudoinverse does not reverse products the way an inverse does: in general $(AB)^+ \neq B^+A^+$ (Exercise 36.17).

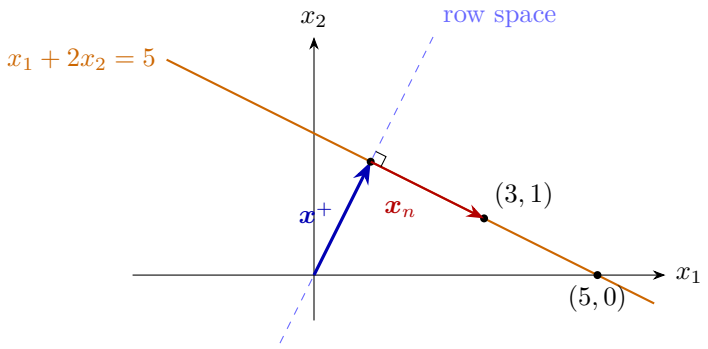


Figure 49: All solutions of $\begin{bmatrix}1 & 2\end{bmatrix}\boldsymbol{x} = 5$ form a line parallel to the nullspace direction $(-2,1)$. The pseudoinverse solution $\boldsymbol{x}^+ = A^+5 = (1,2)$ is the one in the row space, perpendicular to the line, hence the shortest ($\|\boldsymbol{x}^+\| = \sqrt{5}$, against $\sqrt{10}$ for $(3,1)$ and 5 for $(5,0)$).

36.7 Exercises

Exercise 36.15. Find the left inverse $(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ of $A = \begin{bmatrix}1 & 2\\ 0 & 1\\ 1 & 0\end{bmatrix}$, check that it is a left inverse, and identify $A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$.

Solution. $A^\mathsf{T}A = \begin{bmatrix}2 & 2\\ 2 & 5\end{bmatrix}$ has determinant 6, so $(A^\mathsf{T}A)^{-1} = \frac{1}{6}\begin{bmatrix}5 & -2\\ -2 & 2\end{bmatrix}$ and

$$B = (A^\mathsf{T}A)^{-1}A^\mathsf{T} = \frac{1}{6}\begin{bmatrix}5 & -2\\ -2 & 2\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ 2 & 1 & 0\end{bmatrix} = \frac{1}{6}\begin{bmatrix}1 & -2 & 5\\ 2 & 2 & -2\end{bmatrix}.$$

Then $BA = \frac{1}{6}\begin{bmatrix}1+0+5 & 2-2+0\\ 2+0-2 & 4+2+0\end{bmatrix} = I_2$. The product $AB = \frac{1}{6}\begin{bmatrix}5 & 2 & 1\\ 2 & 2 & -2\\ 1 & -2 & 5\end{bmatrix}$ is symmetric with trace 2; it is the projection onto the plane $\mathbf{C}(A)$ (Theorem 36.4). It sends $(1,-2,-1)$, which is orthogonal to both columns, to $\mathbf{0}$. □

Exercise 36.16. Find every right inverse of $A = \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$ and show that $A^\mathsf{T}(AA^\mathsf{T})^{-1}$ from Example 36.6 is the one with the smallest sum of squared entries.

Solution. Write $B = \begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix}$. Then $AB = \begin{bmatrix}a+e & b+f\\ c & d\end{bmatrix} = I$ forces $c = 0$, $d = 1$, $e = 1-a$, $f = -b$:

$$B = \begin{bmatrix}a & b\\ 0 & 1\\ 1-a & -b\end{bmatrix}, \qquad a, b \text{ arbitrary.}$$

The sum of squares is $a^2 + (1-a)^2 + 2b^2 + 1$, smallest at $a = \frac{1}{2}$, $b = 0$, which is exactly $A^\mathsf{T}(AA^\mathsf{T})^{-1}$. The other right inverses differ from it by matrices whose columns are multiples of $(1,0,-1)$, the nullspace of A; the favourite one has its columns in the row space, which is why it is shortest (compare Theorem 36.13). □

Exercise 36.17. Let $A = \begin{bmatrix}1 & 1\end{bmatrix}$ and $B = \begin{bmatrix}1\\0\end{bmatrix}$. Compute $(AB)^+$ and B^+A^+.

Solution. $AB = [1]$, so $(AB)^+ = [1]$. A has full row rank, so $A^+ = A^\mathsf{T}(AA^\mathsf{T})^{-1} = \frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}$; B has full column rank, so $B^+ = (B^\mathsf{T}B)^{-1}B^\mathsf{T} = \begin{bmatrix}1 & 0\end{bmatrix}$. Then $B^+A^+ = \begin{bmatrix}1 & 0\end{bmatrix}\frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{2} \neq 1$. The rule $(AB)^{-1} = B^{-1}A^{-1}$ does not carry over to pseudoinverses. □

Exercise 36.18. Find the pseudoinverse of $A = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$. Use it to find the shortest solution of $A\boldsymbol{x} = (2,2)$ and the shortest least-squares solution of $A\boldsymbol{x} = (1,3)$.

Solution. A is symmetric of rank one: $A = 2\boldsymbol{q}\boldsymbol{q}^\mathsf{T}$ with $\boldsymbol{q} = (1,1)/\sqrt{2}$, so $\sigma_1 = 2$, $\boldsymbol{u}_1 = \boldsymbol{v}_1 = \boldsymbol{q}$ and

$$A^+ = \frac{1}{2}\boldsymbol{q}\boldsymbol{q}^\mathsf{T} = \frac{1}{4}\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}.$$

For $\boldsymbol{b} = (2,2)$: $A^+\boldsymbol{b} = (1,1)$, and indeed $A(1,1) = (2,2)$; every other solution $(1+t, 1-t)$ is longer. For $\boldsymbol{b} = (1,3)$, which is not in $\mathbf{C}(A)$: $A^+\boldsymbol{b} = \frac{1}{4}(4,4) = (1,1)$ again, and $A(1,1) = (2,2)$ is the projection of $(1,3)$ onto the line through $(1,1)$, since $\frac{(1,3)\cdot(1,1)}{2}(1,1) = (2,2)$. □

Exercise 36.19. Using $A^+ = V\Sigma^+U^\mathsf{T}$, prove that $AA^+A = A$ and $A^+AA^+ = A^+$, and that $(A^+)^+ = A$.

Solution. $AA^+A = U\Sigma V^\mathsf{T}V\Sigma^+U^\mathsf{T}U\Sigma V^\mathsf{T} = U(\Sigma\Sigma^+\Sigma)V^\mathsf{T}$. The diagonal entries of $\Sigma\Sigma^+\Sigma$ are $\sigma_i \cdot \frac{1}{\sigma_i} \cdot \sigma_i = \sigma_i$ for $i \le r$ and 0 otherwise, so $\Sigma\Sigma^+\Sigma = \Sigma$ and $AA^+A = A$. In the same way $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$ gives $A^+AA^+ = A^+$. Finally $A^+ = V\Sigma^+U^\mathsf{T}$ is itself an SVD (orthogonal, diagonal with nonnegative entries $1/\sigma_i$, orthogonal), so its pseudoinverse is $U(\Sigma^+)^+V^\mathsf{T} = U\Sigma V^\mathsf{T} = A$, because inverting the nonzero diagonal entries twice returns σ_i. □

37 Exam 3 review

The third exam is about understanding a matrix through its eigenvalues: $d\boldsymbol{u}/dt = A\boldsymbol{u}$ and e^{At} (Section 25), symmetric matrices $A = Q\Lambda Q^\mathsf{T}$ (Section 29), positive definite matrices (Section 31), similar matrices (Section 32) and the SVD (Section 33). Linear transformations wait for the final. This review collects the facts that make eigenvalue questions fast and then works the review problems in full.

37.1 Why elimination is set aside

Elimination was the engine of the first two thirds of the course, but it is the wrong tool now: row operations change the eigenvalues. The operations that preserve eigenvalues are similarity transformations.

Proposition 37.1 (Similar matrices share eigenvalues)**.** *If $B = M^{-1}AM$ with M invertible, then A and B have the same characteristic polynomial, hence the same eigenvalues with the same multiplicities. If $A\boldsymbol{x} = \lambda\boldsymbol{x}$ then $B(M^{-1}\boldsymbol{x}) = \lambda M^{-1}\boldsymbol{x}$, and $B^k = M^{-1}A^kM$ for every k.*

Proof. Since $B - \lambda I = M^{-1}(A - \lambda I)M$,

$$\det(B - \lambda I) = \det M^{-1} \det(A - \lambda I) \det M = \det(A - \lambda I).$$

For the eigenvectors, $BM^{-1}\boldsymbol{x} = M^{-1}AMM^{-1}\boldsymbol{x} = \lambda M^{-1}\boldsymbol{x}$. Finally, writing B^k as k copies of $M^{-1}AM$, each inner pair MM^{-1} cancels and leaves $M^{-1}A^kM$. □

So if powers of A grow, so do powers of B; if one decays, the other does too. In the language of Section 35, similar matrices represent the same transformation in different bases.

Example 37.2 (Elimination moves eigenvalues; similarity does not)**.** $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has trace 4 and determinant 3, so $\lambda = 1, 3$. One elimination step gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 3/2 \end{bmatrix}$, whose eigenvalues are its diagonal entries 2 and 3/2: the determinant survived ($2 \cdot \frac{3}{2} = 3$) but the trace did not. With $M = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $M^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$,

$$M^{-1}AM = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 3 \end{bmatrix},$$

which is triangular with eigenvalues 1 and 3, the same as A.

37.2 Eigenvalues of the special matrices

Most exam questions are answered by knowing where each special family keeps its eigenvalues (Figure 50). The proofs are short and worth knowing.

Proposition 37.3 (Where the eigenvalues live)**.** *Let $A\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$ (complex entries allowed).*

(a) If Q is orthogonal, $|\lambda| = 1$.

(b) If P is a projection ($P^2 = P$), $\lambda = 0$ or 1.

(c) If A is real symmetric, λ is real; if A is real skew-symmetric ($A^{\mathsf{T}} = -A$), λ is purely imaginary.

Proof. (a) Orthogonal matrices preserve length: $\|Q\boldsymbol{x}\|^2 = \bar{\boldsymbol{x}}^{\mathsf{T}}Q^{\mathsf{T}}Q\boldsymbol{x} = \|\boldsymbol{x}\|^2$. Taking lengths in $Q\boldsymbol{x} = \lambda\boldsymbol{x}$ gives $\|\boldsymbol{x}\| = |\lambda|\,\|\boldsymbol{x}\|$, so $|\lambda| = 1$. (b) $\lambda\boldsymbol{x} = P\boldsymbol{x} = P^2\boldsymbol{x} = \lambda^2\boldsymbol{x}$, so $\lambda(\lambda - 1)\boldsymbol{x} = \mathbf{0}$ and $\lambda \in \{0, 1\}$. (c) Let $s = \bar{\boldsymbol{x}}^{\mathsf{T}}A\boldsymbol{x} = \lambda\,\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{x} = \lambda\|\boldsymbol{x}\|^2$. Since A is real, the conjugate of the number s is $\boldsymbol{x}^{\mathsf{T}}A\bar{\boldsymbol{x}}$, and transposing this 1×1 matrix gives $\bar{\boldsymbol{x}}^{\mathsf{T}}A^{\mathsf{T}}\boldsymbol{x}$. If $A^{\mathsf{T}} = A$ this is s itself, so s is real and so is $\lambda = s/\|\boldsymbol{x}\|^2$. If $A^{\mathsf{T}} = -A$ it is $-s$, so s and λ are purely imaginary. □

Remark 37.4 (Which matrices have orthogonal eigenvectors). A real matrix has a full set of orthonormal (possibly complex) eigenvectors exactly when $AA^{\mathsf{T}} = A^{\mathsf{T}}A$. Symmetric matrices pass (A^2 on both sides), skew-symmetric matrices pass ($-A^2$ on both sides) and orthogonal matrices pass (I on both sides). The proof goes through Schur's triangular form and is not needed for the exam; the symmetric case is the spectral theorem of Section 29. When the eigenvectors are complex, "orthogonal" means $\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{y} = 0$: conjugate the first vector.

Two more facts round out the toolkit. Markov matrices (Section 26) have $\lambda = 1$ and all $|\lambda| \leq 1$. Adding cI or multiplying by c keeps the eigenvectors and shifts or scales the eigenvalues: $A + cI$ has $\lambda + c$, and cA has $c\lambda$.

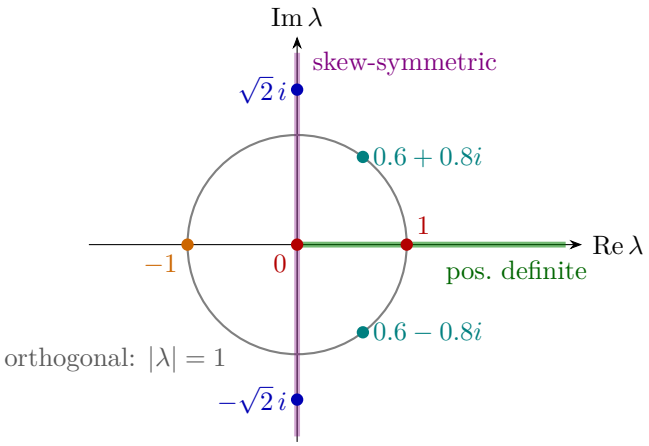


Figure 50: Where eigenvalues live. Symmetric: the real axis; positive definite: the positive real axis; skew-symmetric: the imaginary axis (the matrix of Example 37.5 has $0, \pm\sqrt{2}\,i$); orthogonal: the unit circle (the rotation of Exercise 37.10 has $0.6 \pm 0.8i$); projections: only 0 and 1; reflections and symmetric orthogonal matrices: only ± 1.

37.3 Solved review problems

Example 37.5 (A skew-symmetric system $d\boldsymbol{u}/dt = A\boldsymbol{u}$)**.** Solve $\dfrac{d\boldsymbol{u}}{dt} = A\boldsymbol{u}$ for $A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$.

Eigenvalues. Rows one and three are negatives of each other, so A is singular and $\lambda_1 = 0$. Also $A^{\mathsf{T}} = -A$, so the other eigenvalues are imaginary. Expanding,

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & -1 \\ 0 & 1 & -\lambda \end{vmatrix} = -\lambda^3 - 2\lambda = -\lambda(\lambda^2 + 2),$$

so $\lambda = 0,\ \sqrt{2}\,i,\ -\sqrt{2}\,i$. Solving $(A - \lambda I)\boldsymbol{x} = \boldsymbol{0}$: the first row gives $x_2 = -\lambda x_1$ and the second $x_3 = x_1 - \lambda x_2 = (1 + \lambda^2)x_1$, so

$$\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{x}_2 = \begin{bmatrix} 1 \\ -\sqrt{2}\,i \\ -1 \end{bmatrix} \ (\lambda = \sqrt{2}\,i), \qquad \boldsymbol{x}_3 = \begin{bmatrix} 1 \\ \sqrt{2}\,i \\ -1 \end{bmatrix} \ (\lambda = -\sqrt{2}\,i).$$

They are orthogonal, as Remark 37.4 promises: $\bar{\boldsymbol{x}}_1^{\mathsf{T}}\boldsymbol{x}_2 = 1 + 0 - 1 = 0$ and $\bar{\boldsymbol{x}}_2^{\mathsf{T}}\boldsymbol{x}_3 = 1 + (\sqrt{2}\,i)(\sqrt{2}\,i) + 1 = 1 - 2 + 1 = 0$.

General solution. $\boldsymbol{u}(t) = c_1\boldsymbol{x}_1 + c_2 e^{\sqrt{2}\,it}\boldsymbol{x}_2 + c_3 e^{-\sqrt{2}\,it}\boldsymbol{x}_3$. Each $e^{\pm\sqrt{2}\,it}$ has absolute value 1 and travels around the unit circle, so the solution neither grows nor decays. It is periodic: everything returns to its starting value when $\sqrt{2}\,T = 2\pi$, that is $T = \pi\sqrt{2}$.

The exponential. Since A has three independent eigenvectors, the eigenvector matrix X diagonalizes it and

$$e^{At} = X e^{\Lambda t} X^{-1}, \qquad e^{\Lambda t} = \begin{bmatrix} 1 & & \\ & e^{\sqrt{2}\,it} & \\ & & e^{-\sqrt{2}\,it} \end{bmatrix}.$$

A real form avoids complex arithmetic: the characteristic polynomial gives $A^3 = -2A$ (Cayley–Hamilton), so the power series collapses to

$$e^{At} = I + \frac{\sin\sqrt{2}\,t}{\sqrt{2}}\,A + \frac{1-\cos\sqrt{2}\,t}{2}\,A^2, \qquad A^2 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & -1 \end{bmatrix}.$$

For $\boldsymbol{u}(0) = (1,0,0)$ this gives $\boldsymbol{u}(t) = \left(\frac{1+\cos\sqrt{2}\,t}{2},\ \frac{\sin\sqrt{2}\,t}{\sqrt{2}},\ \frac{1-\cos\sqrt{2}\,t}{2}\right)$, whose length is 1 for all t (Exercise 37.11).

Example 37.6 (A matrix known only by its eigenvectors)**.** A 3×3 matrix has eigenvalues 0, c, 2 with eigenvectors $\boldsymbol{x}_1 = (1,1,1)$, $\boldsymbol{x}_2 = (1,-1,0)$, $\boldsymbol{x}_3 = (1,1,-2)$. For which c is it diagonalizable, symmetric, positive definite, Markov, or twice a projection?

The three vectors are not just independent but orthogonal ($\boldsymbol{x}_1\cdot\boldsymbol{x}_2 = 0$, $\boldsymbol{x}_1\cdot\boldsymbol{x}_3 = 0$, $\boldsymbol{x}_2\cdot\boldsymbol{x}_3 = 0$), and that settles most parts.

- *Diagonalizable*: for every c; three independent eigenvectors are all that matter, even if c repeats 0 or 2.
- *Symmetric*: for every real c. Normalizing gives $A = Q\Lambda Q^{\mathsf{T}}$, which is symmetric when Λ is real; a symmetric matrix cannot have a non-real eigenvalue. Explicitly,
$$A = \begin{bmatrix} \frac{c}{2}+\frac{1}{3} & \frac{1}{3}-\frac{c}{2} & -\frac{2}{3} \\ \frac{1}{3}-\frac{c}{2} & \frac{c}{2}+\frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}.$$
- *Positive definite*: never, because $\lambda = 0$ is not positive. It is positive *semidefinite* for $c \geq 0$.
- *Markov*: never; a Markov matrix has no eigenvalue larger than 1, and here $\lambda = 2$.
- $\frac{1}{2}A$ *a projection*: its eigenvalues 0, $c/2$, 1 must all be 0 or 1, and it is symmetric, so $c = 0$ or $c = 2$. For $c = 2$ the formula gives $\frac{1}{2}A = I - \frac{1}{3}\boldsymbol{x}_1\boldsymbol{x}_1^{\mathsf{T}}$, the projection onto the plane perpendicular to $(1,1,1)$.

Without orthogonal eigenvectors, symmetric, positive definite and projection would all have failed.

Example 37.7 (Reading an SVD)**.** Suppose $A = U\Sigma V^{\mathsf{T}}$ with U and V having two columns each.

(a) $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$: A is 2×2 and invertible, since all three factors are; $|\det A| = 3\cdot 2 = 6$.

(b) $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & -5 \end{bmatrix}$: impossible. Singular values are never negative.

(c) $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$: $A = 3\boldsymbol{u}_1\boldsymbol{v}_1^{\mathsf{T}}$ has rank one, a one-dimensional nullspace, and orthonormal bases of all four subspaces can be read off: $\boldsymbol{v}_1$ for the row space, $\boldsymbol{v}_2$ for $\mathbf{N}(A)$ (since $A\boldsymbol{v}_2 = 0\boldsymbol{u}_2$), $\boldsymbol{u}_1$ for $\mathbf{C}(A)$ and $\boldsymbol{u}_2$ for $\mathbf{N}(A^{\mathsf{T}})$.

To *compute* an SVD, take V from the eigenvectors of $A^{\mathsf{T}}A$ and $\sigma_i = \sqrt{\lambda_i}$, then get each $\boldsymbol{u}_i$ from $A\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$. For $A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$: $A^{\mathsf{T}}A = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$ has $\lambda = 32, 18$ with $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$. Then $A\boldsymbol{v}_1 = (4\sqrt{2},0)$ gives $\sigma_1 = 4\sqrt{2}$, $\boldsymbol{u}_1 = (1,0)$, and $A\boldsymbol{v}_2 = (0,-3\sqrt{2})$ gives $\sigma_2 = 3\sqrt{2}$, $\boldsymbol{u}_2 = (0,-1)$.

Common trap. Taking U from the eigenvectors of AA^{T} independently of V can produce a wrong sign. For the matrix just above, $AA^{\mathsf{T}} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$ suggests $\boldsymbol{u}_2 = (0,1)$, and then $U\Sigma V^{\mathsf{T}} \neq A$. Eigenvectors are only determined up to sign; $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ fixes the sign of $\boldsymbol{u}_i$ once $\boldsymbol{v}_i$ is chosen.

Example 37.8 (Symmetric and orthogonal)**.** Suppose A is both symmetric and orthogonal. Its eigenvalues are real (symmetric) with $|\lambda| = 1$ (orthogonal), so each is 1 or -1. Hence: A need not be positive definite ($\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a counterexample); it must have repeated eigenvalues once $n \geq 3$; it is diagonalizable, even with orthonormal eigenvectors; it is invertible, as every orthogonal matrix is. Finally $P = \frac{1}{2}(A + I)$ is a projection: it is symmetric, and $A^2 = A^{\mathsf{T}}A = I$ gives

$$P^2 = \tfrac{1}{4}(A^2 + 2A + I) = \tfrac{1}{4}(2A + 2I) = \tfrac{1}{2}(A + I) = P.$$

(Or: the eigenvalues ± 1 of A become 1 and 0.) For instance the exchange matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ gives $P = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the projection onto the line through $(1,1)$.

Example 37.9 (Five tests for positive definiteness)**.** For a symmetric matrix the following are equivalent (Sections 29 and 31): all $\lambda_i > 0$; all pivots > 0; all upper-left determinants > 0; $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$; $A = R^{\mathsf{T}}R$ with independent columns in R. For $K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$: the upper-left determinants are $2, 3, 4$; the pivots are their ratios $2, \frac{3}{2}, \frac{4}{3}$; the eigenvalues are $2 - \sqrt{2},\ 2,\ 2 + \sqrt{2}$ (sum $6 = \operatorname{tr} K$, product $4 = \det K$); and

$$\boldsymbol{x}^{\mathsf{T}}K\boldsymbol{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 = x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2,$$

which is zero only when $x_1 = x_2 = x_3 = 0$. Every test agrees: K is positive definite.

37.4 Exercises

Exercise 37.10. Let $\boldsymbol{a} = (3,4)$. Find the eigenvalues and eigenvectors of (a) the projection $P = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$, (b) the rotation $Q = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$, (c) the reflection $R = 2P - I$.

Solution. (a) $P = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$. A projection has $\lambda \in \{0,1\}$. $P\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a})/(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}) = \boldsymbol{a}$, so $\boldsymbol{a} = (3,4)$ has $\lambda = 1$; $\boldsymbol{b} = (-4,3)$ is perpendicular to $\boldsymbol{a}$, so $P\boldsymbol{b} = \boldsymbol{a}(\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b})/25 = \boldsymbol{0}$ and $\lambda = 0$. (b) $0.6^2 + 0.8^2 = 1$, so Q is the rotation with $\cos\theta = 0.6$, $\sin\theta = 0.8$. $\det(Q - \lambda I) = (0.6 - \lambda)^2 + 0.64 = 0$ gives $\lambda = 0.6 \pm 0.8i$, a conjugate pair on the unit circle. For $\lambda = 0.6 + 0.8i$, $Q - \lambda I = \begin{bmatrix} -0.8i & -0.8 \\ 0.8 & -0.8i \end{bmatrix}$, whose second row is i times the first, and $\boldsymbol{x} = (1, -i)$ solves $-0.8i(1) - 0.8(-i) = 0$. The other eigenvector is the conjugate $(1, i)$. (c) $R = \frac{1}{25}\begin{bmatrix} -7 & 24 \\ 24 & 7 \end{bmatrix}$ has the same eigenvectors as P, with $\lambda \mapsto 2\lambda - 1$: $R\boldsymbol{a} = \boldsymbol{a}$ ($\lambda = 1$) and $R\boldsymbol{b} = -\boldsymbol{b}$ ($\lambda = -1$). A reflection keeps the mirror line and flips the perpendicular direction. □

Exercise 37.11. Show that if A is real and skew-symmetric, then e^{At} is an orthogonal matrix, so every solution of $d\boldsymbol{u}/dt = A\boldsymbol{u}$ keeps its length. Check this for the solution $\boldsymbol{u}(t)$ found in Example 37.5.

Solution. Transposing the series term by term gives $(e^{At})^{\mathsf{T}} = e^{A^{\mathsf{T}}t} = e^{-At}$. Since At and $-At$ commute, $e^{At}e^{-At} = e^0 = I$, so $(e^{At})^{\mathsf{T}}e^{At} = I$ and $\|\boldsymbol{u}(t)\| = \|e^{At}\boldsymbol{u}(0)\| = \|\boldsymbol{u}(0)\|$. With $C = \cos\sqrt{2}\,t$ and $S = \sin\sqrt{2}\,t$,

$$\|\boldsymbol{u}(t)\|^2 = \frac{(1+C)^2}{4} + \frac{S^2}{2} + \frac{(1-C)^2}{4} = \frac{2+2C^2}{4} + \frac{S^2}{2} = \frac{1+C^2+S^2}{2} = 1 = \|\boldsymbol{u}(0)\|^2.$$

□

Exercise 37.12. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and I have the same eigenvalues. Are they similar?

Solution. No. Every matrix similar to I is $M^{-1}IM = I$ itself, and $A \neq I$. Same eigenvalues is necessary for similarity but not sufficient: A has only one independent eigenvector, $(1, 0)$, while I has two (Section 32 on Jordan form). □

Exercise 37.13. $A = U\Sigma V^{\mathsf{T}}$ with $U = I$, $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$ and $V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$. Find A and a basis for each of its four subspaces.

Solution. $A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathsf{T}} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{3}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Row space: $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$. Nullspace: $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(-1, 1)$; check $A\boldsymbol{v}_2 = \frac{3}{2}(-1 + 1, 0) = \boldsymbol{0}$. Column space: $\boldsymbol{u}_1 = (1, 0)$. Left nullspace: $\boldsymbol{u}_2 = (0, 1)$, and indeed the second row of A is zero. □

38 Exam 3

The Unit III exam has three questions (34, 33 and 33 points), and each one is about reading a matrix through its eigenvalues or singular values rather than its entries. Question 1 asks what "all singular values equal 1" means and builds a matrix from an orthonormal eigenvector matrix; Question 2 diagonalizes a triangular matrix to get powers and exponentials; Question 3 expands vectors in orthonormal eigenvectors. Each question is stated as an exercise; try it before reading the solution, which follows the exam's own reasoning with every step filled in. Section 37 reviews the tools.

38.1 Question 1: singular values equal to one, and a matrix built from its eigenvectors

Exercise 38.1. (a) A square matrix A has all n of its singular values equal to 1. Which of these classes must A belong to: singular, symmetric, orthogonal, positive definite or semidefinite, diagonal?

(b) The orthonormal columns of

$$H = \frac{1}{2}\begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}, \qquad H^{-1} = H^{\mathsf{T}},$$

are eigenvectors of B, with eigenvalues $\lambda = 0, 1, 2, 3$ in that order. Write B as a product of three specific matrices, and do the same for $C = (B + I)^{-1}$.

(c) Which classes from the list in (a) do B and C belong to?

Solution. (a) With $\Sigma = I$ the SVD reads $A = UIV^{\mathsf{T}} = UV^{\mathsf{T}}$, a product of orthogonal matrices, which is orthogonal: $(UV^{\mathsf{T}})^{\mathsf{T}}(UV^{\mathsf{T}}) = VU^{\mathsf{T}}UV^{\mathsf{T}} = I$. Equivalently, the σ_i^2 are the eigenvalues of the symmetric matrix $A^{\mathsf{T}}A = V\Sigma^{\mathsf{T}}\Sigma V^{\mathsf{T}}$, so all $\sigma_i = 1$ means $A^{\mathsf{T}}A = VIV^{\mathsf{T}} = I$. Nothing else is forced. A is never singular, but it need not be symmetric: $U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $V = I$ give the rotation $A = U$ by 90°, which is not symmetric, not diagonal, and (with eigenvalues $\pm i$) not positive definite or semidefinite.

(b) The eigenvector matrix is H and $H^{-1} = H^{\mathsf{T}}$, so

$$B = H\Lambda H^{-1} = H \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 2 & \\ & & & 3 \end{bmatrix} H^{\mathsf{T}}.$$

Adding I keeps the eigenvectors and adds 1 to each eigenvalue, and inverting keeps the eigenvectors and inverts each eigenvalue:

$$C = (B + I)^{-1} = H(\Lambda + I)^{-1}H^{-1} = H \begin{bmatrix} 1 & & & \\ & 1/2 & & \\ & & 1/3 & \\ & & & 1/4 \end{bmatrix} H^{\mathsf{T}}.$$

(c) $B = H\Lambda H^{\mathsf{T}}$ with real Λ is symmetric. It has $\lambda = 0$, so it is singular, and its eigenvalues are ≥ 0, so it is positive semidefinite (not definite). It is not orthogonal ($|\lambda| \neq 1$) and, as Example 38.2 shows, not diagonal. C is symmetric with eigenvalues $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} > 0$: symmetric positive definite, invertible, not orthogonal, not diagonal. □

Example 38.2 (The matrices B and C written out)**.** Multiplying out $B = \sum_i \lambda_i \boldsymbol{h}_i \boldsymbol{h}_i^{\mathsf{T}} = 1 \cdot \boldsymbol{h}_2\boldsymbol{h}_2^{\mathsf{T}} + 2\,\boldsymbol{h}_3\boldsymbol{h}_3^{\mathsf{T}} + 3\,\boldsymbol{h}_4\boldsymbol{h}_4^{\mathsf{T}}$, where $\boldsymbol{h}_i$ is column i of H, gives

$$B = \frac{1}{4}\begin{bmatrix} 6 & -2 & -4 & 0 \\ -2 & 6 & 0 & -4 \\ -4 & 0 & 6 & -2 \\ 0 & -4 & -2 & 6 \end{bmatrix}, \qquad C = (B + I)^{-1} = \frac{1}{48}\begin{bmatrix} 25 & 7 & 11 & 5 \\ 7 & 25 & 5 & 11 \\ 11 & 5 & 25 & 7 \\ 5 & 11 & 7 & 25 \end{bmatrix}.$$

Checks: B is symmetric; each row of B sums to 0, so $B\boldsymbol{h}_1 = \boldsymbol{0}$ for $\boldsymbol{h}_1 = \frac{1}{2}(1, 1, 1, 1)$; $\operatorname{tr} B = 6 = 0 + 1+2+3$; and $B\boldsymbol{h}_2 = \frac{1}{4}\cdot\frac{1}{2}(6+2-4+0,\ -2-6+0+4,\ -4+0+6+2,\ 0+4-2-6) = \frac{1}{2}(1, -1, 1, -1) = \boldsymbol{h}_2$. For C, $\operatorname{tr} C = \frac{100}{48} = \frac{25}{12} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$. Notice how much easier the three-factor form is to use than these entries: the question asks for the factors because they display everything.

38.2 Question 2: powers and exponentials through diagonalization

Exercise 38.3. Let $A = \begin{bmatrix} -1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Find the three eigenvalues of A and an eigenvector matrix X (called S on the exam).

(b) Explain why $A^{1001} = A$. Is $A^{1000} = I$? Find the three diagonal entries of e^{At}.

(c) The matrix $A^{\mathsf{T}}A = \begin{bmatrix} 1 & -2 & -4 \\ -2 & 4 & 8 \\ -4 & 8 & 42 \end{bmatrix}$. Without computing them, how many of its eigenvalues are positive, zero, negative? Does $A^{\mathsf{T}}A$ have the same eigenvectors as A?

Solution. (a) A is triangular, so its eigenvalues are its diagonal entries $-1, 0, 1$. Eigenvectors:

- $\lambda = -1$: $(A + I)\boldsymbol{x} = \mathbf{0}$ has rows $(0, 2, 4)$, $(0, 1, 5)$, $(0, 0, 2)$, forcing $x_3 = x_2 = 0$: $\boldsymbol{x} = (1, 0, 0)$.
- $\lambda = 0$: $A\boldsymbol{x} = \mathbf{0}$ gives $x_3 = 0$ and $-x_1 + 2x_2 = 0$: $\boldsymbol{x} = (2, 1, 0)$.
- $\lambda = 1$: $(A - I)\boldsymbol{x} = \mathbf{0}$ has rows $(-2, 2, 4)$, $(0, -1, 5)$, $(0, 0, 0)$. With $x_3 = 1$: $x_2 = 5$, then $-2x_1 + 10 + 4 = 0$ gives $x_1 = 7$: $\boldsymbol{x} = (7, 5, 1)$.

So $X = \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$, upper triangular like A, with $X^{-1} = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$.

(b) $A = X\Lambda X^{-1}$ gives $A^k = X\Lambda^k X^{-1}$. Since $(-1)^{1001} = -1$, $0^{1001} = 0$ and $1^{1001} = 1$, we have $\Lambda^{1001} = \Lambda$ and therefore $A^{1001} = A$. But $A^{1000} \neq I$: $\Lambda^{1000} = \mathrm{diag}(1, 0, 1) \neq I$, and in any case a singular matrix has singular powers. (In fact $A^{1000} = A^2 = \begin{bmatrix} 1 & -2 & 10 \\ 0 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix}$.) For the exponential,

$$e^{At} = Xe^{\Lambda t}X^{-1} = X \begin{bmatrix} e^{-t} & & \\ & 1 & \\ & & e^{t} \end{bmatrix} X^{-1}.$$

X and X^{-1} are triangular with ones on the diagonal, so the diagonal of e^{At} is e^{-t}, 1, e^t. A second proof: every term $(At)^k/k!$ of the series is upper triangular with diagonal $(-t)^k/k!$, $0^k/k!$, $t^k/k!$, and these sum to e^{-t}, 1 (only the $k = 0$ term survives) and e^t. Multiplying out completely,

$$e^{At} = \begin{bmatrix} e^{-t} & 2 - 2e^{-t} & 7e^t - 10 + 3e^{-t} \\ 0 & 1 & 5e^t - 5 \\ 0 & 0 & e^t \end{bmatrix},$$

which is I at $t = 0$ as it must be.

(c) $A^{\mathsf{T}}A$ is symmetric, and $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = \|A\boldsymbol{x}\|^2 \geq 0$, so no eigenvalue is negative. Its rank equals the rank of A, which is 2 (one zero eigenvalue, two independent eigenvectors for ± 1), so exactly one eigenvalue is 0 and $3 - 1 = 2$ are positive. Alternatively, the signs of the eigenvalues of a symmetric matrix match the signs of its pivots: elimination subtracts -2 and -4 times row one, leaving

$$\begin{bmatrix} 1 & -2 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 26 \end{bmatrix},$$

nonzero pivots 1 and 26 (from $42-16$) and one zero. The eigenvectors are different: $A^{\mathsf{T}}A$ is symmetric, so its eigenvectors are orthogonal, while those of A are not ($(1,0,0)\cdot(2,1,0)=2$). (Numerically the eigenvalues are 0 and $\frac{1}{2}(47\pm\sqrt{1689})\approx 2.95,\ 44.05$.) □

Common trap. "$\Lambda^{1001}=\Lambda$, so $\Lambda^{1000}=I$" divides by Λ, which is not allowed when $\lambda=0$ is an eigenvalue: $0^{1000}=0$, not 1. Cancelling a factor of A from $A^{1001}=A$ is legitimate only when A is invertible.

38.3 Question 3: expanding in orthonormal eigenvectors

Exercise 38.4. The $n\times n$ matrix A has orthonormal eigenvectors $\boldsymbol{q}_1,\dots,\boldsymbol{q}_n$ and positive eigenvalues $\lambda_1,\dots,\lambda_n$, so $A\boldsymbol{q}_j=\lambda_j\boldsymbol{q}_j$.

(a) What are the eigenvalues and eigenvectors of A^{-1}? Prove it.

(b) Any $\boldsymbol{b}$ is a combination $\boldsymbol{b}=c_1\boldsymbol{q}_1+\cdots+c_n\boldsymbol{q}_n$. Give a quick formula for c_1.

(c) The solution of $A\boldsymbol{x}=\boldsymbol{b}$ is $A^{-1}\boldsymbol{b}=d_1\boldsymbol{q}_1+\cdots+d_n\boldsymbol{q}_n$. Give a quick formula for d_1.

Solution. (a) All $\lambda_j>0$, so A is invertible. Multiply $A\boldsymbol{q}_j=\lambda_j\boldsymbol{q}_j$ by A^{-1} to get $\boldsymbol{q}_j=\lambda_jA^{-1}\boldsymbol{q}_j$, and divide by $\lambda_j\neq 0$: $A^{-1}\boldsymbol{q}_j=\frac{1}{\lambda_j}\boldsymbol{q}_j$. Same eigenvectors, eigenvalues $1/\lambda_j$.

(b) Multiply $\boldsymbol{b}=c_1\boldsymbol{q}_1+\cdots+c_n\boldsymbol{q}_n$ on the left by $\boldsymbol{q}_1^{\mathsf{T}}$. Orthogonality kills every term but the first, and $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_1=1$:

$$c_1=\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b}.$$

(c) By (a), A^{-1} multiplies each $\boldsymbol{q}_i$ by $1/\lambda_i$, so $A^{-1}\boldsymbol{b}=\sum_i\frac{c_i}{\lambda_i}\boldsymbol{q}_i$ and

$$d_1=\frac{c_1}{\lambda_1}=\frac{\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b}}{\lambda_1}.$$

□

Example 38.5 (Question 3 with numbers)**.** Take $A=\begin{bmatrix}2&1\\1&2\end{bmatrix}$, with $\boldsymbol{q}_1=\frac{1}{\sqrt{2}}(1,1)$ for $\lambda_1=3$ and $\boldsymbol{q}_2=\frac{1}{\sqrt{2}}(1,-1)$ for $\lambda_2=1$, and $\boldsymbol{b}=(4,2)$. Then

$$c_1=\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b}=\tfrac{6}{\sqrt{2}}=3\sqrt{2},\qquad c_2=\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{b}=\tfrac{2}{\sqrt{2}}=\sqrt{2},$$

so $\boldsymbol{b}=3\sqrt{2}\,\boldsymbol{q}_1+\sqrt{2}\,\boldsymbol{q}_2=(3,3)+(1,-1)$. Dividing by the eigenvalues, $d_1=3\sqrt{2}/3=\sqrt{2}$ and $d_2=\sqrt{2}/1=\sqrt{2}$:

$$\boldsymbol{x}=A^{-1}\boldsymbol{b}=\sqrt{2}\,\boldsymbol{q}_1+\sqrt{2}\,\boldsymbol{q}_2=(1,1)+(1,-1)=(2,0).$$

Check: $A(2,0)=(4,2)=\boldsymbol{b}$. No elimination and no inverse were needed, only two dot products and two divisions (Figure 51).

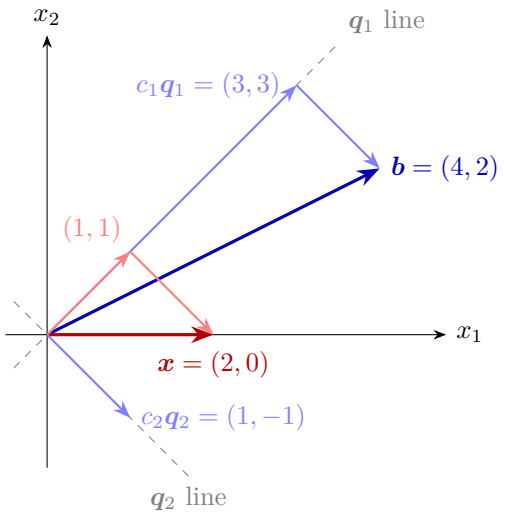


Figure 51: Solving $A\boldsymbol{x} = \boldsymbol{b}$ in the eigenvector basis. $\boldsymbol{b}$ splits along the orthonormal eigenvectors into $(3,3) + (1,-1)$; A^{-1} divides the first piece by $\lambda_1 = 3$ and the second by $\lambda_2 = 1$, giving $\boldsymbol{x} = (1,1) + (1,-1) = (2,0)$.

38.4 More practice in the same style

Exercise 38.6. Every singular value of the square matrix A equals 2. What can you say about A? What are the singular values of A^{-1} and of A^2?

Solution. $\Sigma = 2I$, so $A = U(2I)V^{\mathsf{T}} = 2(UV^{\mathsf{T}}) = 2Q$ with Q orthogonal; equivalently $A^{\mathsf{T}}A = 4I$. Then $A^{-1} = \frac{1}{2}Q^{\mathsf{T}}$ has all singular values $\frac{1}{2}$, and $A^2 = 4Q^2$ with Q^2 orthogonal has all singular values 4. Such an A doubles the length of every vector. □

Exercise 38.7. With B from Exercise 38.1, solve $d\boldsymbol{u}/dt = -B\boldsymbol{u}$ starting from $\boldsymbol{u}(0) = (4,0,0,0)$, and find $\lim_{t\to\infty} \boldsymbol{u}(t)$.

Solution. $-B = H(-\Lambda)H^{\mathsf{T}}$ has eigenvalues $0, -1, -2, -3$ with eigenvectors $\boldsymbol{h}_1, \dots, \boldsymbol{h}_4$, so

$$\boldsymbol{u}(t) = \sum_{i=1}^{4} e^{-\lambda_i t}(\boldsymbol{h}_i^{\mathsf{T}}\boldsymbol{u}(0))\,\boldsymbol{h}_i, \qquad \boldsymbol{h}_i^{\mathsf{T}}\boldsymbol{u}(0) = 4 \times (\text{first entry of } \boldsymbol{h}_i) = 2,\ 2,\ -2,\ -2.$$

Hence $\boldsymbol{u}(t) = 2\boldsymbol{h}_1 + 2e^{-t}\boldsymbol{h}_2 - 2e^{-2t}\boldsymbol{h}_3 - 2e^{-3t}\boldsymbol{h}_4$. Every term but the first decays, so $\boldsymbol{u}(t) \to 2\boldsymbol{h}_1 = (1,1,1,1)$: the initial total of 4 spreads evenly. (Check at $t = 0$: $2(\boldsymbol{h}_1 + \boldsymbol{h}_2 - \boldsymbol{h}_3 - \boldsymbol{h}_4) = (1+1+1+1,\ 1-1+1-1,\ 1+1-1-1,\ 1-1-1+1) = (4,0,0,0)$.) □

Part IV
Review

39 Final course review

The final exam covers the whole course, and its best questions connect the parts: a question about solvability turns into a question about $A^{\mathsf{T}}A$; a Markov matrix is solved by noticing that it is singular. The way to review is to work old final-exam problems, and that is what this section does after a one-page summary of the tools they use. Even the "easy" material from the first third of the course does real work here.

39.1 The toolkit

(1) *Rank and solvability* (Sections 9–11). For $m \times n$ A of rank r: $A\boldsymbol{x} = \boldsymbol{b}$ is solvable for every $\boldsymbol{b}$ iff $r = m$; solutions are unique (when they exist) iff $r = n$, i.e. $\mathbf{N}(A) = \{\mathbf{0}\}$. The dimensions are r, $n - r$, r, $m - r$ for $\mathbf{C}(A^{\mathsf{T}}), \mathbf{N}(A), \mathbf{C}(A), \mathbf{N}(A^{\mathsf{T}})$.

(2) *The matrix* $A^{\mathsf{T}}A$ (Sections 18 and 36). It is square, symmetric and positive semidefinite, and $\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A)$; so it is invertible (indeed positive definite) iff $r = n$.

(3) *Projections and least squares* (Sections 17–18). $A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$, $\boldsymbol{p} = A\hat{\boldsymbol{x}}$, $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$; the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ lies in $\mathbf{N}(A^{\mathsf{T}})$. For a line, $P = \boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}/\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}$.

(4) *Eigenvalue shortcuts* (Sections 23–26). Sum $= \operatorname{tr} A$, product $= \det A$; singular $\Rightarrow \lambda = 0$; Markov $\Rightarrow \lambda = 1$; $A + cI$ has $\lambda + c$; triangular matrices show their eigenvalues on the diagonal; $A = X\Lambda X^{-1}$ builds a matrix from given eigenvalues and eigenvectors.

(5) *Special matrices* (Section 37). Symmetric, skew-symmetric, orthogonal, projection, permutation, Markov, positive definite: each has a signature in its eigenvalues and eigenvectors.

39.2 Solved problems from past finals

Example 39.1 (What one solvable and one unsolvable system reveal)**.** A is $m \times n$ of rank r. $A\boldsymbol{x} = (1, 0, 0)$ has no solution, and $A\boldsymbol{x} = (0, 1, 0)$ has exactly one.

(a) *What are m, n, r?* The right-hand sides have three components, so $m = 3$. Some right-hand side is unreachable, so $\mathbf{C}(A) \neq \mathbb{R}^3$ and $r < 3$. A solution exists and is unique, so $\mathbf{N}(A) = \{\mathbf{0}\}$ and $r = n$; and $(0, 1, 0) \in \mathbf{C}(A)$ gives $r \geq 1$. In short: $m = 3$ and $1 \leq r = n \leq 2$.

(b) *An example.* Put $(0, 1, 0)$ in the column space and stop: $A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $(n = r = 1)$. Or add a column that keeps $(1, 0, 0)$ out: $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $(n = r = 2)$.

(c) *True or false for every such A:* (i) $\det A^{\mathsf{T}}A = \det AA^{\mathsf{T}}$; (ii) $A^{\mathsf{T}}A$ is invertible; (iii) AA^{T} is positive definite. Statement (ii) is true because $r = n$ (toolkit item 2); for the examples, $A^{\mathsf{T}}A = I$. Statement (iii) is false: AA^{T} is 3×3 with rank $n < 3$, so it is always positive

*semi*definite; for the second example $AA^{\mathsf{T}} = \operatorname{diag}(0,1,1)$. Then (i) is false too: $\det A^{\mathsf{T}}A \neq 0$ but $\det AA^{\mathsf{T}} = 0$. The rule $\det(BA) = \det(AB) = \det A \det B$ needs square matrices.

(d) *Prove that $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{c}$ has infinitely many solutions for every $\boldsymbol{c}$.* A^{T} is $n \times 3$ with rank n: full row rank, so a solution exists for every $\boldsymbol{c}$. Its nullspace $\mathbf{N}(A^{\mathsf{T}})$ has dimension $m - r = 3 - n \geq 1$, and adding any multiple of a nonzero nullspace vector to one solution gives another.

Example 39.2 (Columns $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$)**.** The columns of A are $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$. (a) $A\boldsymbol{x} = \boldsymbol{v}_1 - \boldsymbol{v}_2 + \boldsymbol{v}_3$ is solved by $\boldsymbol{x} = (1,-1,1)$: $A\boldsymbol{x}$ is the combination of columns with coefficients x_j, the very first idea of the course. (b) *True or false: if $\boldsymbol{v}_1 - \boldsymbol{v}_2 + \boldsymbol{v}_3 = \boldsymbol{0}$, the solution in (a) is not unique.* True. Then $(1,-1,1)$ is in $\mathbf{N}(A)$, and adding any multiple of it to a solution gives another solution. (c) *If $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are orthonormal, which combination of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ is closest to $\boldsymbol{v}_3$?* The projection of $\boldsymbol{v}_3$ onto the plane of $\boldsymbol{v}_1, \boldsymbol{v}_2$ is $(\boldsymbol{v}_1^{\mathsf{T}}\boldsymbol{v}_3)\boldsymbol{v}_1 + (\boldsymbol{v}_2^{\mathsf{T}}\boldsymbol{v}_3)\boldsymbol{v}_2 = 0\boldsymbol{v}_1 + 0\boldsymbol{v}_2$. Think of $\boldsymbol{v}_3$ as the z-axis and the plane as the xy-plane: the closest point is the origin.

Example 39.3 (A singular Markov matrix)**.** $A = \begin{bmatrix} .2 & .4 & .3 \\ .4 & .2 & .3 \\ .4 & .4 & .4 \end{bmatrix}$; note that column 1 + column 2 $= 2\times$ column 3.

Eigenvalues. The columns are dependent, so $\lambda = 0$. The columns sum to 1, so $\lambda = 1$ (the all-ones vector is an eigenvector of A^{T}). The trace is 0.8, so the third is $\lambda = 0.8 - 0 - 1 = -0.2$.

Steady state from $\boldsymbol{u}_0 = (0,10,0)$. $\boldsymbol{u}_k = A^k\boldsymbol{u}_0 = c_1(0)^k\boldsymbol{x}_1 + c_2(1)^k\boldsymbol{x}_2 + c_3(-0.2)^k\boldsymbol{x}_3$, so only $c_2\boldsymbol{x}_2$ survives as $k \to \infty$. For $\boldsymbol{x}_2$ solve $(A - I)\boldsymbol{x} = \boldsymbol{0}$: the matrix has columns $(-.8,.4,.4)$, $(.4,-.8,.4)$, $(.3,.3,-.6)$, and a lucky guess with $x_1 = x_2$ works: $\boldsymbol{x}_2 = (3,3,4)$, since $A(3,3,4) = (.6 + 1.2 + 1.2,\ 1.2 + .6 + 1.2,\ 1.2 + 1.2 + 1.6) = (3,3,4)$. A Markov step preserves the total of the entries, which is 10 at the start and 10 in $(3,3,4)$, so $c_2 = 1$ and

$$\boldsymbol{u}_\infty = (3,3,4).$$

Figure 52 shows the approach: the third entry is 4 from the first step on, and the first two oscillate around 3 because of the factor $(-0.2)^k$.

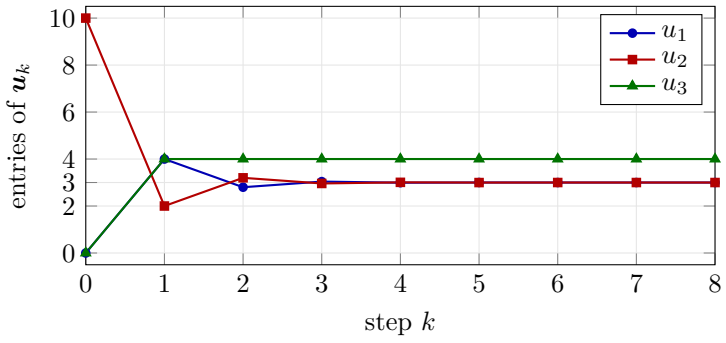


Figure 52: The Markov chain of Example 39.3 from $\boldsymbol{u}_0 = (0,10,0)$. The $\lambda = 0$ part disappears after one step, the $\lambda = -0.2$ part shrinks by a factor 5 and changes sign each step, and $\boldsymbol{u}_k \to (3,3,4)$.

Example 39.4 (Four 2×2 matrices on demand)**.** (a) *The projection onto the line through $\boldsymbol{a} = (4,-3)$:* $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}} = \dfrac{1}{25}\begin{bmatrix} 16 & -12 \\ -12 & 9 \end{bmatrix}$; as a check, $\det P = 0$ and $\operatorname{tr} P = 1$.

(b) *The matrix with* $\lambda = 0$, $\boldsymbol{x} = (1,2)$ *and* $\lambda = 3$, $\boldsymbol{x} = (2,1)$*:*

$$A = X\Lambda X^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \end{bmatrix} = \begin{bmatrix} 0 & 6 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix}.$$

Check: $A(1,2) = (0,0)$ and $A(2,1) = (6,3) = 3(2,1)$.

(c) *A real matrix that is not* $B^{\mathsf{T}}B$ *for any* B*:* $B^{\mathsf{T}}B$ is always symmetric, so any non-symmetric matrix works, e.g. $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$. (So is any symmetric matrix with a negative eigenvalue, since $B^{\mathsf{T}}B$ is semidefinite.)

(d) *Orthogonal eigenvectors but not symmetric:* the skew-symmetric $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ or the rotation $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, both of which commute with their transposes; their eigenvectors $(1, \pm i)$ are complex and orthogonal.

Example 39.5 (Least squares, given the answer)**.** For $A\boldsymbol{x} = \boldsymbol{b}$ with

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix},$$

the least-squares solution is $(\hat{c}, \hat{d}) = (\frac{11}{3}, -1)$. (Check: $A^{\mathsf{T}}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^{\mathsf{T}}\boldsymbol{b} = (8,6)$, and $3 \cdot \frac{11}{3} - 3 = 8$, $3 \cdot \frac{11}{3} - 5 = 6$.)

(a) *The projection of* $\boldsymbol{b}$ *onto* $\mathbf{C}(A)$ is the best combination of the columns: $\boldsymbol{p} = \frac{11}{3}(1,1,1) - 1 \cdot (0,1,2) = (\frac{11}{3}, \frac{8}{3}, \frac{5}{3})$.

(b) *The straight-line problem.* The rows say $C + Dt = b$ at $t = 0, 1, 2$: fit the points $(0,3)$, $(1,4)$, $(2,1)$. The best line is $b = \frac{11}{3} - t$ (Figure 53); the errors $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (-\frac{2}{3}, \frac{4}{3}, -\frac{2}{3})$ are perpendicular to both columns.

(c) *A nonzero* $\boldsymbol{b}$ *whose least-squares solution is* $(0,0)$. We need $\boldsymbol{p} = \mathbf{0}$, i.e. $\boldsymbol{b}$ orthogonal to both columns, i.e. $\boldsymbol{b} \in \mathbf{N}(A^{\mathsf{T}})$: $\boldsymbol{b} = (1,-2,1)$ works ($1 + (-2) + 1 = 0$ and $0 - 2 + 2 = 0$). Notice that $\boldsymbol{e}$ in (b) is $-\frac{2}{3}(1,-2,1)$, in the same line.

Example 39.6 (Eigenvalues, pivots and semidefiniteness without computing)**.** $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ has eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, and its first two pivots are 1 and 1.

(a) *The third eigenvalue and pivot.* The trace is $1+1+0 = 2 = 1+2+\lambda_3$, so $\lambda_3 = -1$. Elimination without row exchanges does not change the determinant, so $d_1 d_2 d_3 = \det A = \lambda_1\lambda_2\lambda_3 = -2$, giving $d_3 = -2$. (No row exchange is needed, since the first two pivots $1, 1$ are nonzero, so the determinant really is the product of the three pivots.)

(b) *The smallest* a_{33} *that makes* A *positive semidefinite.* With a_{33} in the corner, elimination gives pivots 1, 1 and $\det A/(1 \cdot 1)$, where expanding the determinant gives $\det A = a_{33} - 2$. A symmetric matrix factored as LDL^{T} is semidefinite exactly when all pivots are ≥ 0, so the condition is $a_{33} \geq 2$; the smallest choice is $a_{33} = 2$ (the eigenvalues are then $0, 1, 3$).

The smallest c *that makes* $A + cI$ *positive semidefinite.* Shifting adds c to each eigenvalue: $1 + c$, $2 + c$, $-1 + c$ are all ≥ 0 exactly when $c \geq 1$.

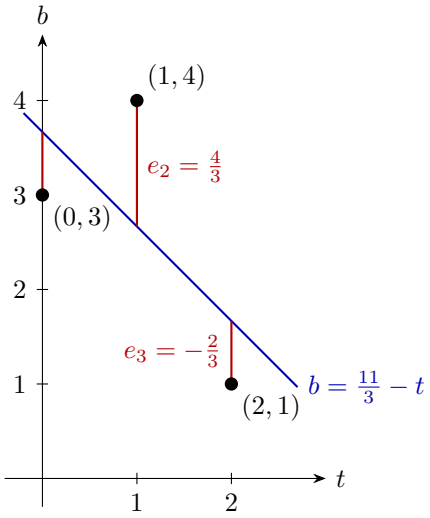


Figure 53: The straight-line fit behind Example 39.5. The vertical errors $e_1 = -\frac{2}{3}$, $e_2 = \frac{4}{3}$, $e_3 = -\frac{2}{3}$ are the components of $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$; they sum to zero and are orthogonal to $(0, 1, 2)$, which is exactly $A^{\mathsf{T}}\boldsymbol{e} = \boldsymbol{0}$.

Common trap. Positive leading determinants are a complete test for a positive definite matrix, but nonnegative leading determinants do not guarantee a semidefinite one: the diagonal matrix $\operatorname{diag}(0, -1)$ has leading determinants 0 and 0 but an eigenvalue -1. In Example 39.6 the argument works because the first two pivots are strictly positive, so the last pivot alone decides.

39.3 Exercises

Exercise 39.7. With A as in Example 39.6, let $\boldsymbol{u}_{k+1} = \frac{1}{2}A\boldsymbol{u}_k$ and start from $(3, 0, 0)$, $(0, 3, 0)$ or $(0, 0, 3)$. Find $\lim_{k\to\infty} \boldsymbol{u}_k$.

Solution. Each column of A sums to 2, so $\frac{1}{2}A$ is a Markov matrix. It has zero entries, so the usual guarantee of a unique steady state (for positive matrices) does not apply directly; check the eigenvalues instead. They are half of $1, 2, -1$: $\frac{1}{2}$, 1, $-\frac{1}{2}$. Only one has absolute value 1, so $\boldsymbol{u}_k$ converges to a multiple of its eigenvector. Each row of A also sums to 2, so $A(1, 1, 1) = (2, 2, 2)$ and the eigenvector is $(1, 1, 1)$. A Markov step preserves the total 3, so in all three cases

$$\lim_{k\to\infty} \boldsymbol{u}_k = (1, 1, 1).$$

□

Exercise 39.8. For the matrix A of Example 39.5, compute the projection matrix $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ and check that $P\boldsymbol{b} = \boldsymbol{p}$ and $P(1, -2, 1) = \boldsymbol{0}$.

Solution. $(A^{\mathsf{T}}A)^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$, and

$$P = \frac{1}{6}\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

Then $P\boldsymbol{b} = \frac{1}{6}(15+8-1,\ 6+8+2,\ -3+8+5) = \frac{1}{6}(22,16,10) = (\frac{11}{3},\frac{8}{3},\frac{5}{3}) = \boldsymbol{p}$, and $P(1,-2,1) = \frac{1}{6}(5-4-1,\ 2-4+2,\ -1-4+5) = \mathbf{0}$. □

Exercise 39.9. Change Example 39.1: $A\boldsymbol{x} = (1,0,0)$ still has no solution, but $A\boldsymbol{x} = (0,1,0)$ now has infinitely many. What can you say about m, n, r? Which of the statements in (c) and (d) change?

Solution. Still $m = 3$ and $1 \le r < 3$. Infinitely many solutions means $\mathbf{N}(A) \neq \{\mathbf{0}\}$, so now $r < n$ (n can be anything larger than r). Then $A^{\mathsf{T}}A$ is singular ($\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A) \neq \{\mathbf{0}\}$), so (ii) becomes false; (iii) stays false since AA^{T} is 3×3 of rank $r < 3$. Statement (d) fails: A^{T} is $n \times 3$ of rank $r < n$, not full row rank, so $A^{\mathsf{T}}\boldsymbol{y} = \boldsymbol{c}$ is solvable only for $\boldsymbol{c}$ in the row space of A. Example: $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$. □

Exercise 39.10. Find the symmetric 2×2 matrix with eigenvalue 0 for $\boldsymbol{x} = (1,2)$ and eigenvalue 3. Why could the matrix in Example 39.4(b) not be symmetric?

Solution. A symmetric matrix has orthogonal eigenvectors, so $\lambda = 3$ goes with $\boldsymbol{q} = (2,-1)/\sqrt{5}$, and $A = 0 \cdot (\dots) + 3\boldsymbol{q}\boldsymbol{q}^{\mathsf{T}} = \frac{3}{5}\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$. Check: $A(1,2) = \frac{3}{5}(4-4,-2+2) = \mathbf{0}$, and $A(2,-1) = \frac{3}{5}(10,-5) = 3(2,-1)$. In Example 39.4(b) the eigenvectors $(1,2)$ and $(2,1)$ have dot product $4 \neq 0$, and eigenvectors of a symmetric matrix for different eigenvalues are always orthogonal. □

40 Final exam

A typical 18.06 final has nine questions in three closed-book hours, and this one runs the whole length of the course: from the special solutions of $A\boldsymbol{x} = \mathbf{0}$ through inverses, ranks, $A^{\mathsf{T}}A$, Gram–Schmidt, projections and least squares, to eigenvalues, determinants and a recursion solved by a 2×2 eigenvalue problem. Most of the questions are posed with letters rather than numbers, so each is followed where useful by an example with numbers. The questions are grouped by theme; each is an exercise with its full solution. The reviews in Sections 37 and 39 cover the tools.

40.1 Subspaces, ranks and one-sided inverses

Exercise 40.1 (Question 1)**.** A is 3×4, and $A\boldsymbol{x} = \mathbf{0}$ has exactly two special solutions, $\boldsymbol{x}_1 = (1,1,1,0)$ and $\boldsymbol{x}_2 = (-2,-1,0,1)$. (a) Find the reduced row echelon form R of A. (b) Find the dimensions of $\mathbf{C}(A)$, $\mathbf{N}(A)$, $\mathbf{C}(A^{\mathsf{T}})$, $\mathbf{N}(A^{\mathsf{T}})$, and bases for those you can.

Solution. (a) A special solution sets one free variable to 1 and the others to 0. Here $\boldsymbol{x}_1$ has $(x_3, x_4) = (1,0)$ and $\boldsymbol{x}_2$ has $(x_3, x_4) = (0,1)$, so x_3, x_4 are free and x_1, x_2 are pivot variables. Two special solutions means $n - r = 2$, so $r = 2$ and the third row of R is zero:

$$R = \begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$R\boldsymbol{x}_1 = \mathbf{0}$ gives $1 + r_{13} = 0$ and $1 + r_{23} = 0$; $R\boldsymbol{x}_2 = \mathbf{0}$ gives $-2 + r_{14} = 0$ and $-1 + r_{24} = 0$. So

$$R = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

whose free columns are the negatives of the pivot entries of the special solutions.

(b) $\dim \mathbf{N}(A) = n - r = 2$, with basis $\boldsymbol{x}_1, \boldsymbol{x}_2$. $\dim \mathbf{C}(A^{\mathsf{T}}) = r = 2$; elimination does not change the row space, so the nonzero rows of R, $(1, 0, -1, 2)$ and $(0, 1, -1, 1)$, are a basis (they are orthogonal to $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, as they must be). $\dim \mathbf{C}(A) = r = 2$, and the first two columns of A (the pivot columns) are a basis, but we cannot write them down without A. $\dim \mathbf{N}(A^{\mathsf{T}}) = m - r = 1$: it is spanned by any nonzero vector orthogonal to the two pivot columns of A, again unknown. □

Exercise 40.2 (Question 3)**.** (a) A and B have the same number of rows. Compare $\operatorname{rank} A$ with $\operatorname{rank} \begin{bmatrix} A & B \end{bmatrix}$. (b) How do the ranks compare if $B = A^2$? (c) If A is $m \times n$ of rank r, what are the dimensions of $\mathbf{N}(A)$ and $\mathbf{N}(\begin{bmatrix} A & A \end{bmatrix})$?

Solution. (a) Only $\operatorname{rank} A \leq \operatorname{rank} \begin{bmatrix} A & B \end{bmatrix}$: the pivot columns of A remain pivot columns of $\begin{bmatrix} A & B \end{bmatrix}$, and B may contribute more. Both cases occur: with $A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, the rank stays 1 for $B = A$ and jumps to 2 for $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

(b) Equal. Column j of A^2 is A times column j of A, a combination of the columns of A. So $\mathbf{C}(\begin{bmatrix} A & A^2 \end{bmatrix}) = \mathbf{C}(A)$ and no new pivot columns appear.

(c) $\dim \mathbf{N}(A) = n - r$. The block matrix $\begin{bmatrix} A & A \end{bmatrix}$ is $m \times 2n$, and its last n columns repeat the first n, so its rank is still r and its nullspace has dimension $2n - r$. (It contains every $(\boldsymbol{y}, -\boldsymbol{y})$, an n-dimensional family, plus $(\boldsymbol{z}, \mathbf{0})$ for $\boldsymbol{z} \in \mathbf{N}(A)$.) □

Exercise 40.3 (Question 4)**.** A is 5×3 and $A\boldsymbol{x} \neq \mathbf{0}$ whenever $\boldsymbol{x} \neq \mathbf{0}$. (a) What can you say about the columns of A? (b) Show that $A^{\mathsf{T}}A\boldsymbol{x} = \mathbf{0}$ only for $\boldsymbol{x} = \mathbf{0}$ by explaining the key step: if $A^{\mathsf{T}}A\boldsymbol{x} = \mathbf{0}$ then $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = 0$, and then (why?) $A\boldsymbol{x} = \mathbf{0}$. (c) So $A^{\mathsf{T}}A$ is invertible. Why is $B = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ a one-sided inverse of A, on which side, and why is it not two-sided?

Solution. (a) $\mathbf{N}(A) = \{\mathbf{0}\}$, so the three columns are independent: full column rank, $r = n = 3$. (b) $\boldsymbol{x}^{\mathsf{T}}A^{\mathsf{T}}A\boldsymbol{x} = (A\boldsymbol{x})^{\mathsf{T}}(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2$, the squared length of $A\boldsymbol{x}$. A vector of length zero is the zero vector, so $A\boldsymbol{x} = \mathbf{0}$, and by (a) $\boldsymbol{x} = \mathbf{0}$. (c) $BA = (A^{\mathsf{T}}A)^{-1}(A^{\mathsf{T}}A) = I_3$: B is a left inverse. It cannot be a right inverse: AB is 5×5 but has rank at most 3, so $AB \neq I_5$. In fact $AB = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ is the projection onto the three-dimensional column space of A (Theorem 36.4). □

40.2 An inverse that stays triangular

Exercise 40.4 (Question 2)**.** (a) Find the inverse of $U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$, all entries nonzero. (b) The columns of U are eigenvectors of a matrix A. Show that A is upper triangular. (c) Explain why this U cannot be the first factor in an SVD $A = U\Sigma V^{\mathsf{T}}$.

Solution. (a) Gauss–Jordan on $\begin{bmatrix} U & I \end{bmatrix}$. Divide the rows by a, d, f:

$$\left[\begin{array}{ccc|ccc} 1 & b/a & c/a & 1/a & 0 & 0 \\ 0 & 1 & e/d & 0 & 1/d & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/f \end{array}\right].$$

Subtract e/d times row 3 from row 2 and c/a times row 3 from row 1, then b/a times the new row 2 from row 1:

$$U^{-1} = \begin{bmatrix} 1/a & -b/(ad) & (be-cd)/(adf) \\ 0 & 1/d & -e/(df) \\ 0 & 0 & 1/f \end{bmatrix}.$$

(The $(1,3)$ entry is $-\frac{c}{af} + \frac{b}{a}\cdot\frac{e}{df} = \frac{be-cd}{adf}$.) Cofactors give the same answer: $U^{-1} = C^{\mathsf{T}}/\det U$ with $\det U = adf$.

(b) Three independent eigenvectors diagonalize A: $A = U\Lambda U^{-1}$. U, Λ and (by part (a)) U^{-1} are all upper triangular, and products of upper triangular matrices are upper triangular.

(c) The U of an SVD has orthonormal columns. Here the first two columns have dot product $ab + 0\cdot d + 0\cdot 0 = ab \neq 0$. □

Example 40.5 (Question 2 with numbers)**.** For $U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$ $(a,b,c,d,e,f = 1,2,3,4,5,6)$ the formula gives

$$U^{-1} = \begin{bmatrix} 1 & -\frac{2}{4} & \frac{2\cdot 5 - 3\cdot 4}{1\cdot 4\cdot 6} \\ 0 & \frac{1}{4} & -\frac{5}{24} \\ 0 & 0 & \frac{1}{6} \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{12} \\ 0 & \frac{1}{4} & -\frac{5}{24} \\ 0 & 0 & \frac{1}{6} \end{bmatrix}.$$

Check the last column of UU^{-1}: row 1 gives $-\frac{1}{12} - \frac{10}{24} + \frac{3}{6} = \frac{-2-10+12}{24} = 0$, row 2 gives $-\frac{20}{24} + \frac{5}{6} = 0$, row 3 gives 1.

40.3 Energy in an eigenvector basis

Exercise 40.6 (Question 5)**.** A is 3×3 symmetric positive definite, $A\boldsymbol{q}_i = \lambda_i \boldsymbol{q}_i$ with orthonormal $\boldsymbol{q}_i$ and $0 < \lambda_1 < \lambda_2 < \lambda_3$. Let $\boldsymbol{x} = c_1\boldsymbol{q}_1 + c_2\boldsymbol{q}_2 + c_3\boldsymbol{q}_3$. (a) Compute $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$ and $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}$ in terms of the c's and λ's. (b) Which c's make the ratio $\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}/\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}$ as large as possible?

Solution. (a) Expanding $(c_1\boldsymbol{q}_1 + c_2\boldsymbol{q}_2 + c_3\boldsymbol{q}_3)^{\mathsf{T}}(c_1\boldsymbol{q}_1 + c_2\boldsymbol{q}_2 + c_3\boldsymbol{q}_3)$ gives nine terms $c_ic_j\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_j$; orthonormality keeps only $i=j$, each with $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{q}_i = 1$:

$$\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x} = c_1^2 + c_2^2 + c_3^2.$$

Since $A\boldsymbol{x} = c_1\lambda_1\boldsymbol{q}_1 + c_2\lambda_2\boldsymbol{q}_2 + c_3\lambda_3\boldsymbol{q}_3$, the same expansion gives

$$\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x} = \lambda_1 c_1^2 + \lambda_2 c_2^2 + \lambda_3 c_3^2.$$

(b) The ratio is a weighted average of the λ's with weights $c_i^2/\sum c_j^2$, so

$$\frac{\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x}}{\boldsymbol{x}^{\mathsf{T}}\boldsymbol{x}} = \frac{\lambda_1 c_1^2 + \lambda_2 c_2^2 + \lambda_3 c_3^2}{c_1^2 + c_2^2 + c_3^2} \leq \frac{\lambda_3(c_1^2 + c_2^2 + c_3^2)}{c_1^2 + c_2^2 + c_3^2} = \lambda_3,$$

with equality exactly when $c_1 = c_2 = 0$. The ratio is largest when $\boldsymbol{x}$ is a multiple of the top eigenvector $\boldsymbol{q}_3$, and its maximum value is λ_3. This ratio is the *Rayleigh quotient*; maximizing it numerically is one way to find the largest eigenvalue. □

Example 40.7 (The Rayleigh quotient of a 2×2 matrix)**.** For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ ($\lambda = 1, 3$; $\boldsymbol{q} = \frac{1}{\sqrt{2}}(1,-1), \frac{1}{\sqrt{2}}(1,1)$) take unit vectors $\boldsymbol{x} = (\cos\theta, \sin\theta)$. Then

$$\boldsymbol{x}^{\mathsf{T}} A \boldsymbol{x} = 2\cos^2\theta + 2\cos\theta\sin\theta + 2\sin^2\theta = 2 + \sin 2\theta,$$

which ranges between 1 and 3. The maximum $3 = \lambda_2$ occurs at $\theta = \pi/4$, the direction of $(1,1)$, and the minimum $1 = \lambda_1$ at $\theta = 3\pi/4$, the direction of $(-1,1)$ (Figure 54).

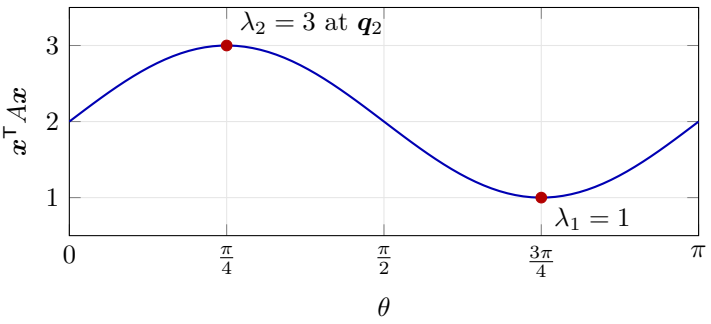


Figure 54: The Rayleigh quotient $\boldsymbol{x}^{\mathsf{T}} A \boldsymbol{x}$ on unit vectors $\boldsymbol{x} = (\cos\theta, \sin\theta)$ for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Its maximum and minimum are the eigenvalues, reached at the eigenvectors.

40.4 Gram–Schmidt, projections and least squares

Exercise 40.8 (Question 6)**.** (a) Find a combination $\boldsymbol{w}$ of the independent vectors $\boldsymbol{v}$ and $\boldsymbol{u}$ that is perpendicular to $\boldsymbol{u}$. (b) For $A = \begin{bmatrix} \boldsymbol{u} & \boldsymbol{v} \end{bmatrix}$ find Q (orthonormal columns) and R (2×2 upper triangular) with $A = QR$. (c) Using only Q, find the projection matrix onto the plane spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$.

Solution. (a) $\boldsymbol{w} = \boldsymbol{0}$ is perpendicular to everything, but the useful answer subtracts from $\boldsymbol{v}$ just enough of $\boldsymbol{u}$: $\boldsymbol{w} = \boldsymbol{v} - c\boldsymbol{u}$ with $0 = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{w} = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{v} - c\,\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u}$, so

$$\boldsymbol{w} = \boldsymbol{v} - \frac{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{v}}{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u}}\,\boldsymbol{u}.$$

(b) Normalize: $\boldsymbol{q}_1 = \boldsymbol{u}/\|\boldsymbol{u}\|$, $\boldsymbol{q}_2 = \boldsymbol{w}/\|\boldsymbol{w}\|$. Then $\boldsymbol{u} = \|\boldsymbol{u}\|\boldsymbol{q}_1$ and $\boldsymbol{v} = \boldsymbol{w} + c\boldsymbol{u} = c\|\boldsymbol{u}\|\boldsymbol{q}_1 + \|\boldsymbol{w}\|\boldsymbol{q}_2$, so

$$A = \begin{bmatrix} \boldsymbol{u} & \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 \end{bmatrix} \begin{bmatrix} \|\boldsymbol{u}\| & c\|\boldsymbol{u}\| \\ 0 & \|\boldsymbol{w}\| \end{bmatrix} = QR, \qquad c = \frac{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{v}}{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u}}.$$

(c) $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ with $A = QR$ and $Q^{\mathsf{T}}Q = I$:

$$P = QR\,(R^{\mathsf{T}}Q^{\mathsf{T}}QR)^{-1}R^{\mathsf{T}}Q^{\mathsf{T}} = QR\,R^{-1}(R^{\mathsf{T}})^{-1}R^{\mathsf{T}}Q^{\mathsf{T}} = QQ^{\mathsf{T}}.$$

□

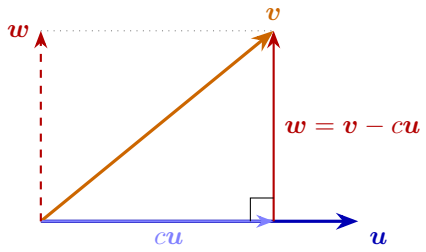


Figure 55: Gram–Schmidt in Question 6: subtract from $\boldsymbol{v}$ its projection $c\boldsymbol{u}$ onto $\boldsymbol{u}$; what remains, $\boldsymbol{w}$, is perpendicular to $\boldsymbol{u}$. Normalizing $\boldsymbol{u}$ and $\boldsymbol{w}$ gives Q.

Example 40.9 (Question 6 with numbers)**.** Take $\boldsymbol{u} = (1,2,2)$ and $\boldsymbol{v} = (5,1,1)$. Then $c = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{v}/\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u} = 9/9 = 1$ and $\boldsymbol{w} = \boldsymbol{v} - \boldsymbol{u} = (4,-1,-1)$, which is perpendicular to $\boldsymbol{u}$ $(4-2-2=0)$. With $\|\boldsymbol{u}\| = 3$ and $\|\boldsymbol{w}\| = \sqrt{18} = 3\sqrt{2}$:

$$Q = \begin{bmatrix} 1/3 & 4/(3\sqrt{2}) \\ 2/3 & -1/(3\sqrt{2}) \\ 2/3 & -1/(3\sqrt{2}) \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 3 \\ 0 & 3\sqrt{2} \end{bmatrix}.$$

Check the second column: $3\boldsymbol{q}_1 + 3\sqrt{2}\,\boldsymbol{q}_2 = (1,2,2) + (4,-1,-1) = (5,1,1) = \boldsymbol{v}$. The projection onto the plane is

$$P = QQ^{\mathsf{T}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix},$$

which makes sense: both $\boldsymbol{u}$ and $\boldsymbol{v}$ have equal second and third entries, so the plane is $\{x_2 = x_3\}$, and P keeps x_1 and averages x_2 with x_3.

Exercise 40.10 (Question 8)**.** A has independent columns. (a) Give formulas for the least-squares solution $\hat{\boldsymbol{x}}$ of $A\boldsymbol{x} = \boldsymbol{b}$ and for $\boldsymbol{p} = A\hat{\boldsymbol{x}}$. (b) In which fundamental subspaces are $\boldsymbol{p}$ and $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$? (c) Find the projection matrix onto the column space of $A = \begin{bmatrix} 1 & 0 \\ 3 & 0 \\ 0 & -1 \\ 0 & -3 \end{bmatrix}$.

Solution. (a) Multiply $A\boldsymbol{x} = \boldsymbol{b}$ by A^{T} to get the normal equations $A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$; $A^{\mathsf{T}}A$ is invertible because the columns are independent, so

$$\hat{\boldsymbol{x}} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}, \qquad \boldsymbol{p} = A\hat{\boldsymbol{x}} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\boldsymbol{b}.$$

(b) $\boldsymbol{p}$ is a combination of the columns, so $\boldsymbol{p} \in \mathbf{C}(A)$. The error is perpendicular to the column space, so $\boldsymbol{e} \in \mathbf{N}(A^{\mathsf{T}})$; the normal equations say exactly $A^{\mathsf{T}}\boldsymbol{e} = \boldsymbol{0}$. (c) The columns are orthogonal: $A^{\mathsf{T}}A = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = 10I$. Hence

$$P = A(10I)^{-1}A^{\mathsf{T}} = \frac{1}{10}AA^{\mathsf{T}} = \frac{1}{10}\begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 9 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 3 & 9 \end{bmatrix},$$

the sum of the projections onto the two orthogonal columns. □

40.5 Permutations, determinants and a recursion

Exercise 40.11 (Question 7)**.** Let $C = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, so $C^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$. (a) Find the eigenvalues of C and C^2. (b) Find C^{-1} and $(C^2)^{-1}$. (c) Find $\det C$, $\det(C+I)$ and $\det(C+2I)$.

Solution. (a) Expanding $\det(C - \lambda I)$ by cofactors gives $\lambda^4 - 1 = (\lambda - 1)(\lambda + 1)(\lambda^2 + 1)$, so $\lambda = 1, -1, i, -i$, the four fourth roots of 1. (A quicker route: C moves each entry down one place cyclically, so $C^4 = I$ and every eigenvalue satisfies $\lambda^4 = 1$; check that all four occur.) The eigenvalues of C^2 are the squares: $1, 1, -1, -1$. The eigenvectors are the Fourier vectors $(1, \bar{\lambda}, \bar{\lambda}^2, \bar{\lambda}^3)$ of Section 30.

(b) Permutation matrices are orthogonal, so $C^{-1} = C^{\mathsf{T}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$. C^2 exchanges entries $1 \leftrightarrow 3$ and $2 \leftrightarrow 4$, so it is its own inverse: $(C^2)^{-1} = C^2$.

(c) $\det C = 1 \cdot (-1) \cdot i \cdot (-i) = -1$. Adding I adds 1 to each eigenvalue, so $\det(C + I) = 2 \cdot 0 \cdot (1 + i)(1 - i) = 0$; adding $2I$ gives $\det(C + 2I) = 3 \cdot 1 \cdot (2 + i)(2 - i) = 3 \cdot 5 = 15$. Equivalently, set $\lambda = -1$ and $\lambda = -2$ in $\det(C - \lambda I) = \lambda^4 - 1$: $1 - 1 = 0$ and $16 - 1 = 15$. □

Exercise 40.12 (Question 9)**.** A_n is the $n \times n$ tridiagonal matrix with 3 on the diagonal, 2 above it and 1 below it; $D_n = \det A_n$. (a) Find D_2 and D_3. (b) Use cofactors of row 1 and column 1 to find a, b in $D_n = aD_{n-1} + bD_{n-2}$. (c) Writing the recursion as $\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} = \begin{bmatrix} a & b \\ 1 & 0 \end{bmatrix} \begin{bmatrix} D_{n-1} \\ D_{n-2} \end{bmatrix}$, how fast does D_n grow? Find D_5.

Solution. (a) $D_1 = 3$, $D_2 = \det \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} = 9 - 2 = 7$, and $D_3 = 3D_2 - 2 \cdot 1 \cdot D_1 = 21 - 6 = 15$ (by the expansion in (b)).

(b) Expand along row 1. The $(1,1)$ cofactor is D_{n-1}. The $(1,2)$ entry is 2 with sign $-$, and its minor (delete row 1, column 2) has first column $(1, 0, \ldots, 0)$; expanding that minor down its first column leaves $1 \cdot D_{n-2}$. So

$$D_n = 3D_{n-1} - 2D_{n-2}, \qquad a = 3,\ b = -2.$$

(c) The matrix $\begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$ has trace 3 and determinant 2, so $\lambda^2 - 3\lambda + 2 = 0$ and $\lambda = 1, 2$. Powers of this matrix grow like 2^n, so D_n grows like 2^n. Matching $D_n = \alpha 2^n + \beta$ to $D_1 = 3$, $D_2 = 7$ gives $\alpha = 2$, $\beta = -1$:

$$D_n = 2^{n+1} - 1, \qquad D_4 = 31, \qquad D_5 = 3 \cdot 31 - 2 \cdot 15 = 63 = 2^6 - 1.$$

□

Common trap. Carry the sign of b through. Writing the recursion as $D_n = 3D_{n-1} + 2D_{n-2}$ gives $D_5 = 207$, which is wrong: a direct computation of $\det A_5$ gives 63, and the sign comes from the checkerboard $(-1)^{1+2}$ in the cofactor of the $(1,2)$ entry. Checking a small case ($D_3 = 15$ from the formula against $\det A_3$ by elimination) catches this kind of slip.

41 Final 18.06 lecture (2023)

On 15 May 2023 Gilbert Strang gave his last 18.06 lecture, at 88, after 61 years on the MIT faculty (more, the department head noted, than anyone in the department's history). Colleagues and former students spoke about what his teaching had done: Alan Edelman about textbooks in which simple examples lead to deep understanding, Pavel Grinfeld about a 3 by 3 matrix being enough to give scientific insight, Michel Goemans about lectures seen by millions on OpenCourseWare. In between, Strang reviewed the course for the final exam with two examples on the board. They are worth working through slowly, because between them they contain the arc of these notes: elimination (Section 3), the matrix form $A\boldsymbol{x} = \boldsymbol{b}$ and $A = LU$ (Section 5), special solutions and rank (Section 8), and the theorem he called the first wonderful fact of the subject, that row rank equals column rank (Section 11).

41.1 Three equations, three unknowns: elimination

The idea that opens the course is very old (Strang traced it to China some thousands of years ago): subtract multiples of equations from other equations to produce zeros, until the system is triangular, then solve from the bottom up.

Example 41.1 (Strang's last elimination)**.** Solve

$$\begin{aligned} 2x + 3y + 4z &= 19 \\ 4x + 11y + 14z &= 55 \\ 2x + 8y + 17z &= 50 \end{aligned} \qquad \text{i.e.} \qquad \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 19 \\ 55 \\ 50 \end{bmatrix}.$$

Column 1. The first pivot is 2. Subtract 2× equation 1 from equation 2 ($\ell_{21} = 2$): $5y + 6z = 17$. Subtract 1× equation 1 from equation 3 ($\ell_{31} = 1$): $5y + 13z = 31$.

Column 2. The second pivot is 5. Subtract 1× the new equation 2 from the new equation 3 ($\ell_{32} = 1$): $7z = 14$.

Back substitution. $z = 2$; then $5y + 12 = 17$ gives $y = 1$; then $2x + 3 + 8 = 19$ gives $x = 4$. So $(x, y, z) = (4, 1, 2)$, and indeed $4 \cdot 4 + 11 + 28 = 55$ and $8 + 8 + 34 = 50$.

In matrix language, elimination turned A into the upper triangular U and $\boldsymbol{b}$ into $\boldsymbol{c}$, and the multipliers fill L:

$$U = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix}, \quad \boldsymbol{c} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad LU = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} = A.$$

Three equations and three unknowns with three nonzero pivots: one solution, as expected. The second board example drops that comfort.

41.2 Three equations, four unknowns: rank and special solutions

Example 41.2 (A 3×4 matrix of rank 2)**.** Solve $A\boldsymbol{x} = \boldsymbol{0}$ for

$$A = \begin{bmatrix} 1 & 2 & 11 & 17 \\ 3 & 7 & 37 & 57 \\ 4 & 9 & 48 & 74 \end{bmatrix}.$$

$\boldsymbol{x} = \mathbf{0}$ is always a solution; the question is what else. With four unknowns and only three equations we expect at least one free variable, but the numbers decide how many.

Spot the dependence. Row 1 + row 2 $= (4, 9, 48, 74) =$ row 3. Subtracting rows 1 and 2 from row 3 leaves a row of zeros: the third equation was a combination of the other two and says nothing new. So there are really two equations in four unknowns, and we expect $4 - 2 = 2$ independent solutions.

Eliminate, downward and then upward. Row 2 $-3\times$row 1 $= (0, 1, 4, 6)$. Then row 1 $-2\times$(new row 2) $= (1, 0, 3, 5)$:

$$R = \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \begin{aligned} x_1 + 3x_3 + 5x_4 &= 0 \\ x_2 + 4x_3 + 6x_4 &= 0. \end{aligned}$$

Special solutions. The pivot variables are x_1, x_2 and the free variables x_3, x_4. Setting $(x_3, x_4) = (1, 0)$ gives $x_1 = -3$, $x_2 = -4$; setting $(x_3, x_4) = (0, 1)$ gives $x_1 = -5$, $x_2 = -6$:

$$\boldsymbol{s}_1 = (-3, -4, 1, 0), \qquad \boldsymbol{s}_2 = (-5, -6, 0, 1).$$

They are independent (look at the last two entries), and every solution is a combination $c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2$, a plane through the origin in $\mathbb{R}^4$. Check one entry: row 2 of A times $\boldsymbol{s}_1$ is $-9 - 28 + 37 + 0 = 0$.

Common trap. No row of A is a multiple of another, yet the rows are dependent. Dependence means some *combination* gives zero (here row 1 + row 2 − row 3), not that one row is a multiple of one other row. The rank is the number of pivots after elimination, not the number of rows that "look different".

41.3 Row rank equals column rank

The matrix A has two independent rows. Look at its columns instead: from R, column 3 $=$ 3 (column 1) $+$4 (column 2) and column 4 $=$ 5 (column 1) $+$6 (column 2); for A itself, $3(1, 3, 4) + 4(2, 7, 9) = (11, 37, 48)$ and $5(1, 3, 4) + 6(2, 7, 9) = (17, 57, 74)$. Columns 1 and 2 are independent, so there are also exactly two independent columns. That the two counts agree is not obvious (the rows live in $\mathbb{R}^4$, the columns in $\mathbb{R}^3$), and it is true for every matrix.

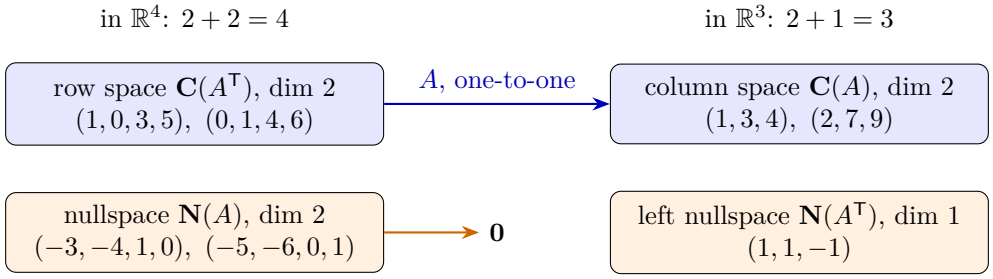


Figure 56: The four subspaces of the board matrix A of Example 41.2, with a basis for each. Row rank and column rank are both 2; the left nullspace vector $(1, 1, -1)$ records the dependence row 1 + row 2 − row 3 $= \mathbf{0}$.

Theorem 41.3 (Row rank equals column rank)**.** *For every $m \times n$ matrix A, the maximum number of independent rows equals the maximum number of independent columns. Both equal the number r of pivots.*

Proof. Let R be the reduced row echelon form of A, with r pivots.

Rows. Each row operation replaces a row by a combination of rows and can be undone, so A and R have the same row space. The r nonzero rows of R are independent: each has a 1 in its pivot column, where every other row of R has a 0, so a combination of them can vanish only if every coefficient is 0. Hence $\dim \mathbf{C}(A^{\mathsf{T}}) = r$.

Columns. Row operations do not change the solutions of $A\boldsymbol{x} = \mathbf{0}$, so $A\boldsymbol{x} = \mathbf{0}$ exactly when $R\boldsymbol{x} = \mathbf{0}$: a combination of columns of A vanishes exactly when the same combination of columns of R does. So any set of columns is independent in A if and only if it is independent in R. In R the r pivot columns are distinct standard basis vectors of $\mathbb{R}^m$, hence independent, and every free column is a combination of them (its entries sit in the pivot rows). So the pivot columns of A form a basis of $\mathbf{C}(A)$, and $\dim \mathbf{C}(A) = r$. □

Figure 56 puts the board example into the picture that organizes the whole course. The two equal dimensions on top are Theorem 41.3; the two at the bottom, $n - r$ and $m - r$, are what is left over; and the matrix is invertible from the top left box to the top right box (Theorem 36.7). Strang closed the mathematics there, on the fact that the number of independent rows always matches the number of independent columns.

41.4 Exercises

Exercise 41.4. Use the factors L and U of Example 41.1 to solve $A\boldsymbol{x} = (9, 29, 27)$ without repeating elimination.

Solution. Forward substitution in $L\boldsymbol{c} = \boldsymbol{b}$: $c_1 = 9$, $c_2 = 29 - 2 \cdot 9 = 11$, $c_3 = 27 - 9 - 11 = 7$. Back substitution in $U\boldsymbol{x} = \boldsymbol{c}$: $7z = 7$ gives $z = 1$; $5y + 6 = 11$ gives $y = 1$; $2x + 3 + 4 = 9$ gives $x = 1$. So $\boldsymbol{x} = (1, 1, 1)$; check: the row sums of A are 9, 29, 27. □

Exercise 41.5. For the 3×4 matrix A of Example 41.2, for which $\boldsymbol{b} = (b_1, b_2, b_3)$ is $A\boldsymbol{x} = \boldsymbol{b}$ solvable? Find the complete solution when $\boldsymbol{b} = (1, 4, 5)$.

Solution. The same row operations act on $\boldsymbol{b}$: row 3 − row 1 − row 2 turns the third equation into $0 = b_3 - b_1 - b_2$. So $A\boldsymbol{x} = \boldsymbol{b}$ is solvable exactly when $b_3 = b_1 + b_2$, i.e. when $\boldsymbol{b}$ is orthogonal to the left nullspace vector $(1, 1, -1)$. For $\boldsymbol{b} = (1, 4, 5)$ this holds. Row $2 - 3\times$row 1 gives $x_2 + 4x_3 + 6x_4 = 4 - 3 = 1$, and row $1 - 2\times$(new row 2) gives $x_1 + 3x_3 + 5x_4 = 1 - 2 = -1$. With the free variables set to zero, $\boldsymbol{x}_p = (-1, 1, 0, 0)$; check $A\boldsymbol{x}_p = -(1, 3, 4) + (2, 7, 9) = (1, 4, 5)$. The complete solution is

$$\boldsymbol{x} = (-1, 1, 0, 0) + c_1(-3, -4, 1, 0) + c_2(-5, -6, 0, 1).$$

□

Exercise 41.6. Without further elimination, find the rank of A^{T} and the dimension of the nullspace of A^{T} for the 3×4 matrix A, and explain the answers using Theorem 41.3.

Solution. The rows of A^{T} are the columns of A, so the row rank of A^{T} is the column rank of A, which is 2; by the theorem the rank of A^{T} is 2. A^{T} is 4×3, so its nullspace has dimension $3 - 2 = 1$, spanned by $(1, 1, -1)$: $A^{\mathsf{T}}(1, 1, -1) =$ row 1 + row 2 − row 3 of A, written as a column, which is $\mathbf{0}$. □

A Formula sheet

The results below are the ones the rest of the course leans on. Each line points to the section where it is derived; none of them should be used without knowing why it is true.

Part I: $A\boldsymbol{x} = \boldsymbol{b}$ and the four subspaces

Result	Meaning	Where
$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$	$A\boldsymbol{x}$ is a combination of the columns	§1
$A\boldsymbol{x} = \boldsymbol{b}$ solvable $\iff \boldsymbol{b} \in \mathbf{C}(A)$	the column picture decides solvability	§7
$(AB)^{-1} = B^{-1}A^{-1}$, $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$	inverses and transposes reverse order	§4, §6
$A = LU$, $PA = LU$	elimination recorded as a factorization	§5, §6
$\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$	complete solution: particular plus nullspace	§9
$\dim \mathbf{C}(A) = \dim \mathbf{C}(A^{\mathsf{T}}) = r$	row rank equals column rank	§11
$\dim \mathbf{N}(A) = n - r$, $\dim \mathbf{N}(A^{\mathsf{T}}) = m - r$	free variables count the nullspace	§8, §11
$r = m$: solvable for every $\boldsymbol{b}$; $r = n$: at most one solution	full row rank / full column rank	§9
$\dim(S + U) = \dim S + \dim U - \dim(S \cap U)$	dimension of a sum of subspaces	§12
$A^{\mathsf{T}}CA\boldsymbol{x} = \boldsymbol{f}$	balance law on a network	§13

Part II: orthogonality, determinants and eigenvalues

Result	Meaning	Where
$\mathbf{N}(A) \perp \mathbf{C}(A^{\mathsf{T}})$, $\mathbf{N}(A^{\mathsf{T}}) \perp \mathbf{C}(A)$	the four subspaces meet at right angles	§16
$\mathbf{N}(A^{\mathsf{T}}A) = \mathbf{N}(A)$	$A^{\mathsf{T}}A$ is invertible iff A has independent columns	§16
$\boldsymbol{p} = \dfrac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}\,\boldsymbol{a}$, $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathsf{T}}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}$	projection onto a line	§17
$A^{\mathsf{T}}A\hat{\boldsymbol{x}} = A^{\mathsf{T}}\boldsymbol{b}$, $P = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$	projection onto $\mathbf{C}(A)$; least squares	§17, §18
$P^2 = P = P^{\mathsf{T}}$	what makes a matrix a projection	§18
$Q^{\mathsf{T}}Q = I$, $A = QR$, $\hat{\boldsymbol{x}} = R^{-1}Q^{\mathsf{T}}\boldsymbol{b}$	orthonormal columns and Gram–Schmidt	§19
$\det AB = \det A \det B$, $\det A^{\mathsf{T}} = \det A$	multiplicative and transpose-invariant	§20
$\det A = \sum_j a_{1j}C_{1j}$, $A^{-1} = \dfrac{C^{\mathsf{T}}}{\det A}$	cofactor expansion and the inverse	§21, §22
$x_j = \det B_j / \det A$	Cramer's rule	§22
$\det(A - \lambda I) = 0$; $\sum \lambda_i = \operatorname{tr} A$, $\prod \lambda_i = \det A$	eigenvalues	§23
$A = X\Lambda X^{-1}$, $A^k = X\Lambda^k X^{-1}$	diagonalization and powers	§24

$\boldsymbol{u}(t) = e^{At}\boldsymbol{u}(0) = Xe^{\Lambda t}X^{-1}\boldsymbol{u}(0)$	solving $\boldsymbol{u}' = A\boldsymbol{u}$; stable iff all $\operatorname{Re}\lambda < 0$	§25
Markov: $\lambda_1 = 1,\ \ \lvert\lambda_i\rvert \le 1$	steady state is the $\lambda = 1$ eigenvector	§26

Part III: symmetric matrices, the SVD and transformations

Result	Meaning	Where
$S = Q\Lambda Q^{\mathsf{T}}$ for $S = S^{\mathsf{T}}$	real eigenvalues, orthonormal eigenvectors	§29
$\bar{\boldsymbol{x}}^{\mathsf{T}}\boldsymbol{y},\ \ A^{\mathsf{H}} = \bar{A}^{\mathsf{T}},\ \ F_n^{-1} = \bar{F}_n/n$	complex inner product; Fourier matrix	§30
$\boldsymbol{x}^{\mathsf{T}}S\boldsymbol{x} > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$	positive definite: pivots, eigenvalues, upper-left determinants all positive	§29, §31
$B = M^{-1}AM$ has the eigenvalues of A	similar matrices; Jordan form	§32
$A = U\Sigma V^{\mathsf{T}},\ \ A\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$	SVD; σ_i^2 are the eigenvalues of $A^{\mathsf{T}}A$	§33
$[T]_{\text{new}} = M^{-1}[T]_{\text{old}}M$	a linear map in a new basis	§34, §35
$A^+ = V\Sigma^+U^{\mathsf{T}}$	pseudoinverse; $A^+\boldsymbol{b}$ is the shortest least-squares solution	§36